

Fundamental Theorem of Matrix Product States

A Formalization Blueprint

Authors

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Chapter 1

Introduction

We study the fundamental theorem for periodic, translation-invariant matrix product vectors. The basic object is a family of matrices $\{A^i\}$, viewed as a tensor that generates vectors $|V^{(N)}(A)\rangle$ on chains of length N .

The standard MPS form of the theorem is the following. Put two tensors in canonical form,

$$A^i = \bigoplus_{j=1}^g \mu_j A_j^i, \quad B^i = \bigoplus_{k=1}^h \nu_k B_k^i,$$

with normal blocks forming bases of normal tensors [PGVWC07, CPGSV21]. If $|V^{(N)}(A)\rangle = |V^{(N)}(B)\rangle$ for every chain length N , then $g = h$ and, after permuting the blocks of B , for each block j there are an invertible matrix X_j and a nonzero scalar ζ_j such that

$$B_j^i = \zeta_j X_j A_j^i X_j^{-1}, \quad \mu_j = \nu_j \zeta_j.$$

Equivalently, after the block permutation and phase absorption, the two weighted block-diagonal tensors are gauge equivalent. In the single injective-block case this reduces to the familiar conclusion $B^i = X A^i X^{-1}$ for all physical indices i .

Chapters 2 and 4 describe the algebraic core. Chapter 2 introduces word evaluation, MPV coefficients, gauge equivalence, injectivity, blocking, overlap, and the block-injective canonical-form data. Chapter 4 proves the single-block fundamental theorem by a purely algebraic argument through trace pairings, linear extension, simplicity of matrix algebras, and Skolem–Noether.

The later chapters develop the spectral and canonical-form machinery needed for the full multi-block theorem. Chapter 5 studies positive maps and quantum channels. Chapters 8 and 9 supply Perron–Frobenius theory and overlap decay. Chapters 10 and 11 connect blocking, primitivity, irreducibility, and the normal/canonical-form reductions. Chapters 12 and 13 carry out block permutation and BNT theory; Chapter 14 proves the blocking-based Fundamental Theorem, while Chapter 15 develops the direct periodic extension.

Chapter 2

Matrix Product Vectors

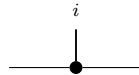
Fix a physical dimension d and a bond dimension D . This chapter studies the tensors that generate matrix product vector (MPV) families. All tensors here are translation-invariant with periodic boundary conditions (PBC).

2.1 Basic definitions

Definition 2.1 (MPS tensor). MPSTensor ✓ Stmt OK A (translation-invariant, PBC) *MPS tensor* with physical dimension d and bond dimension D is a collection of matrices

$$\{A^i\}_{i=0}^{d-1}, \quad A^i \in M_D(\mathbb{C}),$$

indexed by a physical index $i \in \{0, \dots, d-1\}$. Such a tensor defines an MPV family. Diagrammatically,

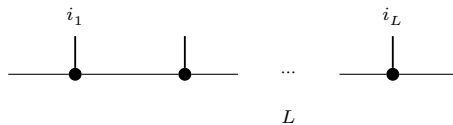


The black node denotes the tensor A , the horizontal legs are virtual, and the upper leg is the physical index i .

Definition 2.2 (Word evaluation). evalWord ✓ Stmt OK Given a word $w = (i_1, \dots, i_L) \in \{0, \dots, d-1\}^L$, the *word evaluation* is the matrix product

$$A^w := A^{i_1} A^{i_2} \dots A^{i_L} \in M_D(\mathbb{C}),$$

with the convention that the empty word evaluates to the identity: $A^\emptyset = \mathbb{1}_D$. In tensor-network notation,



in which each black node denotes the same local tensor A , the virtual legs remain open, and the physical legs record the word $(i_1, \dots, i_L)$.

Lemma 2.3 (Multiplicativity of word evaluation). evalWord_append ✓ Stmt OK For any words w_1, w_2 , $A^{w_1 \cdot w_2} = A^{w_1} A^{w_2}$.

Proof. ✓ Proof OK Immediate from the definition of A^w . □

Definition 2.4 (Matrix product vector). mpv ✓ Stmt OK The *matrix product vector* (MPV) of a tensor A at system size N is the vector

$$|V^{(N)}(A)\rangle = \sum_{i_1, \dots, i_N=0}^{d-1} \text{tr}(A^{i_1} A^{i_2} \dots A^{i_N}) |i_1, \dots, i_N\rangle \in (\mathbb{C}^d)^{\otimes N}.$$

Equivalently, for a configuration $\sigma = (i_1, \dots, i_N) \in \{0, \dots, d-1\}^N$, we write

$$V^{(N)}(A)_\sigma := \text{tr}(A^{i_1} A^{i_2} \dots A^{i_N}).$$

The coefficient function $\sigma \mapsto V^{(N)}(A)_\sigma$ gives the components; the displayed ket is the corresponding vector in $(\mathbb{C}^d)^{\otimes N}$. For a general word w , we also write

$$c_w(A) := \text{tr}(A^w),$$

The *MPV family* generated by A is the collection $\mathcal{V}(A) = \{|V^{(N)}(A)\rangle\}_{N \geq 1}$. The coefficient $V^{(N)}(A)_\sigma$ is the periodic contraction



of N copies of the local tensor A , with the outer virtual legs closed by the trace.

Definition 2.5 (Transfer map). transferMap ✓ Stmt OK The *transfer map* associated to a tensor A is the linear map $\mathcal{E}_A : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ defined by

$$\mathcal{E}_A(X) = \sum_{i=0}^{d-1} A^i X (A^i)^\dagger.$$

Diagrammatically, the transfer map is the double-layer contraction



in which the upper node denotes A , the lower node denotes $A^\dagger$, and the physical index is summed over between them.

Remark 2.6. ✓ Stmt OK The transfer map is automatically positive: if $X \geq 0$ then $\mathcal{E}_A(X) \geq 0$, because each summand $A^i X (A^i)^\dagger$ is positive semidefinite. The abstract positive-map framework appears in Chapter 5.

2.2 Periodic-chain aliases

Definition 2.7 (Periodic-chain tensor alias). PeriodicMPSTensor ✓ Stmt OK For a fixed period m , a *periodic-chain tensor* is a site-dependent tensor family indexed by $\{0, \dots, m-1\}$, with local physical dimension d and bond dimension D .

Definition 2.8 (Periodic same-state and gauge-equivalence relations). SameState GaugeEquiv ✓ Stmt OK For periodic-chain tensors A, B at fixed period m :

- equality of the induced periodic-chain states;
- cyclic gauge equivalence.

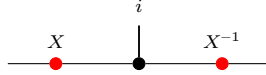
Lemma 2.9 (Equivalence structures for periodic relations). instEquivalenceSameState instEquivalenceGaugeEquiv ✓ Stmt OK *periodic relations are equivalence relations (reflexive, symmetric, transitive).*

2.3 Gauge equivalence and same MPV

Definition 2.10 (Gauge equivalence). GaugeEquiv ✓ Stmt OK Two tensors A and B of the same bond dimension D are *gauge equivalent* if there exists an invertible matrix $X \in \text{GL}_D(\mathbb{C})$ such that, for every $i \in \{0, \dots, d-1\}$,

$$B^i = X A^i X^{-1}$$

In tensor-network notation,



the black node denotes the tensor named in the surrounding formula, the upper leg is the physical index i , and the two red side nodes denote the gauge matrices acting on the virtual legs.

Definition 2.11 (Same MPV). SameMPV ✓ Stmt OK Two tensors A and B (of the same bond dimension) *generate the same MPV family* if $|V^{(N)}(A)\rangle = |V^{(N)}(B)\rangle$ for every system size N .

Definition 2.12 (Same MPV — different bond dimensions). SameMPV2 ✓ Stmt OK For tensors A and B of possibly different bond dimensions D_1 and D_2 , we write $\mathcal{V}(A) = \mathcal{V}(B)$ if $|V^{(N)}(A)\rangle = |V^{(N)}(B)\rangle$ for every N .

Definition 2.13 (Positive-length same MPV — different bond dimensions). SameMPV2Pos ✓ Stmt OK For tensors A and B of possibly different bond dimensions, we write $\mathcal{V}^+(A) = \mathcal{V}^+(B)$ if $|V^{(N)}(A)\rangle = |V^{(N)}(B)\rangle$ for every $N > 0$.

Remark 2.14. ✓ Stmt OK Definition 2.11 is the same-bond-dimension specialization of Definition 2.12. We keep both because later results use both the same-dimension and different-dimension forms.

Definition 2.15 (Proportional MPV). ProportionalMPV2 ✓ Stmt OK Two tensors A and B (of possibly different bond dimensions) *generate proportional MPV families* if for every N there exists $c_N \in \mathbb{C}$ such that $|V^{(N)}(A)\rangle = c_N |V^{(N)}(B)\rangle$.

Lemma 2.16 (Scaling of word evaluation). evalWord_smul ✓ Stmt OK Let ζA denote the tensor with matrices $\{\zeta A^i\}_{i=0}^{d-1}$. Then for any word w ,

$$(\zeta A)^w = \zeta^{|w|} A^w,$$

where $|w|$ is the length of w .

Proof. ✓ Proof OK Induction on w : the base case gives $\zeta^0 = 1$, and each step picks up one factor of ζ . □

Lemma 2.17 (Scaling of MPV components). mpv_smul ✓ Stmt OK Let ζA denote the tensor with matrices $\{\zeta A^i\}_{i=0}^{d-1}$. Then for every N and every configuration $\sigma = (i_1, \dots, i_N)$,

$$V^{(N)}(\zeta A)_\sigma = \zeta^N V^{(N)}(A)_\sigma.$$

Proof. ✓ Proof OK Apply Lemma 2.16 to the word $w = (i_1, \dots, i_N)$ of length N , then take the trace; the scalar ζ^N pulls out by linearity. □

Definition 2.18 (Gauge-phase equivalence). GaugePhaseEquiv ✓ Stmt OK Two tensors A and B are *gauge-phase equivalent* if there exist $X \in \text{GL}_D(\mathbb{C})$ and $\zeta \in \mathbb{C} \setminus \{0\}$ such that, for every physical index i ,

$$B^i = \zeta X A^i X^{-1}$$

Remark 2.19. ✓ Stmt OK After spectral-radius normalization one may restrict ζ to a unit-modulus phase $e^{i\phi}$. We allow an arbitrary nonzero scalar because later canonical-form statements keep the block-scaling factors explicit.

Lemma 2.20 (Gauge covariance of word evaluation). evalWord_gauge ✓ Stmt OK If $B^i = X A^i X^{-1}$ for all i , then for every word w ,

$$B^w = X A^w X^{-1}.$$

Proof. ✓ Proof OK Induction on the word length, using $XX^{-1} = \mathbb{1}$. □

Theorem 2.21 (Gauge invariance of MPVs). sameMPV ✓ Stmt OK If A and B are gauge equivalent, then they generate the same MPV family.

Proof. ✓ Proof OK Let $B^i = X A^i X^{-1}$. By Lemma 2.20, $B^w = X A^w X^{-1}$ for every word w . Then $\text{tr}(B^w) = \text{tr}(X A^w X^{-1}) = \text{tr}(A^w)$ by cyclicity of the trace. □

Theorem 2.22 (Gauge-phase scaling of MPVs). mpv_eq_pow_mul_of_gaugePhase ✓ Stmt OK If, for every physical index i ,

$$B^i = \zeta X A^i X^{-1}$$

then for every system size N and configuration σ one has

$$V^{(N)}(B)_\sigma = \zeta^N V^{(N)}(A)_\sigma.$$

Proof. ✓ Proof OK Combine gauge covariance of word evaluation with scalar scaling, then take traces. □

2.4 Injectivity and normality

Definition 2.23 (Injectivity). IsInjective ✓ Stmt OK A tensor A is *injective* if its matrices span the full matrix algebra: $\text{span}_{\mathbb{C}}\{A^i : i = 0, \dots, d-1\} = M_D(\mathbb{C})$.

Definition 2.24 (L -block injectivity). IsNBlkInjective ✓ Stmt OK A tensor A is *L -block injective* if $\text{span}_{\mathbb{C}}\{A^{i_1} \dots A^{i_L} : (i_1, \dots, i_L) \in \{0, \dots, d-1\}^L\} = M_D(\mathbb{C})$. An injective tensor is 1-block injective.

Lemma 2.25 (One-site injectivity gives one-block injectivity). isNBlkInjective_one_of_isInjective ✓ Stmt OK If the one-site matrices $\{A^i\}_{i=0}^{d-1}$ span $M_D(\mathbb{C})$, then the length-one word products span $M_D(\mathbb{C})$, so A is 1-block injective.

Proof. ✓ Proof OK The length-one words are exactly the letters $i \in \{0, \dots, d-1\}$, and evaluating a length-one word gives the matrix A^i . Thus the length-one word span is the same span as in one-site injectivity. $\square$

Definition 2.26 (Normal tensor). IsNormal ✓ Stmt OK A tensor A is *normal* [CPGSV21] if there exists a finite L_0 such that A is L_0 -block injective.

Remark 2.27. ✓ Stmt OK The algebraic characterization of normality in Definition 2.26 — eventual full matrix span, equivalently eventual block injectivity — is equivalent to the spectral characterization used in [CPGSV21, Definition IV.1]: after TP normalization, the transfer map $\mathcal{E}_A$ is a primitive channel, equivalently irreducible with trivial peripheral spectrum. This equivalence is proved in [SPGWC10, Proposition 3]. In what follows, the directions of the equivalence are supplied by Theorem 10.62, Theorem 10.72, and Theorem 11.30. In particular, every subsequent use of “normal” here refers to this single notion; Definition 11.38 employs the spectral formulation, but it describes the same class of tensors.

2.5 Canonical form

Definition 2.28 (Block-injective canonical form). CanonicalForm ✓ Stmt OK Here, a *block-injective canonical form* for a tensor consists of:

1. a number of blocks $r \geq 1$,
2. bond dimensions $D_1, \dots, D_r$,
3. injective tensors $\{A_k^i\}_{i=0}^{d-1}$ with $A_k^i \in M_{D_k}(\mathbb{C})$ for each block $k \in \{1, \dots, r\}$,
4. scaling factors $\mu_1, \dots, \mu_r \in \mathbb{C}$.

The associated tensor is the block-diagonal tensor

$$A^i = \bigoplus_{k=1}^r \mu_k A_k^i.$$

This is the block-injective version of the canonical-form data; see [PGVWC07] and [CPGSV21].

Remark 2.29. ✓ Stmt OK Here we use the block-injective canonical form: each block is injective in the sense of Definition 2.23. This is stronger than the NT-block canonical form of [CPGSV21], where the blocks are only normal in the spectral sense. Chapter 11 treats the normal canonical form needed for the blocking-based multi-block theory, while Chapter 15 develops the separate periodic irreducible-form extension. The present section isolates the stronger block-injective form used in the injective and block-injective arguments.

Definition 2.30 (Block projection). blockProjection ✓ Stmt OK For a finite direct sum $\bigoplus_k \mathbb{C}^{D_k}$, the block projection P_j is the matrix that is the identity on the j -th summand and zero on all other summands.

Definition 2.31 (Block-diagonal matrix). IsBlockDiagonal' ✓ Stmt OK A matrix on $\bigoplus_k \mathbb{C}^{D_k}$ is block diagonal if it is a direct sum of square matrices, one on each summand.

Lemma 2.32 (Off-block characterization). isBlockDiagonal'_iff_offBlock_zero ✓ Stmt OK A matrix on $\bigoplus_k \mathbb{C}^{D_k}$ is block diagonal if and only if every entry from one summand to a distinct summand is zero.

Proof. ✓ Proof OK The forward direction is immediate from the definition of a direct sum of matrices. Conversely, if all off-block entries vanish, the diagonal blocks reconstruct the original matrix. $\square$

Lemma 2.33 (Commutation with block projections). isBlockDiagonal'_of_commutes_blockProjection ✓ Stmt OK *If a matrix on $\bigoplus_k \mathbb{C}^{D_k}$ commutes with every block projection P_j , then it is block diagonal.*

Proof. ✓ Proof OK Compare the (i, j) block of the identity $XP_j = P_jX$ for $i \neq j$. The right multiplication by P_j keeps that block, while the left multiplication by P_j kills it. Hence every off-block entry is zero, and Lemma 2.32 applies. $\square$

Definition 2.34 (Block-diagonal tensor from blocks). toTensorFromBlocks ✓ Stmt OK Given r blocks with bond dimensions $(D_k)_{k=1}^r$, per-block tensors $\{A_k^i\}$, and scaling factors $(\mu_k)_{k=1}^r$, the associated *block-diagonal tensor* is the tensor of total bond dimension $D = \sum_{k=1}^r D_k$ defined by

$$A^i = \bigoplus_{k=1}^r \mu_k A_k^i.$$

Lemma 2.35 (Word evaluation of a block-diagonal tensor). evalWord_toTensorFromBlocks_eq_reindex_blockDiagonal ✓ Stmt OK *a word w , the word evaluation of a block-diagonal tensor remains block diagonal:*

$$A^w = \bigoplus_k \mu_k^{|w|} A_k^w.$$

Proof. ✓ Proof OK Induct on the word and multiply block-diagonal matrices block by block; every letter contributes one scalar factor μ_k on the k -th block. $\square$

Theorem 2.36 (MPV decomposition). mpv_toTensorFromBlocks_eq_sum ✓ Stmt OK *The MPV of a block-diagonal tensor decomposes as*

$$|V^{(N)}(A)\rangle = \sum_{k=1}^r \mu_k^N |V^{(N)}(A_k)\rangle.$$

Equivalently, for each configuration σ one has

$$V^{(N)}(A)_\sigma = \sum_{k=1}^r \mu_k^N V^{(N)}(A_k)_\sigma.$$

Proof. ✓ Proof OK By Lemma 2.35, the full word evaluation is block diagonal with blocks $\mu_k^N A_k^w$ for a word w of length N . Taking the trace of a block-diagonal matrix gives the sum of the block traces: $\text{tr}(A^w) = \sum_k \mu_k^N \text{tr}(A_k^w)$. $\square$

2.6 Blocking

Definition 2.37 (Blocked tensor). blockTensor ✓ Stmt OK Given a tensor A and a blocking length L , the L -*blocked tensor* is the tensor with physical index set $\{0, \dots, d-1\}^L$ (hence physical dimension d^L) and the same bond dimension D , whose matrices are indexed by words $(i_1, \dots, i_L) \in \{0, \dots, d-1\}^L$:

$$(A^{[L]})^{(i_1, \dots, i_L)} = A^{i_1} A^{i_2} \dots A^{i_L}.$$

Lemma 2.38 (Blocked word evaluation). evalWord_blockTensor ✓ Stmt OK If $w = (J_1, \dots, J_m)$ is a word in the blocked alphabet, where each J_k is an L -tuple in $\{0, \dots, d-1\}^L$, then concatenating the block words gives a word $\tilde{w}$ of length mL in the original alphabet and one has

$$(A^{[L]})^w = A^{\tilde{w}}.$$

Proof. ✓ Proof OK Induct on the blocked word w and split off the first block word. Lemma 2.3 turns concatenation of blocks into multiplication of the corresponding evaluations. $\square$

Lemma 2.39 (Blocking preserves MPV coefficients). mpv_blockTensor_eq_mpv ✓ Stmt OK For each basis state $\sigma = (\sigma_1, \dots, \sigma_N)$ of the blocked system, where each σ_k is an L -tuple in $\{0, \dots, d-1\}^L$, there exists a basis state $\tilde{\sigma} = (\tilde{\sigma}_1, \dots, \tilde{\sigma}_{NL})$ of the original system such that

$$V^{(N)}(A^{[L]})_{\sigma} = V^{(NL)}(A)_{\tilde{\sigma}}.$$

Proof. ✓ Proof OK Concatenate the block words and apply the blocked word evaluation lemma (Lemma 2.38). $\square$

Theorem 2.40 (Blocking preserves same MPV). blockTensor ✓ Stmt OK If A and B generate the same MPV family, then for any blocking length L , the blocked tensors $A^{[L]}$ and $B^{[L]}$ also generate the same MPV family.

Proof. ✓ Proof OK By Lemma 2.39, each MPV coefficient of the blocked tensor $A^{[L]}$ equals a coefficient of A at the corresponding concatenated configuration. The same holds for $B^{[L]}$ (with the same concatenated configuration). Since A and B agree on all configurations, so do $A^{[L]}$ and $B^{[L]}$. $\square$

2.7 MPV overlap

Definition 2.41 (MPV as Hilbert-space vector). mpvState ✓ Stmt OK We regard $|V^{(N)}(A)\rangle$ as an element of $\mathbb{C}^{d^N}$ indexed by configurations $\sigma \in \{0, \dots, d-1\}^N$, with σ -component $V^{(N)}(A)_{\sigma}$ as in Definition 2.4. This allows us to use the standard Euclidean inner product on $\mathbb{C}^{d^N}$.

Definition 2.42 (MPV inner product). mpvInner ✓ Stmt OK The MPV inner product between two tensors A and B at system size N is

$$\langle V^{(N)}(A) | V^{(N)}(B) \rangle := \sum_{\sigma \in \{0, \dots, d-1\}^N} \overline{V^{(N)}(A)_{\sigma}} V^{(N)}(B)_{\sigma},$$

i.e. the Euclidean inner product on $\mathbb{C}^{d^N}$ (conjugate-linear in the first argument).

Definition 2.43 (MPV overlap). mpvOverlap ✓ Stmt OK The MPV overlap between two tensors A (of bond dimension D_1) and B (of bond dimension D_2) at system size N is

$$O_{AB}(N) := \sum_{\sigma \in \{0, \dots, d-1\}^N} V^{(N)}(A)_{\sigma} \overline{V^{(N)}(B)_{\sigma}}.$$

This is the bilinear overlap (without complex conjugation) used later. By Lemma 2.44, it is the complex conjugate of the Hilbert-space inner product from Definition 2.42.

Lemma 2.44 (Overlap equals conjugate inner product). mpvOverlap_eq_star_mpvInner ✓ Stmt OK $O_{AB}(N) = \overline{\langle V^{(N)}(A) | V^{(N)}(B) \rangle}.$

Proof. ✓ Proof OK The overlap sums $V_\sigma \overline{W_\sigma}$ while the inner product sums $\overline{V_\sigma} W_\sigma$. Complex conjugation swaps these. □

Lemma 2.45 (Overlap determined by positive-length MPV). mpvOverlap_eq_of_pos_mpv_eq ✓ Stmt OK *If $V^{(N)}(A)_\sigma = V^{(N)}(A')_\sigma$ and $V^{(N)}(B)_\sigma = V^{(N)}(B')_\sigma$ for all $N > 0$ and all σ , then $O_{AB}(N) = O_{A'B'}(N)$ for all positive N .*

Proof. ✓ Proof OK Direct pointwise rewriting: if the MPV coefficients agree for all positive chain lengths, the overlap sums are identical. □

Chapter 3

Matrix Product Operators and Density Operators

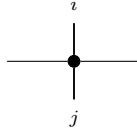
Fix a physical dimension d and a bond dimension D . An MPO tensor carries two physical indices, one ket index and one bra index, together with the virtual left and right indices.

3.1 MPO tensors

Definition 3.1 (MPO tensor). MPOTensor ✓ Stmt OK An *MPO tensor* is a family of matrices

$$\{M^{ij}\}_{i,j=0}^{d-1}, \quad M^{ij} \in M_D(\mathbb{C}),$$

indexed by a ket index i and a bra index j . Diagrammatically,



The black node denotes the tensor M , the horizontal legs are virtual, the upper leg is the ket index i , and the lower leg is the bra index j .

Definition 3.2 (Doubled-index MPS view). toMPSTensor ✓ Stmt OK By combining the pair (i, j) into a single physical index in $\{0, \dots, d^2 - 1\}$, one obtains an MPS tensor with physical dimension d^2 and bond dimension D .

Definition 3.3 (Word evaluation). evalWord ✓ Stmt OK For words $u = (i_1, \dots, i_N)$ and $v = (j_1, \dots, j_N)$ in $\{0, \dots, d-1\}^N$, the *word evaluation* is

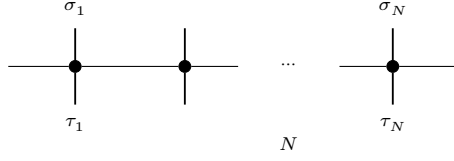
$$M^{u,v} := M^{i_1 j_1} M^{i_2 j_2} \dots M^{i_N j_N} \in M_D(\mathbb{C}).$$

The empty pair of words evaluates to $\mathbb{1}_D$, and word pairs of different lengths evaluate to 0.

Definition 3.4 (MPO operator family). mpo ✓ Stmt OK The operator generated by M on N sites is the matrix $\rho^{(N)}(M)$ indexed by configurations $\sigma, \tau \in \{0, \dots, d-1\}^N$ and given by

$$\rho^{(N)}(M)_{\sigma, \tau} = \text{tr}(M^{\sigma, \tau}).$$

Diagrammatically,



each black node denotes the same local tensor M , and the matrix entry $\rho^{(N)}(M)_{\sigma,\tau}$ is obtained by closing the outer virtual legs cyclically and taking the resulting trace.

Definition 3.5 (Hermitian MPO tensor). IsHermitian ✓ Stmt OK An MPO tensor is *Hermitian* if, for all $i, j \in \{0, \dots, d-1\}$,

$$M^{ij} = (M^{ji})^\dagger$$

3.2 Transfer maps

Definition 3.6 (MPO transfer map). transferMap ✓ Stmt OK The *transfer map* associated to M is the linear map $\mathcal{E}_M : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ defined by

$$\mathcal{E}_M(X) = \sum_{i,j=0}^{d-1} M^{ij} X (M^{ij})^\dagger.$$

Lemma 3.7 (Identification with the doubled-index MPS transfer map). transferMap_eq_toMPSTensor ✓ Stmt OK The *transfer map of an MPO tensor agrees with the transfer map of its doubled-index MPS view*.

Proof. ✓ Proof OK After identifying the single physical index of the doubled tensor with a pair (i, j) , the single sum defining the MPS transfer map is exactly the double sum defining $\mathcal{E}_M$. $\square$

Theorem 3.8 (Positivity of the MPO transfer map). transferMap_pos ✓ Stmt OK If $X \geq 0$, then $\mathcal{E}_M(X) \geq 0$.

Proof. ✓ Proof OK By Lemma 3.7, $\mathcal{E}_M$ is the transfer map of the doubled-index MPS tensor. The corresponding MPS transfer map is positive, so $\mathcal{E}_M(X) \geq 0$ whenever $X \geq 0$. $\square$

3.3 Zero correlation length

Definition 3.9 (Zero correlation length for MPO tensors). IsZCL ✓ Stmt OK An MPO tensor M has *zero correlation length* when its transfer map is idempotent:

$$\mathcal{E}_M \circ \mathcal{E}_M = \mathcal{E}_M.$$

This is the mixed-state analogue of the renormalization fixed-point condition for MPS tensors; see [CPGSV21, Definition 4.2].

Theorem 3.10 (MPO ZCL iff doubled-index MPS RFP). isZCL_iff_toMPSTensor_isRFP ✓ Stmt OK An MPO tensor has zero correlation length if and only if its doubled-index MPS view is a renormalization fixed point.

Proof. ✓ Proof OK By Lemma 3.7, the transfer map of M agrees with the transfer map of its doubled-index MPS view. Thus $\mathcal{E}_M \circ \mathcal{E}_M = \mathcal{E}_M$ holds exactly when the transfer map of that doubled-index MPS tensor is idempotent, which is the RFP condition. $\square$

3.4 MPDOs and LPDOs

Definition 3.11 (MPDO). [isMPDO] ✓ Stmt OK An MPO tensor is an *MPDO* if, for every $N \in \mathbb{N}$,

$$\rho^{(N)}(M) \geq 0$$

This is the global positivity condition of [CPGSV17a, §4.1].

Definition 3.12 (LPDO). [isLPDO] ✓ Stmt OK An MPO tensor is an *LPDO* if there exist an ancilla dimension d_K , an inner bond dimension D' , an identification of bond indices $e : \{0, \dots, D-1\} \simeq \{0, \dots, D'-1\} \times \{0, \dots, D'-1\}$ (equivalently, D and $(D')^2$ have the same cardinality), and matrices $A^{(i,k)} \in M_{D'}(\mathbb{C})$ such that, for all physical indices i, j ,

$$M^{ij} = \left(\sum_{k=0}^{d_K-1} A^{(i,k)} \otimes \overline{A^{(j,k)}} \right)_{e,e}$$

where $\otimes$ denotes the Kronecker product, $\overline{(\cdot)}$ denotes entrywise complex conjugation, and $(\cdot)_{e,e}$ denotes reindexing along e from $\{0, \dots, D'-1\} \times \{0, \dots, D'-1\}$ back to $\{0, \dots, D-1\}$. This is the local purification form of [CPGSV17a, §4.3].

Theorem 3.13 (LPDOs are MPDOs). [isMPDO] ✓ Stmt OK Every LPDO tensor is an MPDO.

Proof. ✓ Proof OK By the mixed-product property $(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$, the N -site product decomposes as a sum of Kronecker products after transporting along the bond-space identification e :

$$M^{\sigma_1 \tau_1} \dots M^{\sigma_N \tau_N} = \left(\sum_{\kappa} P_{\kappa} \otimes \overline{Q_{\kappa}} \right) \Big|_e,$$

where $P_{\kappa} = A^{(\sigma_1, \kappa_1)} \dots A^{(\sigma_N, \kappa_N)}$ and $Q_{\kappa} = A^{(\tau_1, \kappa_1)} \dots A^{(\tau_N, \kappa_N)}$. Since the trace is invariant under the reindexing by e , we apply $\text{tr}(X \otimes Y) = \text{tr}(X) \text{tr}(Y)$ to obtain

$$\rho^{(N)}(M)_{\sigma, \tau} = \sum_{\kappa} \text{tr}(P_{\kappa}) \overline{\text{tr}(Q_{\kappa})} = \sum_{\kappa} \psi_{\kappa}(\sigma) \overline{\psi_{\kappa}(\tau)},$$

where $\psi_{\kappa}(\sigma) = \text{tr}(P_{\kappa})$. Hence $\rho^{(N)}(M) = \sum_{\kappa} |\psi_{\kappa}\rangle \langle \psi_{\kappa}| \geq 0$. □

3.5 Purification RFP

Definition 3.14 (Purifying MPS tensor). [purifyingMPSensor] ✓ Stmt OK Given a purifying family $A^{(i,k)} \in M_{D'}(\mathbb{C})$ for $i \in \{0, \dots, d-1\}$ and $k \in \{0, \dots, d_K-1\}$, the *purifying MPS tensor* is the MPS tensor with physical dimension $d \cdot d_K$ and bond dimension D' obtained by combining the pair (i, k) into a single index.

Definition 3.15 (Purification RFP). [isPRFP] ✓ Stmt OK An MPO tensor M has a *purification RFP* if it admits an LPDO witness whose purifying MPS tensor is a pure-state renormalization fixed point. This is [CPGSV21, Definition 4.3].

Remark 3.16 (PRFP does not imply MPO ZCL). [exists_isPRFP_not_isZCL] ✓ Stmt OK Purification RFP is strictly weaker than MPO zero correlation length: there exists an MPO tensor with purification RFP whose transfer map is not idempotent. Equivalently, purification RFP does not imply MPO zero correlation length.

Theorem 3.17 (Purification RFP implies LPDO). [isLPDO] ✓ Stmt OK A purification-RFP tensor is, in particular, an LPDO.

Proof. ✓ Proof OK The PRFP condition provides an LPDO witness by construction. □

3.6 Pure-state recovery inside the MPO formalism

Definition 3.18 (MPDO renormalization fixed point). [IsRFP] [✓ Stmt OK] An MPO tensor M is an *MPDO renormalization fixed point* if its transfer map is idempotent:

$$\mathcal{E}_M \circ \mathcal{E}_M = \mathcal{E}_M.$$

Theorem 3.19 (MPDO RFP iff MPO ZCL). [isRFP_iff_isZCL] [✓ Stmt OK] *MPDO renormalization fixed points are exactly MPO tensors with zero correlation length.*

Proof. [✓ Proof OK] Both conditions assert $\mathcal{E}_M \circ \mathcal{E}_M = \mathcal{E}_M$; they are definitionally identical. □

Theorem 3.20 (MPDO RFP iff doubled-index MPS RFP). [isRFP_iff_toMPSTensor_isRFP] [✓ Stmt OK] *The MPDO RFP condition is equivalent to the pure-state RFP condition for the doubled-index MPS tensor.*

Proof. [✓ Proof OK] Unfold the MPDO RFP condition to its definitional equivalent, MPO ZCL, then apply Theorem 3.10. □

Definition 3.21 (Diagonal pure-state embedding as an MPO). [toMPOTensor] [✓ Stmt OK] An MPS tensor $A = \{A^i\}_{i=0}^{d-1}$ determines an MPO tensor by placing A^i on the diagonal in the bra and ket indices:

$$M^{ij} = \delta_{ij} A^i.$$

Equivalently, $M^{ii} = A^i$ and $M^{ij} = 0$ for $i \neq j$.

Theorem 3.22 (Transfer map of the diagonal pure-state embedding). [toMPOTensor_transferMap] [✓ Stmt OK] *The transfer map of the diagonal MPO associated to A agrees with the original MPS transfer map:*

$$\mathcal{E}_{A^{\text{MPO}}} = \mathcal{E}_A.$$

Proof. [✓ Proof OK] In the double sum defining $\mathcal{E}_{A^{\text{MPO}}}$, all off-diagonal terms vanish because $M^{ij} = 0$ for $i \neq j$, while the diagonal terms are exactly $A^i X(A^i)^\dagger$. Thus only the sum over i remains, which is the defining formula for $\mathcal{E}_A$. □

Theorem 3.23 (Pure-state recovery of the RFP condition). [toMPOTensor_isRFP_iff_isRFP] [✓ Stmt OK] *For an MPS tensor A , its diagonal MPO embedding is MPDO-RFP if and only if A is an RFP:*

$$A^{\text{MPO}} \text{ is MPDO-RFP} \iff A \text{ is RFP}.$$

Proof. [✓ Proof OK] Both conditions assert idempotence of the corresponding transfer map. Theorem 3.22 identifies those transfer maps, so the two predicates are equivalent. □

Theorem 3.24 (Pure-state recovery of zero correlation length). [toMPOTensor_isRFP_iff_isZCL] [✓ Stmt OK] *For an MPS tensor A , the diagonal MPO embedding is MPDO-RFP if and only if A has zero correlation length.*

Proof. [✓ Proof OK] Combine Theorem 3.23 with the characterization of zero correlation length by idempotence of the MPS transfer map (Theorem 21.26). □

3.7 Simple local structure from SAL and ZCL

For the simple MPDO case in Appendix C.2 of [CPGSV17a], the strong-area-law hypothesis is used locally through the equality case of strong subadditivity for a normalized three-site reduced state.

Definition 3.25 (Local η -structure for a three-site reduced state). [EtaStructure] ✓ Stmt OK For a three-site density operator ρ_{ABC} , the local η -structure is the quantum Markov decomposition on the middle subsystem B produced by equality in strong subadditivity. Concretely, after a unitary change of basis on B , one has a finite direct-sum decomposition

$$B = \bigoplus_k (b_1^{(k)} \otimes b_2^{(k)})$$

such that

$$\rho_{ABC} = \bigoplus_k \rho_{Ab_1}^{(k)} \otimes \rho_{b_2C}^{(k)}.$$

Theorem 3.26 (SAL implies local η -structure). [sal_implies_eta_structure] ✓ Stmt OK *If a normalized three-site reduced state satisfies the equality case of strong subadditivity — the local entropy form of the simple-MPDO strong area law — then it admits the local η -structure of Definition 3.25.*

Proof. ✓ Proof OK This is exactly the forward implication of the Hayashi equality characterization: equality in strong subadditivity yields a quantum Markov decomposition on the middle subsystem. $\square$

Definition 3.27 (Injective inverse-map layer for a simple MPO tensor). [IsInjective] [inverseTensor] [physRealize] [physRealizeLeft] ✓ Stmt OK A simple MPO tensor K is called injective if its doubled-index MPS tensor is injective. For such a tensor, the chosen decomposition map of the doubled-index MPS furnishes a concrete inverse tensor K^{-1} together with right- and left-physical realization maps that turn virtual bond insertions back into local physical operators.

Theorem 3.28 (Inverse-map identities for an injective simple MPO tensor). [inverseTensor_spec] [physRealize_spec] [physRealize_mul] [physRealizeLeft_spec] ✓ Stmt OK *For an injective simple MPO tensor, contracting K^{-1} with K recovers the matrix units on the virtual bond space. Likewise, every right virtual insertion admits a physical realization, this realization is multiplicative, and the analogous left-insertion statement also holds.*

Definition 3.29 (Explicit neighboring η -operator data). [etaOperators] [ExplicitEtaOperators] ✓ Stmt OK Fix a Hayashi decomposition witness h_η . An explicit neighboring η -family is a dependent family $(\eta_{k,h})$ indexed by sector pairs, where $\eta_{k,h}$ acts on the neighboring bond space $B_k^R \otimes B_h^L$. Requiring positivity for each pair gives the corresponding positive η -data.

Definition 3.30 (Trace matrices of explicit η -data). [traceMatrix] [traceMatrix_apply] [traceMatrixRe] [traceMatrixRe_apply] ✓ Stmt OK Every explicit neighboring η -family determines a sector-by-sector trace matrix. We give both its complex-valued form and the real-part version, which is the direct data for the real Perron–Frobenius matrix T used in the rank-one step.

Theorem 3.31 (Positivity of bipartite partial traces). [traceLeft] [traceRight] ✓ Stmt OK *If ρ is positive semidefinite on a bipartite space $A \otimes B$, then both reduced operators $\text{tr}_A \rho$ and $\text{tr}_B \rho$ are positive semidefinite.*

Proof. ✓ Proof OK Write each partial trace as the finite sum, over the traced index, of a principal submatrix of ρ . Principal submatrices of a positive semidefinite matrix are positive semidefinite, and finite sums of positive semidefinite matrices remain positive semidefinite. $\square$

Definition 3.32 (Hayashi–Markov extraction of explicit η -operators). ofHayashiMarkov ✓ Stmt OK Every Hayashi decomposition witness h_η canonically produces an explicit neighboring η -family by Kronecker-multiplying the sector-reduced states on the neighboring bond spaces:

$$\eta_{k,h} := \text{tr}_C(\rho_{b_2C}^{(k)}) \otimes \text{tr}_A(\rho_{Ab_1}^{(h)}).$$

Positive semidefiniteness of each $\eta_{k,h}$ follows from the fact that partial traces preserve positivity and that positive semidefiniteness is closed under Kronecker products. This is the sector-reduced extraction layer; the remaining connection to Appendix C.2 is the identification of this family with the inverse-map construction in the original simple-MPDO tensor coordinates.

Theorem 3.33 (Trace matrix of the Hayashi–Markov η -operators). traceMatrix_ofHayashiMarkov traceMatrixRe_ofHayashiMarkov ✓ Stmt OK *The trace matrix induced by the extracted family is entrywise $T_{k,h} = 1$ in both its complex-valued form and its real-valued form, matching the paper’s normalization $T_{k,h} = a_k b_h$ with $a_k = b_h = 1$.*

Proof. ✓ Proof OK Each $\rho_{b_2C}^{(k)}$ is a density matrix, hence has unit trace; the partial trace tr_C preserves the trace. The analogous statement holds for $\text{tr}_A \rho_{Ab_1}^{(h)}$. Multiplicativity of the trace under Kronecker products gives $T_{k,h} = 1 \cdot 1 = 1$, and taking real parts gives the real-valued statement. $\square$

Theorem 3.34 (Entrywise nonnegativity of the real trace matrix). traceMatrixRe_nonneg ✓ Stmt OK *Positivity of the neighboring operators gives entrywise nonnegativity of the real trace matrix. Matrix-level positive semidefiniteness of T is a separate structural condition.*

Proof. ✓ Proof OK The trace of a positive semidefinite neighboring operator is nonnegative in the complex order. Taking real parts gives the stated entrywise nonnegativity. $\square$

Definition 3.35 (Auxiliary matrix predicates for the rank-one step). TracePowersConstant HasRankOneFactorization PrimitiveTracePowersConstantImpliesRankOne ✓ Stmt OK The auxiliary predicates state three matrix-theoretic ingredients used in the rank-one step of Appendix C.2, Lemma C.4: constant traces of all positive powers, existence of a rank-one factorization $T = ab^\top$, and the conditional Perron–Frobenius implication from primitivity together with constant trace powers to such a factorization.

This conditional implication is not a globally provable theorem. As a universal statement over primitive nonnegative matrices it is false; a concrete 3×3 counterexample appears in the archive. The corrected criterion proved here adds positive semidefiniteness and trace normalization, which provide the diagonalizability needed to turn the trace-power condition into a rank-one factorization.

Theorem 3.36 (Trace square of a Hermitian matrix). trace_sq_eq_sum_eigenvalues_sq ✓ Stmt OK *Let T be a finite Hermitian matrix over a real or complex scalar field, with Hermitian eigenvalues λ_i . Then*

$$\text{tr}(T^2) = \sum_i \lambda_i^2.$$

Proof. ✓ Proof OK The spectral theorem writes T as a unitary conjugate of the diagonal matrix of its Hermitian eigenvalues. Squaring commutes with this conjugation, and cyclicity of trace reduces the identity to the trace of the squared diagonal matrix. $\square$

Theorem 3.37 (PSD trace-power rank-one criterion). [trace_powers_constant_implies_rank_one] [primitive_trace_powers_constant] *Let T be a finite real square matrix. If T is positive semidefinite, normalized by $\text{tr}(T) = 1$, and has constant traces on all positive powers, then T has a rank-one factorization. Consequently, under the same positive-semidefinite and trace-normalization hypotheses, the conditional rank-one criterion used in Lemma C.4 follows.*

Proof. [✓ Proof OK] The spectral theorem diagonalizes the positive semidefinite matrix with non-negative real eigenvalues. The trace gives that their sum is 1, while the second trace moment gives that the sum of their squares is also 1. Hence exactly one eigenvalue is equal to 1 and all others vanish, so the diagonal form is rank one. Unitary conjugation preserves the existence of an outer-product factorization. □

Theorem 3.38 (SAL and ZCL imply rank-one T (conditional form)). [sal_zcl_implies_rank_one_T] [✓ Stmt OK] *Let T be a finite real square matrix. Assume that T is primitive, that $\text{tr}(T) = 1$, and that every positive power of T has the same trace as T . If one further supplies the Perron–Frobenius hypothesis that this particular primitive nonnegative matrix with constant trace powers is rank one, then*

$$T = ab^\top$$

for some vectors a and b , with normalization

$$a \cdot b = 1.$$

Proof. [✓ Proof OK] The Perron–Frobenius hypothesis gives the factorization $T = ab^\top$. Taking traces and using $\text{tr}(ab^\top) = a \cdot b$ converts the normalization $\text{tr}(T) = 1$ into $a \cdot b = 1$. □

Theorem 3.39 (PSD-corrected rank-one T criterion). [sal_zcl_implies_rank_one_T_of_posSemidef] [✓ Stmt OK] *Let T be a finite real square matrix. Assume that T is primitive, positive semidefinite, normalized by $\text{tr}(T) = 1$, and has constant traces on all positive powers. Then*

$$T = ab^\top$$

for some vectors a and b , with $a \cdot b = 1$.

Proof. [✓ Proof OK] The positive-semidefinite matrix criterion diagonalizes T , uses the first two trace moments to show that exactly one eigenvalue is nonzero, and hence supplies the auxiliary rank-one criterion. The conditional Lemma C.4 statement then gives the factorization and trace normalization. □

3.8 Horizontal and Vertical Canonical Forms

Definition 3.40 (Horizontal canonical form for an MPO). [IsHorizontalCF] [Not ready] An MPO tensor is in *horizontal canonical form* if its doubled-index MPS tensor generates the same MPV family as a block-diagonal tensor $\bigoplus_k \mu_k A_k$ whose blocks are injective, left-canonical, and have nonzero weights μ_k .

Definition 3.41 (Vertical canonical form for an MPO). [IsVerticalCF] [Not ready] An MPO tensor is in *vertical canonical form* if its diagonal MPS tensor $\{M^{ii}\}_{i=0}^{d-1}$ generates the same MPV family as a block-diagonal tensor obtained from a basis of normal tensors by repeating each block according to a multiplicity and weighting each copy by a positive scalar. This is the flattened form of the decomposition $\bigoplus_\alpha \mu_\alpha \otimes M_\alpha$ from [CPGSV17a, §4.4].

Remark 3.42 (Finite witnesses for block-injective canonical form).

[WordTupleSpanTop](#) [BlockEntryIndex](#)
[wordEntryFamily](#) [duplicateScalarBlocks_counterexample](#) [wordTupleSpanTop_of_wordEntryFamily_linearIndependent](#)
[hasBlockSelectorWords_of_wordTupleSpanTop](#) [hasBlockSelectorWords_of_wordEntryFamily_linearIndependent](#) [HasBiCF](#)
[hasBiCF_of_wordTupleSpanTop](#) [hasBiCF_of_wordEntryFamily_linearIndependent](#) [HasBlockSelectorWords](#) [PropBlockInjective](#)
[wordTupleSpanTop_of_propBlockInjective](#) [hasBiCF_of_propBlockInjective](#) [duplicateScalarBlocks_not_hasBiCF](#) [horizontalCFData_of_wordTup](#)
[horizontalCFData_of_wordEntryFamily_linearIndependent](#) [horizontalCFData_of_propBlockInjective](#) [✓ Stmt OK](#)

Let (A_k) be a family of block tensors, and fix a length L . Write A_k^w for the length- L word evaluation of A_k along a word w . The finite-length spanning hypothesis says that the tuple-valued evaluations

$$w \mapsto (A_k^w)_k$$

span the full product algebra of block matrices. The corresponding finite-length separation property asserts that there exists a length L such that if, for every word w of length L ,

$$\sum_k \text{tr}(\Delta_k A_k^w) = 0$$

then these pairings vanish on a spanning set for the product algebra, so the induced linear functional on the product algebra is zero; by nondegeneracy of the product trace pairing, every Δ_k vanishes. In other words, a single finite blocking length already separates the blocks by their word evaluations. Combined with injectivity, left-canonicity, and nonzero block weights, this finite-length separation yields the horizontal canonical-form data of the block-diagonal tensor.

The entrywise scalar family of block-matrix coefficients encodes the same finite-length data. Linear independence of that scalar family implies the product-algebra spanning statement, hence a biCF witness. The same criterion therefore gives horizontal canonical-form data directly. The duplicate-scalar counterexample shows that this finite-length witness is strictly stronger than blockwise injectivity, left-canonicity, and nonzero weights alone: two identical scalar blocks satisfy those hypotheses, but the biCF separation condition fails and no blocking length makes the entrywise family linearly independent.

Proposition IV.3 gives a concrete sufficient criterion for the spanning condition: after a common blocking length each block is block-injective, and after a further finite blocking length one can form linear combinations of word evaluations which equal the identity on one chosen block and vanish on all others. Concatenating an injective prefix with this finite block-separation data produces the product-algebra spanning condition, hence the horizontal canonical-form data.

Theorem 3.43 (A finite criterion from Proposition IV.3). [hasBiCF_of_propBlockInjective](#) [✓ Stmt OK](#) *If a block family admits the finite-length data described in Proposition IV.3, then it is in block-injective canonical form.*

Proof. [✓ Proof OK](#) This is the direct implication from the finite-length data to the finite-length trace-separation statement. What remains open in the full Proposition IV.3 is the derivation of this finite witness from separated canonical-form / BNT data. $\square$

Lemma 3.44 (Blockwise equality from agreement on the MPV family). [blockwise_insert_eq_of_mpv_agree](#) [✓ Stmt OK](#) *Let $\bigoplus_k \mu_k A_k$ be a block-diagonal tensor whose blocks are injective, left-canonical, weighted by nonzero scalars, and jointly block-injective in the sense that for some blocking length L the tuple of trace pairings against length- L block words is faithful across all blocks simultaneously (the block-injective canonical form, biCF). If two physical operators induce the same first-site action on every MPV generated by this tensor, then their first-site insertions agree on each block A_k .*

Proof. [✓ Proof OK](#) This is Lemma L from [CPGSV17a, Appendix A], using the finite criterion from Theorem 3.43. Expanding the hypothesis over a word of length $L + 1$ beginning with the

fixed site and folding each block contribution into a trace pairing against the corresponding first-site insertion tensor yields, upon taking the Y - Z difference, a tuple of block matrices whose trace pairings against every length- L block word vanish. The biCF hypothesis then forces each block difference to be zero, and dividing by the nonzero weight μ_k^{L+1} concludes the blockwise equality of the two first-site insertions. $\square$

Theorem 3.45 (Horizontal canonical form implies vertical canonical form). Not ready Every MPDO in horizontal canonical form is also in vertical canonical form.

Proof. Starting from the horizontal canonical-form decomposition, one uses the MPDO positivity condition to compare the first-site action of the doubled-index tensor on the diagonal sector. Lemma 3.44 then shows that the diagonal tensor splits along the same normal blocks, with the coefficients rearranged into positive scalar weights. Flattening those positive diagonal coefficients produces the required vertical canonical-form decomposition. $\square$

3.9 Fusion isometries

Following [CPGSV17a, §4.5], the next definitions describe the transfer-map content of the fusion-isometry picture. For each blocked size $n \geq 1$, the doubled-index blocked tensor of an MPO has a blocked transfer map E_n acting on bond-space matrices, and the support algebra is modeled by a subspace through which E_n factors as a retract.

Definition 3.46 (Transfer-map fusion-isometry datum). FusionIsometryData ✓ Stmt OK A transfer-map fusion-isometry datum at blocked size n for an MPO tensor consists of a subspace $\mathcal{A}_n \subseteq M_D(\mathbb{C})$ together with linear maps

$$T_n : M_D(\mathbb{C}) \rightarrow \mathcal{A}_n, \quad S_n : \mathcal{A}_n \rightarrow M_D(\mathbb{C})$$

such that

$$T_n \circ S_n = \text{id}_{\mathcal{A}_n}, \quad S_n \circ T_n = E_n,$$

where E_n denotes the blocked transfer map.

Definition 3.47 (Fusion-isometry RFP predicate). isRFP_MPDO_via_fusion ✓ Stmt OK An MPO tensor M satisfies the *fusion-isometry formulation* of the renormalization fixed-point condition when for every blocked size $n \geq 1$ there exists transfer-map fusion-isometry data for the blocked transfer map E_n .

Theorem 3.48 (Fusion-isometry data imply the MPDO RFP condition). isRFP_of_isRFP_MPDO_via_fusion ✓ Stmt OK The fusion-isometry formulation implies the MPDO RFP condition.

Proof. ✓ Proof OK Apply the definition at blocked size $n = 1$. If $E_1 = S_1 \circ T_1$ and $T_1 \circ S_1 = \text{id}$, then

$$E_1^2 = S_1 T_1 S_1 T_1 = S_1 (T_1 S_1) T_1 = S_1 T_1 = E_1.$$

Since E_1 is the original transfer map, this is exactly the MPDO RFP condition. $\square$

Theorem 3.49 (Transfer-map fusion criterion for MPDO RFP). isRFP_MPDO_via_fusion_iff_isRFP ✓ Stmt OK The fusion-isometry formulation is equivalent to the MPDO RFP condition.

Proof. ✓ Proof OK The forward implication is Theorem 3.48. Conversely, if the transfer map E is idempotent, then every positive blocked transfer map equals E and therefore factors through its range. This range-factorization gives the required fusion-isometry data. $\square$

Corollary 3.50 (Pure-state recovery). [toMPOTensor_isRFP_MPDO_via_fusion_iff_isRFP](#) ✓ Stmt OK For the diagonal MPO associated to a pure MPS tensor, the fusion-isometry formulation reduces to the usual pure-state RFP condition.

Proof. ✓ Proof OK Combine Theorem 3.49 with the diagonal pure-state recovery theorem for the MPO predicate. □

3.10 Algebra structure

Definition 3.51 (Algebra-structure data for an MPDO). [AlgebraStructureData](#) ✓ Stmt OK A family of support $*$ -algebras $\{\mathcal{A}_n\}_{n \geq 0}$, one at every blocking size, equipped with blocking maps $m_n : \mathcal{A}_n \otimes \mathcal{A}_n \rightarrow \mathcal{A}_{2n}$ and inclusion maps $\iota_n : \mathcal{A}_n \rightarrow \mathcal{A}_{n+1}$ such that, after viewing all elements inside the ambient bond-space matrix algebra, m_n is ordinary matrix multiplication and ι_n is the ambient inclusion.

Definition 3.52 (Algebra-structure RFP predicate). [IsRFP_MPDO_via_algebra](#) ✓ Stmt OK An MPO tensor M satisfies the *algebra-structure formulation* used here when there exist algebra-structure data whose support algebra at every positive blocking size coincides with the fixed-point algebra of the adjoint blocked transfer map. This is a nontrivial algebraic condition, but it is still weaker than the full coefficient formulation of [CPGSV17a, Theorem IV.13 (ii)]; that formulation also includes the coefficient family $c_{\alpha, \beta, \gamma}^{(L)}$.

Theorem 3.53 (RFP yields a stationary algebra tower under a faithful fixed point). [isRFP_MPDO_via_algebra_of_isRFP_of_isT](#) *the doubled tensor is trace-preserving and the transfer map admits a positive-definite fixed point. If the transfer map is idempotent, then there exists a stationary family of support algebras, all equal to the fixed-point algebra of the adjoint transfer map, with multiplication given by matrix multiplication and inclusions given by the identity.*

Proof. ✓ Proof OK Under the trace-preserving normalization and the faithful fixed point, Wolf's fixed-point-algebra theorem gives a $*$ -subalgebra of adjoint fixed points. Idempotence identifies every positive blocked transfer map with the original one, so the same fixed-point algebra works at every blocking size. □

Theorem 3.54 (Fusion yields a stationary algebra tower under a faithful fixed point). [isRFP_MPDO_via_algebra_of_isRFP_MPDO](#) *the doubled tensor is trace-preserving and the transfer map admits a positive-definite fixed point. If the transfer-map fusion criterion holds, then the MPO satisfies the algebra-structure formulation of the renormalization fixed-point condition.*

Proof. ✓ Proof OK Combine Theorem 3.48 with Theorem 3.53. □

Theorem 3.55 (Stabilized blocked adjoint fixed points yield algebra data). [isRFP_MPDO_via_algebra_of_adjointFixedPoints_eq](#) *the doubled tensor is trace-preserving and the transfer map admits a positive-definite fixed point. If, for every positive blocked size n , the adjoint fixed-point space of the blocked transfer map agrees with the adjoint fixed-point space of the original transfer map, then the algebra-structure formulation holds.*

Proof. ✓ Proof OK Use the adjoint fixed-point algebra of the original transfer map as a constant support-algebra tower. The stabilization hypothesis identifies that same algebra with the adjoint fixed-point space of every positive blocked transfer map, so the stationary family satisfies the definition. □

This does not give the full converse direction of [CPGSV17a, Theorem IV.13]: the compatibility condition above only sees stabilized adjoint fixed-point algebras, not the coefficient/BNT data. In particular, Theorem IV.13(ii) also requires the special diagonal matrices $\chi_{\alpha,\beta,\gamma}$. The coefficient data attached to the algebra tower are: a chosen basis of each support algebra, coordinate maps into a finite scalar family, and the corresponding multiplication / inclusion coefficient arrays.

Definition 3.56 (Blocked coefficients for the algebra tower). [toBlockedCoefficients](#) ✓ Stmt OK For each blocked size n , once a basis of $\mathcal{A}_n$ is chosen, every element of $\mathcal{A}_n$ is identified with a finite coefficient family on that basis.

Theorem 3.57 (Reconstruction from blocked coefficients). [reconstructFromBlockedCoefficients_apply](#) ✓ Stmt OK *Reconstructing a coefficient family gives the corresponding basis expansion $\sum_i a_i b_i$ in $\mathcal{A}_n$.*

Proof. ✓ Proof OK This is exactly the finite-basis reconstruction formula for the chosen basis of $\mathcal{A}_n$. □

Definition 3.58 (Blocked multiplication coefficients). [blockedStructureCoefficients](#) ✓ Stmt OK The blocking multiplication map m_n determines a coefficient family for the product of any two chosen basis elements of $\mathcal{A}_n$, now viewed in the basis of $\mathcal{A}_{2n}$.

Theorem 3.59 (Blocked multiplication reconstructs from its coefficients). [coe_mul_eq_sum_blockedStructureCoefficients](#) ✓ Stmt OK *the ambient bond-space matrix algebra, the product of two chosen basis elements of $\mathcal{A}_n$ is the sum of the basis elements of $\mathcal{A}_{2n}$ weighted by these extracted coefficients.*

Proof. ✓ Proof OK Apply the reconstruction formula of Theorem 3.57 to the element $m_n(b_i, b_j) \in \mathcal{A}_{2n}$. □

Theorem 3.60 (Compatible adjoint fixed points reconstruct from blocked coefficients). [reconstructFromBlockedCoefficients_apply](#) ✓ Stmt OK *the algebra tower is compatible with an MPO tensor M , then every adjoint blocked fixed point of the blocked transfer map E_n belongs to $\mathcal{A}_n$ and is recovered exactly from its extracted coefficient family.*

Proof. ✓ Proof OK Compatibility identifies $\mathcal{A}_n$ with the adjoint fixed-point algebra of E_n , so the fixed point first becomes an element of $\mathcal{A}_n$ and then Theorem 3.57 reconstructs it. □

Theorem 3.61 (Adjoint fixed points descend through the algebra tower). [adjoint_transferMap_apply_of_isRFP_MPDO_via_algebraTower](#) ✓ Stmt OK *an MPO tensor satisfies the algebra-structure formulation, then every adjoint fixed point of the blocked transfer map E_n at a positive blocking size n is already an adjoint fixed point of the unblocked transfer map E_1 .*

Proof. ✓ Proof OK Compatibility at size n places X in $\mathcal{A}_n$. The inclusion map ι_n lifts X into $\mathcal{A}_{n+1}$, and since ι_n is just an inclusion it does not change the underlying operator, so we continue to denote the lifted element by X . Thus compatibility at size $n+1$ yields $E_{n+1}^\dagger X = X$. Rewriting $E_{n+1} = E_n \circ E_1$ and applying $(A \circ B)^\dagger = B^\dagger \circ A^\dagger$ together with the hypothesis $E_n^\dagger X = X$ extracts $E_1^\dagger X = X$. □

Lemma 3.62 (Reverse inclusion of adjoint fixed points). [adjoint_blockedTransferMap_apply_of_adjoint_transferMap_apply](#) ✓ Stmt OK *every adjoint fixed point of the unblocked transfer map E_1 is an adjoint fixed point of every blocked transfer map E_n .*

Proof. ✓ Proof OK Induction on n using $E_{n+1} = E_n \circ E_1$ and $(A \circ B)^\dagger = B^\dagger \circ A^\dagger$; the base case is $E_0 = \text{id}$. This step does not require the algebra-structure data. Combined with Theorem 3.61 this yields the fixed-point equality $\text{Fix}(E_n^\dagger) = \text{Fix}(E_1^\dagger)$ at every positive blocking size under the algebra-structure predicate. $\square$

Remark 3.63 (Why algebra data do not imply fusion data). The converse implication 3.52 $\Rightarrow$ 3.47 requires a stronger hypothesis than the algebra-structure formulation above. Theorem 3.61 already gives the strongest available consequence of the algebra-structure data – exclusion of *finite order* (root-of-unity) peripheral eigenvalues of the adjoint transfer map, since any such eigenvalue would produce a new adjoint fixed point at some positive blocking size. This does *not*, however, exclude unit-modulus eigenvalues of irrational phase, nor force idempotence of the transfer map, so ergodic quantum channels with a strict spectral gap on the 1-complement still satisfy the predicate without being RFPs. For instance, on $M_D(\mathbb{C})$ with $0 < \varepsilon < 1$ take

$$E(X) = \varepsilon X + (1 - \varepsilon) \Pi_{\text{diag}}(X),$$

where Π_{diag} is the projection that zeroes the off-diagonal entries of X ; then $\Pi_{\text{diag}} \circ \Pi_{\text{diag}} = \Pi_{\text{diag}}$, the channel E has peripheral spectrum $\{1\}$ and diagonal fixed-point algebra at every blocking size, yet $E \circ E \neq E$. Closing the converse therefore requires strengthening the predicate to the coefficient formulation of [CPGSV17a, Theorem IV.13 (ii)]: the positive diagonal matrices $\chi_{\alpha, \beta, \gamma}$ of Appendix C.3 and the coefficient identity $c_{\alpha, \beta, \gamma}^{(L)} = \text{tr}(\chi_{\alpha, \beta, \gamma}^L)$ of Appendix C.4. The scalar Newton–Girard power-sum identity needed for the latter step is already available as Theorem 13.29.

3.10.1 Diagonal χ -matrices and the trace-power formula

The following statements isolate the form of [CPGSV17a, Theorem IV.13 (ii)]: the structure coefficients of the length- L operator algebra are trace-powers of positive diagonal matrices $\chi_{\alpha, \beta, \gamma}$. They state the coefficient identity and the elementary trace-power identity for diagonal matrices.

Definition 3.64 (Diagonal χ family). DiagonalChiFamily ✓ Stmt OK A family of diagonal complex matrices $\chi_{\alpha, \beta, \gamma}$ indexed by ordered triples drawn from a common index type, each of a specified size. This is the paper’s $\chi_{\alpha, \beta, \gamma}$ in Theorem IV.13(ii).

Theorem 3.65 (Trace of χ^L equals the sum of L -th powers of entries). trace_matrix_pow ✓ Stmt OK For every triple (α, β, γ) , if $\chi_{\alpha, \beta, \gamma}$ has size $r_{\alpha, \beta, \gamma}$, then for every $L \geq 0$,

$$\text{tr}(\chi_{\alpha, \beta, \gamma}^L) = \sum_{k=1}^{r_{\alpha, \beta, \gamma}} \chi_{\alpha, \beta, \gamma, k}^L,$$

where $\chi_{\alpha, \beta, \gamma, k}$ are the diagonal entries.

Proof. ✓ Proof OK Diagonal matrices raise to powers entrywise and the trace of a diagonal matrix is the sum of its entries. $\square$

Definition 3.66 (Trace-power form of a structure-coefficient family). HasChiTracePowerForm ✓ Stmt OK A binary compatibility predicate between an abstract structure-coefficient family $c_{\alpha, \beta, \gamma}^{(L)}$ and a diagonal χ family: the pair (c, χ) is said to be in *trace-power form* when

$$c_{\alpha, \beta, \gamma}^{(L)} = \text{tr}(\chi_{\alpha, \beta, \gamma}^L) = \sum_{k=1}^{r_{\alpha, \beta, \gamma}} \chi_{\alpha, \beta, \gamma, k}^L$$

for every $L \geq 0$ and every triple (α, β, γ) , where $r_{\alpha, \beta, \gamma}$ is the size of $\chi_{\alpha, \beta, \gamma}$. This is the formal analogue of the target identity of [CPGSV17a, Theorem IV.13 (ii)]; the existential version—“there exists χ for which (c, χ) has trace-power form”—is obtained by quantifying over χ .

3.11 Commuting form and GSNNCH

Definition 3.67 (Commuting-form data for an MPDO). [CommutingFormData] ✓ Stmt OK For a fixed chain length $N \geq 2$, this consists of a positive semidefinite two-site operator B such that its translated copies on the periodic chain commute pairwise.

Definition 3.68 (Commuting-form property). [HasCommutingForm] ✓ Stmt OK An MPO tensor has the commuting-form property if for every $N \geq 2$ its N -site operator is a positive scalar multiple of the product of the translated copies of some commuting two-site operator B .

Definition 3.69 (GSNNCH). [IsGSNNCH] ✓ Stmt OK An MPO tensor is a Gibbs state of a nearest-neighbor commuting Hamiltonian if every finite-chain operator admits the equivalent positive-operator presentation $\rho^{(N)} = c \prod_i B_{i, i+1}$ with $c > 0$ and commuting nearest-neighbor factors. The projector-limit convention in [CPGSV17a] identifies this with the exponential Gibbs form.

Theorem 3.70 (GSNNCH iff commuting form). [isGSNNCH_iff_hasCommutingForm] ✓ Stmt OK For an MPO tensor, the GSNNCH condition is equivalent to the existence of commuting-form data on every finite chain.

Proof. ✓ Proof OK A positive normalization constant converts commuting-form data into a GSNNCH witness, and every GSNNCH witness remembers its underlying commuting-form data. □

Definition 3.71 (GSNNCH with zero correlation length). [IsGSNNCHWithZCL] ✓ Stmt OK This is the conjunction of the GSNNCH condition with MPO zero correlation length, matching item (iii) of the simple-MPDO equivalence in [CPGSV17a].

Theorem 3.72 (GSNNCH with zero correlation length iff commuting form with zero correlation length). [isGSNNCHWithZCL_iff_hasCommutingForm_and_isZCL] ✓ Stmt OK The GSNNCH–ZCL branch is equivalent to the conjunction of the commuting-form property with MPO zero correlation length.

Proof. ✓ Proof OK Unfold the definition of GSNNCH with zero correlation length and apply Theorem 3.70. □

Definition 3.73 (η -local commuting-form data). [EtaLocalStructureData] ✓ Stmt OK This defines the post-extraction form of Appendix C.2: a single translation-invariant positive two-site bond whose translated copies commute on every periodic chain and realize the MPO at every finite length.

Theorem 3.74 (η -local structure implies commuting form). [hasCommutingForm_of_etaLocalStructure] ✓ Stmt OK Once the explicit neighboring operators $\eta_{k, h}$ have been assembled into the data above, the global commuting-form property follows.

Proof. ✓ Proof OK For each $N \geq 2$, take the commuting-form datum given by these data. □

Remark 3.75 (Remaining entropy-side construction). The remaining entropy-side theorem is to construct the local data needed for Theorem 3.74 from the SAL and ZCL hypotheses. In particular, one must extract explicit neighboring operators $\eta_{k, h}$ from the Hayashi Markov decomposition. Once these operators are available, Theorem 3.74 yields the commuting-form conclusion and hence the GSNNCH branch of the simple-MPDO equivalence.

3.12 Blocked-RFP construction for simple MPDOs

Definition 3.76 (Local simple-MPDO structure data). SimpleMPDOLocalStructureData ✓ Stmt OK This structure collects the local Appendix C.2 data: a normalized three-site reduced state with equality in strong subadditivity, the resulting local η -structure, and the rank-one conclusion for the auxiliary primitive matrix T .

Definition 3.77 (Simple-MPDO blocked-RFP data). SimpleMPDOBlockedRFPData ✓ Stmt OK For an MPO tensor K , this consists of the local Appendix C.2 data together with a compatibility relation attaching those local data to K , the commuting-form property, and zero correlation length. The local component records the hypotheses used in the entropy-side route to commuting form, while the blocked-RFP consequences below use only commuting form and zero correlation length.

Theorem 3.78 (Zero correlation length gives a blocked fusion-isometry witness). structural_implies_rfp_blocked ✓ Stmt OK *For an MPO tensor K with zero correlation length, then it has a blocked fusion-isometry witness at size 2 in the sense of Definition 3.46.*

Proof. ✓ Proof OK Zero correlation length is the MPDO RFP condition. Applying the converse direction of Theorem 3.49 to that RFP witness yields fusion-isometry data at every positive blocked size; evaluating it at size 2 produces the blocked witness. □

Theorem 3.79 (Simple-MPDO RFP chain from commuting form and ZCL). simple_mpdo_rfp_chain ✓ Stmt OK *From the commuting-form property and zero correlation length one simultaneously recovers the GSNNCH-with-ZCL branch of Theorem 4.9, the blocked fusion-isometry witness at size 2, and the transfer-map fusion formulation of MPDO RFP.*

Proof. ✓ Proof OK The commuting-form and zero-correlation-length hypotheses give the GSNNCH-ZCL branch via Theorem 3.72. The blocked witness is Theorem 3.78. The full transfer-map fusion formulation follows by applying the converse direction of Theorem 3.49 to zero correlation length, which is the MPDO RFP condition. □

Remark 3.80 (Remaining implication for the GSNNCH-ZCL criterion). Not ready The implication from SAL and ZCL alone to the conclusion of Appendix C.2 requires a derivation of the commuting-form property from the local simple-MPDO structure of Lemmas C.3–C.4. Once that derivation is available, Theorem 3.79 gives the GSNNCH-ZCL branch and the blocked fusion-isometry witness.

Chapter 4

The Single-Block Fundamental Theorem

This chapter proves that if an injective MPS tensor A generates the same MPV family as another tensor B of the same bond dimension, then A and B are gauge equivalent. The proof is purely algebraic and does not use the later overlap or normal-form reductions. The trace-pairing maps introduced below are auxiliary devices attached to the chosen generating families $\{A^i\}$ and $\{B^i\}$; they serve only to construct the linear map that is later shown to be a gauge transform.

4.1 Trace pairing

Theorem 4.1 (Nondegeneracy of the trace pairing). trace_mul_right_eq_zero_iff ✓ Stmt OK For $M \in M_D(\mathbb{C})$, $\text{tr}(MN) = 0$ for all $N \in M_D(\mathbb{C})$ if and only if $M = 0$.

Proof. ✓ Proof OK Testing against the matrix units E_{ji} (the matrix with 1 in position (j, i) and 0 elsewhere) gives $\text{tr}(E_{ji}M) = M_{ij}$. Hence if $\text{tr}(MN) = 0$ for all N , then every entry of M vanishes. □

Lemma 4.2 (Same MPV implies trace agreement). trace_evalWord ✓ Stmt OK If A and B generate the same MPV family, then $\text{tr}(A^w) = \text{tr}(B^w)$ for every word w .

Proof. ✓ Proof OK Viewing the word w as a configuration of length $|w|$, Definition 2.4 gives the corresponding MPV coefficient as the trace coefficient $c_w(A) = \text{tr}(A^w)$. The same-MPV hypothesis gives equality at every system size. □

Definition 4.3 (Trace pairing map). traceMulRightPi ✓ Stmt OK For a tensor A , attach to the chosen family $\{A^i\}$ the auxiliary linear map $\Phi_A : M_D(\mathbb{C}) \rightarrow \mathbb{C}^d$ defined by

$$\Phi_A(M)_i = \text{tr}(M \cdot A^i).$$

Theorem 4.4 (Injectivity of the trace pairing map). traceMulRightPi_ker_eq_bot ✓ Stmt OK If A is injective, then $\ker \Phi_A = \{0\}$.

Proof. ✓ Proof OK If $\Phi_A(M) = 0$, then $\text{tr}(M \cdot A^i) = 0$ for all i . Since the $\{A^i\}$ span $M_D(\mathbb{C})$, this gives $\text{tr}(MN) = 0$ for all N , so $M = 0$ by Theorem 4.1. □

Theorem 4.5 (Nonvanishing trace on injective tensors). [trace_ne_zero_of_injective] ✓ Stmt OK If $D \geq 1$, A is injective, A and B generate the same MPV family, and $B^i = 0$ for all i , then we have a contradiction.

Proof. ✓ Proof OK If $B^i = 0$ for all i , then Lemma 4.2 gives $\text{tr}(A^w) = 0$ for every word w , in particular for all length-1 words. Since the $\{A^i\}$ span $M_D(\mathbb{C})$, the trace functional vanishes identically, contradicting $\text{tr}(\mathbb{1}_D) = D \neq 0$. $\square$

4.2 Linear extension

Theorem 4.6 (Existence and uniqueness of the linear extension). [linearExtension_exists_unique] ✓ Stmt OK If A is injective and A, B generate the same MPV family, then there exists a unique linear map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ satisfying $T(A^i) = B^i$ for all i .

Proof. ✓ Proof OK Write $\ell_A : \mathbb{C}^d \rightarrow M_D(\mathbb{C})$ for the map that sends a coefficient vector c to $\sum_i c_i A^i$ (linear combination with the A^i). Applying Lemma 4.2 to all words of length 2 shows $\Phi_A \circ \ell_A = \Phi_B \circ \ell_B$. Since A is injective, ℓ_A is surjective, giving $\text{range}(\Phi_A) \subseteq \text{range}(\Phi_B)$. By Theorem 4.4, $\ker \Phi_A = \{0\}$, so rank-nullity gives $\dim(\text{range}(\Phi_A)) = \dim(M_D(\mathbb{C}))$. Hence $\dim(\text{range}(\Phi_B)) \geq \dim(M_D(\mathbb{C}))$, and rank-nullity again gives $\ker \Phi_B = \{0\}$. A left inverse g of Φ_B gives $T := g \circ \Phi_A$, and uniqueness follows because the $\{A^i\}$ span $M_D(\mathbb{C})$. $\square$

Theorem 4.7 (Multiplicativity of the linear extension). [linearExtension_mul] ✓ Stmt OK If A is injective and A, B generate the same MPV family, and $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is the unique linear map satisfying $T(A^i) = B^i$ for all i , then $T(MN) = T(M)T(N)$ for all $M, N \in M_D(\mathbb{C})$.

Proof. ✓ Proof OK **Step 1** ($\Phi_B \circ T = \Phi_A$). Applying Lemma 4.2 to words of length 2 gives $\Phi_B(T(A^i)) = \Phi_B(B^i) = \Phi_A(A^i)$ for all i . Since the $\{A^i\}$ span $M_D(\mathbb{C})$, $\Phi_B \circ T = \Phi_A$.

Step 2 ($T(A^i A^j) = B^i B^j$). Lemma 4.2 applied to the word (i, j, k) gives $\text{tr}(A^i A^j A^k) = \text{tr}(B^i B^j B^k)$ for all i, j, k . So $\Phi_B(T(A^i A^j))_k = \Phi_A(A^i A^j)_k = \text{tr}(A^i A^j A^k) = \text{tr}(B^i B^j B^k) = \Phi_B(B^i B^j)_k$. The same rank-nullity argument used in the proof of Theorem 4.6 gives $\ker \Phi_B = \{0\}$. Hence $T(A^i A^j) = B^i B^j$.

Step 3 (bilinear extension). For fixed j , the maps $M \mapsto T(M \cdot A^j)$ and $M \mapsto T(M) \cdot B^j$ agree on the spanning set $\{A^i\}$, hence on all of $M_D(\mathbb{C})$. Applying this once more with j replaced by a general matrix gives full multiplicativity. $\square$

4.3 Simplicity and Skolem–Noether

Theorem 4.8 (Simplicity of $M_D(\mathbb{C})$). [linear_mul_endomorphism_bijective] ✓ Stmt OK A nonzero multiplicative $\mathbb{C}$ -linear endomorphism of $M_D(\mathbb{C})$ is bijective.

Proof. ✓ Proof OK The kernel of a multiplicative linear map is a two-sided ideal. Since $M_D(\mathbb{C})$ is a simple ring (it has no nontrivial two-sided ideals), the kernel is either $\{0\}$ or $M_D(\mathbb{C})$. The latter would force the map to be zero, contradicting the hypothesis. Hence the map is injective, and surjectivity follows from finite-dimensionality. $\square$

Theorem 4.9 (Skolem–Noether). [skolemNoether_matrix] ✓ Stmt OK Every $\mathbb{C}$ -algebra automorphism of $M_D(\mathbb{C})$ is inner: for any $f \in \text{Aut}_{\mathbb{C}\text{-alg}}(M_D(\mathbb{C}))$, there exists $X \in \text{GL}_D(\mathbb{C})$ such that $f(M) = XMX^{-1}$ for all M .

Proof. ✓ Proof OK Via the canonical identification $M_D(\mathbb{C}) \cong \text{End}_{\mathbb{C}}(\mathbb{C}^D)$, the automorphism f induces an automorphism of $\text{End}_{\mathbb{C}}(\mathbb{C}^D)$. Every algebra automorphism of $\text{End}(V)$ is conjugation by an invertible linear map (this is a standard result in the theory of central simple algebras). Translating back to matrices gives the invertible X . $\square$

Lemma 4.10 (Promotion to algebra homomorphism). linearMapToAlgHom ✓ Stmt OK If $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is a multiplicative surjective $\mathbb{C}$ -linear map, then $T(\mathbb{1}) = \mathbb{1}$. Equivalently, the same underlying function defines a $\mathbb{C}$ -algebra homomorphism.

Proof. ✓ Proof OK By surjectivity, there exists X with $T(X) = \mathbb{1}$. Then $T(\mathbb{1}) = T(X) \cdot T(\mathbb{1}) = T(X \cdot \mathbb{1}) = T(X) = \mathbb{1}$, using multiplicativity and $T(X) = \mathbb{1}$. With $T(\mathbb{1}) = \mathbb{1}$ established, T preserves the unit and hence is a $\mathbb{C}$ -algebra homomorphism. $\square$

4.4 Proof of the single-block theorem

Theorem 4.11 (Single-block Fundamental Theorem). fundamentalTheorem_singleBlock ✓ Stmt OK Let A be an injective MPV tensor and let B be an MPV tensor of the same bond dimension. If A and B generate the same MPV family, then they are gauge equivalent: there exists $X \in \text{GL}_D(\mathbb{C})$ such that $B^i = XA^iX^{-1}$ for all i .

Proof. ✓ Proof OK By Theorem 4.6, there is a unique linear T with $T(A^i) = B^i$. By Theorem 4.7, T is multiplicative. T is nonzero: if $T = 0$ then $B^i = 0$ for all i , contradicting Theorem 4.5. By Theorem 4.8, T is bijective. By Lemma 4.10, T is a $\mathbb{C}$ -algebra automorphism. Theorem 4.9 gives an invertible X with $T(M) = XMX^{-1}$, and evaluating at $M = A^i$ yields $B^i = XA^iX^{-1}$. $\square$

4.5 Alternative fixed-length non-TI setup

The proof above treats a single translation-invariant tensor and equality of the full MPV family for all system sizes. The alternative algebraic proof of [MGRPG⁺18] instead keeps a fixed periodic chain length and allows the local tensors to vary from site to site. The first step is therefore a separate setup for finite non-translation-invariant chains.

Definition 4.12 (Periodic non-TI chain data). MPSPChainTensor coeff SameState IsInjective GaugeEquiv ✓ Stmt OK Fix a chain length n . A *non-translation-invariant periodic MPS chain* consists of sitewise tensors

$$A_k = \{A_k^i\}_{i=0}^{d-1}, \quad k = 0, \dots, n-1,$$

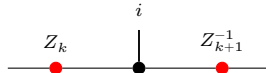
where each $A_k^i \in M_D(\mathbb{C})$. For a basis configuration $\sigma = (\sigma_0, \dots, \sigma_{n-1})$, its coefficient is

$$c_{A,\sigma} := \text{tr}(A_0^{\sigma_0} A_1^{\sigma_1} \dots A_{n-1}^{\sigma_{n-1}}).$$

Two such chains generate the *same chain state* if these coefficients agree for every σ . The chain is *injective* if each local tensor A_k is injective in the usual single-site sense. Finally, two chains are *cyclically gauge equivalent* if there are invertible bond matrices $Z_0, \dots, Z_{n-1}$ such that

$$B_k^i = Z_k A_k^i Z_{k+1}^{-1}$$

for every site k and physical index i , with indices understood modulo n so that $Z_n = Z_0$. Locally, this gauge relation is represented by



where the black node denotes the site tensor named in the surrounding formula and the two red side nodes denote the bond gauges on the neighboring virtual legs.

Lemma 4.13 (Cyclic gauge equivalence is an equivalence relation). refl symm trans ✓ Stmt OK *On fixed-length periodic chains, cyclic gauge equivalence is reflexive, symmetric, and transitive.*

Proof. ✓ Proof OK Reflexivity uses the identity gauge on every bond. Symmetry inverts each bond gauge, and transitivity multiplies the gauges pointwise along the chain. □

Chapter 5

Quantum Channels and Positive Maps

This chapter develops the theory of positive maps and quantum channels on matrix algebras, following [Wol12].

5.1 Positive maps and channels

Definition 5.1 (Positive map). IsPositiveMap ✓ Stmt OK A linear map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is *positive* if $E(X) \geq 0$ whenever $X \geq 0$, where $X \geq 0$ means that X is positive semidefinite.

Remark 5.2. ✓ Stmt OK We use the Loewner order on matrices: $X \leq Y$ means $Y - X \geq 0$.

Definition 5.3 (Trace-preserving map). IsTracePreservingMap ✓ Stmt OK A linear map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is *trace-preserving* if $\text{tr}(E(X)) = \text{tr}(X)$ for all X .

Definition 5.4 (Completely positive map). IsCPMap ✓ Stmt OK A linear map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is *completely positive* (CP) if it admits a *Kraus representation*: there exist operators $\{K_i\}_{i=0}^{r-1}$ with $K_i \in M_D(\mathbb{C})$ such that, for every $X \in M_D(\mathbb{C})$,

$$E(X) = \sum_{i=0}^{r-1} K_i X K_i^\dagger$$

We use the Kraus characterisation because it matches the later Kadison–Schwarz arguments. By Choi’s theorem, this is equivalent to $E \otimes \text{id}_n$ being positive for all n .

Theorem 5.5 (CP maps are positive). isPositiveMap ✓ Stmt OK *Every completely positive map is positive.*

Proof. ✓ Proof OK If $X \geq 0$, then each summand $K_i X K_i^\dagger \geq 0$ (conjugation preserves positive semidefiniteness), so the sum is PSD. □

Theorem 5.6 (Channels are positive). pos ✓ Stmt OK *Every quantum channel is positive.*

Proof. ✓ Proof OK A quantum channel is completely positive by definition, so Theorem 5.5 applies. □

Theorem 5.7 (Positive maps preserve Hermiticity). map_isHermitian ✓ Stmt OK If E is positive and $X = X^\dagger$, then $E(X) = E(X)^\dagger$.

Proof. ✓ Proof OK Decompose $X = X_+ - X_-$ into its positive and negative parts. Positivity makes $E(X_+)$ and $E(X_-)$ positive semidefinite, hence Hermitian, so their difference $E(X)$ is Hermitian. □

Definition 5.8 (Quantum channel). IsChannel ✓ Stmt OK A linear map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is a *quantum channel* if it is both completely positive and trace-preserving (CPTP), following [Wol12].

Definition 5.9 (Density matrices). densityMatrices mem_densityMatrices ✓ Stmt OK The set of *density matrices* in $M_D(\mathbb{C})$ is

$$\mathcal{D}_D = \{\rho \in M_D(\mathbb{C}) : \rho \geq 0, \text{tr}(\rho) = 1\}.$$

Theorem 5.10 (Density matrices are compact). densityMatrices_isCompact ✓ Stmt OK $\mathcal{D}_D$ is compact (in the entry-wise topology on $M_D(\mathbb{C})$).

Proof. ✓ Proof OK The PSD cone is closed (preimage of the closed nonnegative cone under continuous quadratic forms), and if $\rho \geq 0$ with $\text{tr}(\rho) = 1$ then each entry satisfies $|\rho_{ij}|^2 \leq \rho_{ii}\rho_{jj} \leq 1$: the diagonal entries are nonnegative and sum to 1, and the off-diagonal bound is the PSD Cauchy–Schwarz inequality. Hence the matrix entries are uniformly bounded, and Heine–Borel gives compactness. □

Theorem 5.11 (Density matrices are convex). densityMatrices_isConvex ✓ Stmt OK $\mathcal{D}_D$ is convex.

Proof. ✓ Proof OK Convex combination of PSD matrices is PSD, and trace is linear. □

Theorem 5.12 (Density matrices are nonempty). densityMatrices_nonempty ✓ Stmt OK For $D \geq 1$, $\mathcal{D}_D \neq \emptyset$.

Proof. ✓ Proof OK $\frac{1}{D}\mathbb{1}_D$ is a density matrix. □

Theorem 5.13 (Channels preserve density matrices). map_densityMatrices ✓ Stmt OK If E is a quantum channel, then $E(\mathcal{D}_D) \subseteq \mathcal{D}_D$.

Proof. ✓ Proof OK Complete positivity implies positivity (Theorem 5.5), so $E(\rho) \geq 0$; trace preservation gives $\text{tr}(E(\rho)) = \text{tr}(\rho) = 1$. □

5.2 Irreducibility

Definition 5.14 (Orthogonal projection). IsOrthogonalProjection ✓ Stmt OK A matrix $P \in M_D(\mathbb{C})$ is an *orthogonal projection* if $P = P^\dagger$ and $P^2 = P$.

Definition 5.15 (Irreducible map). IsIrreducibleMap ✓ Stmt OK A linear map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is *irreducible* if whenever P is an orthogonal projection satisfying $E(PM_D(\mathbb{C})P) \subseteq PM_D(\mathbb{C})P$, then $P = 0$ or $P = \mathbb{1}$. This is [Wol12, Thm. 6.2, condition (1)]. The definition applies to any linear map; complete positivity is not required.

Theorem 5.16 (Injectivity implies irreducibility). injective_implies_irreducibleCP ✓ Stmt OK If A is an injective MPS tensor, then its transfer map $\mathcal{E}_A$ is irreducible.

Proof. ✓ Proof OK If P is an invariant projection for $\mathcal{E}_A$, then $(\mathbb{1} - P)A^iP = 0$ for all i . Since the $\{A^i\}$ span $M_D(\mathbb{C})$, $(\mathbb{1} - P)MP = 0$ for all M , so $P = 0$ or $P = \mathbb{1}$. $\square$

Remark 5.17 (Transfer map is CP). transferMap_isCPMap ✓ Stmt OK The transfer map $\mathcal{E}_A(X) = \sum_i A^i X (A^i)^\dagger$ is completely positive: it is already written in Kraus form, with Kraus operators $\{A^i\}$.

Theorem 5.18 (Transfer map is a channel). transferMap_isChannel ✓ Stmt OK If $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, then the transfer map $\mathcal{E}_A(X) = \sum_i A^i X (A^i)^\dagger$ is a quantum channel (CPTP).

Proof. ✓ Proof OK Complete positivity is the observation of Remark 5.17. Trace preservation: $\text{tr}(\mathcal{E}_A(X)) = \sum_i \text{tr}(A^i X (A^i)^\dagger) = \text{tr}((\sum_i (A^i)^\dagger A^i)X) = \text{tr}(X)$ by cyclicity and the normalization hypothesis. $\square$

5.3 Cesàro fixed-point theorem

Definition 5.19 (Cesàro mean). cesaroMean ✓ Stmt OK The *Cesàro mean* of a linear map E at order N , applied to a starting matrix ρ_0 , is

$$\sigma_N = \frac{1}{N} \sum_{n=0}^{N-1} E^n(\rho_0).$$

Theorem 5.20 (Cesàro telescope). cesaroMean_telescope ✓ Stmt OK $E(\sigma_N) - \sigma_N = \frac{1}{N}(E^N(\rho_0) - \rho_0)$.

Proof. ✓ Proof OK Telescoping: $E(\sigma_N) = \frac{1}{N} \sum_{n=1}^N E^n(\rho_0)$ and subtracting σ_N cancels all but the first and last terms. $\square$

Theorem 5.21 (Hermitian fixed-point decomposition). posSemidef_parts_of_hermitian_fixedPoint ✓ Stmt OK Let E be a quantum channel and $X = E(X)$ a Hermitian fixed point with decomposition $X = P_+ - P_-$ into orthogonal positive and negative parts ($P_\pm \geq 0$, $\text{tr}(P_+ P_-) = 0$). Then $E(P_+) = P_+$ and $E(P_-) = P_-$. This is [Wol12, Prop. 6.8].

Proof. ✓ Proof OK Let Q be the support projection of P_+ . Trace preservation and positivity give $\text{tr}(QE(P_-)) = 0$ and $E(P_-) \geq 0$, so $QE(P_-) = 0$. The fixed-point equation $E(P_+) - E(P_-) = P_+ - P_-$ then identifies $E(P_\pm) = P_\pm$. $\square$

Theorem 5.22 (Cesàro fixed-point theorem). exists_posSemidef_fixedPoint ✓ Stmt OK Every quantum channel $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ (with $D \geq 1$) has a nonzero positive semidefinite fixed point: there exists $\rho \geq 0$, $\rho \neq 0$, with $E(\rho) = \rho$.

Proof. ✓ Proof OK Start with any $\rho_0 \in \mathcal{D}_D$ (Theorem 5.12). Each Cesàro mean σ_N lies in $\mathcal{D}_D$ (by convexity and channel preservation). By compactness (Theorem 5.10), extract a convergent subsequence $\sigma_{N_k} \rightarrow \rho$. The telescope identity (Theorem 5.20) gives $\|E(\sigma_N) - \sigma_N\| = \frac{1}{N} \|E^N(\rho_0) - \rho_0\| \rightarrow 0$, so $E(\rho) = \rho$ by continuity. In [Wol12, Thm. 6.11], existence is proved via Brouwer's fixed-point theorem; we use the Cesàro argument instead. $\square$

5.4 Fixed-point projection

Definition 5.23 (Fixed-point projection). [fixedPointProj](#) ✓ Stmt OK Given a linear map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ and a fixed point ρ with $E(\rho) = \rho$ and $\text{tr}(\rho) \neq 0$, the *fixed-point projection* is the rank-one map

$$P(X) = \frac{\text{tr}(X)}{\text{tr}(\rho)} \rho.$$

This projects onto the span of ρ along the kernel of the trace functional. When the fixed point is unique up to scaling, this span is the full fixed-point space.

Theorem 5.24 (Power decomposition). [pow_eq_fixedPointProj_add_compl_pow](#) ✓ Stmt OK If E is trace-preserving and $E(\rho) = \rho$, then for $n \geq 1$,

$$E^n = P + (E - P)^n,$$

where P is the fixed-point projection.

Proof. ✓ Proof OK P is idempotent ($P^2 = P$), the fixed-point relation $E(\rho) = \rho$ gives $E \circ P = P$, and trace preservation gives $P \circ E = P$. These imply $(E - P) \circ P = 0$ and $P \circ (E - P) = 0$, so in the binomial expansion of $(P + (E - P))^n$ all cross terms vanish, leaving $E^n = P^n + (E - P)^n = P + (E - P)^n$. □

5.5 Peripheral spectrum and primitivity

Definition 5.25 (Peripheral spectrum). [peripheralSpectrum](#) ✓ Stmt OK The *peripheral spectrum* of a bounded linear operator T is the set of spectral values μ with $|\mu|$ equal to the spectral radius of T .

Definition 5.26 (Peripheral eigenvalues). [peripheralEigenvalues](#) ✓ Stmt OK The *peripheral eigenvalues* of a linear map E on a finite-dimensional space are the eigenvalues λ on the unit circle: $|\lambda| = 1$. For a channel, the spectral radius is 1 [Wol12, Prop. 6.1], so these coincide with the eigenvalues of maximal modulus.

Note: The condition $|\lambda| = 1$ is appropriate for channels, since the spectral radius of a channel is 1. For a general linear map with spectral radius $r \neq 1$ the condition would generalise to $|\lambda| = r$.

Theorem 5.27 (1 is a peripheral eigenvalue). [one_mem_peripheralEigenvalues](#) ✓ Stmt OK If E has a nonzero fixed point $\rho \neq 0$ with $E(\rho) = \rho$, then 1 is a peripheral eigenvalue of E .

Proof. ✓ Proof OK ρ is an eigenvector with eigenvalue 1, and $|1| = 1$. □

Remark 5.28 (Peripheral eigenvalues lie on the unit circle). [peripheralEigenvalues_subset_unit_circle](#) ✓ Stmt OK Every peripheral eigenvalue λ satisfies $|\lambda| = 1$. This is simply the defining condition in Definition 5.26.

Theorem 5.29 (Finiteness of peripheral eigenvalues). [peripheralEigenvalues_finite](#) ✓ Stmt OK On a finite-dimensional space, the set of peripheral eigenvalues is finite.

Proof. ✓ Proof OK Peripheral eigenvalues are a subset of all eigenvalues, which is a finite set in finite dimensions. □

Theorem 5.30 (Power-stable peripheral eigenvalues are roots of unity). [peripheral_isRootOfUnity_of_pow_eigenvalue](#) ✓ Stmt OK If E be a linear endomorphism on a finite-dimensional space. If μ is a peripheral eigenvalue of E and μ^n is an eigenvalue of E for every $n \geq 1$, then μ is a root of unity.

Proof. ✓ Proof OK Since μ^n is an eigenvalue for every n , the pigeonhole principle on the finite eigenvalue set gives $\mu^a = \mu^b$ for some $a \neq b$, hence $\mu^{|a-b|} = 1$. □

Theorem 5.31 (Peripheral powers imply root of unity). peripheral_isRootOfUnity_of_closed_powers ✓ Stmt OK *If the set of peripheral eigenvalues of a linear endomorphism E on a finite-dimensional space is closed under powers ($\mu \in \text{peripheral}(E)$ and $n \geq 1$ imply $\mu^n \in \text{peripheral}(E)$), then every peripheral eigenvalue is a root of unity.*

Proof. ✓ Proof OK The closure hypothesis provides $\mu^n \in \text{peripheral}(E)$ for all n , which in particular means μ^n is an eigenvalue. Theorem 5.30 then gives $\mu^p = 1$ for some $p > 0$. □

5.5.1 Periodicity removal by powering

Remark 5.32. ✓ Stmt OK The results in this subsection are purely spectral: they use only finiteness of a set of complex numbers and the spectral mapping theorem for powers. In particular, they do not rely on complete positivity. For transfer maps, replacing a tensor by its p -blocked tensor replaces $\mathcal{E}_A$ by $\mathcal{E}_A^p$, so this powering step is the operator-theoretic form of blocking.

Lemma 5.33 (Common exponent for a finite set of roots of unity). exists_common_power_eq_one_of_finite ✓ Stmt OK *Let $s \subseteq \mathbb{C}$ be a finite set. Assume that for every $\mu \in s$ there exists an exponent $p_\mu > 0$ with $\mu^{p_\mu} = 1$. Then there exists $p > 0$ such that $\mu^p = 1$ for all $\mu \in s$.*

Proof. ✓ Proof OK Take p to be the least common multiple of the exponents p_μ . □

Lemma 5.34 (Powering collapses peripheral eigenvalues). peripheralEigenvalues_pow_eq_singleton ✓ Stmt OK *Let $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a linear map, and assume that E has a nonzero fixed point $\rho \neq 0$. Let $p > 0$. If every peripheral eigenvalue μ of E satisfies $\mu^p = 1$, then $\text{peripheral}(E^p) = \{1\}$.*

Proof. ✓ Proof OK The fixed point ρ gives $1 \in \text{peripheral}(E^p)$. For the reverse inclusion, let $\nu \in \text{peripheral}(E^p)$. Spectral mapping gives $\nu = \mu^p$ for some $\mu \in \sigma(E)$. From $|\nu| = 1$ and $p > 0$ we obtain $|\mu| = 1$, hence $\mu \in \text{peripheral}(E)$. The hypothesis $\mu^p = 1$ then forces $\nu = 1$. □

For a normalized MPS tensor, the transfer map is naturally trace-preserving but need not be unital. In general one should not expect a single gauge choice to make it both unital and trace-preserving.

Remark 5.35 (Bi-canonical special case). ✓ Stmt OK When a Kraus map happens to be both unital and trace-preserving (the “bi-canonical” case), the Kadison–Schwarz equality at peripheral eigenvectors (Theorem 5.41) shows directly that the peripheral eigenvalues are closed under powers, and hence are roots of unity by Theorem 5.31. This is the simplest argument, but it requires both normalizations simultaneously. The more general adjoint-fixed-point formulation below covers the case where only unitality and a positive definite adjoint fixed point are available.

5.5.2 Kadison–Schwarz input for peripheral and fixed-point arguments

Definition 5.36 (Kraus maps, adjoints, and normalizations). krausMap krausAdjointMap IsUnitalKraus IsTPKraus ✓ Stmt OK *Let operators $\{K_i\}_{i=0}^{d-1} \subseteq M_D(\mathbb{C})$, the associated Kraus map and adjoint Kraus map are*

$$E(X) = \sum_i K_i X K_i^\dagger, \quad E^*(X) = \sum_i K_i^\dagger X K_i.$$

The Kraus family is *unital* when $\sum_i K_i K_i^\dagger = \mathbb{1}$ and *trace-preserving* when $\sum_i K_i^\dagger K_i = \mathbb{1}$.

Theorem 5.37 (Kadison–Schwarz inequality). [kadison_schwarz](#) ✓ Stmt OK Let $E(X) = \sum_i K_i X K_i^\dagger$ be a unital Kraus map. Then for every $X \in M_D(\mathbb{C})$,

$$E(X^\dagger X) \geq E(X)^\dagger E(X).$$

This is [Wol12, Eq. (5.2)].

Proof. ✓ Proof OK Apply the unital completely positive map entrywise to the positive 2×2 block matrix $\begin{pmatrix} X^\dagger X & X^\dagger \\ X & \mathbb{1} \end{pmatrix}$ and take the Schur complement of the lower-right block. $\square$

Theorem 5.38 (Hilbert–Schmidt contraction). [hilbertSchmidt_contraction](#) ✓ Stmt OK If a Kraus map is both unital and trace-preserving, then

$$\mathrm{tr}(E(X)^\dagger E(X)) \leq \mathrm{tr}(X^\dagger X)$$

for all $X \in M_D(\mathbb{C})$.

Proof. ✓ Proof OK The Kadison–Schwarz gap is positive semidefinite by Theorem 5.37. Trace preservation then identifies the traces of its two terms. $\square$

Theorem 5.39 (KS gap decomposition). [ks_gap_eq_sum_squares](#) ✓ Stmt OK For a unital Kraus map,

$$E(X^\dagger X) - E(X)^\dagger E(X) = \sum_i R_i^\dagger R_i, \quad R_i := X K_i^\dagger - K_i^\dagger E(X).$$

Proof. ✓ Proof OK Expand the right-hand side, distribute the products, and use $\sum_i K_i K_i^\dagger = \mathbb{1}$ to recover the Kadison–Schwarz gap. $\square$

Theorem 5.40 (KS equality implies Kraus commutation). [kraus_commute_of_ks_equality](#) ✓ Stmt OK Let E be a unital Kraus map. If $E(X^\dagger X) = E(X)^\dagger E(X)$, then, for every i ,

$$X K_i^\dagger = K_i^\dagger E(X)$$

Proof. ✓ Proof OK The Kadison–Schwarz gap is a sum of positive semidefinite terms $\sum_i R_i^\dagger R_i$. If the gap vanishes, each R_i must vanish. $\square$

Theorem 5.41 (KS equality for peripheral eigenvectors). [ks_equality_of_peripheral_eigenvector](#) ✓ Stmt OK Let E be a Kraus map that is both unital and trace-preserving. If $E(X) = \mu X$ with $|\mu| = 1$, then

$$E(X^\dagger X) = E(X)^\dagger E(X).$$

Proof. ✓ Proof OK The Kadison–Schwarz gap is positive semidefinite by Theorem 5.37. Trace preservation and the relation $E(X) = \mu X$ with $|\mu| = 1$ show that its trace is zero, hence the gap vanishes. $\square$

Theorem 5.42 (Left multiplicative identity from KS equality). [multiplicative_domain_left](#) ✓ Stmt OK Let E be a unital Kraus map. If $E(X^\dagger X) = E(X)^\dagger E(X)$, then for every $Y \in M_D(\mathbb{C})$,

$$E(X^\dagger Y) = E(X)^\dagger E(Y).$$

Proof. ✓ Proof OK The commutation relation of Theorem 5.40 lets one move $X^\dagger$ past each Kraus operator inside the sum. $\square$

Definition 5.43 (Multiplicative domain). [multiplicativeDomain](#) ✓ Stmt OK The *multiplicative domain* of a Kraus map E is the set of matrices that are simultaneously left- and right-multiplicative for E .

Theorem 5.44 (Multiplicative domain is a $*$ -subalgebra). [multiplicativeDomainStarSubalgebra](#) ✓ Stmt OK For a unital Kraus map, the multiplicative domain is a $*$ -subalgebra of $M_D(\mathbb{C})$.

Proof. ✓ Proof OK The multiplicative identities are stable under addition, multiplication, and adjoint, and they contain the identity matrix. □

5.5.3 Peripheral closure via adjoint fixed point

The following results replace the trace-preservation hypothesis by the existence of a positive definite fixed point for the *adjoint* Kraus map, matching the weighted-trace argument in [CPGSV17a, Appendix A].

Lemma 5.45 (Peripheral closure via adjoint fixed point). [peripheralEigenvalues_pow_mem_of_irreducible_unital_of_adjoint_fixedPoint](#)
 $E(X) = \sum_i K_i X K_i^\dagger$ be a Kraus map that is unital ($\sum_i K_i K_i^\dagger = \mathbb{1}$), and assume:

1. the adjoint Kraus map $E^*(X) = \sum_i K_i^\dagger X K_i$ has a positive definite fixed point $\rho > 0$, and
2. E is irreducible.

Then $\text{peripheral}(E)$ is closed under powers: if $\mu \in \text{peripheral}(E)$ then $\mu^n \in \text{peripheral}(E)$ for every $n \in \mathbb{N}$.

Proof. ✓ Proof OK Let $E(X) = \mu X$ with $|\mu| = 1$, and set $G := E(X^\dagger X) - E(X)^\dagger E(X)$. By Kadison–Schwarz, $G \geq 0$. Since $E^*(\rho) = \rho$,

$$\text{tr}(\rho G) = \text{tr}(\rho E(X^\dagger X)) - \text{tr}(\rho E(X)^\dagger E(X)) = \text{tr}(E^*(\rho) X^\dagger X) - \text{tr}(\rho E(X)^\dagger E(X)) = \text{tr}(\rho X^\dagger X) - |\mu|^2 \text{tr}(\rho X^\dagger X) = 0.$$

Thus the trace of the Kadison–Schwarz gap against the adjoint fixed point vanishes. Because $\rho > 0$ and $G \geq 0$, this forces $G = 0$, so $E(X^\dagger X) = E(X)^\dagger E(X)$. Hence $X^\dagger X$ is a PSD fixed point, which is positive definite by irreducibility (Theorem 8.3). So X is invertible. By Theorem 5.40 (KS equality implies Kraus commutation), X commutes with each Kraus operator; iterating via the left multiplicative identity (Theorem 5.42) gives $E(X^n) = \mu^n X^n$, so μ^n is peripheral. □

Theorem 5.46 (Roots of unity via adjoint fixed point). [peripheral_isRootOfUnity_of_irreducible_unital_of_adjoint_fixedPoint](#)
 $E(X) = \sum_i K_i X K_i^\dagger$ be a Kraus map that is unital, and assume that the adjoint Kraus map $E^*(X) = \sum_i K_i^\dagger X K_i$ has a positive definite fixed point and that E is irreducible. Then every peripheral eigenvalue of E is a root of unity.

Proof. ✓ Proof OK Combine the closure-under-powers result (Lemma 5.45) with Theorem 5.31. □

Remark 5.47. ✓ Stmt OK Lemma 5.45 and Theorem 5.46 cover the bi-canonical (unital + TP) case as well: trace preservation gives $E^*(\mathbb{1}) = \mathbb{1}$, which is a positive definite fixed point of the adjoint. The adjoint-fixed-point formulation matches the argument in [CPGSV17a, Appendix A], which uses a faithful invariant state rather than bi-canonicity.

Definition 5.48 (Channel period). [channelPeriod](#) ✓ Stmt OK The *period* of a channel E is the number of peripheral eigenvalues (counted without multiplicity): $p(E) := |\text{peripheral}(E)|$.

Definition 5.49 (Primitive map). IsPrimitive ✓ Stmt OK A linear map is *primitive* if its only peripheral eigenvalue is $\lambda = 1$, i.e. $\text{peripheral}(E) = \{1\}$. For channels this corresponds to [Wol12, Thm. 6.7]. This definition applies to any linear endomorphism, not only to channels.

Theorem 5.50 (Primitivity equals period one). isPrimitive_iff_period_one ✓ Stmt OK A linear map with a nonzero fixed point is primitive if and only if its period is 1.

Proof. ✓ Proof OK If $\text{peripheral}(E) = \{1\}$, then $|\text{peripheral}(E)| = 1$. Conversely, if $|\text{peripheral}(E)| = 1$, then since $1 \in \text{peripheral}(E)$ (Theorem 5.27), the only element is 1. $\square$

Theorem 5.51 (Primitive maps have a spectral gap on the complement). compl_eigenvalue_norm_lt_one_of_primitive ✓ Stmt OK $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be trace-preserving, and let $\rho \neq 0$ satisfy $E(\rho) = \rho$ and $\text{tr}(\rho) \neq 0$. Let

$$P(X) = \frac{\text{tr}(X)}{\text{tr}(\rho)} \rho$$

be the fixed-point projection associated to ρ . Assume that:

1. E is primitive;
2. every eigenvalue μ of E satisfies $|\mu| \leq 1$;
3. whenever $E(X) = X$ and $\text{tr}(X) = 0$, one has $X = 0$.

Then every eigenvalue ν of $E - P$ satisfies $|\nu| < 1$.

Proof. ✓ Proof OK Let $(E - P)(X) = \nu X$ with $X \neq 0$. Then

$$E(X) = \nu X + P(X).$$

If $\text{tr}(X) = 0$, then $P(X) = 0$, so $E(X) = \nu X$ and hence ν is an eigenvalue of E with $|\nu| \leq 1$. If $|\nu| = 1$, primitivity forces $\nu = 1$, so X is a trace-zero fixed point of E ; by assumption this implies $X = 0$, a contradiction. Thus $|\nu| < 1$.

If instead $\text{tr}(X) \neq 0$, trace preservation and $\text{tr}(P(X)) = \text{tr}(X)$ give

$$\text{tr}(X) = \text{tr}(E(X)) = \nu \text{tr}(X) + \text{tr}(P(X)) = \nu \text{tr}(X) + \text{tr}(X),$$

so $\nu \text{tr}(X) = 0$ and therefore $\nu = 0$. In either case, $|\nu| < 1$. $\square$

5.6 Representations

The following definitions support the Choi–Jamiołkowski isomorphism (Theorem 5.57) and the representation theorems of [Wol12, Ch. 2].

Definition 5.52 (Partial traces). traceLeft traceRight ✓ Stmt OK Let $X \in M_{d \times d'}(\mathbb{C})$ be a bipartite matrix indexed by

$$(\{0, \dots, d-1\} \times \{0, \dots, d'-1\})^2.$$

The *left partial trace* $\text{tr}_A(X)$ and *right partial trace* $\text{tr}_B(X)$ are the $d' \times d'$ and $d \times d$ matrices defined by

$$(\text{tr}_A(X))_{ij} = \sum_k X_{(k,i),(k,j)}, \quad (\text{tr}_B(X))_{ij} = \sum_k X_{(i,k),(j,k)}.$$

Definition 5.53 (Maximally entangled state). [omegaProj](#) [✓ Stmt OK](#) The *maximally entangled state* on $\mathbb{C}^d \otimes \mathbb{C}^d$ is the rank-one projector

$$|\Omega\rangle\langle\Omega|, \quad |\Omega\rangle = \frac{1}{\sqrt{d}} \sum_{j=0}^{d-1} |j, j\rangle.$$

Concretely, $(|\Omega\rangle\langle\Omega|)_{(i_1, i_2), (j_1, j_2)} = \frac{1}{d} \delta_{i_1 j_1} \delta_{i_2 j_2}$ and $\text{tr}(|\Omega\rangle\langle\Omega|) = 1$.

Definition 5.54 (Tensor product of a map with identity). [tensorMapId](#) [✓ Stmt OK](#) Given a linear map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$, the *tensor extension* $T \otimes \text{id}$ acts on bipartite matrices $X \in M_{D \times D}(\mathbb{C})$ by applying T to each “slice”:

$$(T \otimes \text{id})(X)_{(i_1, i_2), (j_1, j_2)} = (T(X^{(i_2, j_2)}))_{i_1 j_1},$$

where $X_{ab}^{(i_2, j_2)} = X_{(a, i_2), (b, j_2)}$ is the bipartite slice.

5.7 Choi–Jamiołkowski isomorphism

Definition 5.55 (Choi matrix). [choiMatrix](#) [✓ Stmt OK](#) The *Choi matrix* of a linear map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is

$$\tau = (T \otimes \text{id})(|\Omega\rangle\langle\Omega|) \in M_{D \times D}(\mathbb{C}).$$

Concretely, $\tau_{(i_1, i_2), (j_1, j_2)} = \frac{1}{D} (T(E_{i_2 j_2}))_{i_1 j_1}$ where $E_{i_2 j_2}$ is the matrix unit. This is the Choi–Jamiołkowski convention of [Wol12, Prop. 2.1].

Theorem 5.56 (Choi matrix of the identity map). [choiMatrix_id](#) [✓ Stmt OK](#) The *identity channel* has Choi matrix $|\Omega\rangle\langle\Omega|$.

Proof. [✓ Proof OK](#) Apply the identity map in the definition of the Choi matrix. □

Theorem 5.57 (Complete positivity iff the Choi matrix is positive semidefinite). [cp_iff_choi_posSemidef](#) [✓ Stmt OK](#) A linear map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is *completely positive* (in the Kraus sense) if and only if its Choi matrix $\tau \geq 0$.

Proof. [✓ Proof OK](#) ($\Rightarrow$) If $T(X) = \sum_i K_i X K_i^\dagger$, then $\tau = \sum_i |v_i\rangle\langle v_i|$ where $(v_i)_{(a, b)} = c K_i(a, b)$ and $c = 1/\sqrt{D}$. This is a sum of rank-one PSD matrices.

($\Leftarrow$) If $\tau \geq 0$, write $\tau = \sum_m |v_m\rangle\langle v_m|$ by spectral decomposition, define $K_m(a, b) = v_m(a, b)/c$, and recover $T(X) = \sum_m K_m X K_m^\dagger$ by linearity on matrix units. □

Theorem 5.58 (Trace preservation implies the partial-trace condition). [traceLeft_choiMatrix_of_tp](#) [✓ Stmt OK](#) If T is trace-preserving, then $\text{tr}_A(\tau) = \frac{1}{D} \mathbb{1}_D$.

Proof. [✓ Proof OK](#) $(\text{tr}_A(\tau))_{ij} = \text{tr}(T(\Omega^{(ij)})) = \text{tr}(\Omega^{(ij)})$ by trace preservation, where $\Omega^{(ij)} = \frac{1}{D} E_{ij}$ is the (i, j) -slice of $|\Omega\rangle\langle\Omega|$. Taking the trace gives $\frac{1}{D} \delta_{ij}$. □

Theorem 5.59 (Hermiticity preservation and the Choi matrix). [choiMatrix_isHermitian_iff_hermiticityPreserving](#) [✓ Stmt OK](#) Choi matrix τ is Hermitian if and only if $T(B^\dagger) = T(B)^\dagger$ for all $B \in M_D(\mathbb{C})$.

Proof. [✓ Proof OK](#) The Choi matrix entries satisfy $\tau_{(a, i), (b, j)} = \frac{1}{D} (T(E_{ij}))_{ab}$. Hermiticity of τ says $\overline{\tau_{(b, j), (a, i)}} = \tau_{(a, i), (b, j)}$, which unravels to $T(E_{ji})_{ba} = \overline{T(E_{ij})_{ab}}$, i.e., $T(E_{ij}^\dagger) = T(E_{ij})^\dagger$. By linearity this extends to all B . □

Theorem 5.60 (Trace of the Choi matrix of a trace-preserving map). [trace_choiMatrix_of_tp](#) [✓ Stmt OK](#) If T is trace-preserving, then $\text{tr}(\tau) = 1$.

Proof. [✓ Proof OK](#) $\text{tr}(\tau) = \text{tr}(\text{tr}_A(\tau)) = \text{tr}(\frac{1}{D} \mathbb{1}_D) = 1$. □

5.8 Representation corollaries for channel decompositions

The next three corollaries follow from the Choi/Kraus/Stinespring representations already developed above. They correspond to [Wol12, Props. 2.2–2.4].

Theorem 5.61 (Polarization of sesquilinear sandwiches). [polarization_sandwich](#) ✓ Stmt OK For any $A, B, X \in M_D(\mathbb{C})$,

$$4AXB^\dagger = (A+B)X(A+B)^\dagger - (A-B)X(A-B)^\dagger + i(A+iB)X(A+iB)^\dagger - i(A-iB)X(A-iB)^\dagger.$$

Each of the four summands on the right is a single-Kraus CP map, so every sesquilinear sandwich decomposes into a signed complex linear combination of CP maps.

Proof. ✓ Proof OK Working entrywise, reduces to the scalar polarization identity $4\alpha\bar{\delta} = \sum_{k=0}^3 i^k(\alpha + i^k\beta)(\gamma + i^k\delta)$ in $\mathbb{C}$, which holds after substituting $i^2 = -1$. □

Theorem 5.62 (CP decomposition of sandwich sums). [cp_decomposition_of_sandwich_sum](#) ✓ Stmt OK For any finite Kraus-like families $\{A_i\}, \{B_i\}$, the map $T(X) = \sum_i A_i X B_i^\dagger$ satisfies $4T = T_1 - T_2 + iT_3 - iT_4$ with each T_k a CP map given explicitly by a single-side Kraus sum of the combined families.

Proof. ✓ Proof OK Apply the polarization identity summand-by-summand. □

Theorem 5.63 (No information without disturbance). [linearMap_eq_id_of_fixes_rankOne](#) ✓ Stmt OK Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be linear. If $T(|v\rangle\langle v|) = |v\rangle\langle v|$ for every $v \in \mathbb{C}^D$, then $T = \text{id}$. In particular, a quantum channel preserving every pure state equals the identity.

Proof. ✓ Proof OK Rank-one outer products $|u\rangle\langle v|$ polarize into rank-one self-outer-products: $4|u\rangle\langle v| = \sum_{k=0}^3 i^k |u + i^k v\rangle\langle u + i^k v|$, so T preserves every $|u\rangle\langle v|$. Since the matrix units $E_{ij} = |e_i\rangle\langle e_j|$ span $M_D(\mathbb{C})$, T equals the identity on a spanning set, hence everywhere. □

Definition 5.64 (Pure-state ensemble density). [pureEnsembleDensity](#) ✓ Stmt OK Given a finite family $\{\psi_i\}_{i \in I}$ of (unnormalized) vectors in $\mathbb{C}^D$, its *pure-ensemble density* is $\rho = \sum_i |\psi_i\rangle\langle\psi_i|$. Weights $p_i \geq 0$ with $\sum p_i = 1$ can be absorbed by replacing $\psi_i \mapsto \sqrt{p_i} \psi_i$.

Theorem 5.65 (Isometric mixing preserves the density). [pureEnsembleDensity_eq_of_isometric_mixing](#) ✓ Stmt OK If two pure-state ensembles $\{\psi_i\}_{i \in I_1}$ and $\{\phi_j\}_{j \in I_2}$ are related by an isometric mixing matrix $V \in M_{I_1 \times I_2}(\mathbb{C})$ with $V^\dagger V = \mathbb{1}$ and $\psi_i = \sum_j V_{ij} \phi_j$, then they induce the same pure-ensemble density operator. This is the sufficient direction of the Hughston–Jozsa–Wootters theorem; the converse is Theorem 5.66.

Proof. ✓ Proof OK Expanding and applying the orthogonality relation $\sum_i V_{ij} \overline{V_{ij'}} = \delta_{jj'}$ from $V^\dagger V = \mathbb{1}$:

$$\sum_i |\psi_i\rangle\langle\psi_i| = \sum_{j,j'} \left(\sum_i V_{ij} \overline{V_{ij'}} \right) |\phi_j\rangle\langle\phi_{j'}| = \sum_j |\phi_j\rangle\langle\phi_j|.$$

□

Theorem 5.66 (Hughston–Jozsa–Wootters converse). [exists_isometric_mixing_of_pureEnsembleDensity_eq](#) ✓ Stmt OK If two pure-state ensembles $\{\psi_i\}_{i \in I_1}$ and $\{\phi_j\}_{j \in I_2}$ induce the same pure-ensemble density operator and $|I_2| \leq |I_1|$, then there exists a tall isometric mixing matrix $V \in M_{I_1 \times I_2}(\mathbb{C})$ with $V^\dagger V = \mathbb{1}$ and $\psi_i = \sum_j V_{ij} \phi_j$. The cardinality hypothesis is what makes V a tall isometry; the symmetric case $|I_1| \leq |I_2|$ follows by swapping the roles of the ensembles.

Proof. ✓ Proof OK Embed each vector ψ_i, ϕ_j as the 0-th column of a $D \times D$ matrix with zeros elsewhere; call these $K_i, L_j \in M_D(\mathbb{C})$. A direct entry-wise computation shows that for any $X \in M_D(\mathbb{C})$ both $\sum_i K_i X K_i^\dagger$ and $\sum_j L_j X L_j^\dagger$ collapse to $X_{00} \cdot \rho$, so the density equality $\rho_\psi = \rho_\phi$ forces the two Kraus families to define the same CP map. The rectangular Kraus freedom 5.77 then supplies an isometry V with $V^\dagger V = \mathbb{1}$ and $K_i = \sum_j V_{ij} L_j$; reading the equation off at column 0 recovers the vector relation $\psi_i = \sum_j V_{ij} \phi_j$. □

Theorem 5.67 (Hughston–Jozsa–Wootters equivalence). pureEnsembleDensity_eq_iff_exists_isometric_mixing ✓ Stmt OK *Under the cardinality hypothesis $|\iota_2| \leq |\iota_1|$, two pure-state ensembles induce the same density operator iff they are related by a tall isometric mixing matrix.*

Proof. ✓ Proof OK Combine the two directions above. □

5.9 Kraus representation theorem

Theorem 5.68 (Trace preservation implies Kraus normalization). kraus_sum_conjTranspose_mul_of_tp ✓ Stmt OK *If $T(X) = \sum_i K_i X K_i^\dagger$ is trace-preserving, then $\sum_i K_i^\dagger K_i = \mathbb{1}$.*

Proof. ✓ Proof OK For any N , $\text{tr}((\sum_i K_i^\dagger K_i) N) = \sum_i \text{tr}(K_i^\dagger K_i N) = \sum_i \text{tr}(K_i N K_i^\dagger) = \text{tr}(T(N)) = \text{tr}(N)$. Non-degeneracy of the trace pairing forces $\sum_i K_i^\dagger K_i = \mathbb{1}$. □

Theorem 5.69 (Kraus normalization implies trace preservation). kraus_tp_of_sum_conjTranspose_mul ✓ Stmt OK *If $\sum_i K_i^\dagger K_i = \mathbb{1}$, then $T(X) = \sum_i K_i X K_i^\dagger$ is trace-preserving.*

Proof. ✓ Proof OK $\text{tr}(T(X)) = \sum_i \text{tr}(K_i X K_i^\dagger) = \sum_i \text{tr}(K_i^\dagger K_i X) = \text{tr}((\sum_i K_i^\dagger K_i) X) = \text{tr}(X)$. □

Theorem 5.70 (Unitality implies Kraus unital normalization). kraus_sum_mul_conjTranspose_of_unital ✓ Stmt OK *If $T(X) = \sum_i K_i X K_i^\dagger$ satisfies $T(\mathbb{1}) = \mathbb{1}$, then*

$$\sum_i K_i K_i^\dagger = \mathbb{1}.$$

Proof. ✓ Proof OK Evaluate the Kraus formula at the identity matrix. □

Theorem 5.71 (Unitary freedom in Kraus operators). kraus_same_map_of_unitary_combination ✓ Stmt OK *Let U be a unitary $r \times r$ matrix ($U^\dagger U = \mathbb{1}$) and suppose $K_j = \sum_\ell U_{j\ell} \tilde{K}_\ell$. Then $\{K_j\}$ and $\{\tilde{K}_\ell\}$ define the same Kraus map:*

$$\sum_j K_j X K_j^\dagger = \sum_\ell \tilde{K}_\ell X \tilde{K}_\ell^\dagger \quad \forall X \in M_D(\mathbb{C}).$$

Proof. ✓ Proof OK Substitute the combination formula, expand the triple sum over j, ℓ, ℓ' , swap summation order, and use the entry-wise form of $U^\dagger U = \mathbb{1}$ to collapse to $\ell = \ell'$ diagonal terms. □

Theorem 5.72 (Isometric mixing preserves the Kraus map). kraus_same_map_of_isometry_combination ✓ Stmt OK *Let W be an isometry between two Kraus index spaces, so $W^\dagger W = \mathbb{1}$. If*

$$K_j = \sum_\ell W_{j\ell} \tilde{K}_\ell,$$

then the Kraus families $\{K_j\}$ and $\{\tilde{K}_\ell\}$ define the same completely positive map.

Proof. ✓ Proof OK Expand the Kraus sums, interchange the finite summations, and use $W^\dagger W = \mathbb{1}$ to collapse the coefficient matrix. □

Theorem 5.73 (The transition matrix of orthonormal Kraus families is unitary). kraus_transition_unitary_of_hs_orthonormal ✓ Stmt OK *If $\{K_j\}_{j=0}^{r-1}$ and $\{K'_j\}_{j=0}^{r-1}$ be two Hilbert–Schmidt orthonormal Kraus families, and suppose*

$$K_j = \sum_{\ell=0}^{r-1} U_{j\ell} K'_\ell.$$

Then the transition matrix U is unitary: $U^\dagger U = \mathbb{1}_r$.

Proof. ✓ Proof OK Expand $\text{tr}(K_j^\dagger K_i)$ using the change of basis. Hilbert–Schmidt orthonormality of both Kraus families identifies the Gram matrix with the identity, yielding the matrix identity $UU^\dagger = \mathbb{1}_r$, hence also $U^\dagger U = \mathbb{1}_r$. □

Theorem 5.74 (Dual map equality from primal map equality). kraus_dual_eq_of_map_eq ✓ Stmt OK *If two Kraus families satisfy $\sum_\alpha B_\alpha X B_\alpha^\dagger = \sum_j A_j X A_j^\dagger$ for all X , then $\sum_\alpha B_\alpha^\dagger Y B_\alpha = \sum_j A_j^\dagger Y A_j$ for all Y .*

Proof. ✓ Proof OK Use the trace pairing: for all X , $\text{tr}(X^\dagger (\sum B_\alpha^\dagger Y B_\alpha)) = \text{tr}((\sum B_\alpha X^\dagger B_\alpha^\dagger) Y) = \text{tr}((\sum A_j X^\dagger A_j^\dagger) Y) = \text{tr}(X^\dagger (\sum A_j^\dagger Y A_j))$. Nondegeneracy of the trace pairing gives the result. □

Theorem 5.75 (Equal Stinespring Gramians). kraus_conjTranspose_mul_eq_of_map_eq ✓ Stmt OK *If two Kraus families define the same CPM, then $\sum_\alpha B_\alpha^\dagger B_\alpha = \sum_j A_j^\dagger A_j$.*

Proof. ✓ Proof OK Specialise Theorem 5.74 at $Y = \mathbb{1}$. □

Theorem 5.76 (Rectangular Kraus freedom). kraus_rectangular_freedom ✓ Stmt OK *If $\sum_\alpha B_\alpha X B_\alpha^\dagger = \sum_j A_j X A_j^\dagger$ for all X and $r_2 \leq r_1$ (where $\{B_\alpha\}_{\alpha=0}^{r_1-1}$ is the larger family and $\{A_j\}_{j=0}^{r_2-1}$ the smaller), then there exists a rectangular isometry V ($r_1 \times r_2$, $V^\dagger V = \mathbb{1}_{r_2}$) such that $B_\alpha = \sum_j V_{\alpha j} A_j$.*

Proof. ✓ Proof OK Pad the smaller family to size r_1 , establish Gram matrix equality $M_B^\dagger M_B = M_{A'}^\dagger M_{A'}$, from dual map equality, construct a partial isometry via inner product preservation on the range, and extend to a full isometry V with $V^\dagger V = \mathbb{1}_{r_2}$. □

Theorem 5.77 (Rectangular Kraus freedom with general finite index types). kraus_rectangular_freedom' ✓ Stmt OK *Variant of Theorem 5.76 with general finite index sets ι_1, ι_2 in place of standard index sets of cardinalities r_1 and r_2 .*

Proof. ✓ Proof OK Reindex by choosing enumerations of those finite index sets and apply Theorem 5.76. □

Theorem 5.78 (Isometric freedom of Kraus representations). kraus_isometry_freedom_iff ✓ Stmt OK *Let $\{B_\alpha\}_{\alpha \in \iota_1}$ and $\{A_j\}_{j \in \iota_2}$ be finite Kraus families with $|\iota_2| \leq |\iota_1|$. Then the following are equivalent:*

1. $\sum_\alpha B_\alpha X B_\alpha^\dagger = \sum_j A_j X A_j^\dagger$ for all $X \in M_D(\mathbb{C})$.
2. There exists a matrix $V = (V_{\alpha j})$ with $V^\dagger V = \mathbb{1}$ such that $B_\alpha = \sum_j V_{\alpha j} A_j$ for every α .

Proof. ✓ Proof OK The implication (1) $\Rightarrow$ (2) is Theorem 5.77. Conversely, expand $\sum_\alpha B_\alpha X B_\alpha^\dagger$ using $B_\alpha = \sum_j V_{\alpha j} A_j$, interchange the finite sums, and use $V^\dagger V = \mathbb{1}$ to collapse the coefficient matrix to the diagonal. □

Theorem 5.79 (Unitary freedom of Kraus representations). [kraus_unitary_freedom_iff](#) [✓ Stmt OK](#) Let $\{B_\alpha\}_{\alpha \in \iota}$ and $\{A_j\}_{j \in \iota}$ be two Kraus families with the same finite index set. Then the following are equivalent:

1. $\sum_\alpha B_\alpha X B_\alpha^\dagger = \sum_j A_j X A_j^\dagger$ for all $X \in M_D(\mathbb{C})$.
2. There exists a unitary matrix $U = (U_{\alpha j})$ such that $B_\alpha = \sum_j U_{\alpha j} A_j$ for every α .

Proof. [✓ Proof OK](#) Apply Theorem 5.78 with $|\iota| = |\iota|$. In the square case an isometry matrix is unitary, and the converse is immediate. $\square$

Remark 5.80 (Choi rank and minimal Kraus cardinality). [HasKrausCard](#) [HasKrausRankLE](#) [choiRank](#) [choiRank_le_of_hasKrausCard](#) A completely positive map admits a Kraus representation with exactly r operators, then its Choi matrix is a sum of r rank-one outer products. Consequently,

$$\text{rank}(\tau_E) \leq r.$$

Conversely, diagonalizing the positive semidefinite Choi matrix and keeping only the nonzero eigenvalue summands produces a Kraus family with exactly $\text{rank}(\tau_E)$ operators. Hence the Choi rank is precisely the minimal Kraus cardinality.

Proof. [✓ Proof OK](#) For the forward implication, expand the Choi matrix using the Kraus formula and write $\tau_E = \sum_j v_j v_j^\dagger$ with one vector v_j for each Kraus operator. The column space of this sum lies in the span of the r vectors $\{v_j\}$, so the rank is at most r . For the converse, use the spectral decomposition of the positive semidefinite Choi matrix, discard the zero-eigenvalue terms, and rescale the remaining eigenvectors into Kraus operators. $\square$

5.10 Stinespring dilation

Definition 5.81 (Stinespring isometry). [stinespringV](#) [✓ Stmt OK](#) Given Kraus operators $\{K_j\}_{j=0}^{r-1}$ with $K_j \in M_D(\mathbb{C})$, the *Stinespring isometry* $V : \mathbb{C}^D \rightarrow \mathbb{C}^D \otimes \mathbb{C}^r$ is the $(D \cdot r) \times D$ matrix

$$V_{(i,j),k} = (K_j)_{ik}.$$

This implements $V = \sum_j K_j \otimes |j\rangle$.

Lemma 5.82 (Entry formula for the Stinespring isometry). [stinespringV_apply](#) [✓ Stmt OK](#) For indices $i, k \in \{0, \dots, D-1\}$ and $j \in \{0, \dots, r-1\}$,

$$V_{(i,j),k} = (K_j)_{ik}.$$

Theorem 5.83 (Stinespring Gram identity). [stinespringV_conjTranspose_mul](#) [✓ Stmt OK](#) The Stinespring isometry satisfies

$$V^\dagger V = \sum_{j=0}^{r-1} K_j^\dagger K_j.$$

Theorem 5.84 (Stinespring dual representation). [stinespring_dual_representation](#) [✓ Stmt OK](#) For Kraus operators $\{K_j\}$ and $A \in M_D(\mathbb{C})$,

$$V^\dagger (A \otimes \mathbb{1}_r) V = \sum_j K_j^\dagger A K_j = T^*(A).$$

This is the Stinespring representation of the dual (Heisenberg picture) map T^* .

Proof. ✓ Proof OK Compute $(V^\dagger(A \otimes \mathbb{1}_r)V)_{ab}$ by summing over the product index (i, j) ; the δ_{jl} from $\mathbb{1}_r$ collapses the sum over l , giving $\sum_j K_j^\dagger A K_j$. □

Theorem 5.85 (The Stinespring isometry condition iff trace preservation). stinespringV_isometry_iff_kraus_normalized ✓ S
 $\mathbb{1}_D$ if and only if $\sum_j K_j^\dagger K_j = \mathbb{1}_D$. In particular, V is an isometry precisely when the Kraus map is trace-preserving.

Proof. ✓ Proof OK $(V^\dagger V)_{ab} = \sum_{(i,j)} \overline{V_{(i,j),a}} V_{(i,j),b} = \sum_j \sum_i \overline{(K_j)_{ia}} (K_j)_{ib} = (\sum_j K_j^\dagger K_j)_{ab}$. □

Theorem 5.86 (Stinespring Schrödinger representation). stinespring_schrodinger_representation ✓ Stmt OK The Kraus map $T(\rho) = \sum_j K_j \rho K_j^\dagger$ equals the partial trace over the dilation space:

$$T(\rho)_{ij} = \sum_{k=0}^{r-1} (V \rho V^\dagger)_{(i,k),(j,k)} = \text{tr}_r(V \rho V^\dagger).$$

This is the Schrödinger-picture Stinespring representation.

Proof. ✓ Proof OK Expand $(V \rho V^\dagger)_{(i,k),(j,k)}$ and sum over k to recover $\sum_k K_k \rho K_k^\dagger$. □

Theorem 5.87 (Existence of a Stinespring dilation). exists_stinespring_dilation ✓ Stmt OK Every completely positive map admits an ancilla dimension r , Kraus operators $\{K_j\}_{j=0}^{r-1}$, and the concrete representation $\pi(A) = A \otimes \mathbb{1}_r$ such that

$$E(A) = V^\dagger \pi(A) V.$$

Proof. ✓ Proof OK Choose a Kraus representation of E and apply the explicit Heisenberg-picture Stinespring formula. □

Theorem 5.88 (CPTP maps admit isometric Stinespring dilations). exists_stinespring_isometry_of_cptp ✓ Stmt OK Every quantum channel admits a Stinespring dilation whose matrix V is an isometry, equivalently $V^\dagger V = \mathbb{1}$.

Proof. ✓ Proof OK Choose Kraus operators for the channel and use trace preservation to obtain the Stinespring isometry condition $V^\dagger V = \mathbb{1}$. □

5.11 Ordered CP maps, Radon–Nikodym, and open-system representation

The next three results form the structural part of [Wol12, Thms. 2.3–2.5]: the ordered CP-map theorem, the Radon–Nikodym theorem for completely positive maps, and the open-system representation. They refine the Stinespring dilation by tracking how two CP maps related by domination or decomposition sit inside a common dilation space.

Definition 5.89 (CP partial order). CPDominates ✓ Stmt OK For linear maps $S, T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$, write $T \leq S$ (in the CP order) when $S - T$ is completely positive.

Definition 5.90 (Block-top contraction). blockTopRows ✓ Stmt OK For natural numbers r, s , let $C = C_{r,s} \in M(\mathbb{C})(\mathbb{C})$ be the rectangular $r \times (r+s)$ matrix whose rows are the first r rows of the identity on $\mathbb{C}^{r+s}$: $C_{ij} = 1$ if $j = i < r$ and 0 otherwise.

Lemma 5.91 (C is a co-isometry). [blockTopRows_mul_conjTranspose](#) [✓ Stmt OK](#) The block-top matrix satisfies $CC^\dagger = \mathbb{1}_r$.

Proof. [✓ Proof OK](#) Direct entrywise computation: $(CC^\dagger)_{ii'} = \sum_j C_{ij} \overline{C_{i'j}}$ collapses to $\delta_{ii'}$. $\square$

Lemma 5.92 (C is a contraction). [blockTopRows_conjTranspose_mul_le_one](#) [✓ Stmt OK](#) The block-top matrix satisfies $C^\dagger C \leq \mathbb{1}_{r+s}$.

Proof. [✓ Proof OK](#) The matrix $C^\dagger C$ is diagonal with a 1 on the first r coordinates and a 0 on the last s , so $\mathbb{1} - C^\dagger C$ is positive semidefinite. $\square$

Lemma 5.93 (Intertwining of Stinespring isometries). [stinespringV_eq_kronecker_blockTopRows_mul_append](#) [✓ Stmt OK](#) Let $K : \{0, \dots, r-1\} \rightarrow M_D(\mathbb{C})$ and $L : \{0, \dots, s-1\} \rightarrow M_D(\mathbb{C})$ be two Kraus families. Write $K \mathbin{++} L$ for their concatenation as a family indexed by $\{0, \dots, r+s-1\}$. Then

$$V_K = (\mathbb{1}_D \otimes C_{r,s}) V_{K \mathbin{++} L}.$$

Proof. [✓ Proof OK](#) Entrywise expansion of both sides: the Kronecker product with $C_{r,s}$ picks out the first r coordinates of the dilation space, matching the action of $V_{K \mathbin{++} L}$ on those coordinates, which is precisely V_K . $\square$

Theorem 5.94 (Ordered CP maps via Stinespring contractions). [exists_stinespring_contraction](#) [✓ Stmt OK](#) Let $T_1, T_2 : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be CP maps with $T_1 \leq T_2$. Then there exist an ancilla dimension m , Heisenberg-form Stinespring matrices V_1, V_2 realizing T_1, T_2 via $T_i(A) = V_i^\dagger (A \otimes \mathbb{1}) V_i$, and a contraction $\tilde{C}$ on the dilation space with $\tilde{C}^\dagger \tilde{C} \leq \mathbb{1}$ such that

$$V_1 = (\mathbb{1}_D \otimes \tilde{C}) V_2.$$

This is [Wol12, Thm. 2.3].

Proof. [✓ Proof OK](#) Choose a Heisenberg-form Kraus family K_1 of T_1 with Stinespring matrix $V_1 = V_{K_1}$, and Kraus operators L of the CP map $T_2 - T_1$ in Schrödinger orientation. Let $L^\dagger$ denote the conjugate-transposed family. Set $m = r_1 + s$ and let $K_2 = K_1 \mathbin{++} L^\dagger$. Then $T_2(A) = T_1(A) + (T_2 - T_1)(A) = \sum_i (K_1^i)^\dagger A K_1^i + \sum_j L_j A L_j^\dagger$, and the latter equals $\sum_i (K_1^i)^\dagger A K_1^i + \sum_j (L_j^\dagger)^\dagger A L_j^\dagger$ which is the Heisenberg form for K_2 , yielding $T_2(A) = V_{K_2}^\dagger (A \otimes \mathbb{1}) V_{K_2}$. Taking $\tilde{C} = C_{r_1, s}$ the intertwining $V_{K_1} = (\mathbb{1}_D \otimes \tilde{C}) V_{K_2}$ is Lemma 5.93, and $\tilde{C}^\dagger \tilde{C} \leq \mathbb{1}$ is Lemma 5.92. $\square$

Definition 5.95 (Block-diagonal projectors on the dilation space). [blockDiagTopProj](#) [blockDiagBotProj](#) [✓ Stmt OK](#) For r, s , let $P_{\text{top}} = C_{r,s}^\dagger C_{r,s}$ and $P_{\text{bot}} = \mathbb{1} - P_{\text{top}}$. Both are PSD (for P_{bot} , by Lemma 5.92) and $P_{\text{top}} + P_{\text{bot}} = \mathbb{1}$.

Theorem 5.96 (Radon–Nikodym theorem for CP maps). [exists_radon_nikodym](#) [✓ Stmt OK](#) Let $T_1, T_2 : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be completely positive. There exist an ancilla dimension m , a Kraus family K with Stinespring matrix $V = V_K$, and positive semidefinite operators $P_1, P_2 : \mathbb{C}^m \rightarrow \mathbb{C}^m$ with $P_1 + P_2 = \mathbb{1}_m$ such that, for every $A \in M_D(\mathbb{C})$,

$$T_i(A) = V^\dagger (A \otimes P_i) V, \quad i = 1, 2.$$

This is [Wol12, Thm. 2.4].

Proof. ✓ Proof OK Take Heisenberg-form Kraus families K_1 for T_1 and Schrödinger-form L for T_2 , and form $K = K_1 \mathrel{++} L^\dagger$ on $\mathbb{C}^{r_1+s}$. Choose $P_1 = P_{\text{top}}$, $P_2 = P_{\text{bot}}$ as in Definition 5.95. The Kronecker identity $A \otimes (C^\dagger C) = (\mathbb{1} \otimes C)^\dagger (A \otimes \mathbb{1}) (\mathbb{1} \otimes C)$ rewrites $V^\dagger (A \otimes P_{\text{top}}) V$ as $((\mathbb{1} \otimes C) V)^\dagger (A \otimes \mathbb{1}) ((\mathbb{1} \otimes C) V)$, which equals $V_{K_1}^\dagger (A \otimes \mathbb{1}) V_{K_1} = T_1(A)$ by the intertwining Lemma 5.93. For the complementary block, $A \otimes P_{\text{bot}} = A \otimes \mathbb{1} - A \otimes P_{\text{top}}$ and sandwiching by V yields $(T_1 + T_2)(A) - T_1(A) = T_2(A)$. $\square$

Theorem 5.97 (Open-system representation of quantum channels). exists_stinespring_open_system ✓ Stmt OK Every quantum channel $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ admits a Stinespring isometry V on an ancilla space $\mathbb{C}^r$ such that

$$T(\rho)_{ij} = \sum_{k=0}^{r-1} (V \rho V^\dagger)_{(i,k),(j,k)} = \text{tr}_r(V \rho V^\dagger).$$

Proof. ✓ Proof OK Apply the existence of an isometric Stinespring dilation (Theorem 5.88) and the Schrödinger-picture identity (Theorem 5.86). $\square$

Theorem 5.98 (Open-system representation, partial-trace form). exists_stinespring_open_system_traceRight ✓ Stmt OK Every quantum channel T admits an isometric Stinespring dilation V such that $T(\rho) = \text{tr}_E(V \rho V^\dagger)$.

Proof. ✓ Proof OK Rewrite the componentwise identity of Theorem 5.97 as a partial trace over the ancilla factor. $\square$

Remark 5.99 (Unitary form). The fully unitary open-system form $T(\rho) = \text{tr}_E(U(\rho \otimes |\phi\rangle\langle\phi|)U^\dagger)$ of [Wol12, Thm. 2.5] follows from the isometric form by extending the Stinespring isometry V to a unitary U on $\mathbb{C}^D \otimes \mathbb{C}^r$; this uses orthonormal basis extension and is recorded as an outstanding formalization item.

5.12 POVMs and Naimark dilation

Definition 5.100 (Positive operator-valued measure). POVM ✓ Stmt OK A positive operator-valued measure with n outcomes on $\mathbb{C}^D$ is a family $\{E_i\}_{i=1}^n$ of positive semidefinite operators on $\mathbb{C}^D$ satisfying the resolution of identity

$$\sum_{i=1}^n E_i = \mathbb{1}_D.$$

Definition 5.101 (Naimark Kraus square roots). naimarkKraus naimarkKraus_spec ✓ Stmt OK For a POVM $\{E_i\}$, each effect $E_i \geq 0$ admits a square-root factorisation $E_i = M_i^\dagger M_i$ with $M_i \in M_D(\mathbb{C})$. The operators M_i are the *Naimark Kraus square roots*.

Theorem 5.102 (Naimark square roots satisfy the Kraus normalization). sum_naimarkKraus_conjTranspose_mul ✓ Stmt OK For a POVM $\{E_i\}$ with square roots $E_i = M_i^\dagger M_i$, one has

$$\sum_i M_i^\dagger M_i = \mathbb{1}.$$

Proof. ✓ Proof OK Sum the identities $E_i = M_i^\dagger M_i$ and use the defining resolution of the identity for the POVM. $\square$

Definition 5.103 (Naimark isometry). [naimarkIsometry](#) ✓ Stmt OK The Naimark isometry $V : \mathbb{C}^D \rightarrow \mathbb{C}^D \otimes \mathbb{C}^n$ of a POVM is the Stinespring-type construction

$$V = \sum_{i=1}^n M_i \otimes |i\rangle, \quad V_{(a,i),k} = (M_i)_{a,k}.$$

Definition 5.104 (Naimark projective measurement). [naimarkProjection](#) [naimarkProjection_apply](#) ✓ Stmt OK The Naimark projectors on $\mathbb{C}^D \otimes \mathbb{C}^n$ are

$$P_i = \mathbb{1}_D \otimes |i\rangle\langle i|, \quad (P_i)_{(a,b),(c,d)} = \delta_{a,c} \delta_{b,i} \delta_{d,i}.$$

Theorem 5.105 (Naimark isometry condition). [naimarkIsometry_isometry](#) ✓ Stmt OK The Naimark isometry satisfies $V^\dagger V = \mathbb{1}_D$.

Proof. ✓ Proof OK The Stinespring Gram identity gives $V^\dagger V = \sum_i M_i^\dagger M_i = \sum_i E_i = \mathbb{1}_D$ by the resolution of identity. □

Theorem 5.106 (Naimark projectors form a projective measurement). [naimarkProjection_mul_self](#) [naimarkProjection_hermit](#) Naimark projectors satisfy $P_i^2 = P_i$, $P_i^\dagger = P_i$, $P_i P_j = 0$ for $i \neq j$, and $\sum_i P_i = \mathbb{1}_{\mathbb{C}^D \otimes \mathbb{C}^n}$.

Proof. ✓ Proof OK Each identity follows from direct entrywise computation using the delta-function form of $(P_i)_{(a,b),(c,d)}$. □

Theorem 5.107 (Naimark dilation). [naimark_recovers_povm](#) ✓ Stmt OK Every POVM $\{E_i\}_{i=1}^n$ arises as a projective measurement on a dilation: for the isometry V and projectors P_i above,

$$E_i = V^\dagger P_i V.$$

Proof. ✓ Proof OK Computing entrywise, $(V^\dagger P_i V)_{k,\ell} = \sum_a \overline{(M_i)_{a,k}} (M_i)_{a,\ell} = (M_i^\dagger M_i)_{k,\ell} = (E_i)_{k,\ell}$. □

Theorem 5.108 (Existential Naimark dilation). [exists_naimark_dilation](#) ✓ Stmt OK For every POVM $\{E_i\}$ on $\mathbb{C}^D$ there exist a dilation dimension r , an isometry $V : \mathbb{C}^D \rightarrow \mathbb{C}^D \otimes \mathbb{C}^r$, and a projective measurement $\{P_i\}$ on the dilation satisfying $E_i = V^\dagger P_i V$.

Proof. ✓ Proof OK Take $r = n$ and the explicit witnesses $V = \sum_i M_i \otimes |i\rangle$ and $P_i = \mathbb{1}_D \otimes |i\rangle\langle i|$. □

Definition 5.109 (Naimark dilation as a structure). [IsNaimarkDilation](#) ✓ Stmt OK A Naimark dilation of a POVM $\{E_i\}$ consists of an isometry V into a larger Hilbert space together with a projective measurement $\{P_i\}$ there such that $E_i = V^\dagger P_i V$ for all i .

Theorem 5.110 (Canonical dilation satisfies the Naimark dilation axioms). [isNaimarkDilation_naimark](#) ✓ Stmt OK The explicit isometry $V = \sum_i M_i \otimes |i\rangle$ and projectors $P_i = \mathbb{1}_D \otimes |i\rangle\langle i|$ satisfy the defining axioms of a Naimark dilation.

Proof. ✓ Proof OK Combine the previously proved identities $V^\dagger V = \mathbb{1}_D$, $P_i^2 = P_i$, $P_i^\dagger = P_i$, $P_i P_j = 0$ for $i \neq j$, $\sum_i P_i = \mathbb{1}$, and $V^\dagger P_i V = E_i$. □

Theorem 5.111 (Concrete uniqueness for canonical projectors). [exists_isometry_mul_naimarkIsometry_of_recovery](#) ✓ Stmt OK $V : \mathbb{C}^D \rightarrow \mathbb{C}^D \otimes \mathbb{C}^n$ satisfy $V^\dagger P_i V = E_i$ for the canonical projectors $P_i = \mathbb{1}_D \otimes |i\rangle\langle i|$. Then there exists an isometry W on the dilated space such that $V = W V_0$, where V_0 is the canonical Naimark isometry of $\{E_i\}$.

Proof. ✓ Proof OK For each outcome i , the i -th block of V has Gram matrix E_i . Comparing with the canonical square root M_i gives $V_i^\dagger V_i = M_i^\dagger M_i$, hence $V_i = U_i M_i$ for a unitary U_i on $\mathbb{C}^D$. Assemble the U_i block-diagonally to obtain an isometry $W = \bigoplus_i U_i$ on $\mathbb{C}^D \otimes \mathbb{C}^n$, and then $V = W \sum_i M_i \otimes |i\rangle = W V_0$. □

Definition 5.112 (POVM from a PSD resolution of identity on a dilation). ofPSDResolutionOfIdentity ✓ Stmt OK Given an isometry $V : \mathbb{C}^D \rightarrow \mathbb{C}^{d'}$ with $V^\dagger V = \mathbb{1}_D$ and a family $\{P_i\}_{i=1}^n$ of positive semidefinite operators on $\mathbb{C}^{d'}$ summing to the identity, the pulled-back operators

$$E_i := V^\dagger P_i V$$

form a POVM. In particular, any projective measurement on the dilation (a special case of a PSD resolution of identity) pulls back to a POVM.

Definition 5.113 (Quantum instrument). Instrument ✓ Stmt OK A quantum instrument with n outcomes is a family $\{\Phi_i\}_{i=1}^n$ of completely positive maps on $M_D(\mathbb{C})$ whose sum $\sum_i \Phi_i$ is trace-preserving.

Definition 5.114 (Instrument operations). total update probability posteriorState ✓ Stmt OK Associated to an instrument are the total channel $\sum_i \Phi_i$, the unnormalized update $\rho \mapsto \Phi_i(\rho)$ for each outcome i , the outcome probability $p_i(\rho) = \text{tr}(\Phi_i(\rho))$, and the normalized posterior state $\Phi_i(\rho)/p_i(\rho)$ whenever $p_i(\rho) \neq 0$.

Theorem 5.115 (Instrument total map is a channel). total_isChannel ✓ Stmt OK The total map $\sum_i \Phi_i$ of an instrument is a quantum channel.

Proof. ✓ Proof OK Complete positivity of the sum follows from closure of CP maps under finite addition; trace preservation is built into the definition. □

Theorem 5.116 (Conservation of probability for instruments). sum_probability ✓ Stmt OK For every state $\rho \in M_D(\mathbb{C})$, the outcome probabilities $p_i(\rho) := \text{tr}(\Phi_i(\rho))$ sum to $\text{tr}(\rho)$.

Proof. ✓ Proof OK Linearity of trace and trace preservation of $\sum_i \Phi_i$ give $\sum_i \text{tr}(\Phi_i(\rho)) = \text{tr}((\sum_i \Phi_i)(\rho)) = \text{tr}(\rho)$. □

Theorem 5.117 (Non-negativity of instrument probabilities). probability_nonneg ✓ Stmt OK If $\rho \in M_D(\mathbb{C})$ is positive semidefinite, then each outcome probability $p_i(\rho) = \text{tr}(\Phi_i(\rho))$ is a non-negative real number.

Proof. ✓ Proof OK Each Φ_i is completely positive, hence positive, so $\Phi_i(\rho)$ is positive semidefinite and its trace is a non-negative real. □

5.12.1 Trace-pairing expansion in transfer-matrix form

Definition 5.118 (Trace-self-dual basis). TracePairingSelfDualBasis ✓ Stmt OK A basis $\{\sigma_i\}_i$ of $M_D(\mathbb{C})$ is *trace-self-dual* when its coordinate functionals are given by trace pairing:

$$X = \sum_i \text{tr}(\sigma_i X) \sigma_i.$$

Definition 5.119 (Trace-pairing coefficients). tracePairingCoeff ✓ Stmt OK For a linear map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ and a trace-self-dual basis $\{\sigma_i\}$, define coefficients

$$t_{ij} := \text{tr}(\sigma_i T(\sigma_j)).$$

Theorem 5.120 (Trace-pairing expansion of a linear map). linearMap_expand_tracePairing ✓ Stmt OK If $\{\sigma_i\}$ is trace-self-dual, then every linear map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ admits the expansion

$$T(\rho) = \sum_i \sum_j t_{ij} \operatorname{tr}(\sigma_j \rho) \sigma_i, \quad t_{ij} = \operatorname{tr}(\sigma_i T(\sigma_j)).$$

Proof. ✓ Proof OK Expand ρ and each $T(\sigma_j)$ in the chosen basis and substitute. The trace-self-dual identity replaces coordinates by traces, giving the stated double-sum formula. $\square$

5.12.2 SVD normal form (existence)

Wolf §2.3 discusses two families of normal forms for the transfer-matrix representation of a quantum channel: the *singular value decomposition* (SVD) and, for qubit channels, the *Lorentz normal form* obtained by $\operatorname{SL}(2, \mathbb{C})$ filtering operations. The SVD normal form is proved below. The Lorentz normal form theorem (Theorem 5.129) requires a compactness / minimisation argument over $\operatorname{SL}(2, \mathbb{C})$ filterings (Wolf Prop. 2.9 / Prop. 2.11); see §5.12.3.

Theorem 5.121 (SVD for positive semi-definite matrices). svd_of_posSemidef ✓ Stmt OK Every positive semi-definite matrix $M \in M_D(\mathbb{C})$ admits a decomposition

$$M = U \operatorname{diag}(\sigma) U^\dagger,$$

where $U \in \operatorname{U}(D)$ is unitary and $\sigma : \{1, \dots, D\} \rightarrow \mathbb{R}_{\geq 0}$ has non-negative entries.

Proof. ✓ Proof OK The spectral theorem for Hermitian matrices provides an orthonormal eigenbasis with real eigenvalues. Positive semi-definiteness forces those eigenvalues to be non-negative, and the decomposition is packaged in SVD form with U the eigenvector unitary and σ the eigenvalue sequence. $\square$

Theorem 5.122 (SVD existence for invertible matrices). svd_of_isUnit ✓ Stmt OK Every invertible complex square matrix $M \in M_D(\mathbb{C})$ admits a singular value decomposition

$$M = U \operatorname{diag}(\sigma) V^\dagger,$$

with $U, V \in \operatorname{U}(D)$ unitary and $\sigma_i > 0$ for all i .

Proof. ✓ Proof OK Apply the spectral theorem to the positive definite matrix $M^\dagger M$ to obtain $M^\dagger M = V \operatorname{diag}(\lambda) V^\dagger$ with $\lambda_i > 0$. Set $\sigma_i := \sqrt{\lambda_i}$ and $\Sigma := \operatorname{diag}(\sigma)$; since Σ is a real positive diagonal, it is self-adjoint and invertible. Define $U := M V \Sigma^{-1}$. Then $U^\dagger U = \Sigma^{-1} V^\dagger (M^\dagger M) V \Sigma^{-1} = \Sigma^{-1} \operatorname{diag}(\lambda) \Sigma^{-1} = I$, so U is unitary, and $U \Sigma V^\dagger = M V (\Sigma^{-1} \Sigma) V^\dagger = M V V^\dagger = M$. $\square$

Theorem 5.123 (SVD representation of a transfer matrix). transferMatrix_svd_of_isUnit ✓ Stmt OK Every invertible transfer matrix $\widehat{T}$ of a linear super-operator $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ admits a singular value decomposition $\widehat{T} = U \operatorname{diag}(\sigma) V^\dagger$ with U, V unitary on $\mathbb{C}^D \otimes \mathbb{C}^D$ and $\sigma_i > 0$.

Proof. ✓ Proof OK Direct specialisation of Theorem 5.122 to the index type $\{1, \dots, D\} \times \{1, \dots, D\}$ used by the transfer-matrix representation. $\square$

Remark 5.124 (Sorted / unique singular values). Theorem 5.122 produces the singular values as an unordered family, without the usual convention that σ_i are sorted in non-increasing order or the statement that they are uniquely determined by M . Downstream applications in Wolf §2.3 (trace-norm identities, polar decomposition, Lorentz normal form) will typically require the sorted / unique variant; that refinement is future work.

5.12.3 Lorentz normal form (Wolf §2.3, existence statements)

Definition 5.125 (SL-filtering operation). [SLFiltering](#) [✓ Stmt OK](#) An *SL-filtering* for $D \times D$ matrices is a completely positive map of the form $\Phi(X) = SXS^\dagger$ where $S \in M_D(\mathbb{C})$ satisfies $\det S = 1$. Such maps are invertible and have Kraus rank 1.

Definition 5.126 (Doubly-stochastic map). [DoublyStochastic](#) [✓ Stmt OK](#) A linear map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is *doubly-stochastic* if $T(1) \propto 1$ and the reduced density matrix $\text{tr}_1[\tau]$ of its Choi matrix $\tau = (T \otimes \text{id})(|\Omega\rangle\langle\Omega|)$ is proportional to the identity. This is the target normal form in Wolf Prop 2.8.

Lemma 5.127 (Infimum attainment for SL-filterings). [infimum_is_attained](#) For a positive-definite Choi matrix τ , the infimum of $\text{tr}[(S_2 \otimes_k S_1)\tau(S_2 \otimes_k S_1)^\dagger]$ over $S_1, S_2 \in M_D(\mathbb{C})$ with $\det S_1 = \det S_2 = 1$ is attained. **Not yet proved**—requires compactness of bounded subsets of $\text{SL}(D, \mathbb{C})$ and the extreme-value theorem.

Theorem 5.128 (Generic normal form for CP maps). [exists_normal_form_generic](#) Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a completely positive map whose Choi matrix is positive-definite (equivalently, T has full Kraus rank). Then there exist SL-filterings Φ_1, Φ_2 such that $\Phi_2 \circ T \circ \Phi_1$ is doubly-stochastic. **Not yet proved**—proof blocked on Lemma 5.127.

Theorem 5.129 (Lorentz normal form for qubit channels). [exists_lorentz_normal_form_qubit](#) For every qubit channel $T : M_2(\mathbb{C}) \rightarrow M_2(\mathbb{C})$, there exist $\text{SL}(2, \mathbb{C})$ -filterings Φ_1, Φ_2 such that the filtered channel $T' = \Phi_2 \circ T \circ \Phi_1$ is in one of the three Lorentz normal forms: diagonal, non-diagonal, or singular. **Not yet proved**—proof blocked on Lemma 5.127 and the Lorentz group classification of $\text{SL}(2, \mathbb{C})$ orbits.

5.13 Determinant of a quantum channel

Definition 5.130 (Channel determinant). [channelDet](#) [✓ Stmt OK](#) For a linear map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$, the *channel determinant* $\det T$ is the determinant of the matrix of T with respect to the standard matrix-unit basis of $M_D(\mathbb{C})$. This is the quantity studied in [Wol12, §6.1.1].

Remark 5.131 (Channel determinant as an operator determinant). [channelDet_eq_linearMap_det](#) [✓ Stmt OK](#) The channel determinant agrees with the ordinary determinant of the linear endomorphism $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$.

Definition 5.132 (Unitary channel). [unitaryChannel](#) [unitaryChannel_isChannel](#) [✓ Stmt OK](#) For a unitary matrix $U \in \mathcal{U}(D)$, the *unitary channel* is $T(\rho) = U\rho U^\dagger$. It is automatically a quantum channel.

Theorem 5.133 (Unitary channels have determinant one). [channelDet_unitary_eq_one](#) [channelDet_norm_eq_one_of_unitaryChannel](#) $T(\rho) = U\rho U^\dagger$ is a unitary channel, then $\det T = 1$ and hence $|\det T| = 1$.

Proof. [✓ Proof OK](#) Vectorization identifies T with a Kronecker product $\overline{U} \otimes U$, whose determinant is $\det \overline{U}^D (\det U)^D = 1$. □

Theorem 5.134 (Determinant bound for positive trace-preserving maps). [channelDet_norm_le_one_of_positive_tracePreserving](#) any positive trace-preserving map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$, $|\det T| \leq 1$. This is [Wol12, Thm. 6.1(1)].

Proof. [✓ Proof OK](#) Every eigenvalue μ of T satisfies $|\mu| \leq 1$ (positivity and trace preservation force spectral radius ≤ 1). Since $\det T$ is the product of the eigenvalues (counted with algebraic multiplicity), the bound $|\det T| = \prod_i |\mu_i| \leq 1$ follows by induction. □

Theorem 5.135 (Determinant one iff the channel is unitary).

[channelDet_norm_eq_one_iff_exists_unitaryChannel](#)

[channelDet_no](#)

a CPTP map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$,

$$|\det T| = 1 \iff \exists U \in \mathcal{U}(D), \quad T(\rho) = U\rho U^\dagger.$$

This is [Wol12, Thm. 6.1(2)].

Proof. [✓ Proof OK](#) Reverse direction is Theorem 5.133. Forward direction: $|\det T| = 1$ together with $|\mu| \leq 1$ for every eigenvalue μ (from trace preservation and positivity) forces every eigenvalue to satisfy $|\mu| = 1$. Hence T is bijective. A bijective CPTP map is a $*$ -automorphism: the Heisenberg dual $T^*(Y) = \sum_i K_i^\dagger Y K_i$ is unital (since T is TP), so the Kadison–Schwarz inequality $T^*(Y^\dagger Y) \geq T^*(Y)^\dagger T^*(Y)$ becomes an equality for every Y . Hence T^* , and therefore T , preserves products. By the Skolem–Noether theorem, every $*$ -automorphism of $M_D(\mathbb{C})$ has the form $T(\rho) = U\rho U^\dagger$ for some unitary U . Kraus freedom then gives $K_i = c_i U$ with $\sum_i |c_i|^2 = 1$. $\square$

5.14 Fixed-point algebra

Definition 5.136 (Kraus fixed points).

[fixedPoints](#)

[✓ Stmt OK](#)

The *fixed-point set* of a Kraus map $E(X) = \sum_i K_i X K_i^\dagger$ is $\text{Fix}(E) = \{X \in M_D(\mathbb{C}) : E(X) = X\}$.

Theorem 5.137 (Fixed points lie in the multiplicative domain).

[fixedPoints_in_multiplicativeDomain](#)

[✓ Stmt OK](#)

Let E be a trace-preserving Kraus map whose Schrödinger-picture map has a positive definite fixed point $\rho > 0$. Then every adjoint fixed point belongs to the multiplicative domain of the adjoint Kraus family.

Proof. [✓ Proof OK](#) The Kadison–Schwarz equality for fixed points forces both one-sided multiplicative identities, so the fixed point lies in the full multiplicative domain. $\square$

Theorem 5.138 (Fixed points form a $*$ -subalgebra).

[fixedPointsStarSubalgebra](#)

[✓ Stmt OK](#)

Let E be a unital Kraus map whose adjoint map E^* has a positive definite fixed point $\rho > 0$. Then $\text{Fix}(E)$ is a $*$ -subalgebra of $M_D(\mathbb{C})$. This is [Wol12, Thm. 6.12].

Proof. [✓ Proof OK](#) Closure under addition and scalar multiplication is immediate from linearity. Closure under $\dagger$: if $E(X) = X$ then $E(X^\dagger) = X^\dagger$ because fixed points are stable under adjoint. Closure under multiplication: if $E(X) = X$ and $E(Y) = Y$, the weighted Kadison–Schwarz argument gives $E(X^\dagger X) = E(X)^\dagger E(X)$ and, applied to $X^\dagger$, also $E(X X^\dagger) = E(X) E(X)^\dagger$. Thus X lies in the multiplicative domain, and Theorem 5.42 yields $E(XY) = E(X)E(Y) = XY$. $\square$

Remark 5.139. [✓ Stmt OK](#) Theorem 5.137 records the trace-preserving / Heisenberg-picture counterpart of the same multiplicative-domain argument.

Theorem 5.140 (Projected elements lie in the multiplicative domain).

[map_mem_multiplicativeDomain_of_idempotent](#)

[✓ Stmt OK](#)

Let E be a unital Kraus map whose adjoint map has a positive definite fixed point, and suppose $E^2 = E$. Then $E(X)$ lies in the multiplicative domain of E for every $X \in M_D(\mathbb{C})$.

Proof. [✓ Proof OK](#) Idempotence gives $E(E(X)) = E(X)$, so $E(X)$ is a fixed point. By the argument in Theorem 5.138, the Kadison–Schwarz gap for the fixed point $E(X)$ vanishes, hence

$$E(E(X)^\dagger E(X)) = E(X)^\dagger E(X).$$

Fixed points are closed under $\dagger$, so the same argument applied to $E(X)^\dagger$ gives

$$E(E(X)E(X)^\dagger) = E(X)E(X)^\dagger.$$

Therefore Theorem 7.28 places $E(X)$ in the multiplicative domain. $\square$

Theorem 5.141 (Choi–Effros left absorption). [choi_effros_left](#) [✓ Stmt OK](#) Under the same hypotheses, for all $X, Y \in M_D(\mathbb{C})$,

$$E(E(X)Y) = E(E(X)E(Y)).$$

Proof. [✓ Proof OK](#) The previous theorem places $E(X)$ in the multiplicative domain, so $E(E(X)Y) = E(E(X))E(Y) = E(X)E(Y)$. Applying the same multiplicativity to $E(X)$ and $E(Y)$ gives $E(E(X)E(Y)) = E(E(X))E(E(Y))$, and idempotence rewrites this as $E(X)E(Y)$. $\square$

Theorem 5.142 (Choi–Effros right absorption). [choi_effros_right](#) [✓ Stmt OK](#) Under the same hypotheses, for all $X, Y \in M_D(\mathbb{C})$,

$$E(XE(Y)) = E(E(X)E(Y)).$$

Proof. [✓ Proof OK](#) The previous theorem places $E(Y)$ in the multiplicative domain, so $E(XE(Y)) = E(X)E(E(Y)) = E(X)E(Y)$. Applying the same multiplicativity to $E(X)$ and $E(Y)$ gives $E(E(X)E(Y)) = E(E(X))E(E(Y))$, and idempotence rewrites this as $E(X)E(Y)$. $\square$

Theorem 5.143 (Choi–Effros symmetric absorption). [choi_effros_left_eq_right](#) [✓ Stmt OK](#) Under the same hypotheses, for all $X, Y \in M_D(\mathbb{C})$,

$$E(E(X)Y) = E(XE(Y)).$$

Proof. [✓ Proof OK](#) Both sides equal $E(E(X)E(Y))$ by the two preceding theorems. $\square$

Theorem 5.144 (Choi–Effros projected product). [choi_effros_sandwich](#) [✓ Stmt OK](#) Under the same hypotheses, for all $X, Y \in M_D(\mathbb{C})$,

$$E(E(X)E(Y)) = E(X)E(Y).$$

Proof. [✓ Proof OK](#) The previous theorem places $E(X)$ and $E(Y)$ in the multiplicative domain. Therefore multiplicativity gives

$$E(E(X)E(Y)) = E(E(X))E(E(Y)).$$

Since E is idempotent under the standing hypotheses, $E(E(X)) = E(X)$ and $E(E(Y)) = E(Y)$, so

$$E(E(X)E(Y)) = E(X)E(Y).$$

$\square$

Definition 5.145 (Adjoint fixed points). [adjointFixedPoints](#) [✓ Stmt OK](#) The *adjoint fixed-point set* $\text{Fix}(E^*) = \{X \in M_D(\mathbb{C}) : E^*(X) = X\}$, where $E^*(X) = \sum_i K_i^\dagger X K_i$.

Theorem 5.146 (Fixed points form a $*$ -subalgebra in the Heisenberg picture). [adjointFixedPointsStarSubalgebra](#) [fixedPoints](#) [✓ Stmt OK](#) The *Kraus map* is trace-preserving and has a positive definite fixed point $\rho > 0$ with $E(\rho) = \rho$, then $\text{Fix}(E^*)$ is a $*$ -subalgebra of $M_D(\mathbb{C})$.

Proof. [✓ Proof OK](#) Apply Theorem 5.138 to the adjoint family $\{K_i^\dagger\}$. $\square$

Definition 5.147 (Kraus commutant). [krausCommutant](#) [krausCommutantStarSubalgebra](#) [✓ Stmt OK](#) The *Kraus commutant* is $\{K_i, K_i^\dagger\}' = \{X \in M_D(\mathbb{C}) : [X, K_i] = [X, K_i^\dagger] = 0 \ \forall i\}$. It is a $*$ -subalgebra of $M_D(\mathbb{C})$.

Theorem 5.148 (Adjoint fixed points equal the Kraus commutant). [adjointFixedPointsStarSubalgebra_eq_krausCommutantStarSubalgebra](#)
*the hypotheses of Theorem 5.146, the adjoint fixed-point *-subalgebra equals the Kraus commutant:*

$$\text{Fix}(E^*) = \{K_i, K_i^\dagger\}'.$$

This is [Wol12, Thm. 6.13].

Proof. [✓ Proof OK](#) ($\supseteq$): if X commutes with all K_i and $K_i^\dagger$, then $E^*(X) = \sum_i K_i^\dagger X K_i = X \sum_i K_i^\dagger K_i = X$ by trace preservation. ($\subseteq$): any *-subalgebra inside $\text{Fix}(E^*)$ lies in the Kraus commutant, because for $X \in \text{Fix}(E^*)$ with $X^\dagger X \in \text{Fix}(E^*)$ the Kadison–Schwarz gap vanishes, which forces $XK_i = K_iX$ and $XK_i^\dagger = K_i^\dagger X$. $\square$

Theorem 5.149 (The Kraus commutant is the largest *-subalgebra inside the adjoint fixed points). [krausCommutantStarSubalgebra_isGreatest_adjointFixedPointStarSubalgebra](#) [✓ Stmt OK](#) *Under the hypotheses above, the Kraus commutant is the largest *-subalgebra contained in the adjoint fixed-point set.*

Proof. [✓ Proof OK](#) By Theorem 5.148, every *-subalgebra inside the adjoint fixed points is automatically contained in the Kraus commutant, while the Kraus commutant itself is such a *-subalgebra. $\square$

5.15 Quantum dynamical semigroups

The one-parameter semigroup theory of Wolf Chapter 7 is treated in Chapter 16. That later chapter records the exponential representation of norm-continuous semigroups, perturbation bounds, and the GKSL/Lindblad description of generators. We do not duplicate that material here.

Chapter 6

Quantum Entropy

This chapter develops the von Neumann entropy and its basic properties for finite-dimensional quantum systems.

6.1 Von Neumann entropy

Definition 6.1 (Von Neumann entropy). vonNeumannEntropy ✓ Stmt OK For a Hermitian matrix $\rho \in M_D(\mathbb{C})$ with eigenvalues $\lambda_0, \dots, \lambda_{D-1}$, the *von Neumann entropy* is

$$S(\rho) = - \sum_{i=0}^{D-1} \lambda_i \log \lambda_i,$$

where $0 \log 0 := 0$.

Theorem 6.2 (Entropy is non-negative). vonNeumannEntropy_nonneg ✓ Stmt OK For any density matrix ρ , $S(\rho) \geq 0$.

Proof. ✓ Proof OK Each eigenvalue λ_i of a density matrix satisfies $0 \leq \lambda_i \leq 1$, and $-x \log x \geq 0$ on $[0, 1]$. □

Lemma 6.3 (Eigenvalue sum). densityMatrices_eigenvalues_sum_one ✓ Stmt OK The eigenvalues of a density matrix sum to 1.

Proof. ✓ Proof OK Follows from $\text{tr}(\rho) = \sum_i \lambda_i = 1$. □

Theorem 6.4 (Eigenvalue bound). densityMatrices_eigenvalues_le_one ✓ Stmt OK Each eigenvalue of a density matrix lies in $[0, 1]$.

Proof. ✓ Proof OK Non-negativity comes from positive semidefiniteness. The upper bound follows because the eigenvalues are non-negative and sum to 1. □

Theorem 6.5 (Entropy upper bound). vonNeumannEntropy_le_log_dim ✓ Stmt OK For a density matrix $\rho \in M_D(\mathbb{C})$ with $D \geq 1$,

$$S(\rho) \leq \log D.$$

Proof. ✓ Proof OK By Jensen's inequality applied to the concave function $-x \log x$, the entropy is maximized when all eigenvalues equal $1/D$. □

6.2 Tripartite partial traces

Definition 6.6 (Partial trace over A). traceA_ABC ✓ Stmt OK For a tripartite matrix ρ_{ABC} on $\mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B} \otimes \mathbb{C}^{d_C}$, the partial trace over A is

$$(\text{tr}_A \rho_{ABC})_{(b_1, c_1)(b_2, c_2)} = \sum_{a=0}^{d_A-1} (\rho_{ABC})_{(a, b_1, c_1)(a, b_2, c_2)}.$$

Definition 6.7 (Partial trace over C). traceC_ABC ✓ Stmt OK The partial trace over C :

$$(\text{tr}_C \rho_{ABC})_{(a_1, b_1)(a_2, b_2)} = \sum_{c=0}^{d_C-1} (\rho_{ABC})_{(a_1, b_1, c)(a_2, b_2, c)}.$$

Definition 6.8 (Partial trace over AC). traceAC_ABC ✓ Stmt OK The partial trace over A and C :

$$(\text{tr}_{AC} \rho_{ABC})_{b_1 b_2} = \sum_{a=0}^{d_A-1} \sum_{c=0}^{d_C-1} (\rho_{ABC})_{(a, b_1, c)(a, b_2, c)}.$$

Lemma 6.9 (Partial trace preserves Hermiticity). traceA_ABC_isHermitian ✓ Stmt OK If ρ_{ABC} is Hermitian, then $\text{tr}_A(\rho_{ABC})$, $\text{tr}_C(\rho_{ABC})$, and $\text{tr}_{AC}(\rho_{ABC})$ are all Hermitian. The same holds for bipartite partial traces $\text{tr}_A(\rho_{AB})$ and $\text{tr}_B(\rho_{AB})$.

Proof. ✓ Proof OK Follows from $\overline{\rho_{ji}} = \rho_{ij}$ applied entry-wise inside the summation defining each partial trace. $\square$

6.3 Strong subadditivity

Theorem 6.10 (Strong subadditivity). strong_subadditivity Not ready For a tripartite density matrix ρ_{ABC} ,

$$S(\rho_{ABC}) + S(\rho_B) \leq S(\rho_{AB}) + S(\rho_{BC}).$$

Definition 6.11 (SSA equality). IsSSAEquality ✓ Stmt OK A tripartite density matrix ρ_{ABC} satisfies *SSA equality* if

$$S(\rho_{ABC}) + S(\rho_B) = S(\rho_{AB}) + S(\rho_{BC}).$$

Definition 6.12 (Quantum Markov decomposition). HayashiMarkovDecomposition ✓ Stmt OK A *Hayashi Markov decomposition* of a tripartite state ρ_{ABC} consists of a finite direct-sum decomposition

$$H_B \cong \bigoplus_j H_{B_j^L} \otimes H_{B_j^R},$$

together with a unitary change of basis on B , a probability vector $(p_j)_j$, and density matrices $\rho_{AB_j^L}$ and $\rho_{B_j^R C}$ such that, in the adapted basis, the state becomes

$$\bigoplus_j p_j \rho_{AB_j^L} \otimes \rho_{B_j^R C}.$$

Theorem 6.13 (Hayashi SSA-equality characterization). hayashi_ssa_equality_characterization Not ready For a tripartite density matrix ρ_{ABC} ,

$$S(\rho_{ABC}) + S(\rho_B) = S(\rho_{AB}) + S(\rho_{BC})$$

holds if and only if ρ_{ABC} admits a quantum Markov decomposition on the middle subsystem B .

This equality criterion is cited here as an input for the later MPDO arguments. It is the equality criterion needed by Appendix C of arXiv:1606.00608. We cite Hayashi, J. Phys. A 37 (2004) L205–L208, and Ruskai, J. Math. Phys. 43 (2002) 4358; the structural block-decomposition formulation used in the MPDO paper also appears in Hayden–Jozsa–Petz–Winter, Commun. Math. Phys. 246 (2004) 359–374.

6.4 Mutual information

Definition 6.14 (Mutual information). mutualInformation ✓ Stmt OK The quantum mutual information of a bipartite state ρ_{AB} is

$$I(A:B) = S(\rho_A) + S(\rho_B) - S(\rho_{AB}).$$

Theorem 6.15 (Mutual information is non-negative). mutualInformation_ssa_trivial_B_nonneg ✓ Stmt OK For any bipartite density matrix ρ_{AB} , $I(A:B) \geq 0$.

Proof. ✓ Proof OK Apply strong subadditivity (Theorem 6.10) with trivial B (one-dimensional middle system). The SSA inequality $S(\rho_{ABC}) + S(\rho_B) \leq S(\rho_{AB}) + S(\rho_{BC})$ reduces to subadditivity $S(\rho_{AC}) \leq S(\rho_A) + S(\rho_C)$, which gives $I(A:C) \geq 0$. $\square$

6.5 Entropy formulations

This section states the entropy formulations used later in the development: von Neumann entropy, strong subadditivity, quantum Markov decomposition, and mutual information. These statements are cited from the standard entropy literature and supply the entropy-theoretic input for the later MPDO arguments.

Definition 6.16 (Entropy formulation of von Neumann entropy). vonNeumannEntropy ✓ Stmt OK This formulation has the same value $S(\rho) = -\sum_i \lambda_i \log \lambda_i$ as in Definition 6.1.

Theorem 6.17 (Entropy formulation of strong subadditivity). strongSubadditivity ✓ Stmt OK For any tripartite density matrix ρ_{ABC} on $A \otimes B \otimes C$,

$$S(\rho_{ABC}) + S(\rho_B) \leq S(\rho_{AB}) + S(\rho_{BC}).$$

This formulation introduces no new axiom: it is the same strong-subadditivity statement as Theorem 6.10. The underlying axiom’s proof (Lieb–Ruskai 1973) is deferred to the umbrella entropy issue.

Definition 6.18 (Entropy formulation of quantum Markov decomposition). QuantumMarkovDecomposition ✓ Stmt OK This is the entropy formulation of the Hayashi Markov decomposition from Definition 6.12.

Theorem 6.19 (SSA equality iff quantum Markov decomposition). ssaEquality_iff_exists_quantumMarkovDecomposition ✓ Stmt OK For any tripartite density matrix ρ_{ABC} , equality in strong subadditivity holds if and only if ρ_{ABC} admits a quantum Markov decomposition on the middle subsystem B .

This formulation introduces no new axiom: it is the same equality criterion as Theorem 6.13.

Definition 6.20 (Entropy formulation of mutual information). mutualInformation ✓ Stmt OK This formulation has the same value $I(A:B) = S(\rho_A) + S(\rho_B) - S(\rho_{AB})$ as in Definition 6.14.

Theorem 6.21 (Subadditivity from trivial-middle SSA). [subadditivity_ssa_trivial_B](#) ✓ Stmt OK For a tripartite density matrix ρ_{ABC} with $\dim B = 1$,

$$S(\rho_{ABC}) \leq S(\rho_{AB}) + S(\rho_{BC}).$$

Proof. ✓ Proof OK Apply Theorem 6.17 with one-dimensional middle subsystem. The reduced middle state has trace 1 by Theorem 6.24, hence its entropy vanishes by Theorem 6.25. $\square$

6.6 Trivial-factor corollaries

Theorem 6.22 (Partial trace over A preserves the full trace). [trace_eq_trace_traceA_ABC](#) ✓ Stmt OK For any tripartite matrix ρ_{ABC} ,

$$\text{tr}(\rho_{ABC}) = \text{tr}(\text{tr}_A(\rho_{ABC})).$$

Proof. ✓ Proof OK Unfold the definition of tr_A and reorder the finite sums. $\square$

Theorem 6.23 (Partial trace over C preserves the full trace). [trace_eq_trace_traceC_ABC](#) ✓ Stmt OK For any tripartite matrix ρ_{ABC} ,

$$\text{tr}(\rho_{ABC}) = \text{tr}(\text{tr}_C(\rho_{ABC})).$$

Proof. ✓ Proof OK Unfold the definition of tr_C and rewrite the trace as the same triple sum. $\square$

Theorem 6.24 (Partial trace over AC preserves the full trace). [trace_eq_trace_traceAC_ABC](#) ✓ Stmt OK For any tripartite matrix ρ_{ABC} ,

$$\text{tr}(\rho_{ABC}) = \text{tr}(\text{tr}_{AC}(\rho_{ABC})).$$

Proof. ✓ Proof OK Unfold the definition of tr_{AC} and interchange the outer two finite sums. $\square$

Theorem 6.25 (Entropy vanishes in dimension 1). [vonNeumannEntropy_eq_zero_of_fin_one](#) ✓ Stmt OK If $\rho \in M_1(\mathbb{C})$ is Hermitian and $\text{tr}(\rho) = 1$, then

$$S(\rho) = 0.$$

Proof. ✓ Proof OK The unique eigenvalue equals the trace, hence equals 1, and $-1 \cdot \log 1 = 0$. $\square$

Chapter 7

Schwarz Inequalities and Multiplicative Domains

This chapter develops the Schwarz-inequality theory from Wolf’s Chapter 5 [Wol12]. We begin with the Kadison–Schwarz inequality for Kraus maps, then describe the multiplicative domain and its algebraic structure. We also establish the extension to normal operators for positive subunital maps, the resulting order and spectral contractivity statements, and Wolf’s standard example of a Schwarz map that is not completely positive.

7.1 Kadison–Schwarz inequality

A Kraus map is the concrete realisation of a completely positive map (Definition 5.4). The Kadison–Schwarz inequality and the multiplicative-domain characterisation are proved first for Kraus maps. The transfer map is a Kraus map by construction (Remark 5.17), so these results apply directly to transfer maps.

Definitions 7.1–7.4 below restate the Kraus-map notions from Chapter 5 for the reader’s convenience.

Definition 7.1 (Kraus map). krausMap ✓ Stmt OK Given operators $\{K_i\}_{i=0}^{d-1}$ with $K_i \in M_D(\mathbb{C})$, the *Kraus map* is $E(X) = \sum_{i=0}^{d-1} K_i X K_i^\dagger$. A linear map is completely positive (Definition 5.4) if and only if it can be written in this form.

Definition 7.2 (Adjoint Kraus map). krausAdjointMap ✓ Stmt OK The *adjoint Kraus map* is $E^*(X) = \sum_{i=0}^{d-1} K_i^\dagger X K_i$. When the $\{K_i\}$ are the matrices of an MPS tensor A , this is the transfer map of the conjugate-transposed family $i \mapsto (A^i)^\dagger$.

Definition 7.3 (Unital Kraus map). isUnitalKraus ✓ Stmt OK A Kraus map is *unital* if $\sum_{i=0}^{d-1} K_i K_i^\dagger = \mathbb{1}$. Equivalently, $E(\mathbb{1}) = \mathbb{1}$. In the later gauge language this is the right-canonical condition.

Definition 7.4 (Trace-preserving Kraus map). isTPKraus ✓ Stmt OK A Kraus map is *trace-preserving* if $\sum_{i=0}^{d-1} K_i^\dagger K_i = \mathbb{1}$. Equivalently, the adjoint Kraus map is unital. This is the standard MPS normalization condition. In the later gauge language it is the left-canonical condition, so Kadison–Schwarz arguments are often applied to the adjoint map.

Theorem 7.5 (Kadison–Schwarz inequality). [[kadison_schwarz](#)] ✓ Stmt OK Let $E(X) = \sum_i K_i X K_i^\dagger$ be a unital Kraus map (i.e. $\sum_i K_i K_i^\dagger = \mathbb{1}$). Then for all $X \in M_D(\mathbb{C})$,

$$E(X^\dagger X) \geq E(X)^\dagger E(X).$$

This is [[Wol12](#), Eq. (5.2)].

Proof. ✓ Proof OK The 2×2 block matrix $\begin{bmatrix} X^\dagger X & X^\dagger \\ X & \mathbb{1} \end{bmatrix}$ is PSD ($= vv^\dagger$ with $v = [X^\dagger, \mathbb{1}]^T$). Applying the unital CP map blockwise preserves positive semidefiniteness (each block $K_i(\cdot)K_i^\dagger$ preserves PSD, and unitality ensures the $(2, 2)$ -block maps to $\mathbb{1}$). The Schur complement of the $(2, 2)$ -block $\mathbb{1}$ gives the result. $\square$

Theorem 7.6 (Kadison–Schwarz for the adjoint channel). [[kadison_schwarz_adjoint](#)] ✓ Stmt OK If $\sum_i K_i^\dagger K_i = \mathbb{1}$ (trace-preserving), then the adjoint Kraus map satisfies $E^*(X^\dagger X) \geq E^*(X)^\dagger E^*(X)$.

Proof. ✓ Proof OK The adjoint operators $\{K_i^\dagger\}$ satisfy $\sum_i K_i^\dagger (K_i^\dagger)^\dagger = \sum_i K_i^\dagger K_i = \mathbb{1}$, so they form a unital family. Apply Theorem 7.5 to $\{K_i^\dagger\}$. $\square$

7.1.1 Two-positive maps and the generalized Schwarz inequality

Definition 7.7 (k -positive map). [[IsNPositiveMap](#)] ✓ Stmt OK A linear map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is k -positive if $E \otimes \mathbb{1}_k$ is positive on $M_D(\mathbb{C}) \otimes M_k(\mathbb{C})$.

Definition 7.8 (Two-positive map). [[Is2PositiveMap](#)] ✓ Stmt OK A linear map is *two-positive* if it is 2-positive.

Theorem 7.9 (Completely positive maps are k -positive). [[isNPositiveMap](#)] ✓ Stmt OK Every completely positive map is k -positive for all $k \geq 1$.

Theorem 7.10 ($(k+1)$ -positive implies k -positive). [[mono](#)] ✓ Stmt OK Positivity under amplification is monotone in k .

Theorem 7.11 (Completely positive implies two-positive). [[is2PositiveMap](#)] ✓ Stmt OK Every completely positive map is two-positive.

Theorem 7.12 (Two-positive implies positive). [[isPositiveMap](#)] ✓ Stmt OK Every two-positive map is positive.

Definition 7.13 (Unital linear map). [[IsUnitalMap](#)] ✓ Stmt OK A linear map E is unital if $E(\mathbb{1}) = \mathbb{1}$.

Theorem 7.14 (Kadison–Schwarz for unital two-positive maps). [[kadison_schwarz_2positive](#)] ✓ Stmt OK If E is unital and two-positive, then

$$E(X^\dagger X) \geq E(X)^\dagger E(X)$$

for all $X \in M_D(\mathbb{C})$.

Proof. Form the 2×2 block matrix $\begin{pmatrix} \mathbb{1} & X \\ X^\dagger & X^\dagger X \end{pmatrix} \geq 0$. Apply $E \otimes \text{id}_2$ (positive by 2-positivity) and read off the $(2, 2)$ -block Schur complement: $E(X^\dagger X) - E(X)^\dagger E(X) \geq 0$. $\square$

Theorem 7.15 (Kraus-form Kadison–Schwarz as a special case). [[kadison_schwarz_from_2positive](#)] ✓ Stmt OK The Kraus-map Kadison–Schwarz inequality is recovered from the two-positive formulation.

7.1.2 Douglas-type factorization lemmas

Theorem 7.16 (Range inclusion implies right factorization). [factorization_of_range_mulVecLin_le](#) ✓ Stmt OK If $\text{ran}(A) \subseteq \text{ran}(B)$, then there exists C such that $A = BC$.

Proof. In finite dimensions, $\text{ran}(A) \subseteq \text{ran}(B)$ implies $A = B(B^\dagger B)^\dagger B^\dagger A$, where $(B^\dagger B)^\dagger$ denotes the Moore–Penrose pseudoinverse. Set $C = (B^\dagger B)^\dagger B^\dagger A$. □

Theorem 7.17 (Vectorwise range criterion implies factorization). [factorization_of_forall_mulVec_mem_range](#) ✓ Stmt OK If $Av \in \text{ran}(B)$ for every vector v , then there exists C with $A = BC$.

Theorem 7.18 (Hilbert–Schmidt contraction). [hilbertSchmidt_contraction](#) ✓ Stmt OK If a Kraus map is both unital and trace-preserving, then $\text{tr}(E(X)^\dagger E(X)) \leq \text{tr}(X^\dagger X)$ for all $X \in M_D(\mathbb{C})$.

Proof. ✓ Proof OK By Theorem 7.5, $E(X^\dagger X) - E(X)^\dagger E(X) \geq 0$, so $\text{tr}(E(X^\dagger X)) \geq \text{tr}(E(X)^\dagger E(X))$. Trace preservation gives $\text{tr}(E(X^\dagger X)) = \text{tr}(X^\dagger X)$. □

Lemma 7.19 (Trace pairing for Kraus map and adjoint map). [trace_mul_map_eq_trace_adjointMap_mul](#) ✓ Stmt OK For all $X, Y \in M_D(\mathbb{C})$,

$$\text{tr}(X E(Y)) = \text{tr}(E^*(X) Y).$$

Lemma 7.20 (Positive-definite trace test). [posSemidef_eq_zero_of_posDef_trace_mul_eq_zero](#) ✓ Stmt OK If $A > 0$, $X \geq 0$, and $\text{tr}(AX) = 0$, then $X = 0$.

Theorem 7.21 (Fixed-point peripheral eigenvectors satisfy KS equality). [ks_equality_of_peripheral_eigenvector_of_fixedPoin](#) ✓ Stmt OK Let E be unital and let E^* be trace-preserving with a positive definite fixed point $\rho > 0$. If $E(X) = \mu X$ with $|\mu| = 1$, then

$$E(X^\dagger X) = E(X)^\dagger E(X).$$

7.2 Multiplicative domain

Theorem 7.22 (KS gap decomposition). [ks_gap_eq_sum_squares](#) ✓ Stmt OK For a unital Kraus map E , the Kadison–Schwarz gap decomposes as

$$E(X^\dagger X) - E(X)^\dagger E(X) = \sum_{i=0}^{d-1} R_i^\dagger R_i,$$

where $R_i := XK_i^\dagger - K_i^\dagger E(X)$.

Proof. ✓ Proof OK Expand the right-hand side $\sum_i (XK_i^\dagger - K_i^\dagger E(X))^\dagger (XK_i^\dagger - K_i^\dagger E(X))$, distribute, and use unitality $\sum_i K_i K_i^\dagger = \mathbb{1}$ to identify the result with the KS gap. □

Theorem 7.23 (KS equality implies Kraus commutation). [kraus_commute_of_ks_equality](#) ✓ Stmt OK Let E be a unital Kraus map. If $E(X^\dagger X) = E(X)^\dagger E(X)$ for some X , then $XK_i^\dagger = K_i^\dagger E(X)$ for all i .

Proof. ✓ Proof OK By Theorem 7.22, the KS gap is $\sum_i R_i^\dagger R_i$ with $R_i = XK_i^\dagger - K_i^\dagger E(X)$. If the gap vanishes, $\sum_i R_i^\dagger R_i = 0$. Each summand $R_i^\dagger R_i$ is PSD, so each must be zero, giving $R_i = 0$. □

Theorem 7.24 (KS equality for peripheral eigenvectors). [ks_equality_of_peripheral_eigenvector](#) ✓ Stmt OK Let E be a Kraus map that is both unital and trace-preserving. If $E(X) = \mu X$ with $|\mu| = 1$, then the KS gap vanishes: $E(X^\dagger X) = E(X)^\dagger E(X)$.

Proof. ✓ Proof OK The gap $G := E(X^\dagger X) - E(X)^\dagger E(X)$ is PSD by Theorem 7.5. Its trace satisfies $\text{tr}(G) = \text{tr}(E(X^\dagger X)) - \text{tr}(E(X)^\dagger E(X))$. Trace preservation gives $\text{tr}(E(X^\dagger X)) = \text{tr}(X^\dagger X)$. Using $E(X) = \mu X$ and $|\mu| = 1$, $\text{tr}(E(X)^\dagger E(X)) = |\mu|^2 \text{tr}(X^\dagger X) = \text{tr}(X^\dagger X)$. So $\text{tr}(G) = 0$, and a PSD matrix with zero trace must be zero. $\square$

Theorem 7.25 (Left multiplicative identity from KS equality). multiplicative_domain_left ✓ Stmt OK Let E be a unital Kraus map. If $E(X^\dagger X) = E(X)^\dagger E(X)$, then $E(X^\dagger Y) = E(X)^\dagger E(Y)$ for all $Y \in M_D(\mathbb{C})$.

Proof. ✓ Proof OK From Theorem 7.23, $XK_i^\dagger = K_i^\dagger E(X)$ for all i . Taking conjugate transposes: $K_i X^\dagger = E(X)^\dagger K_i$. Then

$$\begin{aligned} E(X^\dagger Y) &= \sum_i K_i X^\dagger Y K_i^\dagger \\ &= \sum_i E(X)^\dagger K_i Y K_i^\dagger \\ &= E(X)^\dagger E(Y). \end{aligned}$$

$\square$

7.2.1 Full multiplicative-domain structure

Definition 7.26 (Right and left multiplicative domains). rightMultiplicativeDomain leftMultiplicativeDomain ✓ Stmt OK For a Kraus map E , define

$$\mathcal{A}_R(E) := \{X \in M_D(\mathbb{C}) : E(XY) = E(X)E(Y) \text{ for all } Y \in M_D(\mathbb{C})\}$$

and

$$\mathcal{A}_L(E) := \{X \in M_D(\mathbb{C}) : E(YX) = E(Y)E(X) \text{ for all } Y \in M_D(\mathbb{C})\}.$$

We follow the convention that $\mathcal{A}_R(E)$ controls right multiplication and $\mathcal{A}_L(E)$ controls left multiplication. These are the two one-sided multiplicative domains from [Wol12, Eqs. (5.11)–(5.12)].

Definition 7.27 (Full multiplicative domain). multiplicativeDomain ✓ Stmt OK The *multiplicative domain* of E is

$$\mathcal{A}(E) := \mathcal{A}_R(E) \cap \mathcal{A}_L(E).$$

Theorem 7.28 (Multiplicative-domain characterization). mem_rightMultiplicativeDomain_iff mem_leftMultiplicativeDomain_iff Let E be a unital Kraus map. Then

$$X \in \mathcal{A}_R(E) \iff E(XX^\dagger) = E(X)E(X)^\dagger$$

and

$$X \in \mathcal{A}_L(E) \iff E(X^\dagger X) = E(X)^\dagger E(X).$$

Together these are the two one-sided characterizations from [Wol12, Thm. 5.7].

Proof. ✓ Proof OK If X lies in a one-sided multiplicative domain, evaluate the defining identity at $Y = X^\dagger$. Conversely, the left-domain implication is exactly Theorem 7.25, and the right-domain implication follows by applying Theorem 7.25 to $X^\dagger$. $\square$

Theorem 7.29 (One-sided multiplicative domains are subalgebras). rightMultiplicativeDomainSubalgebra leftMultiplicativeDomainSubalgebra Let E be a unital Kraus map. Both $\mathcal{A}_R(E)$ and $\mathcal{A}_L(E)$ are unital subalgebras of $M_D(\mathbb{C})$.

Proof. ✓ Proof OK Closure under addition follows from linearity of E . Closure under multiplication: if $X, Y \in \mathcal{A}_R(E)$, then $E((XY)Z) = E(X)E(YZ) = E(X)E(Y)E(Z)$ for all Z , using Theorem 7.28 iteratively, and the $\mathcal{A}_L(E)$ case is analogous. Containment of $\mathbb{1}$ follows from unitality of E . $\square$

Theorem 7.30 (Multiplicative domain is a $*$ -subalgebra). multiplicativeDomainStarSubalgebra ✓ Stmt OK For a unital Kraus map E , the full multiplicative domain $\mathcal{A}(E)$ is a $*$ -subalgebra of $M_D(\mathbb{C})$. This is the algebraic content of [Wol12, Thm. 5.7].

Proof. ✓ Proof OK If $X \in \mathcal{A}(E) = \mathcal{A}_R(E) \cap \mathcal{A}_L(E)$, then $E(X^\dagger X) = E(X)^\dagger E(X)$ and $E(XX^\dagger) = E(X)E(X)^\dagger$. The characterization of Theorem 7.28 shows $X^\dagger \in \mathcal{A}(E)$. Combined with Theorem 7.29, this gives a $*$ -subalgebra. $\square$

Theorem 7.31 (Restriction to multiplicative domain is a $*$ -homomorphism). krausMapStarAlgHom ✓ Stmt OK If E is unital, then the restricted map $E|_{\mathcal{A}(E)} : \mathcal{A}(E) \rightarrow M_D(\mathbb{C})$ is a $*$ -algebra homomorphism. This is the homomorphism conclusion of [Wol12, Thm. 5.7].

Proof. ✓ Proof OK Multiplicativity on $\mathcal{A}(E)$ is immediate from the definition. For $X \in \mathcal{A}(E)$, the Kraus commutation relation from Theorem 7.23 gives $E(X^\dagger) = E(X)^\dagger$, so the restriction preserves adjoints. $\square$

7.2.2 Schwarz inequality for normal operators

Theorem 7.32 (CP Schwarz inequality for normal operators). kadison_schwarz_normal_cp kadison_schwarz_normal_cp_le ✓ Stmt OK If $E^*(X) = \sum_i K_i^\dagger X K_i$ be the adjoint Kraus map of a trace-preserving Kraus family, and let $A \in M_D(\mathbb{C})$ be normal. Then the Schwarz gap

$$E^*(A^\dagger A) - E^*(A^\dagger)E^*(A)$$

is positive semidefinite. Equivalently,

$$E^*(A^\dagger)E^*(A) \leq E^*(A^\dagger A).$$

This is the completely positive case of [Wol12, Prop. 5.1].

Proof. ✓ Proof OK The positivity statement is exactly Theorem 7.6. The second formulation is just the corresponding Loewner-order reformulation. $\square$

Theorem 7.33 (Schwarz inequality for normal operators). schwarz_inequality_normal_operator ✓ Stmt OK Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a positive linear map with $T(\mathbb{1}) \leq \mathbb{1}$. If $A \in M_D(\mathbb{C})$ is normal, then

$$T(A^\dagger A) - T(A^\dagger)T(A) \geq 0.$$

This is [Wol12, Prop. 5.1].

Proof. ✓ Proof OK Since A is normal, diagonalize $A = UDU^\dagger$ with D diagonal and U unitary. Express $A^\dagger A = U|D|^2U^\dagger$ as a sum $\sum_{i,j} d_i d_j P_i P_j$ where $P_i = U e_i e_i^\dagger U^\dagger$. Each P_i is rank-one PSD, and the images $T(P_i)$ pairwise commute because they are images of mutually orthogonal projectors under a positive map. The Schwarz gap then reduces to a block quadratic form in the images, which is shown nonneg by a simultaneous-diagonalization argument: positivity of T ensures the scalar PSD matrix $(\langle v_k, T(P_i P_j) v_k \rangle)_{i,j}$ dominates the product $(\langle v_k, T(P_i) v_k \rangle \langle v_k, T(P_j) v_k \rangle)_{i,j}$, and combining over a common eigenbasis of the $T(P_i)$ yields the result. $\square$

Remark 7.34. The proof follows the positivity-on-abelian argument of [Wol12, Prop. 1.6]: a positive map restricted to the commutative $*$ -subalgebra generated by the spectral projectors of A satisfies the Schwarz inequality directly, without invoking the Arveson extension theorem.

7.2.3 Schwarz inequality for subnormal and commuting-dominant operators

Theorem 7.35 (Schwarz inequality for subnormal operators). [schwarz_inequality_subnormal_operator](#) ✓ Stmt OK Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a positive linear map with $T(\mathbb{1}) \leq \mathbb{1}$. If $A \in M_D(\mathbb{C})$ is subnormal (i.e. there exists a normal operator on a larger space whose compression to $\mathbb{C}^D$ is A), then

$$T(A^\dagger A) - T(A^\dagger)T(A) \geq 0.$$

This is [Wol12, Thm. 5.5].

Proof. ✓ Proof OK Embed A as the northwest corner of a normal operator N on $\mathbb{C}^D \oplus \mathbb{C}^E$. Extend T to a positive subunitary map $\tilde{T}$ on the larger space (acting as T on the northwest block, zero elsewhere). Apply Theorem 7.33 to $\tilde{T}$ and N , then compress the resulting inequality back to $\mathbb{C}^D$. □

Theorem 7.36 (Schwarz inequality for commuting-dominant operators). [schwarz_inequality_commuting_dominant_operator](#) ✓ Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a positive linear map with $T(\mathbb{1}) \leq \mathbb{1}$. If $A \in M_D(\mathbb{C})$ admits a positive semidefinite dominant $B \geq 0$ with $A^\dagger A \leq B$ and B commuting with A (in the sense $BA = AB$, $BA^\dagger = A^\dagger B$), then

$$T(A^\dagger)T(A) \leq T(B).$$

This is [Wol12, Thm. 5.6].

Proof. ✓ Proof OK Let $B^{1/2}$ be the positive square root of B . Since B commutes with A and $A^\dagger$, the contraction $C = B^{-1/2}A$ (defined on the range of $B^{1/2}$) is subnormal. An ε -regularisation $B_\varepsilon = B + \varepsilon \mathbb{1}$ makes $B_\varepsilon > 0$ and the contraction C_ε globally defined. Apply Theorem 7.35 to C_ε , multiply both sides by $T(B_\varepsilon)$, use positivity and monotonicity, and take $\varepsilon \rightarrow 0$. □

Theorem 7.37 (CP variant: Kadison–Schwarz for commuting-dominant inputs). [kadison_schwarz_commuting_dominant_cp](#) ✓ Let E^* be the adjoint of a trace-preserving Kraus map. If $A \in M_D(\mathbb{C})$ admits a positive semidefinite dominant B commuting with A , $A^\dagger$, and satisfying $A^\dagger A \leq B$, then $E^*(A^\dagger)E^*(A) \leq E^*(B)$.

Proof. ✓ Proof OK Apply Theorem 7.36 to the adjoint Kraus map (which is unital under the trace-preservation hypothesis). □

7.2.4 Positive maps preserve order and spectral intervals

Theorem 7.38 (Positive maps are monotone). [map_le_map](#) ✓ Stmt OK If $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is positive and $A \leq B$, then $T(A) \leq T(B)$.

Proof. ✓ Proof OK Since $B - A \geq 0$, positivity gives $T(B - A) \geq 0$. By linearity, this is $T(B) - T(A) \geq 0$. □

Theorem 7.39 (Positive maps preserve adjoints). [map_conjTranspose](#) ✓ Stmt OK If $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is positive, then $T(A^\dagger) = T(A)^\dagger$ for all $A \in M_D(\mathbb{C})$.

Proof. ✓ Proof OK Write $A = B + iC$ with B and C Hermitian. Positive maps send Hermitian matrices to Hermitian matrices, so they preserve this decomposition and commute with taking adjoints. □

Theorem 7.40 (Order intervals are preserved). image_bounded ✓ Stmt OK Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive with $T(\mathbb{1}) \leq \mathbb{1}$. If $a \leq 0 \leq b$ and $a\mathbb{1} \leq A \leq b\mathbb{1}$, then

$$a\mathbb{1} \leq T(A) \leq b\mathbb{1}.$$

This is the matrix form of [Wol12, Eq. (5.21)].

Proof. ✓ Proof OK By monotonicity, $T(a\mathbb{1}) \leq T(A) \leq T(b\mathbb{1})$. For the lower bound, $T(a\mathbb{1}) = aT(\mathbb{1})$. Since $a \leq 0$ and $T(\mathbb{1}) \leq \mathbb{1}$, we have $aT(\mathbb{1}) \geq a\mathbb{1}$. For the upper bound, $T(b\mathbb{1}) = bT(\mathbb{1}) \leq b\mathbb{1}$ since $b \geq 0$. Therefore $a\mathbb{1} \leq T(A) \leq b\mathbb{1}$. □

Theorem 7.41 (Spectral interval contractivity). spectrum_contractivity ✓ Stmt OK Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive with $T(\mathbb{1}) \leq \mathbb{1}$, and let $A \in M_D(\mathbb{C})$ be Hermitian. If $\text{spec}(A) \subseteq [a, b]$ with $a \leq 0 \leq b$, then

$$\text{spec}(T(A)) \subseteq [a, b].$$

This is [Wol12, Eq. (5.21)].

Proof. ✓ Proof OK For Hermitian matrices, the inclusion $\text{spec}(A) \subseteq [a, b]$ is equivalent to the order bounds $a\mathbb{1} \leq A \leq b\mathbb{1}$. Apply Theorem 7.40 and translate the resulting inequalities back into spectral bounds. □

7.2.5 A positive Schwarz map that is not completely positive

Example 7.42 (Transpose-trace map on $M_2(\mathbb{C})$). wolfExample53 ✓ Stmt OK Consider the linear map $T : M_2(\mathbb{C}) \rightarrow M_2(\mathbb{C})$ defined by

$$T(A) = \frac{1}{2}A^T + \frac{1}{4}\text{tr}(A)\mathbb{1}.$$

This is the map from [Wol12, Ex. 5.3].

Theorem 7.43 (The transpose-trace map is positive). wolfExample53_isPositive ✓ Stmt OK The map from Example 7.42 is positive.

Proof. ✓ Proof OK It is a nonnegative linear combination of the transpose map and the map $A \mapsto \text{tr}(A)\mathbb{1}$, and both of these maps send positive semidefinite matrices to positive semidefinite matrices. □

Theorem 7.44 (The transpose-trace map satisfies the Schwarz inequality). wolfExample53_satisfies_schwarz ✓ Stmt OK For every $A \in M_2(\mathbb{C})$,

$$T(A^\dagger A) - T(A^\dagger)T(A) \geq 0,$$

where T is the map from Example 7.42.

Proof. ✓ Proof OK Write the Schwarz gap as $F(A)$. One checks that $F(A + \lambda\mathbb{1}) = F(A)$ for every scalar λ , so it is enough to treat the case $\text{tr}(A) = 0$. In that case a direct 2×2 computation shows that $F(A)$ dominates $\frac{1}{2}(A^\dagger A)^T$, hence is positive semidefinite. □

Theorem 7.45 (The transpose-trace map is not completely positive). wolfExample53_not_cp ✓ Stmt OK The map from Example 7.42 is not completely positive.

Proof. ✓ Proof OK Compute the Choi matrix of T and test it on the antisymmetric vector $|01\rangle - |10\rangle$. The resulting expectation value is negative, so the Choi matrix is not positive semidefinite. □

7.3 Operator convexity and Jensen inequalities

This section collects the operator convexity/concavity results, trace inequalities, the operator Jensen inequality for positive maps, and the Lieb concavity theorem. These are standard results in matrix analysis (Bhatia Ch. V, Wolf Thm. 5.1, Hansen–Pedersen 2003, Lieb 1973). They are used here as cited inputs, since the corresponding matrix-analysis proofs for real powers $x \mapsto x^p$ on the Loewner order and for the matrix logarithm are not developed in this chapter.

7.3.1 Diagonal Jensen inequality

Lemma 7.46 (Diagonal Jensen inequality). [diagonal_jensen_of_convexOn](#) ✓ Stmt OK Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be convex on $[0, \infty)$, let A be a positive semidefinite matrix indexed by a finite type n , and let $v \in \mathbb{C}^n$ satisfy $\langle v, v \rangle = 1$. Then

$$f(\Re \langle v, Av \rangle) \leq \Re \langle v, f(A)v \rangle,$$

where $f(A)$ is defined by the Hermitian continuous functional calculus.

Proof. ✓ Proof OK Write $A = U \text{diag}(\mu) U^*$ by the spectral theorem and set $w = U^*v$. Since U is unitary, $p_i := |w_i|^2$ satisfies $\sum_i p_i = \|v\|^2 = 1$, so (p_i) is a probability distribution on n . The eigenvalues μ_i lie in $[0, \infty)$ because A is positive semidefinite. A direct computation yields $\Re \langle v, Av \rangle = \sum_i p_i \mu_i$ and $\Re \langle v, f(A)v \rangle = \sum_i p_i f(\mu_i)$, so the scalar Jensen inequality applied to the weights (p_i) and points (μ_i) gives the conclusion. See Bhatia, *Matrix Analysis*, Ch. V. $\square$

7.3.2 Trace concavity and convexity of matrix powers

Theorem 7.47 (Trace concavity of real powers). [trace_rpow_concave](#) [trace_rpow_concave](#) ✓ Stmt OK For $p \in [0, 1]$, PSD matrices A_1, A_2 , and $t \in [0, 1]$:

$$t \Re \text{tr}(A_1^p) + (1-t) \Re \text{tr}(A_2^p) \leq \Re \text{tr}((tA_1 + (1-t)A_2)^p).$$

Follows from operator concavity of $x \mapsto x^p$ composed with trace monotonicity on the Loewner order.

Proof. ✓ Proof OK Let $A = tA_1 + (1-t)A_2$ and let $\{\psi_j\}$ with eigenvalues $\mu_j \geq 0$ be an orthonormal eigenbasis of A . A direct computation gives $\Re \text{tr}(A^p) = \sum_j \mu_j^p$ and $\mu_j = t a_j + (1-t) b_j$ where $a_j = \Re \langle \psi_j, A_1 \psi_j \rangle$, $b_j = \Re \langle \psi_j, A_2 \psi_j \rangle$. Scalar concavity of $x \mapsto x^p$ on $[0, \infty)$ yields $\mu_j^p \geq t a_j^p + (1-t) b_j^p$, and the diagonal Jensen inequality (Lemma 7.46, concave form) gives $a_j^p \leq \Re \langle \psi_j, A_1^p \psi_j \rangle$ (and similarly for b_j). Summing over j and using $\sum_j \Re \langle \psi_j, M \psi_j \rangle = \Re \text{tr} M$ yields the conclusion. See Bhatia, *Matrix Analysis*, Ch. V. $\square$

Theorem 7.48 (Trace convexity of real powers). [trace_rpow_convex](#) [trace_rpow_convex](#) ✓ Stmt OK For $p \in [1, 2]$, PSD matrices A_1, A_2 , and $t \in [0, 1]$:

$$\Re \text{tr}((tA_1 + (1-t)A_2)^p) \leq t \Re \text{tr}(A_1^p) + (1-t) \Re \text{tr}(A_2^p).$$

Follows from operator convexity of $x \mapsto x^p$ composed with trace monotonicity on the Loewner order.

Proof. ✓ Proof OK Identical eigenbasis reduction as in Theorem 7.47, with scalar convexity (rather than concavity) of $x \mapsto x^p$ on $[0, \infty)$ for $p \geq 1$ and the convex form of the diagonal Jensen inequality. $\square$

7.3.3 Operator Jensen inequality for positive maps

Definition 7.49 (Finite-POVM dilation data). [povmIsometry](#) [povmDiagonal](#) [✓ Stmt OK](#) For a finite family of matrices $\{C_i\}_{i \in \iota}$ and a defect block S , one forms an isometric dilation whose rows are the adjoints of the matrices C_i together with the adjoint of S . One also forms the scalar block-diagonal matrix whose ι -blocks carry prescribed weights and whose defect block carries a single scalar.

Lemma 7.50 (Compression auxiliary for Jensen). [inverse_compression_le](#) [✓ Stmt OK](#) Let Y be positive definite and let W be an isometry. Then

$$(W^* Y W)^{-1} \leq W^* Y^{-1} W.$$

Proof. [✓ Proof OK](#) Apply the Schur-complement criterion to the block matrix $\begin{pmatrix} Y^{-1} & W \\ W^* & W^* Y W \end{pmatrix}$. The lower-right Schur complement is zero, hence positive semidefinite, and compressing the resulting upper-left Schur complement by W gives the stated inequality. $\square$

Lemma 7.51 (Finite-POVM compression identities). [povmIsometry_star_mul](#) [povmIsometry_compress_diagonal](#) [povm_sum_add_defect](#) the defect block satisfies

$$S S^* = \mathbb{1} - \sum_{i \in \iota} C_i C_i^*,$$

then the dilation is an isometry, compressing the weighted scalar block-diagonal matrix gives

$$\sum_{i \in \iota} w_i C_i C_i^* + t S S^*,$$

the defect relation rewrites as

$$\sum_{i \in \iota} C_i C_i^* + S S^* = \mathbb{1},$$

and strict positivity of all weights implies positivity of the scalar block-diagonal matrix.

Proof. [✓ Proof OK](#) Expand the block matrix multiplication entrywise. The defect identity gives the isometry relation after summing the ι -blocks and the defect block, and the weighted compression identity follows from the same expansion with the scalar diagonal inserted. Positivity of the scalar block-diagonal matrix is immediate from positivity of its diagonal entries. $\square$

Theorem 7.52 (Jensen for concave real powers). [posMap_rpow_concave_jensen](#) [Not ready](#) For a positive subunitary map T and $p \in [0, 1]$:

$$T(A^p) \leq (T A)^p.$$

This is [Wol12, Thm. 5.1] for concave $f(x) = x^p$.

Theorem 7.53 (Map form of Jensen for concave real powers). [rpow_concave_jensen](#) [Not ready](#) This is the corresponding map-form statement of Theorem 7.52. Its proof is deferred until Theorem 7.52 is proved.

Proof. Direct application of the axiom. $\square$

Theorem 7.54 (Jensen for convex real powers). [posMap_rpow_convex_jensen](#) [Not ready](#) For a positive subunitary map T and $p \in [1, 2]$:

$$(T A)^p \leq T(A^p).$$

This is [Wol12, Thm. 5.1] for convex $f(x) = x^p$.

Theorem 7.55 (Map form of Jensen for convex real powers). [rpow_convex_jensen](#) Not ready This is the corresponding map-form statement of Theorem 7.54. Its proof is deferred until Theorem 7.54 is proved.

Proof. Direct application of the axiom. □

Theorem 7.56 (Jensen for concave log). [posMap_log_concave_jensen](#) Not ready For a positive unital map T and positive-definite A :

$$T(\log A) \leq \log(TA).$$

This is [Wol12, Thm. 5.1] for $f = \log$. Requires unitality ($T(\mathbb{1}) = \mathbb{1}$), not merely subunitarity.

Theorem 7.57 (Map form of Jensen for the concave logarithm). [log_concave_jensen](#) Not ready This is the corresponding map-form statement of Theorem 7.56. Its proof is deferred until Theorem 7.56 is proved.

Proof. Direct application of the axiom. □

7.3.4 Lieb concavity theorem

Theorem 7.58 (Lieb concavity). [lieb_concavity_axiom](#) Not ready For $s \in [0, 1]$, any matrix K , and positive-definite matrices A_1, A_2, B_1, B_2 , the map $(A, B) \mapsto \text{tr}(K^\dagger A^s K B^{1-s})$ is jointly concave:

$$t \Re \text{tr}(K^\dagger A_1^s K B_1^{1-s}) + (1-t) \Re \text{tr}(K^\dagger A_2^s K B_2^{1-s}) \leq \Re \text{tr}(K^\dagger (tA_1 + (1-t)A_2)^s K (tB_1 + (1-t)B_2)^{1-s}).$$

See Lieb (1973) and Ando (1979).

Theorem 7.59 (Map form of Lieb concavity). [lieb_concavity](#) Not ready This is the corresponding map-form statement of Theorem 7.58. Its proof is deferred until Theorem 7.58 is proved.

Proof. Direct application of the axiom. □

Corollary 7.60 (Lieb concavity, $K = \mathbb{1}$ case). [lieb_concavity_id](#) ✓ Stmt OK For $s \in [0, 1]$, $t \in [0, 1]$, and positive-definite matrices A_1, A_2, B_1, B_2 , the map $(A, B) \mapsto \Re \text{tr}(A^s B^{1-s})$ is jointly concave:

$$t \Re \text{tr}(A_1^s B_1^{1-s}) + (1-t) \Re \text{tr}(A_2^s B_2^{1-s}) \leq \Re \text{tr}((tA_1 + (1-t)A_2)^s (tB_1 + (1-t)B_2)^{1-s}).$$

Obtained from Theorem 7.59 by taking $K = \mathbb{1}$.

Proof. ✓ Proof OK Specialize Theorem 7.59 at $K = \mathbb{1}$ and simplify $\mathbb{1}^\dagger = \mathbb{1}$ together with $\mathbb{1} \cdot X = X \cdot \mathbb{1} = X$. □

Corollary 7.61 (Concavity of Lieb's map in the first argument). [lieb_concavity_infst](#) ✓ Stmt OK For $s \in [0, 1]$, $t \in [0, 1]$, matrix K , and positive-definite matrices A_1, A_2, B , the map $A \mapsto \Re \text{tr}(K^\dagger A^s K B^{1-s})$ is concave:

$$t \Re \text{tr}(K^\dagger A_1^s K B^{1-s}) + (1-t) \Re \text{tr}(K^\dagger A_2^s K B^{1-s}) \leq \Re \text{tr}(K^\dagger (tA_1 + (1-t)A_2)^s K B^{1-s}).$$

Proof. ✓ Proof OK Specialize Theorem 7.59 at $B_1 = B_2 = B$, so that the convex combination collapses: $tB + (1-t)B = B$. □

Corollary 7.62 (Concavity of Lieb's map in the second argument). [lieb_concavity_insnd](#) ✓ Stmt OK For $s \in [0, 1]$, $t \in [0, 1]$, matrix K , and positive-definite matrices A, B_1, B_2 , the map $B \mapsto \Re \text{tr}(K^\dagger A^s K B^{1-s})$ is concave:

$$t \Re \text{tr}(K^\dagger A^s K B_1^{1-s}) + (1-t) \Re \text{tr}(K^\dagger A^s K B_2^{1-s}) \leq \Re \text{tr}(K^\dagger A^s K (tB_1 + (1-t)B_2)^{1-s}).$$

Proof. ✓ Proof OK Specialize Theorem 7.59 at $A_1 = A_2 = A$, so that the convex combination collapses: $tA + (1-t)A = A$. □

Chapter 8

Perron–Frobenius Theory for Channels and Transfer Maps

This chapter establishes the Perron–Frobenius theory used in subsequent chapters: existence, positive definiteness, and uniqueness of fixed points for injective and irreducible transfer maps, canonical gauge constructions, ergodicity of irreducible channels, the spectral-radius identification of Perron eigenvalues, and Wolf’s spectral characterization of irreducibility. The conceptual source is Wolf’s treatment of irreducible positive maps [Wol12, Ch. 6, especially Thms. 6.2–6.5 and Cor. 6.3] and [EHK78].

8.1 Positive definiteness

Remark 8.1. ✓ Stmt OK Wolf’s general theory treats irreducible positive maps first ([Wol12, Thm. 6.2]), then derives non-degeneracy and positive definiteness ([Wol12, Thm. 6.3]). The injective-tensor version (Theorem 8.2) uses a shorter direct kernel argument; the irreducible CP-map version (Theorem 8.3) follows via the support-projection argument of [Wol12, Thm. 6.2].

Theorem 8.2 (PSD fixed point is positive definite). posSemidef_fixedPoint_isPosDef ✓ Stmt OK *Let A be an injective MPS tensor and let $\rho \geq 0$, $\rho \neq 0$, be a fixed point of the transfer map $\mathcal{E}_A(\rho) = \rho$. Then ρ is positive definite.*

Proof. ✓ Proof OK Suppose $v^\dagger \rho v = 0$ for some $v \neq 0$. Since $\rho \geq 0$, this implies $\rho v = 0$. The fixed-point equation gives $v^\dagger \rho v = \sum_i ((A^i)^\dagger v)^\dagger \rho ((A^i)^\dagger v)$. Each term is nonneg; if the sum is zero, then $\rho (A^i)^\dagger v = 0$ for all i : the kernel of ρ is invariant under all $(A^i)^\dagger$. Since the $\{A^i\}$ span $M_D(\mathbb{C})$, we have $\rho M v = 0$ for every matrix M . Given any vector w , choose M with $M v = w$. Then $\rho w = 0$ for every w , so $\rho = 0$, contradicting $\rho \neq 0$. In [Wol12, Thm. 6.3, item 2], the same conclusion is reached via the growth condition $(\mathbb{1} + T)^{d-1}(X) > 0$ of [Wol12, Thm. 6.2, condition (2)]; here the direct kernel argument is shorter under injectivity. $\square$

Theorem 8.3 (PSD fixed point is positive definite — irreducible version). posSemidef_fixedPoint_isPosDef_of_irreducible ✓ *$\mathcal{E}_A$ is irreducible and $\rho \geq 0$, $\rho \neq 0$ is a fixed point, then ρ is positive definite.*

Proof. ✓ Proof OK Suppose ρ is not positive definite. Then ρ has a nontrivial kernel; let Q be the orthogonal projection onto the support of ρ (the complement of the kernel). A direct computation using the fixed-point equation $\mathcal{E}_A(\rho) = \rho$ shows that Q is an invariant projection for $\mathcal{E}_A$: it satisfies $Q \mathcal{E}_A(Q X Q) Q = \mathcal{E}_A(Q X Q)$ for all X . By irreducibility of $\mathcal{E}_A$ (Definition 5.15), every

invariant projection is 0 or $\mathbb{1}$. If $Q = 0$ then $\rho = 0$, contradicting $\rho \neq 0$. If $Q = \mathbb{1}$ then ρ is positive definite, contradicting our assumption. Hence ρ must be positive definite. This is the support-projection direction of [Wol12, Thm. 6.2, condition (1)]. $\square$

Theorem 8.4 (Orthogonal trace condition). [orthogonal_trace_pos_of_irreducible_cp](#) ✓ Stmt OK *Let E be an irreducible completely positive map on $M_D(\mathbb{C})$, and let $A, B \geq 0$ be nonzero positive semidefinite matrices with $\text{tr}(BA) = 0$. Then there exists t with $1 \leq t \leq D - 1$ such that*

$$\text{tr}(B \cdot E^t(A)) > 0.$$

Proof. ✓ Proof OK The growth theorem for irreducible CP maps ([Wol12, Thm. 6.2, item 2]) gives $(\mathbb{1} + E)^{D-1}(A) > 0$ (positive definite). Since $B \geq 0$ is nonzero, $\text{tr}(B \cdot (\mathbb{1} + E)^{D-1}(A)) > 0$. Expanding by the binomial theorem:

$$\text{tr}\left(B \cdot \sum_{k=0}^{D-1} \binom{D-1}{k} E^k(A)\right) > 0.$$

Each term $\text{tr}(B \cdot E^k(A)) \geq 0$ since both factors are PSD. The $k = 0$ term vanishes by hypothesis ($\text{tr}(BA) = 0$). Hence at least one $k \in \{1, \dots, D - 1\}$ contributes a strictly positive term. This is [Wol12, Thm. 6.2, item 4]: item (2) (growth condition) implies item (4) by a binomial expansion and PSD positivity argument. $\square$

8.2 Uniqueness

Theorem 8.5 (Uniqueness of PSD fixed point). [posSemidef_fixedPoint_unique](#) ✓ Stmt OK *Let A be an injective MPS tensor. If $\rho, \sigma \geq 0$ are both nonzero fixed points of $\mathcal{E}_A$, then $\sigma = c\rho$ for some $c \in \mathbb{C}$.*

Proof. ✓ Proof OK Both ρ and σ are positive definite by Theorem 8.2. Write $\rho = SS^\dagger$ via a spectral square root and set $H = (S^\dagger)^{-1}\sigma S^{-1}$, which is Hermitian and positive definite. Let c_0 be the minimal eigenvalue of H . Then $\tau := \sigma - c_0\rho$ is a PSD fixed point which is *not* positive definite (it has a zero eigenvalue by construction of c_0). By Theorem 8.2, τ must be zero, giving $\sigma = c_0\rho$. **Relation to Wolf:** In [Wol12, Thm. 6.3, item 2], uniqueness is proved via the growth condition of [Wol12, Thm. 6.2]. The same minimal-eigenvalue argument applies here, using the injective-tensor positive-definiteness (Theorem 8.2) in place of the growth condition. $\square$

Remark 8.6. ✓ Stmt OK The proof produces a positive real scalar $c_0 > 0$: it is the minimal eigenvalue of the positive definite Hermitian matrix $H = (S^\dagger)^{-1}\sigma S^{-1}$.

Theorem 8.7 (Uniqueness of positive eigenvalue for irreducible CP maps). [eigenvalue_unique_of_irreducible_cp](#) ✓ Stmt OK *$D \geq 1$ and let E be an irreducible completely positive map on $M_D(\mathbb{C})$. Suppose $\rho, \sigma \geq 0$ are nonzero positive semidefinite matrices satisfying $E(\rho) = r_1\rho$ and $E(\sigma) = r_2\sigma$ for real scalars $r_1, r_2 > 0$. Then $r_1 = r_2$.*

Proof. ✓ Proof OK Extract Kraus operators K for E . Irreducibility of E passes to the adjoint transfer map $E^\dagger(\cdot) = \sum_i K_i^\dagger(\cdot) K_i$, which is a positive CP map. The Perron–Frobenius existence theorem (Theorem 8.17) together with the irreducible positive-definiteness upgrade (Theorem 8.3) gives a positive definite eigenvector $\tau > 0$ with eigenvalue $t > 0$: $E^\dagger(\tau) = t\tau$. For any nonzero PSD X with $E(X) = sX$, the trace pairing identity yields

$$s \cdot \text{tr}(\tau X) = \text{tr}(\tau \cdot E(X)) = \text{tr}(E^\dagger(\tau) \cdot X) = t \cdot \text{tr}(\tau X).$$

Since $\tau > 0$ and $X \geq 0$, $X \neq 0$, the scalar $\text{tr}(\tau X) > 0$, so $s = t$. Applying this to both (ρ, r_1) and (σ, r_2) gives $r_1 = t = r_2$. **Relation to Wolf:** This is [Wol12, Thm. 6.3, item 3], establishing non-degeneracy of the spectral-radius eigenvalue for irreducible CP maps. The proof follows Wolf's dual-map trace argument. $\square$

8.3 Existence and the Perron–Frobenius theorem

Theorem 8.8 (Existence of PSD fixed point). exists_posSemidef_fixedPoint ✓ Stmt OK Assume $D \geq 1$. Let A be an MPS tensor with $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ (so that $\mathcal{E}_A$ is trace-preserving). Then there exists $\rho \geq 0$, $\rho \neq 0$, with $\mathcal{E}_A(\rho) = \rho$.

Proof. ✓ Proof OK Under the normalization $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, the transfer map is a quantum channel (Theorem 5.18). The Cesàro fixed-point theorem (Theorem 5.22) then provides the fixed point; injectivity is not needed for this step. $\square$

Theorem 8.9 (Quantum Perron–Frobenius). quantum_perron_frobenius ✓ Stmt OK Assume $D \geq 1$. Let A be an injective MPS tensor with $\sum_i (A^i)^\dagger A^i = \mathbb{1}$. Then the transfer map $\mathcal{E}_A$ has a unique PSD fixed point ρ (up to scaling), and ρ is positive definite.

Proof. ✓ Proof OK Combine Theorems 8.8, 8.2, and 8.5. $\square$

8.4 Right- and left-canonical gauges

Theorem 8.10 (Right-canonical (unital) gauge). gauged_unital ✓ Stmt OK Let $\mathcal{E}_A(\rho) = \rho$ and let S be an invertible matrix with $SS^\dagger = \rho$. Define the gauged operators $A'^i = S^{-1}A^iS$. Then $\sum_i A'^i (A'^i)^\dagger = \mathbb{1}$, i.e. the gauged Kraus map is unital. This is the right-canonical normalization used in subsequent chapters.

Proof. ✓ Proof OK Expand each summand: $(S^{-1}A^iS)(S^{-1}A^iS)^\dagger = S^{-1}A^i \underbrace{SS^\dagger}_{\rho} (A^i)^\dagger (S^\dagger)^{-1}$. Summing over i and using $\sum_i A^i \rho (A^i)^\dagger = \rho$ gives $S^{-1}\rho(S^\dagger)^{-1} = S^{-1}SS^\dagger(S^\dagger)^{-1} = \mathbb{1}$. $\square$

Theorem 8.11 (Left-canonical (trace-preserving) gauge). gauged_tracePreserving ✓ Stmt OK Let σ be a fixed point of the adjoint transfer map $\sum_i (A^i)^\dagger \sigma A^i = \sigma$, and let S be invertible with $S^\dagger S = \sigma$. Define $A'^i = SA^iS^{-1}$. Then $\sum_i (A'^i)^\dagger A'^i = \mathbb{1}$, i.e. the gauged Kraus map is trace-preserving. This is the left-canonical normalization used in subsequent chapters.

Proof. ✓ Proof OK Expand each summand: $(SA^iS^{-1})^\dagger (SA^iS^{-1}) = (S^\dagger)^{-1} (A^i)^\dagger \underbrace{S^\dagger S}_{\sigma} A^i S^{-1}$. Summing over i and using $\sum_i (A^i)^\dagger \sigma A^i = \sigma$ gives $(S^\dagger)^{-1} \sigma S^{-1} = (S^\dagger)^{-1} S^\dagger S S^{-1} = \mathbb{1}$. $\square$

Remark 8.12. ✓ Stmt OK The right-canonical gauge of Theorem 8.10 and the left-canonical gauge of Theorem 8.11 are generally different similarity transforms: one is built from a fixed point of $\mathcal{E}_A$, the other from a fixed point of the adjoint transfer map. This section does not claim that a single gauge makes the transfer map simultaneously unital and trace-preserving.

8.5 Similarity preserves irreducibility

Lemma 8.13 (Similarity transform preserves irreducibility). [isIrreducibleMap_full_similarity](#) ✓ Stmt OK *Let E be an irreducible completely positive map on $M_D(\mathbb{C})$, let $C \in M_D(\mathbb{C})$ be invertible ($\det C \neq 0$), and let $c > 0$. Define the similarity-transformed map*

$$E'(X) = c \cdot C^{-1} E(C X C^\dagger) (C^\dagger)^{-1}.$$

Then E' is also irreducible.

Proof. ✓ Proof OK Suppose Q is an invariant projection for E' : $Q \neq 0$, $Q \neq \mathbb{1}$, and $(\mathbb{1} - Q)E'(QXQ)Q = 0$ for all X . Setting $R = CQC^\dagger$, the support projection P of R is an invariant projection for E . Irreducibility of E forces $P = 0$ or $P = \mathbb{1}$. The invertibility of C then forces $Q = 0$ or $Q = \mathbb{1}$, a contradiction. This is the full similarity version of [Wol12, Prop. 6.6]; the scalar case ($C = \mathbb{1}$) gives the scaling result stated next. □

Lemma 8.14 (Similarity is an involution). [similarityMap_similarityMap_inv](#) ✓ Stmt OK *For any invertible $C \in M_D(\mathbb{C})$ and any linear map E on $M_D(\mathbb{C})$, the similarity transforms by C and C^{-1} compose to the identity:*

$$X \mapsto C(C^{-1} E(C X C^\dagger) (C^\dagger)^{-1}) C^\dagger = E(X).$$

Proof. ✓ Proof OK Direct computation: the inner conjugation $C \cdot (C^{-1} X (C^\dagger)^{-1}) \cdot C^\dagger$ collapses to X using $CC^{-1} = \mathbb{1}$ and $(C^\dagger)^{-1}C^\dagger = \mathbb{1}$, and the outer wrapping by $C(\cdot)C^\dagger$ then cancels the inner $C^{-1}(\cdot)(C^\dagger)^{-1}$ on $E(X)$ for the same reason. □

Lemma 8.15 (Irreducibility is invariant under similarity). [isIrreducibleMap_similarity_iff](#) ✓ Stmt OK *For any invertible $C \in M_D(\mathbb{C})$ and any linear map E on $M_D(\mathbb{C})$, the similarity transform $X \mapsto C^{-1} E(C X C^\dagger) (C^\dagger)^{-1}$ is irreducible if and only if E is.*

Proof. ✓ Proof OK The forward direction follows from Lemma 8.13 applied with the inverse C^{-1} (whose determinant is also nonzero): if the similarity transform of E by C is irreducible, then so is its similarity transform by C^{-1} , which equals E by Lemma 8.14. The reverse direction is Lemma 8.13 itself (with $c = 1$). □

Theorem 8.16 (Scaling preserves irreducibility). [isIrreducibleMap_smul](#) ✓ Stmt OK *If E is an irreducible map and $c \neq 0$, then cE is also irreducible. This is the scalar special case ($C = \mathbb{1}$) of [Wol12, Prop. 6.6].*

Proof. ✓ Proof OK An invariant projection for cE is an invariant projection for E (scale out the nonzero constant from the commutation relation). □

8.6 Perron–Frobenius eigenvector existence

This section establishes the existence of a positive definite eigenvector for the adjoint transfer map of an irreducible MPS tensor, and uses it to construct a TP-gauge normalization. The PSD-eigenvector step corresponds to [Wol12, Thm. 6.5] (general positive maps) and [EHK78]; the upgrade to positive definiteness uses the irreducible-map theory of [Wol12, Thm. 6.3]. The TP-gauge application follows [CPGSV17a, Appendix A].

Theorem 8.17 (PSD eigenvector for positive maps). exists_posSemidef_eigenvector ✓ Stmt OK Let $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a positive linear map with $D > 0$, and assume that $E(\sigma) \neq 0$ for every nonzero PSD matrix σ . Then there exist a nonzero PSD matrix ρ and a positive real $r > 0$ such that $E(\rho) = r\rho$.

This is the finite-dimensional Perron–Frobenius theorem for positive maps on matrix algebras ([Wol12, Thm. 6.5]; see also [EHK78]). The proof applies Brouwer’s fixed-point theorem to the normalization map $\rho \mapsto E(\rho)/\text{tr}(E(\rho))$ on the compact convex set of density matrices.

Theorem 8.18 (Adjoint transfer map is nonzero). adjointTransferMap_ne_zero_of_nonzero ✓ Stmt OK If some Kraus operator $A^i \neq 0$, then the adjoint transfer map $\mathcal{E}_A^\dagger(X) = \sum_i (A^i)^\dagger X A^i$ is nonzero.

Proof. ✓ Proof OK If $\mathcal{E}_A^\dagger = 0$ then $\mathcal{E}_A^\dagger(\mathbb{1}) = 0$, so $\sum_i (A^i)^\dagger A^i = 0$. Since each summand $(A^i)^\dagger A^i$ is PSD, each $(A^i)^\dagger = 0$ and hence each $A^i = 0$. □

Theorem 8.19 (Positive-definite adjoint eigenvector for irreducible tensors). exists_posDef_adjoint_eigenvector ✓ Stmt OK Let A be an irreducible MPS tensor with $D > 0$ and some $A^i \neq 0$. Then there exist a positive definite matrix σ and a positive real $r > 0$ such that $\mathcal{E}_A^\dagger(\sigma) = r\sigma$.

This combines the general PF existence ([Wol12, Thm. 6.5]) with the irreducible upgrade to positive definiteness ([Wol12, Thm. 6.3, item 2]); the application to the adjoint transfer map follows [CPGSV17a, Appendix A].

Proof. ✓ Proof OK By Theorem 8.18, $\mathcal{E}_A^\dagger \neq 0$. Since $\mathcal{E}_A^\dagger(X) = \sum_i (A^i)^\dagger X A^i$ is a Kraus map, it is positive. Irreducibility of A transfers to the adjoint transfer map. More precisely: if P is invariant for $\mathcal{E}_A$ (i.e. $(\mathbb{1} - P)K_i P = 0$ for all Kraus operators K_i), then taking adjoints gives $PK_i^\dagger(\mathbb{1} - P) = 0$, which means $\mathbb{1} - P$ is invariant for $\mathcal{E}_A^\dagger$. Hence $\mathcal{E}_A$ irreducible $\iff \mathcal{E}_A^\dagger$ irreducible, and together with $\mathcal{E}_A^\dagger \neq 0$ this implies $\mathcal{E}_A^\dagger(\tau) \neq 0$ for every nonzero PSD matrix τ . Hence Theorem 8.17 applies to $\mathcal{E}_A^\dagger$, giving $\sigma_0 \geq 0$ (PSD, nonzero) and $r > 0$ with $\mathcal{E}_A^\dagger(\sigma_0) = r\sigma_0$. Rescale: set $T^i = r^{-1/2}(A^i)^\dagger$. Then $\mathcal{E}_T(\sigma_0) = \sigma_0$, and $\mathcal{E}_T$ is irreducible (by Theorem 8.16, since irreducibility of $\mathcal{E}_A$ transfers to the adjoint and is preserved under scaling). Since $\mathcal{E}_T$ is irreducible with a PSD fixed point, the positive-definiteness upgrade (Theorem 8.3) gives σ_0 positive definite. □

Theorem 8.20 (TP-gauge existence for irreducible tensors). exists_tp_data_of_irreducible ✓ Stmt OK Let A be an irreducible MPS tensor with $D > 0$ and some $A^i \neq 0$. Then there exist a positive real r , a positive definite matrix σ , and a gauge-equivalent tensor B such that

$$B^i = \sigma^{1/2} \cdot (r^{-1/2} A^i) \cdot \sigma^{-1/2}$$

is trace-preserving: $\sum_i (B^i)^\dagger B^i = \mathbb{1}$.

This is the “spectral rescaling + TP gauge” step of [CPGSV17a, Appendix A]. The similarity is generally non-unitary.

Proof. ✓ Proof OK By Theorem 8.19, obtain σ positive definite and $r > 0$ with $\mathcal{E}_A^\dagger(\sigma) = r\sigma$. Set $c = r^{-1/2}$ and $A'^i = cA^i$. Then $\mathcal{E}_{A'}^\dagger(\sigma) = \sigma$. By the TP gauge construction (Theorem 8.11), $B^i = \sigma^{1/2} A'^i \sigma^{-1/2}$ satisfies $\sum_i (B^i)^\dagger B^i = \mathbb{1}$. □

8.7 Exponential positivity for irreducible CP maps

Theorem 8.21 (Exponential truncation is positive definite). exp_truncation_posDef_of_irreducible_cp ✓ Stmt OK Let E be an irreducible completely positive map on $M_D(\mathbb{C})$, let $A \geq 0$ be nonzero, and let $t > 0$.

Then the finite exponential truncation

$$\sum_{k=0}^{D-1} \frac{t^k}{k!} E^k(A)$$

is positive definite. This is the finite-sum core of [Wol12, Thm. 6.2, item (3)].

Proof. ✓ Proof OK If a nonzero vector v annihilated this quadratic form, then every term $E^k(A)$ with $0 \leq k \leq D-1$ would annihilate v . The growth theorem for irreducible CP maps (that is, $(\mathbb{1} + E)^{D-1}(A) > 0$ for irreducible E and nonzero PSD A , as established in the proof of Theorem 8.4) then forces $(\mathbb{1} + E)^{D-1}(A)$ to annihilate v , contradicting its positive definiteness. $\square$

Theorem 8.22 (Exponential positivity for irreducible CP maps). exp_posDef_of_irreducible_cp ✓ Stmt OK Let E be an irreducible completely positive map on $M_D(\mathbb{C})$, let $A \geq 0$ be nonzero, and let $t > 0$. Then

$$\exp(tE)(A) = \sum_{k=0}^{\infty} \frac{t^k}{k!} E^k(A)$$

is positive definite. This is [Wol12, Thm. 6.2, item (3)].

Proof. ✓ Proof OK By Theorem 8.21, the first D terms already form a positive definite matrix. Every remaining term is positive semidefinite, so the full exponential series is the sum of a positive definite matrix and a positive semidefinite tail. $\square$

8.8 Ergodicity of irreducible channels

Theorem 8.23 (Unique density-matrix fixed point for an irreducible channel). exists_unique_density_fixedPoint_of_irreduc Let E be an irreducible quantum channel on $M_D(\mathbb{C})$ with $D > 0$. Then there exists a unique density matrix σ such that $E(\sigma) = \sigma$. Moreover σ is positive definite. This is the qualitative form of [Wol12, Cor. 6.3].

Proof. ✓ Proof OK Every Cesàro mean of a density matrix stays inside the compact convex set of density matrices, so a subsequence converges. Its limit is a fixed point by the telescoping argument, positive definiteness follows from irreducibility, and uniqueness comes from the uniqueness of positive semidefinite Perron eigenvectors at eigenvalue 1. $\square$

Theorem 8.24 (Cesàro convergence for irreducible channels). cesaroMean_tendsto_of_irreducible ✓ Stmt OK Let E be an irreducible quantum channel on $M_D(\mathbb{C})$ with $D > 0$, and let ρ be any density matrix. Then the Cesàro means

$$\frac{1}{N} \sum_{n=0}^{N-1} E^n(\rho)$$

converge to the unique positive definite density-matrix fixed point of E . This is [Wol12, Cor. 6.3].

Proof. ✓ Proof OK Every subsequential limit of the Cesàro means is again a density-matrix fixed point. By Theorem 8.23, there is only one such fixed point, so every convergent subsequence has the same limit. Compactness then forces the whole sequence to converge to that limit. $\square$

8.9 Spectral radius at the Perron eigenvalue

Theorem 8.25 (Spectral radius is the Perron eigenvalue). spectralRadius_eq_of_posDef_eigenvector_of_irreducible_cp ✓ Stmt OK *Let E be an irreducible completely positive map on $M_D(\mathbb{C})$. Suppose $\rho > 0$ and $E(\rho) = r\rho$ with $r > 0$. Then the spectral radius of E is r . This is [Wol12, Thm. 6.3, item (4)].*

Proof. ✓ Proof OK Choose a positive definite eigenvector for the adjoint map with the same eigenvalue, gauge by its square root, and rescale by r^{-1} . The resulting map is trace-preserving and has a positive definite fixed point, so 1 is an eigenvalue. The spectral radius of a trace-preserving map F satisfies $\rho(F) \leq 1$, since $\text{tr}(F^n(X)) = \text{tr}(X)$ for all n bounds the growth of iterates. Hence its spectral radius is 1. Undoing the similarity and scalar rescaling recovers the spectral radius of E . □

Theorem 8.26 (Real spectral-radius identity). spectralRadius_toReal_eq_of_posDef_eigenvector_of_irreducible_cp ✓ Stmt OK *Under the hypotheses of Theorem 8.25, the real-valued spectral radius of E is also equal to r .*

Proof. ✓ Proof OK This is the real-valued reformulation of Theorem 8.25. □

8.10 Spectral characterization of irreducibility

Definition 8.27 (Spectral properties). HasSpectralProperties ✓ Stmt OK A completely positive map E on $M_D(\mathbb{C})$ has the *spectral properties* of [Wol12, Thm. 6.4] if there exist a positive real number r , a positive definite right eigenvector ρ , and a positive definite left eigenvector σ for the adjoint map such that $E(\rho) = r\rho$, $E^\dagger(\sigma) = r\sigma$, every positive semidefinite right eigenvector for eigenvalue r is a scalar multiple of ρ , and the spectral radius of E is equal to r .

Theorem 8.28 (Irreducibility gives the spectral properties). hasSpectralProperties_of_irreducible_cp ✓ Stmt OK *Every nonzero irreducible completely positive map on $M_D(\mathbb{C})$ has the spectral properties of Definition 8.27.*

Proof. ✓ Proof OK Combine Perron–Frobenius existence for the map and its adjoint with the uniqueness theorem for positive semidefinite Perron eigenvectors and the spectral-radius identity of Theorem 8.25. □

Theorem 8.29 (Spectral properties imply irreducibility). isIrreducibleMap_of_hasSpectralProperties ✓ Stmt OK *If a completely positive map on $M_D(\mathbb{C})$ has the spectral properties of Definition 8.27, then it is irreducible.*

Proof. ✓ Proof OK Gauge by the positive definite left eigenvector and rescale by the Perron eigenvalue to obtain a trace-preserving map with a positive definite fixed point. A nontrivial invariant projection would then produce, by Cesàro averaging in its corner algebra, a second positive semidefinite fixed point not proportional to the Perron vector, contradicting the uniqueness clause in Definition 8.27. □

Theorem 8.30 (Irreducibility iff the spectral properties hold). isIrreducibleMap_iff_spectral_properties ✓ Stmt OK *Let E be a nonzero completely positive map on $M_D(\mathbb{C})$. Then E is irreducible if and only if it has the spectral properties of Definition 8.27. This is [Wol12, Thm. 6.4].*

Proof. ✓ Proof OK Combine Theorems 8.28 and 8.29. □

Chapter 9

Spectral Gap and Block Separation

This chapter proves that the MPV overlaps between distinct (non-gauge-phase-equivalent) MPS blocks decay to zero as system size grows. It also proves the corresponding self-overlap convergence to 1 under the complementary spectral-gap hypothesis that encodes primitivity in the cases of interest. The argument is based on a spectral analysis of the *mixed transfer operator*, following [PGVWC07].

The algebraic mixed-transfer identities in the first two sections do not use normalization. Starting with the spectral-radius estimates, we assume the trace-preserving normalization

$$\sum_i (A^i)^\dagger A^i = \mathbb{1}.$$

9.1 Mixed transfer operator

Definition 9.1 (Mixed transfer operator). mixedTransferMap ✓ Stmt OK The *mixed transfer operator* (or *cross transfer operator*) for two MPS tensors A and B of the same bond dimension D is the linear map $F_{AB} : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ defined by

$$F_{AB}(X) = \sum_{i=0}^{d-1} A^i X (B^i)^\dagger.$$

Definition 9.2 (Rectangular mixed transfer operator). mixedTransferMap2 ✓ Stmt OK For MPS tensors A of bond dimension D_1 and B of bond dimension D_2 (possibly $D_1 \neq D_2$), the *rectangular mixed transfer operator* is $F_{AB}^{\text{rect}} : M_{D_1 \times D_2}(\mathbb{C}) \rightarrow M_{D_1 \times D_2}(\mathbb{C})$ defined by

$$F_{AB}^{\text{rect}}(X) = \sum_{i=0}^{d-1} A^i X (B^i)^\dagger.$$

Remark 9.3 (Self-transfer). mixedTransferMap_self ✓ Stmt OK Setting $B = A$ recovers the ordinary transfer map: $F_{AA} = \mathcal{E}_A$.

Theorem 9.4 (Iterated mixed transfer). mixedTransferMap_pow_apply ✓ Stmt OK For all $X \in M_D(\mathbb{C})$ and $N \geq 0$,

$$F_{AB}^N(X) = \sum_{\sigma \in \{0, \dots, d-1\}^N} A^\sigma X (B^\sigma)^\dagger,$$

where $A^\sigma = A^{\sigma_1} \dots A^{\sigma_N}$ denotes the word evaluation (Definition 2.2).

Proof. ✓ Proof OK Induction on N : the base case is $F^0(X) = X$; the inductive step applies F_{AB} to each summand and re-indexes via $\{0, \dots, d-1\}^{N+1} \cong \{0, \dots, d-1\} \times \{0, \dots, d-1\}^N$. $\square$

Theorem 9.5 (Matrix trace of iterated mixed transfer at the identity). trace_mixedTransferMap_pow_identity ✓ Stmt OK For $N \geq 0$,

$$\text{tr}(F_{AB}^N(\mathbb{1})) = \sum_{\sigma} \text{tr}(A^\sigma (B^\sigma)^\dagger).$$

This is the matrix trace of F_{AB}^N evaluated at the identity matrix. It should not be confused with the operator trace $\text{Tr}(F_{AB}^N)$; its relation to the MPV overlap is proved in the next section.

Proof. ✓ Proof OK Apply the matrix trace to each summand in Theorem 9.4 with $X = \mathbb{1}$: $\text{tr}(F_{AB}^N(\mathbb{1})) = \sum_{\sigma} \text{tr}(A^\sigma \cdot \mathbb{1} \cdot (B^\sigma)^\dagger) = \sum_{\sigma} \text{tr}(A^\sigma (B^\sigma)^\dagger)$. $\square$

9.2 MPV overlap as transfer trace

Recall from Definition 2.43 that

$$O_{AB}(N) = \sum_{\sigma \in \{0, \dots, d-1\}^N} V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma}.$$

This section rewrites that scalar as the operator trace of the mixed transfer power.

Lemma 9.6 (Trace expansion over matrix units). linearMap_trace_eq_sum_apply_single ✓ Stmt OK For any linear endomorphism T on $M_D(\mathbb{C})$, the operator trace can be expanded as

$$\text{Tr}(T) = \sum_{p,q=0}^{D-1} (T(E_{pq}))_{pq},$$

where E_{pq} is the matrix with 1 in position (p, q) and 0 elsewhere.

Proof. ✓ Proof OK The operator trace $\text{Tr}(T)$ equals the sum of $\text{tr}(T(E_{pq}) \cdot E_{qp})$ over all (p, q) , and $\text{tr}(M \cdot E_{qp}) = M_{pq}$. $\square$

Theorem 9.7 (Overlap equals transfer trace). trace_mixedTransferMap_pow_eq_mpvOverlap ✓ Stmt OK For tensors A, B of bond dimension $D \geq 1$,

$$O_{AB}(N) = \text{Tr}(F_{AB}^N).$$

Proof. ✓ Proof OK Expand the operator trace using Lemma 9.6 as a sum over matrix units E_{pq} . Apply Theorem 9.4 to evaluate $F_{AB}^N(E_{pq})$, extract the (p, q) -entry, and reassemble. The resulting double sum over (p, q) and σ equals $\sum_{\sigma} V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma} = O_{AB}(N)$. $\square$

Theorem 9.8 (Rectangular overlap as transfer trace). trace_mixedTransferMap2_pow_eq_mpvOverlap ✓ Stmt OK For tensors A of bond dimension $D_1 \geq 1$ and B of bond dimension $D_2 \geq 1$,

$$O_{AB}(N) = \text{Tr}((F_{AB}^{\text{rect}})^N).$$

Proof. ✓ Proof OK Expand the operator trace over the rectangular matrix-unit basis. Then apply the rectangular iterated-transfer formula and sum over the matrix-unit indices exactly as in Theorem 9.7. $\square$

Lemma 9.9 (Word trace as entrywise inner product). [mpv_inner_product_via_trace](#) ✓ Stmt OK For a word $\sigma \in \{0, \dots, d-1\}^N$,

$$\mathrm{tr}(A^\sigma (B^\sigma)^\dagger) = \sum_{j,k=0}^{D-1} (A^\sigma)_{jk} \overline{(B^\sigma)_{jk}}.$$

Proof. ✓ Proof OK Expand the matrix trace and the (j, j) -entry of $A^\sigma (B^\sigma)^\dagger$. □

9.3 Frobenius norm estimates

The spectral-gap proofs use the Frobenius (Hilbert–Schmidt) norm as the natural norm for the finite-dimensional matrix algebra. We write $\|\cdot\|_F$ for this norm; it coincides with, and we use it interchangeably with, the Hilbert–Schmidt norm $\|\cdot\|_{\mathrm{HS}}$ on finite-dimensional spaces. We record the squared version and an isometric embedding into Euclidean space.

Definition 9.10 (Frobenius norm squared). [frobSq](#) ✓ Stmt OK For a (possibly rectangular) matrix $X \in M_{m \times n}(\mathbb{C})$, the *Frobenius norm squared* is

$$\|X\|_F^2 := \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} |X_{ij}|^2.$$

Lemma 9.11 (Trace formula for Frobenius norm). [frobSq_trace](#) ✓ Stmt OK $\|X\|_F^2 = \mathrm{Re} \, \mathrm{tr}(X^\dagger X)$.

Proof. ✓ Proof OK Expand the matrix product and trace entry by entry. □

Definition 9.12 (Euclidean-space embedding). [matToES](#) ✓ Stmt OK The map $\mathrm{vec} : M_{m \times n}(\mathbb{C}) \rightarrow \mathbb{C}^{mn}$ that flattens matrix entries is an isometric embedding with respect to the Frobenius norm: $\|\mathrm{vec}(X)\|^2 = \|X\|_F^2$.

Lemma 9.13 (Norm of embedded matrix). [norm_matToES_sq](#) ✓ Stmt OK $\|\mathrm{vec}(X)\|^2 = \|X\|_F^2$.

Proof. ✓ Proof OK Expand both sides as sums over entries and use $|z|^2 = \bar{z}z$. □

9.4 Eigenvalue bound and spectral gap

The eigenvalue bound comes from a uniform Frobenius-norm estimate obtained directly from the trace-preserving normalization. For the rigidity theorem, one gauges A and B separately using positive fixed points of their transfer maps to obtain unital Kraus families, and then applies the Kadison–Schwarz equality argument to a block embedding of a mixed-transfer eigenvector.

Theorem 9.14 (Eigenvalue bound). [eigenvalue_norm_le_one](#) ✓ Stmt OK Let A, B be normalized MPS tensors. Every eigenvalue μ of F_{AB} satisfies $|\mu| \leq 1$.

Proof. ✓ Proof OK Expand $F_{AB}^n(X)$ as a sum over words and use Cauchy–Schwarz, together with $\sum_i (A^i)^\dagger A^i = \sum_i (B^i)^\dagger B^i = \mathbb{1}$, to obtain a uniform Frobenius-norm bound $\|F_{AB}^n(X)\|_{\mathrm{HS}} \leq C_D \|X\|_{\mathrm{HS}}$ for all n , where C_D depends only on D . If $F_{AB}(X) = \mu X$ and $X \neq 0$, then $|\mu|^n \|X\|_{\mathrm{HS}} = \|F_{AB}^n(X)\|_{\mathrm{HS}} \leq C_D \|X\|_{\mathrm{HS}}$ for every n . Hence $|\mu| \leq 1$.

The argument uses only the trace-preserving normalization and does not appeal to Perron–Frobenius theory. □

Theorem 9.15 (Spectral radius bound). [spectralRadius_mixedTransfer_le_one](#) ✓ Stmt OK $\rho(F_{AB}) \leq 1$ for normalized tensors.

Proof. ✓ Proof OK The spectral radius is the supremum of $|\mu|$ over all eigenvalues, and each satisfies $|\mu| \leq 1$ by Theorem 9.14. $\square$

Theorem 9.16 (Spectral radius ≥ 1 implies gauge-phase equivalence). modulus_one_eigenvalue_implies_gauge ✓ Stmt OK Let A, B be injective normalized MPS tensors. If $\rho(F_{AB}) \geq 1$, then A and B are gauge-phase equivalent.

Proof. ✓ Proof OK Since $\rho(F_{AB}) \leq 1$ (Theorem 9.15) and $\rho(F_{AB}) \geq 1$ by hypothesis, there exists an eigenvalue μ with $|\mu| = 1$ and eigenvector $X \neq 0$: $F_{AB}(X) = \mu X$. The proof proceeds in five steps.

Step 1 (Separate gauging). Choose positive definite fixed points ρ_A, ρ_B of the transfer maps $\mathcal{E}_A, \mathcal{E}_B$ (Theorem 8.9) and gauge each tensor separately to a unital Kraus family (Theorem 8.10): set $A'^i = \rho_A^{-1/2} A^i \rho_A^{1/2}$ and similarly for B . Let $X' = \rho_A^{-1/2} X \rho_B^{-1/2}$ denote the correspondingly gauged eigenvector.

Step 2 (Block Kraus family and off-diagonal embedding). Form the block Kraus operators $C^i = \begin{pmatrix} A'^i & 0 \\ 0 & B'^i \end{pmatrix}$. These constitute a unital Kraus family on $M_{2D}(\mathbb{C})$. The off-diagonal matrix $Y = \begin{pmatrix} 0 & X' \\ 0 & 0 \end{pmatrix}$ satisfies $E_C(Y) = \mu Y$.

Step 3 (KS equality and Kraus commutation). Because $|\mu| = 1$ and E_C is unital and trace-preserving, the Kadison–Schwarz equality for peripheral eigenvectors (Theorem 7.24) gives $E_C(Y^\dagger Y) = E_C(Y)^\dagger E_C(Y)$. By the Kraus commutation relation (Theorem 7.23), Y commutes with every block Kraus operator: $C^i Y = Y C^i$ for all i .

Step 4 (Intertwining identity and invertibility). Expanding the commutation relation block by block gives $A'^i X' = X' B'^i$ for every i . Since $\{A'^i\}$ span $M_D(\mathbb{C})$ (injectivity of A), X' is a nonzero intertwiner from $M_D(\mathbb{C})$ to $M_D(\mathbb{C})$. The left multiplicative identity (Theorem 7.25) gives $E_C(Y^\dagger Y) = E_C(Y^\dagger) E_C(Y)$, so $X'^\dagger X'$ is a positive semi-definite fixed point of $\mathcal{E}_{B'}$, positive definite by injectivity. Hence X' is invertible.

Step 5 (Skolem–Noether conclusion). The map $M \mapsto X' M (X')^{-1}$ is a $\mathbb{C}$ -algebra automorphism of $M_D(\mathbb{C})$ sending B'^i to A'^i . By Skolem–Noether (Theorem 4.9) this is an inner automorphism. Undoing the separate gauges gives $B^i = \bar{\mu} X A^i X^{-1}$ for all i , establishing gauge-phase equivalence (Definition 2.18).

The crucial point is that one gauges the individual transfer maps to unital form; one does not require the mixed transfer map itself to be simultaneously unital and trace-preserving. $\square$

Theorem 9.17 (Gauged peripheral eigenvector yields Kraus intertwining). gauged_intertwining_core ✓ Stmt OK Let A and B be normalized tensors, and let $X \neq 0$ satisfy $F_{AB}(X) = \mu X$ with $|\mu| = 1$. Choose positive-definite fixed points ρ_A, ρ_B of the transfer maps and gauge to unital families $A'^i = S_A^{-1} A^i S_A$ and $B'^i = S_B^{-1} B^i S_B$ with $S_A S_A^\dagger = \rho_A$ and $S_B S_B^\dagger = \rho_B$. Then the transported matrix $X' = S_A^{-1} X (S_B^\dagger)^{-1}$ is nonzero, the families A' and B' are unital, and for every i one has

$$X' (B'^i)^\dagger = \mu (A'^i)^\dagger X', \quad A'^i X' = \mu X' B'^i.$$

Proof. ✓ Proof OK Block-embed the gauged families and transported eigenvector into a single unital Kraus map, apply Kadison–Schwarz equality for the modulus-one eigenvalue, then extract the intertwining identities from the block structure. $\square$

Theorem 9.18 (Intertwining yields gauge-phase equivalence). gaugePhaseEquiv_of_gauged_intertwining ✓ Stmt OK Let A and B be tensors of the same bond dimension. Suppose there are invertible gauges taking them to families A' and B' , an invertible matrix X' , and a scalar μ with $|\mu| = 1$ such that, for every physical index i ,

$$A'^i X' = \mu X' B'^i$$

Then A and B are gauge-phase equivalent.

Proof. ✓ Proof OK Compose the two gauge matrices with the intertwiner to obtain a single invertible matrix conjugating A to a scalar multiple of B . $\square$

Remark 9.19 (Two gauge conventions). ✓ Stmt OK The right-canonical gauge used here, $A'^i = \rho^{-1/2} A^i \rho^{1/2}$, makes $\{A'^i\}$ a *unital* Kraus family ($\sum_i A'^i (A'^i)^\dagger = \mathbb{1}$). A different convention, $\tilde{A}^i = \rho^{1/2} A^i \rho^{-1/2}$, makes the family *trace-preserving* ($\sum_i (\tilde{A}^i)^\dagger \tilde{A}^i = \mathbb{1}$). Both conventions appear in practice: the unital version via direct matrix square-root inversion (used in the spectral-gap proof above), and the trace-preserving version (used in the canonical-form reduction of Chapter 11).

Theorem 9.20 (Strict spectral gap). spectralRadius_mixedTransfer_lt_one ✓ Stmt OK Let A, B be injective normalized MPS tensors that are not gauge-phase equivalent. Then $\rho(F_{AB}) < 1$.

Proof. ✓ Proof OK $\rho(F_{AB}) \leq 1$ by Theorem 9.15. If $\rho(F_{AB}) = 1$, Theorem 9.16 gives gauge-phase equivalence, contradicting the hypothesis. $\square$

9.5 Spectral gap — rectangular case

Theorem 9.21 (Rectangular intertwining forces equal bond dimension). dim_eq_of_gauged_intertwining ✓ Stmt OK Let A and B be injective tensors of bond dimensions D_1 and D_2 . If there exist $X \in M_{D_1 \times D_2}(\mathbb{C}) \setminus \{0\}$ and $\mu \in \mathbb{C}$ such that, for every physical index i ,

$$X(B^i)^\dagger = \mu (A^i)^\dagger X, \quad A^i X = \mu X B^i$$

then $D_1 = D_2$.

Proof. ✓ Proof OK The intertwining relation shows that $\ker(X)$ is invariant under all generators of B . Since B is injective, its generators span the full matrix algebra, so $\ker(X)$ is trivial and $D_2 \leq D_1$. Applying the same argument to $X^\dagger$ gives $D_1 \leq D_2$. $\square$

Theorem 9.22 (Rectangular spectral gap). mixedTransferSpectralRadius2_lt_one_of_dim_ne ✓ Stmt OK Let A be an injective normalized tensor of bond dimension D_1 and B an injective normalized tensor of bond dimension D_2 , with $D_1 \neq D_2$. Then $\rho(F_{AB}^{\text{rect}}) < 1$.

Proof. ✓ Proof OK The same word-expansion argument as in Theorem 9.14 gives $|\mu| \leq 1$ for every eigenvalue of F_{AB}^{rect} . If $\rho(F_{AB}^{\text{rect}}) = 1$, choose a modulus-one eigenvector $X \in M_{D_1 \times D_2}(\mathbb{C}) \setminus \{0\}$.

Gauge A and B separately to unital Kraus families and apply the block-embedding Kadison–Schwarz argument as in the square case. The resulting intertwining relations satisfy the hypotheses of Theorem 9.21, so $D_1 = D_2$, contradicting the hypothesis. $\square$

Theorem 9.23 (Rectangular overlap decay). mpvOverlap_tendsto_zero_of_dim_ne ✓ Stmt OK If $D_1 \neq D_2$ and both tensors are injective and normalized, then $O_{AB}(N) \rightarrow 0$ as $N \rightarrow \infty$.

Proof. ✓ Proof OK The spectral gap $\rho(F_{AB}^{\text{rect}}) < 1$ (Theorem 9.22) implies $(F_{AB}^{\text{rect}})^N \rightarrow 0$, so the operator trace converges to 0. By Theorem 9.8, $O_{AB}(N) = \text{Tr}((F_{AB}^{\text{rect}})^N) \rightarrow 0$. $\square$

9.6 MPV overlap decay

Theorem 9.24 (Transfer powers converge to zero). mixedTransfer_pow_tendsto_zero ✓ Stmt OK Let A, B be injective normalized tensors that are not gauge-phase equivalent. Then $F_{AB}^n(X) \rightarrow 0$ for every $X \in M_D(\mathbb{C})$.

Proof. ✓ Proof OK The spectral gap $\rho(F_{AB}) < 1$ (Theorem 9.20) implies $\|F_{AB}^n\|_{\text{op}} \rightarrow 0$ by the Gelfand formula, so $F_{AB}^n(X) \rightarrow 0$ for every X . $\square$

Theorem 9.25 (Overlap decay). mpvOverlap_tendsto_zero ✓ Stmt OK *Let A, B be injective normalized tensors of the same bond dimension that are not gauge-phase equivalent. Then $O_{AB}(N) \rightarrow 0$ as $N \rightarrow \infty$.*

Proof. ✓ Proof OK By Theorem 9.7, $O_{AB}(N) = \text{Tr}(F_{AB}^N)$. Expand the operator trace over matrix units (Lemma 9.6): $\text{Tr}(F_{AB}^N) = \sum_{p,q} (F_{AB}^N(E_{pq}))_{pq}$. By Theorem 9.24, each $F_{AB}^N(E_{pq}) \rightarrow 0$ entry-wise. Since the sum is finite, $O_{AB}(N) \rightarrow 0$. $\square$

9.7 Block separation

These asymptotic facts are the ingredients used later in the canonical-form separation arguments.

Remark 9.26 (Cross-correlation decay). cross_correlation_tendsto_zero ✓ Stmt OK If A and B are injective normalized tensors that are not gauge-phase equivalent, then for every $X \in M_D(\mathbb{C})$ one has $\text{tr}(F_{AB}^N(X)) \rightarrow 0$ as $N \rightarrow \infty$. This is the scalar trace consequence of the operator convergence $F_{AB}^N(X) \rightarrow 0$ from Theorem 9.24.

Remark 9.27 (Self-correlation persists). self_correlation_persists ✓ Stmt OK If ρ is a fixed point of $\mathcal{E}_A$, then $\text{tr}(\mathcal{E}_A^N(\rho)) = \text{tr}(\rho)$ for all $N \geq 0$. In particular this applies to the distinguished QPF fixed point from Theorem 8.9.

9.8 Spectral gap under irreducible-TP hypotheses

The preceding spectral-gap results require *injectivity* of both tensors. The following variants weaken this to *irreducibility* combined with trace-preserving normalization ($\sum_i (A^i)^\dagger A^i = \mathbb{1}$), following Cirac et al. [CPGSV17b]. The argument replaces the Kraus-commutation step with a Cauchy–Schwarz rigidity argument for the Perron–Frobenius fixed point.

Theorem 9.28 (Gauge-equivalence from irreducible-TP spectral radius). modulus_one_eigenvalue_implies_gauge_of_irreducibl *A, B be irreducible trace-preserving normalized MPS tensors of bond dimension D . If $\rho(F_{AB}) \geq 1$, then A and B are gauge-phase equivalent.*

Proof. ✓ Proof OK Choose a modulus-one eigenvector as in Theorem 9.16. The proof follows the five-step structure of Theorem 9.16, with one key adaptation. Steps 1–3 (separate gauging, block Kraus family, KS equality and Kraus commutation) proceed identically: the gauging uses the unique positive definite fixed points guaranteed by irreducibility (Theorem 8.9) and the right-canonical gauge (Theorem 8.10). The change is in Step 4: the Kadison–Schwarz equality gives $X'^\dagger X'$ as a nonzero positive semidefinite fixed point of $\mathcal{E}_{B'}$. Under injectivity this is immediately positive definite (the fixed point is unique up to scaling). Under the weaker hypothesis of irreducibility, Theorem 8.3 upgrades it to positive definite, giving invertibility of X' . Formally, Theorem 9.17 combines the separately gauged eigenvector into the required intertwining relations. Once X' is invertible, Theorem 9.18 upgrades the intertwining to gauge-phase equivalence. $\square$

Theorem 9.29 (Strict spectral gap (irreducible-TP)). spectralRadius_mixedTransfer_lt_one_of_irreducible_TP ✓ Stmt OK *Let A, B be irreducible trace-preserving normalized MPS tensors of the same bond dimension that are not gauge-phase equivalent. Then $\rho(F_{AB}) < 1$.*

Proof. ✓ Proof OK Contrapositive of Theorem 9.28, combined with $\rho(F_{AB}) \leq 1$ (Theorem 9.15). $\square$

Theorem 9.30 (Transfer power decay (irreducible-TP)). [mixedTransfer_pow_tendsto_zero_of_irreducible_TP](#) ✓ Stmt OK Under the hypotheses of Theorem 9.29, $F_{AB}^n(X) \rightarrow 0$ for every $X \in M_D(\mathbb{C})$.

Proof. ✓ Proof OK By the Gelfand formula, $\rho(F_{AB}) < 1$ implies $\|F_{AB}^n\|_{\text{op}} \rightarrow 0$. □

Theorem 9.31 (Overlap decay (irreducible-TP)). [mpvOverlap_tendsto_zero_of_irreducible_TP](#) ✓ Stmt OK Let A, B be irreducible trace-preserving normalized tensors of the same bond dimension that are not gauge-phase equivalent. Then $O_{AB}(N) \rightarrow 0$ as $N \rightarrow \infty$.

Proof. ✓ Proof OK Combine the spectral gap (Theorem 9.29) with the overlap–trace identity (Theorem 9.7). □

Theorem 9.32 (Rectangular spectral gap (irreducible-TP)). [mixedTransferSpectralRadius2_lt_one_of_dim_ne_of_irreducible_TP](#) ✓ Stmt OK Let A (bond dimension D_1) and B (bond dimension D_2) be irreducible trace-preserving normalized tensors with $D_1 \neq D_2$. Then $\rho(F_{AB}^{\text{rect}}) < 1$.

Proof. ✓ Proof OK Assume for contradiction that $\rho = 1$. Choose a modulus-one eigenvector $X \in M_{D_1 \times D_2}(\mathbb{C})$. The Cauchy–Schwarz rigidity argument produces an intertwiner X' with $A'^i X' = X' B'^i$ for all i . Since $\ker(X')$ is invariant under all B'^i , irreducibility forces $\ker(X') = \{0\}$; similarly for $X'^\dagger$. This gives $D_1 = D_2$, contradicting the hypothesis. □

Theorem 9.33 (Rectangular overlap decay (irreducible-TP)). [mpvOverlap_tendsto_zero_of_dim_ne_of_irreducible_TP](#) ✓ Stmt OK $D_1 \neq D_2$ and both tensors are irreducible and trace-preserving normalized, then $O_{AB}(N) \rightarrow 0$ as $N \rightarrow \infty$.

Proof. ✓ Proof OK Combine Theorem 9.32 with Theorem 9.8. □

9.9 Conditional expectation from a faithful fixed point

Definition 9.34 (Conditional expectation). [IsConditionalExpectation](#) ✓ Stmt OK Given a $*$ -subalgebra $S \subseteq M_D(\mathbb{C})$, a linear map $P : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is a conditional expectation onto S if it is idempotent, unital, has range contained in S , and satisfies $P(X) = X$ for all $X \in S$.

Definition 9.35 (Scalar conditional expectation). [scalarConditionalExpectation](#) ✓ Stmt OK For $\sigma \geq 0$ with $\text{tr}(\sigma) \neq 0$, define

$$E_\sigma(X) := \frac{\text{tr}(\sigma X)}{\text{tr}(\sigma)} \mathbb{1}.$$

Lemma 9.36 (Evaluation formula). [scalarConditionalExpectation_apply](#) ✓ Stmt OK $E_\sigma(X) = \frac{\text{tr}(\sigma X)}{\text{tr}(\sigma)} \mathbb{1}$.

Theorem 9.37 (Unitality). [scalarConditionalExpectation_unital](#) ✓ Stmt OK $E_\sigma(\mathbb{1}) = \mathbb{1}$.

Theorem 9.38 (Idempotence). [scalarConditionalExpectation_idempotent](#) ✓ Stmt OK $E_\sigma(E_\sigma(X)) = E_\sigma(X)$ for all X .

Lemma 9.39 (Range is scalar). [scalarConditionalExpectation_range_scalar](#) ✓ Stmt OK For every X , there exists $c \in \mathbb{C}$ with $E_\sigma(X) = c \mathbb{1}$.

Lemma 9.40 (Fixes scalar operators). [scalarConditionalExpectation_fixes_scalar](#) ✓ Stmt OK For $c \in \mathbb{C}$, $E_\sigma(c \mathbb{1}) = c \mathbb{1}$.

Theorem 9.41 (Absorbs the adjoint map). [scalarConditionalExpectation_absorbs_adjointMap](#) ✓ Stmt OK If $T(\sigma) = \sigma$, then $E_\sigma \circ T^* = E_\sigma$.

Theorem 9.42 (Adjoint map absorbs scalar projection). [adjointMap_absorbs_scalarConditionalExpectation](#) ✓ Stmt OK If T is trace-preserving, then $T^* \circ E_\sigma = E_\sigma$.

Lemma 9.43 (Image lies in adjoint fixed points). [scalarConditionalExpectation_mem_adjointFixedPoints](#) ✓ Stmt OK For all X , $E_\sigma(X)$ is fixed by T^* .

Theorem 9.44 (Scalar map is a conditional expectation). [scalarConditionalExpectation_isConditionalExpectation](#) ✓ Stmt OK Let K be trace-preserving, let $\rho > 0$ be a positive-definite fixed point of the associated channel $T(\rho) = \sum_i K_i \rho K_i^\dagger$ (so $T(\rho) = \rho$), and assume every adjoint fixed point of T^* is a scalar multiple of $\mathbb{1}$. Then E_ρ is a conditional expectation onto the adjoint fixed-point $*$ -subalgebra.

Theorem 9.45 (Complete positivity of the scalar conditional expectation). [scalarConditionalExpectation_isCPMap](#) ✓ Stmt OK For any positive semidefinite σ , the scalar conditional expectation E_σ is a completely positive map.

Proof. ✓ Proof OK Write $S = \sqrt{\sigma}$ and define the Kraus family $K_{j,i} = |j\rangle\langle i| S$ indexed by $(j, i) \in \{1, \dots, D\}^2$. Using $|j\rangle\langle i| M |i\rangle\langle j| = M_{ii} |j\rangle\langle j|$ and summing first over i and then over j ,

$$\sum_{j,i} K_{j,i} X K_{j,i}^\dagger = \sum_j |j\rangle\langle j| \operatorname{tr}(S X S) = \operatorname{tr}(\sigma X) \mathbb{1},$$

where we used cyclicity of the trace and $S^2 = \sigma$. Hence $E_\sigma = (\operatorname{tr} \sigma)^{-1} \cdot (X \mapsto \sum_{j,i} K_{j,i} X K_{j,i}^\dagger)$ is the Kraus map scaled by the nonnegative real number $(\operatorname{tr} \sigma)^{-1}$, and therefore completely positive. $\square$

Theorem 9.46 (σ -trace preservation of the scalar conditional expectation). [scalarConditionalExpectation_preserves_weighted](#) ✓ Stmt OK If $\operatorname{tr} \sigma \neq 0$, then for every X

$$\operatorname{tr}(\sigma E_\sigma(X)) = \operatorname{tr}(\sigma X).$$

Proof. ✓ Proof OK Direct computation: $\operatorname{tr}(\sigma \cdot \frac{\operatorname{tr}(\sigma X)}{\operatorname{tr} \sigma} \mathbb{1}) = \frac{\operatorname{tr}(\sigma X)}{\operatorname{tr} \sigma} \operatorname{tr} \sigma = \operatorname{tr}(\sigma X)$. $\square$

9.10 Stationary support

This section records the support-projection theory behind stationary states, following [Wol12, §6.4].

Lemma 9.47 (Lower-triangular Kraus condition implies corner invariance). [lowerZero_implies_invariance](#) ✓ Stmt OK Let A be an MPS tensor and let P be an orthogonal projection. If, for every physical index i ,

$$(\mathbb{1} - P)A^i P = 0$$

then

$$P \mathcal{E}_A(PXP) P = \mathcal{E}_A(PXP)$$

for all $X \in M_D(\mathbb{C})$.

Proof. ✓ Proof OK Expand the transfer map and use $(\mathbb{1} - P)A^i P = 0$ to reduce each Kraus term to the P -corner. $\square$

Theorem 9.48 (Support projection invariance). [support_proj_fixed](#) ✓ Stmt OK Let E be a quantum channel and $\rho \geq 0$ a positive semidefinite fixed point ($E(\rho) = \rho$). Let P denote the support projection of ρ . Then

$$P E(PXP) P = E(PXP)$$

for all $X \in M_D(\mathbb{C})$.

Proof. ✓ Proof OK Write E in Kraus form $E(X) = \sum_i K_i X K_i^\dagger$. From $E(\rho) = \rho$ and $\rho \geq 0$ one deduces $(\mathbb{1} - P)K_i P = 0$ for all i . Apply Lemma 9.47. $\square$

Definition 9.49 (Stationary state). stationaryState ✓ Stmt OK For an irreducible channel E on $M_D(\mathbb{C})$ (with $D \geq 1$), the *stationary state* is the unique density-matrix fixed point ρ_∞ satisfying $E(\rho_\infty) = \rho_\infty$, $\rho_\infty > 0$, and $\text{tr}(\rho_\infty) = 1$. Uniqueness follows from the quantum Perron–Frobenius theorem (Theorem 8.9).

Definition 9.50 (Stationary support). stationarySupport ✓ Stmt OK The *stationary support* of an irreducible channel E is the support projection of the stationary state (Definition 9.49).

Theorem 9.51 (Stationary support is full). stationarySupport_eq_one ✓ Stmt OK For an irreducible channel E , the stationary support equals the identity: $\text{supp}(\rho_\infty) = \mathbb{1}$.

Proof. ✓ Proof OK The support projection P of ρ_∞ is an orthogonal projection satisfying $PE(PXP)P = E(PXP)$ for all X (Theorem 9.48). By irreducibility, $P = 0$ or $P = \mathbb{1}$. Since $\rho_\infty \neq 0$, we have $P \neq 0$, so $P = \mathbb{1}$. $\square$

Remark 9.52 (Remaining stationary-support formulations). The non-trivial formulations needed here are:

- **Prop. 6.10** (converse of Prop. 6.9): if the support projection of *every* PSD fixed point is full, then the channel is irreducible.
- **Prop. 6.11**: minimality of the stationary support among invariant projections, without assuming irreducibility.

These statements are deferred in the present chapter.

9.10.1 Faithful compression onto the support sector

This subsection establishes the results on the corner algebra, $*$ -structure, and compression of a PSD fixed point onto its support sector needed for [Wol12, Cor. 6.6]: the corner carries a $\mathbb{C}$ -algebra and $*$ -structure, the two standard presentations of the corner are identified, and the compression of a PSD fixed point onto its support sector is positive definite.

Lemma 9.53 (Support projection annihilates the kernel). supportProj_mulVec_eq_zero_of_mulVec_eq_zero ✓ Stmt OK Let $\rho \succeq 0$ on $\mathbb{C}^D$ with support projection $P = \text{supp}(\rho)$. For every $v \in \mathbb{C}^D$, if $\rho v = 0$, then $Pv = 0$.

Proof. ✓ Proof OK Diagonalize $\rho = U \text{diag}(\lambda) U^\dagger$ with $\lambda_j \geq 0$. Writing $w := U^\dagger v$, the hypothesis $\rho v = 0$ gives $\lambda_j w_j = 0$ for every j ; hence $w_j = 0$ whenever $\lambda_j > 0$. The support projection acts as $\text{diag}(\chi_{\lambda_j > 0})$ in the eigenbasis, so $Pv = U \text{diag}(\chi_{\lambda_j > 0}) w = 0$. $\square$

Definition 9.54 (Corner submodule / idempotent-corner identification). cornerSubmoduleCornerEquiv ✓ Stmt OK For an idempotent $P \in M_D(\mathbb{C})$, the subspace $\{X \mid PXP = X\}$ and the corner $\{PXP \mid X \in M_D(\mathbb{C})\}$ are identified as $\mathbb{C}$ -linear spaces by the identity on underlying matrices.

Remark 9.55 ($\mathbb{C}$ -algebra and $*$ -structure on the corner). instAlgebraComplexCorner cornerStar cornerStarRing cornerStarModuleComplex ✓ Stmt OK The corner algebra associated to an idempotent already carries a ring structure with unit P . In the matrix case it also has $\mathbb{C}$ -module and $\mathbb{C}$ -algebra structures and, under the additional assumption $P^\dagger = P$, star, star ring, and star module structures over $\mathbb{C}$.

Theorem 9.56 (Faithful compression onto the support). [compression_on_support_posDef](#) ✓ Stmt OK Let $\rho \succeq 0$ on $\mathbb{C}^D$ with support projection $P = \text{supp}(\rho)$, and let $V : \mathbb{C}^D \rightarrow \mathbb{C}^k$ be a compression isometry satisfying $V V^\dagger = \mathbb{1}_k$ and $V^\dagger V = P$. Then the compression

$$V \rho V^\dagger \in M_k(\mathbb{C})$$

is positive definite.

Proof. ✓ Proof OK Hermiticity follows from Hermiticity of ρ and the product-star identity. For strict positivity, let $w \neq 0$, set $u := V^\dagger w$, and note that $V V^\dagger = \mathbb{1}_k$ forces $u \neq 0$, while $V^\dagger V = P$ forces $Pu = u$. The quadratic form computes as $w^\dagger (V \rho V^\dagger) w = u^\dagger \rho u$, which is nonnegative by positive semidefiniteness. If it vanishes, then $\rho u = 0$, and by Lemma 9.53 also $Pu = 0$, contradicting $Pu = u \neq 0$. $\square$

9.11 Wedderburn decomposition of the fixed-point algebra

Theorem 9.57 (Fixed-point algebra is semisimple). [fixedPointAlgebra_isSemisimpleRing](#) ✓ Stmt OK The adjoint-fixed-point $\ast$ -subalgebra of a trace-preserving Kraus map on $M_D(\mathbb{C})$ is a semisimple ring.

Proof. ✓ Proof OK The subalgebra is finite-dimensional (as a subalgebra of $M_D(\mathbb{C})$), hence Artinian. By [Wol12, Thm. 6.14], it suffices to show the Jacobson radical J vanishes. If $x \in J$, then $x^\ast x \in J$ (left-ideal closure). The radical of an Artinian ring is nilpotent, so $x^\ast x$ is nilpotent. A self-adjoint nilpotent matrix satisfies $\|x^\ast x\|^{2^k} = \|(x^\ast x)^{2^k}\| = 0$, hence $x^\ast x = 0$, and the $C^\ast$ -identity gives $x = 0$. $\square$

Theorem 9.58 (Abstract Wedderburn–Artin decomposition). [fixedPointAlgebra_wedderburnArtin](#) ✓ Stmt OK There exist $n \in \mathbb{N}$ and dimensions $d_1, \dots, d_n \geq 1$ such that the adjoint-fixed-point $\ast$ -subalgebra is $\mathbb{C}$ -algebra isomorphic to $\prod_{k=1}^n M_{d_k}(\mathbb{C})$.

Proof. ✓ Proof OK Combine semisimplicity (Theorem 9.57) with the Wedderburn–Artin structure theorem for finite-dimensional semisimple algebras over an algebraically closed field. $\square$

Theorem 9.59 (Fixed-point algebra decomposition with multiplicities). [adjointFixedPoints_wedderburnDecomp](#) ✓ Stmt OK There exist integers n , block sizes $d_1, \dots, d_n \geq 1$, multiplicities $m_1, \dots, m_n \geq 1$, and a $\mathbb{C}$ -algebra isomorphism from the adjoint-fixed-point $\ast$ -subalgebra to

$$\prod_{k=1}^n M_{d_k}(\mathbb{C})$$

such that

$$\sum_{k=1}^n d_k m_k \leq D.$$

Proof. ✓ Proof OK Apply the abstract Wedderburn decomposition (Theorem 9.58). Since the adjoint-fixed-point algebra is realized as a subalgebra of the ambient matrix algebra $M_D(\mathbb{C})$, this realization also yields multiplicities m_k and the bound $\sum_k d_k m_k \leq D$. $\square$

Remark 9.60 (Block form of the fixed-point algebra). Wolf’s block form identifies the adjoint-fixed-point algebra, up to a unitary change of basis, with

$$0 \oplus \bigoplus_k M_{d_k}(\mathbb{C}) \otimes \mathbb{1}_{m_k}$$

together with the density-block form $M_{d_k}(\mathbb{C}) \otimes \rho_k$. The theorem above records the product decomposition, multiplicities, and dimension bound needed from this block form.

Theorem 9.61 (Weighted fixed points form a $*$ -subalgebra). [weightedFixedPointsStarSubalgebra](#) ✓ Stmt OK Let $T(X) = \sum_i K_i X K_i^\dagger$ be trace-preserving, and let $\rho > 0$ satisfy $T(\rho) = \rho$. If

$$F_T := \{X \in M_D(\mathbb{C}) \mid T(X) = X\},$$

then $\rho^{-1/2} F_T \rho^{-1/2}$ is a $*$ -subalgebra of $M_D(\mathbb{C})$.

Proof. ✓ Proof OK Set $B_i = \rho^{-1/2} K_i \rho^{1/2}$. The equation $T(\rho) = \rho$ implies that the family $\{B_i\}$ is unital, and trace preservation of T implies that ρ is a positive-definite fixed point of the adjoint map of the gauged family. Apply Theorem 5.138 to B . The identity

$$\sum_i B_i X B_i^\dagger = X \iff T(\rho^{1/2} X \rho^{1/2}) = \rho^{1/2} X \rho^{1/2}$$

identifies the fixed points of the gauged map with $\rho^{-1/2} F_T \rho^{-1/2}$. □

9.12 Further irreducibility and primitivity equivalences

This section records several criteria and equivalences from Wolf's Chapter 6 that are used later.

Theorem 9.62 (Exponential positivity condition). [exp_posDef_of_irreducible_cp](#) [irreducible_iff_exp_posDef_forall](#) ✓ Stmt OK Let E be an irreducible completely positive map, $A \geq 0$ with $A \neq 0$, and $t > 0$. Then $\exp(tE)(A)$ is positive definite. Conversely, if $\exp(tE)(A)$ is positive definite for every such t, A , then E is irreducible (full equivalence [irreducible_iff_exp_posDef_forall](#)).

Theorem 9.63 (Kraus-span equivalence). [wolf_theorem_6_8_kraus_span](#) ✓ Stmt OK Under normalization, primitivity in Wolf's sense is equivalent to eventual full Kraus rank.

Theorem 9.64 (Combined primitivity equivalence). [wolf_theorem_6_8_conjunction](#) ✓ Stmt OK Under normalization, primitivity in Wolf's sense is equivalent to the conjunction of eventual full Kraus rank, normality, and strong irreducibility.

Remark 9.65 (Pairwise primitivity equivalences). [primitivePaper_iff_hasEventuallyFullKrausRank](#) [primitivePaper_iff_stronglyIrreducible](#) [hasEventuallyFullKrausRank_iff_stronglyIrreducible](#) [hasEventuallyFullKrausRank_iff_isNormal](#) ✓ Stmt OK These are the pairwise equivalences appearing in Wolf's Theorem 6.8.

Theorem 9.66 (Scalar fixed-point algebra case). [scalarConditionalExpectation_isConditionalExpectation](#) ✓ Stmt OK In the scalar fixed-point algebra setting, the weighted map $X \mapsto \frac{\text{tr}(\rho X)}{\text{tr}(\rho)} \mathbb{1}$ is a conditional expectation onto the adjoint fixed-point $*$ -subalgebra.

9.13 Multi-cycle block-permutation data

Definition 9.67 (Block-permutation data). [CycleStructure](#) ✓ Stmt OK Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$. A block-permutation structure for T consists of a finite index set ι , a family of orthogonal projections $P_k \in M_D(\mathbb{C})$ for $k \in \iota$, and a permutation $\sigma \in \text{Sym}(\iota)$ (not necessarily a single cycle — in general σ may have multiple disjoint cycles), such that

$$T(P_{\sigma(k)}) = P_k, \quad T(P_k X) = T(P_k) T(X), \quad T(X P_k) = T(X) T(P_k)$$

for every $k \in \iota$ and $X \in M_D(\mathbb{C})$.

Definition 9.68 (Multi-cycle block-permutation data). [MultiCycleDecomposition](#) ✓ Stmt OK A multi-cycle decomposition of $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ records a finite cycle index ι , a per-cycle period $m : \iota \rightarrow \mathbb{N}_{>0}$, a projection family $P_{c,k} \in M_D(\mathbb{C})$ for $c \in \iota$ and $k \in \{0, \dots, m(c)-1\}$ consisting of orthogonal projections, together with the per-cycle cyclic action

$$T(P_{c,k+1}) = P_{c,k} \quad (k \in \{0, \dots, m(c)-1\}, c \in \iota)$$

and the multiplicative-domain factorisations $T(P_{c,k} X) = T(P_{c,k}) T(X)$ and $T(X P_{c,k}) = T(X) T(P_{c,k})$.

Theorem 9.69 (Per-cycle corner preservation). [preserves_corner_pow_period](#) ✓ Stmt OK For every cycle $c \in \iota$ and index $k \in \{0, \dots, m(c)-1\}$, the iterate $T^{m(c)}$ preserves the corner $P_{c,k} M_D(\mathbb{C}) P_{c,k}$.

Proof. ✓ Proof OK By induction on n , the n -th iterate T^n sends $P_{c,k+n} X P_{c,k+n}$ to $P_{c,k} T^n(X) P_{c,k}$, using the multiplicative-domain factorisations and the cyclic action $T(P_{c,j+1}) = P_{c,j}$ at each induction step. Specialising to $n = m(c)$ and using that the indices $k+m(c) = k$ in $\{0, \dots, m(c)-1\}$ yields corner preservation of $T^{m(c)}$. □

Theorem 9.70 (Common-period corner preservation). [preserves_corner_pow_of_dvd](#) ✓ Stmt OK If $N \in \mathbb{N}$ is divisible by every per-cycle period $m(c)$, then T^N preserves every corner $P_{c,k} M_D(\mathbb{C}) P_{c,k}$ of the decomposition.

Proof. ✓ Proof OK Write $N = m(c)q$. Corner preservation is stable under taking powers, so $T^N = (T^{m(c)})^q$ preserves each corner by Theorem 9.69. □

Definition 9.71 (Flattening to single-index block-permutation data). [toCycleStructure](#) ✓ Stmt OK A multi-cycle decomposition gives a single-index block-permutation structure on the sigma index $\Sigma_{c \in \iota} \{0, \dots, m(c)-1\}$: the permutation is the sigma-product of per-cycle cyclic shifts $k \mapsto k+1$ (whose cycle decomposition has one cycle per $c \in \iota$), the projections are $(c, k) \mapsto P_{c,k}$, and the multiplicative-domain factorisations descend componentwise. The resulting map forgets the explicit cycle-indexing.

9.14 Peripheral eigenvalue group structure

Lemma 9.72 (Peripheral eigenvectors are invertible). [isUnit_peripheral_eigenvector](#) ✓ Stmt OK Let E be an irreducible unital Kraus map on $M_D(\mathbb{C})$ with a positive-definite adjoint fixed point. If $X \neq 0$ satisfies $E(X) = \mu X$ with $|\mu| = 1$, then X is invertible.

Proof. ✓ Proof OK The KS equality gives $E(X^\dagger X) = X^\dagger X$, so $X^\dagger X$ is a nonzero PSD fixed point. Irreducibility upgrades it to positive definite, hence invertible, so $\det(X) \neq 0$. □

Lemma 9.73 (Peripheral eigenvalues: product closure). [peripheralEigenvalues_mul_mem_of_irreducible_unital_of_adjoint_fixedPo](#) ✓ Stmt OK Let E be an irreducible unital Kraus map on $M_D(\mathbb{C})$ with a positive-definite adjoint fixed point. If μ, ν are peripheral eigenvalues of E , then so is $\mu\nu$.

Proof. ✓ Proof OK Take eigenvectors X, Y for μ, ν . The KS equality at X puts X in the multiplicative domain, so $E(YX) = E(Y)E(X) = (\nu\mu)(YX)$. Since X, Y are units (Lemma 9.72), $YX \neq 0$ and $|\mu\nu| = 1$. □

Theorem 9.74 (Peripheral eigenvalues: cyclic structure). [peripheral_eigenvalues_cyclic_structure](#) ✓ Stmt OK Let E be an irreducible unital Schwarz map on $M_D(\mathbb{C})$ with a positive-definite adjoint fixed point (trace-preservation is not required). Then there exist $m \geq 1$ and a primitive m -th root of unity γ such that the peripheral eigenvalues of E are exactly $\{1, \gamma, \gamma^2, \dots, \gamma^{m-1}\}$.

Proof. ✓ Proof OK The peripheral eigenvalues form a finite subset of $\mathbb{C}$, contain 1, and are closed under multiplication and inversion, so they form a finite subgroup of $\mathbb{C}^\times$. Any finite subgroup of $\mathbb{C}^\times$ is cyclic. The generator γ has order $m = |\text{peripheral spectrum}|$. □

Theorem 9.75 (Peripheral eigenvalues form a cyclic group). peripheralEigenvalues_eq_range_primitiveRoot ✓ Stmt OK *Let E be an irreducible unital trace-preserving Schwarz map on $M_D(\mathbb{C})$ with a positive-definite adjoint fixed point. Then there exist $m \geq 1$ and a primitive m -th root of unity γ such that $m \mid D$ and the peripheral eigenvalues of E are exactly $\{1, \gamma, \gamma^2, \dots, \gamma^{m-1}\}$.*

Proof. ✓ Proof OK Apply Theorem 9.74 for the cyclic structure. Divisibility $m \mid D$ follows from the cyclic decomposition: m mutually orthogonal projections summing to $\mathbb{1}$ each have equal trace (by trace preservation), so $m \cdot \text{tr}(P_0) = D$ with $\text{tr}(P_0) \in \mathbb{N}$. □

Theorem 9.76 (Period divides bond dimension). channel_period_divides_dim ✓ Stmt OK *Let E be as in Theorem 9.75, and let γ be a primitive m -th root of unity generating the peripheral eigenvalues. Then $m \mid D$.*

Proof. ✓ Proof OK Follows from the cyclic projection decomposition and trace preservation, as in the final step of Theorem 9.75. □

9.14.1 Cyclic decomposition of irreducible Schwarz maps

Definition 9.77 (Corner preservation). PreservesCorner ✓ Stmt OK For an orthogonal projection P and linear map E , the corner $PM_D(\mathbb{C})P$ is preserved if $E(PXP) = PE(PXP)P$ for all X .

Definition 9.78 (Corner subspace). cornerSubmodule ✓ Stmt OK The corner subspace associated with P is $\{PXP : X \in M_D(\mathbb{C})\}$.

Definition 9.79 (Restricted corner map). cornerRestriction ✓ Stmt OK The restriction of E to an invariant corner $PM_D(\mathbb{C})P$.

Definition 9.80 (Irreducible restriction on a corner). IsIrreducibleOnCorner ✓ Stmt OK Irreducibility of the corner-restricted map.

Definition 9.81 (Rank of an orthogonal projection). cornerRank ✓ Stmt OK The rank of an orthogonal projection $P \in M_D(\mathbb{C})$, i.e. the dimension of the corner algebra $PM_D(\mathbb{C})P$.

Theorem 9.82 (Compression isometry for an orthogonal projection). cornerSubmoduleMatrixLinearEquiv ✓ Stmt OK *Let $P \in M_D(\mathbb{C})$ be an orthogonal projection of rank n . The corner algebra $PM_D(\mathbb{C})P$ is linearly isomorphic to the full matrix algebra $M_n(\mathbb{C})$. The isomorphism is built from the spectral diagonalisation of P by conjugating the top-left $n \times n$ block by the unitary diagonalising P .*

Theorem 9.83 (Rank equals trace for an orthogonal projection). cornerRank_eq_trace ✓ Stmt OK *The rank of an orthogonal projection P equals its trace: $n = \text{tr } P$.*

Theorem 9.84 (Peripheral unitary eigenvector). exists_peripheral_unitary_of_irreducible_schwarz ✓ Stmt OK *For an irreducible unital Schwarz map with faithful adjoint fixed point, each peripheral eigenvalue admits a unitary eigenvector.*

Theorem 9.85 (Powers on a peripheral-unitary orbit). map_powers_of_peripheral_unitary ✓ Stmt OK *If $E(U) = \mu U$ with $|\mu| = 1$ and U unitary, then $E(U^n) = \mu^n U^n$ for all n .*

Theorem 9.86 (Normalized peripheral unitary). exists_normalized_peripheral_unitary_of_irreducible_schwarz ✓ Stmt OK *Peripheral unitary eigenvectors can be normalized compatibly with the faithful fixed-point state.*

Theorem 9.87 (Cyclic projections from a peripheral unitary). [exists_cyclic_projections_of_peripheral_unitary](#) ✓ Stmt OK A primitive m -th peripheral root yields m orthogonal projections summing to $\mathbb{1}$ and cyclically permuted by the channel.

Theorem 9.88 (Cyclic decomposition for irreducible Schwarz maps). [exists_cyclic_decomposition_of_irreducible_schwarz](#) ✓ Irreducibility and faithful fixed point hypotheses, there is a full cyclic projection decomposition realizing the peripheral period.

Theorem 9.89 (Power maps preserve each cyclic sector). [preserves_corner_pow_of_cyclic_decomp](#) ✓ Stmt OK Appropriate powers of the channel preserve each sector corner.

Theorem 9.90 (Irreducibility of restricted sector dynamics). [isIrreducible_restriction_of_cyclic_decomp](#) ✓ Stmt OK Each sector restriction is irreducible.

Theorem 9.91 (Primitivity of restricted sector dynamics). [isPrimitive_restriction_of_cyclic_decomp](#) ✓ Stmt OK Each sector restriction is primitive.

Theorem 9.92 (Corner invariance at the period). [preserves_corner_pow_orderOf_of_perm_decomp](#) ✓ Stmt OK The channel raised to the period preserves every cyclic corner.

9.14.2 Group structure of the peripheral spectrum

Theorem 9.93 (Peripheral product closure). [peripheral_eigenvalues_closed_under_mul](#) ✓ Stmt OK The product of two peripheral eigenvalues is peripheral.

Theorem 9.94 (Peripheral inverse closure). [peripheral_eigenvalues_closed_under_inv](#) ✓ Stmt OK The inverse of a peripheral eigenvalue is peripheral.

Theorem 9.95 (Peripheral cyclic group with period divisibility). [peripheral_eigenvalues_form_cyclic_group](#) ✓ Stmt OK The peripheral eigenvalues form a finite cyclic group whose order divides the dimension.

Theorem 9.96 (Peripheral eigenvalue multiplicity one). [peripheral_eigenvalue_multiplicity_one](#) ✓ Stmt OK Each peripheral eigenvalue has algebraic multiplicity one.

Proof. ✓ Proof OK Given a peripheral unitary eigenvector U for γ , the multiplicative-domain identity gives $E(XU^\dagger) = XU^\dagger$ for any γ -eigenvector X . By the scalar fixed-point lemma for irreducible unital maps, $XU^\dagger = c \cdot I$, hence $X = c \cdot U$ and the eigenspace is one-dimensional. $\square$

9.15 Primitive overlap convergence (spectral-gap form)

The theorems in this section are stated directly in terms of the complementary map $E - P$. This is the analytic form needed for the later overlap argument. For primitive normalized transfer maps, Chapter 5 supplies this spectral-gap input under explicit hypotheses.

Theorem 9.97 (Complementary spectral gap implies trace convergence to 1). [linearMap_trace_pow_tendsto_one_of_spectralRa](#) ✓ E be trace-preserving with fixed point ρ ($\text{tr}(\rho) \neq 0$) and let P be the fixed-point projection (Definition 5.23). If $\rho_{\text{spec}}(E - P) < 1$, then $\text{Tr}(E^n) \rightarrow 1$ as $n \rightarrow \infty$.

Proof. ✓ Proof OK By the power decomposition (Theorem 5.24), $E^n = P + (E - P)^n$ for $n \geq 1$. Since $\rho_{\text{spec}}(E - P) < 1$, the Gelfand formula gives $(E - P)^n \rightarrow 0$, so $\text{Tr}(E^n) \rightarrow \text{Tr}(P)$. The operator trace of P equals 1: since $P(X) = \frac{\text{tr}(X)}{\text{tr}(\rho)} \rho$, we have $\text{Tr}(P) = \frac{\text{tr}(\rho)}{\text{tr}(\rho)} = 1$. $\square$

Theorem 9.98 (Self-overlap convergence from a complementary spectral gap). [\[mpvOverlap_tendsto_one_of_transfer_spectral\]](#)
A be a normalized MPS tensor with a nonzero PSD fixed point ρ of the transfer map. If $\rho_{\text{spec}}(\mathcal{E}_A - P) < 1$ (where P is the fixed-point projection), then $O_{AA}(N) \rightarrow 1$ as $N \rightarrow \infty$.

Proof. ✓ Proof OK By Theorem [9.7](#) (with $A = B$), $O_{AA}(N) = \text{Tr}(\mathcal{E}_A^N)$. By Theorem [9.97](#), $\text{Tr}(\mathcal{E}_A^N) \rightarrow 1$. □

Chapter 10

Wielandt Bound

This chapter proves a quantum analog of Wielandt's inequality: for a normal MPS tensor with bond dimension D , the cumulative span of matrix products stabilises after at most D^2 steps. The results follow [SPGWC10].

10.1 Cumulative span

Definition 10.1 (Word span). wordSpan ✓ Stmt OK The *word span* at length n is the subspace

$$S_n(A) = \text{span}_{\mathbb{C}}\{A^w : |w| = n\} \subseteq M_D(\mathbb{C}).$$

This is [SPGWC10, §II].

Definition 10.2 (Cumulative span). cumulativeSpan ✓ Stmt OK The *cumulative span* at level n is

$$T_n(A) = S_0(A) + S_1(A) + \cdots + S_n(A).$$

Lemma 10.3 (Monotonicity of cumulative span). cumulativeSpan_mono ✓ Stmt OK $T_n(A) \leq T_{n+1}(A)$ for all n .

Proof. ✓ Proof OK T_{n+1} includes all words of length $\leq n+1$, which contains all words of length $\leq n$. □

Lemma 10.4 (Dimension bound). cumulativeSpan_finrank_le ✓ Stmt OK $\dim T_n(A) \leq D^2$ for all n .

Proof. ✓ Proof OK $T_n(A) \subseteq M_D(\mathbb{C})$ and $\dim M_D(\mathbb{C}) = D^2$. □

Theorem 10.5 (Cumulative span stabilisation). cumulativeSpan_stable ✓ Stmt OK If $T_n(A) = T_{n+1}(A)$, then $T_m(A) = T_n(A)$ for all $m \geq n$.

Proof. ✓ Proof OK If $T_n = T_{n+1}$, then $S_{n+1} \subseteq T_n$. Left-multiplying any element of S_{n+1} by A^i gives an element of S_{n+2} , so $S_{n+2} \subseteq T_{n+1} = T_n$. By induction, $S_m \subseteq T_n$ for all $m > n$, hence $T_m = T_n$. □

Theorem 10.6 (Dimension growth). cumulativeSpan_finrank_strict_mono ✓ Stmt OK If $T_n(A) \neq T_{n+1}(A)$, then $\dim T_{n+1}(A) > \dim T_n(A)$.

Proof. ✓ Proof OK $T_n \leq T_{n+1}$ (Lemma 10.3) and $T_n \neq T_{n+1}$ imply strict submodule inclusion, hence strict rank increase. □

Lemma 10.7 (Word span characterises block injectivity). wordSpan_eq_top_iff_isNBlkInjective ✓ Stmt OK $S_L(A) = M_D(\mathbb{C})$ if and only if A is L -block injective.

Proof. ✓ Proof OK Both conditions assert that the span of all products of length L equals $M_D(\mathbb{C})$. □

10.2 Fitting decomposition

Definition 10.8 (Fitting decomposition). FittingDecomposition ✓ Stmt OK A *Fitting decomposition* of a linear endomorphism $f : V \rightarrow V$ on a finite-dimensional vector space over an algebraically closed field consists of:

1. f is nilpotent on the generalized 0-eigenspace,
2. f is invertible on each generalized μ -eigenspace for $\mu \neq 0$,
3. the generalized eigenspaces span V ,
4. the generalized eigenspaces are linearly independent.

Theorem 10.9 (Fitting decomposition exists). fittingDecomposition ✓ Stmt OK Every linear endomorphism on a finite-dimensional vector space over an algebraically closed field admits a Fitting decomposition.

Proof. ✓ Proof OK Nilpotency on the zero generalized eigenspace follows from the definition of generalized eigenspaces. Invertibility on nonzero generalized eigenspaces follows because $f - \mu$ is nilpotent there, so $f = \mu(1 - (1 - f/\mu))$ is invertible. Spanning and independence are standard results for generalized eigenspaces over algebraically closed fields. In [SPGWC10] and [Wol12, Lemma 6.3(b)], the Jordan normal form is used directly. The argument is phrased in terms of generalized eigenspaces instead. □

Theorem 10.10 (Nilpotent power bound). nilpotent_pow_eq_zero_of_finrank ✓ Stmt OK If f is nilpotent on a space of dimension n , then $f^n = 0$.

Proof. ✓ Proof OK A nilpotent endomorphism on an n -dimensional space has nilpotency index $\leq n$. □

10.3 Nonzero trace product

Theorem 10.11 (Cumulative span reaches $M_D(\mathbb{C})$ — explicit bound). cumulativeSpan_eq_top_of_isNormal_bound ✓ Stmt OK A is normal, then there exists $n \leq D^2$ with $T_n(A) = M_D(\mathbb{C})$.

Proof. ✓ Proof OK By Theorem 10.6, if $T_n \neq T_{n+1}$ then $\dim T_{n+1} > \dim T_n$. Since $\dim T_n \leq D^2$ (Lemma 10.4), after at most D^2 steps either $T_n = M_D(\mathbb{C})$ or T_n has stabilised. Stabilisation before reaching $M_D(\mathbb{C})$ contradicts normality (which requires $S_{L_0} = M_D(\mathbb{C})$ for some L_0 , hence $T_{L_0} = M_D(\mathbb{C})$). □

Theorem 10.12 (Cumulative span reaches $M_D(\mathbb{C})$). cumulativeSpan_eq_top ✓ Stmt OK If A is normal, then $T_{D^2}(A) = M_D(\mathbb{C})$.

Proof. ✓ Proof OK Take n from Theorem 10.11; since $n \leq D^2$ and T is monotone, $T_{D^2} = M_D(\mathbb{C})$. □

Theorem 10.13 (Nonzero trace product). exists_nonzero_trace_word ✓ Stmt OK If A is normal, then there exists a word w of length $\leq D^2$ such that $\text{tr}(A^w) \neq 0$. This is [SPGWC10, Lemma 1]. For the trace argument below, the nontrivial case is $D \geq 1$; when $D = 0$ the normality hypothesis is vacuous.

Proof. ✓ Proof OK By Theorem 10.12, $\mathbb{1} \in T_{D^2}(A)$, so $\mathbb{1}$ is a linear combination of word products of length $\leq D^2$. In the nontrivial case $D \geq 1$, at least one has nonzero trace (since $\text{tr}(\mathbb{1}) = D \neq 0$). $\square$

Theorem 10.14 (Cumulative span reaches $M_D(\mathbb{C})$ — sharp bound). cumulativeSpan_eq_top_of_isNormal_sharp ✓ Stmt OK If A is normal, then $T_{D^2-d'+1}(A) = M_D(\mathbb{C})$, where $d' = \dim S_1(A) = \text{kr}(A)$. This is the sharp form of [SPGWC10, Lemma 1]: the cumulative span already reaches $M_D(\mathbb{C})$ by index $D^2 - d' + 1 \leq D^2$.

Proof. ✓ Proof OK The key observation is that if $T_n(A) \neq T_{n+1}(A)$, the dimension strictly increases. Since $S_1(A) \subseteq T_1(A)$ and $\dim S_1(A) = d'$, we have $\dim T_1(A) \geq d'$. After at most $D^2 - d'$ additional strict-growth steps the dimension reaches $D^2 = \dim M_D(\mathbb{C})$. Stabilisation before reaching $M_D(\mathbb{C})$ contradicts normality. Hence $T_{D^2-d'+1}(A) = M_D(\mathbb{C})$. $\square$

Theorem 10.15 (Sharp nonzero trace product). exists_nonzero_trace_word_sharp ✓ Stmt OK If A is normal, then there exists a word w of length $\leq D^2 - \text{kr}(A) + 1$ such that $\text{tr}(A^w) \neq 0$. This is [SPGWC10, Lemma 1] with the sharp length bound.

Proof. ✓ Proof OK By Theorem 10.14, the identity $\mathbb{1}$ lies in $T_{D^2-d'+1}(A)$, hence is a linear combination of word products of length $\leq D^2 - d' + 1$. At least one such product has nonzero trace. $\square$

Theorem 10.16 (Sharp nonzero trace product — positive length). exists_nonzero_trace_word_sharp_pos ✓ Stmt OK If A is normal and $D \geq 2$, then there exists a word w of positive length $1 \leq |w| \leq D^2 - \text{kr}(A) + 1$ such that $\text{tr}(A^w) \neq 0$. This strengthens Theorem 10.15 by requiring $|w| \geq 1$, a feature needed for the blocking argument in the general Wielandt bound below.

Proof. ✓ Proof OK Define the *positive-level* cumulative span $V_n = \text{span}\{A^w : 1 \leq |w| \leq n\}$. For $D \geq 2$, the span $S_0(A) = \text{span}\{\mathbb{1}\}$ has dimension 1, so $V_1 \supseteq S_1(A)$ has dimension $\geq d' \geq 1$. The same stabilization-or-strict-growth dichotomy shows $V_{D^2-d'+1} = M_D(\mathbb{C})$: if V_n stabilises and is not $M_D(\mathbb{C})$, then V_n is closed under left-multiplication by each A^i , so $S_N(A) \subseteq V_n \neq M_D(\mathbb{C})$ for all $N \geq 1$, contradicting normality. Since $V_{D^2-d'+1} = M_D(\mathbb{C})$, the identity $\mathbb{1}$ lies in $V_{D^2-d'+1}$ and is therefore a linear combination of word products of positive length. At least one such product has nonzero trace (since $\text{tr}(\mathbb{1}) = D \geq 2 \neq 0$). $\square$

10.4 Eigenvector extraction

Theorem 10.17 (Nonzero trace implies eigenvector). exists_eigenvector_of_trace_ne_zero ✓ Stmt OK If $M \in M_D(\mathbb{C})$ has $\text{tr}(M) \neq 0$, then M has a nonzero eigenvalue $\mu \neq 0$ and a corresponding eigenvector $\varphi \neq 0$ with $M\varphi = \mu\varphi$.

Proof. ✓ Proof OK A matrix with nonzero trace has a nonzero eigenvalue (since trace = sum of eigenvalues counted with multiplicity). The Fitting decomposition (Theorem 10.9) ensures that the corresponding generalized eigenspace is nontrivial. On that generalized eigenspace, $M - \mu\mathbb{1}$ is nilpotent, so its kernel is nontrivial. Any nonzero vector in $\ker(M - \mu\mathbb{1})$ is therefore a genuine eigenvector with eigenvalue μ . $\square$

Theorem 10.18 (Eigenvalue extraction). exists_word_eigenvector ✓ Stmt OK If A is normal, then there exist a word w_0 with $|w_0| \leq D^2$, a nonzero scalar $\mu \neq 0$, and a nonzero vector $\varphi \neq 0$ such that $A^{w_0}\varphi = \mu\varphi$.

Proof. ✓ Proof OK By Theorem 10.13, there is a word w_0 with $\text{tr}(A^{w_0}) \neq 0$ and $|w_0| \leq D^2$. By Theorem 10.17, A^{w_0} has a nonzero eigenvector. $\square$

10.5 Eigenvector spreading

Definition 10.19 (Cumulative vector span). cumulativeVectorSpan ✓ Stmt OK Given an MPS tensor A and a vector $\varphi \in \mathbb{C}^D$, the *cumulative vector span* is

$$K_n(A, \varphi) = \text{span}_{\mathbb{C}}\{A^w\varphi : |w| \leq n\} \subseteq \mathbb{C}^D.$$

This is [SPGWC10, proof of Lemma 2(a)].

Theorem 10.20 (Vector span stabilisation). cumulativeVectorSpan_stable ✓ Stmt OK If $K_n(A, \varphi) = K_{n+1}(A, \varphi)$, then $K_m(A, \varphi) = K_n(A, \varphi)$ for all $m \geq n$.

Proof. ✓ Proof OK Same argument as Theorem 10.5: left-multiplication by A^i cannot escape the stabilised span. $\square$

Lemma 10.21 (Vector span from matrix span). cumulativeVectorSpan_eq_top_of_cumulativeSpan_eq_top ✓ Stmt OK If $T_N(A) = M_D(\mathbb{C})$ and $\varphi \neq 0$, then $K_N(A, \varphi) = \mathbb{C}^D$.

Proof. ✓ Proof OK Every matrix $M \in M_D(\mathbb{C})$ is in $T_N(A)$, so in particular every rank-one matrix sending φ to any target vector v is in T_N . Applying such matrices to φ recovers all of $\mathbb{C}^D$. $\square$

Theorem 10.22 (Eigenvector spreading). eigenvector_spreading ✓ Stmt OK Let A be a normal MPS tensor and let $A^{i_0}\varphi = \mu\varphi$ with $\mu \neq 0$ and $\varphi \neq 0$. Then $K_{D-1}(A, \varphi) = \mathbb{C}^D$. This is [SPGWC10, Lemma 2(a)].

Proof. ✓ Proof OK The vector span K_n is monotone and $\dim K_n \leq D$. If $K_n = K_{n+1}$ for some $n < D - 1$, the span stabilises (Theorem 10.20). But $T_{D^2}(A) = M_D(\mathbb{C})$ by Theorem 10.12, so Lemma 10.21 gives $K_{D^2}(A, \varphi) = \mathbb{C}^D$, contradicting early stabilisation. Hence $\dim K_n$ grows by ≥ 1 at each step, reaching D by step $D - 1$. $\square$

10.6 Proof of the cumulative Wielandt bound

Definition 10.23 (Wielandt analysis). WielandtAnalysis ✓ Stmt OK A *Wielandt analysis* for a normal MPS tensor A consists of: a word w_0 of length $\leq D^2$, a nonzero eigenvalue μ , and a nonzero eigenvector φ of the matrix A^{w_0} , such that $A^{w_0}\varphi = \mu\varphi$.

Theorem 10.24 (Wielandt analysis exists). wielandtAnalysis ✓ Stmt OK Every normal MPS tensor admits a *Wielandt analysis*.

Proof. ✓ Proof OK Immediate from Theorem 10.18. $\square$

Theorem 10.25 (Wielandt chain). wielandt_chain ✓ Stmt OK Let A be a normal MPS tensor with bond dimension $D \geq 1$. Then the following hold simultaneously:

1. $T_{D^2}(A) = M_D(\mathbb{C})$ (cumulative span stabilises);

2. there exists a word w_0 with $|w_0| \leq D^2$ and $\text{tr}(A^{w_0}) \neq 0$;
3. there exist $w_0, \mu \neq 0, \varphi \neq 0$ with $|w_0| \leq D^2$ and $A^{w_0}\varphi = \mu\varphi$;
4. for every $\varphi \neq 0$, $K_{D^2}(A, \varphi) = \mathbb{C}^D$.

Proof.  **Proof OK** Items (1)–(3) follow from earlier results (Theorems 10.12, 10.24). Item (4) combines these with the eigenvector spreading bound (Theorem 10.22). $\square$

Theorem 10.26 (Wielandt bound). isNormal_implies_cumulativeSpan ✓ Stmt OK If A is a normal MPS tensor with bond dimension D , then $T_{D^2}(A) = M_D(\mathbb{C})$. In particular, the cumulative span stabilises at index at most D^2 . This is the cumulative-span part of [SPGWC10, Theorem 1]. The paper’s theorem is stronger: via Lemma 2(b) it upgrades this to a fixed-length conclusion $S_N(A) = M_D(\mathbb{C})$, stated below as a separate statement.

Proof. ✓ Proof OK Immediate from Theorem 10.12.

Remark 10.27 (Explicit blocking-length estimates).  **Stmt OK** Theorem 10.26 gives the D^2 cumulative-span bound. The full-rank index bound $\iota(A) \leq (D^2 - d + 1) \cdot D^2$ (Theorem 1 case (1) of [SPGWC10]) is proved in Theorem 10.40 below. The sharp blocking-length estimates for injectivity (D^4) and bi-canonical form ($3D^5$) from [SPGWC10] are not proved here.

Definition 10.28 (One-step augmentation). oneStepAugment ✓ Stmt OK Let A be an MPS tensor with Kraus family $(A^i)_{i=1}^{d-1}$, and let $X \in M_D(\mathbb{C})$. The *one-step augmentation* of A by X is the tensor $\tilde{A}$ with physical dimension $d + 1$ whose first Kraus operator is X and whose remaining Kraus operators are those of A :

$$\tilde{A}^0 = X, \quad \tilde{A}^{i+1} = A^i.$$

Lemma 10.29 (Kraus operators lie in the one-step span). apply_mem_wordSpan_one ✓ Stmt OK Every Kraus operator A^i belongs to the one-step span $S_1(A)$.

Proof. ✓ Proof OK This is the length-one word with letter i .

Lemma 10.30 (Redundant one-step augmentation). `oneStepAugment_apply_mem_wordSpan_one` `wordSpan_oneStepAugment_one` `wordSpan_oneStepAugment_one`
 $X \in S_1(A)$, and let $\tilde{A}$ be the one-step augmentation of A by X . Then

$$S_n(\tilde{A}) = S_n(A)$$

for every $n \geq 0$.

Proof.  **Proof OK** The new generator X is already in $S_1(A)$, and each original generator remains in the augmented family. Hence the one-step spans agree. Multiplying word spans and arguing by induction on the word length gives equality at every length. $\square$

Lemma 10.31 (Normality and Kraus rank under one-step augmentation). `isNormal_oneStepAugment_of_mem_wordSpan_one` `kra`
 $X \in S_1(A)$, and let $\tilde{A}$ be the one-step augmentation of A by X . If A is normal, then $\tilde{A}$ is normal, and

$$\mathrm{kr}(\tilde{A}) = \mathrm{kr}(A).$$

Proof. ✓ Proof OK Since all exact word spans are unchanged by Lemma 10.30, eventual full word span for A gives eventual full word span for $\tilde{A}$. The equality of Kraus ranks is the equality of the dimensions of the common one-step span. $\square$

Theorem 10.32 (Wielandt case (2), invertible generator). [wordSpan_eq_top_of_isNormal_of_isUnit](#) [iIndex_le_of_isNormal_of_is](#)
A be a normal MPS tensor with bond dimension D , and suppose that some Kraus operator A^{i_0} is invertible. Then

$$S_{D^2 - \text{kr}(A) + 1}(A) = M_D(\mathbb{C}), \quad \iota(A) \leq D^2 - \text{kr}(A) + 1.$$

Proof. [✓ Proof OK](#) The dimensions of the exact word spans cannot decrease, because left multiplication by the invertible Kraus operator is injective. If this dimension growth stopped before the full matrix algebra was reached, the same argument applied at all later lengths would contradict normality. Since $\dim S_1(A) = \text{kr}(A)$ and $\dim M_D(\mathbb{C}) = D^2$, the full algebra is reached by length $D^2 - \text{kr}(A) + 1$. The index bound follows from the definition of $\iota(A)$. $\square$

Theorem 10.33 (Wielandt case (2), one-step invertible element). [wordSpan_eq_top_of_isPrimitivePaper_of_mem_wordSpan_one_of](#)
A be an MPS tensor with bond dimension $D \geq 1$, satisfying $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, and suppose that *A* is primitive with a positive-definite fixed point. If $S_1(A)$ contains an invertible matrix X , then

$$S_{D^2 - \text{kr}(A) + 1}(A) = M_D(\mathbb{C}).$$

In the terminology of Sanz-Pérez-García-Wolf-Cirac, this is the case where the first application space $S_1(A)$ contains an invertible operator. This is the exact word-span form of [SPGWC10, Theorem 1, case (2)].

Proof. [✓ Proof OK](#) Adjoin X as an extra first Kraus operator, obtaining $\tilde{A}$. Primitivity with a positive-definite fixed point and normalization give normality of A , hence normality of $\tilde{A}$; the Kraus rank and all exact word spans are unchanged. The first Kraus operator of $\tilde{A}$ is invertible, so Theorem 10.32 gives $S_{D^2 - \text{kr}(\tilde{A}) + 1}(\tilde{A}) = M_D(\mathbb{C})$. Transfer this equality back to A . $\square$

Corollary 10.34 (Wielandt case (2), index bound). [iIndex_le_of_mem_wordSpan_one_of_isUnit](#) [✓ Stmt OK](#) Under the hypotheses of Theorem 10.33,

$$\iota(A) \leq D^2 - \text{kr}(A) + 1,$$

where $\iota(A) = \min\{n : S_n(A) = M_D(\mathbb{C})\}$.

Proof. [✓ Proof OK](#) The displayed word-span equality gives an admissible value in the defining minimum for $\iota(A)$. $\square$

Corollary 10.35 (Wielandt case (2), invertible Kraus operator). [wordSpan_eq_top_of_isPrimitivePaper_of_isUnit](#) [iIndex_le_of](#)
A be an MPS tensor with bond dimension $D \geq 1$, satisfying $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, and suppose that *A* is primitive with a positive-definite fixed point. If some Kraus operator A^{i_0} is invertible, then

$$S_{D^2 - \text{kr}(A) + 1}(A) = M_D(\mathbb{C}), \quad \iota(A) \leq D^2 - \text{kr}(A) + 1.$$

Proof. [✓ Proof OK](#) Apply the one-step case to $X = A^{i_0}$, which belongs to $S_1(A)$ by Lemma 10.29. $\square$

Lemma 10.36 (Rank-one extraction from a non-invertible eigenvector). [vecMulVec_eigenvector_exact_wordSpan](#) [✓ Stmt OK](#)
A be a normal MPS tensor with bond dimension $D \geq 1$. Suppose that a Kraus operator A^{i_0} is non-invertible and that there exists a vector $\varphi \in \mathbb{C}^D$ and a nonzero scalar μ such that $A^{i_0}\varphi = \mu\varphi$. Then, for every $\psi \in \mathbb{C}^D$,

$$|\varphi\rangle\langle\psi| \in S_{D^2 - D + 1}(A).$$

Proof. ✓ Proof OK Decompose A^{i_0} into its nilpotent and nonzero generalized eigenspaces. On the nonzero part, multiplication by A^{i_0} is injective, so the rectangular spans grow strictly until they fill the whole range. The nilpotent index contributes the remaining length. Since the eigenvalue on φ is nonzero, eigenvector padding moves the resulting cumulative membership to the exact length $D^2 - D + 1$. $\square$

Theorem 10.37 (Wielandt case (3), one-step non-invertible element). wordSpan_eq_top_of_isPrimitivePaper_of_mem_wordSpan_o
A be an MPS tensor with bond dimension $D \geq 1$, satisfying $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, and suppose that A is primitive with a positive-definite fixed point. If $S_1(A)$ contains a non-invertible matrix X with a nonzero eigenvalue, then

$$S_{D^2}(A) = M_D(\mathbb{C}).$$

More explicitly, it is enough to have $X\varphi = \mu\varphi$ with $\varphi \neq 0$ and $\mu \neq 0$. In the terminology of Sanz-Pérez-García-Wolf-Cirac, this is the case where the first application space $S_1(A)$ contains a non-invertible operator with a nonzero eigenvalue. This is the exact word-span form of [SPGWC10, Theorem 1, case (3)].

Proof. ✓ Proof OK Adjoin X as an extra first Kraus operator. The augmented tensor is normal and has the same exact word spans as A . The eigenvector relation for X is now a one-generator eigenvector relation. The eigenvector-spreading argument gives full fixed-length vector span by length $D - 1$, and Lemma 10.36 gives the rank-one operators $|\varphi\rangle\langle\psi|$ in length $D^2 - D + 1$. Multiplying these two fixed-length spans gives S_{D^2} for the augmented tensor, hence for A . $\square$

Corollary 10.38 (Wielandt case (3), index bound). iIndex_le_sq_of_mem_wordSpan_one_of_noninvertible_eigenvector ✓ Stmt OK
the hypotheses of Theorem 10.37,

$$\iota(A) \leq D^2.$$

Proof. ✓ Proof OK The equality $S_{D^2}(A) = M_D(\mathbb{C})$ gives an admissible value in the defining minimum for $\iota(A)$. $\square$

Corollary 10.39 (Wielandt case (3), non-invertible Kraus operator). wordSpan_eq_top_of_isPrimitivePaper_of_noninvertible_e
A be an MPS tensor with bond dimension $D \geq 1$, satisfying $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, and suppose that A is primitive with a positive-definite fixed point. If some Kraus operator A^{i_0} is non-invertible and has an eigenvector $\varphi \neq 0$ with nonzero eigenvalue μ , then

$$S_{D^2}(A) = M_D(\mathbb{C}), \quad \iota(A) \leq D^2.$$

Proof. ✓ Proof OK Apply the one-step case to $X = A^{i_0}$, which belongs to $S_1(A)$ by Lemma 10.29. $\square$

Theorem 10.40 (Wielandt's inequality — general bound). iIndex_le_general_of_isPrimitivePaper ✓ Stmt OK *Let A be a normalized MPS tensor ($\sum_i (A^i)^\dagger A^i = \mathbb{1}$) that is primitive with a positive-definite fixed point. Then*

$$\iota(A) \leq (D^2 - \text{kr}(A) + 1) \cdot D^2,$$

where $\iota(A) = \min\{n : S_n(A) = M_D(\mathbb{C})\}$ is the full-Kraus-rank index. This is [SPGWC10, Theorem 1, case (1)].

Proof. ✓ Proof OK

1. $D = 1$: In this case $\iota(A) = 0$, so the bound holds trivially.

2. $D \geq 2$: By Theorem 10.16, there exists a *positive-length* word w with $|w| \leq D^2 - d' + 1$ and $\text{tr}(A^w) \neq 0$, where $d' = \text{kr}(A)$. The matrix A^w has nonzero trace, hence a nonzero eigenvalue μ and eigenvector φ (Theorem 10.17). Set $n = |w|$ and let $B = A^{[n]}$ be the blocked tensor. Then B is normal (Theorem 10.48), and the encoding of w as a single block index gives $B^{i_0} = A^w$.

- (a) If A^w is invertible (invertible case), $S_{D^2}(B) = M_D(\mathbb{C})$, whence $S_{D^2 \cdot n}(A) = M_D(\mathbb{C})$.
- (b) If A^w is non-invertible (non-invertible case), the eigenvector φ spans a proper subspace, and the Wielandt chain argument gives $S_{D^2}(B) = M_D(\mathbb{C})$, whence $S_{D^2 \cdot n}(A) = M_D(\mathbb{C})$.

In the invertible case, the invertible Kraus operator B^{i_0} allows direct dimension growth, giving $S_{D^2 - k_B + 1}(B) = M_D(\mathbb{C})$ where $k_B = \text{kr}(B)$. In the non-invertible case, the bound $S_{D^2}(B) = M_D(\mathbb{C})$ follows from tracking indices through the rank-one extraction argument of [SPGWC10, Lemma 2(b)]; the separate qualitative fixed-length conclusion stated below does not supply this explicit bound. In both cases $\iota(A) \leq D^2 \cdot n \leq D^2 \cdot (D^2 - d' + 1)$.

□

Remark 10.41 (Quantitative vs. qualitative content of Theorem 10.40). ✓ Stmt OK The quantitative bound $\iota(A) \leq (D^2 - d' + 1) \cdot D^2$ is cited from [SPGWC10, Theorem 1, case (1)]. The blueprint's own lemmas prove the qualitative conclusion used on the fundamental-theorem route, namely Theorem 10.53: there exists N with $S_N(A) = M_D(\mathbb{C})$. The quantitative bound is not on the fundamental-theorem critical path.

10.7 Fixed-length matrix spanning

The preceding sections establish the cumulative-spanning part of the Lemma 2(a) argument from [SPGWC10]: eigenvector spreading gives a cumulative vector span $K_{D-1}(A, \varphi) = \mathbb{C}^D$. The next stage is to pass to a span at one fixed length, namely to find n_0 such that $S_{n_0}(A) = M_D(\mathbb{C})$ (i.e. word products of *exactly* one length span all matrices).

Definition 10.42 (Fixed-length vector span). vectorSpreadSpan_eq_top_of_isPrimitivePaper_of_eigenvector ✓ Stmt OK The *fixed-length vector span* at length n is

$$H_n(A, \varphi) = \text{span}_{\mathbb{C}}\{A^w \varphi : |w| = n\} \subseteq \mathbb{C}^D.$$

This is $S_n(A)|\varphi\rangle$ in the notation of [SPGWC10].

Lemma 10.43 (Word span products). wordSpan_mul_le ✓ Stmt OK $S_m(A) \cdot S_n(A) \subseteq S_{m+n}(A)$: the product of word spans at lengths m and n is contained in the word span at length $m + n$.

Proof. ✓ Proof OK Every product $A^{w_1} A^{w_2}$ with $|w_1| = m$ and $|w_2| = n$ equals $A^{w_1 w_2}$ with $|w_1 w_2| = m + n$. □

Lemma 10.44 (Eigenvector padding). cumulativeVectorSpan_eq_vectorSpreadSpan_of_eigenvector ✓ Stmt OK Suppose $A^{i_0} \varphi = \mu \varphi$ with $\mu \neq 0$. Then for all n ,

$$K_n(A, \varphi) = H_n(A, \varphi).$$

In particular, if $K_n(A, \varphi) = \mathbb{C}^D$ then already $H_n(A, \varphi) = \mathbb{C}^D$.

Proof. ✓ Proof OK Any word w with $|w| \leq n$ can be padded to exact length n by appending $n - |w|$ copies of the index i_0 . The eigenvector relation gives $A^{w \cdot i_0^k} \varphi = \mu^k \cdot A^w \varphi$, so the padded vector is a nonzero scalar multiple of the original. Hence every generator of K_n lies in H_n . The reverse inclusion $H_n \leq K_n$ is immediate. $\square$

Theorem 10.45 (Fixed-length vector spanning implies matrix spanning). wordSpan_eq_top_of_vectorSpreadSpan_eq_top_of_rank

1. $H_n(A, \varphi) = \mathbb{C}^D$ (length- n word products applied to φ span all of $\mathbb{C}^D$), and
2. for each standard basis vector e_j , the rank-one operator $|\varphi\rangle\langle e_j|$ lies in $S_m(A)$.

Then $S_{n+m}(A) = M_D(\mathbb{C})$.

Proof. ✓ Proof OK Fix a matrix unit E_{ij} . By hypothesis (1), there exists $M_i \in S_n(A)$ with $M_i \varphi = e_i$. By hypothesis (2), $|\varphi\rangle\langle e_j| \in S_m(A)$. Their product satisfies

$$M_i \cdot |\varphi\rangle\langle e_j| = |M_i \varphi\rangle\langle e_j| = |e_i\rangle\langle e_j| = E_{ij},$$

and $E_{ij} \in S_{n+m}(A)$ by Lemma 10.43. Since the matrix units span $M_D(\mathbb{C})$, $S_{n+m}(A) = M_D(\mathbb{C})$. $\square$

Lemma 10.46 (Blocking transfer). wordSpan_eq_top_of_blockTensor_wordSpan_eq_top ✓ Stmt OK If the blocked tensor $A^{[L]}$ satisfies $S_n(A^{[L]}) = M_D(\mathbb{C})$, then $S_{nL}(A) = M_D(\mathbb{C})$.

Proof. ✓ Proof OK Each length- n word in the blocked alphabet $\{0, \dots, d^L - 1\}$ decodes to a length- nL word in the original alphabet $\{0, \dots, d - 1\}$, so $S_n(A^{[L]}) \subseteq S_{nL}(A)$. $\square$

Remark 10.47 (Relation with the rank-one extraction lemma). ✓ Stmt OK In [SPGWC10], Lemma 2(b) combines a rank-one extraction for fixed-length spans with a product argument turning fixed-length vector spanning into fixed-length matrix spanning. Theorem 10.45 is the second step. Theorem 10.52 supplies the rank-one hypothesis in blocked form, and Theorem 10.53 then gives a blocked, fixed-length word-span saturation result.

10.8 Blocked tensor and rectangular span

This section extends the Lemma 2(b) argument to the *blocked tensor* $A^{[L]}$ and develops the “rectangular span” used to produce rank-one elements.

Theorem 10.48 (Blocking preserves normality). isNormal_blockTensor_of_isNormal ✓ Stmt OK If A is normal and $L > 0$, then the blocked tensor $A^{[L]}$ is also normal.

Proof. ✓ Proof OK Choose N with $S_N(A) = M_D(\mathbb{C})$. Then $\mathbb{1} \in S_N(A)$. For any $M \in S_N(A)$, write $\mathbb{1} = \sum_r c_r A^{w_r}$ with each $|w_r| = N$. Then

$$M = M \cdot \mathbb{1} = \sum_r c_r M A^{w_r} \in S_N(A) \cdot S_N(A) \subseteq S_{2N}(A)$$

by Lemma 10.43. Iterating this inclusion gives $S_N(A) \subseteq S_{kN}(A)$ for every $k \geq 1$; taking $k = L$ yields $M_D(\mathbb{C}) = S_N(A) \subseteq S_{NL}(A)$. By Lemma 10.49, $S_{NL}(A) \subseteq S_N(A^{[L]})$, so $S_N(A^{[L]}) = M_D(\mathbb{C})$. $\square$

Lemma 10.49 (Chunking: word span embeds into blocked word span). wordSpan_le_wordSpan_blockTensor ✓ Stmt OK $S_{nL}(A) \subseteq S_n(A^{[L]})$ for all n, L .

Proof. ✓ Proof OK Every length- nL word in the original alphabet can be split into n consecutive chunks of length L , each of which is a single blocked Kraus operator. By induction on n , using the word span product lemma (Lemma 10.43). $\square$

Theorem 10.50 (Word eigenvector gives blocked eigenvector). blockTensor_single_eigenvector ✓ Stmt OK *If $A^{w_0}\varphi = \mu\varphi$ for a word w_0 of length L , then $(A^{[L]})^{j_0}\varphi = \mu\varphi$, where j_0 is the encoded blocked index corresponding to w_0 .*

Proof. ✓ Proof OK By definition, $(A^{[L]})^{j_0} = A^{w_0}$. $\square$

Theorem 10.51 (Blocked fixed-length matrix spanning). wielandt_blocked_assembly_complete ✓ Stmt OK *Suppose A is normal and $L > 0$, and we have:*

1. *a word w_0 of length L with eigenvector $A^{w_0}\varphi = \mu\varphi$ ($\mu \neq 0, \varphi \neq 0$),*
2. *a word τ_0 of length L with transpose eigenvector $(A^{\tau_0})^T\psi = \nu\psi$ ($\nu \neq 0, \psi \neq 0$),*
3. *a rank-one element $\varphi\psi^T \in S_m(A^{[L]})$ for some m .*

Then there exists N such that $S_N(A) = M_D(\mathbb{C})$.

Remark. The proof gives the explicit bound $N = ((D-1) + m + (D-1)) \cdot L$; the statement above gives only the existential conclusion.

Proof. ✓ Proof OK Apply Theorem 10.45 to the blocked tensor $A^{[L]}$, which is normal by Theorem 10.48. The eigenvector and transpose eigenvector are transferred to single-index eigenvectors of $A^{[L]}$ by Theorem 10.50. Transfer back to the original tensor via Lemma 10.46. $\square$

Theorem 10.52 (Rank-one extraction for blocked tensor). exists_rankOne_mem_wordSpan_blockTensor ✓ Stmt OK *Suppose A is normal and $N_0 \geq 1$ with $S_{N_0}(A) = M_D(\mathbb{C})$. Then there exist words σ_0, τ_0 of length N_0 , nonzero vectors φ, ψ , nonzero scalars μ, ν , and $m \in \mathbb{N}$ such that:*

1. $A^{\sigma_0}\varphi = \mu\varphi$,
2. $(A^{\tau_0})^T\psi = \nu\psi$,
3. $\varphi\psi^T \in S_m(A^{[N_0]})$.

Proof. ✓ Proof OK Set $B = A^{[N_0]}$. Since $S_{N_0}(A) = M_D(\mathbb{C})$, we have $S_1(B) = M_D(\mathbb{C})$ by definition of the blocked tensor. Hence $\mathbb{1} \in S_1(B)$, so in the nontrivial case $D \geq 1$ not all blocked generators can have trace zero. Choose j_0 with $\text{tr}(B^{j_0}) \neq 0$. By Theorem 10.17, B^{j_0} has a nonzero eigenvector φ with eigenvalue $\mu \neq 0$. Writing j_0 as the blocked index corresponding to a word σ_0 of length N_0 gives $A^{\sigma_0}\varphi = \mu\varphi$. Applying the same argument to the transpose blocked tensor gives a word τ_0 of length N_0 , a nonzero vector ψ , and $\nu \neq 0$ with $(A^{\tau_0})^T\psi = \nu\psi$. Since A is normal and $N_0 \geq 1$, the blocked tensor B is normal by Theorem 10.48. Therefore there exists M with $S_M(B) = M_D(\mathbb{C})$. In particular $\varphi\psi^T \in S_M(B)$. Taking $m = M$ proves (3). $\square$

Theorem 10.53 (Fixed-length word-span saturation via blocking). wielandt_lemma2b ✓ Stmt OK *If A is a normal MPS tensor with bond dimension D , then there exists N such that $S_N(A) = M_D(\mathbb{C})$. This gives the fixed-length spanning conclusion corresponding to [SPGWC10, Lemma 2(b)].*

Proof. ✓ Proof OK By normality, some N_0 has $S_{N_0}(A) = M_D(\mathbb{C})$. If $N_0 = 0$ we are done. Otherwise, Theorem 10.52 produces eigenvectors and a rank-one element $\varphi\psi^T \in S_m(A^{[N_0]})$. Theorem 10.51 then gives $S_{((D-1)+m+(D-1)) \cdot N_0}(A) = M_D(\mathbb{C})$. $\square$

Remark 10.54 (Blocking argument for fixed-length spanning). ✓ Stmt OK Theorem 10.53 is obtained by first extracting a rank-one operator in a blocked tensor (Theorem 10.52) and then applying the fixed-length matrix-spanning theorem (Theorem 10.51). Thus the fixed-length span of A is reached through a blocked auxiliary tensor rather than by a direct unblocked argument.

10.9 Primitive MPS tensors

Definition 10.55 (Primitive MPS tensor). IsPrimitiveMPS ✓ Stmt OK Assume $D > 0$. A tensor A together with a matrix $\rho \in M_D(\mathbb{C})$ is *primitive* if

$$\sum_i (A^i)^\dagger A^i = \mathbb{1}, \quad \rho \geq 0, \quad \rho \neq 0, \quad \mathcal{E}_A(\rho) = \rho,$$

and, if P is the fixed-point projection associated to ρ , then $\rho_{\text{spec}}(\mathcal{E}_A - P) < 1$. When the choice of ρ is irrelevant, we simply say that A is a primitive MPS tensor. This is weaker than the paper's notion, which requires $\rho > 0$; see Remark 10.61. The stronger positive-definite hypothesis is imposed separately in Theorems 10.62 and 10.73.

Theorem 10.56 (Primitive overlap convergence). overlap_tendsto_one ✓ Stmt OK If A is a primitive MPS tensor, then $O_{AA}(N) \rightarrow 1$.

Proof. ✓ Proof OK Immediate from the self-overlap convergence under a complementary spectral gap (Theorem 9.98). □

10.10 Primitivity and normality

Throughout this section, primitivity means the condition of Definition 10.55: the tensor is in TP gauge, it has a nonzero PSD fixed point, and the complementary map has spectral radius < 1 .

Theorem 10.57 (Primitive MPS tensors have nonzero word products). exists_nonzero_evalWord_of_isPrimitiveMPS ✓ Stmt OK If A is a primitive MPS tensor, then for every $n \geq 0$ there exists a word w of length exactly n with $A^w \neq 0$.

Proof. ✓ Proof OK The PSD fixed point ρ of $\mathcal{E}_A$ satisfies $\mathcal{E}_A^n(\rho) = \rho \neq 0$. Since $\mathcal{E}_A^n(\rho) = \sum_{|w|=n} A^w \rho (A^w)^\dagger$, at least one summand is nonzero, hence some $A^w \neq 0$. □

Theorem 10.58 (Iterated transfer does not vanish). transferMap_pow_ne_zero_of_isPrimitiveMPS ✓ Stmt OK If A is a primitive MPS tensor, then $\mathcal{E}_A^n \neq 0$ for all $n \geq 0$.

Proof. ✓ Proof OK By Theorem 10.57, there exists a word w of length n with $A^w \neq 0$, so $\mathcal{E}_A^n(\rho)$ has a nonzero summand. □

10.10.1 From primitivity to normality

The primitivity condition of Definition 10.55 combines TP normalization, a nonzero PSD fixed point, and a complementary spectral gap. It is connected to normality through the following results.

Theorem 10.59 (Fixed-point uniqueness from spectral gap). fixedPoint_unique ✓ Stmt OK If A is a primitive MPS tensor with PSD fixed point ρ , then every fixed point σ of $\mathcal{E}_A$ is proportional to ρ : $\sigma = \frac{\text{tr}(\sigma)}{\text{tr}(\rho)} \rho$.

Proof. ✓ Proof OK Set $\sigma' = \sigma - \frac{\text{tr}(\sigma)}{\text{tr}(\rho)}\rho$. Then $\text{tr}(\sigma') = 0$ and $(E - P_\rho)(\sigma') = \sigma'$. If $\sigma' \neq 0$, then 1 is an eigenvalue of $E - P_\rho$, contradicting the spectral gap $\rho_{\text{spec}}(E - P_\rho) < 1$. $\square$

Theorem 10.60 (Complement powers tend to zero). complement_pow_tendsto_zero ✓ Stmt OK If A is a primitive MPS tensor with PSD fixed point ρ , then $(\mathcal{E}_A - P_\rho)^n \rightarrow 0$ in operator norm.

Proof. ✓ Proof OK The spectral radius of $\mathcal{E}_A - P_\rho$ is strictly less than 1 by the spectral-gap hypothesis. Apply the general result that powers of an operator with spectral radius < 1 tend to zero. $\square$

Remark 10.61 (Primitivity: spectral gap vs. positive definiteness). ✓ Stmt OK The primitivity condition used in §10.10.1 has three parts: TP normalization $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, a nonzero PSD fixed point ρ , and the spectral gap $\rho_{\text{spec}}(\mathcal{E}_A - P_\rho) < 1$. The notion of “primitive” in [SPGWC10, Proposition 3] additionally requires ρ to be positive definite (full rank). Without positive definiteness, the spectral-gap condition alone does *not* imply irreducibility or normality. A counterexample is $D = 2$, $\rho = \text{diag}(1, 0)$.

With the additional hypothesis $\rho > 0$, Theorem 10.62 gives irreducibility. Together with Burnside’s theorem (Chapter 11, §11.4.1) and Theorem 10.71, one obtains

$$\text{primitive} + \rho \text{ positive definite} \Rightarrow \text{irreducible} \Rightarrow \text{alg}(A) = M_D(\mathbb{C}) \Rightarrow \text{normal}.$$

The last step uses the aperiodicity condition $\mathbb{1} \in S_1(A)$ from Remark 10.66. After restoring the positive-definite hypothesis, [SPGWC10, Proposition 3] identifies the paper’s primitive/strongly irreducible condition with eventual full Kraus rank, i.e. with our notion of normality.

10.10.2 Primitive implies irreducible

Theorem 10.62 (Primitive with positive-definite fixed point implies irreducible tensor). isIrreducibleTensor_of_isPrimitiv A is a primitive MPS tensor with PSD fixed point ρ and ρ is positive definite, then A is an irreducible tensor. This is part of [SPGWC10, Proposition 3] (see also [CPGSV17a, §2.3]).

Proof. ✓ Proof OK By contrapositive. Assume A has an invariant projection P ($P \neq 0$, $P \neq \mathbb{1}$, $(\mathbb{1} - P)A^i P = 0$ for all i). Set $\sigma = P\rho P$.

1. *Invariance of σ under $\mathcal{E}_A^n$:* the projection relation $(\mathbb{1} - P)A^i P = 0$ implies $(\mathbb{1} - P) \cdot \mathcal{E}_A^n(\sigma) = 0$ for all n (by induction on n).
2. *Convergence:* By Theorem 10.60, $\mathcal{E}_A^n(\sigma) \rightarrow \frac{\text{tr}(\sigma)}{\text{tr}(\rho)}\rho$. Hence $(\mathbb{1} - P) \cdot \frac{\text{tr}(\sigma)}{\text{tr}(\rho)}\rho = 0$.
3. *Nonzero scalar:* Since $P \neq 0$ and ρ is positive definite, $\text{tr}(P\rho P) > 0$, so $\frac{\text{tr}(\sigma)}{\text{tr}(\rho)} \neq 0$. Therefore $(\mathbb{1} - P)\rho = 0$.
4. *Contradiction:* Since ρ is positive definite, it is a unit in $M_D(\mathbb{C})$. From $(\mathbb{1} - P)\rho = 0$ we get $\mathbb{1} - P = 0$, i.e. $P = \mathbb{1}$, contradicting $P \neq \mathbb{1}$.

$\square$

10.10.3 From primitivity to strong irreducibility and normality

The results in this subsection remove the aperiodicity assumption from the normality conclusions. They connect the spectral-gap primitivity of Definition 10.55 directly to normality (eventually full Kraus rank), using the convergence of the transfer-map iterates to the projection onto the fixed-point subspace.

Theorem 10.63 (Primitive MPS with positive-definite fixed point has irreducible transfer map). [isIrreducibleMap_of_isPrimitiveMPS_of_posDef](#) ✓ Stmt OK If A is a primitive MPS tensor with positive-definite fixed point ρ , then the transfer map $\mathcal{E}_A$ is irreducible.

The proof uses fixed-point uniqueness (Theorem 10.59): every PSD fixed point is proportional to ρ , so Wolf's criterion for irreducibility (positive-definite fixed point implies irreducibility) applies directly.

Proof. ✓ Proof OK By Theorem 10.59, if $\sigma \geq 0$ and $\mathcal{E}_A(\sigma) = \sigma$, then $\sigma = \frac{\text{tr } \sigma}{\text{tr } \rho} \rho$. Since $\rho > 0$, Wolf's uniqueness criterion for irreducibility applies. $\square$

Theorem 10.64 (Primitive MPS with positive-definite fixed point is strongly irreducible). [isStronglyIrreduciblePaper_of_isPrimitiveMPS_of_posDef](#) ✓ Stmt OK If A is a primitive MPS tensor with positive-definite fixed point ρ , then A is strongly irreducible in the sense of [SPGWC10, Proposition 3(c)]: $\mathcal{E}_A$ is irreducible, the fixed point ρ is positive definite, and the peripheral spectrum of $\mathcal{E}_A$ is $\{1\}$.

Proof. ✓ Proof OK Combine Theorem 10.63 (irreducibility) with the peripheral-spectrum primitivity from the spectral-gap hypothesis (Theorem 10.56). $\square$

Theorem 10.65 (Primitive MPS with positive-definite fixed point is normal). [isNormal_of_isPrimitiveMPS_with_posDef](#) ✓ Stmt OK If A is a primitive MPS tensor with positive-definite fixed point ρ , then A is normal (has eventually full Kraus rank). No aperiodicity assumption is needed.

This is the key connection between the spectral-gap primitivity of Definition 10.55 and normality; it removes the need for the aperiodicity-based argument.

Proof. ✓ Proof OK By Theorem 10.64, A is strongly irreducible, i.e. primitive in Wolf's sense. Under normalization, Theorem 9.64 identifies this primitivity condition directly with normality (equivalently, eventual full Kraus rank). $\square$

10.11 Cumulative span to word span

Cumulative spanning ($T_N(A) = M_D(\mathbb{C})$, the union of word spans of all lengths $\leq N$) is weaker than normality ($S_L(A) = M_D(\mathbb{C})$ for a single length L). An aperiodicity hypothesis upgrades cumulative spanning to a single-length word span.

Remark 10.66 (Aperiodicity condition). ✓ Stmt OK We say A satisfies the *aperiodicity condition* if $\mathbb{1} \in S_1(A)$, i.e. the identity matrix lies in the span of the Kraus operators $\{A^i\}$. This is an additional hypothesis. The one-sided TP normalization $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ does *not* imply $\mathbb{1} \in S_1(A)$; it only gives a quadratic identity among the Kraus operators. The aperiodicity condition is kept explicit in this section for generality. For the paper-style primitive condition (peripheral-spectrum primitivity together with a positive-definite fixed point), normality follows directly from Wolf's primitivity equivalence (Theorem 9.64) without passing through aperiodicity.

Remark 10.67 (Counterexample: cumulative spanning does not imply normality). ✓ Stmt OK In $M_2(\mathbb{C})$, let $A^0 = E_{12}$ and $A^1 = E_{21}$ (the off-diagonal matrix units). Then $\text{alg}(A) = M_2(\mathbb{C})$ and $T_2(A) = M_2(\mathbb{C})$, but the word spans alternate: $S_n(A)$ contains only off-diagonal matrices for odd n and only diagonal matrices for even n . Hence $S_n(A) \neq M_2(\mathbb{C})$ for every n , and A is not normal. The aperiodicity condition $\mathbb{1} \in S_1(A)$ fails: $S_1(A) = \text{span}_{\mathbb{C}}\{E_{12}, E_{21}\}$.

Theorem 10.68 (Word span monotonicity from aperiodicity). [wordSpan_mono_succ_of_one_mem_wordSpan_one](#) ✓ Stmt OK If $\mathbb{1} \in S_1(A)$, then $S_n(A) \leq S_{n+1}(A)$ for all n .

Proof. ✓ Proof OK For any $M \in S_n(A)$, $M = M \cdot \mathbb{1} \in S_n(A) \cdot S_1(A) \subseteq S_{n+1}(A)$ by Lemma 10.43. □

Theorem 10.69 (Cumulative span equals word span under aperiodicity). cumulativeSpan_eq_wordSpan_of_one_mem_wordSpan_one
 $\mathbb{1} \in S_1(A)$, then $T_n(A) = S_n(A)$ for all n .

Proof. ✓ Proof OK By Theorem 10.68, the word spans are monotone: $S_0 \leq S_1 \leq \dots$. The supremum $T_n = S_0 + \dots + S_n$ therefore equals the largest term S_n . □

Theorem 10.70 (Cumulative spanning plus aperiodicity implies normality). isNormal_of_cumulativeSpan_eq_top_of_aperiodic
 $T_N(A) = M_D(\mathbb{C})$ and $\mathbb{1} \in S_1(A)$, then A is normal.

Proof. ✓ Proof OK By Theorem 10.69, $S_N(A) = T_N(A) = M_D(\mathbb{C})$. □

Theorem 10.71 (Algebra span plus aperiodicity implies normality). isNormal_of_algSpan_eq_top_of_aperiodic ✓ Stmt OK If
 $\text{alg}(A) = M_D(\mathbb{C})$ and $\mathbb{1} \in S_1(A)$, then A is normal.

Proof. ✓ Proof OK Since $\text{alg}(A) = M_D(\mathbb{C})$, there exists N with $T_N(A) = M_D(\mathbb{C})$ (by the ascending chain condition on finite-dimensional subspaces). Apply Theorem 10.70. □

Theorem 10.72 (Irreducible tensor plus aperiodicity implies normality). isNormal_of_isIrreducibleTensor_of_aperiodic ✓ S
 A is an irreducible tensor, $D > 0$, and $\mathbb{1} \in S_1(A)$, then A is normal. This assembles the full chain:

$$\text{irreducible tensor} \xrightarrow{11.21} \text{irreducible action} \xrightarrow{\text{Burnside}} \text{alg}(A) = M_D(\mathbb{C}) \xrightarrow{10.71} \text{normal}.$$

This corresponds to the irreducibility-to-normality direction of [SPGWC10, Proposition 3]. For the Burnside step, we appeal forward to Theorems 11.21 and 11.22; this is a presentation-level forward dependency, but the mathematical dependency is acyclic.

Proof. ✓ Proof OK By Theorem 11.21, A acts irreducibly on $\mathbb{C}^D$. By Theorem 11.22, $\text{alg}(A) = M_D(\mathbb{C})$. By Theorem 10.71 with the aperiodicity hypothesis, A is normal. □

Theorem 10.73 (Primitive + positive definite fixed point imply normality). isNormal_of_isPrimitiveMPS_of_posDef ✓ Stmt OK
 A is a primitive MPS tensor with fixed point ρ and ρ is positive definite, then A is normal. No aperiodicity assumption is needed.

Proof. ✓ Proof OK By Theorem 10.64, A is strongly irreducible, i.e. primitive in Wolf's sense. Under normalization, Theorem 9.64 therefore gives normality directly, hence eventual full Kraus rank. □

Remark 10.74 (Normal canonical form avoids the aperiodicity step). The aperiodicity results in §10.11 are needed only for the block-injective argument through Definition 2.26, where one upgrades cumulative spanning to eventual block injectivity. The normal-canonical theory used later in Chapters 11–13 does not pass through this step: it works directly with irreducible TP-normalized blocks whose blocked transfer maps are primitive and, once the corresponding weighted block data are available, places them in normal canonical form in the sense of [CPGSV17a]. There are analogous cumulative-span lemmas for blocked irreducible tensors and eigenvector-spreading arguments, but they are auxiliary here and we do not list them separately.

Chapter 11

Canonical Form Reduction

This chapter reduces a general MPS tensor to block-diagonal data via iterated invariant-subspace decomposition. Two reduction schemes are relevant. The block-injective reduction aims at injective TP-normalized blocks and the canonical form conditions of Chapter 12. The normal-canonical reduction uses a weighted family of irreducible TP-normalized blocks with primitive transfer maps and pairwise distinct nonzero weight moduli, in the sense of [CPGSV17a]. The Perron–Frobenius eigenvector existence and TP-gauge construction are in Chapter 8; this chapter uses them to produce left-canonical normalization data and the normal canonical form data. The reduction draws on [PGVWC07], [CPGSV21], and [Wol12, Ch. 6].

11.1 Block-diagonal construction

Definition 11.1 (Block-diagonal tensor from blocks). [toTensorFromBlocks](#) ✓ Stmt OK Recall the block-diagonal tensor of Definition 2.34. Given r blocks with bond dimensions $(D_k)_{k=1}^r$, per-block MPS tensors $\{A_k^i\}$, and scaling factors $(\mu_k)_{k=1}^r$, we write $\bigoplus_{k=1}^r \mu_k A_k$ for the tensor of total bond dimension $D = \sum_k D_k$ whose Kraus operators are $A^i = \bigoplus_{k=1}^r \mu_k A_k^i$, obtained by placing the scaled block Kraus operators on the diagonal.

Theorem 11.2 (Per-block single-block FT). [fundamentalTheorem_multiBlock_blocks](#) ✓ Stmt OK *If all block tensors A_k are injective and the per-block same-MPV condition holds (A_k and B_k generate the same MPV family for each k), then each pair (A_k, B_k) is gauge equivalent.*

Proof. ✓ Proof OK Apply the single-block fundamental theorem (Theorem 4.11) to each block independently. □

Theorem 11.3 (Per-block same MPV implies global same MPV). [sameMPV_toTensorFromBlocks_of_blockSameMPV](#) ✓ Stmt OK *If A_k and B_k generate the same MPV family for each block k , then the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ also generate the same MPV family.*

Proof. ✓ Proof OK By the MPV decomposition (Theorem 2.36), each MPV is $\sum_k \mu_k^N V^{(N)}(A_k)_\sigma$. Per-block equality of MPV coefficients gives global equality. □

Theorem 11.4 (Global multi-block fundamental theorem). [fundamentalTheorem_multiBlock_global](#) ✓ Stmt OK *If each block A_k is injective and each pair (A_k, B_k) generates the same MPV family, then the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ are gauge equivalent.*

Proof. ✓ Proof OK Apply the per-block single-block fundamental theorem (Theorem 11.2) to obtain blockwise gauge equivalences from injectivity and equality of MPV families, then combine them into a global gauge for the block-diagonal tensors. $\square$

11.2 Transfer map normalization

Theorem 11.5 (Scaling preserves injectivity). isInjective_smul ✓ Stmt OK If A is injective and $\zeta \neq 0$, then ζA is injective.

Proof. ✓ Proof OK Scaling by a nonzero scalar preserves linear independence and hence spanning. $\square$

Theorem 11.6 (Transfer map under scaling). transferMap_smul ✓ Stmt OK $\mathcal{E}_{\zeta A}(X) = |\zeta|^2 \mathcal{E}_A(X)$.

Proof. ✓ Proof OK Each summand: $(\zeta A^i)X(\zeta A^i)^\dagger = |\zeta|^2 A^i X(A^i)^\dagger$. $\square$

Theorem 11.7 (Phase-scaling preserves normalization). leftCanonical_smul_of_norm_one ✓ Stmt OK If $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ and $|\zeta| = 1$, then $\sum_i (\zeta A^i)^\dagger (\zeta A^i) = \mathbb{1}$.

Proof. ✓ Proof OK $(\zeta A^i)^\dagger (\zeta A^i) = \bar{\zeta} \zeta (A^i)^\dagger A^i = (A^i)^\dagger A^i$, since $\bar{\zeta} \zeta = |\zeta|^2 = 1$. $\square$

Theorem 11.8 (MPV under block normalization). mpv_toTensorFromBlocks_normalize ✓ Stmt OK For any nonzero scaling factors μ_k , set $\eta_k := \mu_k / |\mu_k|$ and $B_k := |\mu_k| A_k$. Then $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \eta_k B_k$ generate the same MPV family. Equivalently, the k -th block contribution factors as

$$\mu_k^N V^{(N)}(A_k)_\sigma = \eta_k^N |\mu_k|^N V^{(N)}(A_k)_\sigma,$$

so the modulus contributes only through $|\mu_k|^N$, while the remaining factor η_k is a unit-modulus phase.

Proof. ✓ Proof OK By the MPV decomposition (Theorem 2.36), the k -th contribution of $\bigoplus_k \mu_k A_k$ is $\mu_k^N V^{(N)}(A_k)_\sigma$. For $B_k = |\mu_k| A_k$, the MPV-scaling lemma (Lemma 2.17) gives $V^{(N)}(B_k)_\sigma = |\mu_k|^N V^{(N)}(A_k)_\sigma$. Hence $\eta_k^N V^{(N)}(B_k)_\sigma = \eta_k^N |\mu_k|^N V^{(N)}(A_k)_\sigma = \mu_k^N V^{(N)}(A_k)_\sigma$ for every block, so the total MPV families agree. $\square$

Remark 11.9 (Why blockwise cancellation cannot be separated without normalization). The implication

$$\mathcal{V}(\bigoplus_{k=1}^r \mu_k A_k) = \mathcal{V}(\bigoplus_{k=1}^r \mu_k B_k) \implies \forall k, \mathcal{V}(A_k) = \mathcal{V}(B_k)$$

is false if one assumes only that the weights μ_k are pairwise distinct and nonzero. Already for $r = 2$, with $\mu = (1, 2)$ and scalar 1×1 blocks,

$$A = (1, 3/2), \quad B = (3, 1/2),$$

one has

$$1^N(1^N - 3^N) + 2^N((3/2)^N - (1/2)^N) = 0$$

for every $N \geq 1$, so the two direct sums generate the same MPV family, even though the first blocks already disagree at $N = 1$. The reason is exactly the rescaling freedom described above: a nonzero scalar may be moved from the weight μ_k into the block tensor A_k , or conversely, without changing the contribution $\mu_k^N V^{(N)}(A_k)_\sigma$. Thus distinct nonzero weights alone do not identify the individual blocks.

The canonical-form hypotheses remove this ambiguity. If each block is first placed in the normalization $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$, then further rescaling is restricted to unit-modulus phases, and the strict ordering $|\mu_1| > \dots > |\mu_r|$ used in [PGVWC07, CPGSV21] prevents trading a nontrivial modulus between the weight and the block tensor.

11.3 Invariant subspace decomposition

Theorem 11.10 (Unitary conjugation preserves the MPV family). [sameMPV_conj_unitary](#) [✓ Stmt OK](#) If U is a unitary matrix, then A and $U^\dagger A U$ generate the same MPV family.

Proof. [✓ Proof OK](#) By cyclicity of the trace: $\text{tr}((U^\dagger A^{i_1} U) \cdots (U^\dagger A^{i_N} U)) = \text{tr}(A^{i_1} \cdots A^{i_N})$. $\square$

Definition 11.11 (Support projection). [supportProj](#) [✓ Stmt OK](#) For a positive semi-definite matrix $\rho \geq 0$, the *support projection* P is the orthogonal projection onto the range of ρ . It is constructed via the spectral decomposition: if $\rho = U \text{diag}(\lambda_1, \dots, \lambda_D) U^\dagger$, then $P = U \text{diag}(\mathbf{1}_{\lambda_1 > 0}, \dots, \mathbf{1}_{\lambda_D > 0}) U^\dagger$, where $\mathbf{1}_{\lambda_j > 0}$ is 1 if $\lambda_j > 0$ and 0 otherwise.

Definition 11.12 (Invariant projection predicate). [HasInvariantProj](#) [✓ Stmt OK](#) An MPS tensor A has an *invariant projection* if there exists an orthogonal projection P with $P \neq 0$, $P \neq \mathbb{1}$, and $(\mathbb{1} - P)A^i P = 0$ for all i . A tensor is *irreducible* if it has no nontrivial invariant projection (Definition 11.15).

Theorem 11.13 (PSD fixed point gives invariant projection). [lowerZero_of_posSemidef_fixedPoint](#) [✓ Stmt OK](#) Let $\rho \geq 0$ be a fixed point of $\mathcal{E}_A$, and let P be the support projection of ρ . Then for all i , $(\mathbb{1} - P)A^i P = 0$: the range of P is invariant under all A^i .

Proof. [✓ Proof OK](#) From $\mathcal{E}_A(\rho) = \rho$ and positivity of each summand $A^i \rho (A^i)^\dagger$, $(\mathbb{1} - P)A^i \rho (A^i)^\dagger (\mathbb{1} - P) = 0$ for each i . Since ρ is supported on the range of P , this forces $(\mathbb{1} - P)A^i P = 0$.

This is the finite-dimensional version of [Wol12, Prop. 6.10] (“Stationary subspaces”: the support projection of any stationary state is sub-harmonic, i.e. invariant under the channel). The equivalence between irreducibility and absence of nontrivial invariant projections is [Wol12, Thm. 6.2]; see also [CPGSV21, §IV.A.1]. $\square$

Theorem 11.14 (Two-block decomposition). [exists_twoBlock_decomp_of_posSemidef_fixedPoint_strict](#) [✓ Stmt OK](#) If the transfer map has a PSD fixed point ρ that is not positive definite (i.e. ρ has a nontrivial kernel), then there exist MPS tensors A_1, A_2 of smaller bond dimensions such that the two-block tensor $A_1 \oplus A_2$ generates the same MPV family as A : $\mathcal{V}(A) = \mathcal{V}(A_1 \oplus A_2)$.

Proof. [✓ Proof OK](#) The support projection P of ρ is neither 0 nor $\mathbb{1}$ (since $\rho \neq 0$ and ρ is not PD). Theorem 11.13 gives $(\mathbb{1} - P)A^i P = 0$ for all i . A unitary change of basis that block-diagonalizes P puts A^i in upper-triangular block form, with diagonal blocks B_{11}^i and B_{22}^i . Theorem 11.10 ensures the MPV is preserved by the conjugation. The trace of the upper-triangular product equals the sum of the diagonal-block traces, so the block-diagonal tensor obtained by discarding the off-diagonal blocks has the same MPV coefficients as the conjugated tensor, and hence as A . $\square$

11.4 Iterated reduction

Definition 11.15 (Irreducible tensor). [IsIrreducibleTensor](#) [✓ Stmt OK](#) An MPS tensor A is *irreducible* if it has no nontrivial invariant projection.

Theorem 11.16 (Irreducible block decomposition). [exists_irreducible_blockDecomp](#) [✓ Stmt OK](#) Every MPS tensor A admits a block-diagonal tensor $\bigoplus_{k=1}^r A_k$ where each A_k is irreducible and $\mathcal{V}(A) = \mathcal{V}(\bigoplus_k A_k)$.

Proof. ✓ Proof OK By strong induction on the bond dimension D . If A is already irreducible, take $r = 1$. Otherwise, by Definition 11.15, A has a nontrivial invariant projection P . Diagonalizing P yields the same two-block upper-triangular splitting used in Theorem 11.14; the block-diagonal tensor obtained by discarding the off-diagonal part has the same MPV family as A , and the two diagonal blocks have strictly smaller bond dimensions. Iterating this splitting produces a block-diagonal tensor $\bigoplus_k A_k$ whose blocks are irreducible. Apply the induction hypothesis to each block and concatenate. (If one prefers a positive-eigenvector argument without assuming TP normalization, the correct replacement for the channel-specific Cesàro theorem is Theorem 8.17; the induction itself needs only the invariant projection.)

Both [PGVWC07] (Theorem 4, “TI canonical form”) and [CPGSV21] (§IV.A.1) obtain the block decomposition by the same invariant-projection splitting used here. An alternative derivation, developed in [Wol12, Thm. 6.14], characterises the fixed-point set of a quantum channel as a $*$ -algebra and then applies the Wedderburn–Artin decomposition ([Wol12, Eq. (1.39)]) to read off the block structure directly. The block decomposition is obtained by iterating the invariant-projection splitting and using strong induction on D to ensure termination. $\square$

Theorem 11.17 (Unitary diagonal positive fixed point in TP gauge). exists_unitary_diag_posDef_fixedPoint_of_TP_of_isIrredu
A be an irreducible MPS tensor satisfying $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, and assume $D > 0$. Then there exist a unitary U and a diagonal positive definite matrix Λ such that, for $B^i := U^\dagger A^i U$,

$$\sum_i (B^i)^\dagger B^i = \mathbb{1} \quad \text{and} \quad \mathcal{E}_B(\Lambda) = \Lambda.$$

Proof. ✓ Proof OK The TP hypothesis makes $\mathcal{E}_A$ a channel, so it has a nonzero PSD fixed point. The irreducible tensor condition implies that $\mathcal{E}_A$ is irreducible as a CP map, and Theorem 8.3 upgrades the fixed point to a positive definite one. The spectral theorem diagonalizes this matrix by a unitary conjugation, and unitary conjugation preserves the TP normalization. $\square$

11.4.1 From irreducibility to full spanning via Burnside’s theorem

The irreducibility condition (Definition 11.15) is formulated in terms of invariant *projections*. For Burnside-style arguments it is more natural to work with invariant *subspaces* of $\mathbb{C}^D$ under the Kraus operators.

Definition 11.18 (Algebra span). algSpan ✓ Stmt OK For an MPS tensor A , let $\text{alg}(A)$ be the unital $\mathbb{C}$ -subalgebra of $M_D(\mathbb{C})$ generated by the matrices $\{A^i\}$:

$$\text{alg}(A) = \mathbb{C}\langle A^i : i = 0, \dots, d-1 \rangle \subseteq M_D(\mathbb{C}).$$

Definition 11.19 (Invariant submodule). IsInvariantSubmodule ✓ Stmt OK A subspace $W \leq \mathbb{C}^D$ is *A-invariant* if $A^i W \subseteq W$ for every i .

Definition 11.20 (Irreducible action). IsIrreducibleAction ✓ Stmt OK The Kraus operators of A act *irreducibly* on $\mathbb{C}^D$ if every A -invariant subspace is trivial, i.e. equal to 0 or to $\mathbb{C}^D$.

Theorem 11.21 (Irreducible tensor implies irreducible action). isIrreducibleAction_of_isIrreducibleTensor ✓ Stmt OK *If A is irreducible as a tensor (Definition 11.15), then the Kraus operators act irreducibly on $\mathbb{C}^D$ (Definition 11.20).*

Proof. ✓ Proof OK Let $W \leq \mathbb{C}^D$ be an A -invariant subspace. In finite dimension the orthogonal projection P onto W exists. Invariance of W implies $(\mathbb{1} - P)A^i P = 0$ for all i . Thus P is a nontrivial invariant projection unless $W = 0$ or $W = \mathbb{C}^D$, contradicting irreducibility of the tensor. $\square$

Theorem 11.22 (Burnside’s theorem for Kraus algebras). [burnside_matrix](#) [✓ Stmt OK](#) Assume $D > 0$. If A acts irreducibly on $\mathbb{C}^D$, then the generated unital subalgebra is the full matrix algebra:

$$\text{alg}(A) = M_D(\mathbb{C}).$$

This is the complex finite-dimensional case of Burnside’s theorem (equivalently, Jacobson’s density theorem); see [\[Jac09\]](#).

Proof. [✓ Proof OK](#) Transport $\text{alg}(A)$ along the algebra equivalence $M_D(\mathbb{C}) \simeq_{\mathbb{C}} \text{End}_{\mathbb{C}}(\mathbb{C}^D)$. The irreducible-action hypothesis makes $\mathbb{C}^D$ a simple module for this algebra. Over the algebraically closed field $\mathbb{C}$, Schur’s lemma identifies the endomorphisms commuting with the action with scalars, and Jacobson density then forces the action map to be surjective onto all of $\text{End}_{\mathbb{C}}(\mathbb{C}^D)$. $\square$

Remark 11.23. [✓ Stmt OK](#) This Burnside/Jacobson-density step is a substantial external algebraic ingredient in the chapter. We use it here as the finite-dimensional complex Burnside theorem, namely the statement that an irreducible matrix algebra action already generates the full endomorphism algebra.

Theorem 11.24 (Irreducible action implies eventual cumulative spanning). [exists_cumulativeSpan_eq_top_of_isIrreducibleA](#) $D > 0$. If A acts irreducibly on $\mathbb{C}^D$, then there exists N such that $T_N(A) = M_D(\mathbb{C})$.

Proof. [✓ Proof OK](#) By Theorem 11.22, $\text{alg}(A) = M_D(\mathbb{C})$. Every element of the algebra span is a linear combination of word products, hence belongs to $T_n(A)$ for some n . Since $M_D(\mathbb{C})$ is finite-dimensional, the ascending chain $T_0(A) \leq T_1(A) \leq \dots$ stabilizes, so one can choose a uniform N with $T_N(A) = M_D(\mathbb{C})$. $\square$

11.5 Proportional single-block theorem

Theorem 11.25 (Proportional MPV implies gauge-phase equivalence). [gaugePhaseEquiv_of_proportionalMPV2_of_overlap_tends](#) A and B be injective MPS tensors satisfying $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ and $\sum_i (B^i)^\dagger B^i = \mathbb{1}$. If A and B generate proportional MPV families $(V^{(N)}(A))_\sigma = c_N V^{(N)}(B)_\sigma$ for some $c_N \in \mathbb{C}$ and $O_{AA}(N) \rightarrow 1$, $O_{BB}(N) \rightarrow 1$, then A and B are gauge-phase equivalent.

Proof. [✓ Proof OK](#) By contradiction. If A and B are not gauge-phase equivalent, then $O_{AB}(N) \rightarrow 0$ (Theorem 9.25). From the proportionality hypothesis, $O_{AB}(N) = c_N O_{BB}(N)$. Since $O_{BB}(N) \rightarrow 1$, it follows that $c_N \rightarrow 0$. But $O_{AA}(N) = |c_N|^2 O_{BB}(N) \rightarrow 0$, contradicting $O_{AA}(N) \rightarrow 1$.

In [\[PGVWC07\]](#), the proportional case is handled as part of the canonical form argument. Here it is obtained directly from the overlap-decay theorem. $\square$

Theorem 11.26 (Proportional MPV + primitive implies gauge-phase equivalence). [gaugePhaseEquiv_of_proportionalMPV2_of](#) A and B be MPS tensors with primitive transfer maps E_A, E_B and positive definite fixed points ρ_A, ρ_B . If A and B generate proportional MPV families, then A and B are gauge-phase equivalent. This strengthens Theorem 11.25: primitivity supplies left-canonicity, irreducibility, and overlap convergence, reducing the hypothesis count from seven to four.

Proof. [✓ Proof OK](#) From primitivity, extract a left-canonical normalization ($\sum_i (A^i)^\dagger A^i = \mathbb{1}$), overlap convergence ($O_{AA}(N) \rightarrow 1$), and irreducibility (Theorem 10.62). Apply Theorem 11.25. $\square$

11.6 Canonical form from primitivity

The canonical form conditions (Definition 12.16) include an explicit hypothesis that the self-overlap $O_{A_k A_k}(N) \rightarrow 1$ for each block. The first theorem below derives this from blockwise primitive fixed-point data. The second derives the same conclusion from per-block peripheral-spectrum primitivity.

Theorem 11.27 (Canonical form from primitive fixed-point data). isCanonicalForm_of_primitive ✓ Stmt OK Let $(\mu_k, A_k)_{k=1}^r$ be a collection of block tensors and scaling factors satisfying:

1. each A_k is injective,
2. each A_k satisfies the TP normalization $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$,
3. $|\mu_1| > \dots > |\mu_r|$ (strictly decreasing),
4. all $\mu_k \neq 0$,
5. each block has positive bond dimension,
6. each block admits a primitive fixed point.

Then (μ_k, A_k) satisfies the canonical form conditions.

Proof. ✓ Proof OK All canonical-form clauses except the overlap condition are already assumptions. For each block, the primitive fixed-point hypothesis gives $O_{A_k A_k}(N) \rightarrow 1$. □

Theorem 11.28 (Canonical form from peripheral primitivity). isCanonicalForm_of_peripheralPrimitive ✓ Stmt OK Let $(\mu_k, A_k)_{k=1}^r$ be a collection of block tensors and scaling factors satisfying:

1. each A_k is injective,
2. each A_k satisfies the TP normalization $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$,
3. $|\mu_1| \geq \dots \geq |\mu_r|$ (non-increasing),
4. all $\mu_k \neq 0$,
5. each transfer map $\mathcal{E}_{A_k}$ is primitive, i.e. has peripheral spectrum $\{1\}$.

Then (μ_k, A_k) satisfies the canonical form conditions (Definition 12.16).

Proof. ✓ Proof OK Fix a block k and write $E := \mathcal{E}_{A_k}$. We check that each primitive block already satisfies the defining properties of the canonical form conditions. By injectivity and TP normalization, Theorem 8.9 provides a nonzero positive definite fixed point for E . To apply the spectral-gap criterion (Theorem 5.51), we check its three hypotheses:

1. E is primitive, since the peripheral spectrum is $\{1\}$ by assumption;
2. every eigenvalue of E has modulus at most 1, because $E = F_{A_k A_k}$ is a normalized mixed transfer operator and Theorem 9.14 applies;
3. any trace-zero fixed point of E is zero; this is the trace-zero uniqueness consequence of the PSD fixed-point uniqueness theorem (Theorem 8.5).

Hence E has a spectral gap on the complement of its fixed-point projection. Theorem 9.98 then gives $O_{A_k A_k}(N) \rightarrow 1$. Since this holds for every block, the overlap clause required in Definition 12.16 follows from the other hypotheses. □

Remark 11.29. ✓ Stmt OK Theorem 11.28 shows that the overlap convergence hypothesis in the canonical form conditions is redundant once peripheral-spectrum primitivity is known. The argument factors through the spectral-gap formulation of Chapter 9: peripheral-spectrum primitivity is first converted to that spectral-gap condition, and the overlap limit is then deduced from it. In the paper [CPGSV21], primitivity is obtained from irreducibility plus periodicity removal by blocking to period one. Theorem 11.30 is precisely this periodicity-removal step under irreducibility and TP normalization.

Theorem 11.30 (Blocking yields a peripheral-spectrum primitive transfer map). exists_blockTensor_isPrimitive_of_TP_of_i
A be an irreducible MPS tensor with $D > 0$. Assume that A satisfies the TP normalization $\sum_i (A^i)^\dagger A^i = \mathbb{1}$. Then there exists a blocking length $p > 0$ such that the transfer map $\mathcal{E}_{A^{[p]}}$ of the blocked tensor $A^{[p]}$ is primitive, i.e. its peripheral spectrum is $\{1\}$.

Proof. ✓ Proof OK The period- p blocking step is the diagrammatic move of grouping the chain into consecutive blocks of length p . Pass to the conjugate-transposed Kraus family $K_i := (A^i)^\dagger$. The TP normalization of A makes K unital, and irreducibility of A gives irreducibility of the associated Kraus map. The TP fixed-point theorem for $\mathcal{E}_A$ provides a positive definite fixed point for the adjoint map of K , so Theorem 5.46 shows that every peripheral eigenvalue is a root of unity. Since $\text{peripheral}(\mathcal{E}_A)$ is finite (Theorem 5.29), Lemma 5.33 gives a common exponent p such that $\mu^p = 1$ for all $\mu \in \text{peripheral}(\mathcal{E}_A)$. Lemma 5.34 then implies $\text{peripheral}(\mathcal{E}_A^p) = \{1\}$. Finally, blocking corresponds to taking a power of the transfer map, so $\mathcal{E}_{A^{[p]}}$ is primitive. $\square$

Remark 11.31. ✓ Stmt OK Theorem 11.30 is the “periodicity removal by blocking” step in [CPGSV17a, Appendix A] for an irreducible block already placed in TP gauge. The later existence theorems then reuse this step.

11.6.1 Adjoint transport for blocking

Lemma 11.32 (Characteristic polynomial under conjugate transpose). charpoly_conjTranspose ✓ Stmt OK *For a complex matrix M , the characteristic polynomial of $M^\dagger$ is obtained from the characteristic polynomial of M by coefficientwise complex conjugation.*

Proof. ✓ Proof OK Write $M^\dagger$ as the transpose of the coefficientwise conjugate matrix. The characteristic polynomial is invariant under transpose and commutes with coefficientwise conjugation. $\square$

Theorem 11.33 (Eigenvalues of the adjoint are conjugated). hasEigenvalue_adjoint_iff ✓ Stmt OK *Let E be a linear endomorphism of a finite-dimensional complex inner-product space, and let $\mu \in \mathbb{C}$. Then μ is an eigenvalue of E if and only if $\bar{\mu}$ is an eigenvalue of the adjoint $E^\dagger$.*

Proof. ✓ Proof OK Express eigenvalues as roots of the characteristic polynomial, convert the adjoint to the conjugate-transposed matrix in an orthonormal basis, and apply Lemma 11.32. $\square$

Theorem 11.34 (Primitivity is invariant under adjoint). adjoint_iff ✓ Stmt OK *A linear endomorphism is primitive if and only if its adjoint is primitive.*

Proof. ✓ Proof OK The peripheral eigenvalues of the adjoint are the complex conjugates of those of the original map, so both maps have the same peripheral spectrum on the unit circle. $\square$

Lemma 11.35 (Transfer map of the conjugate-transposed Kraus family). transferMap_conjTranspose_eq_adjoint ✓ Stmt OK *an MPS tensor A , the transfer map of the conjugate-transposed Kraus family $\{(A^i)^\dagger\}$ is the adjoint of $\mathcal{E}_A$ with respect to the Frobenius inner product.*

Proof. ✓ Proof OK Compare both sides against the Frobenius pairing $\langle X, Y \rangle = \text{tr}(YX^\dagger)$ and rewrite the Kraus sums. $\square$

Theorem 11.36 (Blocking primitivity with a positive-dimension hypothesis). exists_blockTensor_isPrimitive ✓ Stmt OK *A be an irreducible MPS tensor with positive bond dimension and TP normalization $\sum_i (A^i)^\dagger A^i = \mathbb{1}$. Then there exists a blocking length $p > 0$ such that $\mathcal{E}_{A^{[p]}}$ is primitive.*

Proof. ✓ Proof OK This is Theorem 11.30 with the positive-dimension assumption expressed as a typeclass hypothesis. $\square$

Theorem 11.37 (Blocking primitivity in left-canonical gauge). exists_blockTensor_isPrimitive_of_leftCanonical_of_isIrreducible *A be a left-canonical irreducible MPS tensor with positive bond dimension. Then there exists a blocking length $p > 0$ such that $\mathcal{E}_{A^{[p]}}$ is primitive.*

Proof. ✓ Proof OK This is the left-canonical reformulation of Theorem 11.30. $\square$

11.7 Normal canonical form

Definition 11.38 (Normal canonical form conditions). IsNormalCanonicalForm ✓ Stmt OK A collection of scaling factors $(\mu_k)_{k=1}^r$ and block tensors $(A_k)_{k=1}^r$ satisfies the *normal canonical form conditions* if:

1. each A_k is irreducible,
2. each A_k satisfies the TP normalization $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$,
3. each transfer map $\mathcal{E}_{A_k}$ is primitive (equivalently, has peripheral spectrum $\{1\}$),
4. the moduli are non-increasing: $|\mu_1| \geq |\mu_2| \geq \dots \geq |\mu_r|$,
5. all $\mu_k \neq 0$,
6. all bond dimensions are positive.

Here “normal” is used in the spectral sense of [CPGSV17a]: irreducible blocks whose TP-normalized transfer maps are primitive. As noted in the remark after Definition 2.26, this is equivalent to the eventual block-injectivity formulation of Definition 2.26; see [SPGWC10, Proposition 3].

Theorem 11.39 (Self-overlap is derived in normal canonical form). overlap_tendsto_one ✓ Stmt OK *If (μ_k, A_k) satisfies the normal canonical form conditions, then $O_{A_k A_k}(N) \rightarrow 1$ for every block k .*

Proof. ✓ Proof OK Peripheral-spectrum primitivity is part of the data. Together with irreducibility and TP normalization, it yields the spectral-gap primitive theorem needed for overlap convergence, so the self-overlap condition follows from the other hypotheses rather than appearing as a separate assumption. $\square$

Theorem 11.40 (Modulus-one eigenvalue rigidity for irreducible TP blocks). modulus_one_eigenvalue_implies_gauge_of_irreducible *A and B be irreducible MPS tensors satisfying $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ and $\sum_i (B^i)^\dagger B^i = \mathbb{1}$. If the mixed transfer map has spectral radius at least 1 (equivalently, it has a peripheral eigenvalue of modulus 1), then A and B are gauge-phase equivalent.*

Proof. ✓ Proof OK The argument is the same as in Theorem 9.16 (with the gauging formula given by Theorem 8.10 directly), but the fixed-point theorem used here comes from irreducibility rather than injectivity. Take a modulus-one eigenvector of the mixed transfer map. Because A and B are TP-normalized, each transfer map has a nonzero PSD fixed point by Theorem 8.8. Irreducibility upgrades these fixed points to positive definite ones by Theorem 8.3. Theorem 8.10 therefore gauges A and B separately to unital Kraus families. In this unital gauge, the embedded modulus-one eigenvector saturates the Kadison–Schwarz inequality, and the same equality / multiplicative-domain argument as in Theorem 9.16 produces an invertible intertwiner between the two Kraus families. That intertwiner is exactly the required gauge-phase equivalence. $\square$

11.8 Steps toward canonical form existence

This section combines the components of the canonical-form construction following [CPGSV17a, §2.3 and Appendix A]. The approach here requires blocking to remove non-trivial periodicity. A separate reduction for periodic blocks without blocking is developed in [DICCSPG17]; see Chapter 15 for that direct periodic extension.

Theorem 11.41 (Irreducible block decomposition). exists_irreducible_blockDecomp ✓ Stmt OK Every MPS tensor A generates the same MPV family as a block-diagonal tensor $\bigoplus_{k=1}^r A_k$ with each block irreducible.

Proof. ✓ Proof OK Apply Theorem 11.16. $\square$

Theorem 11.42 (Left-canonical normalization for irreducible TP blocks). exists_CFII_data_of_TP_of_isIrreducibleTensor ✓ A be an irreducible MPS tensor already in TP gauge ($\sum_i (A^i)^\dagger A^i = \mathbb{1}$) with $D > 0$. Then there exist a unitary U and a diagonal positive definite matrix Λ such that:

1. $B^i := U^\dagger A^i U$ is TP,
2. $\mathcal{E}_B(\Lambda) = \Lambda$,
3. Λ is diagonal and positive definite,
4. A and B generate the same MPV family.

This is the left-canonical normalization step (Canonical Form II) of [CPGSV17a, Appendix A]. The unitary conjugation occurs only after TP normalization has already been achieved; it is a second step inside TP gauge, not a unitary normalization of an arbitrary irreducible block.

Proof. ✓ Proof OK Apply Theorem 11.17 to obtain the unitary and the diagonal PD fixed point. Unitary conjugation preserves trace-preservation and the MPV family (Theorem 11.10). $\square$

Theorem 11.43 (Irreducible block decomposition with left-canonical normalization). exists_irreducible_blockDecomp_with any MPS tensor A , there exist irreducible blocks $(A_k)_{k=1}^r$ that generate the same MPV family as A . Moreover, for each block k , if that block has already been placed in TP gauge and $D_k > 0$, one obtains a left-canonical normalization (unitary, diagonal PD fixed point, MPV preservation). This yields the left-canonical normalization; the further reduction to normal canonical form is carried out in the subsequent spectral analysis.

Proof. ✓ Proof OK Combine Theorems 11.41 and 11.42. $\square$

Theorem 11.44 (Normal canonical form from a primitive block decomposition). exists_normalCanonicalForm_of_primitive_b A generates the same MPV family as a weighted block-diagonal tensor $\bigoplus_k \mu_k A_k$, and assume that:

1. each A_k is irreducible,
2. each A_k satisfies the TP normalization $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$,
3. each transfer map $\mathcal{E}_{A_k}$ is primitive (equivalently, has peripheral spectrum $\{1\}$),
4. the moduli $|\mu_k|$ are pairwise distinct,
5. all $\mu_k \neq 0$,
6. all bond dimensions are positive.

Then there exist a blocking length $p > 0$, weights (ν_k) , and blocked tensors (B_k) such that $A^{[p]}$ generates the same MPV family as $\bigoplus_k \nu_k B_k$, and (ν_k, B_k) satisfies the normal canonical form conditions.

Proof. ✓ Proof OK No new blocking is needed here: the given blocks are already primitive, so one may take $p = 1$. The theorem is stated with a blocking length because earlier reduction steps may require a common blocking period; under these hypotheses that period can be chosen to be 1. Since the moduli $|\mu_k|$ are pairwise distinct, there is a reordering of the blocks for which they become strictly decreasing. After this reindexing, set $\nu_k := \mu_k$ and $B_k := A_k$ in the new order. Irreducibility, TP normalization, primitivity, nonzero weights, and positive bond dimensions are unchanged by the reordering, so the reordered family satisfies Definition 11.38 and generates the same MPV family as A . □

11.9 Zero-block separation

In the irreducible block decomposition, some blocks may have all-zero Kraus operators. Such blocks contribute to the MPV only at word length $N = 0$ (where their contribution equals their bond dimension). The following results separate the zero blocks from the nonzero blocks.

Definition 11.45 (Zero MPS tensor). zeroMPSTensor ✓ Stmt OK The zero MPS tensor of physical dimension d and bond dimension D , denoted $\mathbf{0}_{d,D}$, is the tensor whose Kraus operators are all zero.

Theorem 11.46 (MPV of zero tensor). mpv_zeroMPSTensor ✓ Stmt OK For any spin configuration σ of length N ,

$$V^{(N)}(\mathbf{0}_{d,D})_\sigma = \begin{cases} D & \text{if } N = 0, \\ 0 & \text{if } N \geq 1. \end{cases}$$

Proof. ✓ Proof OK For $N = 0$, $(\mathbf{0}_{d,D})^\emptyset = \mathbb{1}_D$, so the MPV equals $\text{tr}(\mathbb{1}_D) = D$. For $N \geq 1$, every word product of a zero tensor is zero. □

Theorem 11.47 (Zero-block separation). exists_irreducible_blockDecomp_nonzeroBlocks ✓ Stmt OK Every MPS tensor A admits a block decomposition into:

1. a trivial block of bond dimension D_0 with all Kraus operators zero (accumulating all all-zero irreducible blocks), and
2. a finite family of nontrivial blocks $(B_k)_{k=1}^r$, each irreducible, each with at least one nonzero Kraus operator, and each with positive bond dimension,

such that

$$V^{(N)}(A)_\sigma = V^{(N)}(\mathbf{0}_{d,D_0})_\sigma + V^{(N)}(\bigoplus_{k=1}^r B_k)_\sigma$$

for every spin configuration σ of length N . At $N = 0$ this reads $D = D_0 + \sum_{k=1}^r D_k$. For $N \geq 1$ the zero-tail term vanishes.

Proof. ✓ Proof OK Apply the irreducible block decomposition (Theorem 11.16) and partition the blocks by whether each has a nonzero Kraus operator. The all-zero blocks satisfy a dimension bound via irreducibility; their total bond dimension is D_0 . The MPV identity follows from the block structure. □

11.10 TP-gauge reduction for arbitrary tensors

Theorem 11.48 (TP-gauge reduction for arbitrary tensors with zero-block separation). exists_tp_gauge_from_arbitrary_wi
any MPS tensor A , there exist:

1. a zero-tail bond dimension D_0 ,
2. nontrivial blocks $(B_k)_{k=1}^r$ with nonzero weights $(\mu_k)_{k=1}^r$, each block irreducible and left-canonical ($\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$), with positive bond dimension,

such that

$$V^{(N)}(A)_\sigma = V^{(N)}(\mathbf{0}_{d,D_0})_\sigma + V^{(N)}(\bigoplus_{k=1}^r \mu_k B_k)_\sigma.$$

This is the furthest unconditional reduction step available for arbitrary tensors before periodicity removal.

Proof. ✓ Proof OK Apply Theorem 11.47 to separate the trivial block and nontrivial blocks. Then apply the blockwise TP-gauge theorem to the nontrivial blocks. The MPV identity chains through both steps. □

11.11 Blocking and primitivity reduction

Theorem 11.49 (Blocking distributes over MPV equivalence). sameMPV2_blockTensor_of_sameMPV2_toTensorFromBlocks ✓ Stmt OK
 A generates the same MPV family as $\bigoplus_k \mu_k B_k$, then for any $p \geq 0$,

$$A^{[p]} \sim \bigoplus_k \mu_k^p B_k^{[p]}.$$

Proof. ✓ Proof OK The identity $V^{(N)}(A^{[p]})_\sigma = V^{(Np)}(A)_{\sigma_{\text{flat}}}$ holds for any tensor by definition of blocking. The MPV equivalence of A and $\bigoplus_k \mu_k B_k$ at length Np then gives the equivalence at length N with weights μ_k^p . □

Theorem 11.50 (Blocking assembled weighted blocks). sameMPV2_blockTensor_toTensorFromBlocks ✓ Stmt OK For any weighted block tensor $\bigoplus_k \mu_k B_k$ and any blocking length $p \geq 0$, its blocked MPV family agrees with the weighted block tensor formed from the blocked blocks $B_k^{[p]}$ and the powered weights μ_k^p .

Proof. ✓ Proof OK Apply Theorem 11.49 to the identity MPV relation for the assembled weighted block tensor. □

Theorem 11.51 (Positive-length MPV equality after blocking). sameMPV2Pos_blockTensor ✓ Stmt OK If two tensors agree on all positive system sizes, then their p -blocked tensors agree on all positive blocked system sizes for every $p \geq 1$.

Proof. ✓ Proof OK A positive blocked chain of length N corresponds to an original chain of length Np , which is positive when both N and p are positive. □

Theorem 11.52 (Positive-length equality for powered weighted nonzero blocks). sameMPV2Pos_toTensorFromBlocks_blockPower ✓ Stmt OK *two weighted nonzero-block tensors agree on all positive system sizes, then after any positive common blocking their blocked nonzero-block tensors still agree on all positive system sizes, with weights transported from μ_k to μ_k^p .*

Proof. ✓ Proof OK Use Theorem 11.51 for the assembled nonzero parts, then rewrite each blocked assembled tensor by Theorem 11.50. □

Lemma 11.53 (Nonzero weights after blocking powers). blockWeights_ne_zero ✓ Stmt OK *If every μ_k is nonzero, then every powered weight μ_k^p is nonzero.*

Proof. ✓ Proof OK This is the elementary fact that powers of a nonzero complex number are nonzero. □

Lemma 11.54 (Nonzero weights after sector replication). replicatedWeights_pow_ne_zero ✓ Stmt OK *Suppose a nonzero-weight block k is replaced by a finite family of cyclic sectors. If $\mu_k \neq 0$, then every replicated sector weight $\mu_k^{p_k}$ remains nonzero.*

Proof. ✓ Proof OK Apply Lemma 11.53 to the parent block weight and the blocking power assigned to that block. □

Lemma 11.55 (Nonzero replicated weights with sector phases). replicatedWeights_pow_mul_phase_ne_zero ✓ Stmt OK *If the replicated sector weights are additionally multiplied by nonzero sector phases, then the resulting sector weights are still nonzero.*

Proof. ✓ Proof OK Combine Lemma 11.54 with nonvanishing of the phase factor. □

Theorem 11.56 (Primitive channels remain primitive under powers). isPrimitive_pow_of_isPrimitive ✓ Stmt OK *If $\mathcal{E}$ has peripheral spectrum $\{1\}$ (is primitive) and $m \geq 1$, then $\mathcal{E}^m$ is also primitive.*

Proof. ✓ Proof OK The peripheral spectrum of $\mathcal{E}^m$ is the set of m -th powers of that of $\mathcal{E}$. Since $\{1\}^m = \{1\}$, the result follows. □

Theorem 11.57 (Blocking-period divisibility preserves primitivity). isPrimitive_transferMap_blockTensor_of_dvd ✓ Stmt OK *the transfer map of $A^{[p]}$ is primitive, $p \mid q$, and $q \geq 1$, then the transfer map of $A^{[q]}$ is also primitive.*

Proof. ✓ Proof OK Write $q = p \cdot m$. Then $\mathcal{E}_{A^{[q]}} = \mathcal{E}_{A^{[p]}}^m$. Apply Theorem 11.56 with $\mathcal{E} = \mathcal{E}_{A^{[p]}}$. □

Theorem 11.58 (Iterated blocking multiplies physical dimensions). blockPhysDim_blockPhysDim ✓ Stmt OK *The physical alphabet obtained by first blocking m sites and then blocking n of those sites has the same cardinality as blocking mn original sites:*

$$\text{blockPhysDim}(\text{blockPhysDim}(d, m), n) = \text{blockPhysDim}(d, mn).$$

Proof. ✓ Proof OK Both sides are powers of d : the left-hand side is $(d^m)^n$, and the right-hand side is d^{mn} . □

Theorem 11.59 (Iterated blocking and direct blocking yield the same MPV family after relabeling). sameMPV2_blockTensor_blockTensor_mul_reindex ✓ Stmt OK *If one first blocks m sites and then blocks n of the resulting sites, the resulting tensor has the same MPV family as the tensor obtained by blocking mn original sites, after the physical labels are identified by flattening each iterated block.*

Proof. ✓ Proof OK For each iterated blocked index, decode the outer block, decode each inner block, and concatenate the resulting words. This is exactly the word decoded from the corresponding directly blocked index, so the word evaluations, and hence all MPV coefficients, agree. □

Theorem 11.60 (Physical-dimension transport and MPV equality). sameMPV2_cast_physDim ✓ Stmt OK *If two physical alphabets have the same cardinality, then the transported tensors have the same MPV family if and only if the original tensors do.*

Proof. ✓ Proof OK After substituting the equality of physical dimensions, the two MPV equality statements are the same. □

Theorem 11.61 (Physical-dimension transport for block-diagonal tensors). toTensorFromBlocks_cast_physDim ✓ Stmt OK *If equal physical alphabets commutes with forming a weighted block-diagonal tensor from a finite block family.*

Proof. ✓ Proof OK Substitute the equality of physical dimensions; both sides are then the same block-diagonal tensor. □

Theorem 11.62 (Physical-dimension transport and trace preservation). leftCanonical_cast_physDim ✓ Stmt OK *After identifying equal physical alphabets, the transported tensor is trace-preserving if and only if the original tensor is trace-preserving.*

Proof. ✓ Proof OK Substitute the equality of physical dimensions; the two normalization sums are identical. □

Theorem 11.63 (Physical-dimension transport and primitive transfer maps). isPrimitive_transferMap_cast_physDim ✓ Stmt OK *After identifying equal physical alphabets, the transported transfer map is primitive if and only if the original transfer map is primitive.*

Proof. ✓ Proof OK Substitute the equality of physical dimensions; the transfer maps coincide. □

Theorem 11.64 (Physical-dimension transport and tensor irreducibility). isIrreducibleTensor_cast_physDim ✓ Stmt OK *After identifying equal physical alphabets, the transported tensor is irreducible if and only if the original tensor is irreducible.*

Proof. ✓ Proof OK Substitute the equality of physical dimensions; the irreducibility predicates are identical. □

Theorem 11.65 (Iterated blocking multiplies periods at the transfer-map level). transferMap_blockTensor_mul ✓ Stmt OK *any MPS tensor A and integers $m, n \geq 0$,*

$$\mathcal{E}_{(A^{[m]})^{[n]}} = \mathcal{E}_{A^{[mn]}}.$$

Proof. ✓ Proof OK Rewrite the outer blocked transfer map as the n -th power of $\mathcal{E}_{A^{[m]}}$, then rewrite $\mathcal{E}_{A^{[m]}}$ as $\mathcal{E}_A^m$. This gives $(\mathcal{E}_A^m)^n$, which simplifies to $\mathcal{E}_A^{mn}$. Finally rewrite $\mathcal{E}_A^{mn}$ back as the transfer map of $A^{[mn]}$. □

Theorem 11.66 (Common blocking period for a block family). exists_common_blocking_all_primitive ✓ Stmt OK *Let $(B_k)_{k=1}^r$ be a finite family of MPS tensors with positive bond dimensions, each admitting a blocking period p_k making its transfer map primitive. Then $p = \text{lcm}(p_1, \dots, p_r)$ is a common period: $\mathcal{E}_{B_k^{[p]}}$ is primitive for every k .*

Proof. ✓ Proof OK Each $p_k \mid p$, so Theorem 11.57 applies. □

Definition 11.67 (Common-period blocking of a block family). [commonPeriodBlocking](#) ✓ Stmt OK Given a finite family of tensors (B_k) and positive integers (p_k) , the common-period blocking is the family obtained by blocking every B_k at the least common multiple of the periods p_k .

Theorem 11.68 (Primitivity of the common-period blocking). [isPrimitive_transferMap_commonPeriodBlocking](#) ✓ Stmt OK *Sup* each B_k has positive bond dimension and each blocked transfer map $\mathcal{E}_{B_k^{[p_k]}}$ is primitive. Then every member of the common-period blocked family also has primitive transfer map.

Proof. ✓ Proof OK Each p_k divides the common blocking length, so Theorem 11.57 applies to every block. □

Theorem 11.69 (Formula for common-period blocking). [commonPeriodBlocking_apply](#) ✓ Stmt OK Each block of the common-period blocking is exactly the corresponding tensor blocked at the common least common multiple.

Proof. ✓ Proof OK This is the definition of the common-period blocking. □

11.12 Cyclic sector decomposition

Theorem 11.70 (Compression to a supported projection sector). [exists_compressedTensor_of_supported_projection](#) ✓ Stmt OK *Sup* A be an MPS tensor satisfying $\sum_i (A^i)^\dagger A^i = P$ for an orthogonal projection P , and suppose $PA^iP = A^i$ for every i . Then there exist a left-canonical tensor C of bond dimension $n = \text{tr}(P)$ and a $*$ -algebra isomorphism $\varphi : M_n(\mathbb{C}) \xrightarrow{\sim} PM_D(\mathbb{C})P$ (preserving both multiplication and adjoint) with

$$V^{(N)}(C)_\sigma = \text{tr}(P \cdot A^\sigma), \quad \varphi(\mathcal{E}_{C^\dagger}(X)) = \mathcal{E}_{A^\dagger}(\varphi(X))$$

for every σ and every $X \in M_n(\mathbb{C})$, together with $\varphi(XY) = \varphi(X)\varphi(Y)$ and $\varphi(X^\dagger) = \varphi(X)^\dagger$.

Proof. ✓ Proof OK Diagonalize P via a unitary U (spectral theorem). The eigenvalues are in $\{0, 1\}$. Set $n = \text{tr} P$; let C^i be the top-left $n \times n$ block of $U^\dagger A^i U$ (restricted to eigenvalue-1 indices). TP follows from the projection normalization; the MPV identity follows from trace cyclicity. □

Corollary 11.71 (Compression preserves positive-length MPV coefficients). [exists_compressedTensor_of_supported_projection](#) ✓ Stmt OK *Sup* the hypotheses of Theorem 11.70, the compressed tensor C additionally satisfies $V^{(N)}(C)_\sigma = V^{(N)}(A)_\sigma$ for all positive chain lengths $N > 0$.

Proof. ✓ Proof OK For $N > 0$ the word $A^{i_1} \cdots A^{i_N}$ is nonempty, so P acts as the identity on the left by idempotence and the support hypothesis. The MPV identity then follows from trace cyclicity via Theorem 11.70. □

Theorem 11.72 (Block decomposition from commuting projections). [exists_blockDecomp_of_commuting_projections](#) ✓ Stmt OK *Sup* A be a left-canonical MPS tensor, and let $(P_k)_{k=1}^m$ be orthogonal projections summing to $\mathbb{1}$ that commute with every Kraus operator: $P_k A^i = A^i P_k$ for all i, k . Then there exist left-canonical blocks $(C_k)_{k=1}^m$ with unit weights such that A generates the same MPV family as $\bigoplus_{k=1}^m C_k$. Moreover, each sector satisfies the per-sector trace relation $V^{(N)}(C_k)_\sigma = \text{tr}(P_k \cdot A^{i_1} \cdots A^{i_N})$ for every word $\sigma = (i_1, \dots, i_N)$, and each block carries a compression linear equivalence $\varphi_k : M_{\dim C_k}(\mathbb{C}) \xrightarrow{\sim} P_k M_D(\mathbb{C}) P_k$ intertwining the compressed adjoint transfer map with the sector adjoint transfer map.

Proof. ✓ Proof OK For each k , the tensor $P_k A^i$ is supported on P_k (using commutativity and $P_k^2 = P_k$). The TP condition $\sum_i (P_k A^i)^\dagger (P_k A^i) = P_k$ follows from $\sum_i (A^i)^\dagger A^i = \mathbb{1}$. Apply Theorem 11.70 to each sector. The MPV identity uses $\sum_k P_k = \mathbb{1}$ and linearity of trace. $\square$

Theorem 11.73 (Block decomposition from adjoint-fixed projections). exists_blockDecomp_of_adjoint_fixed_projections ✓ Stmt OK
A be a left-canonical MPS tensor, and let $(P_k)_{k=1}^m$ be orthogonal projections summing to $\mathbb{1}$ that are fixed by the adjoint transfer map: $\mathcal{E}_A^\dagger(P_k) = P_k$ for every k . Then the same block decomposition conclusion holds as in Theorem 11.72, including the per-sector trace relation.

Proof. ✓ Proof OK The Kadison–Schwarz equality condition for a TP map (Theorem 7.23) implies that a projection fixed by $\mathcal{E}_A^\dagger$ commutes with every Kraus operator. Apply Theorem 11.72. $\square$

Theorem 11.74 (Conjugate-transposed Kraus family for irreducible TP tensors). conjTranspose_kraus_setup ✓ Stmt OK
A be a left-canonical irreducible tensor. Then the conjugate-transposed family $K_i := (A^i)^\dagger$ is unital and irreducible, and there exists a positive-definite matrix ρ fixed by the adjoint map associated to K .

Proof. ✓ Proof OK Left-canonicity gives $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, so the conjugate-transposed family is unital. Irreducibility passes to the adjoint map, and the quantum Perron–Frobenius theorem provides the required positive-definite fixed point. $\square$

Theorem 11.75 (One full cyclic turn fixes the sector projections). adjointTransferMap_pow_fixes_cyclic_projection ✓ Stmt OK
a family of projections $(P_k)_{k=1}^m$ satisfies $\mathcal{E}^\dagger(P_{k+1}) = P_k$, then the m -th iterate of $\mathcal{E}^\dagger$ fixes each projection P_k .

Proof. ✓ Proof OK Iterate the cyclic relation m times. Because the indices are taken modulo m , the projection returns to its starting position. $\square$

Theorem 11.76 (Blocked adjoint transfer map as an iterate). transferMap_adjoint_blocked_eq_pow ✓ Stmt OK *For every tensor A , blocking length m , and matrix X ,*

$$\mathcal{E}_{(A^{[m]})^\dagger}(X) = (\mathcal{E}_{A^\dagger})^m(X).$$

Proof. ✓ Proof OK Write both transfer maps as Frobenius adjoints of ordinary blocked transfer maps. Then use Theorem 11.65. $\square$

Theorem 11.77 (Cyclic sector decomposition after blocking). exists_cyclic_sector_decomp_after_blocking ✓ Stmt OK *Let A be a left-canonical irreducible tensor of period m . Then the blocked tensor $A^{[m]}$ admits:*

1. left-canonical sector tensors $(C_k)_{k=1}^m$,
2. orthogonal projections $(P_k)_{k=1}^m$ summing to $\mathbb{1}$ and satisfying $\mathcal{E}_{A^\dagger}(P_{k+1}) = P_k$,
3. compression linear equivalences $\varphi_k : M_{\dim C_k}(\mathbb{C}) \xrightarrow{\sim} P_k M_D(\mathbb{C}) P_k$,
4. a direct-sum MPV decomposition $A^{[m]} \sim \bigoplus_{k=1}^m C_k$.

Proof. ✓ Proof OK The cyclic decomposition of the adjoint transfer map gives projections (P_k) with $\mathcal{E}_{A^\dagger}(P_{k+1}) = P_k$. Theorem 11.75 upgrades this to $(\mathcal{E}_{A^\dagger})^m(P_k) = P_k$, and Theorem 11.76 identifies $(\mathcal{E}_{A^\dagger})^m$ with the adjoint transfer map of the blocked tensor. Apply Theorem 11.73. $\square$

Theorem 11.78 (Cyclic sector decomposition for irreducible TP tensors). exists_cyclic_sector_decomp_of_TP_of_isIrreducible
A be a left-canonical irreducible tensor. Then there exist $m \geq 1$ and left-canonical blocks $(C_k)_{k=1}^m$ such that the m -blocked tensor $A^{[m]}$ decomposes as $\bigoplus_{k=1}^m C_k$ (with unit weights).

Proof. ✓ Proof OK Apply the extra-blocking theorem to each period-removal sector, substitute the equality between the iterated and common physical alphabets, and use that a positive power of a nonzero complex number is nonzero. $\square$

Theorem 11.87 (Reindexed blocks decompose into common cyclic sectors). nestedBlock_sameMPV2_commonReindexedBlock com
each block, the iterated block $(B_k^{[m_k]})^{[e_k]}$ agrees, after the explicit reindexing of blocked physical words, with the block of length $m_k e_k$. Combining this with the given cyclic-sector decomposition gives an MPV equality between the reindexed block and the corresponding common-sector tensor. Summing over the original blocks transports the weights to μ_k^p and flattens the two-level block/sector sum into one sector family.

Proof. ✓ Proof OK First use the iterated-blocking comparison theorem and substitute the common physical alphabet. Then compose with the per-block cyclic-sector MPV equivalence supplied by the family. Finally expand both block-diagonal tensors as sums and reindex the double sum over pairs (k, s) by the flattened sector index. $\square$

Definition 11.88 (Coordinate grouping for common blocked words). groupedBlockCastAgrees ✓ Stmt OK For one original block k in a common blocked cyclic-sector family, this predicate says that the canonical identification of the common blocked alphabet with the iterated alphabet agrees with the following grouping of words: first read the direct word of length $p = m_k e_k$, then group it into e_k consecutive words of length m_k .

Theorem 11.89 (Comparison of direct and iterated blocking for weighted nonzero blocks).

wordOfBlock_eq_iteratedBlockIndex_of_groupedBlockCastAgrees blockTensor_eq_commonReindexedBlock_of_word_eq blockTensor_eq_commonReinde
comparison has two forms. First, assume that each directly blocked nonzero block has the same MPV family as its corresponding iterated-blocking tensor. Then the weighted direct sum of the directly blocked nonzero blocks agrees with the weighted direct sum built from the iterated-blocking tensors, and hence with the common-sector family. Second, this blockwise MPV equality follows from the stronger pointwise condition that the length- p word read from the common blocked alphabet is the same word as the one obtained by viewing the index as an iterated (m_k, e_k) block and flattening it; under this condition the two individual block tensors are equal.

Proof. ✓ Proof OK The word equality gives equality of the corresponding block tensors, hence blockwise MPV equality. The coordinate-grouping predicate supplies this word equality by comparing the canonical identification with the explicit consecutive grouping of a direct blocked word. Expanding both weighted block-diagonal tensors as finite sums over the original block index and applying the blockwise equality term by term gives the weighted comparison. Composing with Theorem 11.87 gives the common-sector statement. $\square$

Theorem 11.90 (Canonical blocked nonzero part after reindexing blocked words). sameMPV2_weightedCanonicalBlock_commonF
that the canonical blocked nonzero part agrees, as an MPV family, with the reindexed cyclic-sector tensor coming from iterated blocking. Then the canonical weighted nonzero part agrees with the weighted common-sector tensor. The prescribed-length form gives the same conclusion after substituting an equality between the family length and the chosen common length.

Proof. ✓ Proof OK Compose the assumed MPV equality with the reindexed sector decomposition of Theorem 11.87. $\square$

Theorem 11.91 (Common reblocking at a prescribed common multiple). exists_commonBlockedCyclicSectorFamily_of_commonMu
every nonzero-weight block has primitive irreducible cyclic sectors and a positive integer p is a common multiple of all period-removal lengths, then the nonzero-weight family admits a common reblocked cyclic-sector family whose common length is that prescribed p .

Proof. ✓ Proof OK For each block with period m_k , write the prescribed common multiple as $p = m_k e_k$. Then block every cyclic sector by e_k , substitute the equality between the iterated and common physical alphabets, and transport the MPV and structural properties exactly as in the least-common-multiple construction. $\square$

Theorem 11.92 (Common reblocking of per-block cyclic sectors). exists_commonBlockedCyclicSectorFamily_of_hasPrimitiveIrre
every nonzero-weight block in a finite family has primitive irreducible cyclic sectors, then the family admits a common reblocked cyclic-sector family. The proof chooses the least common multiple of the per-block periods, blocks each sector by the corresponding quotient, transports all sectors to the common physical alphabet, and preserves TP, primitivity, tensor irreducibility, positive bond dimensions, and the unit-weight MPV equivalences.

Proof. ✓ Proof OK Choose the least common multiple p of the per-block periods and apply Theorem 11.91. $\square$

11.12.1 Sector irreducibility helpers

Lemma 11.93 (Pairwise orthogonality from projection sum). pairwise_mul_zero_of_orthogonalProjection_sum_one ✓ Stmt OK
 $(P_k)_{k=1}^m$ be orthogonal projections summing to the identity: $\sum_k P_k = \mathbb{1}$. Then for $i \neq j$ we have $P_i P_j = 0$.

Proof. ✓ Proof OK Sandwich the sum between P_i to get $\sum_k P_i P_k P_i = P_i$. Since the i -th summand equals P_i by idempotence, all other summands vanish. Positivity ($BB^* = 0 \Rightarrow B = 0$) then gives $P_i P_k = 0$ for each $k \neq i$. $\square$

Lemma 11.94 (Corner preservation from adjoint-fixed projection). preservesCorner_of_adjoint_fixed_projection ✓ Stmt OK
A be a left-canonical MPS tensor and let P be an orthogonal projection fixed by $\mathcal{E}_A^\dagger$. Then $\mathcal{E}_A^\dagger$ preserves the corner algebra $PM_D(\mathbb{C})P$.

Proof. ✓ Proof OK The fixed-point condition and the Kadison–Schwarz equality imply $[P, A_i] = 0$ for every Kraus operator. Thread the idempotent relation $P^2 = P$ through the corner sandwich. $\square$

Lemma 11.95 (Orbit-sum fixed point). orbitSumProjection_fixed_of_pow_fix ✓ Stmt OK Let T be a linear endomorphism and Q a matrix satisfying $T^m(Q) = Q$. Then the orbit sum $\sum_{l=0}^{m-1} T^l(Q)$ is a fixed point of T .

Proof. ✓ Proof OK Telescoping: T shifts each summand $T^l(Q) \mapsto T^{l+1}(Q)$ and the periodicity condition $T^m(Q) = Q$ wraps the last term $T^{m-1}(Q) \mapsto T^m(Q) = Q$ back to the first. $\square$

Lemma 11.96 (Orbit iterate supported on shifted sector). orbit_iterate_supported_on_shifted_sector ✓ Stmt OK Let $(P_k)_{k=1}^m$ be orthogonal projections with cyclic shift $T(P_{k+1}) = P_k$, and suppose T is multiplicative on each sector corner. If Q is supported on sector P_k (i.e. $QP_k = Q = P_k Q$), then for each $0 \leq l < m$ the iterate $T^l(Q)$ is supported on the shifted sector P_{k-l} .

Proof. ✓ Proof OK Induction on l . The base case is the hypothesis $QP_k = Q = P_k Q$. For the inductive step, the multiplicative-domain identities and the cyclic shift relation $T(P_j) = P_{j-1}$ transfer the support from sector P_j to sector P_{j-1} . $\square$

Lemma 11.97 (Orbit iterate preserves orthogonal projection). orbit_iterate_isOrthogonalProjection ✓ Stmt OK Under the same cyclic shift and multiplicative-domain hypotheses as Lemma 11.96, suppose additionally that T maps every orthogonal projection supported on a single sector to an orthogonal projection. If Q is an orthogonal projection supported on P_k , then $T^l(Q)$ is an orthogonal projection for each $0 \leq l < m$.

Proof. ✓ Proof OK Induction on l . The base case is the hypothesis on Q . At each step, Lemma 11.96 guarantees that $T^n(Q)$ is supported on a single sector, so the one-step preservation hypothesis applies. $\square$

Lemma 11.98 (Orbit sum of a full sector projection). orbitSumProjection_eq_one_of_full_sector ✓ Stmt OK Let $(P_k)_{k=1}^m$ be orthogonal projections with $\sum_k P_k = \mathbb{1}$ and cyclic shift $T(P_{k+1}) = P_k$. Then for every sector index k ,

$$\sum_{l=0}^{m-1} T^l(P_k) = \mathbb{1}.$$

Proof. ✓ Proof OK By induction, $T^l(P_k) = P_{k-l}$ for each l . The orbit sum therefore runs over all distinct sectors, and the reindexing $l \mapsto k-l$ recovers $\sum_j P_j = \mathbb{1}$. $\square$

Theorem 11.99 (Sector irreducibility from orbit-sum lift). isIrreducibleOnCorner_of_cyclic_decomp_mps_of_hLift ✓ Stmt OK Let A be a left-canonical MPS tensor with irreducible adjoint transfer map, and let $(P_k)_{k=1}^m$ be the cyclic sector projections. Suppose the orbit-sum lift hypothesis holds: for every sector k and every subprojection $Q \leq P_k$ fixed by $(\mathcal{E}_A^\dagger)^m$, there exists a global projection R that is invariant under $\mathcal{E}_A^\dagger$ and faithfully reflects triviality ($Q = 0 \Leftrightarrow R = 0$, $Q = P_k \Leftrightarrow R = \mathbb{1}$). Then $(\mathcal{E}_A^\dagger)^m$ is irreducible on each corner P_k .

Proof. ✓ Proof OK Direct application of the cyclic-decomposition irreducibility theorem from spectral theory. $\square$

Lemma 11.100 (Fixed-point upgrade from irreducibility and trace preservation). hFixUpgrade_of_peripheral ✓ Stmt OK Let A be a left-canonical MPS tensor whose transfer map $\mathcal{E}_A$ is irreducible, and let Q be an orthogonal projection such that the corner $Q\mathcal{M}_D Q$ is preserved by $((\mathcal{E}_A^\dagger)^m)$. Then $((\mathcal{E}_A^\dagger)^m)(Q) = Q$.

Proof. ✓ Proof OK Let $\rho > 0$ be the distinguished fixed point of $\mathcal{E}_A$. Since $Q\mathcal{M}_D Q$ is preserved, the difference $Q - ((\mathcal{E}_A^\dagger)^m)(Q)$ can be rewritten as $Q(\mathcal{E}_A^\dagger)^m(\mathbb{1} - Q)Q$, hence it is positive semidefinite. The weighted trace pairing with ρ is invariant under $(\mathcal{E}_A^\dagger)^m$, so this positive semidefinite matrix has weighted trace zero. Faithfulness of the ρ -weighted trace therefore forces it to vanish, giving the claim. $\square$

Definition 11.101 (Sector fixed-point-algebra rigidity). SectorFixedPointAlgebraRigidity ✓ Stmt OK Let T be the adjoint transfer map and $(P_k)_{k=1}^m$ a cyclic family of orthogonal sector projections. We say that the sector dynamics satisfies fixed-point-algebra rigidity if, for each k , the one-step map T is multiplicative on the T^m -fixed operators supported on the corner $P_k\mathcal{M}_D P_k$: whenever

$$X = XP_k = P_k X, \quad Y = YP_k = P_k Y, \quad T^m(X) = X, \quad T^m(Y) = Y,$$

one has

$$T(XY) = T(X)T(Y).$$

Proof. ✓ Proof OK This is the abstract multiplicativity property used in the orbit-sum construction. Theorem 11.103 shows that it is automatic in the irreducible trace-preserving cyclic setting. $\square$

Lemma 11.102 (Projection step from sector fixed-point-algebra rigidity). hProjStep_of_sectorFixedPointAlgebraRigidity ✓ Let X be an orthogonal projection supported on P_k and fixed by T^m , then $T(X)$ is again an orthogonal projection.

Proof. ✓ Proof OK Kraus maps preserve conjugate transpose, so $T(X)$ is Hermitian. The rigidity hypothesis gives

$$T(X)^2 = T(X^2) = T(X),$$

since $X^2 = X$ for an orthogonal projection. Therefore $T(X)$ is again an orthogonal projection. $\square$

Theorem 11.103 (Sector fixed-point-algebra rigidity from irreducibility).

[sectorFixedPointAlgebraRigidity_of_irreducible_cy](#)

A be a left-canonical MPS tensor with irreducible adjoint transfer map, and let $(P_k)_{k=1}^m$ be cyclic orthogonal sector projections with $\sum_k P_k = \mathbb{1}$. Then the sector fixed-point-algebra rigidity property holds automatically for $\mathcal{E}_A^\dagger$.

Proof. ✓ Proof OK Fix k and let X, Y be operators supported on P_k and fixed by $(\mathcal{E}_A^\dagger)^m$. For such an X , the orbit sum

$$R_X := \sum_{l=0}^{m-1} (\mathcal{E}_A^\dagger)^l(X)$$

is fixed by $\mathcal{E}_A^\dagger$ by Lemma 11.95. Since the trace-preserving hypothesis makes $\mathcal{E}_A^\dagger$ unital, the irreducible unital fixed-point theorem implies that R_X is a scalar multiple of $\mathbb{1}$. Compressing by P_k recovers X , so $X = cP_k$; the same argument gives $Y = c'P_k$. Therefore

$$\mathcal{E}_A^\dagger(XY) = \mathcal{E}_A^\dagger((cc')P_k) = (cc')P_{k-1} = \mathcal{E}_A^\dagger(X)\mathcal{E}_A^\dagger(Y),$$

using the cyclic relation $\mathcal{E}_A^\dagger(P_k) = P_{k-1}$. $\square$

Theorem 11.104 (Sector fixed-point-algebra rigidity from scalar blocked fixed points).

[sectorFixedPointAlgebraRigidity_of](#)

A be a left-canonical MPS tensor and let $(P_k)_{k=1}^m$ be cyclic orthogonal sector projections with $\sum_k P_k = \mathbb{1}$. Suppose that for each k and each element $X \in P_k M_D(\mathbb{C}) P_k$ satisfying $\mathcal{E}_{C_k}^\dagger(X) = X$ (where C_k is the blocked sector tensor), there exists $c \in \mathbb{C}$ with $X = c \cdot \mathbb{1}$. Then the sector fixed-point-algebra rigidity property holds for $\mathcal{E}_A^\dagger$.

Proof. ✓ Proof OK Fix k and let X, Y be operators supported on P_k and fixed by $(\mathcal{E}_A^\dagger)^m$. Lift each to the corresponding compressed sector via the intertwining equivalence φ_k ; the scalar fixed-point hypothesis yields $X = c_X P_k$ and $Y = c_Y P_k$. Then

$$\mathcal{E}_A^\dagger(XY) = \mathcal{E}_A^\dagger((c_X c_Y)P_k) = (c_X c_Y)P_{k-1} = \mathcal{E}_A^\dagger(X)\mathcal{E}_A^\dagger(Y),$$

using the cyclic relation $\mathcal{E}_A^\dagger(P_k) = P_{k-1}$ and the linearity of $\mathcal{E}_A^\dagger$. $\square$

Lemma 11.105 (Orbit-sum construction of the sector lift).

[hLift_cyclicDecomp_mps_of_fixUpgrade](#)

✓ Stmt OK Let

A be a left-canonical MPS tensor, $(P_k)_{k=1}^m$ a cyclic family of orthogonal sector projections with $\sum_k P_k = \mathbb{1}$, and set $T = \mathcal{E}_A^\dagger$. Assume two auxiliary facts, both discharged from the block-diagonal canonical-form analysis of [DICCSPG17]:

1. (Projection step.) *For each sector index k and each orthogonal projection X supported on P_k , the image $T(X)$ is again an orthogonal projection.*
2. (Fixed-point upgrade.) *For each sector index k and each orthogonal subprojection $Q \leq P_k$, if Q is a corner fixed point of T^m then $T^m(Q) = Q$.*

Then for every sector k and every orthogonal subprojection $Q \leq P_k$ which is a corner fixed point of T^m , the orbit sum

$$R := \sum_{l=0}^{m-1} T^l(Q)$$

is an orthogonal projection, preserved by T , with $Q = 0 \Leftrightarrow R = 0$ and $Q = P_k \Leftrightarrow R = \mathbb{1}$. In particular R witnesses the orbit-sum lift hypothesis of Theorem 11.99.

Proof. ✓ Proof OK The iterates $T^l(Q)$ are each orthogonal projections (repeatedly using the projection-step hypothesis), supported on the shifted sector P_{k-l} , and are therefore pairwise orthogonal because distinct sectors are. This gives both the Hermitian and idempotent conjuncts for R . Fixedness under T follows from $T^m(Q) = Q$ and Lemma 11.95, and corner preservation from the adjoint trace-preserving relation. The zero equivalence uses $R \cdot Q = Q$ (the diagonal term $l = 0$ is Q , off-diagonal terms vanish by pairwise orthogonality); the full-sector equivalence uses $P_k \cdot R \cdot P_k = Q$ and Lemma 11.98. □

Lemma 11.106 (Orbit-sum construction, projection-step variant). hLift_cyclicDecomp_mps_of_projStep ✓ Stmt OK Let A be a left-canonical MPS tensor with irreducible adjoint transfer map, and let $(P_k)_{k=1}^m$ be cyclic orthogonal sector projections. Assume the projection-step hypothesis of Lemma 11.105. Then the orbit-sum lift holds: for every orthogonal subprojection $Q \leq P_k$ whose corner is preserved by $(\mathcal{E}_A^\dagger)^m$, there is a global projection R invariant under $\mathcal{E}_A^\dagger$ with the same zero/full-sector tests.

Proof. ✓ Proof OK Combine the original orbit-sum construction with Lemma 11.100, which supplies the fixed-point upgrade from irreducibility and trace preservation. The irreducibility hypothesis needed there is obtained by transporting irreducibility from the adjoint transfer map to the tensor and then back to the primal transfer map. □

Lemma 11.107 (Orbit-sum construction, fixed-point-algebra variant). hLift_cyclicDecomp_mps_of_sectorFixedPointAlgebraRig Let A be a left-canonical MPS tensor with irreducible adjoint transfer map, and let $(P_k)_{k=1}^m$ be cyclic orthogonal sector projections. Assume the single fixed-point-algebra rigidity hypothesis from Definition 11.101. Then the orbit-sum lift holds: for every orthogonal subprojection $Q \leq P_k$ whose corner is preserved by $(\mathcal{E}_A^\dagger)^m$, there is a global projection R invariant under $\mathcal{E}_A^\dagger$ with the same zero/full-sector tests.

Proof. ✓ Proof OK First apply Lemma 11.100 to upgrade corner preservation of Q under $(\mathcal{E}_A^\dagger)^m$ to an actual fixed-point relation. The rigidity hypothesis then yields the one-step projection preservation needed along the orbit of Q , via Lemma 11.102. With those two facts in hand, the original orbit-sum construction from Lemma 11.105 applies verbatim. In the irreducible trace-preserving cyclic case, the rigidity hypothesis here is discharged by Theorem 11.103. □

Theorem 11.108 (Orbit-sum construction, unconditional variant). hLift_cyclicDecomp_mps ✓ Stmt OK Let A be a left-canonical MPS tensor with irreducible adjoint transfer map, and let $(P_k)_{k=1}^m$ be cyclic orthogonal sector projections. Then for every orthogonal subprojection $Q \leq P_k$ whose corner is preserved by $(\mathcal{E}_A^\dagger)^m$, there is a global projection R invariant under $\mathcal{E}_A^\dagger$ with the same zero/full-sector tests.

Proof. ✓ Proof OK Apply Lemma 11.107 with the rigidity hypothesis supplied by Theorem 11.103. □

Theorem 11.109 (Sector irreducibility from the orbit-sum construction). isIrreducibleOnCorner_of_cyclic_decomp_mps_of_fix Let A be a left-canonical MPS tensor with irreducible adjoint transfer map, and let $(P_k)_{k=1}^m$ be cyclic orthogonal sector projections. Assume the projection-step and fixed-point-upgrade hypotheses of Lemma 11.105. Then $(\mathcal{E}_A^\dagger)^m$ is irreducible on each corner P_k .

Proof. ✓ Proof OK Apply Lemma 11.105 to obtain the orbit-sum lift, then apply Theorem 11.99. □

Theorem 11.110 (Sector irreducibility from the projection-step hypothesis). isIrreducibleOnCorner_of_cyclic_decomp_mps_of_sec
A be a left-canonical MPS tensor with irreducible adjoint transfer map, and let $(P_k)_{k=1}^m$ be cyclic orthogonal sector projections. Assume only the projection-step hypothesis from Lemma 11.106. Then $(\mathcal{E}_A^\dagger)^m$ is irreducible on each corner P_k .

Proof. ✓ Proof OK Apply Lemma 11.106 and then Theorem 11.99. □

Theorem 11.111 (Sector irreducibility from fixed-point-algebra rigidity). isIrreducibleOnCorner_of_cyclic_decomp_mps_of_sec
A be a left-canonical MPS tensor with irreducible adjoint transfer map, and let $(P_k)_{k=1}^m$ be cyclic orthogonal sector projections. Assume only the fixed-point-algebra rigidity hypothesis from Definition 11.101. Then $(\mathcal{E}_A^\dagger)^m$ is irreducible on each corner P_k .

Proof. ✓ Proof OK Apply Lemma 11.107 and then Theorem 11.99. □

Theorem 11.112 (Unconditional sector irreducibility). isIrreducibleOnCorner_of_cyclic_decomp_mps ✓ Stmt OK *Let A be a left-canonical MPS tensor with irreducible adjoint transfer map, and let $(P_k)_{k=1}^m$ be cyclic orthogonal sector projections. Then $(\mathcal{E}_A^\dagger)^m$ is irreducible on each corner P_k .*

Proof. ✓ Proof OK Apply Theorem 11.111 with the rigidity hypothesis supplied by Theorem 11.103. □

11.13 Reduction to primitive blocks

Theorem 11.113 (Projection-step hypothesis yields primitive blocked sectors). primitive_and_irreducible_sectorBlocks_of_
A be a left-canonical irreducible tensor whose adjoint transfer map has peripheral spectrum $\{\gamma^j : 0 \leq j < m\}$ for a primitive m -th root of unity γ . Let $(C_u)_{u=1}^m$, $(P_u)_{u=1}^m$, and $\varphi_u : M_{\dim C_u}(\mathbb{C}) \xrightarrow{\sim} P_u M_D(\mathbb{C}) P_u$ be the blocked cyclic-sector data. If the one-step projection-preservation hypothesis of Lemma 11.106 holds, then every blocked sector tensor C_u has primitive transfer map and is irreducible.

Proof. ✓ Proof OK Theorem 11.110 gives irreducibility of $(\mathcal{E}_A^\dagger)^m$ on each corner P_u . The blocked cyclic-sector data provide compression equivalences φ_u intertwining the adjoint transfer map of C_u with the corresponding corner restriction of $(\mathcal{E}_A^\dagger)^m$. The same cyclic peripheral-spectrum argument that produces the sector decomposition shows that each corner restriction is primitive. Transport primitivity and irreducibility across φ_u , then use Theorem 11.34 to pass from the adjoint blocked map to the primal transfer map of C_u . □

Theorem 11.114 (Fixed-point-algebra rigidity yields primitive blocked sectors). primitive_and_irreducible_sectorBlocks_of_
A be a left-canonical irreducible tensor whose adjoint transfer map has peripheral spectrum $\{\gamma^j : 0 \leq j < m\}$ for a primitive m -th root of unity γ . Let $(C_u)_{u=1}^m$, $(P_u)_{u=1}^m$, and $\varphi_u : M_{\dim C_u}(\mathbb{C}) \xrightarrow{\sim} P_u M_D(\mathbb{C}) P_u$ be the blocked cyclic-sector data. If the fixed-point-algebra rigidity hypothesis from Definition 11.101 holds, then every blocked sector tensor C_u has primitive transfer map and is irreducible.

Proof. ✓ Proof OK Theorem 11.111 gives irreducibility of $(\mathcal{E}_A^\dagger)^m$ on each corner P_u . The blocked cyclic-sector data provide compression equivalences φ_u intertwining the adjoint transfer map of C_u with the corresponding corner restriction of $(\mathcal{E}_A^\dagger)^m$. The same cyclic peripheral-spectrum argument that produces the sector decomposition shows that each corner restriction is primitive.

Transport primitivity and irreducibility across φ_u , then use Theorem 11.34 to pass from the adjoint blocked map to the primal transfer map of C_u . $\square$

Theorem 11.115 (Scalar blocked fixed points yield primitive blocked sectors). [primitive_and_irreducible_sectorBlocks_of_cy](#)
A be a left-canonical irreducible tensor whose adjoint transfer map has peripheral spectrum $\{\gamma^j : 0 \leq j < m\}$ for a primitive m -th root of unity γ . Let $(C_u)_{u=1}^m$, $(P_u)_{u=1}^m$, and $\varphi_u : M_{\dim C_u}(\mathbb{C}) \xrightarrow{\sim} P_u M_D(\mathbb{C}) P_u$ be the blocked cyclic-sector data. If every $(\mathcal{E}_{C_u}^\dagger)$ -fixed element in the compressed sector $P_u M_D(\mathbb{C}) P_u$ is a scalar multiple of the identity, then every blocked sector tensor C_u has primitive transfer map and is irreducible.

Proof. [✓ Proof OK](#) Apply Theorem 11.104 to obtain sector fixed-point-algebra rigidity from the scalar fixed-point hypothesis, then apply Theorem 11.114. $\square$

Theorem 11.116 (Unconditional primitive blocked sectors). [primitive_and_irreducible_sectorBlocks_of_cyclic_decomp_after_bl](#)
A be a left-canonical irreducible tensor whose adjoint transfer map has peripheral spectrum $\{\gamma^j : 0 \leq j < m\}$ for a primitive m -th root of unity γ . Let $(C_u)_{u=1}^m$, $(P_u)_{u=1}^m$, and $\varphi_u : M_{\dim C_u}(\mathbb{C}) \xrightarrow{\sim} P_u M_D(\mathbb{C}) P_u$ be the blocked cyclic-sector data. Then every blocked sector tensor C_u has primitive transfer map and is irreducible.

Proof. [✓ Proof OK](#) Theorem 11.112 gives irreducibility of $(\mathcal{E}_A^\dagger)^m$ on each corner P_u . The blocked cyclic-sector data provide compression equivalences φ_u intertwining the adjoint transfer map of C_u with the corresponding corner restriction of $(\mathcal{E}_A^\dagger)^m$. The same cyclic peripheral-spectrum argument that produces the sector decomposition shows that each corner restriction is primitive. Transport primitivity and irreducibility across φ_u , then use Theorem 11.34 to pass from the adjoint blocked map to the primal transfer map of C_u . $\square$

Theorem 11.117 (Arbitrary tensor to primitive block decomposition). [exists_tp_primitive_blockDecomp_after_blocking](#) [✓ S](#)
any MPS tensor A , there exist a period $p \geq 1$, a zero-tail bond dimension D_0 , nonzero weights $(\mu_k)_{k=1}^r$, and left-canonical blocks $(B_k)_{k=1}^r$ with primitive transfer maps and positive bond dimensions, such that

$$V^{(N)}(A^{[p]})_\sigma = V^{(N)}(\mathbf{0}_{d^{[p]}, D_0})_\sigma + V^{(N)}(\bigoplus_k \mu_k B_k)_\sigma.$$

Proof. [✓ Proof OK](#) Apply Theorem 11.48 to get the trivial block and left-canonical nontrivial blocks. Apply Theorem 11.66 to find a common blocking period p making all transfer maps primitive. Lemma 11.53 keeps the blocked weights nonzero, and Theorem 11.165 transports the zero-tail/nonzero MPV identity through the blocking step. $\square$

Theorem 11.118 (TP-primitive irreducible tensor is normal). [isNormal_of_tp_primitive_irreducible](#) [✓ Stmt OK](#) *Let A be a left-canonical MPS tensor with $D > 0$, irreducible transfer map, and peripheral spectrum $\{1\}$. Then A is normal.*

Proof. [✓ Proof OK](#) Peripheral primitivity and irreducibility yield the spectral-gap condition. The PSD fixed point is positive definite by Theorem 8.3. Apply Theorem 10.65. $\square$

Theorem 11.119 (Blocking preserves normality). [isNormal_blockTensor_of_isNormal](#) [✓ Stmt OK](#) *If A is normal and $p \geq 1$, then $A^{[p]}$ is normal.*

Proof. [✓ Proof OK](#) Since A is normal, there exists N with $S_N(A) = M_D(\mathbb{C})$. Then $S_{Np}(A) = M_D(\mathbb{C})$ (word products at multiples still span), and $S_N(A^{[p]}) \supseteq S_{Np}(A) = M_D(\mathbb{C})$. $\square$

11.13.1 Basis of Normal Tensors

Theorem 11.120 (Sorted reindexing by strictly decreasing weight norms). [exists_sorted_reindexing](#) ✓ Stmt OK If nonzero weights $(\mu_k)_{k=1}^r$ have pairwise distinct norms, then there is a permutation of $\{1, \dots, r\}$ under which $\|\mu_k\|$ is strictly decreasing.

Theorem 11.121 (Sorted decomposition preserves the represented MPS family). [exists_sorted_blockDecomp_of_distinct_norms](#) If the same distinct-norm hypothesis, the sorted block family is gauge equivalent to the original decomposition and has strictly decreasing norms.

Theorem 11.122 (Normal canonical form after sorting). [exists_sortedNCF_of_distinct_norms](#) ✓ Stmt OK If each block is irreducible, trace-preserving, primitive, and has positive dimension, then sorting by strictly decreasing weight norms produces a normal canonical form.

Definition 11.123 (Granular sector decomposition). [trivialSectorDecomp](#) ✓ Stmt OK Given nonzero weights $(\mu_k)_{k=1}^r$ and blocks $(B_k)_{k=1}^r$, form a sector decomposition with one basis block for each original block, one copy in each sector, and sector weight μ_k on the k th sector.

Lemma 11.124 (Granular sector decomposition preserves the represented family). [sameMPV2_trivialSectorDecomp](#) ✓ Stmt OK granular sector decomposition associated to $(\mu_k, B_k)_k$ represents the same MPS family as the original weighted block sum $\bigoplus_k \mu_k B_k$.

Proof. ✓ Proof OK Expanding the sector decomposition gives the sum $\sum_k \mu_k^N V^{(N)}(B_k)_\sigma$, which is exactly the finite block-sum expression for $\bigoplus_k \mu_k B_k$. $\square$

Theorem 11.125 (Trivial sector decomposition for the sorted distinct-norm case). [exists_trivialSectorDecomp_of_sorted_distinct_norms](#) If all weights are nonzero and strictly decreasing in norm, one obtains a sector decomposition with one copy per sector that preserves the represented MPS family.

Definition 11.126 (Norm-class partition). [NormClassGroupingData](#) ✓ Stmt OK The grouping data consists of: norm classes, one representative value per class, class multiplicities, an enumeration of members in each class, and a regrouping identity recovering the full finite sum.

Definition 11.127 (Norm-class enumeration in decreasing order). [normClassGroupingData](#) ✓ Stmt OK From any finite weight family, construct norm classes ordered by strictly decreasing norm values.

Theorem 11.128 (BNT grouping theorem). [exists_bnt_grouping](#) ✓ Stmt OK If equal-norm blocks represent the same MPS family, then one obtains a sector decomposition whose sector norms are strictly decreasing.

11.13.2 Equal-Norm Blocks

Theorem 11.129 (Gauge equivalence from non-decaying cross-overlap). [gaugePhaseEquiv_of_nonDecaying_overlap](#) ✓ Stmt OK trace-preserving irreducible blocks, a non-decaying cross-overlap forces equal bond dimensions and gauge equivalence up to a unit-modulus phase.

Theorem 11.130 (Gauge phases have unit modulus for primitive irreducible blocks). [norm_gaugePhase_eq_one_of_irr_TP_primitive_blocks](#) trace-preserving primitive irreducible blocks, every gauge-phase equivalence has a phase factor of modulus 1.

Proof. ✓ Proof OK Primitivity forces the peripheral spectrum to consist of $\{1\}$ alone, so the gauge phase must have modulus one. $\square$

Theorem 11.131 (BNT partition from gauge equivalence). [exists_bnt_grouping_of_gaugePhaseEquiv](#) ✓ Stmt OK If equal-norm blocks are related by gauge equivalence with unit-modulus phase, the phases can be absorbed into sector weights to obtain a BNT grouping with strictly decreasing sector norms.

Theorem 11.132 (Sector decomposition from TP primitive irreducible blocks). [exists_sectorDecomp_of_tp_primitive_irr_blocks](#) a trace-preserving primitive irreducible block decomposition with nonzero weights and non-decaying cross-overlaps for equal-norm blocks, one obtains a sector decomposition with strict norm ordering.

Theorem 11.133 (Existential BNT independence from overlap limits). [exists_eventually_linearIndependent_of_overlap_tends](#) the self-overlaps of a finite block family tend to 1 and the cross-overlaps of distinct blocks tend to 0, then the MPV states are linearly independent for all sufficiently large system sizes.

Proof. ✓ Proof OK This is Theorem 13.2, restated with an explicit threshold N_0 . □

Theorem 11.134 (BNT independence for separated primitive normal blocks). [exists_eventually_linearIndependent_of_tp_primitive_normal_blocks](#) trace-preserving primitive irreducible blocks of positive bond dimension that are pairwise not gauge-phase equivalent, the block MPV states are linearly independent for all sufficiently large system sizes.

Proof. ✓ Proof OK Primitivity and trace preservation give self-overlap convergence to 1. BNT separation gives cross-overlap convergence to 0 for distinct blocks. The result follows from Theorem 11.133. □

Theorem 11.135 (Granular sector decomposition from eventual BNT independence). [exists_bnt_sectorDecomp_of_linearIndependent_blocks](#) the MPV states of the blocks are linearly independent for all sufficiently large system sizes and all weights are nonzero, then the granular sector decomposition represents the same MPS family and satisfies the BNT linear-independence condition.

Proof. ✓ Proof OK The equality of represented families is Lemma 11.124. The assumed eventual linear independence is exactly the BNT condition for the basis blocks of the granular sector decomposition. □

Theorem 11.136 (Separated primitive normal sector construction). [exists_bnt_sectorDecomp_of_tp_primitive_irr_blocks_of_bond_dimension](#) trace-preserving primitive irreducible blocks of positive bond dimension with nonzero weights, if distinct blocks are pairwise not gauge-phase equivalent, then the granular sector decomposition represents the same MPS family and satisfies the BNT linear-independence condition. Equal-modulus separated blocks remain as distinct basis blocks.

Proof. ✓ Proof OK The granular sector decomposition preserves the represented family by Lemma 11.124. The separated primitive normal hypotheses give the eventual linear independence by Theorem 11.134. □

Definition 11.137 (MPV phase class data). [MPVPhaseEquiv](#) [MPVPhaseClassData](#) [mpvPhaseClassData](#) ✓ Stmt OK For a finite family of blocks, put $j \sim k$ when there is a nonzero scalar factor ζ such that $|V^{(N)}(B_k)\rangle = \zeta^N |V^{(N)}(B_j)\rangle$ for every length N . The class data enumerate the finite quotient, choose one representative per class, give the scalar relating the representative to each member, and prove that distinct representatives are not gauge-phase equivalent.

Theorem 11.138 (Phase-class BNT sector construction). [exists_bnt_sectorDecomp_of_tp_primitive_irr_blocks](#) ✓ Stmt OK For trace-preserving primitive irreducible blocks of positive bond dimension with nonzero weights, there is a sector decomposition representing the same weighted block sum and satisfying the BNT linear-independence condition. The construction chooses one representative per MPV phase class and absorbs the nonzero scalar factors into the sector weights.

Proof. ✓ Proof OK Finite phase classes partition the original block indices. Expanding the sector decomposition gives the original finite sum after replacing each class member by its representative and moving the factor ζ^N into $(\zeta\mu_k)^N$. If two chosen representatives were gauge-phase equivalent, they would lie in the same MPV phase class, contradicting their being distinct classes. Therefore the representatives satisfy the separated primitive-normal hypothesis, and Theorem 11.134 gives the BNT linear-independence condition. $\square$

Theorem 11.139 (Phase-class BNT sector construction with one-sided overlap data). exists_bnt_sectorDecomp_of_tp_primitive_irr_blocks_with_overlapData_and_basisSpan
trace-preserving primitive irreducible blocks of positive bond dimension with nonzero weights, there is a sector decomposition, obtained by choosing one block per MPV phase class, which represents the same weighted block sum, satisfies the BNT linear-independence condition, and has a basis satisfying the single-family overlap-orthogonality data. If the original blocks are one-site injective, then the chosen basis blocks are one-site injective.

Proof. ✓ Proof OK The proof is the phase-class construction of Theorem 11.138. The chosen basis blocks keep positive bond dimension, normalization, and any supplied one-site injectivity from the original trace-preserving primitive irreducible blocks. Primitivity gives the self-overlap limit, while the separation of distinct chosen phase-class blocks gives off-overlap decay. $\square$

Theorem 11.140 (Phase-class BNT sector construction with overlap data). exists_bnt_sectorDecomp_of_tp_primitive_irr_blocks_with_overlapData_and_basisSpan
trace-preserving primitive irreducible blocks of positive bond dimension, with nonzero weights, suppose moreover that the blocks are injective. Then there is a sector decomposition, obtained by choosing one block per MPV phase class, which represents the same weighted block sum, satisfies the BNT linear-independence condition, and has chosen basis blocks of positive bond dimension which are injective and trace-preserving, have self-overlap tending to 1, and have mutual overlaps tending to 0 for distinct basis blocks.

Proof. ✓ Proof OK Apply Theorem 11.139 and use the supplied injectivity of the original blocks to fill the injectivity field required by this theorem's explicit conclusion. $\square$

Theorem 11.141 (Chosen phase representatives preserve finite-length spans). representative_mpv_span_eq
exists_bnt_sectorDecomp_of_tp_primitive_irr_blocks_with_overlapData_and_basisSpan ✓ Stmt OK *For any finite block family, the chosen MPV phase-class representatives span the same finite-length MPV subspace as the original blocks at every system size. Consequently, for trace-preserving primitive irreducible injective blocks with nonzero weights, the one-sided BNT sector construction can be chosen with all overlap data from Theorem 11.140 and with this representative-span identity.*

Proof. ✓ Proof OK Each representative is one of the original blocks. Conversely, every original block appears in a phase class and differs from that class representative by a nonzero scalar power at each length, so its MPV state lies in the representative span. The one-sided sector theorem then gives this identity for the chosen sector basis. $\square$

Theorem 11.142 (TP primitive irreducible version with eventual BNT independence). exists_bnt_sectorDecomp_of_tp_primitive_irr_blocks_with_overlapData_and_basisSpan
trace-preserving primitive irreducible blocks of positive bond dimension, if the block MPV states are linearly independent for all sufficiently large system sizes and the weights are nonzero, then there is a sector decomposition representing the same MPS family and satisfying the BNT linear-independence condition.

Proof. ✓ Proof OK Apply Theorem 11.135; the additional trace-preserving, primitive, irreducible, and positivity assumptions are carried along for compatibility with the one-sided BNT construction. $\square$

11.13.3 Cyclic Sectors

Definition 11.143 (Left sector tensor). [leftSectorTensor](#) ✓ Stmt OK For a projection P , the left sector tensor is obtained by left multiplication of each Kraus operator by P .

Lemma 11.144 (Commuting letters imply commuting words). [commutes_evalWord_of_commutes_letters](#) ✓ Stmt OK If a projection commutes with every Kraus operator, it commutes with every finite word product.

Lemma 11.145 (Left multiplication identity for sector words). [left_mul_evalWord_leftSectorTensor_of_commutes](#) ✓ Stmt OK If a projection commutes with all Kraus operators, then word evaluation for the left sector tensor equals left multiplication by the projection.

Lemma 11.146 (Sector support). [leftSectorTensor_supported](#) ✓ Stmt OK The left sector tensor is supported on the chosen projection.

Theorem 11.147 (Adjoint-fixed projections commute with Kraus operators). [commutes_letters_of_adjoint_fixed_projection](#) ✓ Stmt OK If a projection is fixed by the adjoint transfer map of a trace-preserving tensor, then it commutes with each Kraus operator.

Theorem 11.148 (Left-canonical primitive blocking from irreducibility). [exists_blockTensor_leftCanonical_isPrimitive](#) ✓ Stmt OK If a trace-preserving irreducible tensor, a positive blocking period yields a blocked tensor that is left-canonical and primitive.

Theorem 11.149 (Normality from primitivity and a positive-definite fixed point). [isNormal_of_isPrimitiveMPS](#) ✓ Stmt OK If a primitive tensor admitting a positive-definite fixed point of its transfer map is normal.

Theorem 11.150 (Zero-word evaluation for the zero tensor). [evalWord_eq_zero_of_all_zero](#) ✓ Stmt OK Every nonempty finite word evaluates to zero when all Kraus operators are zero.

Theorem 11.151 (Zero MPV for the zero tensor). [mpv_eq_zero_of_all_zero](#) ✓ Stmt OK For a nonempty spin chain, the MPS value of every spin word is zero for the zero tensor.

Theorem 11.152 (Nonzero Kraus operator in irreducible dimension at least two). [exists_nonzero_kraus_of_isIrreducibleTensor](#) ✓ Stmt OK If an irreducible tensor of bond dimension at least two has at least one nonzero Kraus operator.

Theorem 11.153 (All-zero tensors are irreducible only in dimension at most one). [isIrreducibleTensor_allZero_dim_le_one](#) ✓ Stmt OK If every Kraus operator vanishes and the tensor is irreducible, then the bond dimension is at most one.

Theorem 11.154 (Irreducible block decomposition with trace-preserving gauge). [exists_irreducible_blockDecomp_with_tpGauge](#) ✓ Stmt OK If a tensor is irreducible, removing a trivial block, one obtains a block decomposition into irreducible blocks with a trace-preserving gauge.

11.13.4 Reduction and structural after-blocking results

Theorem 11.155 (Blockwise TP-gauge normalization of irreducible blocks). [exists_tp_gauge_blockwise](#) ✓ Stmt OK Suppose A generates the same MPV family as a block-diagonal tensor $\bigoplus_{k=1}^r B_k$ with each block B_k irreducible and containing at least one nonzero Kraus operator. Then, after rescaling the sector weights, one obtains blocks (C_k) such that $A \sim \bigoplus_k \mu_k C_k$, each C_k is irreducible, each C_k is left-canonical, all $\mu_k \neq 0$, and all bond dimensions are positive.

Proof. ✓ Proof OK The Perron–Frobenius TP-gauge construction is applied separately to each irreducible block. Gauge equivalence preserves irreducibility and the MPV family, and the positive scaling factors are absorbed into the new weights. $\square$

Theorem 11.156 (Primitive nonzero blocks are normal). isNormal_live_block_of_primitive ✓ Stmt OK A is left-canonical irreducible tensor with primitive transfer map and positive bond dimension is normal.

Proof. ✓ Proof OK Apply Theorem 11.118. □

Theorem 11.157 (Blocking preserves irreducibility for primitive irreducible blocks). isIrreducibleTensor_blockTensor_of_tp A is left-canonical, irreducible, primitive, and has positive bond dimension, then every positive blocking length P preserves irreducibility: $A^{[P]}$ is again irreducible.

Proof. ✓ Proof OK A positive-definite fixed point of $\mathcal{E}_A$ remains fixed by the blocked transfer map. Primitivity gives uniqueness of positive semidefinite fixed points for the blocked map, hence the blocked transfer map is irreducible; this is exactly the irreducibility criterion for the blocked tensor. □

Theorem 11.158 (Extra blocking of primitive irreducible sector blocks). tp_primitive_irreducible_extra_blocking ✓ Stmt OK A is a sector block is already trace-preserving, has primitive transfer map, and is tensor-irreducible, then every later positive blocking length k preserves all three properties. This k is the subsequent common-refinement or injectivity length, distinct from the period-removal length that produced the sector block.

Proof. ✓ Proof OK Trace preservation follows from blocking a trace-preserving tensor, primitivity from taking a positive power of a primitive transfer map, and irreducibility from Theorem 11.157. □

Theorem 11.159 (Sorted TP-primitive irreducible data already forms a normal canonical form). isNormalCanonicalForm_of_tp_primitive_irr_sorted ✓ Stmt OK Let (μ_k, A_k) be a finite block family with strictly decreasing norms $|\mu_1| > \dots > |\mu_r|$. If every block A_k is left-canonical, primitive, irreducible, has positive bond dimension, and every $\mu_k \neq 0$, then (μ_k, A_k) is a normal canonical form.

Proof. ✓ Proof OK This is the defining constructor for normal canonical form with all required data supplied by the hypotheses. □

Theorem 11.160 (Normal canonical form from an already primitive block decomposition). exists_normalCanonicalForm_of_primitive_input ✓ Stmt OK If A already admits a block decomposition into irreducible left-canonical primitive blocks with pairwise distinct weight norms, nonzero weights, and positive bond dimensions, then some blocked tensor $A^{[p]}$ is MPV-equivalent to a normal canonical form.

Proof. ✓ Proof OK Apply Theorem 11.44 to the given primitive block decomposition. □

Theorem 11.161 (Conditional block matching after blocking). weakFundamentalTheorem_conditional ✓ Stmt OK Consider two block families in normal canonical form, with no two distinct blocks of the same dimension related by gauge-phase equivalence. If two tensors decompose into these families with coefficient sequences having nonzero limits, and the two tensors are asymptotically proportional coefficient-wise, then the numbers of blocks agree and the blocks match up to permutation and gauge-phase equivalence.

Proof. ✓ Proof OK This is the block-matching statement from the multi-block fundamental theorem applied to two decompositions that already satisfy the normal-canonical-form and BNT-separation hypotheses. □

Theorem 11.162 (Common blocking period for two primitive normal tensors). [bilateral_commonPeriod_blocking_tp_primitiv](#)

A and B be left-canonical normal tensors of positive bond dimensions. Suppose $A^{[p_A]}$ and $B^{[p_B]}$ already have primitive transfer maps for some positive periods p_A, p_B . Then there exists a common positive period p such that both $A^{[p]}$ and $B^{[p]}$ are left-canonical, primitive, and normal.

Proof. [✓ Proof OK](#) Take p to be the least common multiple of p_A and p_B . Primitivity follows from divisibility, and left-canonical normalization and normality are both preserved under blocking. $\square$

Theorem 11.163 (Structural after-blocking form of the fundamental theorem). [fundamentalTheorem_after_blocking_structur](#)

two tensors generate the same MPV family, then each tensor admits some blocking after which it decomposes into nonzero left-canonical blocks with primitive transfer maps and positive bond dimensions.

Proof. [✓ Proof OK](#) Apply Theorem 11.117 separately to the two tensors. $\square$

Theorem 11.164 (Zero-tail decomposition after blocking). [zeroTail_mpv_decomp_blockTensor](#) [✓ Stmt OK](#) Suppose

A is expressed as a zero-tail tensor plus a nonzero part L at all system sizes. For every positive blocking length p , the blocked tensor $A^{[p]}$ is expressed as the same zero-tail bond dimension plus $L^{[p]}$. The zero-tail term still contributes only at blocked length 0.

Proof. [✓ Proof OK](#) Flatten a blocked word of length N to an original word of length Np . Since $p > 0$, this length is zero exactly when $N = 0$, so the zero-tail MPV is unchanged by the reblocking convention. $\square$

Theorem 11.165 (Zero-tail weighted block decomposition after blocking). [zeroTail_toTensorFromBlocks_blockPower](#) [✓ Stmt](#)

A is expressed as a zero-tail tensor plus the weighted nonzero-block tensor $\bigoplus_k \mu_k B_k$. For every positive blocking length p , $A^{[p]}$ is expressed as the same zero-tail contribution plus the weighted blocked nonzero tensor $\bigoplus_k \mu_k^p B_k^{[p]}$.

Proof. [✓ Proof OK](#) Apply Theorem 11.164 to the assembled weighted nonzero tensor and rewrite the blocked assembled tensor using Theorem 11.50. $\square$

Theorem 11.166 (Zero-tail identity for nonzero block tensors). [nonzeroBlock_positive_sameMPV2_and_zeroTail_identity_of_sameM](#)

two tensors have the same MPV family and each is expressed as a zero-tail tensor plus a weighted nonzero-block tensor. Then the two nonzero-block tensors agree on every positive system size, and at size 0 the equality is exactly the equality of the two zero-tail contributions plus the two nonzero bond-dimension traces.

Proof. [✓ Proof OK](#) For $N > 0$, the MPV of a zero-tail tensor is zero, so the two decompositions can be compared directly through the common MPV family. For $N = 0$, the zero-tail MPV is its bond dimension, giving the stated length-zero identity. $\square$

Theorem 11.167 (Reblocked nonzero-part equality with zero-tail identity). [nonzeroBlock_blockPower_positive_sameMPV2_and_2](#)

the setting of Theorem 11.166, after any positive common blocking length p , the powered weighted nonzero-block tensors agree on all positive blocked system sizes. At blocked length 0, the same zero-tail identity holds with weights transported from μ_k to μ_k^p .

Proof. [✓ Proof OK](#) Use Theorem 11.165 on both zero-tail decompositions, block the original MPV equality, and apply Theorem 11.166 at the blocked physical dimension. $\square$

Theorem 11.168 (Equal zero tails give full nonzero-part MPV equality). [nonzeroBlock_sameMPV2_of_sameMPV2_of_zeroTail_eq](#)

the setting of Theorem 11.166, if the two zero-tail dimensions are equal, then the two nonzero-block tensors have the same MPV family, including the length-zero case.

Proof. ✓ Proof OK The positive-length cases are exactly the first conclusion of Theorem 11.166. At $N = 0$, substitute the equality of zero-tail dimensions into the length-zero identity and cancel the common scalar term. $\square$

Theorem 11.169 (Structural after-blocking form with zero-tail equations). [afterBlocking_tpPrimitiveBlockDecompositions_of](#)
two tensors generate the same MPV family, then each tensor admits a blocked decomposition into a zero-tail tensor plus nonzero left-canonical blocks with primitive transfer maps and positive bond dimensions. The theorem also retains, for each blocking period, the relation that the blocked tensors generate the same MPV family.

Proof. ✓ Proof OK Apply Theorem 11.117 separately to the two tensors and keep the zero-tail MPV equations returned by that theorem. The property that the blocked tensors generate the same MPV family follows by blocking the original equality. $\square$

Theorem 11.170 (Per-block cyclic-sector decomposition with zero-tail identity). [fundamentalTheorem_after_blocking_perBlo](#)
two tensors generate the same MPV family, then after the zero-tail split and trace-preserving gauge each side has irreducible nonzero-weight blocks. The nonzero-block tensors agree at all positive system sizes, while the length-zero case is included as the exact zero-tail identity. Moreover each nonzero-weight block has primitive irreducible cyclic sectors. The period-removal length for those sectors is kept separate from any later common blocking or injectivity length.

Proof. ✓ Proof OK Apply the arbitrary-tensor TP-gauge theorem separately to A and B . The zero-tail identity theorem compares the two resulting nonzero parts at positive lengths and gives the $N = 0$ identity. Finally apply Theorem 11.81 to every trace-preserving irreducible nonzero-weight block. $\square$

Theorem 11.171 (Common reblocked nonzero cyclic sectors with zero-tail identity). [fundamentalTheorem_after_blocking_co](#)
two tensors generate the same MPV family, then the zero-tail/TP-gauge reduction for nonzero-weight blocks can be combined with a common reblocking of all per-block cyclic sectors. Each side obtains a finite flattened sector family at one physical blocking length, with trace preservation, primitive transfer maps, tensor irreducibility, positive bond dimensions, and nonzero unit weights. The theorem keeps the positive-length equality of the nonzero parts and the exact $N = 0$ zero-tail identity for the original nonzero-block tensors.

Proof. ✓ Proof OK Apply Theorem 11.170. Then apply Theorem 11.92 to the per-block cyclic-sector data on each side. The nonzero unit-weight conclusion is Theorem 11.84. $\square$

The next results put the cyclic-sector decompositions on a common blocked alphabet. They compare direct blocking with iterated cyclic-sector blocking and transport the zero-tail identity through this comparison.

Theorem 11.172 (Common cyclic sectors after reblocking). [fundamentalTheorem_after_blocking_reindexed_commonSector_live_wit](#)
two tensors generate the same MPV family, then after reblocking the nonzero blocks one obtains cyclic-sector families over a common alphabet. These families preserve the MPV of the reblocked nonzero part, with nonzero transported weights μ_k^p , and retain the zero-tail identity.

Proof. ✓ Proof OK Start from the common reblocked cyclic-sector families on the two sides. Apply the zero-tail reblocking theorem to the original nonzero-block decomposition, then use the reindexed sector decomposition and the derived structural properties of the common-sector family. $\square$

Theorem 11.173 (Common blocking length for cyclic sectors).

[fundamentalTheorem_after_blocking_commonLength_commonSector](#)

two tensors generate the same MPV family, then the cyclic-sector decompositions on the two sides can be expressed at one common positive blocking length. At this length the nonzero parts still agree at positive system sizes, and the length-zero identity is exactly the zero-tail identity.

Proof. [✓ Proof OK](#) Start from the per-block cyclic-sector theorem. Let p_A and p_B be the least common multiples of the period-removal lengths on the two sides, and use $p = \text{lcm}(p_A, p_B)$. The prescribed-common-multiple theorem constructs both one-sided sector families at this same length. The zero-tail and positive-length equalities are the existing reblocking statements, and the relabeled sector conclusions are the reindexed common-sector MPV equalities. $\square$

Theorem 11.174 (Zero-tail identity under compatible blocking).

[zeroTail_commonFlat_of_blockwise](#)

[zeroTail_commonFlat_of_w](#)

a zero-tail/nonzero decomposition on one side. If direct blocking of each nonzero block agrees with the corresponding iterated cyclic-sector blocking, then the zero-tail identity can be rewritten using the common cyclic-sector family. The coordinate-grouping condition supplies this agreement at any prescribed common blocking length.

Proof. [✓ Proof OK](#) Transport the zero-tail/nonzero decomposition to the common blocking length. The blockwise comparison then replaces the directly blocked nonzero part by the weighted common-sector tensor. $\square$

Theorem 11.175 (Zero-tail identity under word reindexing).

[zeroTail_commonFlat_of_reindexed](#)

[zeroTail_commonFlatAt_of_rein](#)

a zero-tail/nonzero decomposition on one side. If the blocked nonzero tensor agrees, after reindexing blocked words, with the common cyclic-sector family, then the same zero-tail identity can be written in that common alphabet. At positive system sizes this says that the blocked tensor and the weighted common-sector tensor have the same MPV coefficients.

Proof. [✓ Proof OK](#) Transport the zero-tail/nonzero decomposition to the common blocking length. The reindexing hypothesis then replaces the blocked nonzero tensor by the weighted common-sector tensor. At positive lengths the zero-tail tensor has zero MPV coefficients. $\square$

Theorem 11.176 (Primitive block decompositions at a common length).

[afterBlocking_commonPrimitiveIrreducibleBlocks_of](#)

that the nonzero part can be reindexed into the common blocked alphabet. Then two tensors with the same MPV family admit one positive blocking length at which both nonzero parts are weighted sums of trace-preserving, primitive, tensor-irreducible blocks with positive bond dimensions and nonzero weights.

Proof. [✓ Proof OK](#) Apply Theorem 11.173 to choose the common blocking length and the two common cyclic-sector families. The assumed word reindexing converts the blocked nonzero part into the weighted common-sector family on each side. The zero-tail equations then give the displayed decompositions. The structural properties and nonzero weights come from the common-sector families. $\square$

Theorem 11.177 (Transport of MPV and length-zero identities).

[sameMPV2Pos_and_zeroTail_identity_of_sameMPV2](#)

[✓ Stmt OK](#)

L_A, L_B be two nonzero parts whose MPV families agree at positive lengths and whose zero-tail dimensions satisfy the length-zero identity. If L'_A and L'_B generate the same MPV families as L_A and L_B , respectively, then L'_A and L'_B satisfy the same positive-length equality and the same length-zero zero-tail identity.

Proof. [✓ Proof OK](#) Use the two MPV equalities to replace L'_A by L_A and L_B by L'_B in the positive-length equality and in the $N = 0$ identity. $\square$

Theorem 11.178 (Common-sector zero-tail identity under reindexing). [zeroTail_commonFlat_transport_of_reindexed](#) ✓ Stmt OK *the blocked nonzero part agrees with the reindexed common-sector tensor, then the zero-tail identity can be written with that common-sector tensor. Nonzero original weights remain nonzero after the blocking power.*

Proof. ✓ Proof OK Combine Theorem 11.175 with the common-sector MPV equality and the nonzero-weight statement. □

Theorem 11.179 (Common-sector zero-tail identity from grouped coordinates). [zeroTail_commonFlat_transport_of_groupedB](#) *a zero-tail/nonzero decomposition with nonzero weights. If the common blocked alphabet is identified with each iterated alphabet by grouping consecutive words, then the zero-tail identity can be written with the weighted common-sector family.*

Proof. ✓ Proof OK The coordinate-grouping comparison identifies the directly blocked nonzero part with the weighted common-sector family. Combine this with the zero-tail identity and the nonzero-weight statement. □

Theorem 11.180 (Common sectors at a common length after reindexing). [fundamentalTheorem_after_blocking_commonLength_c](#) *two tensors generate the same MPV family, choose a common blocking length for their cyclic-sector decompositions. If each blocked nonzero part agrees with its common-sector form after reindexing words, then all zero-tail identities and positive-length MPV equalities can be stated using the common sector families.*

Proof. ✓ Proof OK Apply Theorem 11.173. The two reindexing hypotheses identify the blocked nonzero parts with the weighted common-sector tensors. Substituting these MPV equalities into the zero-tail equations gives the rewritten zero-tail identities, the positive-length MPV equalities, and the $N = 0$ identity. □

11.14 Open directions

Remark 11.181 (Canonical-form reduction before the equal-case comparison). ✓ Stmt OK The canonical-form proof in the MPS literature first separates the nilpotent part, which contributes only to the length-zero trace, from the nonzero part. The nonzero part is decomposed into irreducible blocks, and each block is put in trace-preserving gauge. A block of period m is then blocked over m sites and restricted to its cyclic spectral sectors. These sector tensors are the normal blocks used in the comparison theorem: they are trace-preserving, primitive, irreducible, and have positive bond dimension. For a finite nonzero block family, the paper chooses one common multiple of all periods so that every sector is expressed over the same blocked physical alphabet. This is the construction of [CPGSV17a, §2.3 and Appendix A], and it is the canonical-form information used before the basis-of-normal-tensors argument in [CPGSV21, Proposition IV.4 and Corollary IV.5].

The statements in this chapter follow this reduction. Theorem 11.48 gives the zero-tail block and the trace-preserving irreducible nonzero block family. Theorem 11.170 applies the cyclic-sector decomposition to each nonzero block, preserving the positive-length MPV equality and the zero-tail identity at $N = 0$. Theorem 11.91 constructs the common reblocked sector family at any common multiple of the period-removal lengths. Consequently Theorem 11.173 chooses one physical blocking length for both tensors, transports the zero-tail identities to that length, and supplies cyclic-sector families on a common blocked alphabet. The supporting blocking and transport results, including Theorem 11.59, identify iterated blocking with one blocking by the product length, after relabeling the blocked physical words.

The results above now separate the finite-sum comparison from the remaining blocked-word assertion. Theorem 11.89 shows that, once the length- p word read directly from the common blocked alphabet equals the word obtained by flattening an iterated (m_k, e_k) block, the weighted direct sum of directly blocked nonzero blocks agrees with the common-sector family. Theorem 11.174 then rewrites the one-sided zero-tail identity with the weighted common-length cyclic sectors under this blockwise word equality.

The remaining mathematical assertion is the corresponding equality, after relabeling blocked physical words, between the blocked nonzero part and the cyclic-sector tensor for the whole weighted family. Theorem 11.175 gives the one-sided form of this conversion, and Theorem 11.180 records it simultaneously for the two common-length sector families. The auxiliary transport statements, Theorems 11.178, 11.179, and 11.177, record the nonzero transported weights and the preservation of the positive-length and length-zero identities, while Theorem 11.176 collects the resulting primitive irreducible sector decompositions needed for comparison. The remaining blocked-word assertion is the equality of these words for the chosen coordinates, or an equivalent final comparison with the relabeling made explicit. Theorem 14.67 records the intermediate span consequence for these common-sector families: a common MPV phase cover or proportional-decomposition conclusion supplies the finite-length nonzero-block span equality. Theorem 14.68 states the exact remaining inputs after this point: equality of the zero-tail dimensions, injectivity at the chosen common length, and a common MPV phase cover (or the equivalent finite-length span equality) for the two nonzero-sector families. Thus the remaining canonical-form assertions are the blocked-word relabeling equality for the whole weighted nonzero family, the zero-tail and injectivity refinements, and the common-phase or proportional-decomposition comparison for the common-length nonzero sectors.

The sorted normal-canonical-form theorem, Theorem 11.122, assumes pairwise distinct nonzero weight norms. This is a useful special form after sorting. Equal-norm families require the basis-of-normal-tensors and multiplicity comparison developed later, rather than an artificial separation of equal weight norms.

Chapter 12

Block Permutation and Separation

This chapter records the algebra of block permutations, the product-algebra maps built from per-block same-MPV data, the invariant-projection splitting step that reduces a tensor to smaller blocks, and the block-separation argument that recovers the individual blocks from a block-diagonal same-MPV hypothesis. The results correspond to the canonical-form uniqueness of [PGVWC07] and the block-separation results of [CPGSV21].

12.1 Block permutation algebra

Definition 12.1 (Block ideal). blockIdeal ✓ Stmt OK Let $\{R_k\}_{k=1}^r$ be simple rings. The k -th block ideal in $\prod_{j=1}^r R_j$ is

$$I_k := \{x \in \prod_{j=1}^r R_j : x_j = 0 \text{ for } j \neq k\}.$$

Theorem 12.2 (Each block ideal is an atom). isAtom_blockIdeal ✓ Stmt OK Each block ideal I_k is an atom in the lattice of two-sided ideals of $\prod_{j=1}^r R_j$.

Proof. ✓ Proof OK Under the order isomorphism between two-sided ideals of $\prod_{j=1}^r R_j$ and products of two-sided ideal lattices, I_k corresponds to the tuple with $\top$ in the k -th coordinate and $\perp$ elsewhere. This tuple is an atom, hence so is I_k . □

Theorem 12.3 (Ring isomorphisms permute block ideals). ringEquiv_pi_simple_permutes_blockIdeals ✓ Stmt OK Let $\{R_k\}_{k=1}^r$ be simple rings. Every ring isomorphism $T : \prod_k R_k \rightarrow \prod_k R_k$ permutes the block ideals: there exists a permutation π such that T maps the k -th block ideal to the $\pi(k)$ -th block ideal.

Proof. ✓ Proof OK By Theorem 12.2, each block ideal is an atom in the lattice of two-sided ideals of $\prod_k R_k$. A ring isomorphism induces an order automorphism of this lattice, so it permutes the atoms. Since the atoms are exactly the block ideals, this gives the required permutation π . □

Remark 12.4 (Ring level versus algebra level). Theorem 12.3 is a ring-level statement: it only uses the ideal structure of $\prod_k R_k$. Theorem 12.6 stays at this ring level: once the block ideals are permuted, one can extract the corresponding single-block maps without using scalar multiplication. Theorems 12.5 and 12.7 require an upgrade to $\mathbb{C}$ -algebra automorphisms. The reason

is threefold: dimension preservation compares the blocks as $\mathbb{C}$ -vector spaces; Skolem–Noether applies to the resulting central simple $\mathbb{C}$ -algebras; and the final per-block maps must commute with scalar multiplication. Thus the argument separates the ring-level permutation step from the later algebra-level decomposition.

Theorem 12.5 (Dimension preservation). dim_preserved ✓ Stmt OK Assume $D_k \geq 1$ for all k . If T is an algebra automorphism of $\prod_{k=1}^r M_{D_k}(\mathbb{C})$, and π is the induced permutation, then $D_{\pi(k)} = D_k$ for all k .

Proof. ✓ Proof OK The restriction of T to the k -th block ideal induces a $\mathbb{C}$ -algebra isomorphism $M_{D_k}(\mathbb{C}) \cong M_{D_{\pi(k)}}(\mathbb{C})$, which forces $D_k = D_{\pi(k)}$ by comparing dimensions as $\mathbb{C}$ -vector spaces. $\square$

Theorem 12.6 (Per-block component maps are bijections). componentMap_bijective ✓ Stmt OK Assume $D_k \geq 1$ for all k . Let T be a ring automorphism of $\prod_{k=1}^r M_{D_k}(\mathbb{C})$, and let π be the induced permutation from Theorem 12.3. For each block index k , the map

$$M \mapsto (T(0, \dots, 0, M, 0, \dots, 0))_{\pi(k)}$$

from $M_{D_k}(\mathbb{C})$ to $M_{D_{\pi(k)}}(\mathbb{C})$ is bijective.

Proof. ✓ Proof OK Theorem 12.3 forces a single-support element in the k -th block ideal to remain supported on the $\pi(k)$ -th block after applying T . Injectivity follows from injectivity of T . For surjectivity, apply the same argument to T^{-1} and the inverse permutation. $\square$

Theorem 12.7 (Decomposition of automorphisms of $\prod M_{D_k}(\mathbb{C})$). algEquiv_pi_matrix_decomposition ✓ Stmt OK Assume $D_k \geq 1$ for all k . Every $\mathbb{C}$ -algebra automorphism T of $\prod_{k=1}^r M_{D_k}(\mathbb{C})$ decomposes as a block permutation π followed by per-block inner automorphisms: there exist $\pi \in S_r$ with $D_{\pi(k)} = D_k$ and invertible matrices $X_k \in \text{GL}_{D_k}(\mathbb{C})$ such that, for every tuple $M = (M_1, \dots, M_r)$, $T(M)_k = X_k M_{\pi(k)} X_k^{-1}$, where $T(M)_k$ denotes the k -th block component of $T(M)$.

Proof. ✓ Proof OK Theorem 12.3 gives the permutation π . Theorem 12.5 identifies the matrix sizes $D_{\pi(k)} = D_k$. For each k , Theorem 12.6 gives a bijective multiplicative map from $M_{D_k}(\mathbb{C})$ to $M_{D_{\pi(k)}}(\mathbb{C})$ extracted from the single-support inputs of T ; after reindexing by the dimension equality, this is a $\mathbb{C}$ -algebra automorphism of $M_{D_k}(\mathbb{C})$. Skolem–Noether (Theorem 4.9) makes each such automorphism inner.

This is a standard result in the theory of semisimple algebras; see [CPGSV21, §IV.A] for the MPS context. $\square$

12.2 Product algebra linear extension

Definition 12.8 (Per-block linear extension). perBlockLinearExtension ✓ Stmt OK Given injective tensors A_k, B_k of bond dimension D_k with the same MPV for each block k , the *per-block linear extension* T_k is the unique linear map $M_{D_k}(\mathbb{C}) \rightarrow M_{D_k}(\mathbb{C})$ satisfying $T_k(A_k^i) = B_k^i$ for all i . It exists (and is unique) because $\{A_k^i\}$ spans $M_{D_k}(\mathbb{C})$ by injectivity.

Definition 12.9 (Product algebra automorphism). piAlgEquiv ✓ Stmt OK The *product algebra automorphism* is the blockwise map $T : \prod_k M_{D_k}(\mathbb{C}) \rightarrow \prod_k M_{D_k}(\mathbb{C})$ defined by $T(M)_k := T_k(M_k)$. Theorem 12.10 shows that each T_k is a $\mathbb{C}$ -algebra homomorphism and that this map is a $\mathbb{C}$ -algebra automorphism.

Theorem 12.10 (Per-block linear extensions define a product algebra automorphism). piAlgEquiv_decomposition ✓ Stmt OK If A_k and B_k be injective tensors of bond dimension D_k for $k = 1, \dots, r$, and suppose the per-block same-MPV condition holds. Then the per-block linear extensions $T_k : A_k^i \mapsto B_k^i$ define an algebra automorphism of $\prod_k M_{D_k}(\mathbb{C})$, which decomposes as a permutation π with $D_{\pi(k)} = D_k$ and per-block conjugation matrices X_k .

Proof. ✓ Proof OK Each T_k exists uniquely and is multiplicative by Theorems 4.6 and 4.7. Promotion to algebra homomorphisms (Lemma 4.10) and blockwise composition give an algebra automorphism of $\prod_k M_{D_k}(\mathbb{C})$. Theorem 12.7 then decomposes this automorphism as a block permutation together with per-block inner automorphisms. □

Theorem 12.11 (Per-block same MPV gives global gauge equivalence). fundamentalTheorem_multiBlock_full ✓ Stmt OK If all block tensors A_k are injective, and B_k are block tensors of the same bond dimensions such that per-block same-MPV holds, then each pair (A_k, B_k) is gauge equivalent, and the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ are globally gauge equivalent.

Proof. ✓ Proof OK Per-block gauge equivalence follows from the single-block FT (Theorem 4.11) applied to each block. Global gauge equivalence follows by placing the per-block gauge matrices on the block diagonal (Theorem 11.4). □

12.3 Burnside and invariant projections

Theorem 12.12 (The algebra span reaches a finite cumulative span). exists_cumulativeSpan_eq_top_of_algSpan_eq_top ✓ Stmt OK If the generated unital algebra satisfies $\text{alg}(A) = M_D(\mathbb{C})$, then there exists N such that $T_N(A) = M_D(\mathbb{C})$.

Proof. ✓ Proof OK Every element of $\text{alg}(A)$ lies in some cumulative span $T_n(A)$: the generators lie in $T_1(A)$, scalar multiples of the identity lie in $T_0(A)$, and the family $\{T_n(A)\}_{n \geq 0}$ is closed under addition and under multiplication with lengths adding. Since $M_D(\mathbb{C})$ is finite-dimensional, the ascending chain $T_0(A) \subseteq T_1(A) \subseteq \dots$ stabilizes, so one can choose a single N with $T_N(A) = M_D(\mathbb{C})$. □

Theorem 12.13 (Irreducible action gives an irreducible tensor). isIrreducibleTensor_of_isIrreducibleAction ✓ Stmt OK If the letters of A act irreducibly on $\mathbb{C}^D$, then A has no nontrivial invariant orthogonal projection.

Proof. ✓ Proof OK If P were a nontrivial invariant orthogonal projection, then its range would be invariant under every letter of A . Irreducibility of the action forces this range to be either 0 or all of $\mathbb{C}^D$, so P must be 0 or $\mathbb{1}$, a contradiction. □

Theorem 12.14 (Invariant projection gives a two-block decomposition). exists_twoBlock_decomp_of_lowerZero ✓ Stmt OK If P is an orthogonal projection satisfying, for every physical index i ,

$$(\mathbb{1} - P)A^iP = 0.$$

Then there exist n, m with $n + m = D$ and MPS tensors A_1, A_2 of bond dimensions n and m such that A and the two-block tensor $A_1 \oplus A_2$ generate the same MPV family in the sense of Definition 2.12.

Proof. ✓ Proof OK Diagonalize P by a unitary U . Since $P^2 = P$, the diagonal form of P has only 0- and 1-eigenvalues, so the basis splits as $n + m = D$. Conjugating A by U preserves the MPV family by Theorem 11.10. In this basis the relation $(\mathbb{1} - P)A^iP = 0$ makes each letter upper block triangular. Discarding the off-diagonal blocks does not change the trace of any word product, so the resulting block-diagonal tensor has the same MPV coefficients. Reindex the two diagonal blocks by $\{0, \dots, n-1\}$ and $\{0, \dots, m-1\}$. □

Theorem 12.15 (Nontrivial invariant projection gives strict two-block reduction). [exists_twoBlock_decomp_of_lowerZero_str](#)
the hypotheses of Theorem 12.14, assume in addition that $P \neq 0$ and $P \neq \mathbb{1}$. Then there exist n, m with $n + m = D$, $n < D$, and $m < D$, together with block tensors A_1, A_2 , such that A and $A_1 \oplus A_2$ generate the same MPV family in the sense of Definition 2.12.

Proof. [✓ Proof OK](#) Apply Theorem 12.14. In the diagonal form of P , the 1-eigenspace is nonempty because $P \neq 0$, and the 0-eigenspace is nonempty because $P \neq \mathbb{1}$. Hence both block sizes are positive. Since $n + m = D$, this implies $n < D$ and $m < D$. $\square$

12.4 Block separation

Recall from Definition 2.43 and Theorem 9.7 that the bilinear overlap is

$$O_{AB}(N) = \sum_{\sigma \in \{0, \dots, d-1\}^N} V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma} = \text{Tr}(F_{AB}^N).$$

We use this notation throughout the block-separation argument.

Definition 12.16 (Canonical form conditions). [IsCanonicalForm](#) [✓ Stmt OK](#) A collection of scaling factors $(\mu_k)_{k=1}^r$ and block tensors $(A_k)_{k=1}^r$ satisfies the *canonical form conditions* if:

1. each A_k is injective,
2. each A_k satisfies the TP normalization $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$,
3. the moduli are non-increasing: $|\mu_1| \geq |\mu_2| \geq \dots \geq |\mu_r|$,
4. all $\mu_k \neq 0$,
5. the self-overlap converges: $O_{A_k A_k}(N) \rightarrow 1$ for each k .

Only this one-sided normalization is assumed; no unital condition is assumed here. Condition (5) is a primitivity hypothesis: for a primitive channel ([Wol12, Thm. 6.7, item 3]), $T^n(\rho) \rightarrow \rho_\infty$ for every initial state, which forces the self-overlap to converge to 1. See also [Wol12, Thm. 6.8] for the completely-positive characterisation and Chapter 9 for the spectral-gap proof.

Remark 12.17 (Normalization convention). The convention here differs from [PGVWC07, Thm. 4], which uses the unital gauge $\sum_i A_k^i (A_k^i)^\dagger = \mathbb{1}$. This chapter instead works in the dual TP gauge $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$. A positive-definite fixed point of the adjoint transfer map provides a gauge transformation between these two normalizations (Theorem 8.11).

Remark 12.18 (Overlap hypothesis). In the literature [CPGSV21], condition (5) is derived from primitivity of the transfer map (peripheral spectrum = $\{1\}$). Theorem 11.28 gives this implication: per-block primitivity implies overlap convergence, so condition (5) is redundant when primitivity holds. The canonical form conditions retain condition (5) explicitly because it can be verified independently of primitivity.

Remark 12.19 (Normal-canonical companion). Definition 12.16 gives the block-injective conditions. The normal-canonical analogue instead uses Definition 11.38: injectivity and the explicit self-overlap hypothesis are replaced by irreducibility, TP normalization, and primitive transfer maps. Theorem 11.39 then supplies the self-overlap limit automatically.

Theorem 12.20 (Vandermonde separation). [vandermonde_separation](#) ✓ Stmt OK Let C be a canonical form with block weights $\mu_1, \dots, \mu_r$, and assume these weights are pairwise distinct. If $c_1, \dots, c_r \in \mathbb{C}$ satisfy, for $N = 0, 1, \dots, r-1$,

$$\sum_{k=1}^r c_k \mu_k^N = 0$$

then $c_k = 0$ for all k .

Proof. ✓ Proof OK The coefficient vector c satisfies $Vc = 0$, where V is the Vandermonde matrix of the weights μ_k . Pairwise distinctness makes V invertible, so $c = 0$. □

Lemma 12.21 (Repeated-word evaluation). [evalWord_flatten_replicate](#) ✓ Stmt OK For any word w and any $L \geq 0$, the evaluation of A on the L -fold concatenation w^L is $(A^w)^L$.

Proof. ✓ Proof OK By induction on L , using the multiplicativity of word evaluation under concatenation. □

Theorem 12.22 (Repeated-word trace identity). [sameMPV2_repeated_word](#) ✓ Stmt OK If the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ generate the same MPV family, then for every word w and every $L \geq 0$,

$$\sum_k \mu_k^{|w|L} (\text{tr}((A_k^w)^L) - \text{tr}((B_k^w)^L)) = 0.$$

Proof. ✓ Proof OK Apply the block-diagonal same-MPV identity to the word w^L obtained by repeating w exactly L times. The block-diagonal coefficient is a weighted sum over the blocks. Lemma 12.21 rewrites each block evaluation on w^L as a matrix power, giving the displayed trace identity. □

Remark 12.23. Theorems 12.20 and 12.22 record an alternative Newton/Vandermonde route. The checked proof chain in this chapter instead continues with the mixed-transfer peeling argument of Lemma 12.24, and the later BNT weight-matching step uses the linear-independence results from Chapter 13 rather than Newton–Girard identities.

Lemma 12.24 (Block separation core). [block_separation_core](#) ✓ Stmt OK Let $((\mu_k), (A_k))$ satisfy the canonical form conditions, and let (B_k) be injective tensors satisfying the same TP normalization $\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$. If $\sum_k \mu_k^N (V^{(N)}(A_k)_\sigma - V^{(N)}(B_k)_\sigma) = 0$ for all N, σ , then each pair (A_k, B_k) generates the same MPV family.

Proof. ✓ Proof OK By induction on the number of blocks r . In the inductive step, take overlaps with the leading block A_0 ; after division by μ_0^N , the contributions from $k \geq 1$ decay because $|\mu_k/\mu_0| < 1$ (strict weight ordering) and the bilinear overlaps $O_{A_0 A_k}(N)$ and $O_{A_0 B_k}(N)$ are bounded (Theorems 9.23 and 9.25 supply the relevant bounds), leaving $O_{A_0 A_0}(N) - O_{A_0 B_0}(N) \rightarrow 0$. Condition (5) gives $O_{A_0 A_0}(N) \rightarrow 1$, hence $O_{A_0 B_0}(N) \rightarrow 1$. Since $O_{A_0 B_0}(N) \rightarrow 1 \neq 0$, the bond dimensions must match: $D_{A_0} = D_{B_0}$; otherwise Theorem 9.23 would give $O_{A_0 B_0}(N) \rightarrow 0$, a contradiction. With equal bond dimensions established, the contrapositive of Theorem 9.25 forces A_0 and B_0 to be gauge-phase equivalent; the limit $O_{A_0 B_0}(N) \rightarrow 1$ then makes the phase trivial, so A_0 and B_0 generate the same MPV family. Peeling off the leading block and dividing by μ_0^N , the induction hypothesis applies. □

Theorem 12.25 (Block separation from canonical form). [per_block_sameMPV_of_canonical_form](#) ✓ Stmt OK Let $((\mu_k), (A_k))$ satisfy the canonical form conditions, and let (B_k) be injective tensors satisfying the same TP normalization $\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$. If the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ generate the same MPV family, then each pair (A_k, B_k) generates the same MPV family.

Proof. ✓ Proof OK Apply Lemma 12.24 to the identity obtained from the equality of the two block-diagonal MPV families. The induction peels off the dominant block and treats the remainder as a decaying tail. Concretely, one isolates the leading term $\mu_1^N V(A_1)$ and bounds the remaining contribution by $O(|\mu_2/\mu_1|^N)$ after dividing by μ_1^N . The canonical form conditions supply injectivity, TP normalization, strict nonzero weights, and self-overlap convergence for (A_k) , while the hypotheses on (B_k) supply the comparison injectivity and TP normalization. $\square$

Theorem 12.26 (Block separation from normal canonical form). per_block_sameMPV_of_normal_canonical_form ✓ Stmt OK Let $((\mu_k), (A_k))$ satisfy the normal canonical form conditions, and let (B_k) be irreducible tensors of the same bond dimensions, satisfying the same TP normalization $\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$. If the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ generate the same MPV family, then each pair (A_k, B_k) generates the same MPV family.

Proof. ✓ Proof OK The proof follows the same peeling argument as Theorem 12.25, with Theorem 11.40 (modulus-one eigenvalue rigidity for irreducible tensors) replacing the injective overlap-decay argument at the identification step, since the blocks are irreducible rather than injective. Theorem 11.39 replaces the explicit self-overlap hypothesis on the leading block. Since both block families already use the same dimension function, the conclusion is direct per-block same MPV. $\square$

Theorem 12.27 (Canonical form fundamental theorem, same-structure version). fundamentalTheorem_canonicalForm ✓ Stmt OK If the hypotheses of Theorem 12.25, each pair (A_k, B_k) is gauge equivalent. Hence the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ are gauge equivalent.

Proof. ✓ Proof OK Theorem 12.25 gives per-block same MPV. Theorem 12.11 then converts this to per-block gauge equivalence and then to global gauge equivalence of the block-diagonal tensors. $\square$

Remark 12.28 (Relation to the literature). The block-permutation part follows the decomposition of automorphisms of semisimple matrix algebras used in [CPGSV21, §IV.A]. The separation part follows the induction on blocks from [PGVWC07, Appendix E]; the checked proof here uses overlap bounds and mixed-transfer decay rather than the alternative Newton/Vandermonde route recorded above.

Chapter 13

Basis of Normal Tensors

This chapter introduces the *basis of normal tensors* (BNT) notion, the canonical-form variants with BNT separation, the cross-overlap decay theorem, and the permutation-rigidity arguments used in the proof of the Fundamental Theorem. The chapter concludes with auxiliary results on coefficient convergence and Newton–Girard identities. The main references are [PGVWC07] and [CPGSV21].

13.1 Basis of normal tensors

Definition 13.1 (Basis of Normal Tensors (BNT)). IsBNT ✓ Stmt OK A *basis of normal tensors* (BNT) for a total tensor A_{tot} consists of g block tensors $(A_1, \dots, A_g)$ with bond dimensions $(D_1, \dots, D_g)$ such that:

1. each A_j is normal (Definition 2.26),
2. at each system size N , the MPV of A_{tot} lies in the span of the block MPVs $\{|V^{(N)}(A_j)\}\}_{j=1}^g$,
3. for sufficiently large N , the block MPV vectors $\{|V^{(N)}(A_j)\}\}_{j=1}^g$ are linearly independent.

This is [CPGSV21, Definition 4.2].

Theorem 13.2 (Overlap-orthonormality implies eventual linear independence). eventually_linearIndependent_of_overlap_t
 $(A_j)_{j=1}^g$ be a finite family of tensors such that $O_{A_j A_j}(N) \rightarrow 1$ for each j and $O_{A_i A_j}(N) \rightarrow 0$ for $i \neq j$. Then the MPV states $\{|V^{(N)}(A_j)\}\}_{j=1}^g$ are linearly independent for all sufficiently large N .

Proof. ✓ Proof OK The Gram matrix of the family $\{|V^{(N)}(A_j)\}\}_{j=1}^g$ converges entrywise to the identity matrix. Hence for all sufficiently large N it is invertible, so the MPV states are linearly independent. □

Definition 13.3 (No gauge-phase-equivalent distinct blocks). BlocksNotGaugePhaseEquiv ✓ Stmt OK A finite block family has no gauge-phase-equivalent distinct blocks when, for any two distinct indices with the same bond dimension, the corresponding blocks are not gauge-phase equivalent.

Definition 13.4 (Canonical form with BNT separation). IsCanonicalFormBNT ✓ Stmt OK A family is in *canonical form with BNT separation* if it satisfies Definition 12.16 and, in addition:

1. the moduli are *strictly* decreasing: $|\mu_1| > |\mu_2| > \dots > |\mu_r|$ (strengthened from the non-increasing ordering of Definition 12.16),

2. distinct blocks of the same bond dimension are *not* gauge-phase equivalent.

The strictly decreasing condition is justified by the BNT grouping step, which merges gauge-phase-equivalent blocks with equal moduli.

Theorem 13.5 (Distinct bond dimensions force BNT separation). toIsCanonicalFormBNT_of_distinct_dims ✓ Stmt OK *If a canonical-form family has strictly decreasing weight moduli and pairwise distinct bond dimensions, then it is in canonical form with BNT separation.*

Proof. ✓ Proof OK The only new clause in Definition 13.4 is the prohibition on gauge-phase equivalence between distinct blocks of the same bond dimension. When the bond dimensions are pairwise distinct, this clause is vacuous. □

Definition 13.6 (Normal canonical form with BNT separation). IsNormalCanonicalFormBNT ✓ Stmt OK A family is in *normal canonical form with BNT separation* if it satisfies Definition 11.38 and, in addition:

1. the moduli are strictly decreasing (strengthened from non-increasing),
2. distinct blocks of the same bond dimension are not gauge-phase equivalent.

Theorem 13.7 (Equal-norm grouping step). exists_bnt_grouping ✓ Stmt OK *Let $(\mu_k, A_k)_{k=1}^r$ be a weighted block family with $\mu_k \neq 0$ for every k . Assume that whenever $\|\mu_j\| = \|\mu_k\|$, the tensors A_j and A_k generate the same MPV family. Then one can regroup the blocks into a sector decomposition whose associated total tensor generates the same MPV family as $\bigoplus_k \mu_k A_k$, and whose sector-weight norms are strictly decreasing.*

Proof. ✓ Proof OK Order the distinct norms of the weights in decreasing order, choose one representative block from each norm class, and retain the multiplicities inside each class as sector copies. The equal-MPV hypothesis lets every block in a class be replaced by the chosen representative, so the regrouped decomposition has the same MPV family. By construction, the sector-weight norms are strictly decreasing. □

Theorem 13.8 (Cross-overlap decay from explicit BNT hypotheses). cross_overlap_tendsto_zero_of_separated_CFBNT_data ✓ *$(A_j)_{j=1}^r$ be injective and trace-preserving normalized. Assume that distinct equal-dimension blocks are not gauge-phase equivalent. Then $O_{A_j A_k}(N) \rightarrow 0$ for every pair $j \neq k$.*

Proof. ✓ Proof OK If the bond dimensions differ, Theorem 9.23 gives the decay. If the bond dimensions agree, the non-equivalence hypothesis excludes gauge-phase equivalence, so Theorem 9.25 gives the decay. □

Theorem 13.9 (Cross-overlap decay for distinct blocks in canonical form with BNT separation). cross_overlap_tendsto_zero ✓ Stmt OK *If $((\mu_k), (A_k))$ is in canonical form with BNT separation and $j \neq k$, then $O_{A_j A_k}(N) \rightarrow 0$.*

Proof. ✓ Proof OK Definition 13.4 supplies injectivity, left-canonical normalization, and the BNT separation hypothesis, so Theorem 13.8 applies directly. □

Theorem 13.10 (BNT from explicit blockwise hypotheses). isBNT_of_separated_CFBNT_data ✓ Stmt OK *Let $(\mu_k, A_k)_{k=1}^r$ be a weighted block family. Assume that the blocks are injective and trace-preserving normalized, that $O_{A_k A_k}(N) \rightarrow 1$ for each k , and that distinct equal-dimension blocks are not gauge-phase equivalent. Then the blocks form a basis of normal tensors for the block-diagonal tensor $\bigoplus_k \mu_k A_k$.*

Proof. ✓ Proof OK Injective blocks are normal. The block-diagonal decomposition theorem gives the MPV expansion of $\bigoplus_k \mu_k A_k$ in terms of the block MPVs. The self-overlap limits are part of the hypotheses, and Theorem 13.8 supplies the off-diagonal decay. Therefore Theorem 13.2 gives the eventual linear independence required in Definition 13.1. $\square$

Theorem 13.11 (Canonical form with BNT separation yields a basis of normal tensors). isBNT ✓ Stmt OK *If $((\mu_k), (A_k))$ is in canonical form with BNT separation, then the block tensors form a basis of normal tensors for the block-diagonal tensor $\bigoplus_k \mu_k A_k$.*

Proof. ✓ Proof OK Definition 13.4 provides injective blocks, left-canonical normalization, normalized self-overlaps, and BNT separation. Hence Theorem 13.10 applies. $\square$

Remark 13.12 (Comparing the two normality conditions). Definition 13.6 references the normal canonical form conditions (Definition 11.38), which encode normality via the spectral characterization (primitive transfer map). Definition 13.1 requires normality via the algebraic characterization (eventual block injectivity, Definition 2.26). These are equivalent by Remark 2.20, but the equivalence must be invoked explicitly: the Wielandt theory of Chapter 10 provides the implication from primitivity to eventual block injectivity needed here.

Theorem 13.13 (Normal canonical form with BNT separation yields a BNT after algebraic normality is supplied). isBNT ✓ Stmt OK *If a family is in normal canonical form with BNT separation and each block is also known to be normal in the algebraic sense, then the blocks form a basis of normal tensors for the associated block-diagonal tensor.*

Proof. ✓ Proof OK Normal-canonical BNT data already give the block-diagonal MPV expansion, the self-overlap normalization, and the eventual linear independence coming from irreducible trace-preserving overlap decay. Supplying algebraic normality for each block adds the remaining clause in Definition 13.1. $\square$

13.2 Permutation rigidity for bases of normal tensors

Theorem 13.14 (Invertible change of basis). exists_invertible_changeBasis ✓ Stmt OK *Let $v_1, \dots, v_g$ and $w_1, \dots, w_g$ be two linearly independent families in a complex vector space with the same span. Then there exists an invertible matrix $U \in \text{GL}_g(\mathbb{C})$ such that*

$$w_j = \sum_{i=1}^g U_{ij} v_i$$

for every j .

Proof. ✓ Proof OK Because the two families span the same subspace, each w_j has a unique expansion in the basis $\{v_i\}_{i=1}^g$; these coefficients form the columns of U . If $Ua = 0$, then the corresponding linear relation among the w_j is trivial because the w_j are linearly independent. Thus U is injective, hence invertible. $\square$

Theorem 13.15 (Eventual change of basis from equal spans). eventually_exists_invertible_changeBasis ✓ Stmt OK *If two finite families of blocks have asymptotically orthonormal overlaps within each family and their MPV spans agree for every system size N , then for all sufficiently large N there exists an invertible matrix U_N expressing the MPV states of one family as linear combinations of the MPV states of the other.*

Proof. ✓ Proof OK By Theorem 13.2, both families are linearly independent for all sufficiently large N . At such an N , the span equality lets us apply Theorem 13.14. $\square$

Theorem 13.16 (Permutation rigidity for equal block counts). exists_perm_dimEq_gaugePhaseEquiv_of_overlapOrtho ✓ Stmt OK
 $(A_j)_{j=1}^g$ and $(B_j)_{j=1}^g$ be injective families satisfying trace-preserving normalization, with self-overlaps tending to 1 and off-diagonal overlaps tending to 0 within each family. If the spans of the MPV states agree for every system size N , then there is a permutation π of $\{1, \dots, g\}$ such that A_j and $B_{\pi(j)}$ have the same bond dimension and are gauge-phase equivalent.

Proof. ✓ Proof OK For large N , Theorem 13.15 gives an invertible matrix U_N relating the two MPV families. Fix j . If every mixed overlap $O_{A_i B_j}(N)$ tended to 0, then taking inner products of the relation with the A_i and using that the Gram matrix of the A -family tends to the identity would force the j th column of U_N to vanish asymptotically, contradicting the self-overlap limit of B_j . Thus each B_j has a matched block $A_{f(j)}$ with non-decaying mixed overlap. Theorems 9.23 and 9.25 then force equal bond dimension and gauge-phase equivalence for each match. Off-diagonal decay within the B -family makes f injective, hence bijective, and its inverse is the required permutation. $\square$

Theorem 13.17 (Permutation rigidity for proportional MPV decompositions). exists_eq_numBlocks_and_equiv_gaugePhase_of
 $(A_j)_{j=1}^{g_A}$ and $(B_k)_{k=1}^{g_B}$ be injective families satisfying the TP normalization $\sum_i (A_j^i)^\dagger A_j^i = \mathbb{1}$ and $\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$, whose self-overlaps converge to 1 and whose cross-overlaps within each family converge to 0. Assume moreover that there exist MPS tensors A_{tot} and B_{tot} , coefficient arrays $a_{N,j}$ and $b_{N,k}$, and a sequence c_N with nonzero limits a_j^∞ , b_k^∞ , c^∞ such that for every N ,

$$|V^{(N)}(A_{\text{tot}})\rangle = \sum_{j=1}^{g_A} a_{N,j} |V^{(N)}(A_j)\rangle, \quad |V^{(N)}(B_{\text{tot}})\rangle = \sum_{k=1}^{g_B} b_{N,k} |V^{(N)}(B_k)\rangle,$$

with $a_{N,j} \rightarrow a_j^\infty \neq 0$ and $b_{N,k} \rightarrow b_k^\infty \neq 0$, and also

$$|V^{(N)}(A_{\text{tot}})\rangle = c_N |V^{(N)}(B_{\text{tot}})\rangle$$

with $c_N \rightarrow c^\infty \neq 0$. Then $g_A = g_B$, and there is a permutation π of the common block index set such that $A_{\pi(k)}$ and B_k have the same bond dimension and are gauge-phase equivalent for each k . The canonical-form application of these hypotheses is described in the following remark.

Proof. ✓ Proof OK The proportional decomposition implies that, for each block on the B -side, the mixed overlaps with A_{tot} are linear combinations of the overlaps with the A -family. If every mixed overlap with a fixed B_k decayed to 0, then the coefficient limits would force the limiting coefficient b_k^∞ to vanish, contrary to the hypotheses. Thus every B_k matches some A_j with non-decaying mixed overlap. Theorems 9.23 and 9.25 then force equal bond dimension and gauge-phase equivalence for each match. Repeating the same argument in the opposite direction gives an injective matching from the A -family to the B -family. Comparing the two injective matchings yields the permutation. $\square$

Remark 13.18. In the canonical-form application, the abstract data above come from block-diagonal tensors

$$A_{\text{tot}} = \bigoplus_{j=1}^{g_A} \mu_j A_j, \quad B_{\text{tot}} = \bigoplus_{k=1}^{g_B} \nu_k B_k,$$

and Theorem 2.36 gives the coefficient arrays $a_{N,j} = \mu_j^N$ and $b_{N,k} = \nu_k^N$. To fit the convergence hypotheses of Theorem 13.17, one first normalizes by the dominant weight on each side: the

relevant arrays are $(\mu_j/\mu_0)^N$ and $(\nu_k/\nu_0)^N$, with $(\mu_j/\mu_0)^N \rightarrow 0$ for $j \geq 1$ and $(\nu_k/\nu_0)^N \rightarrow 0$ for $k \geq 1$ by Theorem 13.27, while the overall factors μ_0^N and ν_0^N are absorbed into the proportionality constant c_N . Definition 13.4 supplies injectivity, TP normalization, and the within-family overlap limits, so the theorem isolates the abstract permutation argument from this canonical-form setting.

Theorem 13.19 (Permutation rigidity with asymptotically orthonormal overlaps). exists_eq_numBlocks_and_equiv_gaugePhase
 $(A_j)_{j=1}^{g_A}$ and $(B_k)_{k=1}^{g_B}$ be two families of injective tensors satisfying the TP normalization $\sum_i (A_j^i)^\dagger A_j^i = \mathbb{1}$ and $\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$, whose self-overlaps converge to 1 and whose cross-overlaps within each family converge to 0. If for every system size N ,

$$\text{span}_{\mathbb{C}}\{|V^{(N)}(A_j)\rangle : 1 \leq j \leq g_A\} = \text{span}_{\mathbb{C}}\{|V^{(N)}(B_k)\rangle : 1 \leq k \leq g_B\},$$

then $g_A = g_B$, and there is a permutation π of the common block index set such that A_j and $B_{\pi(j)}$ have the same bond dimension and are gauge-phase equivalent for each j .

Proof. ✓ Proof OK By Theorem 13.2, both families are linearly independent for all sufficiently large N . At such an N , the common span has the same dimension when computed from the A -family or the B -family, so $g_A = g_B$. After identifying the two index sets, Theorem 13.16 yields the permutation conclusion. □

Theorem 13.20 (Permutation rigidity for proportional decompositions, irreducible TP version). exists_eq_numBlocks_and_equiv_gaugePhase_of_proportional_decomp_of_irreducible_TP ✓ Stmt OK Under the hypotheses of Theorem 13.17, replace injectivity of each block by irreducibility while keeping trace-preserving normalization and the same overlap limits. Then the same permutation conclusion holds.

Proof. ✓ Proof OK The same matching argument applies. The only change is at the two contradiction steps: the overlap-decay inputs for injective blocks are replaced by the corresponding irreducible trace-preserving decay theorems. □

Theorem 13.21 (Canonical-form proportional decomposition theorem). fundamentalTheorem_of_IsCanonicalFormBNT ✓ Stmt OK
two families in canonical form with BNT separation admit proportional MPV decompositions with convergent nonzero coefficients, then they have the same number of blocks and match blockwise up to permutation, bond-dimension equality, and gauge-phase equivalence.

Proof. ✓ Proof OK Project the two canonical-form hypotheses to blockwise injectivity, left-canonical normalization, self-overlap convergence, and BNT separation. Then Theorem 13.17 applies to the resulting proportional decomposition data. □

Theorem 13.22 (Normal-canonical proportional decomposition theorem). fundamentalTheorem_of_IsNormalCanonicalFormBNT
two families in normal canonical form with BNT separation admit proportional MPV decompositions with convergent nonzero coefficients, then they have the same number of blocks and match blockwise up to permutation, bond-dimension equality, and gauge-phase equivalence.

Proof. ✓ Proof OK Project the two normal-canonical hypotheses to irreducibility, left-canonical normalization, self-overlap convergence, and BNT separation, and then apply Theorem 13.20. □

Remark 13.23. Theorem 13.19 is retained for completeness but is not used in the proofs given here; the proportional version (Theorem 13.17) suffices for the Fundamental Theorem.

13.2.1 Fixed-size separation by Vandermonde inversion

Theorem 13.24 (Vandermonde separation). [vandermonde_separation](#) ✓ Stmt OK Let C be a canonical form with r blocks and pairwise distinct weights μ_k . If coefficients c_k satisfy

$$\sum_k c_k \mu_k^N = 0$$

for $N = 0, \dots, r-1$, then every c_k vanishes.

Proof. ✓ Proof OK The Vandermonde matrix with entries $V_{Nk} = \mu_k^N$ is invertible when the weights are pairwise distinct. Therefore the displayed linear system has only the zero solution. $\square$

Theorem 13.25 (Fixed-size linear independence of block MPVs). [block_mpv_lin_indep_at_fixed_size](#) ✓ Stmt OK Let C be a canonical form with r blocks and pairwise distinct weights μ_k , and fix a system size N_0 . Assume that each block MPV $|V^{(N_0)}(A_k)\rangle$ is nonzero. If

$$\sum_k c_k V^{(N_0)}(A_k)_\sigma \mu_k^i = 0$$

for every word σ of length N_0 and every $i = 0, \dots, r-1$, then $c_k = 0$ for all k .

Proof. ✓ Proof OK Apply Theorem 13.24 pointwise in the word σ . This shows that every component of $c_k |V^{(N_0)}(A_k)\rangle$ vanishes. Since the MPV of each block is nonzero at size N_0 , some component survives, so $c_k = 0$. $\square$

Remark 13.26. These fixed-size Vandermonde theorems remain available for alternative finite-length arguments, but the present BNT proof chain uses the eventual linear independence from Theorem 13.2 instead.

13.3 Coefficient convergence

Theorem 13.27 (Coefficient ratio decay). [coeff_ratio_tendsto](#) ✓ Stmt OK Under the canonical form with BNT separation of Definition 13.4, indexed so that μ_0 is the (strictly) dominant weight, one has $(\mu_k/\mu_0)^N \rightarrow 0$ for every $k \geq 1$ as $N \rightarrow \infty$.

Proof. ✓ Proof OK By the BNT separation condition in Definition 13.4, $|\mu_k| < |\mu_0|$ strictly, so $|\mu_k/\mu_0| < 1$ and the geometric sequence converges to 0. $\square$

Remark 13.28 (Strict-dominance regime). Not ready Theorem 13.27 applies when one modulus is strictly dominant: $|\mu_0| > |\mu_k|$ for $k \geq 1$. In the more general periodic setting of [CPGSV21, Eq. (72a)], the coefficient attached to block j can take the form $\sum_q \mu_{j,q}^N$, where the $\mu_{j,q}$ are the individual weights contributed by that block. The oscillatory case with $|\mu_{j,q}| = 1$ is treated separately in the full paper.

13.4 Newton–Girard identities

Theorem 13.29 (Newton–Girard trace recursion). [charpoly_eq_of_forall_trace_pow_eq](#) ✓ Stmt OK If $A, B \in M_n(\mathbb{C})$ satisfy $\text{tr}(A^k) = \text{tr}(B^k)$ for all $k \geq 1$, then A and B have the same characteristic polynomial.

Proof. ✓ Proof OK Let e_m denote the m th elementary symmetric polynomial in the eigenvalues, with $e_0 = 1$. The Newton–Girard recursion

$$me_m = \sum_{i=1}^m (-1)^{i-1} e_{m-i} \operatorname{tr}(A^i)$$

expresses the coefficients of the characteristic polynomial in terms of the power-sum traces $\operatorname{tr}(A^i)$. Equal power sums for A and B therefore give equal coefficients, hence equal characteristic polynomials. □

Theorem 13.30 (Equal power sums imply equal multisets). sum_pow_eq_implies_multiset_eq ✓ Stmt OK Let $\alpha, \beta : \{1, \dots, n\} \rightarrow \mathbb{C}$. If $\sum_i \alpha_i^k = \sum_i \beta_i^k$ for all $k \geq 1$, then the multisets $\{\alpha_i\}$ and $\{\beta_i\}$ are equal.

Proof. ✓ Proof OK Construct diagonal matrices $\operatorname{diag}(\alpha)$ and $\operatorname{diag}(\beta)$. The power-sum hypothesis gives $\operatorname{tr}(\operatorname{diag}(\alpha)^k) = \operatorname{tr}(\operatorname{diag}(\beta)^k)$. By Theorem 13.29, the characteristic polynomials agree: $\prod_i (X - \alpha_i) = \prod_i (X - \beta_i)$. Extracting roots gives multiset equality. □

Remark 13.31. Theorems 13.29 and 13.30 remain relevant for the general equal-MPV Theorem 14.5, where one expects a power-sum step to recover the multiplicity-weight data. The present explicit-coefficient special case, Theorem 14.6, instead uses linear independence of the block MPV family supplied by a basis of normal tensors.

Chapter 14

Proof of the Fundamental Theorem

This chapter proves the Fundamental Theorem of Matrix Product States. We state the Fundamental Theorem in two versions: one for block-injective tensors in canonical form with BNT separation, and one for the normal-canonical variant. We then treat the equal-MPV case, including the common-block-structure specialization that deduces gauge equivalence. The results combine the canonical form reduction (Chapter 11), block separation (Chapter 12), spectral gap (Chapter 9), and the BNT permutation arguments of Chapter 13.

The theorems in this chapter follow the approach of [CPGSV17a]: they first block the tensor to remove non-trivial periodicity, then apply the aperiodic fundamental theorem. Chapter 15 develops the stronger irreducible-form framework of [DCCSPG17], which gives a fundamental theorem valid for periodic tensors directly, with an extra Z -gauge degree of freedom.

14.1 Conditional Fundamental Theorem

The following theorem combines the reduction to canonical form (zero-block separation, TP gauge, blocking, primitivity) with the Fundamental Theorem for normal canonical form. It is still conditional on the two post-blocking hypotheses left open in Chapter 11: irreducibility for the blocked sectors and pairwise distinct nonzero weight norms. The direct periodic route of Chapter 15 is not used in this chapter.

Theorem 14.1 (Conditional Fundamental Theorem). weakFundamentalTheorem_conditional ✔ Stmt OK *Let $(\mu_j^A, A_j)_{j=1}^{g_A}$ and $(\mu_k^B, B_k)_{k=1}^{g_B}$ be two families in normal canonical form, each satisfying the non-degeneracy condition that no two distinct blocks in the same family are gauge-phase equivalent. Let A_{tot} and B_{tot} be tensors with MPV decompositions*

$$V^{(N)}(A_{\text{tot}})_\sigma = \sum_j a_{N,j} V^{(N)}(A_j)_\sigma, \quad V^{(N)}(B_{\text{tot}})_\sigma = \sum_k b_{N,k} V^{(N)}(B_k)_\sigma,$$

with $a_{N,j} \rightarrow a_j \neq 0$ and $b_{N,k} \rightarrow b_k \neq 0$. Suppose moreover that $V^{(N)}(A_{\text{tot}})_\sigma = c_N V^{(N)}(B_{\text{tot}})_\sigma$ with $c_N \rightarrow c \neq 0$. Then $g_A = g_B$ and there is a permutation π such that $D_j^A = D_{\pi(j)}^B$ and A_j is gauge-phase equivalent to $B_{\pi(j)}$ for every j .

Proof. ✔ Proof OK This is exactly the normal-canonical-form Fundamental Theorem (Theorem 14.4). The normal canonical form data provide the BNT separation and overlap convergence hypotheses; the non-degeneracy condition (no two distinct blocks in the same family are gauge-phase

equivalent) ensures that the permutation π produced by Theorem 14.4 is uniquely determined, so that the per-block gauge equivalences can be assembled consistently. $\square$

14.2 Proportional Fundamental Theorem

Theorem 14.2 (Fundamental Theorem — proportional MPV case). [fundamentalTheorem_proportionalMPV_CFBNT](#) ✓ Stmt OK *Let A_{tot} and B_{tot} be MPS tensors in canonical form with BNT separation, with associated families $(\mu_j^A, A_j)_{j=1}^{g_A}$ and $(\mu_k^B, B_k)_{k=1}^{g_B}$. Suppose there exist coefficient arrays $a_{N,j}$, $b_{N,k}$ and nonzero limits a_j^∞ , b_k^∞ , c^∞ such that*

$$V^{(N)}(A_{\text{tot}})_\sigma = \sum_{j=1}^{g_A} a_{N,j} V^{(N)}(A_j)_\sigma, \quad V^{(N)}(B_{\text{tot}})_\sigma = \sum_{k=1}^{g_B} b_{N,k} V^{(N)}(B_k)_\sigma,$$

with $a_{N,j} \rightarrow a_j^\infty \neq 0$ and $b_{N,k} \rightarrow b_k^\infty \neq 0$, and suppose also that

$$V^{(N)}(A_{\text{tot}})_\sigma = c_N V^{(N)}(B_{\text{tot}})_\sigma$$

for a sequence $c_N \rightarrow c^\infty \neq 0$. Then $g_A = g_B$, and there is a permutation π such that $D_j^A = D_{\pi(j)}^B$ and A_j is gauge-phase equivalent to $B_{\pi(j)}$ for each j .

Proof. ✓ Proof OK Apply Theorem 13.17 to the two canonical-form decompositions with BNT separation. This yields the permutation π that reindexes the separated sectors before comparing them one by one. These hypotheses provide injectivity, TP normalization, diagonal overlap limits, and off-diagonal decay via Theorem 13.9. With the supplied decomposition identities and convergence hypotheses, Theorem 13.17 yields the equality of block counts, the permutation, and the per-block gauge-phase equivalence. $\square$

Remark 14.3 (Coefficient arrays for canonical-form decompositions). In applications the coefficient arrays in Theorem 14.2 come from the block-diagonal decomposition itself. If

$$A_{\text{tot}} = \bigoplus_{j=1}^{g_A} \mu_j^A A_j, \quad B_{\text{tot}} = \bigoplus_{k=1}^{g_B} \mu_k^B B_k,$$

then Theorem 2.36 gives

$$V^{(N)}(A_{\text{tot}})_\sigma = \sum_{j=1}^{g_A} (\mu_j^A)^N V^{(N)}(A_j)_\sigma, \quad V^{(N)}(B_{\text{tot}})_\sigma = \sum_{k=1}^{g_B} (\mu_k^B)^N V^{(N)}(B_k)_\sigma.$$

Thus one takes $a_{N,j} = (\mu_j^A)^N$ and $b_{N,k} = (\mu_k^B)^N$, while the sequence c_N is the global proportionality factor from the hypothesis $V^{(N)}(A_{\text{tot}}) = c_N V^{(N)}(B_{\text{tot}})$. Theorem 14.2 keeps these arrays explicit because Theorem 13.17 is formulated in that abstract form.

Theorem 14.4 (Fundamental Theorem for normal canonical form in the proportional case). [fundamentalTheorem_proportionalMPV_normalCFBNT](#) ✓ Stmt OK *Let A_{tot} and B_{tot} be MPS tensors in normal canonical form with BNT separation, with associated families $(\mu_j^A, A_j)_{j=1}^{g_A}$ and $(\mu_k^B, B_k)_{k=1}^{g_B}$. Suppose there exist coefficient arrays $a_{N,j}$, $b_{N,k}$ and nonzero limits a_j^∞ , b_k^∞ , c^∞ such that*

$$V^{(N)}(A_{\text{tot}})_\sigma = \sum_{j=1}^{g_A} a_{N,j} V^{(N)}(A_j)_\sigma, \quad V^{(N)}(B_{\text{tot}})_\sigma = \sum_{k=1}^{g_B} b_{N,k} V^{(N)}(B_k)_\sigma,$$

with $a_{N,j} \rightarrow a_j^\infty \neq 0$ and $b_{N,k} \rightarrow b_k^\infty \neq 0$, and suppose also that

$$V^{(N)}(A_{\text{tot}})_\sigma = c_N V^{(N)}(B_{\text{tot}})_\sigma$$

for a sequence $c_N \rightarrow c^\infty \neq 0$. Then $g_A = g_B$, and there is a permutation π such that $D_j^A = D_{\pi(j)}^B$ and A_j is gauge-phase equivalent to $B_{\pi(j)}$ for each j .

Proof. ✓ Proof OK Normal canonical form with BNT separation provides irreducibility, TP normalization, and primitive self-overlap convergence via Theorem 11.39. Theorem 11.40 supplies the leading-block identification step: a nondecaying mixed overlap forces the corresponding comparison block to be gauge-phase equivalent. The same peeling argument as in the block-injective case therefore yields the block permutation and gauge-phase equivalence. Remark 13.12 explains why the argument stops at this separated normal-canonical level. $\square$

14.3 Equal-MPV Fundamental Theorem

Theorem 14.5 (Fundamental Theorem — equal MPV case). Not ready Let A_{tot} and B_{tot} be MPS tensors in canonical form with BNT separation, with associated families $(\mu_j^A, A_j)_{j=1}^{g_A}$ and $(\mu_k^B, B_k)_{k=1}^{g_B}$. If

$$V^{(N)}(A_{\text{tot}})_\sigma = V^{(N)}(B_{\text{tot}})_\sigma$$

for every N and σ , then the block-diagonal tensors $\bigoplus_{j=1}^{g_A} \mu_j^A A_j$ and $\bigoplus_{k=1}^{g_B} \mu_k^B B_k$ are gauge equivalent.

Proof. Begin by applying Theorem 14.2 with $c_N = 1$, which yields a permutation of the matched blocks and blockwise gauge-phase equivalences. One then compares the resulting MPV expansions from Theorem 2.36. The explicit-coefficient version (Theorem 14.6) proves gauge equivalence of the reindexed weighted block-diagonal tensors using BNT linear independence. The remaining step combines this with the coefficient-array construction of Remark 14.3 and the weight-recovery theorem (Theorem 14.16). After the weights are identified, the phase factors are absorbed into them and the blockwise conjugacies combine to a global gauge equivalence. $\square$

Theorem 14.6 (Fundamental Theorem — equal MPV case with explicit coefficients). fundamentalTheorem_equalMPV_full ✓ Let A_{tot} and B_{tot} be MPS tensors in canonical form with BNT separation, with associated families $(\mu_j^A, A_j)_{j=1}^{g_A}$ and $(\mu_k^B, B_k)_{k=1}^{g_B}$. Suppose there exist coefficient arrays $a_{N,j}$, $b_{N,k}$ and nonzero limits a_j^∞ , b_k^∞ such that

$$V^{(N)}(A_{\text{tot}})_\sigma = \sum_{j=1}^{g_A} a_{N,j} V^{(N)}(A_j)_\sigma, \quad V^{(N)}(B_{\text{tot}})_\sigma = \sum_{k=1}^{g_B} b_{N,k} V^{(N)}(B_k)_\sigma,$$

with $a_{N,j} \rightarrow a_j^\infty \neq 0$ and $b_{N,k} \rightarrow b_k^\infty \neq 0$. If

$$V^{(N)}(A_{\text{tot}})_\sigma = V^{(N)}(B_{\text{tot}})_\sigma$$

for every N and σ , then $g_A = g_B$, and there is a permutation π such that $D_j^A = D_{\pi(j)}^B$ and, after reindexing the B -blocks by π , the weighted block-diagonal tensors $\bigoplus_{j=1}^{g_A} \mu_j^A A_j$ and $\bigoplus_{j=1}^{g_A} \mu_{\pi(j)}^B B_{\pi(j)}$ are gauge equivalent.

Proof. ✓ Proof OK Apply Theorem 14.2 with $c_N = 1$. This yields $g_A = g_B$, a permutation π , equal block dimensions, and for each j a gauge-phase equivalence

$$B_{\pi(j)}^i = \zeta_j X_j A_j^i X_j^{-1}.$$

By Theorem 2.36,

$$V^{(N)}(A_{\text{tot}})_\sigma = \sum_{j=1}^{g_A} (\mu_j^A)^N V^{(N)}(A_j)_\sigma,$$

and similarly for B_{tot} . Reindex the B -sum by π and use the gauge-phase identity to rewrite

$$V^{(N)}(B_{\pi(j)})_\sigma = \zeta_j^N V^{(N)}(A_j)_\sigma.$$

Then equality of MPVs implies

$$\sum_{j=1}^{g_A} \left((\mu_j^A)^N - (\mu_{\pi(j)}^B \zeta_j)^N \right) V^{(N)}(A_j)_\sigma = 0.$$

By Theorem 13.11 and Definition 13.1, the family $\{V^{(N)}(A_j)\}_{j=1}^{g_A}$ is linearly independent for all sufficiently large N , so

$$(\mu_j^A)^N = (\mu_{\pi(j)}^B \zeta_j)^N$$

for every j and all large N . Comparing two consecutive values of N and using that the weights are nonzero gives

$$\mu_j^A = \mu_{\pi(j)}^B \zeta_j.$$

Thus the phase factors can be absorbed into the weights, and the resulting blockwise conjugacies combine to give a gauge equivalence of the reindexed weighted block-diagonal tensors. $\square$

Remark 14.7. Theorem 14.6 recovers [CPGSV21, Corollary IV.5] for tensors in canonical form with BNT separation, without assuming common block structure and without Newton–Girard identities. The weight matching uses BNT linear independence (Theorem 13.11 and Definition 13.1) directly.

Theorem 14.8 (Fundamental Theorem — equal MPV case, common block structure).

[fundamentalTheorem_equalMPV_CFBNT](#)

a common block structure: the same number of blocks g , the same bond dimensions $(D_k)_{k=1}^g$, and the same weights $(\mu_k)_{k=1}^g$. Let $(\mu_k, A_k)_{k=1}^g$ and $(\mu_k, B_k)_{k=1}^g$ be two families in canonical form with BNT separation on this common data. If the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ generate the same MPV family, then each pair (A_k, B_k) is gauge equivalent, and hence the corresponding block-diagonal tensors are gauge equivalent.

Proof. ✓ Proof OK Since both families share the canonical form data, the BNT separation condition is not needed. The canonical-form fundamental theorem (Theorem 12.27) then applies directly to this common (g, D_k, μ_k) structure, yielding per-block gauge equivalence and hence global gauge equivalence. $\square$

14.3.1 Multiplicity structure for repeated blocks

This subsection presents the multiplicity data in which repeated copies of the same basis block are kept separate rather than merged into a single weighted block.

Definition 14.9 (Sector data and sector decomposition). [SectorWeightData](#) [SectorDecomposition](#) [✓ Stmt OK](#) A sector datum over g basis blocks specifies: multiplicities $r_j > 0$, sector weights $\mu_{j,q} \neq 0$ for $q \in \{1, \dots, r_j\}$, and the sector coefficients

$$c_{S,j}^{(N)} := \sum_{q=1}^{r_j} \mu_{j,q}^N$$

for a sector datum S . A sector decomposition consists of the basis blocks $(A_j)_{j=1}^g$ together with their sector data.

Theorem 14.10 (Sector decomposition formula). [mpv_toTensor_eq_sum_coeff](#) [✓ Stmt OK](#) For a sector decomposition S with associated tensor A_{tot} , the MPV satisfies

$$V^{(N)}(A_{\text{tot}})_\sigma = \sum_{j=1}^g c_{S,j}^{(N)} V^{(N)}(A_j)_\sigma.$$

Theorem 14.11 (Eventual coefficient comparison from BNT linear independence). [coeff_eventually_eq_of_sameMPV](#) [✓ Stmt OK](#) If two sector data S, T over the same basis, with coefficient arrays $c_{S,j}^{(N)}$ and $c_{T,j}^{(N)}$, produce equal total MPVs and the basis is eventually linearly independent (the BNT property), then $c_{S,j}^{(N)} = c_{T,j}^{(N)}$ for all sufficiently large N .

Theorem 14.12 (Heterogeneous geometric sequence extrapolation). [power_sums_eq_of_eventually_eq_hetero](#) [✓ Stmt OK](#) If two finite families of nonzero complex numbers, possibly with different cardinalities, have equal power sums $\sum_i \alpha_i^k = \sum_j \beta_j^k$ for all sufficiently large exponents k , then the same equality holds for every exponent $k \geq 0$.

Proof. [✓ Proof OK](#) Concatenate the two families with coefficients $+1$ and -1 . The resulting finite linear combination of nonzero geometric sequences vanishes for all sufficiently large exponents, so the telescoping induction for such combinations forces it to vanish identically. $\square$

Theorem 14.13 (Geometric sequence extrapolation). [power_sums_eq_of_eventually_eq](#) [✓ Stmt OK](#) If two finite families of nonzero complex numbers with the same cardinality have equal power sums $\sum_i \alpha_i^k = \sum_i \beta_i^k$ for all $k > N_0$, then equality holds for all $k \geq 0$.

Proof. [✓ Proof OK](#) This is the equal-cardinality specialization of Theorem 14.12. $\square$

Theorem 14.14 (Copy-count recovery from eventual coefficient equality). [copies_eq_of_eventually_coeff_eq](#) [✓ Stmt OK](#) Let S and T be sector data over the same basis. If their coefficient arrays satisfy $c_{S,j}^{(N)} = c_{T,j}^{(N)}$ for every basis block j and all sufficiently large N , then their copy counts agree: $r_j^S = r_j^T$ for every j .

Proof. [✓ Proof OK](#) Apply Theorem 14.12 to the two weight families in a fixed sector and then evaluate the resulting power-sum equality at exponent 0. Since each summand is then 1, the equality is exactly equality of the two copy counts. $\square$

Theorem 14.15 (BNT recovery of copy counts and sector weights). [exists_copies_eq_and_weight_multiset_eq_of_sameMPV_bnt](#) [✓ Stmt OK](#) Let S and T be two sector data over the same basis, without assuming their multiplicities agree. If the basis is eventually linearly independent and the total MPVs of S and T agree, then there are copy-count equalities $r_j^S = r_j^T$ for every basis block j , and, after transporting along these equalities, the sector weight multisets $\{\mu_{j,q}^S\}$ and $\{\mu_{j,q}^T\}$ are equal for every j .

Proof. ✓ Proof OK BNT linear independence and equality of total MPVs give eventual coefficient equality by Theorem 14.11. Then Theorem 14.14 recovers the copy-count equalities. After identifying the two index sets via these copy-count equalities, the positive-power coefficient equalities extend to all positive exponents by Theorem 14.13, and Newton–Girard (Theorem 13.30) recovers the weight multisets. $\square$

Theorem 14.16 (Weight multiset recovery from multiplicity data). weight_multiset_eq_of_sameMPV_bnt ✓ Stmt OK Let S and T be two sector data over the same basis with the same multiplicities ($r_j^S = r_j^T$ for all j). If the basis is eventually linearly independent and the total MPVs of S and T agree, then for each basis block j the sector weight multisets $\{\mu_{j,q}^S\}$ and $\{\mu_{j,q}^T\}$ are equal.

The argument proceeds as follows:

1. BNT LI + equal MPVs $\Rightarrow$ eventual coefficient equality (Theorem 14.11);
2. Geometric extrapolation $\Rightarrow$ full coefficient equality for all positive k (Theorem 14.13);
3. Newton–Girard $\Rightarrow$ equal weight multisets (Theorem 13.30).

At the level of block contributions, one compares the weighted sector sums $\sum_j c_{A,j}^{(N)} V^{(N)}(A_j)$ and $\sum_j c_{B,j}^{(N)} V^{(N)}(B_j)$, with the coefficients now aggregated over the multiplicity layers.

Theorem 14.17 (Two-basis sector comparison after phase matching). fundamentalTheorem_equalMPV_sectorDecomposition_hetero P and Q be sector decompositions. Assume there is a permutation π matching the basis blocks, the multiplicities satisfy $r_j^P = r_{\pi(j)}^Q$, and for each j there is a nonzero complex number ζ_j such that

$$V^{(N)}(Q_{\pi(j)})_\sigma = \zeta_j^N V^{(N)}(P_j)_\sigma$$

for every N and σ . If the total MPV families of P and Q agree and the basis blocks of P are eventually linearly independent, then for each j the sector weight multiset of P_j equals the multiset obtained from that of $Q_{\pi(j)}$ by multiplication by ζ_j .

Proof. ✓ Proof OK Absorb the factors ζ_j into the sector weights of Q and keep the basis blocks of P . The phase-matching hypothesis and the sector decomposition formula show that the resulting decomposition has the same total MPV family as Q , hence also as P . One is then in the shared-basis situation of Theorem 14.16, which yields the claimed equality of multisets. $\square$

Theorem 14.18 (Gauge-phase paired bases imply phase-matched sector comparison). fundamentalTheorem_equalMPV_sectorL P and Q be sector decompositions. Assume there is a permutation π matching the basis blocks, the multiplicities satisfy $r_j^P = r_{\pi(j)}^Q$, and each matched pair of basis blocks is gauge-phase equivalent. If the total MPV families of P and Q agree and the basis blocks of P are eventually linearly independent, then for each j the sector weight multiset of P_j equals the multiset obtained from that of $Q_{\pi(j)}$ after multiplication by the corresponding gauge phase.

Proof. ✓ Proof OK Gauge-phase equivalence gives the required phase-scaling identity for the MPVs by Theorem 2.22. The conclusion then follows immediately from Theorem 14.17. $\square$

Definition 14.19 (Sector basis matching witness). SectorBasisMatching ✓ Stmt OK A sector basis matching between two sector decompositions P and Q consists of

1. a basis permutation (bijection) $\pi : \{1, \dots, g_P\} \simeq \{1, \dots, g_Q\}$,
2. per-block multiplicity equalities $r_j^P = r_{\pi(j)}^Q$,

3. per-block bond-dimension equalities $D_j^P = D_{\pi(j)}^Q$, and
4. per-block gauge-phase equivalences $P_j \simeq_{\text{gp}} Q_{\pi(j)}$ after transporting P_j across the bond-dimension equality.

The hypotheses consumed by Theorem 14.18 are exactly the fields of this matching datum.

Definition 14.20 (Sector basis pre-matching). [SectorBasisPreMatching](#) ✓ Stmt OK A *sector basis pre-matching* between sector decompositions P and Q consists of a permutation of their basis blocks, equality of the matched bond dimensions, and gauge-phase equivalence of each matched pair. It does not include the copy-count equality; that equality is recovered from total MPV equality and BNT linear independence in Theorem 14.25.

Definition 14.21 (Single-family overlap-orthogonality data for sector bases). [SectorBasisOverlapOrthoHypotheses](#) ✓ Stmt OK For one sector decomposition P , this condition states positive basis dimensions, trace-preserving normalization of all basis blocks, self-overlap limits equal to 1, and off-overlap limits equal to 0.

Definition 14.22 (Primitive overlap-span hypotheses for sector bases). [SectorBasisOverlapSpanHypotheses](#) ✓ Stmt OK For sector decompositions P and Q , this condition states the primitive overlap-rigidity hypotheses for their basis blocks: nonzero bond dimensions, injectivity, trace-preserving normalization, self-overlap limits equal to 1, off-overlap limits equal to 0, and equality of the finite-length spans of the basis MPV states for every system size.

Theorem 14.23 (Single-family overlap data plus span give overlap-span hypotheses). [to_overlapSpan](#) ✓ Stmt OK If two sector bases each satisfy the single-family overlap-orthogonality data, and one also supplies one-site injectivity of all their basis blocks together with equality of the finite-length MPV spans, then the two bases satisfy the primitive overlap-span hypotheses.

Proof. ✓ Proof OK This is just collecting the one-family fields on the left and right, and adding the remaining injectivity and span-comparison hypotheses. □

Theorem 14.24 (Phase-matched comparison recovers copy counts). [fundamentalTheorem_equalMPV_sectorDecomposition_hetero_](#) P and Q be sector decompositions whose basis blocks are matched by a permutation and whose matched MPVs differ by nonzero phase powers. If the total MPV families agree and the basis MPVs of P are eventually linearly independent, then the matched copy counts agree. After absorbing the phases into the weights of Q , the corresponding sector-weight multisets agree.

Proof. ✓ Proof OK Absorb the phase powers into the sector weights of Q and write both decompositions over the basis of P . Linear independence gives eventual equality of the coefficient power sums, and Theorem 14.15 recovers both the copy-count equalities and the phase-adjusted weight-multiset equalities. □

Theorem 14.25 (A pre-matching gives a sector basis matching). [exists_sectorBasisMatching_of_sameMPV](#) ✓ Stmt OK Let M be a sector basis pre-matching between P and Q . If the total MPV families agree and the basis MPVs of P are eventually linearly independent, then the copy-count equalities recovered by Theorem 14.24 turn M into a full sector basis matching.

Proof. ✓ Proof OK Gauge-phase equivalence gives the required phase-scaling identities for the corresponding basis MPVs. Theorem 14.24 recovers the missing copy-count equalities, which are precisely the extra fields needed for Definition 14.19. □

Theorem 14.26 (Two-basis sector comparison from a pre-matching). [fundamentalTheorem_equalMPV_sectorDecomposition_hetero_](#) M be a sector basis pre-matching between sector decompositions P and Q . If the total MPV families agree and the basis MPVs of P are eventually linearly independent, then the matched copy

counts agree, and after absorbing the gauge phases into the weights of Q the corresponding sector-weight multisets agree.

Proof. ✓ Proof OK The gauge-phase equivalences in the pre-matching give the phase-scaling identities for the corresponding basis MPVs. Theorem 14.24 then gives the copy-count equalities and the phase-adjusted multiset equalities. □

Theorem 14.27 (Overlap rigidity gives a sector basis matching). exists_sectorBasisMatching_of_overlapOrtho_span_sameMPV ✓ *the basis blocks of two sector decompositions have nonzero bond dimensions, are trace-preserving and injective, have asymptotically orthonormal overlaps within each family, and span the same MPV subspace at every system size. If the total MPV families also agree, then there exists a sector basis matching between the two decompositions.*

Proof. ✓ Proof OK The permutation-rigidity theorem for bases of normal tensors gives a permutation, the matched bond dimensions, and the gauge-phase equivalences of the basis blocks, hence a sector basis pre-matching. Asymptotic orthonormality gives the BNT linear-independence condition, and Theorem 14.25 recovers the copy-count equalities from equality of the total MPV families. □

Theorem 14.28 (Primitive overlap-span hypotheses give a sector basis matching). exists_sectorBasisMatching ✓ Stmt OK *two sector decompositions satisfy the primitive overlap-span hypotheses and their total MPV families agree, then there exists a sector basis matching between them.*

Proof. ✓ Proof OK The collected hypotheses are exactly the nonzero-dimension, injectivity, normalization, overlap, and span assumptions of Theorem 14.27. Applying that theorem gives the matching witness. □

Theorem 14.29 (Nonzero-block span equality gives phase-class representative overlap-span data). exists_bnt_sectorDecomp_pair_with_overlapSpan_of_block_span_eq ✓ Stmt OK *Let two nonzero-weight block families be trace-preserving, primitive, irreducible, injective, and carry nonzero weights. If their finite-length MPV spans agree for every system size, then there exist sector decompositions P and Q whose total tensors have the same MPVs as the two weighted nonzero-block tensors, whose bases satisfy BNT linear independence, and whose sector bases satisfy the primitive overlap-span hypotheses.*

Proof. ✓ Proof OK Apply the one-sided construction with the representative-span identity on both sides. The one-sided positive-dimension, injectivity, normalization, and overlap fields fill all fields of Definition 14.22 except the two-family span equality. That span equality follows by transporting the assumed nonzero-block span equality through the two representative-span identities. □

Definition 14.30 (MPV phase equivalence for nonzero-weight blocks). MPVBlockPhaseEquiv ✓ Stmt OK Two blocks A and B are MPV-phase equivalent if there is a nonzero scalar ζ such that, for every system size N and every physical word, the MPV coefficient of B equals ζ^N times the MPV coefficient of A . The two bond dimensions are allowed to differ.

Definition 14.31 (Common MPV phase-cover data). MPVCommonPhaseCover ✓ Stmt OK For two finite nonzero-weight block families, a common MPV phase cover consists of a third finite family of blocks, a map from each nonzero-weight block family onto that common family, and MPV-phase equivalences between each block and its common representative. Both maps are required to be surjective, so the common family contains no unused representative.

Lemma 14.32 (Common MPV phase cover gives nonzero-block span equality). [mpv_span_eq_of_common_phase_cover](#) ✓ Stmt OK *two finite nonzero-weight block families map onto a third finite family of common blocks. Suppose both maps are surjective, and each block has finite-length MPV vectors equal to a nonzero scalar power times the MPV vectors of its common block. Then, for every system size, the two nonzero-weight block families span the same MPV subspace.*

Proof. ✓ Proof OK Each nonzero-block MPV vector lies in the span of the common family because it is a scalar multiple of its common block's MPV vector. Surjectivity gives the reverse inclusion, so each nonzero-block span equals the common span; transitivity gives the equality of the two nonzero-block spans. □

Lemma 14.33 (Common phase-cover data give nonzero-block span equality). [span_eq](#) ✓ Stmt OK *A common MPV phase cover for two nonzero-weight block families gives equality of their finite-length MPV spans at every system size.*

Proof. ✓ Proof OK The common cover data specify the common family, the two surjective class maps, and the MPV-phase equivalences. The span lemma then applies directly. □

Definition 14.34 (Common phase cover from a sector basis pre-matching). [commonPhaseCover](#) ✓ Stmt OK *A sector basis pre-matching between sector decompositions P and Q canonically gives a common MPV phase cover of their basis block families. The common family is the basis of P , the class map from P is the identity, and the class map from Q is the inverse of the pre-matching permutation.*

Lemma 14.35 (Sector basis pre-matching gives basis-span equality). [span_eq](#) ✓ Stmt OK *A sector basis pre-matching between P and Q implies equality, for every system size, of the finite-length MPV spans of the two sector bases.*

Proof. ✓ Proof OK Apply the common-cover span lemma to the common phase cover constructed from the pre-matching. □

Theorem 14.36 (Pre-matching supplies primitive overlap-span hypotheses). [to_overlapSpan](#) ✓ Stmt OK *If two sector bases satisfy the one-family overlap-orthogonality data and their basis blocks are one-site injective, then a sector basis pre-matching between them establishes the finite-length span equality and hence the full primitive overlap-span hypotheses.*

Proof. ✓ Proof OK The pre-matching gives the basis-span equality. Combining that equality with the one-family overlap data and injectivity is exactly the construction of Theorem 14.23. □

Theorem 14.37 (Common phase cover from a BNT proportional-decomposition conclusion). [nonempty_mpvCommonPhaseCover_of_proportionalDecompositionConclusion](#) ✓ Stmt OK *If a BNT proportional-decomposition comparison has produced a permutation of the two basis families together with matched bond dimensions and gauge-phase equivalences, then the two basis families have a common MPV phase cover.*

Proof. ✓ Proof OK Take the left basis family as the common family. The comparison permutation and its inverse give the two class maps, while each gauge-phase equivalence gives the required MPV-phase equivalence. □

Theorem 14.38 (Proportional-decomposition conclusion gives nonzero-block span equality). [mpv_span_eq_of_proportionalDecompositionConclusion](#) ✓ Stmt OK *A BNT proportional-decomposition comparison gives equality, for every system size, of the finite-length MPV spans of the two nonzero-weight block families.*

Proof. ✓ Proof OK First form the common MPV phase cover from the proportional-decomposition comparison. The common-cover span lemma then gives the stated span equality. $\square$

Theorem 14.39 (Common phase cover gives phase-class representative overlap-span data).

exists_bnt_sectorDecomp_pair_with_overlapSpan_of_commonPhaseCover ✓ Stmt OK *Let two block families be trace-preserving, primitive, irreducible, injective, and carry nonzero weights. If they have common MPV phase-cover data, then there exist BNT sector decompositions P and Q for the two weighted nonzero parts, and the sector bases satisfy the primitive overlap-span hypotheses.*

Proof. ✓ Proof OK The common cover gives finite-length span equality for the two nonzero-weight block families. The phase-class representative construction then transports this span equality to the BNT sector bases and supplies the remaining one-family overlap data. $\square$

Theorem 14.40 (Proportional-decomposition data give phase-class representative overlap-span data).

exists_bnt_sectorDecomp_pair_with_overlapSpan_of_proportionalDecompositionConclusion ✓ Stmt OK *Let two block families be trace-preserving, primitive, irreducible, injective, and carry nonzero weights. If a BNT proportional-decomposition comparison supplies a permutation, equal bond dimensions, and gauge-phase equivalences between the two families, then the phase-class representative BNT sector decompositions satisfy the primitive overlap-span hypotheses.*

Proof. ✓ Proof OK First turn the proportional-decomposition data into common MPV phase-cover data. The preceding theorem then converts that cover into the overlap-span hypotheses for the constructed sector bases. $\square$

Theorem 14.41 (Separated normal canonical-form data produce a common phase cover).

nonempty_mpvCommonPhaseCover_of_separated_normalCFBNT_data ✓ Stmt OK *For two normal canonical-form block families with no gauge-phase-equivalent distinct equal-dimension blocks, proportional decompositions whose coefficient limits are nonzero give a common MPV phase cover of the two basis families. This is the BNT route to the cover data; it does not assume equality of finite-length spans.*

Proof. ✓ Proof OK The irreducible trace-preserving BNT permutation theorem gives the permutation matching the basis blocks, bond-dimension equalities, and gauge-phase equivalences. Applying the previous theorem to that conclusion yields the common phase cover data. $\square$

Theorem 14.42 (Separated normal canonical-form data give nonzero-block span equality).

mpv_span_eq_of_separated_normalCFBNT_data ✓ Stmt OK *For two separated normal canonical-form block families as in Theorem 14.41, the BNT comparison data imply equality, at every system size, of the finite-length MPV spans of the two basis families.*

Proof. ✓ Proof OK The separated normal-form comparison first gives a common MPV phase cover. Applying the common-cover span lemma to that cover gives the span equality. $\square$

Theorem 14.43 (Sector comparison via a sector-basis matching).

fundamentalTheorem_equalMPV_sectorDecomposition_hetero_of_ *two sector decompositions P and Q , a sector basis matching (Definition 14.19) between them, eventual linear independence of the basis MPV states of P , and equality of the total MPV families of P and Q , one obtains nonzero phases $(\zeta_j)_j$ such that for each j the sector weight multiset of P_j equals the multiset obtained from that of $Q_{\pi(j)}$ by multiplication by ζ_j .*

Proof. ✓ Proof OK The sector-basis matching contains exactly the permutation, copy, and gauge-phase hypotheses of Theorem 14.18, so the conclusion follows directly from that theorem. $\square$

Theorem 14.44 (After-blocking sector comparison from overlap-span hypotheses). [fundamentalTheorem_after_blocking_sect](#)

A and B have the same MPV family. Suppose that, after a common blocking period $p > 0$, there are sector decompositions P and Q whose total tensors have the same MPVs as the blocked tensors $A^{[p]}$ and $B^{[p]}$, that both decompositions have BNT linear independence, and that P and Q satisfy the primitive overlap-span hypotheses. Then the matched sector-weight conclusion holds: there is a basis permutation, equality of the matched copy counts, and nonzero scalars ζ_j such that the weight multiset over P_j equals the multiset over $Q_{\pi(j)}$ multiplied by ζ_j .

Proof. [✓ Proof OK](#) Blocking preserves the equality of the MPV families of A and B at the common period. Composing this equality with the two sector equivalences gives equality of the total MPV families of P and Q . The primitive overlap-span hypotheses produce a sector basis matching by Theorem 14.28; the two-basis sector comparison theorem then gives the sector-weight multiset equalities. $\square$

Theorem 14.45 (Common nonzero-weight blocks with overlap-span data give sector comparison). [fundamentalTheorem_after_blocking_sector_of_common_blocks_overlapSpan](#) [✓ Stmt OK](#) *Let A and B have the same MPV family. Fix a common blocking period $p > 0$ and exact nonzero-block decompositions of $A^{[p]}$ and $B^{[p]}$ as weighted block tensors whose blocks have nonzero bond dimensions, are trace-preserving, primitive, irreducible, and have nonzero weights. Assume moreover that any pair of BNT sector decompositions equivalent to these two nonzero-block tensors satisfies the primitive overlap-span hypotheses. Then the after-blocking sector comparison conclusion of Theorem 14.44 holds for the sector decompositions obtained from the nonzero-weight block families.*

Proof. [✓ Proof OK](#) Apply the one-sided phase-class BNT construction (Theorem 11.138) to the two nonzero-weight block families. The resulting sector decompositions satisfy the overlap-span hypotheses; Theorem 14.28 then produces a sector basis matching, and Theorem 14.43 gives the matched sector-weight conclusion. $\square$

Theorem 14.46 (Common nonzero-weight blocks with derived overlap data and span comparison). [fundamentalTheorem_after_blocking_sector_of_common_blocks_injectiveSpan](#) [✓ Stmt OK](#) *Let A and B have the same MPV family and fix exact nonzero-block decompositions at a common blocking period as in Theorem 14.45. Assume in addition that the nonzero-weight blocks are one-site injective, and that the two chosen BNT sector bases have equal finite-length MPV spans. Then the same matched sector-weight conclusion holds.*

Proof. [✓ Proof OK](#) The phase-class BNT construction gives BNT linear independence and the single-family overlap-orthogonality data for both sides, and carries one-site injectivity from the nonzero-weight blocks to the chosen basis blocks. The assumed span comparison combines with those one-family data by Theorem 14.23; the matching and sector-weight comparison then proceed as before. $\square$

Theorem 14.47 (Common nonzero-weight blocks with nonzero-block span equality give sector comparison). [fundamentalTheorem_after_blocking_sector_of_common_blocks_blockSpan](#) [✓ Stmt OK](#) *Let A and B have the same MPV family, and fix a common blocking period $p > 0$ with exact nonzero-block decompositions by trace-preserving, primitive, irreducible, injective blocks with nonzero weights. If the two nonzero-weight block families have equal finite-length MPV spans at every system size, then there exist a positive blocking period p' and sector decompositions P, Q over the blocked physical dimension such that $A^{[p']}$ and $B^{[p']}$ have the same MPV families as P and Q , both sector bases satisfy BNT linear independence, and there are a basis permutation, matched copy-count equalities, and nonzero scalars ζ_j making the corresponding sector-weight multisets equal after multiplication by ζ_j .*

Proof. [✓ Proof OK](#) Theorem 14.29 converts the nonzero-block span equality into the primitive overlap-span hypotheses for the chosen sector bases. Equality of the blocked total MPVs gives equality of the sector tensors, so Theorem 14.28 produces the sector basis matching and Theorem 14.43 gives the sector-weight conclusion. $\square$

Theorem 14.48 (Common nonzero-weight blocks with a common MPV phase cover give sector comparison). [fundamentalTheorem_after_blocking_sector_of_common_blocks_phaseCover](#) [✓ Stmt OK](#) *Let A and B have the same MPV family, and fix exact nonzero-block decompositions at a common blocking period as in Theorem 14.47. Assume the nonzero blocks are trace-preserving, primitive, irreducible, injective, and have nonzero weights. Suppose moreover that both nonzero-weight block families map surjectively to one common family of blocks, and that every block is MPV-phase equivalent to its image in the common family. Then the same sector comparison conclusion as in Theorem 14.47 holds.*

Proof. [✓ Proof OK](#) Lemma 14.32 converts the common MPV phase cover into equality of the nonzero-block finite-length spans. Theorem 14.47 then applies without any additional sector-basis assumptions. $\square$

Theorem 14.49 (Common nonzero-weight blocks with common MPV phase-cover data give sector comparison). [fundamentalTheorem_after_blocking_sector_of_common_blocks_commonPhaseCover](#) [✓ Stmt OK](#) *Let A and B have the same MPV family, and fix exact nonzero-block decompositions at a common blocking period as in Theorem 14.48. Assume the nonzero-weight blocks are trace-preserving, primitive, irreducible, injective, and have nonzero weights. If the two nonzero-weight block families are equipped with common MPV phase-cover data, then the same sector comparison conclusion holds.*

Proof. [✓ Proof OK](#) The common cover data specify the common representative family, the two surjective maps, and the MPV-phase equivalences. The common-cover sector theorem then applies with the same nonzero-part decompositions and structural hypotheses. $\square$

Theorem 14.50 (Common nonzero block decompositions with BNT proportional-decomposition data). [afterBlocking_sectorComparison_of_proportionalDecompositionConclusion](#) [✓ Stmt OK](#) *Let A and B have the same MPV family, and fix exact decompositions of the nonzero parts at a common blocking period as in Theorem 14.49. Assume the blocks in the nonzero parts are trace-preserving, primitive, irreducible, injective, and have nonzero weights. If a BNT proportional-decomposition comparison supplies a permutation, equal bond dimensions, and gauge-phase equivalences between the two block families, then the same sector comparison conclusion holds.*

Proof. [✓ Proof OK](#) The BNT proportional-decomposition data give common MPV phase-cover data. Apply Theorem 14.49 to those data. $\square$

Lemma 14.51 (Zero-tail cancellation). [sameMPV2_live_of_sameMPV2_with_zeroTail_eq](#) [✓ Stmt OK](#) *Let A and B have the same MPV family, each decomposed as a zero-tail tensor plus a nonzero part. If the zero-tail dimensions agree, then the nonzero parts themselves have the same MPV family, including at length zero.*

Proof. [✓ Proof OK](#) At positive lengths the zero-tail term vanishes, so the nonzero parts agree. At length zero the zero-tail values are equal by hypothesis, and cancellation gives equality of the nonzero parts there too. $\square$

Theorem 14.52 (Common nonzero-weight blocks with zero-tail agreement and overlap-span data). [fundamentalTheorem_after_blocking_sector_of_common_blocks_overlapSpan_zeroTail](#) [✓ Stmt OK](#) *Let A and*

B have the same MPV family. Fix $p > 0$ and suppose $A^{[p]}$ and $B^{[p]}$ each decompose as a zero-tail tensor plus a weighted nonzero-block tensor, with all blocks trace-preserving, primitive, irreducible, and with nonzero weights. If the two zero-tail dimensions agree and any pair of BNT sector decompositions equivalent to the two nonzero-block tensors satisfies the primitive overlap-span hypotheses, then there exist positive p' and sector decompositions P, Q such that $A^{[p']}$ equals P 's tensor at all positive lengths, $B^{[p']}$ equals Q 's tensor at all positive lengths, P and Q have the same full MPV family, both have BNT data, and the matched sector-weight conclusion of Theorem 14.44 holds.

Proof. ✓ Proof OK Apply the zero-tail cancellation lemma to recover full equality of the nonzero-block tensors. Then proceed as in Theorem 14.45: apply the BNT construction to each block family, compose the sector equivalences to obtain equality of the total MPV families of P and Q , apply the overlap-span matching theorem, and conclude with the two-basis sector comparison theorem. □

Theorem 14.53 (Zero-tail nonzero block decompositions with nonzero-block span equality).

afterBlocking_sectorComparison_zeroTail_of_blockSpan ✓ Stmt OK Let A and B have the same MPV family. Fix $p > 0$ and suppose $A^{[p]}$ and $B^{[p]}$ each decompose as a zero-tail tensor plus a weighted nonzero part, with all blocks in the nonzero part trace-preserving, primitive, irreducible, injective, and with nonzero weights. If the two zero-tail dimensions agree and the two nonzero-weight block families have equal finite-length MPV spans at every system size, then there exist a period $p' > 0$ and sector decompositions P, Q such that $A^{[p']}$ and $B^{[p']}$ agree with the corresponding sector tensors at every positive length, P and Q have the same full MPV family, both have BNT data, and their sector weights satisfy the matched sector-weight conclusion.

Proof. ✓ Proof OK The finite-length span equality gives the primitive overlap-span hypotheses for the BNT sector bases of the two nonzero parts. Zero-tail cancellation identifies the two nonzero parts as full MPV families, while the zero-tail summands vanish at positive lengths, giving the positive-length comparison of $A^{[p]}$ and $B^{[p]}$ with the two sector decompositions. The sector-basis matching and two-basis comparison theorems then give the sector-weight conclusion. □

Definition 14.54 (Remaining span hypotheses for common primitive sectors).

CommonPrimitiveSpanHypotheses ✓ Stmt OK If two common primitive nonzero-sector families at one blocking length, this condition records the remaining hypotheses needed before applying the zero-tail sector comparison: equality of the two zero-tail dimensions, one-site injectivity of all blocks on both sides, and equality of the finite-length MPV spans of the two block families for every system size.

Theorem 14.55 (Common phase cover supplies the remaining span hypotheses).

of_commonPhaseCover ✓ Stmt OK If the two common primitive nonzero-sector families have equal zero-tail dimensions, are one-site injective, and are equipped with a common MPV phase cover, then they satisfy the remaining span hypotheses of Definition 14.54.

Proof. ✓ Proof OK The zero-tail and injectivity fields are exactly the supplied hypotheses. The common phase cover gives the finite-length span equality by Lemma 14.33. □

Definition 14.56 (Remaining phase-cover hypotheses for common primitive sectors).

CommonPrimitivePhaseCoverHypotheses If two common primitive nonzero-sector families at one blocking length, this condition records the phase-cover version of Definition 14.54: equality of the two zero-tail dimensions, one-site injectivity of all blocks on both sides, and a common MPV phase cover for the two block families.

Theorem 14.57 (Phase-cover hypotheses imply the remaining span hypotheses).

toSpanHypotheses ✓ Stmt OK If two common primitive nonzero-sector families satisfy the phase-cover hypotheses of Definition 14.56, then they satisfy the remaining span hypotheses of Definition 14.54.

Proof. ✓ Proof OK The zero-tail equality and the two injectivity assertions are already part of the phase-cover hypotheses. Apply Theorem 14.55 to a common MPV phase cover for the two block families. □

Definition 14.58 (Remaining proportional-comparison hypotheses for common primitive sectors). CommonPrimitiveProportionalHypotheses ✓ Stmt OK For two common primitive nonzero-sector families at one blocking length, this condition records equality of the two zero-tail dimensions, one-site injectivity of all blocks on both sides, and a BNT proportional-decomposition comparison for the two block families. It is the BNT-comparison version of Definition 14.56.

Theorem 14.59 (Proportional-comparison hypotheses imply phase-cover hypotheses). toPhaseCoverHypotheses ✓ Stmt OK *For two common primitive nonzero-sector families satisfy the proportional-comparison hypotheses, then they satisfy the phase-cover hypotheses of Definition 14.56.*

Proof. ✓ Proof OK The zero-tail equality and injectivity assumptions are already present. The BNT proportional-decomposition comparison gives a common MPV phase cover by Theorem 14.37. □

Theorem 14.60 (Proportional-comparison hypotheses imply the remaining span hypotheses). toSpanHypotheses ✓ Stmt OK *If two common primitive nonzero-sector families satisfy the proportional-comparison hypotheses, then they satisfy the remaining span hypotheses of Definition 14.54.*

Proof. ✓ Proof OK First convert the BNT proportional-decomposition comparison into common phase-cover hypotheses. The phase-cover-to-span theorem then gives the finite-length span equality together with the zero-tail and injectivity hypotheses. □

Theorem 14.61 (Zero-tail nonzero block decompositions with a common MPV phase cover). afterBlocking_sectorComparison_zeroTail_of_commonPhaseCover ✓ Stmt OK *Under the hypotheses of Theorem 14.53, a common MPV phase cover for the two nonzero-weight block families supplies the finite-length span equality. Hence there exist a period $p' > 0$ and sector decompositions P, Q such that $A^{[p']}$ and $B^{[p']}$ agree with the corresponding sector tensors at every positive length, P and Q have the same full MPV family, both have BNT data, and their sector weights satisfy the matched sector-weight conclusion.*

Proof. ✓ Proof OK Apply the common-cover span lemma to obtain the nonzero-block span equality at each system size, then invoke the zero-tail block-span theorem. □

Theorem 14.62 (Zero-tail nonzero block decompositions with BNT proportional-decomposition data). afterBlocking_sectorComparison_zeroTail_of_proportionalDecompositionConclusion ✓ Stmt OK *Let A and B have the same MPV family. Fix $p > 0$ and suppose $A^{[p]}$ and $B^{[p]}$ each decompose as a zero-tail tensor plus a weighted nonzero part, with all blocks in the nonzero part trace-preserving, primitive, irreducible, injective, and with nonzero weights. If the two zero-tail dimensions agree and a BNT proportional-decomposition comparison supplies a permutation, equal bond dimensions, and gauge-phase equivalences between the two nonzero-weight block families, then there exist a period $p' > 0$ and sector decompositions P, Q such that $A^{[p']}$ and $B^{[p']}$ agree with the corresponding sector tensors at every positive length, P and Q have the same full MPV family, both have BNT data, and their sector weights satisfy the matched sector-weight conclusion.*

Proof. ✓ Proof OK First convert the proportional-decomposition conclusion into common MPV phase-cover data. The preceding zero-tail theorem then applies. □

Theorem 14.63 (Common primitive sectors with the remaining span hypotheses). [afterBlocking_sectorComparison_zeroTail](#)

A and B have the same MPV family and that the canonically blocked nonzero parts agree with the same families written using the reindexing of blocked physical words. The common primitive theorem then produces one positive blocking length, zero-tail equations, trace-preserving primitive irreducible nonzero-sector families with positive bond dimensions and nonzero weights, and the positive-length and length-zero identities for those families. If these produced families also satisfy Definition 14.54, then the zero-tail sector comparison conclusion holds: after a positive blocking, there are sector decompositions P, Q agreeing with A and B at all positive lengths, with P and Q generating the same full MPV family, carrying BNT data, and with their sector weights matched up to a permutation and nonzero phases.

Proof. [✓ Proof OK](#) Apply the common primitive theorem to obtain the nonzero-sector families and their structural properties. The remaining span hypotheses provide equal zero-tail dimensions, one-site injectivity, and finite-length nonzero-sector span equality. These are exactly the additional hypotheses required by the zero-tail block-span comparison theorem. $\square$

Definition 14.64 (Blocked-word relabeling hypothesis for common sectors). [CommonSectorRelabelingHypothesis](#) [✓ Stmt OK](#)

is the one-sided assertion that the canonically blocked weighted nonzero part agrees, as an MPV family, with the same common cyclic-sector family after the explicit relabeling of blocked physical words. It is the formal hypothesis isolated by the remaining coordinate equality for direct and iterated blocking.

Definition 14.65 (Global coordinate-grouping hypothesis for common sectors). [CommonGroupedBlockCastHypothesis](#) [✓ Stmt OK](#)

is the corresponding global assertion at the level of blocked words: every common blocked cyclic-sector family satisfies the coordinate-grouping condition for each original nonzero block. Equivalently, the canonical identification with each iterated blocked alphabet agrees with grouping the direct blocked word into consecutive words of the period-removal length.

Theorem 14.66 (Coordinate grouping implies common-sector relabeling). [toRelabelingHypothesis](#) [✓ Stmt OK](#) *If the global coordinate-grouping hypothesis holds, then the blocked-word relabeling hypothesis used by the common-sector structural theorem holds. The proof is exactly the weighted finite-sum comparison for direct and iterated blocking applied to each common cyclic-sector family.*

Proof. [✓ Proof OK](#) Apply the coordinate-grouping comparison to each one-sided family and sum over the original nonzero blocks with the transported powers of the weights. $\square$

Theorem 14.67 (Common-length sectors with conditional block-span consequences). [afterBlocking_commonSector_blockSpan](#)

two tensors generate the same MPV family and assume the blocked-word relabeling hypothesis used in the common-sector construction. Then there is a common blocking length at which both tensors have zero-tail decompositions whose nonzero parts are weighted families of trace-preserving, primitive, tensor-irreducible blocks with positive bond dimension and nonzero weights. The theorem further records the three positive-length MPV agreements $\mathcal{V}^+(\cdot) = \mathcal{V}^+(\cdot)$: the two agreements comparing each blocked tensor with its own nonzero-sector part and the agreement comparing the two nonzero-sector parts, together with the length-zero zero-tail identity, as in Theorem 11.176. For these particular common-sector families, any common MPV phase cover gives the finite-length nonzero-block span equality needed by the zero-tail block-span theorem. A BNT proportional-decomposition conclusion for the same families gives the same span equality.

Proof. [✓ Proof OK](#) Apply Theorem 11.176 to obtain the common-sector decompositions. The implication from a common MPV phase cover is Lemma 14.33; the implication from proportional-decomposition data is Theorem 14.38. $\square$

Theorem 14.68 (Relabeled common sectors with a common MPV phase cover). [afterBlocking_sectorComparison_zeroTail_of](#)
A and B have the same MPV family. Assume the blocked-word relabeling hypothesis for common cyclic sectors. The common-sector structural theorem then produces a common blocking length, zero-tail identities, and trace-preserving, primitive, irreducible nonzero-sector families on both sides. If those families satisfy the phase-cover hypotheses of Definition 14.56, then the sector decompositions obtained from the two sides have BNT data, agree as full MPV families, and their sector weights satisfy the matched sector-weight conclusion.

Proof. ✓ Proof OK Use Theorem 14.57 to form the remaining span hypotheses of Definition 14.54. Then apply the relabeled common-sector theorem with those span hypotheses. □

Theorem 14.69 (Relabeled common sectors with proportional-comparison data). [afterBlocking_sectorComparison_zeroTail_of](#)
A and B have the same MPV family, and assume the blocked-word relabeling hypothesis for common cyclic sectors. The common-sector structural theorem produces the common primitive nonzero-sector families. If those families satisfy the proportional-comparison hypotheses of Definition 14.58, then the same sector-weight comparison conclusion holds.

Proof. ✓ Proof OK Convert the BNT proportional-decomposition comparison for the produced families to common phase-cover hypotheses, and then apply the relabeled common-sector comparison theorem with common phase-cover data. □

Definition 14.70 (After-blocking hypotheses for the CPSV formulation). [AfterBlockingFundamentalTheoremHypotheses](#) ✓ Stm
 named hypotheses record the comparison data still needed for the periodic-theorem-free after-blocking formulation of the CPSV Fundamental Theorem. They contain the blocked-word relabeling hypothesis for common cyclic sectors. For the common primitive nonzero-sector families produced from two tensors with the same MPV family, the comparison field is supplied with their isometry, primitivity, and irreducibility evidence before returning the remaining proportional-comparison hypotheses for those blocks. The hypotheses are tied to the produced families rather than to an arbitrary chosen decomposition.

Definition 14.71 (CPSV after-blocking Fundamental Theorem conclusion). [AfterBlockingFundamentalTheoremConclusion](#) ✓
 conclusion consists of a positive blocking length and sector decompositions P, Q at that length. The blocked tensors agree with P and Q at every positive length, while P and Q generate the same full MPV family. Both sector decompositions carry BNT data. Their basis sectors are matched by a permutation, the corresponding multiplicities agree, and the sector-weight multisets agree after multiplying the weights on one side by nonzero sector phases.

Theorem 14.72 (Conditional after-blocking CPSV formulation). [fundamentalTheorem_afterBlocking_of_comparisonHypotheses](#)
A and B generate the same MPV family. Assume the after-blocking comparison hypotheses of Definition 14.70. Then the the CPSV conclusion of Definition 14.71 is nonempty.

The conclusion matches the sector-decomposition part of the Cirac–Pérez-García–Schuch–Verstraete Fundamental Theorem: after blocking, the tensor families are represented by BNT sector decompositions whose basis sectors and weights match up to permutation and nonzero phases, as in [CPGSV16] and [CPGSV21]. The remaining hypotheses are explicit: the statement does not assume away the blocked-word relabeling, zero-tail, isometry, primitivity, irreducibility, or proportional-comparison inputs.

Proof. ✓ Proof OK Apply Theorem 14.69 with the relabeling and proportional-comparison data from Definition 14.70. The proof then records the resulting blocking length, sector decompositions, BNT data, permutation, multiplicity equalities, phases, and sector-weight multiset equalities as the conclusion structure. □

Theorem 14.73 (Cast-compatible MPV scaling implies phase-matched sector comparison).

[fundamentalTheorem_equalMPV_sectorDecomposition_hetero_of_mpvScaling_matched_basis](#) ✓ Stmt OK Let P and Q be sector decompositions. Assume there is a permutation π matching the basis blocks, the multiplicities satisfy $r_j^P = r_{\pi(j)}^Q$, and for each j the corresponding basis blocks have the same auxiliary dimension and satisfy

$$V^{(N)}(Q_{\pi(j)})_{\sigma} = \zeta_j^N V^{(N)}(P_j)_{\sigma}$$

for some nonzero complex number ζ_j , after identifying the two basis tensors along that dimension equality. If the total MPV families of P and Q agree and the basis blocks of P are eventually linearly independent, then for each j the sector weight multiset of P_j equals the multiset obtained from that of $Q_{\pi(j)}$ by multiplication by ζ_j .

Proof. ✓ Proof OK After transporting the corresponding basis tensor of P_j across the dimension identity, this is exactly the hypothesis of Theorem 14.17. The conclusion is therefore the same multiset equality. $\square$

Remark 14.74 (Equal-case comparison after canonical reduction). Not ready The equal-case proof in the literature begins with both tensors in canonical form. One chooses a basis of normal tensors for each tensor, so that every canonical-form block is a phase times a gauge transform of exactly one basis tensor; this is the basis characterization of [CPGSV21] and [CPGSV16]. Equality, or proportionality, of the MPV families then pairs the two bases by phase and gauge, as in the Fundamental Theorem of Matrix Product Vectors. In the equal case, one expands each tensor in its basis of normal tensors and compares, for every basis tensor and every length N , the power sums

$$\sum_q \mu_{j,q}^N = \sum_q (\nu_{j,q} e^{i\phi_j})^N.$$

Newton identities, equivalently the elementary power-sum lemma used in the paper, recover the multiplicities and the weights. The blockwise gauge matrices are then placed on the diagonal to obtain the global conjugating matrix; this is the equal-case corollary of the Fundamental Theorem in [CPGSV16], restated in [CPGSV21].

The comparison theorems in this chapter cover this latter part of the argument. Theorem 14.6 proves the equal case when the coefficient identities are given explicitly, Theorem 14.8 proves the common block-structure special case, and Theorem 14.16 recovers the sector-weight multisets from the multiplicity data. The basis-of-normal-tensors construction is available for trace-preserving primitive irreducible nonzero blocks with positive bond dimensions and nonzero weights in Theorem 11.138. When these blocks are injective, Theorems 11.140 and 11.141 give the asymptotic orthogonality and finite-length span preservation needed to separate coefficients. Theorem 14.29 then turns equality of nonzero-sector spans into the hypotheses used by the sector comparison theorem.

The same comparison can also be expressed through common phases. A common MPV phase cover gives equality of finite-length nonzero-block spans by Lemma 14.33. A sector basis pre-matching gives the same span equality and the primitive overlap-span hypotheses by Lemma 14.35 and Theorem 14.36. The BNT comparison theorems produce such common phases from a proportional-decomposition conclusion in Theorems 14.37 and 14.41.

Chapter 11 supplies the canonical-form reduction needed before these comparison theorems can be applied: a common physical blocking length, the zero-tail identities at that length, and relabeled cyclic-sector families on both sides. The unproved assertion is the exact equality, after relabeling blocked physical words, between the canonical blocked nonzero part and the relabeled cyclic-sector tensor for the whole weighted family. With that equality, Theorem 11.176 gives

the common weighted nonzero-sector decompositions needed for the comparison theorems. Theorem 14.63 then records the remaining hypotheses in their present form: equality of the zero-tail dimensions, injectivity of the common nonzero-sector blocks, and finite-length MPV span equality. Theorem 14.67 records the corresponding conditional span consequence for the common-sector families themselves: a common MPV phase cover or a proportional-decomposition conclusion supplies the finite-length nonzero-block span equality. Theorem 14.68 adds the zero-tail and injectivity data to the common phase-cover route and therefore gives the same sector-weight conclusion. Establishing these hypotheses is the mathematical passage from equality of total MPV vectors to the componentwise power-sum identities used in the paper proof of Corollary IV.5.

Chapter 15

Periodic MPS: irreducible form and the periodic Fundamental Theorem

This chapter develops the periodic equal-case Fundamental Theorem of MPS (Theorem 15.25) following [DICCSPG17], and gives its two applications: the symmetry corollary lifting a physical on-site symmetry of a tensor in irreducible form to a virtual Z -gauge, and the equivalence between p -refinability of the matrix-product-vector family and p -divisibility of the transfer map.

15.1 Periodic irreducible form and physical-index rotation

Definition 15.1 (Periodic tensor). IsPeriodic ✓ Stmt OK An irreducible MPS tensor A of bond dimension D is m -periodic (for some $m \geq 1$) if the peripheral spectrum of its transfer map $\mathcal{E}_A$ is exactly the m -th roots of unity $\{e^{2\pi i k/m} : 0 \leq k < m\}$. A 1-periodic tensor is exactly an irreducible tensor with primitive transfer map (Definition 5.49).

Definition 15.2 (Irreducible form). IsIrreducibleForm ✓ Stmt OK A tensor A is in *irreducible form* if it is block-diagonal

$$A^i = \bigoplus_{j \in J} \mu_j A_j^i,$$

where each block A_j is an m_j -periodic tensor, each weight $\mu_j \neq 0$, and the blocks are pairwise non-repeated (no two distinct j, j' have A_j gauge-equivalent to $A_{j'}$). This is the standard form introduced in [DICCSPG17, §II]. It extends the normal canonical form (Definition 11.38), which requires the stronger condition that every block is 1-periodic.

Remark 15.3 (Relation to normal canonical form). Not ready Every tensor in normal canonical form (Definition 11.38) is in irreducible form with all periods equal to 1. The irreducible form is strictly more general: it accommodates blocks such as the antiferromagnet, which are 2-periodic.

Definition 15.4 (Physical-index rotation). rotatePhysical ✓ Stmt OK Given a matrix $u \in M_d(\mathbb{C})$ and an MPS tensor A , the *physical-index rotation* is the tensor $(A_u)^i := \sum_j u_{ij} A^j$.

Theorem 15.5 (Transfer map preserved by physical rotation). transferMap_rotatePhysical ✓ Stmt OK If $uu^\dagger = I$, then the transfer map is invariant under physical-index rotation: $\mathcal{E}_{A_u} = \mathcal{E}_A$.

Proof. ✓ Proof OK Unitary freedom of the Kraus representation (Theorem 2.18, Wolf *Quantum Channels & Operations*). □

Theorem 15.6 (Left-canonical preserved by physical rotation). isLeftCanonical_rotatePhysical ✓ Stmt OK If $uu^\dagger = I$ and A is left-canonical, then A_u is left-canonical.

Proof. ✓ Proof OK Expand $\sum_i B_i^\dagger B_i$ where $B_i = \sum_j u_{ij} A_j$, swap sums, apply orthogonality $u^\dagger u = I$, and collapse to $\sum_j A_j^\dagger A_j = I$. □

Theorem 15.7 (Irreducibility preserved by physical rotation). isIrreducibleTensor_rotatePhysical ✓ Stmt OK If $uu^\dagger = I$ and A is irreducible, then A_u is irreducible.

Proof. ✓ Proof OK Contrapositive: any nontrivial invariant projection P for the rotated tensor yields $(1 - P) A_k P = 0$ for all k via the orthogonality of u , contradicting irreducibility of A . □

Theorem 15.8 (Periodicity preserved by physical rotation). isPeriodic_rotatePhysical ✓ Stmt OK If $uu^\dagger = I$ and A is periodic with period m , then A_u is periodic with the same period.

Proof. ✓ Proof OK Assembles transfer-map invariance, left-canonical preservation, and irreducibility preservation. □

Theorem 15.9 (Equality of MPV families preserved by physical rotation). sameMPV2_rotatePhysical ✓ Stmt OK If $\mathcal{V}(A) = \mathcal{V}(B)$, then $\mathcal{V}(A_u) = \mathcal{V}(B_u)$.

Proof. ✓ Proof OK Induction on the word length with a strengthened hypothesis carrying an arbitrary prefix matrix, reducing each step to the hypothesis via word concatenation. □

Theorem 15.10 (Physical rotation distributes over block-diagonal composition). rotatePhysical_toTensorFromBlocks ✓ Stmt OK If A is a rotated block-diagonal tensor satisfies $(\bigoplus_k \mu_k A_k)_u = \bigoplus_k \mu_k (A_k)_u$.

Proof. ✓ Proof OK Pointwise matrix-entry computation using linearity of the block-diagonal embedding and reindexing. □

Theorem 15.11 (Irreducible form preserved by physical rotation). isIrreducibleForm_rotatePhysical ✓ Stmt OK If $uu^\dagger = I$ and A is in irreducible form II, then A_u is in irreducible form II with the same block structure and rotated blocks.

Proof. ✓ Proof OK Each block remains periodic by Theorem 15.8, equality of MPV families transfers via Theorem 15.9, and block-diagonal compatibility follows from Theorem 15.10. □

15.1.1 The Z -gauge degree of freedom

Definition 15.12 (Entrywise periodic Z -gauge ratio). zGaugeEntry ✓ Stmt OK Given complex weights μ, ν , define the entrywise Z -gauge ratio by

$$z(\mu, \nu) := \mu / \nu.$$

Lemma 15.13 (Matched powers imply root-of-unity ratio). zGaugeEntry_pow_eq_one_of_pow_eq ✓ Stmt OK If $\mu^m = \nu^m$ and $\nu \neq 0$, then

$$z(\mu, \nu)^m = 1.$$

Lemma 15.14 (Ratio rescales denominator back to numerator). zGaugeEntry_mul_right ✓ Stmt OK If $\nu \neq 0$, then

$$z(\mu, \nu) \nu = \mu.$$

Definition 15.15 (Diagonal periodic Z -gauge matrix). zGaugeDiagonal ✓ Stmt OK For matched weight families $(\mu_i)_i$ and $(\nu_i)_i$ on a finite index type, define the diagonal matrix

$$Z := \text{diag}(z(\mu_i, \nu_i))_i.$$

Lemma 15.16 (Diagonal Z -gauge has finite order under matched powers). zGaugeDiagonal_pow_eq_one ✓ Stmt OK If $\mu_i^m = \nu_i^m$ and $\nu_i \neq 0$ for all i , then

$$Z^m = \mathbb{1}.$$

Lemma 15.17 (Diagonal Z -gauge transports diagonal weights). zGaugeDiagonal_mul_diagonal ✓ Stmt OK Under $\nu_i \neq 0$ for all i ,

$$Z \text{ diag}(\nu) = \text{diag}(\mu).$$

Definition 15.18 (Z -gauge equivalence). ZGaugeEquiv ✓ Stmt OK Let A be an m -periodic irreducible tensor of bond dimension D . A Z -gauge transformation of A is a diagonal unitary matrix $Z \in M_D(\mathbb{C})$ satisfying:

1. Z commutes with every Kraus operator: $[A^i, Z] = 0$ for all physical indices i , and
2. $Z^m = \mathbb{1}_D$ (all diagonal entries are m -th roots of unity).

Replacing A by ZA (i.e. $A^i \mapsto ZA^i$) leaves the MPV family invariant for lengths N that are multiples of m , because the factor $\text{tr}(Z^N) = \text{tr}(\mathbb{1}) = D$ for $N \equiv 0 \pmod{m}$. Two tensors A, B are Z -gauge equivalent if there exists an invertible Y and a Z -gauge transformation Z for A such that $ZA^i = YB^iY^{-1}$ for all i .

Remark 15.19 (Z -gauge is trivial for period-1 blocks). of_period_one ✓ Stmt OK When $m = 1$ the condition $Z^1 = \mathbb{1}$ forces $Z = \mathbb{1}$, so Z -gauge equivalence reduces to ordinary gauge equivalence (Definition 2.10). The Z -gauge is therefore a genuine new degree of freedom that appears only for $m > 1$.

Theorem 15.20 (Blocking a $\mathbb{Z}_m$ -gauge gives an ordinary gauge). blockTensor_gaugeEquiv ✓ Stmt OK If A and B are $\mathbb{Z}_m$ -gauge equivalent, then their m -blocked tensors $A^{[m]}$ and $B^{[m]}$ are gauge equivalent in the ordinary sense.

Proof. ✓ Proof OK Write the $\mathbb{Z}_m$ -gauge relation as $ZA^i = YB^iY^{-1}$ with $Z^m = \mathbb{1}$ and $[A^i, Z] = 0$. For a blocked letter, commute one factor of Z past each physical letter in the associated length- m word. The total prefactor becomes $Z^m = \mathbb{1}$, so the entire blocked word is related by the same ordinary gauge Y . □

Corollary 15.21 (Blocking a $\mathbb{Z}_m$ -gauge preserves the MPV family). blockTensor_sameMPV ✓ Stmt OK If A and B are $\mathbb{Z}_m$ -gauge equivalent, then the blocked tensors $A^{[m]}$ and $B^{[m]}$ have the same MPV family.

Proof. ✓ Proof OK Apply Theorem 15.20 and then the gauge invariance of MPV families. □

15.2 Periodic Fundamental Theorem

The theorems in Chapter 14 all require that each block has a *primitive* transfer map, i.e. peripheral spectrum exactly $\{1\}$. This is achieved by blocking the original tensor by a common period. The paper [DCCSPG17] develops a framework that avoids this blocking step, giving a fundamental theorem that applies directly to periodic tensors.

15.2.1 Common-period blocking lemmas

Definition 15.22 (LCM period for a finite block family). lcmPeriod ✓ Stmt OK Given a finite family of blocking periods (p_i) indexed by a finite set, their least common multiple serves as the shared blocking period.

Definition 15.23 (Common-period blocked family). commonPeriodBlocking ✓ Stmt OK Given a finite family of tensors (B_i) with designated periods (p_i) , block each B_i to the shared period $\text{lcm}(p_i)$, so every block lives in the same physical space.

Theorem 15.24 (Common-period blocking preserves transfer-map primitivity). isPrimitive_transferMap_commonPeriodBlock
each tensor B_i has a primitive transfer map after blocking by its own positive period p_i , then blocking the entire family to the common LCM period preserves primitivity of each member's transfer map.

Proof. ✓ Proof OK Each p_i divides $\text{lcm}(p_i)$. Apply the divisibility monotonicity theorem for primitive blocked transfer maps to each family member separately. $\square$

15.2.2 Statement of the periodic Fundamental Theorem

The following is Theorem 3.8 (“equal case”) of DICCSPG17.

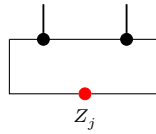
Theorem 15.25 (Periodic Fundamental Theorem). Not ready zgauge_construction equalCase_zgauge_of_power_sums fundament
Let A and B be MPS tensors in irreducible form, with bases of periodic blocks $\{A_j\}_{j \in J}$ and $\{B_k\}_{k \in K}$ respectively. Suppose that $\mathcal{V}(A) = \mathcal{V}(B)$ (equal MPV families for all lengths N).

Then $|J| = |K|$ and there is a bijection $\pi : J \rightarrow K$ such that A_j and $B_{\pi(j)}$ are m_j -periodic for the same period m_j . Moreover, for each $j \in J$ there exists:

1. *an invertible matrix Y_j of size $D_j \times D_{\pi(j)}$, and*
2. *a diagonal unitary matrix Z_j of size $D_j \times D_j$ satisfying $Z_j^{m_j} = \mathbb{1}$ and $[A_j^i, Z_j] = 0$ for all i ,*
such that, for every physical index i ,

$$Z_j A_j^i = Y_j B_{\pi(j)}^i Y_j^{-1}$$

For each periodic block, the extra freedom is encoded by



rather than by an ordinary aperiodic gauge alone: the operator Z_j sits on the closing virtual bond and satisfies $Z_j^{m_j} = \mathbb{1}$. Equivalently, setting $Y = \bigoplus_j Y_j \otimes \mathbb{1}_{r_j}$ and $Z = \bigoplus_j Z_j \otimes \mathbb{1}_{r_j}$, one has $ZA^i = YB^iY^{-1}$ for all i , with $[A^i, Z] = 0$ and $\mathcal{V}(A) = \mathcal{V}(ZA)$.

Remark. The proof is conditional on the periodic-overlap dichotomy (Proposition 3.3 of DICCSPG17), which is left here as a separate hypothesis.

15.2.3 Periodic overlap dichotomy

Definition 15.26 (Cyclic sector decomposition). [IsCyclicSectorDecomp] ✓ Stmt OK A family of compressed sector tensors $(C_k)_{k=0}^{m-1}$ on corner bond spaces forms a *cyclic sector decomposition* of the blocked tensor $A^{(m)}$ if there exist orthogonal projections $P_0, \dots, P_{m-1}$ on the ambient bond space such that $\sum_k P_k = \mathbb{1}$, the projections satisfy the cyclic-shift relation $\mathcal{E}_A^\dagger(P_{k+1}) = P_k$ with the index $k+1$ taken cyclically modulo m , each P_k commutes with every blocked letter $P_k A_i^{(m)} = A_i^{(m)} P_k$, and $V^{(N)}(C_k)_\sigma = \text{tr}(P_k(A^{(m)})^\sigma)$ for every word σ , and each block carries a $*$ -algebra isomorphism $\varphi_k : M_{\dim C_k}(\mathbb{C}) \xrightarrow{\sim} P_k \cdot M_D(\mathbb{C}) \cdot P_k$ (multiplicative and adjoint-preserving) intertwining the compressed adjoint transfer map with the sector adjoint transfer map on the corner of P_k .

Definition 15.27 (One-site corner transition). [cornerTransition] ✓ Stmt OK For a cyclic projection family $(P_k)_{k=0}^{m-1}$ on the ambient bond algebra of an MPS tensor A , the *one-site corner transition* at sector k is the family of per-site matrices

$$\tau_k(A)^i := P_k \cdot A^i \cdot P_{k+1}, \quad i = 0, \dots, d-1,$$

where the index $k+1$ is taken cyclically modulo m .

Definition 15.28 (Staircase evaluation). [cornerEvalWord] ✓ Stmt OK For a word $w = (w_0, \dots, w_{L-1})$ on the physical index alphabet and a starting sector u , the *staircase evaluation* at u is the matrix

$$P_u \cdot A^{w_0} \cdot P_{u+1} \cdot A^{w_1} \cdot P_{u+2} \cdots A^{w_{L-1}} \cdot P_{u+L},$$

characterized recursively by the empty-word value P_u and the rule that prepending a letter i inserts the factor $P_u \cdot A^i$ and advances the starting sector from u to $u+1$.

Definition 15.29 (Cyclic staircase product on blocked indices (Eq. A.8)). [blockedCornerTransitionTensor] ✓ Stmt OK The *blocked corner transition tensor* at sector u is the MPS tensor on blocked physical indices defined by sending a blocked index i to the staircase evaluation along the associated length- m physical word w_i . For a blocked index i with word $(i_0, \dots, i_{m-1})$ this returns the ambient-algebra matrix

$$P_u \cdot A^{i_0} \cdot P_{u+1} \cdot A^{i_1} \cdot P_{u+2} \cdots P_{u+m-1} \cdot A^{i_{m-1}} \cdot P_u.$$

This is the ambient matrix-algebra representative of the compressed blocked sector tensor C_u in a cyclic sector decomposition; the identification with C_u through the compression isometry φ_u is Equation A.8 of [DICCSPG17].

Theorem 15.30 (Self-overlap along period multiples). [periodicSelfOverlap_tendsto] Not ready If A is m -periodic, then the self-overlap along multiples of m converges to m (the period, viewed as a scalar).

Theorem 15.31 (Different periods imply asymptotic orthogonality). [periodicOverlap_tendsto_zero_of_ne_period] ✓ Stmt OK A and B are periodic with different periods, then $\langle V^{(N)}(A) | V^{(N)}(B) \rangle \rightarrow 0$ as $N \rightarrow \infty$.

Proof. ✓ Proof OK Split on whether the bond dimensions agree. If not, the claim follows from Theorem 15.37. If $D_1 = D_2$, suppose for contradiction that A and B are gauge-phase equivalent; Perron–Frobenius eigenvalue uniqueness [Wol12, Theorem 6.3] then forces equal periods, contradicting the hypothesis. Hence the tensors are not gauge-phase equivalent, and the overlap decays. $\square$

Lemma 15.32 (Sector-block normality for periodic tensors). sectorBlocked_isNormal_of_isPeriodic ✓ Stmt OK For a periodic tensor equipped with a cyclic sector decomposition, each nonzero blocked sector tensor is normal.

Theorem 15.33 (No sector match implies asymptotic orthogonality). periodicOverlap_tendsto_zero_of_no_sector_match Not ready For two periodic tensors with the same period admit no gauge-phase matching sector pair after blocking, then their overlap tends to zero.

Lemma 15.34 (Sector-match propagation under translation). sectorMatch_propagation Not ready A single gauge-phase match between blocked sectors propagates to all cyclic translates of that sector pair.

Lemma 15.35 (Per-site proportionality from blocked sector match). sectorTensor_proportional_of_blockedMatch Not ready If all aligned blocked sectors match up to gauge and phase, then the original tensors are related by repeated-block proportionality.

Theorem 15.36 (Sector match yields repeated-block equivalence). periodicOverlap_gaugeEquiv_of_sector_match Not ready For periodic tensors of the same period, existence of one matching sector pair implies repeated-block equivalence.

Theorem 15.37 (Different bond dimensions imply asymptotic orthogonality). periodicOverlap_tendsto_zero_of_ne_dim ✓ S For two periodic tensors have different bond dimensions, then $\langle V^{(N)}(A) | V^{(N)}(B) \rangle \rightarrow 0$ as $N \rightarrow \infty$.

Theorem 15.38 (Periodic overlap dichotomy). periodicOverlapDichotomy Not ready Let A and B be periodic tensors. Then at least one of the following alternatives holds:

1. $\langle V^{(N)}(A) | V^{(N)}(B) \rangle \rightarrow 0$ as $N \rightarrow \infty$;
2. A and B are equivalent up to repeated blocks.

Theorem 15.39 (Eventual linear independence of a periodic basis). periodicBasis_eventuallyLinearlyIndependent Not ready For a finite family of pairwise non-equivalent periodic tensors whose periods divide a common integer p , there is N_0 such that for every $N \geq N_0$ the vectors $\{|V^{(pN)}(A_j)\rangle\}_j$ are linearly independent.

Remark 15.40 (Comparison with the 1606.00608 approach). Not ready Theorem 15.25 and the equal-MPV Fundamental Theorem (Theorem 14.5) both characterize the gauge freedom between two tensors with equal MPV families, but they do so at different levels:

1. *Blocking approach* (Theorem 14.5, following [CPGSV17a]): Block both A and B by a common period p so that all blocks of $A^{[p]}$ and $B^{[p]}$ are primitive (1-periodic). Apply the aperiodic fundamental theorem to the blocked tensors. The conclusion is that $A^{[p]}$ and $B^{[p]}$ are gauge equivalent. The Z -gauge freedom does not appear because blocking absorbs the periodicity.
2. *Periodic approach* (Theorem 15.25, following [DICCSPG17]): Work directly with the periodic blocks of A and B . The conclusion includes an extra Z -gauge transformation reflecting the non-trivial phase ambiguity that arises for periodic blocks. The Z matrix satisfies $Z^m = \mathbb{1}$ and commutes with all Kraus operators; it is invisible to the MPV family because it only introduces m -th roots of unity in the trace.

Chapters 11 and 14 follow only the blocking route: remove periodicity by blocking, then apply the aperiodic theory. The present chapter develops the separate periodic extension, in which the periodic blocks are treated directly.

Theorem 15.41 (Peripheral proportional case from exact MPV equality). [peripheralProportionalCase_periodicFT_of_sameMPV](#)

A and B be periodic tensors with the same matrix product vectors. Then their bond dimensions agree, and after identifying the bond spaces, A and B are equivalent up to repeated blocks.

Proof. Not ready By Theorem 15.38, either the overlap $\langle V^{(N)}(A) | V^{(N)}(B) \rangle$ tends to 0 or the bond dimensions agree and A and B are equivalent up to repeated blocks. In the decay case, equality of the MPV families identifies the cross-overlap with the self-overlap of A for every N. But Theorem 15.30 gives $\langle V^{(m_a N)}(A) | V^{(m_a N)}(A) \rangle \rightarrow m_a$ along multiples of the period m_a , contradiction. $\square$

Theorem 15.42 (Peripheral proportional case from phase rescaling). [peripheralProportionalCase_periodicFT_of_rootFromRescal](#)

that whenever two periodic tensors have proportional MPV families, one can rescale one tensor by a unit-modulus phase so that the MPV families agree exactly. Then proportional periodic MPV families already force the tensors to agree up to repeated blocks.

Proof. Not ready Apply the phase-rescaling hypothesis to replace A by a periodic tensor A' with the same MPV family as B. Theorem 15.41 then yields repeated-block equivalence between A' and B. Since A' differs from A only by a unit-modulus phase, A and A' are themselves equivalent up to repeated blocks, and transitivity gives the claim. $\square$

Remark 15.43 (Proportional periodic FT). Stmt OK

[fundamentalTheorem_periodic_proportional_ofProportionalDecomp](#)

There is also a proportional-case version of the periodic Fundamental Theorem (Theorem 3.4, “proportional case”, in [DICCSPG17]). In the form stated here, if one has periodic block families whose distinct blocks are pairwise non-repeating, together with explicit decompositions of two total MPV families into those blocks, coefficient arrays converging to nonzero limits, and proportionality $\mathcal{V}(A_{\text{tot}}) = \lambda_N \mathcal{V}(B_{\text{tot}})$ with $\lambda_N \rightarrow \lambda \neq 0$, then the bases of periodic tensors match up to the same repeated-block/gauge freedom as above. This extends Theorems 14.2 and 14.4 to the periodic setting while isolating the remaining coefficient-extraction input from irreducible-form assembly.

15.3 Physical symmetries on periodic MPS

Theorem 15.44 (Physical symmetry yields a virtual Z-gauge). [cor_4_1_physical_symmetry_zgauge](#)

Stmt OK *Let*

A be an MPS tensor in irreducible form II and let $U : G \rightarrow \text{GL}_d(\mathbb{C})$ be a representation of a group G on the physical leg, acting unitarily ($U(g)U(g)^\dagger = \mathbb{1}$). Assume moreover the periodic equal-case Fundamental Theorem of MPS (Theorem 15.25) as an explicit hypothesis on the pair (d, D) , expressed by the hypothesis [PeriodicEqualCaseFT](#) d D. If A is on-site symmetric under U — that is, the twisted tensor $\tilde{A}_g$ has the same MPV family as A for every $g \in G$ — then for each $g \in G$ there exists a positive period m_g and a $\mathbb{Z}_{m_g}$ -gauge equivalence between A and $\tilde{A}_g$.

Concretely there are matrices Z_g and Y_g , with $Z_g^{m_g} = \mathbb{1}$ and $[A^i, Z_g] = 0$, satisfying, for every physical index i,

$$Z_g A^i = Y_g \tilde{A}_g^i Y_g^{-1}$$

Proof. Proof OK The twisted tensor $\tilde{A}_g$ coincides with the physical-index rotation $A_{U(g)}$ of Definition 15.4. By Theorem 15.11 the rotated tensor remains in irreducible form II, and the on-site symmetry hypothesis supplies the equality $\mathcal{V}(A) = \mathcal{V}(A_{U(g)})$. Apply the periodic equal-case Fundamental Theorem (Theorem 15.25) to obtain the asserted $\mathbb{Z}_{m_g}$ -gauge equivalence. $\square$

Definition 15.45 (Periodic projective-rigidity hypothesis). [PeriodicProjectiveRigidity](#) ✓ Stmt OK Let A be an MPS tensor with physical dimension d and bond dimension D , and assume it is on-site symmetric under a group action $U : G \rightarrow \text{GL}_d(\mathbb{C})$. The tensor A satisfies the *periodic projective-rigidity hypothesis* if one may choose a family of virtual gauges $Y : G \rightarrow \text{GL}_D(\mathbb{C})$ and a scalar 2-cochain $u : G \times G \rightarrow \mathbb{C}^\times$ such that

1. every Y_g realises the intertwining from [DICCSPG17, Corollary 4.1] with the twisted tensor $\tilde{A}_g$ for some accompanying Z_g (with $Z_g^{m_g} = \mathbb{1}$ and $[A^i, Z_g] = 0$); and
2. the family obeys the projective multiplication law on the bond space: $Y_h Y_g = u(g, h) Y_{gh}$.

The injective case reduces the rigidity to scalar uniqueness of gauges (cf. §19.2); in the genuinely periodic setting the bond-space commutant of A is generated by the Z_g matrices, so coherence of the factor system u is an independent analytic input.

Theorem 15.46 (Projective-representation upgrade). [cor_4_1_projective_rep](#) ✓ Stmt OK Under the hypotheses of Corollary 15.44 together with the periodic projective-rigidity hypothesis (Definition 15.45), the Y_g family assembles into a projective representation of G on the bond space: there exist a scalar 2-cochain $\omega : G \times G \rightarrow \mathbb{C}^\times$ and virtual gauges $X : G \rightarrow \text{GL}_D(\mathbb{C})$ satisfying

$$X(g) X(h) = \omega(g, h) X(gh),$$

while for each g the matrix $X(g^{-1})$ still realises the $\mathbb{Z}_{m_g}$ -intertwining from [DICCSPG17, Corollary 4.1] between A and the twisted tensor $\tilde{A}_g$.

Proof. ✓ Proof OK Unpack the rigidity hypothesis to extract the family (Y, u) together with the Corollary-4.1 intertwiners. Set $X(g) := Y(g^{-1})$ and $\omega(g, h) := u(h^{-1}, g^{-1})$. The projective multiplication law $X(g) X(h) = \omega(g, h) X(gh)$ is then exactly the (h^{-1}, g^{-1}) -instance of the factor-system identity, using $(gh)^{-1} = h^{-1}g^{-1}$. The intertwining relation between A and $\tilde{A}_g$ transports unchanged. $\square$

Theorem 15.47 (Cocycle identity for the upgraded factor system). [cor_4_1_projective_rep_cocycle](#) ✓ Stmt OK If the bond dimension D is positive, the factor system ω produced by Theorem 15.46 satisfies the multiplicative 2-cocycle identity $\omega(g, h) \omega(gh, k) = \omega(g, hk) \omega(h, k)$.

Proof. ✓ Proof OK Associativity of matrix multiplication applied to $X(g) X(h) X(k)$, combined with the projective multiplication law, gives

$$\omega(g, h) \omega(gh, k) X((gh)k) = \omega(g, hk) \omega(h, k) X(g(hk)).$$

Since $(gh)k = g(hk)$, both sides are scalar multiples of the same matrix $X((gh)k)$. The matrix lies in $\text{GL}_D(\mathbb{C})$ and is therefore invertible, but the scalar cancellation leading from an equality of scalar multiples of a matrix to equality of the scalar coefficients additionally requires $D > 0$ so that the underlying index set is inhabited. Cancelling then yields $\omega(g, h) \omega(gh, k) = \omega(g, hk) \omega(h, k)$ in $\mathbb{C}^\times$, hence in $\text{Units}(\mathbb{C})$. $\square$

Remark 15.48 (SPT classification). The cohomology class of ω in $H^2(G, U(1))$ classifies the SPT phase of the symmetric MPS (cf. Chen–Gu–Wen and Pérez-García et al.). This classification statement is not established here; only the underlying 2-cocycle identity of Theorem 15.47 is available.

Theorem 15.49 (Projective representation and cocycle from symmetry). [projectiveRep_of_symmetry](#) ✓ Stmt OK Theorem. and 15.47 combine into a single statement: from the periodic projective-rigidity hypothesis (Definition 15.45), one obtains a projective representation of G on the bond space whose factor system is a genuine 2-cocycle whenever $D > 0$.

15.4 Refinement and divisibility

The next theorem characterises when the matrix-product-vector family of a periodic tensor admits a finer description in terms of a smaller-period tensor whose blocks have been merged. This was the original motivation for the irreducible-form framework in [DICCSPG17].

Definition 15.50 (p -divisible CP map). IsPDivisibleChannel ✓ Stmt OK A linear endomorphism $\mathcal{E}$ of $M_D(\mathbb{C})$ is p -divisible if there exists a CPTP map $\mathcal{E}'$ of $M_D(\mathbb{C})$ such that $\mathcal{E} = (\mathcal{E}')^p$.

Definition 15.51 (p -refinable MPS tensor). IsPRefinable ✓ Stmt OK An MPS tensor B with physical dimension d and bond dimension D is p -refinable if there exist another such tensor A and an isometry $W : \mathbb{C}^d \rightarrow (\mathbb{C}^d)^{\otimes p}$ (encoded as a matrix $W \in M_{d^p \times d}(\mathbb{C})$ with $W^\dagger W = \mathbb{1}$) such that the p -blocked tensor $A^{[p]}$ matches the W -image of B at the level of MPV coefficients:

$$V^{(N)}(A^{[p]})_\tau = \sum_{\sigma \in \{0, \dots, d-1\}^N} \left(\prod_k W_{\tau(k), \sigma(k)} \right) V^{(N)}(B)_\sigma$$

for every $N \in \mathbb{N}$ and every $\tau \in \{0, \dots, d^p - 1\}^N$.

In paper notation, this encodes $|V_{pN}(A)\rangle = W^{\otimes N} |V_N(B)\rangle$ for all N .

Theorem 15.52 (Pullback of a refinement witness). pRefinementCanonicalization_pullback_of_irreducibleForm ✓ Stmt OK Let B be an MPS tensor in irreducible form II. If B is p -refinable, then there exist a tensor A , an isometry $W : \mathbb{C}^d \rightarrow (\mathbb{C}^d)^{\otimes p}$, and the pullback tensor

$$C^\alpha := \sum_{i=0}^{d-1} W_{\alpha, i} B^i \quad (\alpha \in \{0, \dots, d^p - 1\})$$

such that C is again in irreducible form II, $\mathcal{E}_C = \mathcal{E}_B$, and C has the same MPV family as $A^{[p]}$.

Proof. ✓ Proof OK Choose (A, W) from the refinement data. The pullback construction gives $\mathcal{E}_C = \mathcal{E}_B$ and equality of MPV families between C and $A^{[p]}$. To see that C is still in irreducible form II, transport each periodic block B_j of B through the same physical isometry W . The left-canonical normalization is preserved because $W^\dagger W = \mathbb{1}$, while irreducibility and the peripheral spectrum are preserved because the transfer map of each transported block agrees with that of B_j . Reassembling these transported blocks with the same weights gives an irreducible-form decomposition of C . □

Definition 15.53 (Blocked equal-case Z-gauge extraction hypothesis). PeripheralEqualCaseZGaugeOfSameMPV ✓ Stmt OK Fix (d, D, p) . This hypothesis says that whenever a blocked tensor C in irreducible form II has the same MPV family as a blocked root $A^{[p]}$, there exists an integer $m \geq 1$ and a $\mathbb{Z}_m$ -gauge equivalence between C and $A^{[p]}$.

Definition 15.54 (Blocked-to-root reconstruction hypothesis from Z-gauge). PeripheralEqualCaseRootFromZGauge ✓ Stmt OK Fix (d, D, p) . This hypothesis says that if $m \geq 1$ and a blocked tensor C in irreducible form II is $\mathbb{Z}_m$ -gauge equivalent to $A^{[p]}$, then there exists a tensor A' with $\sum_i (A'^i)^\dagger A'^i = \mathbb{1}$ and $\mathcal{E}_C = \mathcal{E}_{A'^{[p]}}$.

Definition 15.55 (Blocked equal-case continuation hypothesis). PeripheralEqualCasePeriodicFTOfSameMPV ✓ Stmt OK Fix (d, D, p) . This states the combined continuation statement: whenever a blocked tensor C in irreducible form II has the same MPV family as a blocked root $A^{[p]}$, one can replace A by a left-canonical tensor A' with $\mathcal{E}_C = \mathcal{E}_{A'^{[p]}}$.

Theorem 15.56 (Blocked equal-case continuation from Z-gauge extraction). peripheralEqualCase_periodicFT_of_sameMPV ✓ the blocked equal-case Z-gauge extraction hypothesis and the blocked-to-root reconstruction hypothesis both hold, then the combined blocked equal-case continuation hypothesis holds.

Proof. ✓ Proof OK Apply the Z-gauge extraction hypothesis to obtain a blocked $\mathbb{Z}_m$ -gauge witness between C and $A^{[p]}$, and then apply the reconstruction hypothesis to that witness. $\square$

Theorem 15.57 (Reduction of forward canonicalization to the blocked equal-case stage). pRefinementCanonicalization_of_
the blocked equal-case continuation hypothesis. Then every p -refinable tensor B in irreducible form II admits a left-canonical root A' with $\mathcal{E}_B = \mathcal{E}_{A'^{[p]}}$.

Proof. ✓ Proof OK Apply Theorem 15.52 to the refinement witness of B . This produces a blocked tensor C in irreducible form II with $\mathcal{E}_C = \mathcal{E}_B$ and the same MPV family as $A^{[p]}$. The blocked equal-case continuation hypothesis then gives a left-canonical tensor A' with $\mathcal{E}_C = \mathcal{E}_{A'^{[p]}}$. Combining the two transfer-map identities yields $\mathcal{E}_B = \mathcal{E}_{A'^{[p]}}$. $\square$

Theorem 15.58 (Forward direction from the blocked equal-case continuation). thm_4_1_p_refinement_forward_of_peripheralE
the blocked equal-case continuation hypothesis. Then for every tensor B in irreducible form II, p -refinability of B implies p -divisibility of $\mathcal{E}_B$.

Proof. ✓ Proof OK Combine Theorem 15.57 with the witness-level forward implication: once $\mathcal{E}_B$ is identified with $\mathcal{E}_{A'^{[p]}}$ for a left-canonical A' , the blocking identity gives $\mathcal{E}_B = (\mathcal{E}_{A'})^p$. $\square$

The following conditional statements isolate the refinability–divisibility equivalence from [DICCSPG17, Theorem 4.1].

Theorem 15.59 (p -refinability iff transfer-map p -divisibility). thm_4_1_p_refinement ✓ Stmt OK *Let B be an MPS tensor in irreducible form II and let $p \geq 1$. Assume both canonicalization hypotheses used in the one-way statements: one for refinability $\Rightarrow$ divisibility and one for divisibility $\Rightarrow$ refinability. Then B is p -refinable iff its transfer map $\mathcal{E}_B$ is p -divisible.*

Proof. ✓ Proof OK Bundle the forward implication (Theorem 15.60) with the reverse implication (Theorem 15.63) into a single biconditional. The forward and reverse directions each assume the corresponding canonicalization hypothesis, and together they give the full equivalence under those hypotheses. $\square$

Theorem 15.60 (p -refinability implies transfer-map p -divisibility). thm_4_1_p_refinement_forward ✓ Stmt OK *Let B be an MPS tensor in irreducible form II. Assume a canonicalization hypothesis providing the remaining analytic assumptions (left-canonical witness with matching transfer map). Then p -refinability of B implies p -divisibility of $\mathcal{E}_B$.*

Proof. ✓ Proof OK The canonicalization hypothesis supplies a witness A with $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ (left-canonical) and $\mathcal{E}_B = \mathcal{E}_{A^{[p]}}$. The left-canonical property makes $\mathcal{E}_A$ a CPTP channel, and the blocking identity $\mathcal{E}_{A^{[p]}} = (\mathcal{E}_A)^p$ then identifies $\mathcal{E}_B$ with the p -th power of the channel $\mathcal{E}_A$. $\square$

Definition 15.61 (Blocked-to-root channel-root hypothesis from Z-gauge). PeripheralEqualCaseRootChannelOfZGauge ✓ Stmt
 (d, D, p) . This sharpened reconstruction hypothesis asks for a CPTP map $\mathcal{E}'$ such that $\mathcal{E}_C = (\mathcal{E}')^p$ and the Choi/Kraus rank of $\mathcal{E}'$ is at most d , whenever C is related to $A^{[p]}$ by a blocked $\mathbb{Z}_m$ -gauge witness.

Theorem 15.62 (Channel-root criterion for blocked-to-root reconstruction). peripheralEqualCaseRootFromZGauge_of_rootChan
the blocked-to-root channel-root hypothesis holds, then the tensor-level blocked-to-root reconstruction hypothesis holds as well.

Proof. ✓ Proof OK Choose a tensor A' realizing the bounded Kraus family of the root channel. Because the channel is trace-preserving, the Kraus operators of A' satisfy $\sum_i (A'^i)^\dagger A'^i = \mathbb{1}$. The identity $\mathcal{E}_C = (\mathcal{E}_{A'})^p$ is then equivalent to $\mathcal{E}_C = \mathcal{E}_{A'^{[p]}}$ by the blocking formula. $\square$

Theorem 15.63 (Transfer-map p -divisibility implies p -refinability). [thm_4_1_p_refinement_reverse](#) ✓ Stmt OK *Let B be an MPS tensor in irreducible form Π and let $p \geq 1$. Assume an inverse canonicalization hypothesis providing a witness $A : \text{MPSTensor } d \ D$ with $\mathcal{E}_B = \mathcal{E}_{A^{[p]}}$ whenever $\mathcal{E}_B$ is p -divisible. Then p -divisibility of $\mathcal{E}_B$ implies p -refinability of B .*

Proof. ✓ Proof OK From the inverse canonicalization hypothesis, extract A with $\mathcal{E}_B = \mathcal{E}_{A^{[p]}}$. This matches two finite Kraus families of the same completely positive map: the blocked family $A^{[p]}$ with d^p operators and the original family B with d operators. The cardinality comparison $d \leq d^p$ holds whenever $p \geq 1$, so [Wol12, Theorem 2.18] supplies an isometry $V \in M_{d^p \times d}(\mathbb{C})$ with $V^\dagger V = \mathbb{1}$ and $(A^{[p]})^\alpha = \sum_j V_{\alpha,j} B^j$. Expanding $\text{coeff}(A^{[p]})(\text{offn } \tau)$ by linearity of matrix multiplication and of the trace, the contributions reorganise into the V -weighted sum over length- N words of B required by Definition 15.51, exhibiting V as the isometry W in the statement of p -refinability. $\square$

Remark 15.64 (Continuum-limit application). Not ready The original motivation for Theorem 15.59 is a continuum-limit construction in the spirit of [DICCSPG17]: refinability along arbitrarily large p is equivalent to infinite divisibility of the transfer map, which selects out continuum limits of translationally invariant MPS. The further consequences are explored elsewhere.

Chapter 16

Quantum Dynamical Semigroups

This chapter develops the theory of one-parameter semigroups of quantum channels and characterizes their generators via the GKSL (Gorini–Kossakowski–Sudarshan–Lindblad) theorem, following [Wol12, Ch. 7].

Section 16.1 establishes definitions and the basic exponential form (Proposition 16.14). Section 16.2 gives the Duhamel formula and perturbation bound. Section 16.3 develops the generator theory and proves the GKSL theorem.

16.1 Dynamical semigroups and the exponential form

Definition 16.1 (Dynamical semigroup). IsDynSemigroup ✓ Stmt OK A family of linear maps $T : \mathbb{R} \rightarrow (M_D(\mathbb{C}) \rightarrow_\ell M_D(\mathbb{C}))$ is a *dynamical semigroup* if

1. (semigroup law) $T_{t+s} = T_t \circ T_s$ for all $t, s \geq 0$; and
2. (initial condition) $T_0 = \text{id}$.

This is [Wol12, Eq. (7.1)].

Definition 16.2 (Continuous dynamical semigroup). IsContinuousDynSemigroup ✓ Stmt OK A dynamical semigroup T is *norm-continuous* if the map $t \mapsto T_t$ is continuous in the operator norm topology. In finite dimension this coincides with strong continuity.

Definition 16.3 (Exponential semigroup). expSemigroup expSemigroupCLM ✓ Stmt OK Given a generator $L \in \text{End}_{\mathbb{C}}(M_D(\mathbb{C}))$, the *exponential semigroup* is

$$T_t := e^{tL} = \sum_{k=0}^{\infty} \frac{(tL)^k}{k!}.$$

Theorem 16.4 (Semigroup law for the exponential). expSemigroupCLM_add expSemigroup_comp ✓ Stmt OK For all $t, s \in \mathbb{R}$, $e^{(t+s)L} = e^{tL} \cdot e^{sL}$.

Proof. ✓ Proof OK Since $(t \cdot L)$ commutes with $(s \cdot L)$, this follows from the exponential addition formula for commuting elements in a Banach algebra. □

Theorem 16.5 (Exponential semigroup at time 0). expSemigroupCLM_zero ✓ Stmt OK The continuous-linear-map exponential satisfies $e^{0L} = \mathbb{1}$.

Proof. ✓ Proof OK The exponential of the zero endomorphism is the identity. □

Theorem 16.6 (Continuity of the exponential semigroup). expSemigroupCLM_continuous expSemigroup_isContinuousDynSemigroup
map $t \mapsto e^{tL}$ is continuous in the operator norm topology.

Proof. ✓ Proof OK The exponential is analytic (convergence radius $= \infty$), hence continuous. □

Theorem 16.7 (Derivative of the exponential semigroup). hasDerivat_expSemigroupCLM ✓ Stmt OK *For all $t \in \mathbb{R}$,*

$$\frac{d}{dt}e^{tL} = e^{tL} \cdot L.$$

This is [Wol12, Eq. (7.2)].

Proof. ✓ Proof OK Chain rule applied to the composition $\mathbb{R} \xrightarrow{t \mapsto t \cdot L} \text{End}(M_D(\mathbb{C})) \xrightarrow{\exp} \text{End}(M_D(\mathbb{C}))$. □

Theorem 16.8 (Derivative at the origin). hasDerivat_expSemigroupCLM_zero ✓ Stmt OK *One has*

$$\left. \frac{d}{dt}e^{tL} \right|_{t=0} = L.$$

Proof. ✓ Proof OK This is Theorem 16.7 at $t = 0$ together with Theorem 16.5. □

Theorem 16.9 (Generator uniqueness). generator_unique ✓ Stmt OK *If $e^{tL} = e^{tL'}$ for all $t \geq 0$, then $L = L'$.*

Proof. ✓ Proof OK Both semigroups agree on $[0, \infty)$, so their derivatives within $[0, \infty)$ at $t = 0$ agree. Since $[0, \infty)$ has unique differentials at 0, $L = L'$. □

Theorem 16.10 (CLM and linear-map forms coincide). expSemigroup_toCLM ✓ Stmt OK *The linear-map exponential semigroup and the continuous-linear-map exponential semigroup agree under the canonical identification of endomorphisms with continuous linear endomorphisms.*

Proof. ✓ Proof OK This is a direct unpacking of the definition of the linear-map exponential semigroup. □

Theorem 16.11 (Pointwise derivative formula). hasDerivat_expSemigroup_apply ✓ Stmt OK *For every $X \in M_D(\mathbb{C})$ and every $t \in \mathbb{R}$,*

$$\frac{d}{dt}(e^{tL}(X)) = e^{tL}(L(X)).$$

Proof. ✓ Proof OK Evaluate the derivative of the continuous-linear-map semigroup at X and transport the statement through Theorem 16.10. □

Theorem 16.12 (Linear-map exponential semigroup at time 0). expSemigroup_zero ✓ Stmt OK *The linear-map exponential semigroup satisfies $e^{0L} = \mathbb{1}$.*

Proof. ✓ Proof OK Transport Theorem 16.5 through Theorem 16.10. □

Theorem 16.13 (Exponential semigroup as a dynamical semigroup). expSemigroup_isDynSemigroup ✓ Stmt OK *The family $t \mapsto e^{tL}$ satisfies the semigroup law and the initial condition, hence is a dynamical semigroup.*

Proof. ✓ Proof OK Transport the continuous-linear-map semigroup law from Theorem 16.4 through Theorem 16.10; the initial condition is Theorem 16.12. □

Theorem 16.14 (Every continuous semigroup is exponential). [continuousDynSemigroup_eq_exp](#) ✓ Stmt OK Every norm-continuous dynamical semigroup T on $M_D(\mathbb{C})$ is of the form

$$T_t = e^{tL}$$

for some generator $L \in \text{End}_{\mathbb{C}}(M_D(\mathbb{C}))$ and every $t \geq 0$. This is [Wol12, Prop. 7.1].

Proof. ✓ Proof OK Since $T_0 = \mathbb{1}$ and T is continuous, the integral $M_\varepsilon = \int_0^\varepsilon T_s ds$ is invertible for small $\varepsilon > 0$. Using the semigroup property, $T_t = M_\varepsilon^{-1}(M_{t+\varepsilon} - M_t)$ is differentiable for $t \geq 0$. ODE uniqueness then gives $T_t = e^{tL}$ for all $t \geq 0$. □

16.2 Perturbation theory

Theorem 16.15 (Duhamel formula). [duhamel_formula](#) ✓ Stmt OK Let $\Delta = L' - L$. For $t \geq 0$,

$$e^{tL'} - e^{tL} = \int_0^t e^{(t-s)L} \Delta e^{sL'} ds.$$

This is [Wol12, Lem. 7.1].

Proof. ✓ Proof OK Define $f(s) = e^{(t-s)L} e^{sL'}$. Then $f'(s) = e^{(t-s)L} \Delta e^{sL'}$ (Theorem 16.7), and $e^{tL'} - e^{tL} = f(t) - f(0) = \int_0^t f'(s) ds$. □

Theorem 16.16 (Perturbation bound). [perturbation_bound](#) ✓ Stmt OK For $t \geq 0$,

$$\|e^{tL'} - e^{tL}\| \leq t \|\Delta\| \cdot \sup_{s \in [0, t]} \|e^{sL}\| \cdot \sup_{s \in [0, t]} \|e^{sL'}\|,$$

where $\Delta = L' - L$. This is [Wol12, Cor. 7.1].

Proof. ✓ Proof OK Apply the triangle inequality and submultiplicativity of the operator norm to the Duhamel integral (Theorem 16.15). □

Corollary 16.17 (Perturbation bound under unit-norm control). [perturbation_bound_unit_norm](#) ✓ Stmt OK For $t \geq 0$, if $\|e^{sL}\| \leq 1$ and $\|e^{sL'}\| \leq 1$ for every $s \in [0, t]$, then

$$\|e^{tL'} - e^{tL}\| \leq t \|L' - L\|.$$

Proof. ✓ Proof OK Apply Theorem 16.16 and bound both suprema by 1. □

Lemma 16.18 (Dyson-series summability from factorial majorant). [summable_dysonTerm_of_factorial_bound](#) ✓ Stmt OK Assume a factorial majorant of Dyson–Phillips iterates:

$$\|D_n(t)\| \leq M \frac{(t \|L' - L\| M)^n}{n!}.$$

Then the Dyson series is summable in operator norm.

Proof. ✓ Proof OK This is the Weierstrass M-test step, using the scalar convergence of $\sum_n a^n/n!$ and lifting via norm-bounded summability. □

Theorem 16.19 (Factorial norm bound for Dyson iterates). [norm_dysonTerm_le](#) ✓ Stmt OK For $s \in [0, t]$ and $M = \sup_{u \in [0, t]} \|e^{uL}\|$,

$$\|\tilde{T}_s^{(n)}\| \leq M \frac{(s \|\Delta\| M)^n}{n!}.$$

Proof. ✓ Proof OK Induction on n . The base case uses the semigroup supremum bound. The inductive step bounds the integrand pointwise by $M\|\Delta\| \cdot M(u\|\Delta\|M)^n/n!$ and integrates $\int_0^s u^n du = s^{n+1}/(n+1)$, recovering the factorial. $\square$

Corollary 16.20 (Dyson series summability). summable_dysonTerm ✓ Stmt OK *The Dyson–Phillips series $\sum_n \tilde{T}_t^{(n)}$ converges in operator norm for every $t \geq 0$.*

Proof. ✓ Proof OK Compose the factorial norm bound (Theorem 16.19) with the Weierstrass M-test step (Lemma 16.18). $\square$

Corollary 16.21 (Factorial norm bound at the final time). norm_dysonTerm_le_at ✓ Stmt OK *For $t \geq 0$, setting $s = t$ in Theorem 16.19 gives*

$$\|\tilde{T}_t^{(n)}\| \leq M \frac{(t\|\Delta\|M)^n}{n!},$$

where $M = \sup_{u \in [0, t]} \|e^{uL}\|$.

Proof. ✓ Proof OK This is the special case $s = t$ of Theorem 16.19. $\square$

Theorem 16.22 (Continuity of Dyson iterates). dysonTerm_continuous ✓ Stmt OK *Each Dyson iterate $t \mapsto \tilde{T}_t^{(n)}$ is continuous in the time parameter.*

Proof. ✓ Proof OK By induction on n . The base case is continuity of the matrix exponential. For the inductive step, the parametric integral $t \mapsto \int_0^t e^{(t-s)L} \Delta \tilde{T}_s^{(n)} ds$ is continuous by the continuous-parameter primitive theorem; the pasting lemma extends this to all of $\mathbb{R}$ since $\tilde{T}_t^{(n+1)} = 0$ for $t \leq 0$. $\square$

Theorem 16.23 (Factorial bound on the Dyson remainder). norm_dysonRemainder_le ✓ Stmt OK *For $s \in [0, t]$, $M = \sup_{u \in [0, t]} \|e^{uL}\|$, and $M' = \sup_{u \in [0, t]} \|e^{uL'}\|$:*

$$\|T'_s - \sum_{n < N} \tilde{T}_s^{(n)}\| \leq M' \frac{(s\|\Delta\|M)^N}{N!}.$$

Proof. ✓ Proof OK Induction on N . The base case is the semigroup supremum bound. The inductive step uses the telescoping remainder identity $R_{N+1}(s) = \int_0^s e^{(s-u)L} \Delta R_N(u) du$ (derived from the Duhamel formula), bounds the integrand pointwise, and integrates to recover the factorial. $\square$

Theorem 16.24 (Dyson series identity [Wol12, Eq. (7.13)]). dyson_series_eq ✓ Stmt OK *For $t \geq 0$, the Dyson–Phillips series equals the perturbed semigroup:*

$$\sum_{n=0}^{\infty} \tilde{T}_t^{(n)} = e^{tL'}.$$

Proof. ✓ Proof OK The partial-sum remainder $\|e^{tL'} - \sum_{n < N} \tilde{T}_t^{(n)}\| \leq M' (t\|\Delta\|M)^N/N!$ tends to zero as $N \rightarrow \infty$, since $c^N/N! \rightarrow 0$ for any constant c . Together with the summability of the series (Corollary 16.20), this identifies the sum via uniqueness of limits. $\square$

Corollary 16.25 (Tsum form of the Dyson series). tsum_dysonTerm_eq ✓ Stmt OK $\sum_{n=0}^{\infty} \tilde{T}_t^{(n)} = e^{tL'}$ (tsum form).

Proof. ✓ Proof OK Immediate from Theorem 16.24. $\square$

16.3 GKSL/Lindblad generators

This section characterizes generators of semigroups of completely positive and trace-preserving maps. The central result is the GKSL theorem (Theorem 16.54), which shows that such generators have the standard Lindblad form.

16.3.1 Generator decomposition and conditional complete positivity

Definition 16.26 (Generator decomposition). GeneratorDecomp ✓ Stmt OK A *generator decomposition* consists of a completely positive map $\phi : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})$ and a matrix $\kappa \in M_d(\mathbb{C})$, defining the linear map

$$L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger.$$

This is [Wol12, Eq. (7.14)].

Definition 16.27 (Conditional complete positivity). IsCCP ✓ Stmt OK A linear map $L : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})$ is *conditionally completely positive* (CCP) if it admits a generator decomposition, i.e. there exist a CP map ϕ and a matrix κ such that $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$. This is condition 1 of [Wol12, Prop. 7.2].

Definition 16.28 (Trace-annihilating map). IsTraceAnnihilating ✓ Stmt OK A linear map $L : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})$ is *trace-annihilating* if $\text{tr}(L(\rho)) = 0$ for all $\rho \in M_d(\mathbb{C})$. This is the infinitesimal version of trace preservation.

Definition 16.29 (Trace constraint). isTraceConstraint ✓ Stmt OK A generator decomposition (ϕ, κ) satisfies the *trace constraint* if $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$, where ϕ^* is the Hilbert–Schmidt adjoint. This is the condition in [Wol12, Eq. (7.20)].

Theorem 16.30 (Trace constraint implies trace-annihilating). traceAnnihilating_of_traceConstraint ✓ Stmt OK If (ϕ, κ) satisfies $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$, then $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$ is trace-annihilating.

Proof. ✓ Proof OK Let $\phi(\rho) = \sum_i K_i \rho K_i^\dagger$. By the cyclic property of trace, $\text{tr}(L(\rho)) = \text{tr}((\sum_i K_i^\dagger K_i)\rho) - \text{tr}(\kappa\rho) - \text{tr}(\kappa^\dagger\rho) = \text{tr}((\kappa + \kappa^\dagger)\rho) - \text{tr}(\kappa\rho) - \text{tr}(\kappa^\dagger\rho) = 0$. $\square$

16.3.2 The Lindblad form

Definition 16.31 (Lindblad form). LindbladForm ✓ Stmt OK A *Lindblad form* consists of a Hermitian matrix $H = H^\dagger$ (the Hamiltonian) and a family of matrices $\{L_j\}_{j=0}^{r-1}$ (the Lindblad operators), defining the linear map

$$L(\rho) = i[\rho, H] + \sum_j \left(L_j \rho L_j^\dagger - \frac{1}{2} \{L_j^\dagger L_j, \rho\}_+ \right),$$

where $[A, B] = AB - BA$ and $\{A, B\}_+ = AB + BA$. This is [Wol12, Eq. (7.21)].

Theorem 16.32 (Lindblad form is trace-annihilating). isTraceAnnihilating ✓ Stmt OK Every Lindblad form defines a trace-annihilating linear map.

Proof. ✓ Proof OK The Hamiltonian part satisfies $\text{tr}(i[\rho, H]) = 0$ by the cyclic property. Each dissipator term satisfies $\text{tr}(L_j \rho L_j^\dagger) = \text{tr}(L_j^\dagger L_j \rho) = \text{tr}(\rho L_j^\dagger L_j)$, so the three terms sum to zero. $\square$

Theorem 16.33 (Lindblad form equals generator decomposition). [toLinearMap_eq_generatorDecomp](#) ✓ Stmt OK *A Lindblad form with Hamiltonian H and operators $\{L_j\}$ defines the same linear map as the generator decomposition (ϕ, κ) with $\phi(\rho) = \sum_j L_j \rho L_j^\dagger$ and $\kappa = iH + \frac{1}{2} \sum_j L_j^\dagger L_j$. This is [Wol12, Eq. (7.24)].*

Proof. ✓ Proof OK Compute $\kappa^\dagger = -iH + \frac{1}{2} \sum_j L_j^\dagger L_j$ using $H^\dagger = H$ and $(A^\dagger A)^\dagger = A^\dagger A$. Setting $S = \sum_j L_j^\dagger L_j$, expand $\phi(\rho) - \kappa\rho - \rho\kappa^\dagger = \sum_j L_j \rho L_j^\dagger - iH\rho - \frac{1}{2}S\rho + i\rho H - \frac{1}{2}\rho S$, which is $i[\rho, H] + \sum_j (L_j \rho L_j^\dagger - \frac{1}{2}L_j^\dagger L_j \rho - \frac{1}{2}\rho L_j^\dagger L_j)$. □

Theorem 16.34 (Lindblad form is CCP). [isCCP](#) ✓ Stmt OK *Every Lindblad form defines a conditionally completely positive map.*

Proof. ✓ Proof OK By Theorem 16.33, the Lindblad form equals a generator decomposition with ϕ CP. □

Definition 16.35 (Commutator form of Lindblad equation). [commutatorForm](#) ✓ Stmt OK *The commutator form of the Lindblad equation writes the generator as*

$$L(\rho) = i[\rho, H] + \frac{1}{2} \sum_j ([L_j, \rho L_j^\dagger] + [L_j \rho, L_j^\dagger]),$$

where $[A, B] = AB - BA$. This is [Wol12, Eq. (7.22)].

Theorem 16.36 (Commutator form equals standard Lindblad form). [commutatorForm_eq_toLinearMap](#) ✓ Stmt OK *The commutator form (Definition 16.35) defines the same linear map as the standard Lindblad form (Definition 16.31).*

Proof. ✓ Proof OK Expanding the commutator brackets $[L_j, \rho L_j^\dagger]$ and $[L_j \rho, L_j^\dagger]$ yields the standard dissipator terms. □

16.3.3 Characterization of conditional complete positivity

Theorem 16.37 (Conditional complete positivity implies projected Choi positivity). [ccp_implies_choi_projected_posSemidef](#) *L is CCP and $P = \mathbb{1} - |\Omega\rangle\langle\Omega|$, then*

$$P((L \otimes \text{id})(|\Omega\rangle\langle\Omega|))P \geq 0.$$

This is [Wol12, Prop. 7.2].

Proof. ✓ Proof OK Write $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$. The left- and right-multiplication terms are annihilated by P , so the projected Choi matrix reduces to the projected Choi matrix of ϕ , which is positive. □

Theorem 16.38 (Projected Choi positivity implies conditional complete positivity). [choi_projected_posSemidef_implies_ccp](#) *L is Hermiticity-preserving and $P((L \otimes \text{id})(|\Omega\rangle\langle\Omega|))P \geq 0$ where $P = \mathbb{1} - |\Omega\rangle\langle\Omega|$, then L is CCP. This is [Wol12, Prop. 7.2].*

Proof. ✓ Proof OK Decompose the Hermitian Choi matrix as $\tau_L = Q - |\psi\rangle\langle\Omega| - |\Omega\rangle\langle\psi|$ with $Q \geq 0$ supported on $|\Omega\rangle^\perp$. The Choi–Jamiołkowski isomorphism applied to Q gives the CP map ϕ , and $(\kappa \otimes \mathbb{1})|\Omega\rangle = |\psi\rangle$ defines κ . □

16.3.4 Completely positive semigroups and conditionally completely positive generators

Theorem 16.39 (A completely positive semigroup has a conditionally completely positive generator). [cp_semigroup_implies_ccp_generator](#) ✓ Stmt OK If e^{tL} is CP for all $t \geq 0$, then L is CCP. This is [Wol12, Prop. 7.3].

Proof. ✓ Proof OK From $(e^{tL} \otimes \text{id})(|\Omega\rangle\langle\Omega|) \geq 0$, expand at infinitesimal t and project onto P to get $P(L \otimes \text{id})(|\Omega\rangle\langle\Omega|)P \geq 0$. Then Theorem 16.38 gives CCP. □

Theorem 16.40 (A conditionally completely positive generator yields a completely positive semigroup). [ccp_generator_implies_cp_semigroup](#) ✓ Stmt OK If L is CCP, then e^{tL} is CP for all $t \geq 0$. This is [Wol12, Prop. 7.3].

Proof. ✓ Proof OK Approximate e^{tL} by the completely positive Euler steps

$$\rho \mapsto (\mathbb{1} - h\kappa)\rho(\mathbb{1} - h\kappa)^\dagger + h\phi(\rho), \quad h = \frac{t}{n+1}.$$

Each step is CP, hence so are its powers. These powers converge in operator norm to e^{tL} , and the cone of CP maps is closed. □

Theorem 16.41 (A semigroup is completely positive iff its generator is conditionally completely positive). [cp_semigroup_iff_ccp_generator](#) ✓ Stmt OK e^{tL} is CP for all $t \geq 0$ if and only if L is CCP.

Proof. ✓ Proof OK Combine Theorems 16.39 and 16.40. □

16.3.5 Freedom in generator representation

Theorem 16.42 (Traceless Kraus operators exist). [exists_traceless_kraus_shift](#) ✓ Stmt OK Assume $d > 0$. Given any Kraus operators $\{K_j\}$, there exist $c_j \in \mathbb{C}$ such that $K'_j = K_j + c_j\mathbb{1}$ is traceless. This is part of [Wol12, Prop. 7.4].

Proof. ✓ Proof OK Set $c_j = -\text{tr}(K_j)/d$. □

Theorem 16.43 (Shift invariance of generators). [generator_shift_invariance](#) ✓ Stmt OK If $K'_j = K_j + c_j\mathbb{1}$ and κ' is adjusted per [Wol12, Eq. (7.19)], then (ϕ', κ') and (ϕ, κ) define the same generator.

Proof. ✓ Proof OK Direct algebraic expansion. □

16.3.6 Uniqueness of the traceless Lindblad form

Definition 16.44 (Traceless Kraus operators). [HasTracelessKraus](#) ✓ Stmt OK A Lindblad form has traceless Kraus operators if $\text{tr}(L_j) = 0$ for every j .

Theorem 16.45 (Uniqueness of the completely positive part). [generatorDecomp_traceless_unique_phi](#) ✓ Stmt OK If two Lindblad forms F, F' induce the same generator and both have traceless Kraus operators, then their CP parts agree: $\phi = \phi'$. This is part of [Wol12, Prop. 7.4, item 2].

Proof. ✓ Proof OK Compare projected Choi matrices and use Choi injectivity. □

Theorem 16.46 (Uniqueness of the drift term modulo phase). [generatorDecomp_traceless_unique_kappa_modPhase](#) ✓ Stmt OK If two Lindblad forms F, F' induce the same generator and both have traceless Kraus operators, then their drift matrices differ by an imaginary scalar: $\kappa' = \kappa + i\lambda\mathbb{1}$ for some $\lambda \in \mathbb{R}$. This is part of [Wol12, Prop. 7.4, item 2].

Proof. ✓ Proof OK After identifying $\phi = \phi'$ (Theorem 16.45), the generator equality forces $\Delta\kappa \cdot \rho + \rho \cdot (\Delta\kappa)^\dagger = 0$ for all ρ . Setting $\rho = \mathbb{1}$ gives $\Delta\kappa + (\Delta\kappa)^\dagger = 0$ (skew-Hermiticity), which converts the equation to $[\Delta\kappa, \rho] = 0$ for all ρ . The scalar commutant lemma gives $\Delta\kappa = c \cdot \mathbb{1}$, and skew-Hermiticity forces $c = i\lambda$ for some $\lambda \in \mathbb{R}$. □

Theorem 16.47 (Combined uniqueness of the traceless Lindblad form). generatorDecomp_traceless_unique ✓ Stmt OK *If two Lindblad forms F, F' induce the same generator and both have traceless Kraus operators, then $\phi = \phi'$ and $\kappa' = \kappa + i\lambda \mathbb{1}$ for some $\lambda \in \mathbb{R}$.*

Proof. ✓ Proof OK Combine Theorems 16.45 and 16.46. □

16.3.7 Trace-annihilation and trace preservation

Theorem 16.48 (Trace-annihilating generators yield trace-preserving semigroups). isTracePreservingMap_expSemigroup_of_i *L is trace-annihilating, then e^{tL} is trace-preserving for every $t \in \mathbb{R}$.*

Proof. ✓ Proof OK For fixed ρ , the function $f(t) = \text{tr}(e^{tL}(\rho))$ has derivative $f'(t) = \text{tr}(L(e^{tL}(\rho))) = 0$, so f is constant and $\text{tr}(e^{tL}(\rho)) = f(t) = f(0) = \text{tr}(\rho)$. □

Theorem 16.49 (Trace-preserving semigroups have trace-annihilating generators). isTraceAnnihilating_of_isTracePreservi *e^{tL} is trace-preserving for every $t \geq 0$, then L is trace-annihilating.*

Proof. ✓ Proof OK Differentiate the identity $\text{tr}(e^{tL}(\rho)) = \text{tr}(\rho)$ at $t = 0$ from the right. □

16.3.8 The GKSL/Lindblad theorem

Definition 16.50 (GKSL generator). IsGKSLGenerator ✓ Stmt OK A linear map $L : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})$ is a *GKSL generator* if e^{tL} is a quantum channel (CPTP) for all $t \geq 0$.

Theorem 16.51 (A trace-constrained generator decomposition gives a GKSL generator). gksl_of_generatorDecomp_with_tr *$L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$ with ϕ CP and $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$, then L is a GKSL generator. This is [Wol12, Thm. 7.1, Eq. (7.20)].*

Proof. ✓ Proof OK CP follows from Theorem 16.40. The trace constraint implies trace-annihilation by Theorem 16.30, so Theorem 16.48 gives trace preservation. □

Theorem 16.52 (GKSL generators admit a trace-constrained generator decomposition). generatorDecomp_of_gksl ✓ Stmt *L is a GKSL generator, then there exist a CP map ϕ and a matrix κ such that*

$$L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$$

and $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$.

Proof. ✓ Proof OK Apply Theorem 16.39 to obtain a CCP decomposition. Then apply Theorem 16.49 to the trace-preserving part of the semigroup and rewrite the resulting trace-annihilation condition as $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$. □

Theorem 16.53 (GKSL iff conditional complete positivity and trace annihilation). gksl_iff_ccp_and_traceAnnihilating ✓ *L is a GKSL generator if and only if L is CCP and trace-annihilating.*

Proof. ✓ Proof OK Forward: apply Theorem 16.39 to the CP part and Theorem 16.49 to the TP part. Reverse: combine Theorem 16.40 with Theorem 16.48. □

Theorem 16.54 (GKSL iff Lindblad form). [gksl_iff_lindbladForm](#) ✓ Stmt OK L is a GKSL generator if and only if

$$L(\rho) = i[\rho, H] + \sum_j \left(L_j \rho L_j^\dagger - \frac{1}{2} \{ L_j^\dagger L_j, \rho \}_+ \right)$$

with $H = H^\dagger$. This is [Wol12, Thm. 7.1].

Proof. ✓ Proof OK Forward: apply Theorem 16.52 to write $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$ with trace constraint. Then choose traceless Kraus operators by Theorem 16.42 and rewrite κ as $iH + \frac{1}{2}\phi^*(\mathbb{1})$; Theorem 16.33 gives the Lindblad formula. Reverse: Theorems 16.34 and 16.32 show that every Lindblad form satisfies the right-hand side of Theorem 16.53. □

16.3.9 Kossakowski matrix form

Definition 16.55 (Kossakowski form). [KossakowskiForm](#) ✓ Stmt OK A Kossakowski form consists of $H = H^\dagger$, a finite family of matrices $\{F_k\}_{k=0}^{n-1}$, and a positive semidefinite matrix $C \in M_n(\mathbb{C})$, defining

$$L(\rho) = i[\rho, H] + \frac{1}{2} \sum_{k,l} C_{lk} ([F_k, \rho F_l^\dagger] + [F_k \rho, F_l^\dagger]).$$

In Wolf's formula one takes $\{F_k\}$ to be a basis of traceless matrices; here we keep only the algebraic data needed for the conversion to Lindblad form.

Theorem 16.56 (Kossakowski form iff Lindblad form). [kossakowski_iff_lindblad](#) ✓ Stmt OK A linear map admits a Kossakowski form iff it admits a Lindblad form.

Proof. ✓ Proof OK Forward: diagonalize $C = B^\dagger B$ and set $L_j = \sum_k B_{jk} F_k$. Reverse: keep the same Hamiltonian and family $F_k = L_k$, and take $C = \mathbb{1}$. □

16.4 Primitivity and irreducibility of QDS

16.4.1 Auxiliary spectral and semigroup lemmas

Definition 16.57 (Quantum dynamical semigroup). [IsQuantumDynSemigroup](#) ✓ Stmt OK A quantum dynamical semigroup is a norm-continuous dynamical semigroup whose time slices are quantum channels for every $t \geq 0$.

Theorem 16.58 (Semigroup power identity). [semigroup_pow](#) ✓ Stmt OK For $t \geq 0$ and $n \in \mathbb{N}$, one has $T_{nt} = (T_t)^n$.

Theorem 16.59 (Eigenvector propagation under powers). [pow_apply_eigenvector](#) ✓ Stmt OK If $E(v) = \mu v$, then $E^n(v) = \mu^n v$ for all $n \in \mathbb{N}$.

Lemma 16.60 (Density matrices are nonzero). [ne_zero_of_mem_densityMatrices'](#) ✓ Stmt OK Every density matrix is nonzero.

Theorem 16.61 (Compression invariance is stable under powers). [compression_preserved_by_pow](#) ✓ Stmt OK If a corner compression is invariant under E , it remains invariant under every power E^n .

Theorem 16.62 (Generator eigenvectors along the exponential semigroup). [expSemigroup_apply_eigenvector](#) ✓ Stmt OK If $L(X) = \mu X$, then $e^{tL}(X) = e^{t\mu} X$ for all $t \in \mathbb{R}$.

Theorem 16.63 (Spectral mapping for semigroup generators). [eigenvalue_exp_of_eigenvalue_generator](#) ✓ Stmt OK If μ is in the spectrum of L , then $e^{t\mu}$ is in the spectrum of e^{tL} .

Theorem 16.64 (Root-of-unity rigidity for phases). [eq_zero_of_exp_mul_I_isRootOfUnity](#) ✓ Stmt OK If $e^{it\theta}$ is a root of unity for every $t > 0$, then $\theta = 0$.

Theorem 16.65 (Peripheral generator eigenvalues are purely imaginary). [re_eq_zero_of_peripheral_generator](#) ✓ Stmt OK If E is a peripheral generator, then all peripheral spectral points of the generator have vanishing real part.

Theorem 16.66 (Peripheral cardinality bound). [peripheral_card_le_finrank](#) ✓ Stmt OK The number of peripheral eigenvalues is bounded by $\dim(M_D(\mathbb{C}))$.

Theorem 16.67 (Bounded order under peripheral power-closure). [bounded_root_of_peripheral_closed_powers](#) ✓ Stmt OK If E is peripheral, then all powers of a peripheral eigenvalue remain peripheral, then some positive power not exceeding $\dim(M_D(\mathbb{C}))$ equals 1.

Theorem 16.68 (Peripheral powers stay peripheral under irreducibility). [peripheral_powers_closed_of_irreducible_channel_w](#) ✓ Stmt OK If E is an irreducible channel with faithful fixed point, every power of a peripheral eigenvalue is again peripheral.

Theorem 16.69 (Continuous-linear-map power evaluation). [toContinuousLinearMap_pow_apply](#) ✓ Stmt OK Evaluating powers of a linear map agrees with powers in its continuous linear realization.

Theorem 16.70 (Spectral-radius criterion from eigenvalue bounds). [spectralRadius_lt_one_of_eigenvalues_lt_one](#) ✓ Stmt OK If all eigenvalues satisfy $|\lambda| < 1$, then the spectral radius is < 1 .

Theorem 16.71 (Primitive powers annihilate trace-zero matrices). [primitive_channel_pow_tendsto_zero_of_trace_zero](#) ✓ Stmt OK If E is an irreducible primitive channel with faithful fixed density σ , then $E^n(X) \rightarrow 0$ for every matrix X with $\text{tr}(X) = 0$.

Proof. ✓ Proof OK Decompose E^n into the fixed-point projection and its complementary part. On the complement, every eigenvalue has modulus < 1 , so the complementary powers tend to 0; the trace-zero hypothesis removes the fixed-point component. □

Theorem 16.72 (Fixed points of an irreducible channel are trace multiples). [fixedPoint_eq_trace_smul_of_irreducible_channe](#) ✓ Stmt OK If E is an irreducible channel with faithful fixed density σ , then every nonzero fixed point V of E satisfies $V = \text{tr}(V)\sigma$.

Proof. ✓ Proof OK Subtract the trace multiple $\text{tr}(V)\sigma$ and apply the vanishing theorem for trace-zero fixed points of an irreducible channel. □

Corollary 16.73 (Nonzero fixed points have nonzero trace). [trace_ne_zero_of_nonzero_fixedPoint_of_irreducible_channel](#) ✓ Stmt OK Every nonzero fixed point of an irreducible channel has nonzero trace.

Proof. ✓ Proof OK Otherwise the preceding theorem would force the fixed point to vanish. □

Theorem 16.74 (Peripheral eigenvectors become periodic fixed points). [exists_power_fixed_eigenvector_of_peripheral](#) ✓ Stmt OK For every positive time t , a quantum dynamical semigroup irreducible at every positive time has the following property: every peripheral eigenvalue μ of T_t admits a nonzero eigenvector V and a positive integer p such that $T_t(V) = \mu V$ and $T_{pt}(V) = V$.

Proof. ✓ Proof OK Peripheral powers remain peripheral for an irreducible channel with faithful fixed point. The bounded-order lemma then gives $\mu^p = 1$, and the semigroup power identity turns the eigenvector equation for T_t into a fixed-point equation for T_{pt} . □

Theorem 16.75 (Peripheral eigenvectors with nonzero trace). [exists_trace_ne_zero_eigenvector_of_peripheral](#) ✓ Stmt OK For every positive time t , under the same hypotheses every peripheral eigenvalue of T_t has a nonzero eigenvector with nonzero trace.

Proof. ✓ Proof OK Apply Theorem 16.74 and then use the previous corollary at the irreducible time pt . □

Theorem 16.76 (Trace-preserving eigenvectors force eigenvalue 1). eigenvalue_eq_one_of_trace_preserving_eigenvector ✓ Stmt OK *If E is trace-preserving, $E(V) = \mu V$, and $\text{tr}(V) \neq 0$, then $\mu = 1$.*

Proof. ✓ Proof OK Take traces in $E(V) = \mu V$ and use trace preservation. □

Theorem 16.77 (Peripheral collapse under irreducibility at all times). peripheral_eq_one_of_irreducible_all ✓ Stmt OK *If a quantum dynamical semigroup is irreducible at every positive time, then every peripheral eigenvalue of every positive-time slice equals 1.*

Proof. ✓ Proof OK Combine Theorem 16.75 with Theorem 16.76. □

Theorem 16.78 (Primitivity from irreducibility at all times). primitive_of_irreducible_all ✓ Stmt OK *If a quantum dynamical semigroup is irreducible at every positive time, then every positive-time slice is primitive.*

Proof. ✓ Proof OK The unique faithful fixed density gives the one-dimensional peripheral spectrum criterion, and Theorem 16.77 collapses the peripheral set to $\{1\}$. □

Remark 16.79 (Semigroup irreducible/primitive equivalence). ✓ Stmt OK The semigroup analogue of the irreducible/primitive equivalence starts from one irreducible time slice, constructs a primitive smaller time step, and lifts the conclusion to the full semigroup (Theorems 16.95 and 16.96).

Remark 16.80 (Reduction to a smaller time step for semigroups). ✓ Stmt OK Starting from one irreducible time t_0 , we construct a positive time step $u = t_0/N$ with $T_{t_0} = (T_u)^N$, prove peripheral eigenvalue collapse at time u , and deduce primitivity of T_u .

Theorem 16.81 (The stationary density is fixed for all times). fixed_density_fixed_for_all_times_of_irreducible_time ✓ Stmt OK *$t_0 > 0$, T_{t_0} is irreducible, and σ is its unique fixed density matrix, then $T_u(\sigma) = \sigma$ for every $u \geq 0$.*

Proof. ✓ Proof OK By semigroup commutativity, $T_{t_0}(T_u(\sigma)) = T_u(T_{t_0}(\sigma)) = T_u(\sigma)$, and uniqueness of the fixed density at time t_0 identifies $T_u(\sigma)$ with σ . □

Theorem 16.82 (Fixed points at the irreducible time are trace multiples). fixedPoint_eq_trace_smul_at_irreducible_time ✓ Stmt OK *T_{t_0} is irreducible with faithful fixed density σ , then every fixed point V of T_{t_0} satisfies $V = \text{tr}(V)\sigma$.*

Proof. ✓ Proof OK This is the fixed-point trace-scalar theorem for the irreducible channel T_{t_0} . □

Definition 16.83 (Residual slice index). residualSliceIndex ✓ Stmt OK For $u \geq 0$ and $s > 0$, the residual slice index is

$$m_n = \left\lfloor \frac{nu}{s} \right\rfloor.$$

Definition 16.84 (Residual slice time). residualSliceTime ✓ Stmt OK For $u \geq 0$ and $s > 0$, the residual slice time is the remainder

$$r_n = s \text{ fract} \left(\frac{nu}{s} \right).$$

Theorem 16.85 (Residual slice decomposition). residualSliceTime_mem_Icc residualSlice_decomp ✓ Stmt OK *For $u \geq 0$ and $s > 0$, one has $r_n \in [0, s]$ and*

$$nu = m_n s + r_n.$$

Proof. ✓ Proof OK This is the floor-plus-fraction decomposition of nu/s , multiplied by s . □

Theorem 16.86 (A residual slice can vanish). exists_residual_time_eq_zero_of_fixedPoint ✓ Stmt OK Suppose $u \geq 0$ and $s > 0$, and that $T_s(\delta) = \delta$ and $T_{nu}(\delta) \rightarrow 0$ as $n \rightarrow \infty$. Then there exists $a \in [0, s]$ such that $T_a(\delta) = 0$.

Proof. ✓ Proof OK The residual slice times lie in the compact interval $[0, s]$, so a subsequence converges to some $a \in [0, s]$. The residual decomposition rewrites $T_{nu}(\delta)$ in terms of $T_{r_n}(\delta)$, and continuity of the semigroup identifies the limit with $T_a(\delta)$. □

Theorem 16.87 (Reduced-time eigenvector has nonzero trace). exists_trace_ne_zero_eigenvector_of_fraction_slice ✓ Stmt OK Under the reduced-time hypotheses $T_{t_0} = (T_u)^{(\dim M_D(\mathbb{C}))!}$, if μ is a peripheral eigenvalue of T_u , then T_u admits a μ -eigenvector with nonzero trace.

Theorem 16.88 (Peripheral collapse at the reduced time step). peripheral_eq_one_of_fraction_slice ✓ Stmt OK Under the same hypotheses, every peripheral eigenvalue of T_u equals 1.

Theorem 16.89 (Primitivity at the reduced time step). primitive_of_fraction_slice ✓ Stmt OK Under the same hypotheses, T_u is primitive.

Theorem 16.90 (Irreducibility descends to a reduced time step). irreducible_of_fraction_slice ✓ Stmt OK If $T_{t_0} = (T_u)^N$ and T_{t_0} is irreducible, then T_u is irreducible.

Proof. ✓ Proof OK Any invariant compression for T_u remains invariant under powers, hence also for T_{t_0} ; irreducibility of T_{t_0} rules this out. □

Theorem 16.91 (Existence of an irreducible reduced time step). exists_irreducible_fraction_slice ✓ Stmt OK If $t_0 > 0$, T_{t_0} is irreducible, and σ is fixed for all times, then there exists a positive time step u such that

$$T_{t_0} = (T_u)^{(\dim M_D(\mathbb{C}))!},$$

the slice T_u is a channel, T_u is irreducible, and $T_u(\sigma) = \sigma$.

Proof. ✓ Proof OK Take $u = t_0/N$ with $N = (\dim M_D(\mathbb{C}))!$. The semigroup power identity gives the factorization of T_{t_0} , channelity comes from the semigroup hypotheses, and Theorem 16.90 gives irreducibility of T_u . □

Theorem 16.92 (Fixed points of the primitive reduced time step are trace-scalars). fixedPoint_eq_trace_smul_of_primitive T_u is primitive and σ is the common fixed density, then every fixed point of any positive-time slice T_s has the form $\tau = \text{tr}(\tau)\sigma$.

Theorem 16.93 (Existence of a primitive reduced time step). exists_primitive_fraction_slice ✓ Stmt OK If T_{t_0} is irreducible for some $t_0 > 0$, then there exists a positive reduced time step u such that T_u is primitive.

Theorem 16.94 (Irreducibility propagates to all positive times). irreducible_all_of_irreducible_time ✓ Stmt OK If T_{t_0} is irreducible for some $t_0 > 0$, then every positive-time slice T_s is irreducible.

Proof. ✓ Proof OK Let σ be the unique faithful fixed density of T_{t_0} . By Theorem 16.81, it is fixed for all times, and Theorem 16.82 makes the fixed-point space of T_{t_0} one-dimensional. Choose a primitive reduced time step by Theorem 16.93; then the trace-scalar description of fixed points from Theorem 16.92 gives the uniqueness criterion for irreducibility at every positive time. □

Theorem 16.95 (An irreducible semigroup is primitive). irreducible_semigroup_implies_primitive ✓ Stmt OK Let $T_t = e^{tL}$ be a quantum dynamical semigroup. If T_{t_0} is irreducible for some $t_0 > 0$, then T_t is primitive (and irreducible) for every $t > 0$. This is [Wol12, Prop. 7.5].

Proof. ✓ Proof OK First apply Theorem 16.94 to propagate irreducibility from the single time t_0 to every positive time. Then apply Theorem 16.78 to conclude primitivity of every positive-time slice. □

Theorem 16.96 (Irreducibility iff primitivity for a quantum dynamical semigroup). qds_irreducible_iff_primitive ✓ Stmt OK *a quantum dynamical semigroup $T_t = e^{tL}$:*

$$(\exists t_0 > 0, T_{t_0} \text{ irreducible}) \iff (\forall t > 0, T_t \text{ primitive and irreducible}).$$

This is [Wol12, Prop. 7.5].

Proof. ✓ Proof OK Forward: apply Theorem 16.95. Reverse: take $t_0 = 1$ (or any positive time); the right-hand side already includes irreducibility of T_t for every $t > 0$. □

16.5 Kernel of the adjoint Liouvillian

Definition 16.97 (Faithful stationary state). HasFaithfulStationaryState ✓ Stmt OK A generator L has a faithful stationary state if there exists a positive definite density matrix ρ_0 with $L(\rho_0) = 0$.

Definition 16.98 (Adjoint dissipator). adjointDissipator ✓ Stmt OK For a Lindblad operator L_j , the adjoint dissipator on observables is

$$\mathcal{D}_{L_j}^*(A) = L_j^\dagger A L_j - \frac{1}{2} L_j^\dagger L_j A - \frac{1}{2} A L_j^\dagger L_j.$$

Definition 16.99 (Adjoint generator). toAdjointLinearMap ✓ Stmt OK The adjoint generator of a Lindblad form is

$$L^*(A) = i[H, A] + \sum_j \mathcal{D}_{L_j}^*(A).$$

Theorem 16.100 (Trace-pairing characterization of the adjoint generator). trace_mul_toAdjointLinearMap_eq_trace_toLinearM *every density matrix ρ and every observable A ,*

$$\text{tr}(\rho L^*(A)) = \text{tr}(L(\rho) A).$$

Proof. ✓ Proof OK Expand the Schrödinger and Heisenberg formulas and apply cyclicity of the trace term by term. □

Definition 16.101 (Commutant of the Lindblad data). commutant ✓ Stmt OK The commutant of a Lindblad form is the set of matrices commuting with H , with every Lindblad operator L_j , and with every adjoint $L_j^\dagger$.

Definition 16.102 (Adjoint kernel). adjointKernel ✓ Stmt OK The adjoint kernel is the set of observables A satisfying $L^*(A) = 0$.

Theorem 16.103 (Commutant lies in the adjoint kernel). mem_adjointKernel_of_mem_commutant commutant_subset_adjointKernel *an observable commutes with H , with every L_j , and with every $L_j^\dagger$, then it lies in $\ker(L^*)$.*

Proof. ✓ Proof OK The Hamiltonian commutator term vanishes, and each adjoint dissipator $\mathcal{D}_{L_j}^*(A)$ is zero under the commutation hypotheses. □

Theorem 16.104 (The adjoint kernel is contained in the commutant under a faithful stationary state). [mem_commutant_of_mem_adjointKernel_of_hasFaithfulStationaryState](#) ✓ Stmt OK Let L be a GKSL generator in Lindblad form with Hamiltonian H and Lindblad operators $\{L_j\}$. Define the adjoint generator L^* by $\text{tr}(\rho L^*(A)) = \text{tr}(L(\rho) A)$. If L has a faithful stationary state $\rho_0 > 0$ with $L(\rho_0) = 0$, then for any A :

$$L^*(A) = 0 \implies [A, H] = [A, L_j] = [A, L_j^\dagger] = 0 \quad \forall j.$$

That is, $\ker(L^*) \subseteq \{H, L_j, L_j^\dagger\}'$. This is [Wol12, Thm. 7.2].

Proof. ✓ Proof OK From $L^*(A) = 0$ and $L^*(A^\dagger) = 0$, compute $L^*(A^\dagger A) = \sum_j [A, L_j]^\dagger [A, L_j]$. Faithfulness of ρ_0 and $\text{tr}(\rho_0 L^*(A^\dagger A)) = 0$ force $[A, L_j] = 0$ for all j . Similarly $[A, L_j^\dagger] = 0$, and then $L^*(A) = 0$ directly gives $[A, H] = 0$. $\square$

Theorem 16.105 (The adjoint kernel equals the commutant under a faithful stationary state). [adjointKernel_eq_commutant_of_hasFaithfulStationaryState](#) ✓ Stmt OK Under the same hypotheses as Theorem 16.104,

$$\ker(L^*) = \{H, L_j, L_j^\dagger\}',$$

i.e. the adjoint kernel equals the commutant of $\{H, L_j, L_j^\dagger\}$. This is [Wol12, Thm. 7.2].

Proof. ✓ Proof OK The inclusion $\supseteq$ is the easy direction (commutant elements are in the kernel by a direct calculation). The reverse $\subseteq$ is Theorem 16.104. $\square$

16.6 Reducibility of quantum dynamical semigroups

The four equivalent reducibility conditions of [Wol12, Prop. 7.6] are as follows.

Definition 16.106 (Nontrivial projection). [IsNontrivialProjection](#) ✓ Stmt OK A projection is nontrivial if it is orthogonal and neither 0 nor $\mathbb{1}$.

Definition 16.107 (Rank-deficient fixed density). [HasRankDeficientFixedDensity](#) ✓ Stmt OK Condition (1): there exists a density matrix ρ_0 with nontrivial kernel such that $e^{tL}(\rho_0) = \rho_0$ for all $t \geq 0$.

Definition 16.108 (Rank-deficient kernel element). [HasRankDeficientKernelElement](#) ✓ Stmt OK Condition (2): there exists a density matrix ρ_0 with nontrivial kernel satisfying $L(\rho_0) = 0$.

Definition 16.109 (Invariant compression). [HasInvariantCompression](#) ✓ Stmt OK Condition (3): there exists a nontrivial orthogonal projector P such that $e^{tL}(PM_D(\mathbb{C})P) \subseteq PM_D(\mathbb{C})P$ for all $t \geq 0$.

Definition 16.110 (Block-upper-triangular Lindblad form). [HasBlockUpperTriangularLindblad](#) ✓ Stmt OK Condition (4): there exists a nontrivial projector P and a Lindblad form $(H, \{L_j\})$ for L such that $(1-P)L_j P = 0$ and $(1-P)\kappa P = 0$ for all j , where $\kappa = iH + \frac{1}{2} \sum_j L_j^\dagger L_j$.

Definition 16.111 (Generator-preserved compression). [GeneratorPreservesCompression](#) ✓ Stmt OK For a projection P , the generator preserves the compression $PM_d(\mathbb{C})P$ if

$$P L(PXP) P = L(PXP)$$

for every $X \in M_d(\mathbb{C})$.

Definition 16.112 (Reducible quantum dynamical semigroup). [IsReducibleQDS](#) ✓ Stmt OK A quantum dynamical semigroup is reducible if it has a nontrivial invariant compression.

Theorem 16.113 (Semigroup invariance implies generator invariance). [generatorPreservesCompression_of_semigroupPreservesC](#)
the semigroup preserves $PM_d(\mathbb{C})P$ for all nonnegative times, then the generator preserves the same compression.

Proof. [✓ Proof OK](#) Differentiate the identity $Pe^{tL}(PXP)P = e^{tL}(PXP)$ at $t = 0$. □

Theorem 16.114 (Generator invariance implies semigroup invariance). [semigroup_preserves_compression_of_generator](#) [✓ Stmt OK](#)
the generator preserves $PM_d(\mathbb{C})P$, then so does the exponential semigroup e^{tL} for every $t \geq 0$.

Proof. [✓ Proof OK](#) Apply the compression map termwise to the exponential series. Every iterate of L preserves the compression, so the whole series does as well. □

Theorem 16.115 (Reducibility as a generator-level criterion). [isReducibleQDS_iff_generator_preserves_compression](#) [✓ Stmt OK](#)
quantum dynamical semigroup is reducible if and only if its generator preserves some nontrivial compression.

Proof. [✓ Proof OK](#) Combine Theorems 16.113 and 16.114. □

Theorem 16.116 (Rank-deficient fixed densities are equivalent to rank-deficient kernel elements). [wolf_prop_7_6_one_iff_two](#) [✓ Stmt OK](#) *A rank-deficient density matrix is a fixed point of e^{tL} for all $t \geq 0$ if and only if it lies in $\ker L$. This is [Wol12, Prop. 7.6, (1) $\leftrightarrow$ (2)].*

Proof. [✓ Proof OK](#) The equivalence $L(\rho) = 0 \iff e^{tL}(\rho) = \rho$ for all $t \geq 0$ follows from the generator–semigroup correspondence: differentiate at $t = 0$ for the forward direction, and exponentiate for the reverse. □

Theorem 16.117 (A rank-deficient fixed density yields an invariant compression). [wolf_prop_7_6_one_implies_three](#) [✓ Stmt OK](#)
 L is a GKSL generator and there exists a rank-deficient fixed density matrix, then the semigroup preserves a nontrivial compression. This is [Wol12, Prop. 7.6, (1) $\rightarrow$ (3)].

Proof. [✓ Proof OK](#) Let ρ_0 be a rank-deficient fixed density. Let P be the support projection of ρ_0 . Since ρ_0 is rank-deficient, $P \neq 1$; since $\rho_0 \neq 0$, $P \neq 0$. Channel positivity and trace preservation force e^{tL} to preserve $PM_D(\mathbb{C})P$. □

Theorem 16.118 (Invariant compression yields block-upper-triangular Lindblad data). [hasBlockUpperTriangularLindblad_c](#)
a GKSL generator preserves a nontrivial compression, then its Lindblad data can be chosen block-upper-triangular with respect to that compression.

Proof. [✓ Proof OK](#) First pass from semigroup invariance to generator invariance by Theorem 16.113. The generator-level vanishing condition forces $(1 - P)L_jP = 0$ and $(1 - P)\kappa P = 0$ through the sum-of-squares identity. □

Theorem 16.119 (Block-upper-triangular Lindblad data yield invariant compression). [hasInvariantCompression_of_hasBlockUpperTriangularLindblad_c](#)
the Lindblad data are block-upper-triangular with respect to a nontrivial projection, then the associated semigroup preserves that compression.

Proof. [✓ Proof OK](#) The block-upper-triangular relations imply generator-level compression invariance; then apply Theorem 16.114. □

Theorem 16.120 (Invariant compression is equivalent to block-upper-triangular Lindblad data). [wolf_prop_7_6_three_implies_four](#) [wolf_prop_7_6_four_implies_three](#) [✓ Stmt OK](#) *For a GKSL generator L , the semigroup preserves a nontrivial compression if and only if the Lindblad data can be chosen block-upper-triangular. This is [Wol12, Prop. 7.6, (3) $\leftrightarrow$ (4)].*

Proof. [✓ Proof OK](#) Combine Theorems 16.119 and 16.118. □

Theorem 16.121 (Block-upper-triangular Lindblad data yield a rank-deficient kernel element). [hasRankDeficientKernelElement_of_hasBlockUpperTriangularLindblad](#) ✓ Stmt OK *If the Lindblad data of a GKSL generator are block-upper-triangular with respect to some nontrivial projector P , then there exists a density matrix ρ_0 with nontrivial kernel satisfying $L(\rho_0) = 0$. This is [Wol12, Prop. 7.6, (4)→(2)].*

Proof. ✓ Proof OK The block-upper-triangular structure implies L preserves the compression $PM_d(\mathbb{C})P$. Therefore the semigroup e^{tL} also preserves $PM_d(\mathbb{C})P$ for all $t \geq 0$. By the Brouwer fixed-point theorem, $e^{(1/m)L}$ has a fixed density matrix ρ_m supported in $PM_d(\mathbb{C})P$ for each m . Passing to a subsequential limit gives a density matrix ρ_0 with $P\rho_0P = \rho_0$ and $L(\rho_0) = 0$. Since $P \neq \mathbb{1}$, the matrix ρ_0 has nontrivial kernel. $\square$

Theorem 16.122 (Equivalent formulations of reducibility). [wolf_prop_7_6_full_equivalence](#) ✓ Stmt OK *For a GKSL generator $L : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$, the four conditions*

1. rank-deficient fixed density,
2. rank-deficient kernel element,
3. invariant compression, and
4. block-upper-triangular Lindblad form

are equivalent. This is [Wol12, Prop. 7.6].

Proof. ✓ Proof OK The cycle $(1) \leftrightarrow (2)$, $(1) \rightarrow (3)$, $(3) \leftrightarrow (4)$, and $(4) \rightarrow (2)$ closes the equivalence chain. $\square$

16.6.1 Sufficient conditions for non-reducibility

Definition 16.123 (Lindblad span). [lindbladSpan](#) ✓ Stmt OK The $\mathbb{C}$ -linear span of the Lindblad operators $\{L_j\}$.

Definition 16.124 (Hermitian-closed Lindblad span). [IsLindbladSpanHermitianClosed](#) ✓ Stmt OK The span of the Lindblad operators is closed under Hermitian conjugation, i.e. it forms a $*$ -subspace of $M_d(\mathbb{C})$.

Definition 16.125 (Trivial Lindblad commutant). [HasLindbladSpanTrivialCommutant](#) ✓ Stmt OK The commutant of the Lindblad family $\{L_j\}'$ equals $\mathbb{C} \cdot \mathbb{1}$, i.e. the Lindblad operators act irreducibly on $M_d(\mathbb{C})$.

Definition 16.126 (Kossakowski rank). [kossakowskiRank](#) ✓ Stmt OK The minimal number of Lindblad operators across all GKSL representations of L , which equals the rank of the Kossakowski matrix in the orthonormal-basis representation.

Theorem 16.127 (κ block vanishing from generator compression). [kappa_block_of_generatorPreservesCompression](#) ✓ Stmt OK *$(H, \{L_j\})$ be a Lindblad form with $\kappa = iH + \frac{1}{2} \sum_j L_j^\dagger L_j$, and let P be an orthogonal projection such that the generator preserves the compression $PM_d(\mathbb{C})P$. If $(1 - P)L_jP = 0$ for every j , then $(1 - P)\kappa P = 0$.*

Proof. ✓ Proof OK From $L = \varphi - \kappa(\cdot) - (\cdot)\kappa^\dagger$ and $(1 - P)L(P) = 0$: the $(1 - P)\varphi(P)$ term vanishes because each $(1 - P)L_jP = 0$, and the $(1 - P)(P\kappa^\dagger)$ term vanishes because $(1 - P)P = 0$. What remains is $(1 - P)\kappa P = 0$. $\square$

Theorem 16.128 (Full algebra generation forbids block-upper-triangular form). [full_algebra_generation_implies_no_blockUpperTriangular](#)
($H, \{L_j\}$) be a Lindblad form. If the algebra generated by $\{L_j\}$ and κ is the entire matrix algebra $M_d(\mathbb{C})$, then the Lindblad data do not admit a block-upper-triangular decomposition.

Proof. [✓ Proof OK](#) Assume for contradiction that a block-upper-triangular decomposition exists with nontrivial projection P . By Theorem 16.127, $(1 - P)\kappa P = 0$. Induction on elements of the subalgebra shows $(1 - P)AP = 0$ for every generator. Since the generated algebra is all of $M_d(\mathbb{C})$, every matrix satisfies $(1 - P)AP = 0$, forcing $P = 0$ or $P = \mathbb{1}$ — a contradiction. $\square$

Theorem 16.129 (Hermitian span + trivial commutant forbids block-upper-triangular form). [hermitian_span_trivial_commutant_implies_no_blockUpperTriangular](#) [✓ Stmt OK](#)
If the Lindblad span is closed under Hermitian conjugation and its commutant is $\mathbb{C} \cdot \mathbb{1}$, then the Lindblad data do not admit a block-upper-triangular decomposition.

Proof. [✓ Proof OK](#) Assume for contradiction that a block-upper-triangular decomposition exists with nontrivial projection P . The block vanishing $(1 - P)L_j P = 0$ and Hermitian closure give $PL_j(1 - P) = 0$ as well, so $[P, L_j] = 0$ for all j . But then P lies in the commutant of the Lindblad span, contradicting triviality. $\square$

Theorem 16.130 (Large Kossakowski rank forbids block-upper-triangular form). [large_kossakowski_rank_implies_no_blockUpperTriangular](#)
the Kossakowski rank satisfies

$$\text{rank}(C) + d \geq d^2 + 1,$$

then no block-upper-triangular Lindblad form exists.

Proof. [✓ Proof OK](#) A block-upper-triangular traceless Lindblad family lies in a proper subspace whose dimension is at most $d^2 - d$. The formal rank minimization argument then contradicts the large-rank bound. $\square$

Theorem 16.131 (No block-upper-triangular Lindblad form implies non-reducibility). [not_isReducibleQDS_of_no_blockUpperTriangular](#)
a GKSL generator admits no block-upper-triangular Lindblad form, then the associated quantum dynamical semigroup is not reducible.

Proof. [✓ Proof OK](#) A reducible semigroup would have an invariant compression, and Theorem 16.118 would then produce a block-upper-triangular Lindblad form. $\square$

Theorem 16.132 (Full algebra generation implies non-reducibility). [not_isReducible_of_generates_full_algebra](#) [✓ Stmt OK](#)
the algebra generated by $\{L_j\}$ and κ is the entire matrix algebra $M_d(\mathbb{C})$, then the QDS is not reducible. This is [Wol12, Cor. 7.2, condition 1].

Proof. [✓ Proof OK](#) By Theorem 16.128, no block-upper-triangular decomposition exists. Then apply Theorem 16.131. $\square$

Theorem 16.133 (Hermitian span and trivial commutant imply non-reducibility). [not_isReducible_of_hermitian_span_trivial_commutant](#)
the Lindblad span is Hermitian-closed and its commutant is trivial, then the QDS is not reducible. This is [Wol12, Cor. 7.2, condition 2].

Proof. [✓ Proof OK](#) By Theorem 16.129, no block-upper-triangular decomposition exists. Then apply Theorem 16.131. $\square$

Theorem 16.134 (Large Kossakowski rank implies non-reducibility). [not_isReducible_of_kossakowski_rank_ge](#) [✓ Stmt OK](#)
If $\text{rank}(C) > d^2 - d$, then the QDS is not reducible. This is [Wol12, Cor. 7.2, condition 3].

Proof. [✓ Proof OK](#) The rank hypothesis rules out block-upper-triangular Lindblad data by Theorem 16.130. Then apply Theorem 16.131. $\square$

Chapter 17

The Algebraic Fundamental Theorem

This chapter develops the one-dimensional MPS results of [MGRPG⁺18]: the Fundamental Theorem for injective and normal MPS on a finite periodic chain of fixed length. The argument is purely algebraic, working directly with $M_D(\mathbb{C})$. Two preparatory lemmas—the virtual bond gauge (Lemma 17.11) and the proportionality lemma (Lemma 17.12)—carry the entire argument, yielding the main theorem and its corollaries for translation-invariant chains, non-translation-invariant descriptions of translation-invariant states, and blocked chains.

Throughout, D is the bond dimension and d is the physical dimension with $d \geq D^2$. The extensions to injective and normal PEPS on arbitrary graphs and the improved system-size bound ($2L + 1$ in place of $3L$) are not pursued in this chapter. The chain data (Definition 4.12), including same chain state, injectivity, and cyclic gauge equivalence, was introduced in Chapter 4.

17.1 One-sided inverse and decomposition

If A is injective, the linear combination map $\Phi_A(c) := \sum_i c_i A^i : \mathbb{C}^d \rightarrow M_D(\mathbb{C})$ is surjective and admits a right inverse.

Lemma 17.1 (Existence of a right inverse). exists_rightInverse ✓ Stmt OK *If A is injective, then there exists a $\mathbb{C}$ -linear map $\Psi : M_D(\mathbb{C}) \rightarrow \mathbb{C}^d$ satisfying $\Phi_A \circ \Psi = \text{id}_{M_D(\mathbb{C})}$.*

Proof. ✓ Proof OK Surjectivity of Φ_A yields a linear section by choice. □

Definition 17.2 (Decomposition map). decompositionMap ✓ Stmt OK Let A be an injective MPS tensor. The *decomposition map* $\Psi_A : M_D(\mathbb{C}) \rightarrow \mathbb{C}^d$ is a fixed choice of right inverse of Φ_A .

Lemma 17.3 (Decomposition property). exists_decomposition ✓ Stmt OK *If A is injective, then for every $X \in M_D(\mathbb{C})$ there exist coefficients $c \in \mathbb{C}^d$ such that*

$$X = \sum_{i=0}^{d-1} c_i A^i.$$

Proof. ✓ Proof OK Take $c = \Psi_A(X)$. Then $\Phi_A(c) = \Phi_A(\Psi_A(X)) = X$. □

Remark 17.4 (Non-uniqueness of the decomposition). When $d > D^2$, the map Φ_A has a nontrivial kernel, so the section Ψ_A is not unique. The statements that follow depend only on $\Phi_A \circ \Psi_A = \text{id}$, so the choice does not affect the conclusions.

17.2 Physical realisation of virtual insertions

For an injective tensor A , the physical realisation map encodes the effect of inserting a virtual matrix X on the bond as a $d \times d$ operation on the physical indices.

Definition 17.5 (Physical realisation map). physRealize ✓ Stmt OK Let A be an injective tensor. For $X \in M_D(\mathbb{C})$, define the $d \times d$ matrix $O_A(X) \in M_d(\mathbb{C})$ by

$$O_A(X)_{ij} := (\Psi_A(A^i X))_j,$$

so that row i of $O_A(X)$ contains the decomposition coefficients of $A^i X$ in the spanning set $\{A^j\}$. Diagrammatically, the inserted virtual matrix is re-expressed as a physical operation on the same local tensor:

The map $X \mapsto O_A(X)$ is $\mathbb{C}$ -linear.

Lemma 17.6 (Defining property of the physical realisation map). physRealize_spec ✓ Stmt OK For every $X \in M_D(\mathbb{C})$ and every physical index i ,

$$A^i X = \sum_{j=0}^{d-1} O_A(X)_{ij} A^j.$$

Proof. ✓ Proof OK By definition, $O_A(X)_{ij} = (\Psi_A(A^i X))_j$. Applying Φ_A gives $\sum_j O_A(X)_{ij} A^j = \Phi_A(\Psi_A(A^i X)) = A^i X$, where the last step uses $\Phi_A \circ \Psi_A = \text{id}$. $\square$

Theorem 17.7 (Physical realisation is multiplicative). physRealize_mul ✓ Stmt OK Let A be an injective MPS tensor. For all $X, Y \in M_D(\mathbb{C})$,

$$O_A(XY) = O_A(X) O_A(Y).$$

Proof. ✓ Proof OK Applying Lemma 17.6 to X yields

$$A^i X = \sum_{j=0}^{d-1} O_A(X)_{ij} A^j.$$

Multiply by Y on the right and apply the linear map Ψ_A :

$$\Psi_A(A^i XY) = \sum_{j=0}^{d-1} O_A(X)_{ij} \Psi_A(A^j Y).$$

Taking the k th component of both sides gives

$$O_A(XY)_{ik} = \sum_{j=0}^{d-1} O_A(X)_{ij} O_A(Y)_{jk} = (O_A(X) O_A(Y))_{ik}.$$

Since this holds for all i and k , the matrices are equal. $\square$

Remark 17.8 (General injective case). When $d > D^2$, the decomposition map Ψ_A need not be unique. Nevertheless, Theorem 17.7 is already the general injective statement: for the fixed choice of Ψ_A in Definition 17.2, the associated physical realisation map O_A is multiplicative. This supersedes the earlier basis-only formulation.

17.3 Intermediate lemmas

Lemma 17.9 (Blocking preserves injectivity). chainCombinedTensor_isInjective ✓ Stmt OK Let $A_0, \dots, A_{m-1}$ be injective MPS tensors with compatible bond dimensions. Then the blocked tensor $(A_0 \otimes \dots \otimes A_{m-1})^{(i_0, \dots, i_{m-1})} := A_0^{i_0} A_1^{i_1} \dots A_{m-1}^{i_{m-1}}$ is again injective.

Proof. ✓ Proof OK The span of the blocked matrices contains all products $A_0^{i_0} \dots A_{m-1}^{i_{m-1}}$. Since each $\{A_k^i\}_i$ spans $M_D(\mathbb{C})$, these products span all of $M_D(\mathbb{C})$. $\square$

Lemma 17.10 (Centre of $M_D(\mathbb{C})$). isScalar_of_commute_span_eq_top ✓ Stmt OK If $Z \in M_D(\mathbb{C})$ commutes with every element of $M_D(\mathbb{C})$, then $Z = \lambda \mathbb{1}_D$ for some $\lambda \in \mathbb{C}$.

Proof. ✓ Proof OK Standard: Z commutes with every matrix unit E_{ij} , which forces all off-diagonal entries to vanish and all diagonal entries to agree. $\square$

17.4 Same state forces a virtual bond gauge

This section presents the 3-site virtual-bond statement used later in the paper proof. Here the hypothesis is already the all-length MPV equality of the combined tensors, and the conclusion is the conjugacy of the middle-bond insertion data.

Lemma 17.11 (Virtual bond gauge – 3-site case). virtual_bond_gauge ✓ Stmt OK Let $A = (A_0, A_1, A_2)$ and $B = (B_0, B_1, B_2)$ be two 3-site injective periodic chains with common bond dimension $D \geq 1$ and $\mathcal{V}(A) = \mathcal{V}(B)$ (same MPV family). Then at the middle bond there exists an invertible matrix $Z \in \text{GL}_D(\mathbb{C})$ such that the virtual-insertion trace coefficients agree after conjugation:

$$\text{tr}(A_0^{i_0} X A_1^{i_1} A_2^{i_2}) = \text{tr}(B_0^{i_0} Z^{-1} X Z B_1^{i_1} B_2^{i_2})$$

for all $X \in M_D(\mathbb{C})$ and all physical indices i_0, i_1, i_2 .

In the paper [MGRPG⁺18], Lemma 1 is stated for arbitrary chain length $n \geq 3$ and any bond position, with uniqueness of Z up to scalar. The chain-level generalization to arbitrary $n \geq 3$ and any bond position is not pursued here.

Proof. ✓ Proof OK The $\mathcal{V}(A) = \mathcal{V}(B)$ hypothesis gives, for every $X \in M_D(\mathbb{C})$, equality of the virtual-insertion trace coefficients: $\text{tr}(A_0^{i_0} X A_1^{i_1} A_2^{i_2}) = \text{tr}(B_0^{i_0} Y B_1^{i_1} B_2^{i_2})$ for some Y depending on X . Since all three site tensors on each side are injective, the physical realisation maps O_A, O_B are algebra homomorphisms by Theorem 17.7. Setting $T(X) = Y$ defines a $\mathbb{C}$ -linear multiplicative map, hence a nonzero $\mathbb{C}$ -algebra endomorphism of $M_D(\mathbb{C})$, which is bijective by Theorem 4.8. Skolem–Noether (Theorem 4.9) gives $Z \in \text{GL}_D(\mathbb{C})$ with $T(X) = Z^{-1} X Z$. $\square$

17.5 Aligned gauges force proportionality

After applying Lemma 17.11 to each bond and absorbing the gauge matrices, one obtains two chains whose virtual-insertion operations agree on every bond. Lemma 2 of [MGRPG⁺18] shows that this agreement forces the local tensors to be proportional.

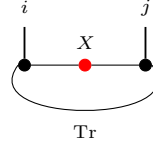
Lemma 17.12 (Aligned gauges force proportionality – Lemma 2 of [MGRPG⁺18]). tensor_proportional ✓ Stmt OK Let A_1, A_2 and B_1, B_2 be injective MPS tensors with the same bond dimension D . Suppose that for

all $X \in M_D(\mathbb{C})$ (on the internal bond), all $Y \in M_D(\mathbb{C})$ (on the external bond), and all physical indices i, j ,

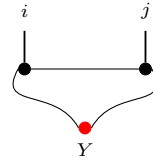
$$\text{tr}(A_1^i X A_2^j) = \text{tr}(B_1^i X B_2^j) \quad (17.1)$$

$$\text{tr}(A_2^j Y A_1^i) = \text{tr}(B_2^j Y B_1^i) \quad (17.2)$$

Diagrammatically, Eq. (17.1) is the equality obtained by inserting X on the bond between the two sites,



while Eq. (17.2) is the corresponding equality with the insertion Y on the outer trace bond,



so the two hypotheses distinguish the internal and external virtual loops of the two-site periodic chain.

Then there exists a nonzero scalar $\lambda \in \mathbb{C}^\times$ such that

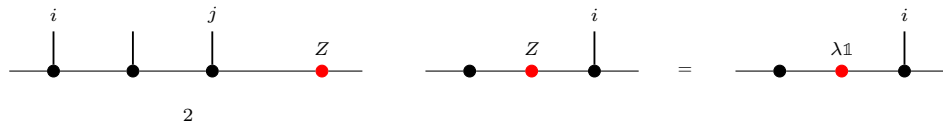
$$A_1^i = \lambda B_1^i \quad \text{and} \quad A_2^j = \lambda^{-1} B_2^j$$

for all physical indices i, j .

Proof. ✓ Proof OK By cyclicity and non-degeneracy of the trace pairing, the conditions in Eqs. (17.1) and (17.2) give, for all physical indices i, j ,

$$\begin{aligned} A_2^j A_1^i &= B_2^j B_1^i, \\ A_1^i A_2^j &= B_1^i B_2^j. \end{aligned}$$

Write $\mathbb{1}_D = \sum_j c_j A_2^j$ using the decomposition map Ψ_{A_2} , and define $Z := \sum_j c_j B_2^j \in M_D(\mathbb{C})$. This is the point where the two-site comparison collapses to a single boundary matrix:



The first product identity gives $A_1^i = Z B_1^i$, and the second gives $A_1^i = B_1^i Z$. Hence Z commutes with every B_1^i ; since $\{B_1^i\}$ spans $M_D(\mathbb{C})$, Z is central. By Lemma 17.10, $Z = \lambda \mathbb{1}_D$ for some $\lambda \in \mathbb{C}^\times$ (nonzero since the tensors are nonzero). Thus $A_1^i = \lambda B_1^i$, and substituting into the first product identity forces $A_2^j = \lambda^{-1} B_2^j$. $\square$

17.6 The Fundamental Theorem for injective chains

The chain result below combines the site and physical indices into a single physical index and applies the single-block Fundamental Theorem to the resulting tensor. The fixed-length same-state statement requires a separate same-state reduction hypothesis.

Theorem 17.13 (Fundamental Theorem for injective chains (combined-tensor MPV version)). fundamentalTheorem_injective_chain ✓ Stmt OK Let $\mathbf{A} = (A_0, \dots, A_{n-1})$ and $\mathbf{B} = (B_0, \dots, B_{n-1})$ be two n -site chains with common bond dimension D , and assume that $\mathbf{A}$ is injective. Suppose that, after combining the site index and physical index into a single MPS tensor, the resulting two tensors generate the same MPV family. Then $\mathbf{A}$ and $\mathbf{B}$ are cyclically gauge equivalent.

Proof. ✓ Proof OK Choose one site of $\mathbf{A}$; its injectivity implies injectivity of the combined tensor by Lemma 17.9. Apply the single-block Fundamental Theorem (Theorem 4.11) to the two combined tensors. This yields one invertible matrix X such that every combined matrix of $\mathbf{B}$ equals the corresponding combined matrix of $\mathbf{A}$ conjugated by X . Reading this identity back at each site gives $B_k^i = X A_k^i X^{-1}$ for all k and i , so the constant choice $Z_k = X$ is a cyclic gauge. $\square$

Lemma 17.14 (Uniform site rescaling rescales the combined tensor). chainCombinedTensor_smul_chain ✓ Stmt OK Let $\mathbf{A} = (A_0, \dots, A_{n-1})$ be an n -site chain and let $\zeta \in \mathbb{C}$. If $\zeta \mathbf{A}$ denotes the chain obtained by replacing each site tensor A_k with ζA_k , and if $\widetilde{\mathbf{A}}$ denotes the combined tensor of $\mathbf{A}$, then

$$\widetilde{\zeta \mathbf{A}} = \zeta \widetilde{\mathbf{A}}.$$

Proof. ✓ Proof OK Bundling the site and physical indices does not change which local matrix is read at the component (k, i) , so both sides evaluate to ζA_k^i there. $\square$

Theorem 17.15 (Injective-chain Fundamental Theorem up to a global phase). fundamentalTheorem_injective_chain_gaugePhase $\mathbf{A} = (A_0, \dots, A_{n-1})$ and $\mathbf{B} = (B_0, \dots, B_{n-1})$ be two n -site chains with common bond dimension D , and assume that $\mathbf{A}$ is injective. Suppose that the combined tensors $\widetilde{\mathbf{A}}$ and $\widetilde{\mathbf{B}}$ are gauge-phase equivalent. Then there exist invertible matrices $Z_0, \dots, Z_{n-1} \in \text{GL}_D(\mathbb{C})$ and a nonzero scalar $\zeta \in \mathbb{C}^\times$ such that

$$B_k^i = \zeta Z_k A_k^i Z_{k+1}^{-1}$$

for all sites k and physical indices i , where $k+1$ is interpreted cyclically modulo n .

Proof. ✓ Proof OK If the combined tensor of $\mathbf{B}$ differs from that of $\mathbf{A}$ by a gauge and a nonzero scalar ζ , rescale each site of $\mathbf{B}$ by ζ^{-1} . Lemma 17.14 removes this common scalar from the combined tensor, so Theorem 17.13 applies to the rescaled chain. Multiplying the resulting sitewise relations by ζ gives the displayed conclusion. $\square$

17.6.1 From fixed-length chain states to combined-tensor MPV agreement

Definition 17.16 (Same-state reduction hypothesis). SameStateBridgeHyp ✓ Stmt OK A same-state reduction hypothesis for dimensions (d, D) is the assertion that whenever $\mathbf{A}$ is an injective n -site chain with $n \geq 3$ and $\mathbf{A}, \mathbf{B}$ generate the same chain state, then the combined tensors of $\mathbf{A}$ and $\mathbf{B}$ generate the same MPV family.

Lemma 17.17 (Same state gives combined-tensor MPV agreement). sameMPV_chainCombined_of_sameState ✓ Stmt OK Assume a same-state reduction hypothesis for (d, D) . Let $\mathbf{A}$ and $\mathbf{B}$ be n -site chains with common bond dimension D , with $n \geq 3$, and assume that $\mathbf{A}$ is injective. If $\mathbf{A}$ and $\mathbf{B}$ generate the same chain state, then the combined tensors of $\mathbf{A}$ and $\mathbf{B}$ generate the same MPV family.

Proof. ✓ Proof OK This is exactly the defining property of the same-state reduction hypothesis. □

Theorem 17.18 (Fundamental Theorem for injective chains from same state). fundamentalTheorem_injective_chain_of_sameSt
a same-state reduction hypothesis for (d, D) . Let $\mathbf{A}$ and $\mathbf{B}$ be n -site chains with common bond dimension D , with $n \geq 3$, and assume that $\mathbf{A}$ is injective. If $\mathbf{A}$ and $\mathbf{B}$ generate the same chain state, then $\mathbf{A}$ and $\mathbf{B}$ are cyclically gauge equivalent.

Proof. ✓ Proof OK Apply Lemma 17.17 to convert the fixed-length chain-state hypothesis into equality of the combined-tensor MPV families, then invoke Theorem 17.13. □

Remark 17.19 (Relationship to the translation-invariant all-sizes theorem). Theorem 17.13 treats a fixed chain length and assumes equality of the combined-tensor MPV families. The single-block Fundamental Theorem (Theorem 4.11) instead studies one tensor at all system sizes and concludes exact gauge equivalence $B^i = XA^iX^{-1}$. The chain theorem is recovered by converting the chain into one combined tensor; the price is that the hypothesis is all-length MPV equality for that combined tensor rather than equality of the chain state at one fixed length.

17.6.2 Finite-length MPV agreement

Definition 17.20 (Finite-length MPV agreement). SameMPVFrom ✓ Stmt OK Two tensors A and B of the same bond dimension satisfy *finite-length MPV agreement from N_0 onward* if they generate the same MPV family for all system sizes $N \geq N_0$, meaning that for every $N \geq N_0$ and $\sigma \in [d]^N$,

$$V^{(N)}(A)_\sigma = V^{(N)}(B)_\sigma$$

Theorem 17.21 (Finite-length agreement implies full agreement). sameMPV_of_sameMPVFrom_of_injective ✓ Stmt OK *Let A be an injective tensor of bond dimension $D \geq 1$, and let B be a tensor of the same bond dimension. If A and B generate the same MPV family for every system size $N \geq N_0$, then they generate the same MPV family for every system size.*

Proof. ✓ Proof OK Write $1 = \sum_i c_i A^i$ (by injectivity) and set $S = \sum_i c_i B^i$.

1. Show that the length- N_0 word span of B is all of $M_D(\mathbb{C})$ via the linear relation between the length- N_0 word-evaluation maps of A and B induced by trace agreement at length $N_0 + 1$, together with a finrank transfer argument.
2. Derive $S = 1$ by trace nondegeneracy: for every $M \in M_D(\mathbb{C})$, $\text{tr}(M(S - 1)) = 0$.
3. Downward induction on $k = N_0 - |w|$: decompose $\text{tr}(w_A)$ and $\text{tr}(w_B)$ using $1 = \sum c_i A^i$ and $S = 1$ respectively, then apply the inductive hypothesis at length $|w| + 1$.

□

Theorem 17.22 (Strengthened single-block FT (finite-length)). fundamentalTheorem_singleBlock_finiteLength ✓ Stmt OK *Let A be an injective tensor of bond dimension $D \geq 1$, and let B be a tensor of the same bond dimension. If A and B agree on all MPV coefficients for system sizes $N \geq N_0$ (for any threshold N_0), then A and B are gauge equivalent.*

Proof. ✓ Proof OK Apply Theorem 17.21 to obtain equality of the MPV families for all system sizes, then invoke the single-block Fundamental Theorem (Theorem 4.11). □

17.7 Corollaries

17.7.1 Translation-invariant chains

Theorem 17.23 (Single gauge for translation-invariant chains). [ti_tensors_single_gauge] [✓ Stmt OK] Let A and B be MPS tensors of bond dimension D , and fix a chain length $n \geq 1$. Assume that A is injective and that the combined tensors of the two constant chains $(A, \dots, A)$ and $(B, \dots, B)$ generate the same MPV family. Then there exists an invertible matrix $X \in \text{GL}_D(\mathbb{C})$ such that

$$B^i = X A^i X^{-1}$$

for every physical index i .

Proof. [✓ Proof OK] Bundle the constant chains into one tensor. Injectivity of A implies injectivity of the combined tensor by Lemma 17.9, and the combined-tensor MPV equality is exactly the hypothesis needed for the single-block Fundamental Theorem. The resulting gauge matrix is independent of the site because every site carries the same local tensor. $\square$

Corollary 17.24 (Translation-invariant collapse to one gauge). [ti_tensors_collapse_to_single_gauge] [✓ Stmt OK] Let A and B be MPS tensors of bond dimension D , and fix a chain length $n \geq 1$. Assume that A is injective and that the combined tensors of the two constant chains $(A, \dots, A)$ and $(B, \dots, B)$ generate the same MPV family. Then there exist an invertible matrix $Z \in \text{GL}_D(\mathbb{C})$ and a scalar $\lambda \in \mathbb{C}^\times$ with $\lambda^n = 1$ such that

$$B^i = \lambda Z^{-1} A^i Z$$

for every physical index i .

Proof. [✓ Proof OK] Apply Theorem 17.23 to obtain $B^i = X A^i X^{-1}$. Set $Z = X^{-1}$ and $\lambda = 1$. Then $\lambda^n = 1$ and the displayed identity is exactly the same gauge relation rewritten in the form used by the corollary. $\square$

17.7.2 Non-translation-invariant descriptions of translation-invariant states

Corollary 17.25 (Non-translation-invariant realisation of a translation-invariant state). [Not ready] Assume a same-state reduction hypothesis for (d, D) . Let $\mathbf{A} = (A_0, \dots, A_{n-1})$ be an injective non-translation-invariant chain of length $n \geq 3$. Suppose the state generated by $\mathbf{A}$ is translation-invariant, i.e. it equals the state generated by the cyclically shifted chain $(A_1, \dots, A_{n-1}, A_0)$. Then $\mathbf{A}$ admits a translation-invariant description with the same bond dimension: there exists an injective tensor B such that the constant chain $(B, \dots, B)$ generates the same state as $\mathbf{A}$. Explicitly, $B^i = A_0^i Z_0^{-1}$ (right multiplication, not conjugation), where Z_0 is extracted from the cyclic shift.

Proof. Apply Theorem 17.18 to $\mathbf{A}$ and its cyclic shift. This gives invertible gauges $Z_0, \dots, Z_{n-1}$ with $A_{k+1}^i = Z_k^{-1} A_k^i Z_{k+1}$. Iterating these identities yields $A_k^i = L_k^{-1} A_0^i R_k$, where the products L_k, R_k satisfy $R_k L_{k+1}^{-1} = Z_0^{-1}$. Substituting into the chain trace shows that the same state is generated by the constant tensor $B^i := A_0^i Z_0^{-1}$. Since A_0 is injective and Z_0 is invertible, B is injective. $\square$

Symmetry-theoretic consequences of the Fundamental Theorem, including virtual representations and string order, are developed in Chapter 19.

17.7.3 Blocked chains

Lemma 17.26 (Block injectivity and blocked-tensor injectivity). [isNBlkInjective_iff_blockTensor_isInjective](#) ✓ Stmt OK *A tensor A is L -block injective if and only if its L -blocked tensor $A^{[L]}$ is injective.*

Proof. ✓ Proof OK The blocked physical alphabet is just a reindexing of words of length L . Decoding blocked indices therefore identifies the two spanning families, so the two injectivity conditions are equivalent. $\square$

Definition 17.27 (Constant blocked chain). [blockedChain](#) ✓ Stmt OK For an MPS tensor A , a blocking length L , and a chain length n , the *blocked chain* is the constant n -site chain whose local tensor is the blocked tensor $A^{[L]}$.

Lemma 17.28 (Blocked chains are injective). [blockedChain_isInjective](#) ✓ Stmt OK *If A is L -block injective, then the constant blocked chain built from $A^{[L]}$ is injective at every site.*

Proof. ✓ Proof OK By Lemma 17.26, the local blocked tensor $A^{[L]}$ is injective, and therefore every site of the constant blocked chain is injective. $\square$

Theorem 17.29 (Fundamental Theorem for blocked chains). [fundamentalTheorem_blockedChain](#) ✓ Stmt OK *Let A and B be MPS tensors of bond dimension D , let L be a blocking length, and let n be a chain length. Assume that A is L -block injective and that the combined tensors of the two constant blocked chains built from $A^{[L]}$ and $B^{[L]}$ generate the same MPV family. Then the two blocked chains are gauge equivalent.*

Proof. ✓ Proof OK By Lemma 17.28, the blocked chain built from $A^{[L]}$ is injective. Apply Theorem 17.13 to the two constant blocked chains. $\square$

Remark 17.30. The theorem above is the exact blocked-chain result established here. Recovering sitewise gauges for the original unblocked tensors, and hence the paper's normal-MPS corollary, is not established here.

17.8 Product algebra construction and per-block gauge

The chain Fundamental Theorem (Theorem 17.13) handles non-translation-invariant periodic chains. A parallel line of argument addresses block-diagonal tensors: given a family of injective block tensors $\{A_k^i\}_{k=0}^{r-1}$ with block dimensions D_k and a second family $\{B_k^i\}$ with per-block $\mathcal{V}(A_k) = \mathcal{V}(B_k)$, one seeks per-block gauge equivalence $B_k^i = X_k A_k^i X_k^{-1}$ for each k independently, and a global block-permutation plus inner-automorphism decomposition.

17.8.1 Per-block linear extension

Definition 17.31 (Per-block linear extension). [perBlockLinearExtension](#) ✓ Stmt OK Let A_k, B_k be MPS tensors of bond dimension D_k with A_k injective and $\mathcal{V}(A_k) = \mathcal{V}(B_k)$ for each block $k \in \{0, \dots, r-1\}$. The *per-block linear extension* $T_k : M_{D_k}(\mathbb{C}) \rightarrow M_{D_k}(\mathbb{C})$ is the unique $\mathbb{C}$ -linear map satisfying $T_k(A_k^i) = B_k^i$ for every physical index i .

Lemma 17.32 (Per-block multiplicativity). [perBlockLinearExtension_mul](#) ✓ Stmt OK *For each block k , the per-block linear extension T_k is multiplicative: $T_k(MN) = T_k(M)T_k(N)$ for all $M, N \in M_{D_k}(\mathbb{C})$.*

Proof. ✓ Proof OK Since $\{A_k^i\}$ spans $M_{D_k}(\mathbb{C})$, it suffices to check on spanning elements, where the result follows from the MPV identity. $\square$

Lemma 17.33 (Per-block bijectivity). perBlockLinearExtension_bijective ✓ Stmt OK Each T_k is bijective.

Proof. ✓ Proof OK A nonzero multiplicative linear endomorphism of $M_{D_k}(\mathbb{C})$ is automatically bijective: injectivity follows from the simplicity of $M_{D_k}(\mathbb{C})$ (the kernel is a proper ideal, hence zero), and surjectivity from finite dimension. Non-vanishing of T_k is established via the trace pairing: if $T_k = 0$, then all $B_k^i = 0$, contradicting the non-vanishing of the trace pairing for injective tensors. $\square$

17.8.2 Product algebra automorphism

Definition 17.34 (Product algebra automorphism). piAlgEquiv ✓ Stmt OK The *product algebra automorphism* is the $\mathbb{C}$ -algebra equivalence

$$T : \prod_{k=0}^{r-1} M_{D_k}(\mathbb{C}) \xrightarrow{\sim} \prod_{k=0}^{r-1} M_{D_k}(\mathbb{C})$$

defined by $T(M)_k = T_k(M_k)$, where each T_k is the per-block linear extension. It is an algebra automorphism since each T_k is a multiplicative, unital, bijective linear map. Diagrammatically, this is the blockwise map on the product algebra obtained by applying the corresponding per-block extension to each summand.

Theorem 17.35 (Block-permutation decomposition). piAlgEquiv_decomposition ✓ Stmt OK The product algebra automorphism T decomposes as a block permutation composed with per-block inner automorphisms: there exist a permutation $\sigma \in S_r$ (with $D_{\sigma(k)} = D_k$ for all k) and invertible matrices $X_k \in \text{GL}_{D_k}(\mathbb{C})$ such that

$$(T(M))_{\sigma(k)} = X_k M X_k^{-1}$$

for all $M \in M_{D_k}(\mathbb{C})$ (viewed as supported in block k) and all k . Equivalently, the product-algebra automorphism first permutes the simple matrix blocks and then acts inside each block by an inner conjugation.

Proof. ✓ Proof OK The automorphism T permutes the block ideals of $\prod_k M_{D_k}(\mathbb{C})$ (each block ideal is simple, and an automorphism sends simple ideals to simple ideals of the same dimension). This gives the permutation σ . On each block, the restricted map is a $\mathbb{C}$ -algebra automorphism of $M_{D_k}(\mathbb{C})$; by Skolem–Noether (Theorem 4.9), it is inner. Thus T splits algebraically into the permutation of block ideals followed by inner conjugation on each surviving block. $\square$

17.8.3 Nondegeneracy of the product trace pairing

Lemma 17.36 (Nondegeneracy of the product trace pairing). piTrace_mul_right_eq_zero ✓ Stmt OK Let $M = (M_0, \dots, M_{r-1}) \in \prod_k M_{D_k}(\mathbb{C})$. If $\sum_k \text{tr}(M_k N_k) = 0$ for all $N = (N_0, \dots, N_{r-1})$, then $M = 0$.

Proof. ✓ Proof OK Choose N supported on a single block k (all other components zero). The hypothesis reduces to $\text{tr}(M_k N_k) = 0$ for all N_k , so $M_k = 0$ by nondegeneracy of the matrix trace pairing on $M_{D_k}(\mathbb{C})$. Since k was arbitrary, $M = 0$. $\square$

Lemma 17.37 (Injectivity of the product Gram map). [piTraceMulRightPi_ker_eq_bot](#) ✓ Stmt OK Let $\{A_k\}$ be a family of injective block tensors. The product Gram map $M \mapsto (k, i \mapsto \text{tr}(M_k \cdot A_k^i))$ is injective.

Proof. ✓ Proof OK Reduces to per-block injectivity of the Gram map $M_k \mapsto (i \mapsto \text{tr}(M_k \cdot A_k^i))$, which holds by injectivity of A_k (the matrices $\{A_k^i\}$ span $M_{D_k}(\mathbb{C})$, so the trace pairing against them separates points). $\square$

17.9 The multi-block Fundamental Theorem

Theorem 17.38 (Multi-block Fundamental Theorem). [fundamentalTheorem_multiBlock_full](#) ✓ Stmt OK Let $\{A_k\}_{k=0}^{r-1}$ and $\{B_k\}_{k=0}^{r-1}$ be two families of MPS tensors with block dimensions D_k , each A_k injective, and $\mathcal{V}(A_k) = \mathcal{V}(B_k)$ for all k . Then:

1. Per-block gauge equivalence: $B_k^i = X_k A_k^i X_k^{-1}$ for some $X_k \in \text{GL}_{D_k}(\mathbb{C})$, for every k and i .
2. Global gauge equivalence: the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ are gauge equivalent.

The second clause is obtained from the first by assembling the per-block gauges into one block-diagonal gauge.

Proof. ✓ Proof OK Per-block gauge equivalence follows from applying the single-block Fundamental Theorem (existing) to each block, using injectivity of A_k and $\mathcal{V}(A_k) = \mathcal{V}(B_k)$. Global gauge equivalence is assembled from the per-block gauges via the block-diagonal structure. $\square$

Theorem 17.39 (Explicit per-block gauges). [fundamentalTheorem_multiBlock_explicit](#) ✓ Stmt OK Under the hypotheses of Theorem 17.38, there exist invertible matrices $X_k \in \text{GL}_{D_k}(\mathbb{C})$ such that

$$B_k^i = X_k A_k^i X_k^{-1}$$

for every block k and every physical index i .

Proof. ✓ Proof OK Extract the gauge matrix from the per-block part of Theorem 17.38 for each block. $\square$

Theorem 17.40 (Product-algebra decomposition of the multi-block theorem). [fundamentalTheorem_multiBlock_decomposition](#) that every block dimension is nonzero. Under the hypotheses of Theorem 17.38, the product-algebra automorphism determined by the per-block linear extensions decomposes into a block permutation together with inner conjugation on each block: there exist a permutation σ of the blocks, compatible block-dimension equalities, and invertible matrices X_k such that, after reindexing the target block $\sigma(k)$ back to block k , the action on the k th matrix block is $M \mapsto X_k M X_k^{-1}$.

Proof. ✓ Proof OK Apply Theorem 17.35 to the product-algebra automorphism built from the per-block linear extensions. $\square$

Theorem 17.41 (Per-block equality of MPV families and per-block gauge equivalence). [perBlock_sameMPV_iff_gaugeEquiv](#) $\{A_k\}$ be a family of injective block tensors. Then $\mathcal{V}(A_k) = \mathcal{V}(B_k)$ for all k if and only if A_k and B_k are gauge equivalent for all k .

Proof. ✓ Proof OK The forward direction is the single-block Fundamental Theorem applied block-wise. The reverse direction holds because gauge equivalence preserves all traces of word evaluations. $\square$

17.10 Block separation from canonical form

The multi-block Fundamental Theorem (Theorem 17.38) assumes per-block $\mathcal{V}(A_k) = \mathcal{V}(B_k)$. In practice, one starts from the weaker global hypothesis that the block-diagonal tensors satisfy the same MPV family: $\mathcal{V}(\bigoplus_k \mu_k A_k) = \mathcal{V}(\bigoplus_k \mu_k B_k)$. Block separation extracts the per-block identities from the global one under canonical-form normalization. The first step is the weighted block-sum identity

$$\sum_k \mu_k^N V^{(N)}(A_k)_\sigma = \sum_k \mu_k^N V^{(N)}(B_k)_\sigma,$$

which is then combined with overlap estimates and the leading-block peel.

Definition 17.42 (Canonical form). IsCanonicalForm ✓ Stmt OK A family of block tensors $\{A_k\}_{k=0}^{r-1}$ with scaling weights $\mu_0, \dots, \mu_{r-1} \in \mathbb{C}^\times$ is in *canonical form* if:

1. each A_k is injective,
2. each A_k is left-canonical: $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}_{D_k}$,
3. the weights are strictly ordered by norm: $|\mu_0| > |\mu_1| > \dots > |\mu_{r-1}| > 0$,
4. the self-overlaps are normalised: $O_{A_k A_k}(N) \rightarrow 1$ as $N \rightarrow \infty$.

Definition 17.43 (Normal canonical form). IsNormalCanonicalForm ✓ Stmt OK A family is in *normal canonical form* if it satisfies the canonical-form conditions with each block additionally *irreducible* (no nontrivial invariant projection) and *primitive* (the transfer map has spectral gap).

Theorem 17.44 (Block separation from weighted word identities). block_separation_all_words ✓ Stmt OK Let $\{A_k\}$ and $\{B_k\}$ be block families with strictly ordered, nonzero weights μ_k . Assume that every A_k and B_k is injective and left-canonical, that each self-overlap $O_{A_k A_k}(N)$ tends to 1, and that for every system size N and every word σ one has the weighted identity

$$\sum_k \mu_k^N (V^{(N)}(A_k)_\sigma - V^{(N)}(B_k)_\sigma) = 0.$$

Then $\mathcal{V}(A_k) = \mathcal{V}(B_k)$ for every block k .

Proof. ✓ Proof OK The proof is the peeling argument described below for canonical form. The strict weight ordering isolates the leading block, the overlap bounds show that the tail decays geometrically after dividing by the dominant weight, and one then iterates on the remaining blocks. □

Theorem 17.45 (Block separation from canonical form). per_block_sameMPV_of_canonical_form ✓ Stmt OK Let $\{A_k\}$ be in canonical form with weights $\{\mu_k\}$, and let $\{B_k\}$ be another family with each B_k injective and left-canonical. If $\mathcal{V}(\bigoplus_k \mu_k A_k) = \mathcal{V}(\bigoplus_k \mu_k B_k)$, then $\mathcal{V}(A_k) = \mathcal{V}(B_k)$ for each block k individually. The proof isolates the leading block and treats the remaining blocks as an exponentially decaying tail with $\rho = |\mu_1|/|\mu_0| < 1$.

Proof. ✓ Proof OK The proof proceeds by strong induction on the number of blocks r . For $r = 1$, the weighted sum degenerates and the result is immediate after cancelling $\mu_0^N \neq 0$.

For $r \geq 2$, the global identity $\sum_k \mu_k^N (V^{(N)}(A_k)_\sigma - V^{(N)}(B_k)_\sigma) = 0$ holds for all N and σ .

$$\mu_0^N (V^{(N)}(A_0)_\sigma - V^{(N)}(B_0)_\sigma) = - \sum_{k \geq 1} \mu_k^N (V^{(N)}(A_k)_\sigma - V^{(N)}(B_k)_\sigma).$$

Dividing by μ_0^N shows that the right-hand side is $O(\rho^N)$ with $\rho = |\mu_1|/|\mu_0| < 1$. Multiplying by the MPV components of A_0 and summing over σ converts this to an overlap identity. Dividing by $|\mu_0|^{2N}$ and using the strict norm ordering $|\mu_1|/|\mu_0| < 1$, the tail terms ($k \geq 1$) decay geometrically with ratio $\rho = |\mu_1|/|\mu_0|$. The leading term ($k = 0$) therefore converges: $O_{A_0 B_0}(N) \rightarrow 1$ as $N \rightarrow \infty$. Combined with the normalised self-overlap $O_{A_0 A_0}(N) \rightarrow 1$, this gives $\mathcal{V}(A_0) = \mathcal{V}(B_0)$ via the gauge-phase equivalence criterion. Subtracting the leading block and applying the inductive hypothesis yields $\mathcal{V}(A_k) = \mathcal{V}(B_k)$ for all remaining k . $\square$

Theorem 17.46 (Block separation from separated canonical data). [per_block_sameMPV_of_separated_canonical_data](#) ✓ Stmt OK ✓ *the weights are nonzero with strictly decreasing norms, the A -blocks are injective and left-canonical with normalized self-overlaps, the B -blocks are injective and left-canonical, and the assembled block-diagonal tensors generate the same MPV family. Then $\mathcal{V}(A_k) = \mathcal{V}(B_k)$ for every block k .*

Proof. ✓ Proof OK Equality of the assembled MPV families gives the weighted word identity required by Theorem 17.44. The separated hypotheses supply exactly the injectivity, left-canonicity, strict-weight, and self-overlap assumptions needed there. $\square$

Theorem 17.47 (Fundamental Theorem from separated canonical data). [fundamentalTheorem_of_separated_canonical_data](#) ✓ *the hypotheses of Theorem 17.46, every block pair A_k, B_k is gauge equivalent, and the assembled block-diagonal tensors are globally gauge equivalent.*

Proof. ✓ Proof OK First apply Theorem 17.46 to get $\mathcal{V}(A_k) = \mathcal{V}(B_k)$ blockwise. Then apply Theorem 17.38. $\square$

Theorem 17.48 (Explicit gauge from separated canonical data). [fundamentalTheorem_of_separated_canonical_data_explicit](#) ✓ *the same hypotheses, there exist invertible matrices $X_k \in \text{GL}_{D_k}(\mathbb{C})$ such that*

$$B_k^i = X_k A_k^i X_k^{-1}$$

for every block k and physical index i .

Proof. ✓ Proof OK Combine blockwise MPV equality from Theorem 17.46 with the explicit-gauge form of the multi-block Fundamental Theorem. $\square$

Theorem 17.49 (Fundamental Theorem from canonical form). [fundamentalTheorem_canonicalForm](#) ✓ Stmt OK ✓ *Under the hypotheses of Theorem 17.45, every block pair A_k, B_k is gauge equivalent, and the block-diagonal tensors are globally gauge equivalent.*

Proof. ✓ Proof OK Theorem 17.45 gives per-block $\mathcal{V}(A_k) = \mathcal{V}(B_k)$. Theorem 17.38 converts this to per-block and global gauge equivalence. $\square$

Theorem 17.50 (Explicit gauge from canonical form). [fundamentalTheorem_canonicalForm_explicit](#) ✓ Stmt OK ✓ *Under the same hypotheses, there exist invertible matrices $X_k \in \text{GL}_{D_k}(\mathbb{C})$ such that $B_k^i = X_k A_k^i X_k^{-1}$ for every block k and physical index i .*

Proof. ✓ Proof OK Extract the gauge matrices from Theorem 17.49 by choice. $\square$

17.11 Heterogeneous equal-case result

Theorem 17.51 (Heterogeneous equal-case Fundamental Theorem). [fundamentalTheorem_equalMPV_CFBNT_hetero](#) ✓ Stmt OK

$\{\mu_j A_j\}_{j=0}^{r_A-1}$ and $\{\nu_k B_k\}_{k=0}^{r_B-1}$ be two canonical-form BNT families, possibly with different block counts and different bond dimensions in corresponding blocks. If the assembled block-diagonal tensors generate the same MPV family, then the two families have the same number of blocks, the blocks can be matched by a permutation preserving the bond dimensions, and each matched pair is gauge-phase equivalent.

Proof. ✓ Proof OK The proof is the block-matching argument carried out in the supporting modules for the heterogeneous equal case: one first produces a non-decaying cross-overlap for each block, applies the overlap dichotomy to show that the matched blocks must have equal dimension and differ only by a phase gauge, and finally upgrades the resulting matching to a permutation of all blocks. □

17.12 Translation-invariant reduction

Theorem 17.52 (Translation-invariant reduction corollary). [ti_reduction_corollary](#) ✓ Stmt OK Let A and B be MPS tensors of bond dimension D with A injective. Consider the two constant translation-invariant chains $(A, \dots, A)$ and $(B, \dots, B)$ of length $n \geq 1$. If the combined tensors of the two chains generate the same MPV family, then:

1. B is gauge equivalent to A : there exists $X \in \text{GL}_D(\mathbb{C})$ with $B^i = X A^i X^{-1}$ for all i .
2. B is injective.

Proof. ✓ Proof OK Theorem 17.23 gives a single gauge matrix X with $B^i = X A^i X^{-1}$ for all i . Since gauge equivalence preserves injectivity, B is injective as well. □

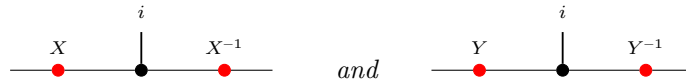
Theorem 17.53 (Translation-invariant reduction from same state). [ti_reduction_of_sameState](#) ✓ Stmt OK Assume a same-state reduction hypothesis for (d, D) . Let A and B be MPS tensors of bond dimension D , and let $n \geq 3$. If the two constant chains $(A, \dots, A)$ and $(B, \dots, B)$ generate the same chain state, with A injective, then the same conclusion as in Theorem 17.52 holds: B is gauge equivalent to A , and B is injective.

Proof. ✓ Proof OK The same-state reduction hypothesis upgrades equality of the two constant chain states to equality of their combined-tensor MPV families. Then Theorem 17.52 applies. □

17.12.1 Global symmetry via gauge composition

The virtual representation theorem (Theorem 19.15) relies on the fact that composing two gauge transformations yields another gauge up to a scalar factor.

Lemma 17.54 (Gauge ratio commutes with tensor matrices). [gauge_ratio_commutates](#) ✓ Stmt OK Diagrammatically, the hypothesis is that the two local gauges



produce the same target tensor B^i from the same source tensor A^i . If $X, Y \in \text{GL}_D(\mathbb{C})$ both conjugate an MPS tensor A to the same tensor B (i.e. $B^i = X A^i X^{-1} = Y A^i Y^{-1}$ for all i), then $Y^{-1}X$ commutes with every A^i .

Proof. ✓ Proof OK Direct matrix calculation: conjugating both gauge relations and composing gives $(Y^{-1}X)A^i = A^i(Y^{-1}X)$. $\square$

Lemma 17.55 (Gauge ratio is scalar for injective tensors). gauge_ratio_isScalar ✓ Stmt OK Under the same hypotheses, if A is additionally injective, then $Y^{-1}X = \lambda \mathbb{1}_D$ for some scalar $\lambda \in \mathbb{C}$.

Proof. ✓ Proof OK The same two-gauge picture shows that the ratio $Y^{-1}X$ acts invisibly on the local tensor. By Lemma 17.54, $Y^{-1}X$ commutes with all A^i . Since $\{A^i\}$ spans $M_D(\mathbb{C})$, the matrix $Y^{-1}X$ commutes with all of $M_D(\mathbb{C})$. By Lemma 17.10, it is a scalar matrix. $\square$

Theorem 17.56 (Projective multiplication law for gauge matrices). gaugeMatrix_projective_mul ✓ Stmt OK Let A be an injective MPS tensor with on-site symmetry under a group representation $U : G \rightarrow \text{GL}_d(\mathbb{C})$. For each $g \in G$, let $X(g) \in \text{GL}_D(\mathbb{C})$ satisfy $\widetilde{A}_g^i = X(g) A^i X(g)^{-1}$ for all i . Then for all $g, h \in G$, there exists $c(g, h) \in \mathbb{C}$ such that

$$X(h)X(g) = c(g, h)X(g \cdot h).$$

In fact $c(g, h) \neq 0$ since both sides are invertible, so $c(g, h) \in \mathbb{C}^\times$. Diagrammatically, the two candidate gauges are competing local replacements for the same twisted tensor:

and both sides conjugate A^i to the same twisted tensor $\widetilde{A}_{gh}^i$.

Proof. ✓ Proof OK By the composition law $\widetilde{A}_{gh} = \widetilde{(\widetilde{A}_h)_g}$ (Lemma 19.5), both $X(h)X(g)$ and $X(gh)$ conjugate A to $\widetilde{A}_{gh}$. Lemma 17.55 then gives the scalar factor. $\square$

17.13 Closing remarks

Remark 17.57. Appendix A of [MGRPG⁺18] strengthens the normal-MPS bound from $n \geq 3L$ to $n \geq 2L + 1$ by a more refined blocking argument. The injective PEPS theorem of Chapter 18 and the normal PEPS theorem on arbitrary graphs both rest on the same two core lemmas (Lemma 17.11 and Lemma 17.12); the graph geometry enters only through the blocking step that reduces to a three-site chain.

Chapter 18

Fundamental Theorem for Injective PEPS

The following definitions and results extend the Fundamental Theorem to injective PEPS on simple graphs, following Theorem 2 of [MGRPG⁺18].

Definition 18.1 (Graph edge for PEPS). Edge ✓ Stmt OK For a finite simple graph G , an *edge* is an ordered pair (u, v) with $u < v$ and u adjacent to v . The ordering is only used to avoid double counting.

Definition 18.2 (Incident edges at a vertex). IncidentEdge ✓ Stmt OK For a vertex v , an *incident edge* is an edge of G having v as one of its endpoints.

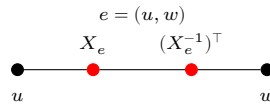
Definition 18.3 (PEPS tensor on a graph). Tensor ✓ Stmt OK A PEPS tensor on G with physical dimension d consists of a bond dimension D_e for every edge e , together with a local tensor at each vertex v whose virtual indices are indexed by the edges incident to v and whose physical index lies in $\{0, \dots, d-1\}$.

Definition 18.4 (Virtual configuration). VirtualConfig ✓ Stmt OK A *virtual configuration* assigns to every edge e one basis index in the corresponding bond space of dimension D_e .

Definition 18.5 (PEPS state coefficient). stateCoeff ✓ Stmt OK For a physical configuration σ on the vertices, the PEPS state coefficient is obtained by summing over all virtual configurations and, for each such configuration, multiplying the local tensor entries over all vertices.

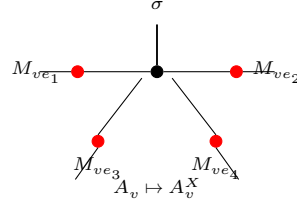
Definition 18.6 (Equality of PEPS states). SameState ✓ Stmt OK Two PEPS tensors represent the same state if all their state coefficients agree for every physical configuration.

Definition 18.7 (Edge gauge at a vertex). edgeGaugeAt ✓ Stmt OK For an edge $e = (u, w)$ with $u < w$ and gauge $X_e \in \text{GL}(D_e)$, the gauge matrix applied at vertex v is X_e if $v = u$ and $(X_e^{-1})^\top$ if $v = w$. Diagrammatically, the ordered edge carries opposite endpoint actions:



Definition 18.8 (Gauge vertex action). gaugeVertex ✓ Stmt OK The gauged tensor at vertex v is A_v^X . For each incident edge ie , write M_{ie} for the matrix specified by Definition 18.7, applied at the

vertex v to the gauge on the underlying edge of ie . Then $A_v^X(\eta, \sigma) = \sum_{\eta'} (\prod_{ie} M_{ie}(\eta(ie), \eta'(ie))) A_v(\eta', \sigma)$. In tensor-network notation, one inserts the appropriate gauge matrix on each incident virtual leg of the local tensor:



Definition 18.9 (Apply gauge). applyGauge ✓ Stmt OK The PEPS tensor obtained by applying edge-gauge matrices to A , with vertex tensors A_v^X .

Definition 18.10 (Gauge equivalence for PEPS). GaugeEquiv ✓ Stmt OK Two PEPS tensors A, B are gauge-equivalent if they have the same bond dimensions and B is obtained from A by applying invertible gauge matrices on each edge.

Theorem 18.11 (Gauge invariance of PEPS state). applyGauge_stateCoeff ✓ Stmt OK Applying a gauge to a PEPS tensor preserves the state coefficients. Along each internal edge, the two endpoint insertions cancel:

$$\begin{array}{ccc}
 X_e(j, i) ((X_e^{-1})^\top)(j, k) & \sum_j X_e(j, i) ((X_e^{-1})^\top)(j, k) = \delta(i, k) & \\
 \bullet \text{---} \bullet \text{---} \bullet \text{---} \bullet & = & \bullet \text{---} \bullet
 \end{array}$$

Theorem 18.12 (Gauge equivalence implies same state). sameState ✓ Stmt OK Gauge equivalence implies that the two PEPS tensors represent the same state.

Definition 18.13 (Local tensor evaluation). localTensorEval ✓ Stmt OK The local tensor evaluated at vertex v with virtual-index weighting f , computing $\sum_{\eta} (\prod_{ie} f(ie)(\eta(ie))) \cdot A_v(\eta, \sigma)$. This is multilinear in the components of f , not linear in the full tuple.

Definition 18.14 (Vertex injectivity). IsVertexInjective ✓ Stmt OK A PEPS tensor A is *vertex-injective* if, at every vertex v , the family of physical vectors

$$(A_v(\eta, \cdot) \in \mathbb{C}^d)_\eta$$

indexed by virtual configurations η on the edges incident to v is linearly independent in $\mathbb{C}^d$. Equivalently, extending A_v by linearity to a map from the free vector space on virtual configurations into the physical space $\mathbb{C}^d$ gives a linear map with trivial kernel. This matches the paper's formulation that "all tensors interpreted as maps from the virtual space to the physical one are injective" ([MGRPG⁺18], §3).

Definition 18.15 (Local tensor map). localTensorMap ✓ Stmt OK For a fixed vertex v , the family of local physical vectors $\eta \mapsto A_v(\eta, \cdot)$ extends by linearity to a map from the coefficient space on local virtual configurations at v into $\mathbb{C}^d$.

Definition 18.16 (Local left inverse). localLeftInverse ✓ Stmt OK If A is vertex-injective, one may choose a linear map Ψ_v from $\mathbb{C}^d$ to the coefficient space on local virtual configurations at v such that $\Psi_v \circ \Phi_v = \mathbb{1}$.

Definition 18.17 (Physical realization of a local virtual operator). [physRealizeLocalOp](#) ✓ Stmt OK For an endomorphism T of the coefficient space on local virtual configurations at v , define the local physical operator

$$O_T = \Phi_v \circ T \circ \Psi_v.$$

Theorem 18.18 (Local virtual operations on the image). [physRealizeLocalOp_spec](#) ✓ Stmt OK For every coefficient function c on the local virtual configurations at v ,

$$O_T(\Phi_v(c)) = \Phi_v(Tc).$$

Proof. ✓ Proof OK This is immediate from $\Psi_v \circ \Phi_v = \mathbb{1}$. □

Definition 18.19 (Split at one incident edge). [localVirtualConfigSplitAt](#) ✓ Stmt OK Choosing one incident edge at v identifies the set of local virtual configurations with the product of the bond index on that edge and the residual indices on the remaining incident edges.

Definition 18.20 (Middle region of an edge). [edgeMiddleVertices](#) ✓ Stmt OK For an edge $e = (u, v)$, the middle region is the complement $V \setminus \{u, v\}$.

Theorem 18.21 (Edge-centered vertex partition). [edgeVertices_union](#) ✓ Stmt OK For an edge $e = (u, v)$, the three regions $\{u\}$, $V \setminus \{u, v\}$, and $\{v\}$ cover the whole vertex set.

Proof. ✓ Proof OK Every vertex is either u , v , or different from both. □

Theorem 18.22 (Product splitting at an edge). [prod_univ_splitAtEdge](#) ✓ Stmt OK For any edge $e = (u, v)$ and any function f on the vertex set with values in a commutative monoid M ,

$$\prod_{x \in V} f(x) = f(u) \left(\prod_{x \in V \setminus \{u, v\}} f(x) \right) f(v).$$

Proof. ✓ Proof OK Isolate the two distinguished endpoints in the product over V and group the remaining factors into the middle region. □

Theorem 18.23 (PEPS coefficient splitting at an edge). [stateCoeff_splitAtEdge](#) ✓ Stmt OK For any edge $e = (u, v)$, each summand in the PEPS coefficient factors into the tensor contribution at u , the product over the middle region $V \setminus \{u, v\}$, and the tensor contribution at v .

Proof. ✓ Proof OK Expand the state coefficient and apply the product splitting at the chosen edge to the per-vertex product inside each summand. □

Definition 18.24 (Edge-centered boundary data). [EdgeBoundaryConfig](#) ✓ Stmt OK For an edge $e = (u, v)$, the boundary data consist of the virtual index on e , the remaining virtual indices incident to u , and the remaining virtual indices incident to v . These are the indices over which the endpoint tensors couple to the blocked middle region.

Definition 18.25 (Boundary data read from a virtual configuration). [edgeBoundaryOfVirtualConfig](#) ✓ Stmt OK A global virtual configuration determines edge-centered boundary data by reading the index on the distinguished edge and the residual endpoint indices.

Definition 18.26 (Boundary compatibility at an edge). [edgeBoundaryMatches](#) ✓ Stmt OK A global virtual configuration is compatible with fixed edge-centered boundary data if it has the prescribed index on the distinguished edge and the prescribed residual indices at both endpoint stars.

Definition 18.27 (Middle configurations over fixed boundary data). [EdgeMiddleConfig](#) ✓ Stmt OK For fixed boundary data β , the middle configurations are the subtype of global virtual configurations satisfying the compatibility condition with β .

Definition 18.28 (Virtual configurations fibred by edge boundary data). [virtualConfigEquivEdgeBoundary](#) ✓ Stmt OK A global virtual configuration is equivalently a choice of edge-centered boundary data together with a middle configuration in the corresponding fibre.

Definition 18.29 (Left endpoint local configuration from boundary data). [edgeLeftLocalConfig](#) ✓ Stmt OK The left endpoint local virtual configuration is reconstructed from the edge index and the residual data on the left endpoint star.

Definition 18.30 (Right endpoint local configuration from boundary data). [edgeRightLocalConfig](#) ✓ Stmt OK The right endpoint local virtual configuration is reconstructed from the edge index and the residual data on the right endpoint star.

Theorem 18.31 (Left endpoint reconstruction for compatible boundary data). [edgeLeftLocalConfig_eq_of_boundaryMatches](#) ✓ Stmt OK *a global virtual configuration is compatible with boundary data, then its restriction to the left endpoint star is the local configuration reconstructed from those data.*

Theorem 18.32 (Right endpoint reconstruction for compatible boundary data). [edgeRightLocalConfig_eq_of_boundaryMatches](#) ✓ Stmt OK *a global virtual configuration is compatible with boundary data, then its restriction to the right endpoint star is the local configuration reconstructed from those data.*

Definition 18.33 (Blocked middle tensor at an edge). [edgeMiddleWeight](#) ✓ Stmt OK Fixing the edge-centered boundary data, the blocked middle tensor is the sum over global virtual configurations compatible with those data of the product of local tensor entries over the middle region $V \setminus \{u, v\}$.

Definition 18.34 (Edge-blocked PEPS coefficient). [edgeBlockedCoeff](#) ✓ Stmt OK The edge-blocked coefficient is the three-block contraction obtained from the left endpoint tensor, the blocked middle tensor, and the right endpoint tensor at the chosen edge.

Theorem 18.35 (Edge-blocked coefficient equals the PEPS coefficient). [edgeBlockedCoeff_eq_stateCoeff](#) ✓ Stmt OK *For every edge e , the edge-blocked three-block coefficient is exactly the original PEPS state coefficient.*

Proof. ✓ Proof OK Fibre the global virtual sum by the edge-centered boundary data. The remaining fibre sum is the blocked middle tensor, and the two endpoint factors are reconstructed from the same boundary data. □

Theorem 18.36 (Same state gives the same edge-blocked coefficients). [edgeBlockedCoeff_eq](#) ✓ Stmt OK *If two PEPS tensors have the same state, then their edge-blocked coefficients agree at every edge and every physical configuration.*

Proof. ✓ Proof OK Rewrite both edge-blocked coefficients as the corresponding PEPS state coefficients and apply the same-state hypothesis. □

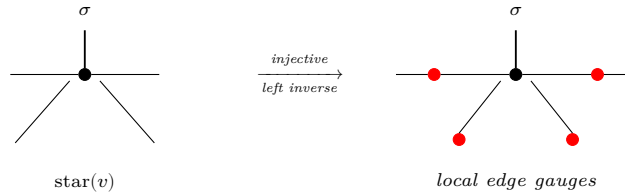
Definition 18.37 (Sharpened local gauge lift datum). [HasLocalGaugeLift](#) ✓ Stmt OK For fixed A , B , a bond-dimension equality, and a vertex v , this states the two local conclusions still needed from the blocked-middle reduction in [MGRPG⁺18], §3: (i) each local component of B lies in the image of the local tensor map of A , and (ii) the canonical pullback through the chosen left inverse of A factorizes as one invertible matrix on each incident edge.

Definition 18.38 (Blocked-middle local gauge data). [BlockedMiddleGaugeHyp](#) ✓ Stmt OK This states the explicit local gauge formula that the edge-centered blocked-middle reduction is expected to produce: a family of invertible matrices on the incident edges of v whose product kernel reconstructs each local component of B from those of A .

Theorem 18.39 (Blocked-middle local gauge data implies sharpened lift). [hasLocalGaugeLift_of_blockedMiddleGaugeHyp](#) ✓ Stmt OK *explicit local gauge formula from the blocked-middle reduction yields the sharpened local lift. The remaining open step is to derive this blocked-middle formula from equality of PEPS states.*

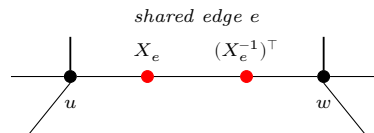
Proof. ✓ Proof OK The explicit local gauge formula already places each local component of B in the image of the local tensor map of A . Applying the chosen left inverse of A identifies the canonical candidate with the same factorized coefficient kernel, so both parts of the sharpened lift follow. $\square$

Theorem 18.40 (Local gauge extraction). [localGauge_exists](#) ✓ Stmt OK *Under the sharpened local lift hypothesis, the local gauge formula follows. What remains open is to upgrade equality of the edge-blocked coefficients to the blocked-middle formula using the corresponding three-site injective-chain argument. Under that hypothesis, injectivity on the star neighbourhood provides a left inverse that isolates the local edge gauges:*



Proof. ✓ Proof OK The sharpened local lift datum already supplies the incident-edge invertible matrices and the factorized local coefficient identity. $\square$

Theorem 18.41 (Gauge consistency). [gaugeConsistency](#) Not ready *Local gauges extracted at adjacent vertices are consistent across shared edges, yielding a global gauge family. On a shared edge, the two neighbouring local extractions match as one edge gauge:*



Theorem 18.42 (Fundamental Theorem for injective PEPS, conditional on bond-dimension equality). [fundamentalTheorem_PEPS_of_bondDim](#) Not ready *Given matching bond dimensions, two vertex-injective PEPS that generate the same state are gauge-equivalent. This conditional form isolates the bond-dimension obligation from the edge-centred consistency obligation of the global gauge construction.*

Theorem 18.43 (Fundamental Theorem for injective PEPS). [fundamentalTheorem_PEPS](#) Not ready *If two PEPS tensors on a simple graph are both vertex-injective and generate the same state, then they are gauge-equivalent (Theorem 2 of [MGRPG⁺ 18]).*

Definition 18.44 (Gauge equivalence modulo balanced edge scalars). [GaugeEquivModEdgeScalars](#) ✓ Stmt OK Two gauge families are equivalent modulo balanced edge scalars if there is a nonzero scalar c_e on each edge such that the corresponding endpoint actions rescale the oriented edge gauges, and the oriented product of those endpoint scalars is 1 at every vertex.

Theorem 18.45 (Balanced edge scalars preserve the local gauge action). [gaugeVertex_eq](#) ✓ Stmt OK *If two gauge families differ by a balanced edge-scalar reweighting, then they produce the same gauged tensor at every vertex.*

Proof. ✓ Proof OK In the defining sum for the gauged tensor, each incident edge contributes its endpoint scalar factor in addition to the original gauge entry. The product of these endpoint scalars is 1 by the balancing condition, so every summand is unchanged. $\square$

Theorem 18.46 (Gauge uniqueness modulo balanced edge scalars). [gauge_unique_mod_edge_scalars](#) Not ready *If two gauge families relate the same injective PEPS tensor to the same target tensor, then they define the same class modulo balanced edge scalars. This replaces the earlier global-scalar claim, which is false on a triangle with bond dimension 1.*

Remark 18.47. The local left inverse Ψ_v , the realization map $T \mapsto O_T$, the split at one incident edge, and the partition $V = \{u\} \sqcup (V \setminus \{u, v\}) \sqcup \{v\}$ isolate the local data needed for the edge-centered reduction of [\[MGRPG⁺18\]](#), §3. The middle-region blocked coefficient is constructed by the preceding definitions; the remaining local-gauge step is the comparison with the corresponding three-site injective chain. The uniqueness statement has now been repaired to a quotient by balanced edge scalars, but its proof still requires the same blocking construction together with the corresponding tensor-factor uniqueness statement at a single vertex.

Chapter 19

Symmetries and String Order

This chapter collects the symmetry-theoretic consequences of the MPS Fundamental Theorem. For injective tensors, physical on-site symmetries give virtual gauges, those gauges define projective representations on the bond space, and the same virtual data controls string-order phenomena. The algebraic theorem supplies the gauge mechanism, and this chapter develops its symmetry consequences.

19.1 Physical Symmetries Give Virtual Gauges

Corollary 19.1 (Physical symmetry implies virtual gauge). Not ready Assume a same-state reduction hypothesis for (d, D) . Let A be an injective MPS tensor and let $U \in \text{GL}_d(\mathbb{C})$ be a unitary acting on the physical index. Fix $n \geq 3$ and suppose the n -site translation-invariant state satisfies

$$U^{\otimes n} |\psi_n(A)\rangle = |\psi_n(A)\rangle.$$

Then there exist $V \in \text{GL}_D(\mathbb{C})$ and a scalar $\lambda \in \mathbb{C}^\times$ with $\lambda^n = 1$ such that

$$\sum_{j=0}^{d-1} U_{ij} A^j = \lambda V^{-1} A^i V$$

for every physical index i .

Proof. Define $B^i := \sum_j U_{ij} A^j$. Since U is invertible, $\{B^i\}$ spans $M_D(\mathbb{C})$, so B is injective. The symmetry condition means the two constant chains $(A, \dots, A)$ and $(B, \dots, B)$ of length n generate the same chain state. Theorem 17.53 then gives the required virtual gauge relation. $\square$

For periodic tensors, the physical-index rotation track is developed in Chapter 15: Definition 15.4 and Theorems 15.5, 15.6, 15.7, 15.8, 15.9, 15.10, and 15.11. We use these results here when deriving symmetry consequences, without restating their proofs.

Corollary 19.2 (Periodic symmetry yields $\mathbb{Z}$ -gauge equivalence). zGaugeEquiv_of_isIrreducibleForm_sameMPV_rotatePhysical ✓
A be in irreducible form II and let u be a matrix such that $B := A_u$ is also in irreducible form II and $\mathcal{V}(A) = \mathcal{V}(B)$. Then there exists $m > 0$ such that A and B are $\mathbb{Z}_m$ -gauge equivalent.

Proof. ✓ Proof OK Direct application of the periodic equal-case fundamental theorem (Theorem 15.25) to A and B . $\square$

19.2 Virtual Representation Theorem

With a full group of on-site symmetries (rather than just a single unitary), the resulting gauges determine a projective representation on the bond space.

Definition 19.3 (Twisted tensor). twistedTensor ✓ Stmt OK Given a group representation $U : G \rightarrow \text{GL}_d(\mathbb{C})$ on the physical index, the g -twisted tensor is defined, for each $i \in \{0, \dots, d-1\}$, by

$$\tilde{A}_g^i := \sum_{j=0}^{d-1} U(g)_{ij} A^j$$

Definition 19.4 (On-site symmetry). IsOnSiteSymmetric ✓ Stmt OK A tensor A is *on-site symmetric* under a group representation $U : G \rightarrow \text{GL}_d(\mathbb{C})$ if $\mathcal{V}(A) = \mathcal{V}(\tilde{A}_g)$ for every $g \in G$.

Lemma 19.5 (Functoriality of twisting). twistedTensor_mul ✓ Stmt OK The twisted tensor satisfies the composition law

$$\tilde{A}_{gh} = (\tilde{A}_h)_g,$$

i.e. twisting by gh is the same as first twisting by h and then by g .

Proof. ✓ Proof OK Expanding both sides using $U(gh) = U(g)U(h)$, then swapping the order of summation. □

Lemma 19.6 (Identity twist). twistedTensor_one ✓ Stmt OK Twisting by the identity group element is trivial: $\tilde{A}_1 = A$.

Proof. ✓ Proof OK Since $U(1) = I_d$, each component reduces to $\sum_j \delta_{ij} A^j = A^i$. □

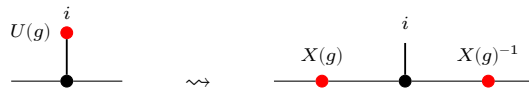
Lemma 19.7 (Symmetric twists are gauge equivalent). gaugeEquiv_twistedTensor_of_injective ✓ Stmt OK If A is injective and on-site symmetric under U , then for every $g \in G$ the twisted tensor $\tilde{A}_g$ is gauge equivalent to A .

Proof. ✓ Proof OK On-site symmetry gives $\mathcal{V}(A) = \mathcal{V}(\tilde{A}_g)$, and the single-block Fundamental Theorem turns this equality of MPV families into gauge equivalence. □

Remark 19.8 (Fundamental-Theorem route for injective symmetry). The injective virtual-representation theorem is not proved from string-order rigidity. The on-site symmetry condition first identifies A and $\tilde{A}_g$ as the same MPV family; the single-block Fundamental Theorem gives the gauge in Lemma 19.7. Gauge uniqueness and the twist composition law then assemble these gauges into Theorem 19.15. The string-order results later in this chapter use this virtual representation as an input.

Theorem 19.9 (Virtual symmetry equation). virtual_symmetry_eq ✓ Stmt OK If A is injective and on-site symmetric under U , then for every $g \in G$ there exist an invertible matrix $X(g) \in \text{GL}_D(\mathbb{C})$ and a nonzero scalar $\varphi(g) \in \mathbb{C}^\times$ (in fact $\varphi = 1$ in the single-block case) such that $\tilde{A}_g^i = \varphi(g) X(g) A^i X(g)^{-1}$ for all i .

Proof. ✓ Proof OK The physical action is transferred to the virtual level by the same local move that replaces the twisted tensor by a gauge-conjugated one:



Apply Lemma 19.7 to obtain $X(g)$, then set $\varphi(g) = 1$. $\square$

Lemma 19.10 (Gauge uniqueness up to scalar). [gauge_unique_up_to_scalar] [✓ Stmt OK] Let A be an injective tensor. If $X, Y \in \text{GL}_D(\mathbb{C})$ both satisfy $B^i = X A^i X^{-1} = Y A^i Y^{-1}$ for all i , then there exists a nonzero scalar $u \in \mathbb{C}^\times$ such that $Y = u \cdot X$.

Proof. [✓ Proof OK] The matrix $Z = Y^{-1}X$ commutes with every A^i . Since A is injective, $\{A^i\}$ spans $M_D(\mathbb{C})$, so Z lies in the centre of $M_D(\mathbb{C})$. By Lemma 17.10, Z is a scalar matrix. Since Z is invertible, that scalar is nonzero, and $Y = u \cdot X$. $\square$

Theorem 19.11 (Gauge uniqueness from equality of MPV families). [gauge_unique_up_to_scalar_of_sameMPV] [✓ Stmt OK] Let A be an injective tensor and assume that A and B generate the same MPV family. If $X, Y \in \text{GL}_D(\mathbb{C})$ both satisfy $B^i = X A^i X^{-1} = Y A^i Y^{-1}$ for all i , then there is a nonzero scalar $u \in \mathbb{C}^\times$ such that $Y = u \cdot X$.

Proof. [✓ Proof OK] This is exactly Lemma 19.10; the extra MPV hypothesis records the setting in which the two gauges are usually produced. $\square$

Definition 19.12 (Scalar 2-cochain). [ScalarCocycle] [✓ Stmt OK] A scalar 2-cochain on a group G is a function $\omega : G \times G \rightarrow \mathbb{C}^\times$.

Definition 19.13 (Projective representation). [ProjectiveRepresentation] [✓ Stmt OK] Given a scalar 2-cochain ω , a projective representation of G on $\mathbb{C}^D$ is a map $\rho : G \rightarrow \text{GL}_D(\mathbb{C})$ satisfying, for all $g, h \in G$,

$$\rho(g) \rho(h) = \omega(g, h) \rho(gh)$$

Lemma 19.14 (Associativity forces the cocycle condition). [cocycle_of_assoc] [✓ Stmt OK] If ρ is a projective representation with factor system ω and $D > 0$, then ω satisfies the 2-cocycle identity:

$$\omega(g, h) \omega(gh, k) = \omega(g, hk) \omega(h, k).$$

Proof. [✓ Proof OK] Expanding $\rho(g) (\rho(h) \rho(k))$ and $(\rho(g) \rho(h)) \rho(k)$ using the projective multiplication law and equating via matrix associativity. The hypothesis $D > 0$ allows cancellation of the invertible matrix factor. $\square$

Theorem 19.15 (Virtual representation theorem for injective MPS). [virtual_rep_of_symmetric_injective] [✓ Stmt OK] Let A be an injective MPS tensor that is on-site symmetric under a group representation $U : G \rightarrow \text{GL}_d(\mathbb{C})$. Then there exist:

1. a scalar 2-cochain $\omega : G \times G \rightarrow \mathbb{C}^\times$, and
2. a map $\rho : G \rightarrow \text{GL}_D(\mathbb{C})$ satisfying, for all $g, h \in G$,

$$\rho(g) \rho(h) = \omega(g, h) \rho(gh)$$

such that, defining the gauge matrices $X(g) := \rho(g^{-1})$, one has, for every $g \in G$ and $i \in \{0, \dots, d-1\}$,

$$\tilde{A}_g^i = X(g) A^i X(g)^{-1}$$

In tensor-network notation, for each fixed g this is the local gauge relation



with the black node denoting the tensor named in the surrounding formula and the two red side nodes acting on the virtual legs. In particular, ρ is a projective representation of G on the bond space $\mathbb{C}^D$.

Proof. ✓ Proof OK Since A is injective and on-site symmetric, each twisted tensor $\tilde{A}_g$ is gauge equivalent to A (Lemma 19.7). Choose $X(g) \in \text{GL}_D(\mathbb{C})$ with $\tilde{A}_g^i = X(g) A^i X(g)^{-1}$ for all i .

By functoriality (Lemma 19.5), the product $X(h)X(g)$ also intertwines A with $\tilde{A}_{gh}$. Since $X(gh)$ does the same, gauge uniqueness (Lemma 19.10) gives a nonzero scalar $\omega'(h, g)$ with $X(h)X(g) = \omega'(h, g)X(gh)$.

Defining $\rho(g) := X(g^{-1})$ converts this to the standard law: $\rho(g)\rho(h) = \omega(g, h)\rho(gh)$ where $\omega(g, h) := \omega'(h^{-1}, g^{-1})$. □

Corollary 19.16 (2-cocycle identity for the factor system). virtual_rep_of_symmetric_injective_cocycle ✓ Stmt OK If $D > 0$, the factor system ω from Theorem 19.15 satisfies the 2-cocycle identity for all $g, h, k \in G$:

$$\omega(g, h)\omega(gh, k) = \omega(g, hk)\omega(h, k)$$

Proof. ✓ Proof OK Theorem 19.15 gives ρ and ω . By Lemma 19.14, the cocycle identity follows from associativity of matrix multiplication and invertibility of the representation matrices. □

19.3 Cohomology Classes of Virtual Symmetry Data

Two cocycles may differ by a coboundary yet represent the same projective-representation class. The following definitions and results make this precise and establish the quotient $H^2(G, \mathbb{C}^\times)$.

Definition 19.17 (Coboundary). IsCoboundary ✓ Stmt OK A scalar 2-cocycle ω is a *coboundary* if there exists $\varphi: G \rightarrow \mathbb{C}^\times$ such that, for all $g, h \in G$,

$$\omega(g, h) = \varphi(g)\varphi(h)\varphi(gh)^{-1}$$

Definition 19.18 (Cohomologous cocycles). CohomologousTo ✓ Stmt OK Two cocycles ω_1, ω_2 are *cohomologous* (written $\omega_1 \sim \omega_2$) if there exists $\varphi: G \rightarrow \mathbb{C}^\times$ such that, for all $g, h \in G$,

$$\omega_1(g, h) = \varphi(g)\varphi(h)\varphi(gh)^{-1}\omega_2(g, h)$$

Theorem 19.19 (Cohomologous-to is an equivalence relation). equivalence ✓ Stmt OK The relation "cohomologous to" is reflexive, symmetric, and transitive on the set of scalar 2-cocycles.

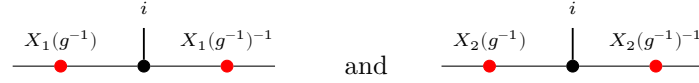
Proof. ✓ Proof OK Reflexivity: take $\varphi = 1$. Symmetry: replace φ by φ^{-1} . Transitivity: compose the witnesses φ and ψ pointwise. □

Lemma 19.20 (Coboundary iff cohomologous to the trivial cocycle). isCoboundary_iff_cohomologousTo_one ✓ Stmt OK A cocycle ω is a coboundary if and only if $\omega \sim \mathbf{1}$, where $\mathbf{1}(g, h) = 1$ for all g, h .

Proof. ✓ Proof OK Both directions follow by absorbing or introducing the factor $\mathbf{1}(g, h) = 1$ via $x \cdot 1 = x$. □

Theorem 19.21 (Gauge independence of the cocycle class). cohomologousTo_of_isInjective ✓ Stmt OK If the bond dimension satisfies $D \geq 1$ and two projective representations ρ_1, ρ_2 (with cocycles ω_1, ω_2) both arise as virtual representations of the same injective MPS tensor A under an on-site symmetry U , then $\omega_1 \sim \omega_2$.

Proof. ✓ Proof OK For each g , gauge uniqueness gives a nonzero scalar $f(g)$ with $X_2(g^{-1}) = f(g) X_1(g^{-1})$. Equivalently, for each fixed g the two local gauges



describe the same twisted tensor, so they differ by a scalar. Substituting into the projective law and cancelling $X_1(g \cdot h)$ yields $\omega_2(g, h) = f(g) f(h) f(gh)^{-1} \omega_1(g, h)$. $\square$

Definition 19.22 (Multiplicative 2-cocycle condition). IsCocycle ✓ Stmt OK A scalar 2-cochain $\omega: G \times G \rightarrow \mathbb{C}^\times$ is a *2-cocycle* if it satisfies, for all $g, h, k \in G$,

$$\omega(g, h)\omega(gh, k) = \omega(g, hk)\omega(h, k)$$

Definition 19.23 (Second cohomology quotient). H2 ✓ Stmt OK The second cohomology set is the quotient

$$H^2(G, \mathbb{C}^\times) := Z^2(G, \mathbb{C}^\times)/\sim,$$

where $\sim$ is the coboundary-equivalence relation.

Definition 19.24 (Projective equivalence). ProjectivelyEquivalent ✓ Stmt OK Two projective representations are *projectively equivalent* when their factor systems are cohomologous.

Theorem 19.25 (Projective equivalence and cohomologous cocycles). projRep_equiv_iff_cohomologous ✓ Stmt OK Let ρ_1, ρ_2 be projective representations with factor systems ω_1, ω_2 . Then

$$\rho_1 \sim_{\text{proj}} \rho_2 \iff \omega_1 \sim \omega_2.$$

Proof. ✓ Proof OK This is immediate from the definition of projective equivalence as cohomology-class equality for factor systems. $\square$

19.4 Permutation-Based Physical Symmetry

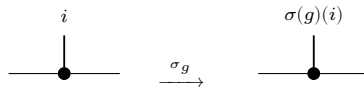
The linear-combination twist $\tilde{A}_g^i = \sum_j U(g)_{ij} A^j$ is the standard form used in the virtual representation theorem. An alternative formulation replaces the matrix action by a permutation of the physical index; this is useful when the symmetry acts by reordering basis states rather than by a general linear map.

Definition 19.26 (Permutation symmetry data). OnSiteSymmetry ✓ Stmt OK An *on-site symmetry datum* for a group G on a physical space of dimension d consists of a matrix-valued map $u: G \rightarrow M_d(\mathbb{C})$ and a physical-index action $\sigma: G \rightarrow \text{End}(\{0, \dots, d-1\})$.

Definition 19.27 (Permutation-twisted tensor). TwistedTensor ✓ Stmt OK Given on-site symmetry data (u, σ) , the *permutation-twisted tensor* at group element g is

$$(A^{\sigma_g})^i := A^{\sigma(g)(i)}.$$

In tensor-network notation the local tensor is unchanged and only the physical leg is relabelled:



Lemma 19.28 (Identity permutation twist). TwistedTensor_one ✓ Stmt OK If $\sigma(1) = \text{id}$, then $(A^{\sigma_1})^i = A^i$ for all i .

Proof. ✓ Proof OK Immediate from $\sigma(1)(i) = i$ for all i . □

Composition order. The linear-combination twist (Lemma 19.5) composes as $\widetilde{(\widetilde{A}_h)}_g = \widetilde{A}_{gh}$, applying h first then g . The permutation twist below uses the same left-action convention $\sigma(gh) = \sigma(g) \circ \sigma(h)$, so $(A^{\sigma_g})^{\sigma_h} = A^{\sigma_{gh}}$; the inner permutation $\sigma(h)$ acts first on the index.

Lemma 19.29 (Composition of permutation twists). TwistedTensor_mul ✓ Stmt OK If $\sigma(gh) = \sigma(g) \circ \sigma(h)$ for all $g, h \in G$, then

$$A^{\sigma_{gh}} = (A^{\sigma_g})^{\sigma_h}.$$

Proof. ✓ Proof OK Expanding both sides: $(A^{\sigma_{gh}})^i = A^{\sigma(g)(\sigma(h)(i))} = ((A^{\sigma_g})^{\sigma_h})^i$. Diagrammatically this is just two consecutive relabellings of the same local tensor:



□

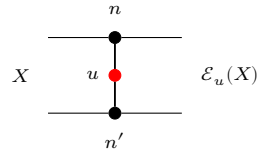
19.5 String Order and Local Symmetry Equivalence

The string order parameter [PGWS⁺08] detects whether an on-site unitary u acts as a local symmetry on the state generated by an injective MPS tensor.

Definition 19.30 (Twisted transfer map). twistedTransferMap ✓ Stmt OK Given an MPS tensor A and a physical-index matrix u , the *twisted transfer map* is

$$\mathcal{E}_u(X) := \sum_{n, n'} \langle n' | u | n \rangle A_n X A_{n'}^\dagger.$$

Diagrammatically one inserts the physical operator u on the doubled local transfer cell:



The upper and lower black nodes denote A_n and $A_{n'}^\dagger$, and the red node labelled u couples the physical legs.

Definition 19.31 (String order parameter). stringOrderParam ✓ Stmt OK The *string order parameter* for twist u and stationary state Λ at length L is

$$R_L(u) := \text{tr}(\Lambda \cdot \mathcal{E}_u^L(\mathbb{1})).$$

In tensor-network notation this is the length- L doubled chain with a copy of u inserted on each physical rung:



The boundary matrix Λ is contracted into the virtual boundary, the red nodes labelled u sit on the physical rungs, and the right boundary is traced.

Definition 19.32 (Boundary string order parameter). [stringOrderBoundaryParam](#) [✓ Stmt OK](#) For a boundary state Λ and boundary matrices $X, Y \in M_D(\mathbb{C})$, the *boundary string order parameter* is

$$R_L(u; X, Y) := \text{tr}(\Lambda X \mathcal{E}_u^L(Y)).$$

The ordinary string order parameter is the special case $X = Y = \mathbb{1}$.

Definition 19.33 (Local symmetry). [IsLocalSymmetry](#) [✓ Stmt OK](#) An MPS tensor A has *local symmetry* under a unitary u with respect to a boundary state Λ if there exist a unitary matrix V and a phase μ with $|\mu| = 1$ such that $V^\dagger \Lambda V = \Lambda$ and, for every physical index i ,

$$\sum_j u_{ij} A^j = \mu V A^i V^\dagger$$

Informally, this means $\|R_L(u)\| = \|R_L(\mathbb{1})\|$ for all L .

Definition 19.34 (String order). [HasStringOrder](#) [✓ Stmt OK](#) String order *exists* for u with respect to a boundary state Λ if there exist boundary matrices X, Y and a positive constant $c > 0$ such that $c \leq \|R_L(u; X, Y)\|$ for all L , where $R_L(u; X, Y)$ is the boundary-refined string-order parameter. Informally, this means the scalar string-order parameter $R_L(u)$ does not decay to zero.

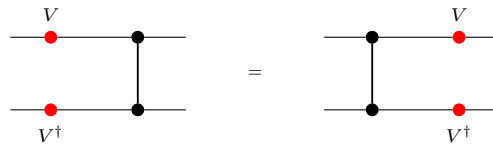
Lemma 19.35 (String order prevents decay to zero). [not_tendsto_zero_of_hasStringOrder](#) [✓ Stmt OK](#) If A has string order for u with respect to a boundary state Λ , then for some boundary matrices X, Y the sequence $R_L(u; X, Y)$ does not tend to 0 as $L \rightarrow \infty$.

Proof. [✓ Proof OK](#) A uniform lower bound $c \leq \|R_L(u; X, Y)\|$ is incompatible with convergence to 0. $\square$

Definition 19.36 (Local physical intertwining). [CondC1](#) [✓ Stmt OK](#) The local physical intertwining relation is $\sum_j u_{ij} A^j = V A^i V^\dagger$. Locally, the physical action of u can therefore be pushed to the virtual legs:



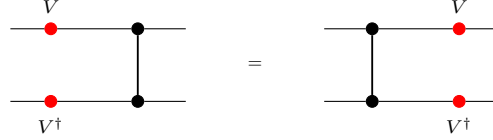
Definition 19.37 (Transfer-map covariance). [CondC2](#) [✓ Stmt OK](#) Transfer-map covariance is the identity that the transfer map commutes with virtual conjugation: $\mathcal{E}(V X V^\dagger) = V \mathcal{E}(X) V^\dagger$. In doubled-layer notation, the virtual operators slide through the local transfer cell:



Definition 19.38 (Commutation in the doubled transfer picture). CondC3 ✓ Stmt OK In the doubled transfer picture, one asks that the doubled transfer matrix commutes with $V \otimes \bar{V}$, expressed as $V \mathcal{E}_1(X) V^\dagger = \mathcal{E}_1(V X V^\dagger)$.

Theorem 19.39 (Transfer-map covariance and doubled commutation are equivalent). condC2_iff_condC3 ✓ Stmt OK *Transfer-map covariance and commutation in the doubled transfer picture are equivalent for any matrix V .*

Proof. ✓ Proof OK Both sides express the same local picture with opposite orientation:



The two formulas are literally the same identity read from opposite sides. □

Lemma 19.40 (Unitary Kraus mixing). unitary_kraus_mixing ✓ Stmt OK *If u is unitary, then replacing Kraus operators $\{A_i\}$ by their unitary mixtures $\{\sum_j u_{ij} A_j\}$ preserves the channel: $\sum_i (\sum_j u_{ij} A_j) Y (\sum_j u_{ij} A_j)^\dagger = \sum_i A_i Y A_i^\dagger$.*

Proof. ✓ Proof OK This is exactly the usual invariance of a Kraus decomposition under a unitary change of Kraus operators; see Theorem 5.71. □

Theorem 19.41 (Local physical intertwining implies transfer-map covariance). condC1_imp_condC2 ✓ Stmt OK *If V and u are unitary and the local physical intertwining relation holds, then the transfer map is covariant.*

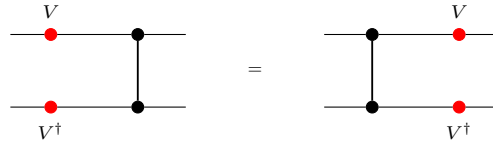
Proof. ✓ Proof OK Starting from



one inserts $V^\dagger V = \mathbb{1}$ so that each local transfer cell is written in terms of conjugated Kraus operators, then replaces those operators using the intertwining relation. After that, the physical u -boxes disappear by the unitary Kraus-mixing identity. □

Theorem 19.42 (Transfer-map covariance implies local physical intertwining). condC2_imp_condC1_of_injective ✓ Stmt OK *If an injective MPS, if V is unitary and the transfer map is covariant, then there exists a unitary u satisfying the local physical intertwining relation.*

Proof. ✓ Proof OK Set $B^i := V A^i V^\dagger$. The transfer-map covariance relation



implies that the two Kraus families B and A define the same transfer map. The missing step — supplied by Kraus freedom for equal channels — converts equality of transfer maps $\mathcal{E}_B = \mathcal{E}_A$ into a unitary Kraus mixing matrix u with $B^i = \sum_j u_{ij} A^j$. □

Definition 19.43 (Twisted companion tensor). [twistedMixedCompanion](#) ✓ Stmt OK For an MPS tensor A and a physical-index matrix u , define the companion tensor B_u by

$$B_u^n := \sum_{n'=0}^{d-1} \overline{u_{n'n}} A^{n'}.$$

Lemma 19.44 (Twisted transfer as a mixed transfer operator). [twistedTransferMap_eq_mixedTransfer](#) ✓ Stmt OK The twisted transfer map of A is the mixed transfer operator of the pair (A, B_u) :

$$\mathcal{E}_u = F_{A, B_u}.$$

Proof. ✓ Proof OK Expand both definitions and rearrange the double sum. □

Theorem 19.45 (Twisted-transfer eigenvalues have modulus at most one). [twistedTransfer_spectralRadius_le_one](#) ✓ Stmt OK Let A be an injective MPS tensor with $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, and let u satisfy $uu^\dagger = \mathbb{1}$. If

$$\mathcal{E}_u(X) = \lambda X$$

for some nonzero $X \in M_D(\mathbb{C})$, then $|\lambda| \leq 1$.

Proof. ✓ Proof OK By Lemma 19.44, the twisted transfer map is the mixed transfer operator F_{A, B_u} . Because B_u is obtained from A by unitary mixing of the Kraus operators, the transfer maps of A and B_u coincide, so the two adjoint channels share a common positive definite fixed point. Gauging both families by that fixed point puts them in a common TP gauge. Theorem 9.14 then applies to the mixed transfer operator in that gauge, and similarity preserves the eigenvalue λ . □

Theorem 19.46 (Modulus-one twisted eigenvalue implies gauge-phase equivalence). [twistedTransfer_modulus_one_implies_gaugePhaseEquivalence](#) Let A be an injective MPS tensor with $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, and let u satisfy $uu^\dagger = \mathbb{1}$. If

$$\mathcal{E}_u(X) = \lambda X$$

for some nonzero $X \in M_D(\mathbb{C})$ with $|\lambda| = 1$, then A is gauge-phase equivalent to the companion tensor B_u .

Proof. ✓ Proof OK By Lemma 19.44, the twisted transfer map is the mixed transfer operator F_{A, B_u} . Because B_u is obtained from A by unitary mixing of the Kraus operators, the transfer maps of A and B_u coincide, so the two adjoint channels share a common positive definite fixed point. Gauging both families by that fixed point puts them in a common TP gauge. In that gauge the eigenvalue equation becomes a modulus-one peripheral eigenvalue for an irreducible mixed transfer operator, so Theorem 9.28 yields gauge-phase equivalence. Undoing the common TP gauge gives the claim for A and B_u . □

Lemma 19.47 (Spectral radius < 1 forces boundary-string decay). [stringOrderBoundaryParam_tendsto_zero_of_spectralRadius_lt_one](#) $\rho(\mathcal{E}_u) < 1$, then for all boundary matrices $X, Y, \Lambda \in M_D(\mathbb{C})$ one has

$$R_L(u; X, Y) \rightarrow 0 \quad \text{as } L \rightarrow \infty.$$

Proof. ✓ Proof OK In finite dimension, $\rho(\mathcal{E}_u) < 1$ implies $\mathcal{E}_u^L \rightarrow 0$ in operator norm. Composing with the fixed trace pairing $Z \mapsto \text{tr}(\Lambda X Z)$ gives the stated scalar convergence. □

Theorem 19.48 (String order implies gauge-phase equivalence with the companion tensor). [gaugePhaseEquiv_twisted_of_hasStringOrder](#) ✓ Stmt OK Let A be injective, let $\Lambda \in M_D(\mathbb{C})$, and assume $uu^\dagger = \mathbb{1}$ and $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$. If string order exists for u with boundary parameter Λ , then A is gauge-phase equivalent to the companion tensor B_u .

Proof. ✓ Proof OK If $\rho(\mathcal{E}_u) < 1$, Lemma 19.47 forces every boundary string-order sequence to decay to zero, contradicting Definition 19.34. Hence $\rho(\mathcal{E}_u) \geq 1$. Theorem 19.45 shows that every eigenvalue has modulus at most 1, so some peripheral eigenvalue has modulus exactly 1. Applying Theorem 19.46 then gives the gauge-phase equivalence. $\square$

Theorem 19.49 (Gauge-phase equivalence with the companion tensor yields a virtual unitary).

virtualUnitary_of_gaugePhaseEquiv_twisted ✓ Stmt OK Let A be injective, assume $uu^\dagger = \mathbb{1}$ and $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$. If A is gauge-phase equivalent to the companion tensor B_u , then there exist a unitary V and a phase μ with $|\mu| = 1$ such that, for every physical index i ,

$$\sum_j u_{ij} A^j = \mu V A^i V^\dagger$$

Proof. ✓ Proof OK Start from an invertible intertwiner X between A and B_u . The positive matrix $Q := XX^\dagger$ is a positive fixed point of $\mathcal{E}_A$. By uniqueness of positive fixed points for an injective normalized tensor, Q is a positive scalar multiple of the identity. Rescaling X by the square root of that scalar produces a unitary V , and the original intertwining relation becomes the phased covariance equation above, with $|\mu| = 1$. $\square$

Theorem 19.50 (Virtual unitary gives a twisted-transfer eigenmatrix).

$\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$ and that, for every physical index i ,

twistedTransfer_eigen_of_virtualUnitary

✓ Stmt OK

$$\sum_j u_{ij} A^j = \mu V A^i V^\dagger$$

with V unitary. Then V is an eigenmatrix of the twisted transfer map:

$$\mathcal{E}_u(V) = \mu V.$$

Proof. ✓ Proof OK Expand $\mathcal{E}_u(V)$, insert the intertwining relation into each local term, and use $V^\dagger V = \mathbb{1}$ together with $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$. $\square$

Theorem 19.51 (Boundary-state invariance of the virtual unitary).

boundaryState_invariant_of_virtualUnitary

✓ Stmt OK

Let A be injective, assume $uu^\dagger = \mathbb{1}$, and let Λ be a positive definite boundary state with $\text{tr}(\Lambda) = 1$ and $\mathcal{E}_A^\dagger(\Lambda) = \Lambda$. If V is a unitary and μ a phase satisfying, for every physical index i ,

$$\sum_j u_{ij} A^j = \mu V A^i V^\dagger$$

then

$$V^\dagger \Lambda V = \Lambda.$$

Proof. ✓ Proof OK The covariance relation shows that $V^\dagger \Lambda V$ is another positive fixed point of the adjoint transfer map. Since Λ is the normalized positive fixed point, uniqueness forces $V^\dagger \Lambda V = \Lambda$. $\square$

Lemma 19.52 (Local symmetry implies string order).

hasStringOrder_of_localSymmetry

✓ Stmt OK

Assume $\text{tr}(\Lambda) = 1$ and $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$. If u is a local symmetry of A with respect to Λ , then string order exists for u .

Proof. ✓ Proof OK Choose the boundary matrices $X = V^\dagger$ and $Y = V$, where V is the unitary from the local-symmetry datum. By Theorem 19.50, one has $\mathcal{E}_u^L(V) = \mu^L V$. Therefore

$$R_L(u; V^\dagger, V) = \mu^L \text{tr}(\Lambda) = \mu^L,$$

whose norm is constantly 1. $\square$

Theorem 19.53 (Local symmetry iff a unitary peripheral eigenmatrix exists). [localSymmetry_iff_spectralRadius_one](#) ✓ Stmt OK

A be injective, and assume $uu^\dagger = \mathbb{1}$, $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, Λ is positive definite, $\text{tr}(\Lambda) = 1$, and $\mathcal{E}_A^\dagger(\Lambda) = \Lambda$. Then u is a local symmetry if and only if there exist a unitary $V \in M_D(\mathbb{C})$ and a phase μ with $|\mu| = 1$ such that

$$\mathcal{E}_u(V) = \mu V.$$

By Theorem 19.45, this witness form is equivalent to the peripheral-spectrum condition $\rho(\mathcal{E}_u) = 1$.

Proof. ✓ Proof OK The forward direction is Theorem 19.50. For the reverse direction, start from a unitary peripheral eigenmatrix. Theorem 19.46 turns it into a gauge-phase equivalence with the companion tensor; then Theorem 19.49 yields a virtual unitary and phase satisfying the local covariance relation, and Theorem 19.51 gives the boundary-state invariance needed to assemble the local-symmetry data. □

Theorem 19.54 (String order $\Leftrightarrow$ local symmetry). [stringOrder_iff_localSymmetry](#) ✓ Stmt OK *For an injective MPS, assume $uu^\dagger = \mathbb{1}$, $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, Λ is positive definite, $\text{tr}(\Lambda) = 1$, and $\mathcal{E}_A^\dagger(\Lambda) = \Lambda$. Then string order for u exists iff u is a local symmetry.*

Proof. ✓ Proof OK If string order exists, Theorem 19.48 gives a gauge-phase equivalence with the companion tensor. Theorem 19.49 extracts a virtual unitary and phase from that equivalence, and Theorem 19.51 shows that the same unitary preserves the boundary state, so one obtains local symmetry. Conversely, Lemma 19.52 turns local symmetry into a non-decaying boundary string-order sequence. □

Theorem 19.55 (Virtual unitary from string order). [virtualUnitary_of_stringOrder](#) ✓ Stmt OK *For an injective MPS, let $\Lambda \in M_D(\mathbb{C})$, and assume $uu^\dagger = \mathbb{1}$ and $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$. If string order exists for u with boundary parameter Λ , then there exists a virtual unitary V and a unit-modulus scalar μ such that $\sum_j u_{ij} A^j = \mu V A^i V^\dagger$ for all i . Diagrammatically this is the same local intertwining relation, but only up to an overall unit scalar on the virtual side:*

Proof. ✓ Proof OK String order first gives gauge-phase equivalence with the companion tensor by Theorem 19.48. Theorem 19.49 then normalizes the resulting intertwiner to a unitary and produces the phase μ . □

19.6 SPT Phase Classification

Two injective symmetric tensors belong to the same symmetry-protected topological phase when their virtual representation cocycles are cohomologous [CGW11].

Definition 19.56 (Same SPT phase). [IsSameSPTPhase](#) ✓ Stmt OK Injective tensors A and B , both symmetric under an on-site representation $U: G \rightarrow \text{GL}_d(\mathbb{C})$, are in the *same SPT phase* if there exist virtual cocycles ω_A, ω_B with corresponding projective representations that intertwine the respective twisted tensors and satisfy $\omega_A \sim \omega_B$ (cohomologous).

Theorem 19.57 (String order holds for injective symmetric MPS). [hasStringOrder_of_symmetric_injective](#) ✓ Stmt OK *Let A be an injective tensor, symmetric under a unitary representation U . Assume $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, Λ is positive definite, $\text{tr}(\Lambda) = 1$, and $\mathcal{E}_A^\dagger(\Lambda) = \Lambda$. Then string order exists for every group element g .*

Proof. ✓ Proof OK The virtual representation theorem produces $\rho(g^{-1}) \in \text{GL}_D(\mathbb{C})$ satisfying $\sum_j U(g)_{ij} A^j = \rho(g^{-1}) A^i \rho(g^{-1})^{-1}$. A direct computation using $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$ shows that $\mathcal{E}_{U(g)}(\rho(g^{-1})) = \rho(g^{-1})$, so $\rho(g^{-1})$ is a nonzero eigenvector with eigenvalue 1. Theorem 19.46 upgrades this to gauge-phase equivalence with the companion tensor, and Theorem 19.49 extracts a virtual unitary and phase. By Theorem 19.51, this unitary preserves the boundary state, so Lemma 19.52 gives string order. $\square$

Theorem 19.58 (String order is an SPT invariant). stringOrder_invariant_of_samePhase ✓ Stmt OK *If A and B are injective, both on-site symmetric under a unitary representation U , and if $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, $\mathcal{E}_B(\mathbb{1}) = \mathbb{1}$, Λ_A and Λ_B are positive definite, $\text{tr}(\Lambda_A) = 1$, $\text{tr}(\Lambda_B) = 1$, $\mathcal{E}_A^\dagger(\Lambda_A) = \Lambda_A$, and $\mathcal{E}_B^\dagger(\Lambda_B) = \Lambda_B$, then for every $g \in G$, string order exists for A with twist $U(g)$ if and only if it exists for B . In particular, since Definition 19.56 implies on-site symmetry for both tensors, this theorem applies to tensors in the same SPT phase.*

Proof. ✓ Proof OK Both directions follow from Theorem 19.57, which shows that string order holds universally for injective symmetric tensors with canonical normalisations. $\square$

Chapter 20

Parent Hamiltonians

This chapter introduces the local ground-space construction and the parent interaction for translation-invariant periodic MPS. We follow the standard setup from [CPGSV21, PGVWC07], with the N -site Hilbert space modelled as coefficient functions

$$\{0, \dots, d-1\}^N \rightarrow \mathbb{C},$$

rather than as an explicit tensor product.

Definition 20.1 (N -site coefficient space). NSiteSpace ✓ Stmt OK For local dimension d and chain length N , the ambient coefficient space is

$$\mathcal{H}_N^{(d)} := \{0, \dots, d-1\}^N \rightarrow \mathbb{C}.$$

This is the concrete model used throughout the chapter for periodic-chain states.

20.1 Local ground space $G_L(A)$

Fix an MPS tensor $A = \{A^i\}_{i=0}^{d-1}$ with $A^i \in M_D(\mathbb{C})$ and a block length $L \geq 1$. For a word $w = (i_1, \dots, i_L)$, write $A^w := A^{i_1} \dots A^{i_L}$. The local ground-space map is

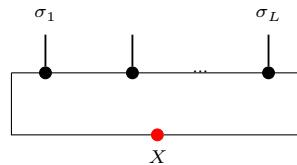
$$\Gamma_L : M_D(\mathbb{C}) \rightarrow (\{0, \dots, d-1\}^L \rightarrow \mathbb{C}), \quad \Gamma_L(X)(\sigma) := \text{tr}(A^\sigma X),$$

where A^σ abbreviates $A^{\sigma(0)} \dots A^{\sigma(L-1)}$.

Definition 20.2 (Ground-space map). groundSpaceMap ✓ Stmt OK The map Γ_L above is a $\mathbb{C}$ -linear map

$$\Gamma_L : M_D(\mathbb{C}) \rightarrow (\{0, \dots, d-1\}^L \rightarrow \mathbb{C}).$$

In tensor-network notation,



the upper chain denotes the word tensor A^σ with open physical indices $\sigma_1, \dots, \sigma_L$, and the lower red node denotes the boundary matrix X inserted on the closing virtual bond before taking the trace. The chain length L is fixed by the surrounding formula, not by an extra label inside the diagram.

Lemma 20.3 (Evaluation formula). groundSpaceMap_apply ✓ Stmt OK For every boundary matrix $X \in M_D(\mathbb{C})$ and every length- L word σ ,

$$\Gamma_L(X)(\sigma) = \text{tr}(A^\sigma X).$$

Definition 20.4 (Local ground space). groundSpace ✓ Stmt OK The *local ground space* is the image

$$G_L(A) := \text{im}(\Gamma_L) \subseteq (\{0, \dots, d-1\}^L \rightarrow \mathbb{C}).$$

Lemma 20.5 (Dimension bound). groundSpace_finrank_le ✓ Stmt OK For every L ,

$$\dim G_L(A) \leq D^2.$$

Proof. ✓ Proof OK Since $G_L(A)$ is the range of a linear map from $M_D(\mathbb{C})$,

$$\dim G_L(A) \leq \dim M_D(\mathbb{C}) = D^2.$$

□

Lemma 20.6 (Ambient-space dimension). nSiteSpace_finrank ✓ Stmt OK The coefficient-function model satisfies

$$\dim(\{0, \dots, d-1\}^L \rightarrow \mathbb{C}) = d^L.$$

Lemma 20.7 (Nontriviality criterion). groundSpace_ne_top ✓ Stmt OK If $d^L > D^2$, then $G_L(A)$ is a proper subspace of the local Hilbert space:

$$G_L(A) \neq (\{0, \dots, d-1\}^L \rightarrow \mathbb{C}).$$

Proof. ✓ Proof OK If $G_L(A)$ were the whole space, its dimension would be d^L by Lemma 20.6; Lemma 20.5 gives the contradiction $d^L \leq D^2$. □

20.2 Parent interaction and chain Hamiltonian

The parent interaction at range L is the orthogonal projector onto $G_L(A)^\perp$ in the L -site Euclidean space.

Definition 20.8 (Euclidean transport of the local ground space). groundSpaceES ✓ Stmt OK Via the canonical identification between coefficient functions and the Euclidean space $\ell^2(\{0, \dots, d-1\}^L)$, one transports the local ground space to a Hilbert-space subspace

$$G_L^{\text{ES}}(A) \subseteq \ell^2(\{0, \dots, d-1\}^L).$$

Lemma 20.9 (Membership in the transported local ground space). mem_groundSpaceES_iff ✓ Stmt OK A Euclidean vector lies in $G_L^{\text{ES}}(A)$ exactly when transporting it back through the canonical coefficient-space equivalence gives an element of the original local ground space $G_L(A)$.

Proof. ✓ Proof OK This is just the definition of $G_L^{\text{ES}}(A)$ as the image of $G_L(A)$ under the inverse of the canonical equivalence. □

Definition 20.10 (Parent interaction). parentInteraction ✓ Stmt OK The parent interaction at range L is the orthogonal projector

$$h_L(A) := \Pi_{G_L(A)^\perp},$$

on the local L -site space.

Lemma 20.11 (Parent interaction kills ground-space elements). parentInteraction_apply_mem_groundSpace ✓ Stmt OK For any $v \in G_L(A)$, we have $h_L(A)v = 0$.

Proof. ✓ Proof OK Since $h_L(A)$ is the orthogonal projector onto $G_L(A)^\perp$, it annihilates every element of $G_L(A)$. □

Definition 20.12 (Periodic window extraction). extractWindow ✓ Stmt OK For a configuration $\sigma \in \{0, \dots, d-1\}^N$ on the periodic chain and a starting site i , the extracted word

$$\text{extractWindow}_{L,i}(\sigma) \in \{0, \dots, d-1\}^L$$

reads the entries of σ on the cyclic interval $i, i+1, \dots, i+L-1$ modulo N .

Definition 20.13 (Periodic window replacement). replaceWindow ✓ Stmt OK If $L \leq N$, the configuration

$$\text{replaceWindow}_{L,i}(\sigma, \tau)$$

is obtained from σ by replacing the cyclic window starting at i by the length- L word τ .

Lemma 20.14 (Extracting a replaced window). extractWindow_replaceWindow ✓ Stmt OK If $L \leq N$, then extracting the same cyclic window after replacement recovers the inserted word:

$$\text{extractWindow}_{L,i}(\text{replaceWindow}_{L,i}(\sigma, \tau)) = \tau.$$

Lemma 20.15 (Replacing an extracted window). replaceWindow_extractWindow ✓ Stmt OK If $L \leq N$, then replacing the cyclic window at i in σ by the values extracted from that same σ leaves σ unchanged:

$$\text{replaceWindow}_{L,i}(\sigma, \text{extractWindow}_{L,i}(\sigma)) = \sigma.$$

Proof. ✓ Proof OK Case-split on whether the offset $(k-i+N) \bmod N$ lies in $[0, L)$: inside the window the value comes from extractWindow , which returns σ at the same position; outside the window both sides already agree with σ . □

For a periodic chain of length N , the parent Hamiltonian is the sum of translated copies of the local interaction.

Definition 20.16 (Translated local term). localTerm ✓ Stmt OK For each site index $i \in \{0, \dots, N-1\}$, the translated local term h_i embeds the parent interaction into the full N -site space at position i (with periodic boundary conditions). Thus h_i acts on the cyclic interval $[i, i+L-1]$ and as the identity outside that interval.

Definition 20.17 (Parent Hamiltonian). parentHamiltonian ✓ Stmt OK The chain Hamiltonian is

$$H_N(A, L) := \sum_{i=0}^{N-1} h_i.$$

Definition 20.18 (Frustration-free condition). `IsFrustrationFree` ✓ Stmt OK A state ψ is frustration-free if each local term annihilates it:

$$\forall i \in \{0, \dots, N-1\}, \quad h_i \psi = 0.$$

Lemma 20.19 (MPV window membership). `mpv_window_mem_groundSpace` ✓ Stmt OK For any window of L consecutive sites starting at position i in an N -site periodic chain ($L \leq N$), the restriction of the MPV state to that window lies in $G_L(A)$.

Proof. ✓ Proof OK The witness is the product of A -matrices on the complement sites. Trace cyclicity rotates the full N -site product so that window indices come first, matching the ground-space map definition. □

Lemma 20.20 (Local term annihilates the MPV). `localTerm_annihilates_mpv` ✓ Stmt OK For each site i , $h_i V^{(N)}(A) = 0$.

Proof. ✓ Proof OK Combines MPV window membership with the fact that the parent interaction kills ground-space elements. □

Lemma 20.21 (Parent Hamiltonian annihilates the MPV). `parentHamiltonian_annihilates` ✓ Stmt OK For every N , the MPV state satisfies

$$H_N(A, L) V^{(N)}(A) = 0.$$

Proof. ✓ Proof OK Sum of zeros, since each local term annihilates the MPV. □

Lemma 20.22 (Frustration-freeness of the MPV). `parentHamiltonian_frustrationFree` ✓ Stmt OK The MPV state is frustration-free for the parent model.

Proof. ✓ Proof OK Immediate from Lemma 20.20. □

20.3 Intersection property and unique ground state

This section develops the intersection property of local ground spaces for injective MPS and the resulting uniqueness of the ground state on the periodic chain.

20.3.1 Restriction maps

Given a state ψ on $L + 1$ sites, we define two restriction maps.

Remark 20.23 (Word and restriction auxiliary formulas). `ofFn_snoc` `evalWord_ofFn_snoc` `evalWord_ofFn_cons` `restrictLast1` `restrictFirst1` `restrictLast_apply` `restrictFirst_apply` ✓ Stmt OK Appending a last letter to a word corresponds to right multiplication by the corresponding tensor letter, while prepending a first letter corresponds to left multiplication. The restriction operators are the associated linear maps on coefficient spaces, together with their pointwise formulas.

Definition 20.24 (Left restriction). `restrictLast` ✓ Stmt OK Fix the last physical index j :

$$\sigma \mapsto \psi(\sigma_1, \dots, \sigma_L, j).$$

Definition 20.25 (Right restriction). `restrictFirst` ✓ Stmt OK Fix the first physical index i :

$$\sigma \mapsto \psi(i, \sigma_1, \dots, \sigma_L).$$

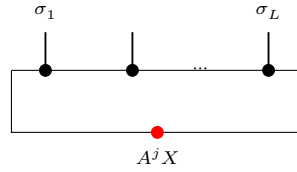
Definition 20.26 (Left ground membership). InLeftGround ✓ Stmt OK A state ψ on $L + 1$ sites satisfies the left ground condition if for every j , the state obtained by fixing the last index to j lies in $G_L(A)$.

Definition 20.27 (Right ground membership). InRightGround ✓ Stmt OK A state ψ on $L + 1$ sites satisfies the right ground condition if for every i , the state obtained by fixing the first index to i lies in $G_L(A)$.

20.3.2 Forward direction: ground space restricts

Lemma 20.28 (Left restriction). groundSpace_inLeftGround ✓ Stmt OK If $\psi \in G_{L+1}(A)$, i.e. $\psi(\sigma) = \text{tr}(A^\sigma X)$ for some X , then for each j , the state obtained by fixing the last index to j lies in $G_L(A)$ with witness $A^j X$.

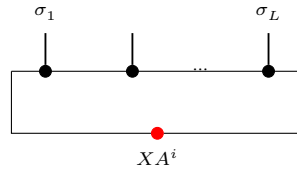
Proof. ✓ Proof OK The restriction simply absorbs the last tensor into the boundary matrix:



Direct computation: $\psi(\sigma_1, \dots, \sigma_L, j) = \text{tr}(A^{\sigma_1} \dots A^{\sigma_L} \cdot A^j \cdot X) = \Gamma_L(A^j X)(\sigma)$. □

Lemma 20.29 (Right restriction). groundSpace_inRightGround ✓ Stmt OK If $\psi \in G_{L+1}(A)$ with $\psi(\sigma) = \text{tr}(A^\sigma X)$, then for each i , the state obtained by fixing the first index to i lies in $G_L(A)$ with witness $X A^i$.

Proof. ✓ Proof OK On the right restriction one first rotates the trace and then reads the resulting boundary matrix on the shortened window:



By trace cyclicity: $\psi(i, \sigma_1, \dots, \sigma_L) = \text{tr}(A^i A^\sigma X) = \text{tr}(A^\sigma \cdot X A^i) = \Gamma_L(X A^i)(\sigma)$. □

20.3.3 Injectivity of the ground-space map

Theorem 20.30 (Ground-space map injectivity). groundSpaceMap_injective ✓ Stmt OK If A is injective and $L \geq 1$, then Γ_L is injective.

Proof. ✓ Proof OK If $\Gamma_L(X) = 0$ then $\text{tr}(A^\sigma X) = 0$ for every length- L word σ . Since A is injective, the products $\{A^\sigma\}_\sigma$ span $M_D(\mathbb{C})$, so the trace pairing gives $X = 0$. □

Theorem 20.31 (Ground-space map injectivity from word span). groundSpaceMap_injective_of_wordSpan_eq_top ✓ Stmt OK If the length- L products $\{A^\sigma : |\sigma| = L\}$ span $M_D(\mathbb{C})$, then Γ_L is injective.

Proof. ✓ Proof OK If $\Gamma_L(X) = 0$, then $\text{tr}(A^\sigma X) = 0$ for every word σ of length L . The spanning hypothesis turns this into $\text{tr}(MX) = 0$ for every $M \in M_D(\mathbb{C})$, and the trace pairing gives $X = 0$. □

Lemma 20.32 (Ground-space map injectivity for block-injective tensors). *A is L_0 -block-injective, then Γ_{L_0} is injective.*

[groundSpaceMap_injective_of_isNBlkInjective](#) ✓ S

Proof. ✓ Proof OK By definition, L_0 -block injectivity says precisely that the length- L_0 products span $M_D(\mathbb{C})$. Apply Theorem 20.31. □

Theorem 20.33 (Ground-space dimension). *If A is injective and $L \geq 1$, then $\dim G_L(A) = D^2$.*

[groundSpace_finrank_eq](#)

✓ Stmt OK

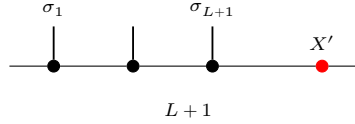
Proof. ✓ Proof OK The upper bound $\dim G_L(A) \leq D^2$ is Lemma 20.5. Injectivity of Γ_L gives $\dim G_L(A) = \dim M_D(\mathbb{C}) = D^2$. □

20.3.4 The intersection property

Theorem 20.34 (Intersection property). *If A is injective and $L \geq 2$, then a state ψ on $L + 1$ sites satisfying both the left and right ground conditions lies in $G_{L+1}(A)$:*

$$\psi \in G_L^{\text{left}} \cap G_L^{\text{right}} \implies \psi \in G_{L+1}(A).$$

Proof. ✓ Proof OK The “invert-and-regrow” argument compares the left and right restrictions on their common $(L - 1)$ -site overlap and then reconstructs the full $(L + 1)$ -site word from a single boundary matrix:



1. From the right ground condition, for each physical index i there exists a unique $Y_i \in M_D(\mathbb{C})$ (by injectivity of Γ_L) such that $\psi(i, \sigma) = \text{tr}(A^\sigma Y_i)$.
2. Similarly, from the left ground condition, for each j there exists a unique Z_j with $\psi(\sigma, j) = \text{tr}(A^\sigma Z_j)$.
3. Matching on the $(L - 1)$ -site overlap and using nondegeneracy of the trace pairing: $A^j Y_i = Z_j A^i$ for all i, j .
4. Since A is injective, the matrices $\{A^j\}$ span $M_D(\mathbb{C})$ (Definition 2.23), so $\mathbb{1} = \sum_j c_j A^j$ for some coefficients $c_j \in \mathbb{C}$. Then $Y_i = \mathbb{1} \cdot Y_i = \sum_j c_j A^j Y_i = \sum_j c_j Z_j A^i = X' A^i$, where $X' := \sum_j c_j Z_j$.
5. By trace cyclicity, $\psi(\sigma) = \text{tr}(A^\sigma X')$, so $\psi \in G_{L+1}(A)$. □

Theorem 20.35 (Intersection characterization). *For injective A and $L \geq 2$: $\psi \in G_{L+1}(A) \iff \psi \in G_L^{\text{left}} \cap G_L^{\text{right}}$.*

[groundSpace_iff_left_right](#)

✓ Stmt OK

Proof. ✓ Proof OK Immediate from the two forward-direction lemmas and Theorem 20.34. □

20.3.5 Unique ground state

Contiguous and cyclic window operators

Remark 20.36 (Contiguous window operators). [contiguousCfg](#) [contiguousRestrictl](#) [contiguousRestrictl_apply](#) [contiguousCfg_zero_full](#) [contiguousRestrictl_restrictLast](#) [contiguousRestrictl_restrictFirst](#) [tailRestrictl_contiguousRestrictl](#) [S](#) On a non-wrapping interval $[s, s + M - 1] \subseteq \{0, \dots, N - 1\}$, one can insert a length- M word into that interval and keep the complementary sites fixed. The associated restriction map evaluates an N -site state on those configurations. The auxiliary lemmas state the full-window case $s = 0$, $M = N$, the compatibility of this restriction with deleting the last or first site, and the fact that fixing the first K entries of a contiguous $(K + L)$ -window leaves the contiguous L -window beginning at $s + K$.

Remark 20.37 (Cyclic window operators). [cyclicCfg](#) [cyclicRestrictl](#) [cyclicRestrictl_apply](#) [eq_cyclic_site_of_offset_eq](#) [cyclicRestrictl_restrictLast](#) [cyclicCfg_eq_contiguousCfg](#) [cyclicRestrictl_eq_contiguousRestrictl](#) [Stmt OK](#) On the periodic chain, a cyclic window starting at i reads the sites $i, i + 1, \dots, i + L - 1$ modulo N . Deleting the final site of a cyclic $(L + 1)$ -window reads the value at cyclic offset L in the outside configuration and gives the L -window starting at the same site. When this window does not wrap around the ring, the cyclic construction agrees with the contiguous one, and so do the two restriction maps.

Lemma 20.38 (Iterated contiguous-window intersection). [contiguous_mem_groundSpace](#) [Stmt OK](#) If an N -site state satisfies the L -site ground condition on every non-wrapping contiguous window (with $L \geq 2$), then it lies in $G_N(A)$.

Proof. [Stmt OK](#) The induction repeatedly merges two neighbouring admissible windows $[i, i + L - 1]$ and $[i + 1, i + L]$ into the larger window $[i, i + L]$ by applying Theorem 20.34 to their common overlap $[i + 1, i + L - 1]$. Strong induction on window size, combining adjacent windows via Theorem 20.34 at each step. $\square$

Definition 20.39 (Suffix restriction). [tailRestrictl](#) [Stmt OK](#) Fix a prefix $u : \{0, \dots, K - 1\} \rightarrow \{0, \dots, d - 1\}$. The suffix restriction of an $(K + L)$ -site state ψ is the L -site state given by

$$(\text{tailRestrict}_u \psi)(\sigma) = \psi(u, \sigma).$$

Definition 20.40 (Suffix ground membership). [InTailGround](#) [Stmt OK](#) A state ψ on $K + L$ sites lies in the suffix ground condition if every fixed prefix u gives a suffix state in $G_L(A)$.

Lemma 20.41 (Ground-space suffix restriction). [restrictLast_groundSpaceMap](#) [tailRestrictl_groundSpaceMap](#) [groundSpace_inTail](#) $\psi \in G_{K+L}(A)$, then every fixed-prefix suffix restriction of ψ lies in $G_L(A)$.

Proof. [Stmt OK](#) Write $\psi(\rho) = \text{tr}(A^\rho X)$ for a boundary matrix X . Fixing a prefix u gives

$$\psi(u, \sigma) = \text{tr}(A^u A^\sigma X) = \text{tr}(A^\sigma X A^u),$$

so the suffix restriction again has the form of a ground-space vector. $\square$

Lemma 20.42 (Left-tail compatibility). [exists_left_tail_compatibility](#) [Stmt OK](#) Assume ψ on $K + L_0 + 1$ sites satisfies the left ground condition on the first $K + L_0$ sites and the suffix ground condition on every fixed prefix of length K . Then there exist matrices $(Z_j)_j$ and $(Y_u)_u$ such that

$$\psi(\sigma, j) = \text{tr}(A^\sigma Z_j), \quad \psi(u, \tau) = \text{tr}(A^\tau Y_u),$$

and for all j and u ,

$$Z_j A^u = A_j Y_u.$$

Proof. ✓ Proof OK Choose Z_j from the long left-window condition and Y_u from the suffix ground condition. Restricting the suffix state at its last site gives the boundary matrix $A_j Y_u$, while first fixing the last site and then taking the suffix gives $Z_j A^u$. The two restrictions are the same because appending the fixed prefix and then the last letter gives the same configuration as appending the last letter to the suffix and then placing the prefix in front. Injectivity of Γ_{L_0} , obtained from L_0 -block injectivity, yields $Z_j A^u = A_j Y_u$. $\square$

Theorem 20.43 (Common right factor from word compatibility). *A is L_0 -block injective with $L_0 > 0$. If matrices $(Z_j)_j$ satisfy*

exists_right_factor_of_evalWord_compatibility

✓ Stmt OK

$$Z_j A^\sigma = A_j Y_\sigma$$

for every word σ of length K , then there exists a matrix X such that $Z_j = A_j X$ for all j .

Proof. ✓ Proof OK For $K = 1$, the hypothesis gives $Z_j A_i = A_j Y_i$ for every letter i . Every blocked word of length L_0 is nonempty, so this identity extends to the coefficients of the blocked tensor $A^{[L_0]}$. Since $A^{[L_0]}$ is injective, the identity matrix is a linear combination of those blocked coefficients, and substituting this decomposition yields $Z_j = A_j X$ for a common matrix X . For larger K , strip the first letter: for each i , the matrices $Z_j A_i$ satisfy the same compatibility hypothesis with words of length $K - 1$, so the induction hypothesis reduces the problem to the case $K = 1$. $\square$

Theorem 20.44 (Open-chain right extension at the injectivity length).

groundSpace_extend_right_of_isNBkInjective

✓ Stmt OK

A is L_0 -block injective with $L_0 > 0$. If an element ψ on $K + L_0 + 1$ sites satisfies the left ground condition on the first $K + L_0$ sites and the suffix ground condition on every prefix of length K , then $\psi \in G_{K+L_0+1}(A)$.

Proof. ✓ Proof OK Lemma 20.42 produces matrices $(Z_j)_j$ and $(Y_u)_u$ with $Z_j A^u = A_j Y_u$ for every prefix word u of length K . Theorem 20.43 gives a common right factor X such that $Z_j = A_j X$ for all j . Therefore each last-site restriction of ψ agrees with the corresponding last-site restriction of the ground-space vector determined by X . Since every configuration is obtained by adjoining its final letter to its initial $(K + L_0)$ -tuple, the two states coincide, so ψ lies in $G_{K+L_0+1}(A)$. $\square$

Theorem 20.45 (Open-chain normal range reduction).

contiguous_mem_groundSpace_of_isNBkInjective

✓ Stmt OK

Let A be L_0 -block injective with $D \geq 1$ and $L_0 > 0$. If an N -site state satisfies the ground-space constraint on every non-wrapping contiguous window of length $L_0 + 1$, for every choice τ of the complementary outside configuration, with $N \geq L_0 + 1$, then it lies in $G_N(A)$.

Proof. ✓ Proof OK Induct on the extra length beyond $L_0 + 1$. The induction hypothesis gives the long left-window ground condition after deleting the last site, while Definition 20.39 and the contiguous-window transport above identify every fixed-prefix suffix with one of the assumed length- $L_0 + 1$ windows. Theorem 20.44 then regrows the rightmost site. $\square$

Lemma 20.46 (Cyclic-window range antitonicity).

chainGroundSpace_le_chainGroundSpace_succ

chainGroundSpace_le_chainGroundSpace_succ

a nonempty periodic chain, longer cyclic-window constraints imply all shorter cyclic-window constraints: if $L' \leq L \leq N$, then $\mathcal{G}_{N,L}(A) \subseteq \mathcal{G}_{N,L'}(A)$.

Proof. ✓ Proof OK Fix a cyclic window of length $L + 1$ and restrict its last site. The resulting length- L vector is again in the ground space, so one-site peeling gives the inclusion for consecutive ranges. Iteration gives the general antitonicity. $\square$

Theorem 20.47 (Cyclic-window normal open-chain reduction).

[chainGroundSpace_le_groundSpace_of_isNBlkInjective](#)

✓ Stmt OK

A be L_0 -block injective with $L_0 > 0$ and $D \geq 1$. If an N -site periodic-chain state satisfies all cyclic ground-space constraints at some range $L_0 + 1 \leq L \leq N$, then it lies in the open-chain ground space $G_N(A)$.

Proof. ✓ **Proof OK** Lemma 20.46 reduces the cyclic hypotheses to range $L_0 + 1$. On every non-wrapping interval, the cyclic and contiguous restriction maps agree, so Theorem 20.45 applies. $\square$

Lemma 20.48 (Wrapped-window one-sided compatibility).

[wrapping_window_compatibility_of_isNBlkInjective](#)

✓ Stmt OK

A is L_0 -block injective with $L_0 > 0$ and $M \geq L_0$. If the reduced wrapped window at the last site has boundary matrices Y_τ , then for every physical letter j one has

$$C_\tau^+ A^j X = Y_\tau A^j,$$

where C_τ^+ is the complementary word seen by that wrapped window.

Proof. ✓ **Proof OK** The trace identity for the wrapped window is tested against all words of length L_0 . Injectivity of Γ_{L_0} , which follows from L_0 -block injectivity, strips the test word and gives the displayed matrix identity. $\square$

Lemma 20.49 (Mirror wrapped-window one-sided compatibility).

[wrapping_window_mirror_compatibility_of_isNBlkInjective](#)

A is L_0 -block injective with $L_0 > 0$ and $M \geq L_0$. If the opposite reduced wrapped window has boundary matrices Y_τ , then

$$X A^j C_\tau^- = A^j Y_\tau$$

for every physical letter j , where C_τ^- is the complementary word seen from the opposite boundary-crossing position.

Proof. ✓ **Proof OK** Factor the cyclic word at the opposite wrapped position, rotate the trace, and use injectivity of Γ_{L_0} to remove the length- L_0 test word. $\square$

Theorem 20.50 (Reduced wrapped-boundary compatibilities).

[chainGroundSpace_wrapped_boundary_compatibilities_of_isNBlkInjective](#)

A be L_0 -block injective with $L_0 > 0$ and $D \geq 1$. Suppose $\psi \in \mathcal{G}_{N,L}(A)$, $N \geq 2$, $L_0 < L \leq N$, and $\psi = \Gamma_N(X)$ for a boundary matrix X . Then there are two families of boundary matrices Y_τ^+ and Y_τ^- , indexed by outside configurations τ , such that for every physical letter j ,

$$C_\tau^+ A^j X = Y_\tau^+ A^j, \quad X A^j C_\tau^- = A^j Y_\tau^-,$$

where C_τ^+ and C_τ^- are the complementary words exposed by the two opposite wrapped windows of reduced length $L_0 + 1$.

Proof. ✓ **Proof OK** First use Lemma 20.46 to reduce the cyclic constraints from range L to range $L_0 + 1$. Then apply the wrapped window and mirror wrapped window compatibility lemmas at the two boundary-crossing positions to obtain the displayed one-sided identities. $\square$

Lemma 20.51 (Middle-word backgrounds for the two wrapped complements).

[wrappedMiddleBackground](#)

[mirrorMiddleBackground](#)

a dummy physical letter η and a middle word μ of length $N - (L_0 + 1)$. Write $\tau_\eta^+(\mu)$ for the last-site wrapped background and $\tau_\eta^-(\mu)$ for the opposite wrapped background. Their exposed complements are both exactly μ .

Proof. ✓ **Proof OK** With the same dummy letter η on the remaining sites, fill the physical sites $L_0, \dots, N - 2$ with μ for the last-site wrapped window, and fill the sites $1, \dots, N - L_0 - 1$ with μ for the opposite wrapped window. The two complement-extraction identities are then immediate from the index arithmetic. $\square$

Lemma 20.52 (Same-witness identities from compared wrapped witnesses). [two_sided_middle_compatibility_of_wrapped_witnesses](#)
a dummy physical letter η . Suppose the wrapped and mirror one-sided witnesses Y^+ and Y^- have been extracted. If, for every middle word μ , the witnesses agree on the backgrounds built from η ,

$$Y_{\tau_\eta^+(\mu)}^+ = Y_{\tau_\eta^-(\mu)}^-,$$

then there is a single family Y_μ satisfying

$$A^\mu A^b X = Y_\mu A^b, \quad X A^a A^\mu = A^a Y_\mu$$

for all letters a, b .

Proof. [✓ Proof OK](#) Define Y_μ to be the wrapped witness evaluated on the background $\tau_\eta^+(\mu)$. Lemma 20.51 rewrites the wrapped complement as μ , while the assumed comparison at $\tau_\eta^-(\mu)$ rewrites the mirror witness to the same Y_μ . $\square$

Lemma 20.53 (Same-witness common-middle commutation). [commutes_words_of_two_sided_middle_compatibility](#) [✓ Stmt OK](#)
pose a single family of boundary matrices Y_μ , indexed by middle words μ , satisfies the two one-sided identities

$$A^\mu A^b X = Y_\mu A^b, \quad X A^a A^\mu = A^a Y_\mu$$

for every pair of physical letters a, b . Then X commutes with every word product $A^a A^\mu A^b$ obtained by adjoining one letter on each side of the common middle.

Proof. [✓ Proof OK](#) Multiply the second identity on the right by A^b and the first identity on the left by A^a :

$$X A^a A^\mu A^b = A^a Y_\mu A^b = A^a A^\mu A^b X.$$

$\square$

Theorem 20.54 (Boundary matrix commutation from the wrapping window). [boundary_matrix_commutates](#) [✓ Stmt OK](#) *Let A be injective with $D \geq 1$, and let $\psi = \Gamma_N(X)$ for some boundary matrix $X \in M_D(\mathbb{C})$. If every cyclic restriction of length L of ψ lies in $G_L(A)$, with $N \geq 2$ and $1 < L \leq N$, then, for every $j = 0, \dots, d-1$,*

$$X A^j = A^j X$$

Proof. [✓ Proof OK](#) The wrapping window across the boundary produces matrix identities of the form $X M W = M Y$ for arbitrary test matrices M and complementary words W . Spanning first over the length- $(L-1)$ tails and then over the complementary words, injectivity forces $(X M - M X) W = 0$ for every W , hence $X M = M X$ for every matrix M . In particular, X commutes with each letter A^j . $\square$

Definition 20.55 (MPV submodule). [mpvSubmodule](#) [✓ Stmt OK](#) The subspace spanned by the MPV state $V^{(N)}(A)(\sigma) = \text{tr}(A^{\sigma_0} \dots A^{\sigma_{N-1}})$.

Lemma 20.56 (The MPV is the ground-space map at the identity). [mpv_eq_groundSpaceMap_one](#) [✓ Stmt OK](#) *For every chain length N ,*

$$V^{(N)}(A) = \Gamma_N(\mathbb{1}).$$

Lemma 20.57 (MPV lies in the ground space). [mpv_mem_groundSpace](#) [✓ Stmt OK](#) $V^{(N)}(A) \in G_N(A)$ *for every N , with boundary matrix $X = I$.*

Proof. [✓ Proof OK](#) $V^{(N)}(A)(\sigma) = \text{tr}(A^\sigma) = \text{tr}(A^\sigma \cdot I) = \Gamma_N(I)(\sigma)$. $\square$

Definition 20.58 (Unique ground state). HasUniqueGroundState ✓ Stmt OK A submodule S has a unique ground state if $\dim S = 1$.

Theorem 20.59 (One-dimensionality criterion). hasUniqueGroundState_iff_proportional ✓ Stmt OK For finite-dimensional S , $\dim S = 1$ iff there exists a nonzero $\psi_0 \in S$ such that every $\psi \in S$ is of the form $c\psi_0$.

Proof. ✓ Proof OK Apply the standard finite-dimensional criterion that a nonzero space has dimension one exactly when all of its vectors are proportional to a single nonzero vector. $\square$

Definition 20.60 (Periodic chain ground space). chainGroundSpace ✓ Stmt OK For $N > 0$ and $L \leq N$, define the *periodic chain ground space*

$$\mathcal{G}_{N,L}(A) := \{\psi \in (\{0, \dots, d-1\}^N \rightarrow \mathbb{C}) : \forall i \in \{0, \dots, N-1\}, \psi|_{[i, i+L-1]} \in G_L(A)\},$$

where the condition $\psi|_{[i, i+L-1]} \in G_L(A)$ means that for every choice of physical indices outside the window, the restriction of ψ to that window lies in $G_L(A)$. When $N = 0$ or $L > N$, set $\mathcal{G}_{N,L}(A) := \top$ by convention.

Lemma 20.61 (The MPV satisfies every cyclic window constraint). mpv_mem_chainGroundSpace ✓ Stmt OK If $0 < N$ and $L \leq N$, then

$$V^{(N)}(A) \in \mathcal{G}_{N,L}(A).$$

Theorem 20.62 (MPV nonvanishing for block-injective tensors). mpv_ne_zero_of_isNBlkInjective ✓ Stmt OK If A is L_0 -block-injective with $L_0 > 0$, $D \geq 1$, and $N \geq L_0 + 1$, then $V^{(N)}(A) \neq 0$ in the N -site space.

Proof. ✓ Proof OK Suppose for contradiction that $V^{(N)}(A) = 0$, i.e. $\text{tr}(A^\sigma) = 0$ for every word σ of length N . For any words w and u with $|w| = N - L_0$ and $|u| = L_0$, we have $\text{tr}(A^w A^u) = 0$. Since A is L_0 -block-injective, $\{A^u : |u| = L_0\}$ spans $M_D(\mathbb{C})$. By nondegeneracy of the trace pairing, $A^w = 0$ for every $|w| = N - L_0$. Repeating the argument with words of length $N - 2L_0$ (since $|w'| + L_0 = N - L_0$, the product $A^{w'} A^{u_1}$ has length $N - L_0$, so $A^{w'} A^{u_1} = 0$ by the previous step) and iterating, we eventually reach $\text{tr}(A^u) = 0$ for $|u| = L_0$, which gives $I = 0$, a contradiction. $\square$

20.3.6 Block-word stripping

Theorem 20.63 (Common right factor from block-word compatibility). exists_right_factor_of_block_word_compatibility ✓ A is L_0 -block-injective with $L_0 > 0$. Let $\{Z_u\}_{|u|=L_0}$ be a family indexed by words of length L_0 . If for some $K \geq 0$ and every suffix word w of length K there exists $Y_w \in M_D(\mathbb{C})$ such that, for every word u of length L_0 ,

$$Z_u A^w = A^u Y_w$$

then there exists $X \in M_D(\mathbb{C})$ with $Z_u = A^u X$ for every word u of length L_0 .

Proof. ✓ Proof OK Induct on K . For $K = 0$ the empty suffix gives $Z_u = A^u Y_\emptyset$. For the successor step, fix the first physical index and apply the induction hypothesis to the shortened suffix. This reduces the statement to letter compatibility on the right. Extending that letter compatibility to all length- L_0 words and decomposing $\mathbb{1}$ in the spanning family $\{A^u : |u| = L_0\}$ yields a common factor X . $\square$

Theorem 20.64 (Block-word commutation from long-word commutation). commutes_block_words_of_commutes_long_words_of_ A is L_0 -block-injective with $L_0 > 0$, $m \geq L_0$, and $X \in M_D(\mathbb{C})$ commutes with every length- m product A^ω , then X already commutes with every length- L_0 product A^u .

Proof. ✓ Proof OK Write each length- m word as a concatenation uw with $|u| = L_0$ and $|w| = m - L_0$. The hypothesis gives

$$XA^u A^w = A^u A^w X = A^u (A^w X).$$

Thus the family $Z_u := XA^u$ satisfies the hypothesis of Theorem 20.63. One obtains $XA^u = A^u R$ for all $|u| = L_0$. Decomposing $\mathbb{1}$ in the span of the length- L_0 words shows $R = X$, hence $XA^u = A^u X$. □

Theorem 20.65 (Centrality from long-word commutation). commutes_all_of_commutes_long_words_of_isNBlkInjective ✓ Stmt OK *If A is L_0 -block-injective with $L_0 > 0$, $m \geq L_0$, and $X \in M_D(\mathbb{C})$ commutes with every length- m word product, then X commutes with every matrix in $M_D(\mathbb{C})$.*

Proof. ✓ Proof OK By Theorem 20.64, X commutes with every length- L_0 word product. These products span $M_D(\mathbb{C})$ by block injectivity, so linearity gives $XM = MX$ for every $M \in M_D(\mathbb{C})$. □

Lemma 20.66 (Amplifying fixed-length word commutation). commutes_words_mul_of_commutes_words ✓ Stmt OK *If a boundary matrix X commutes with every word product of length m , then it commutes with every word product whose length is a multiple qm .*

Proof. ✓ Proof OK Proceed by induction on q . For $q = 0$, the claim is trivial. For the inductive step, let a word of length $(q + 1)m$ be given and write it as a length- m prefix followed by a length- qm suffix. The prefix commutes with X by hypothesis, and the suffix commutes by the induction hypothesis. □

Lemma 20.67 (Padded complement annihilation). eq_zero_of_mul_evalWord_eq_zero_of_isNBlkInjective_of_le_mul ✓ Stmt OK *If A is L_0 -block-injective and $q \geq 1$, then an operator Z that annihilates every length- k word product on the left is zero whenever $k \leq qL_0$:*

$$ZA^w = 0 \quad (|w| = k) \quad \implies \quad Z = 0.$$

Proof. ✓ Proof OK The hypothesis $\text{span}\{A^u : |u| = L_0\} = M_D(\mathbb{C})$ implies the same spanning statement at every positive multiple qL_0 . Each length- qL_0 word factors as a length- k prefix followed by padding, so Z annihilates all generators of the full span and hence annihilates the identity. □

Lemma 20.68 (Generator commutation from long-word commutation). boundary_matrix_commutes_of_isNBlkInjective_of_long *the hypotheses of Theorem 20.65, one has $XA^i = A^i X$ for every physical index i .*

Proof. ✓ Proof OK Apply Theorem 20.65 with $M = A^i$. □

Theorem 20.69 (MPV-line endgame from long-word commutation). groundSpaceMap_mem_mpvSubmodule_of_isNBlkInjective_of_1 *Let A be L_0 -block-injective with $L_0 > 0$ and let $m \geq L_0$. If a boundary matrix X commutes with every length- m word product, then $\Gamma_N(X)$ lies in the MPV line $\text{span}\{V^{(N)}(A)\}$.*

Proof. ✓ Proof OK Theorem 20.65 makes X commute with the whole matrix algebra. The center of $M_D(\mathbb{C})$ consists of scalar matrices, so $X = c\mathbb{1}$, and therefore $\Gamma_N(X) = cV^{(N)}(A)$. □

Theorem 20.70 (MPV-line endgame from positive-length commutation). groundSpaceMap_mem_mpvSubmodule_of_isNBlkInjective_of_1 *Let A be L_0 -block-injective with $L_0 > 0$. If X commutes with every word product of some positive length m , then $\Gamma_N(X)$ lies in the MPV line.*

Proof. ✓ Proof OK Lemma 20.66 amplifies the length- m commutation hypothesis to length $L_0 m \geq L_0$. Then Theorem 20.69 applies. □

Theorem 20.71 (MPV-line endgame from a same-witness common middle). groundSpaceMap_mem_mpvSubmodule_of_isNBlkInjective
A be L_0 -block-injective with $L_0 > 0$. If the common-middle same-witness identities of Lemma 20.53 hold for X , then $\Gamma_N(X)$ lies in the MPV line.

Proof. ✓ Proof OK The common-middle identities give commutation with all words of length $|\mu| + 2$, a positive length. The positive-length MPV-line endgame then finishes. □

Theorem 20.72 (MPV-line endgame from compared wrapped witnesses). groundSpaceMap_mem_mpvSubmodule_of_isNBlkInjective
A be L_0 -block-injective with $L_0 > 0$, and fix a dummy physical letter η . If the wrapped and mirror one-sided witnesses satisfy

$$Y_{\tau_\eta^+(\mu)}^+ = Y_{\tau_\eta^-(\mu)}^-$$

for every middle word μ , then $\Gamma_N(X)$ lies in the MPV line.

Proof. ✓ Proof OK Lemma 20.52 uses the comparison at the two η -backgrounds to turn the wrapped witnesses into a single same-witness common-middle family. Then Theorem 20.71 applies. □

Theorem 20.73 (Conditional normal-range reduction from witness comparison). chainGroundSpace_le_mpvSubmodule_of_isNBlkInjective
A be L_0 -block-injective with $L_0 > 0$ and $D \geq 1$. Assume $N \geq 2$ and $L_0 < L \leq N$. Suppose that, for every dummy letter η and every boundary matrix obtained from an N -site chain ground state, the wrapped and mirror witnesses supplied by Theorem 20.50 agree at $\tau_\eta^+(\mu)$ and $\tau_\eta^-(\mu)$ for every middle word μ . Then the range- L chain ground space is contained in the MPV line.

Proof. ✓ Proof OK The cyclic-to-open-chain reduction writes any chain ground state as $\Gamma_N(X)$. The reduced wrapped-boundary compatibility theorem supplies the two one-sided witness families for this X . The comparison hypothesis, applied to a dummy letter η , says that these witnesses agree at $\tau_\eta^+(\mu)$ and $\tau_\eta^-(\mu)$ for every middle word μ . Theorem 20.72 finishes. □

Theorem 20.74 (Chain ground space equals the MPV line in the injective case). chainGroundSpace_eq_mpvSubmodule ✓ Stmt
A is injective, $D \geq 1$, $N \geq 2$, and $1 < L \leq N$, then

$$\mathcal{G}_{N,L}(A) = \text{span}\{V^{(N)}(A)\}.$$

Proof. ✓ Proof OK Let $\psi \in \mathcal{G}_{N,L}(A)$. The non-wrapping window conditions and Lemma 20.38 imply $\psi \in G_N(A)$, so $\psi = \Gamma_N(X)$ for some boundary matrix X . The wrapping window condition now applies Theorem 20.54 and forces X to commute with every letter A^j . Since A is injective, the letters generate the full matrix algebra, hence X is scalar. Therefore ψ is proportional to $V^{(N)}(A)$, while the reverse inclusion is Lemma 20.61. □

Theorem 20.75 (Normal-range reduction: containment in the MPV line). chainGroundSpace_le_mpvSubmodule_of_normal_range
A is normal, $D \geq 1$, L_0 -block-injective, with $N \geq 2$ and $L_0 < L \leq N$, then

$$\mathcal{G}_{N,L}(A) \subseteq \text{span}\{V^{(N)}(A)\}.$$

Proof. Not ready The remaining step is the closure property from [CPGSV21, § IV.C]: Theorem 20.50 supplies the two one-sided wrapped-boundary identities for the open-chain boundary matrix X . The padded-annihilation lemma states that exact complement-span length mismatches are no longer the obstruction. The new conditional reduction theorem shows that the proof would finish as soon as those actual wrapped-window witnesses are compared after reindexing their complements to a common middle word. What is still missing is precisely that comparison of the witnesses themselves. □

Theorem 20.76 (Chain ground space equals the MPV line in the normal case). [chainGroundSpace_eq_mpvSubmodule_normal](#) [N](#)
A is normal, $D \geq 1$, L_0 -block-injective, with $N \geq 2$ and $L_0 < L \leq N$, then

$$\mathcal{G}_{N,L}(A) = \text{span}\{V^{(N)}(A)\}.$$

Theorem 20.77 (Unique ground state on the periodic chain). [groundSpace_unique_periodic](#) [✓ Stmt OK](#) *For an injective tensor A with $D \geq 1$, if $N \geq 2$ and $1 < L \leq N$, then the periodic chain ground space is one-dimensional.*

Proof. [✓ Proof OK](#) By Theorem 20.74, the ground space is the line $\text{span}\{V^{(N)}(A)\}$. It remains to show that the MPV is nonzero. If $V^{(N)}(A) = 0$, then Lemma 20.56 gives

$$\Gamma_N(\mathbb{1}) = \Gamma_N(0).$$

Since A is injective and $N > 0$, Theorem 20.30 forces $\mathbb{1} = 0$, a contradiction. Therefore the spanning line is one-dimensional. $\square$

Theorem 20.78 (Unique ground state for block-injective tensors). [parentHamiltonian_unique_gs_injective](#) [Not ready](#) *If A is L_0 -block-injective with $L_0 > 0$ and $D \geq 1$, the parent Hamiltonian with interaction range $2L_0$ has a unique ground state on the periodic chain.*

Theorem 20.79 (Optimal unique ground state for normal tensors). [parentHamiltonian_unique_gs_normal](#) [Not ready](#) *If A is normal (hence L_0 -block-injective for some $L_0 > 0$), $D \geq 1$, and in normal form, the interaction range can be reduced to $L_0 + 1$.*

20.4 Commuting parent Hamiltonians

A parent Hamiltonian has *commuting* local terms when all translated interaction projectors commute with each other.

Definition 20.80 (Commuting parent Hamiltonian). [IsCommutingParentHam](#) [✓ Stmt OK](#) The parent Hamiltonian with block length L on N sites has *commuting* local terms if $h_i h_j = h_j h_i$ for all $i, j \in \{0, \dots, N-1\}$. See Ref. [CPGSV21, Definition 3.9].

Definition 20.81 (Nearest-neighbor commuting parent Hamiltonian). [IsNNCPH](#) [✓ Stmt OK](#) A *nearest-neighbor commuting parent Hamiltonian* (NNCPH) is a commuting parent Hamiltonian with block length $L = 2$. See Ref. [CPGSV21, Definition 3.9].

Remark 20.82 (Elementary consequences of the commuting definition). [isCommutingParentHam](#) [symm](#) [ham_comm_localTerm](#) [✓ Stmt OK](#) The nearest-neighbor condition is simply the length-2 specialization of the general commuting condition; the latter is symmetric in the two sites, and it implies that the full parent Hamiltonian commutes with each local interaction term.

Theorem 20.83 (RFP implies NNCPH). [rfp_implies_nneph](#) [✓ Stmt OK](#) *If A is a normal renormalization fixed point, then for every chain length $N \geq 2$ the parent Hamiltonian with $L = 2$ has commuting local terms. This implication is cited here from the product-of-entangled-pairs description in Appendix B of Theorem 3.10 in [CPGSV21].*

Remark 20.84 (Product-pair data for NNCPH). [productPairWindow](#) [productPairWindow_one](#) [productPairState](#) [productPairState_one](#) [HasProductPairMPV](#) [HasProductPairLocalProjectors](#) [isNNCPH](#) [ProductPairBridge](#) [localTerm_idempotent](#) [commuting_twoSite_localTerms](#) [isNNCPH](#) [AppendixBStructuralData](#) [exists_ofRFP](#) [ofRFP](#) [twoSiteAmplitude](#) [coreTensor](#) [coreTensor_apply](#) [gaugeEquiv_coreTensor](#) [mpv_eq_coreTensor](#) [twoSiteAmplitude_eq_mpv](#) [twoSiteAmplitude_eq_coreTensor_mpv](#)

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[commuting_twoSite_localTerms_of_rfp_of_appendixBExtraction](#)
[rfp_implies_nncph_of_appendixBExtraction](#)

an even chain, one can extract the p -th adjacent two-site word of a configuration and compare the MPS coefficients with a repeated two-site pair state. The one-pair case recovers the original two-site amplitude. For an Appendix B structural witness, the gauge-removed tensor ΛU_i has the same MPV coefficients as A , and the structural two-site amplitude is exactly the two-site MPV coefficient. Hence the coefficient part of the extraction reduces to proving the repeated-pair identity for this gauge-removed tensor. The corresponding data also include a family of idempotent chain-level projectors whose values are the nearest-neighbor local terms. The final theorem in these data shows that these local terms commute. Consequently the same data yields the nearest-neighbor commuting condition on every finite chain. The Appendix B construction extracts the already-proved structural decomposition from normal left-canonical RFP data and identifies the remaining internal step: construct product-pair extraction through the two-site amplitude determined by that structural witness.

Theorem 20.85 (NNCPH implies RFP). [nncph_implies_rfp](#) ✓ Stmt OK *If A is normal, in left-canonical form, and its parent Hamiltonian with $L = 2$ has commuting local terms for every chain length $N \geq 2$, then A is a renormalization fixed point. This implication is cited here from Beigi's ground-space characterization of one-dimensional commuting nearest-neighbor Hamiltonians; the left-canonical hypothesis fixes the normalization of the transfer map.*

20.5 Decorrelation and commuting parent Hamiltonians

This section develops the abstract operator-algebraic formulation of decorrelation and its connection to commuting parent Hamiltonians, following Appendix D, §D.2 of [CPGSV21].

Definition 20.86 (Decorrelation). [IsDecorrelated](#) ✓ Stmt OK Given an idempotent endomorphism P_K and sets of observables $\mathcal{O}_A, \mathcal{O}_B$, regions A and B are *decorrelated* w.r.t. K when $P_K \circ O_A \circ (1 - P_K) \circ O_B \circ P_K = 0$ for all $O_A \in \mathcal{O}_A, O_B \in \mathcal{O}_B$.

Definition 20.87 (Commuting parent Hamiltonian structure). [HasCommutingParentHam](#) ✓ Stmt OK A subspace K (given by its idempotent endomorphism P_K) has a *commuting parent Hamiltonian structure* if there exist idempotent endomorphisms P_{AX} and P_{XB} that commute and satisfy $P_{AX} \circ P_{XB} = P_K$.

Theorem 20.88 (Idempotence gives a trivial commuting parent Hamiltonian). [ofIdem](#) ✓ Stmt OK *If*

$$P_K \circ P_K = P_K,$$

then P_K has a commuting parent Hamiltonian structure.

Proof. ✓ Proof OK Take $P_{AX} = P_{XB} = P_K$. Their commutativity is immediate, and the product identity is exactly the assumed idempotence of P_K . □

Lemma 20.89 (Cancellation for commuting idempotents). [comp_complement_comm_zero](#) ✓ Stmt OK *If P and Q are commuting idempotents, then*

$$Q \circ (1 - P \circ Q) \circ P = 0.$$

Proof. ✓ Proof OK Expand the middle factor, use commutativity to rewrite $QPQP$ as $QPPQ$, and then apply idempotence of P and Q . □

Theorem 20.90 (Idempotence of the product projector). pK_idem ✓ Stmt OK If $P_K = P_{AX} \circ P_{XB}$ is a commuting parent Hamiltonian decomposition, then

$$P_K \circ P_K = P_K.$$

Proof. ✓ Proof OK Since P_{AX} and P_{XB} are idempotent and commute, their product $P_{AX} \circ P_{XB}$ is idempotent. Substituting $P_K = P_{AX} \circ P_{XB}$ gives the claim. □

Theorem 20.91 (Absorption by the left projector). pAX_comp_pK ✓ Stmt OK If $P_K = P_{AX} \circ P_{XB}$ is a commuting parent Hamiltonian decomposition, then

$$P_{AX} \circ P_K = P_K.$$

Proof. ✓ Proof OK Using $P_K = P_{AX} \circ P_{XB}$ and the idempotence of P_{AX} ,

$$P_{AX} \circ P_K = P_{AX} \circ P_{AX} \circ P_{XB} = P_{AX} \circ P_{XB} = P_K.$$

□

Theorem 20.92 (Absorption by the right projector). pK_comp_pXB ✓ Stmt OK If $P_K = P_{AX} \circ P_{XB}$ is a commuting parent Hamiltonian decomposition, then

$$P_K \circ P_{XB} = P_K.$$

Proof. ✓ Proof OK Using $P_K = P_{AX} \circ P_{XB}$ and the idempotence of P_{XB} ,

$$P_K \circ P_{XB} = P_{AX} \circ P_{XB} \circ P_{XB} = P_{AX} \circ P_{XB} = P_K.$$

□

Theorem 20.93 (Cross-absorption by the right projector). pXB_comp_pK ✓ Stmt OK If $P_K = P_{AX} \circ P_{XB}$ is a commuting parent Hamiltonian decomposition, then

$$P_{XB} \circ P_K = P_K.$$

Proof. ✓ Proof OK By commutativity, $P_{XB} \circ P_K = P_{XB} \circ P_{AX} \circ P_{XB} = P_{AX} \circ P_{XB} \circ P_{XB}$, and idempotence of P_{XB} yields $P_{XB} \circ P_K = P_K$. □

Theorem 20.94 (Cross-absorption by the left projector). pK_comp_pAX ✓ Stmt OK If $P_K = P_{AX} \circ P_{XB}$ is a commuting parent Hamiltonian decomposition, then

$$P_K \circ P_{AX} = P_K.$$

Proof. ✓ Proof OK By commutativity, $P_K \circ P_{AX} = P_{AX} \circ P_{XB} \circ P_{AX} = P_{AX} \circ P_{AX} \circ P_{XB}$, and idempotence of P_{AX} yields $P_K \circ P_{AX} = P_K$. □

Theorem 20.95 (Reverse product identity). reverse_product ✓ Stmt OK If $P_K = P_{AX} \circ P_{XB}$ is a commuting parent Hamiltonian decomposition, then

$$P_{XB} \circ P_{AX} = P_K.$$

Proof. ✓ Proof OK This is immediate from commutativity of P_{AX} and P_{XB} together with the identity $P_{AX} \circ P_{XB} = P_K$. □

Theorem 20.96 (Commuting complementary projectors). complement_comm ✓ Stmt OK If $P_K = P_{AX} \circ P_{XB}$ is a commuting parent Hamiltonian decomposition and

$$Q_{AX} = 1 - P_{AX}, \quad Q_{XB} = 1 - P_{XB},$$

then

$$Q_{AX} \circ Q_{XB} = Q_{XB} \circ Q_{AX}.$$

Proof. ✓ Proof OK Expanding both products and using $P_{AX} \circ P_{XB} = P_{XB} \circ P_{AX}$ shows that the complementary projectors commute. □

Theorem 20.97 (Commuting parent Hamiltonian implies decorrelation). commutingHam_isDecorrelated ✓ Stmt OK If a subspace K has a commuting parent Hamiltonian decomposition $P_K = P_{AX} \circ P_{XB}$ with $[P_{AX}, P_{XB}] = 0$, and A -observables commute with P_{XB} while B -observables commute with P_{AX} , then regions A and B are decorrelated w.r.t. K . This is the backward direction of Proposition D.3 from [CPGSV21], Appendix D, §D.2.

Proof. ✓ Proof OK The key identity is $P_{XB} \circ (1 - P_{AX} \circ P_{XB}) \circ P_{AX} = 0$. Using the commutation hypotheses to slide O_A past P_{XB} and O_B past P_{AX} , the five-fold composition factors through this zero term. □

Theorem 20.98 (Commuting parent Hamiltonian implies decorrelation). isDecorrelated ✓ Stmt OK Corollary of Theorem 20.97: if P_K has a commuting parent Hamiltonian decomposition and observables respect locality, then regions A and B are decorrelated w.r.t. K .

Lemma 20.99 (Frustration-free identity). frustration_free_ham_eq ✓ Stmt OK For any endomorphisms P and Q ,

$$(1 - P) + (1 - Q) - (1 - P) \circ (1 - Q) = 1 - P \circ Q.$$

Proof. ✓ Proof OK Expanding the left-hand side and cancelling gives the result. □

Theorem 20.100 (Commuting parent Hamiltonian identity). hamiltonian_eq ✓ Stmt OK If $P_K = P_{AX} \circ P_{XB}$ is a commuting parent Hamiltonian decomposition, then

$$Q_{AX} + Q_{XB} - Q_{AX} \circ Q_{XB} = 1 - P_K,$$

where $Q_{AX} = 1 - P_{AX}$ and $Q_{XB} = 1 - P_{XB}$ are the complementary projectors. This is the frustration-free form of the parent Hamiltonian from [CPGSV21], Appendix D, §D.2.

Proof. ✓ Proof OK Applying Lemma 20.99 to P_{AX} and P_{XB} , and substituting $P_{AX} \circ P_{XB} = P_K$. □

Theorem 20.101 (Ground-space membership). mem_ground_iff ✓ Stmt OK If $P_K = P_{AX} \circ P_{XB}$ is a commuting parent Hamiltonian decomposition, then

$$P_K v = v \iff P_{AX} v = v \text{ and } P_{XB} v = v.$$

This is the algebraic form of equation (D.2) from [CPGSV21]: $K_{AXB} = (K_{AX} \otimes H_B) \cap (H_A \otimes K_{XB})$.

Proof. ✓ Proof OK ($\Rightarrow$) If $P_K v = v$, then $P_{AX}(P_K v) = (P_{AX} \circ P_K) v = P_K v = v$ by left absorption, and similarly for P_{XB} .

($\Leftarrow$) If $P_{AX} v = v$ and $P_{XB} v = v$, then $P_K v = (P_{AX} \circ P_{XB}) v = P_{AX}(P_{XB} v) = P_{AX} v = v$. □

20.6 Spectral gap

A frustration-free Hamiltonian is *gapped* if the first excited eigenvalue is bounded away from zero uniformly in the chain length N . For MPS parent Hamiltonians, the spectral gap follows from a martingale criterion due to Kastoryano and Lucia [KL18].

Definition 20.102 (Transported parent ground space). [parentHamiltonianGroundSpaceES](#) [✓ Stmt OK](#) Denote by $\mathcal{G}_{N,L}^{\text{ES}}(A)$ the parent ground space after identifying the coefficient space with its Euclidean ℓ^2 -model.

Definition 20.103 (Transported parent Hamiltonian). [parentHamiltonianES](#) [✓ Stmt OK](#) Denote by $H_{N,L}^{\text{ES}}(A)$ the conjugate of $H_N(A, L)$ by the canonical identification between the coefficient space and its Euclidean avatar.

Theorem 20.104 (Euclidean transport of the parent ground space). [parentHamiltonianGroundSpaceES_eq_ker_parentHamiltonianES](#) [✓ Stmt OK](#) *Under the canonical ℓ^2 transport of the N -site state space, the transported parent ground space is exactly the kernel of the transported parent Hamiltonian.*

Proof. [✓ Proof OK](#) This is an immediate unraveling of the definitions: both constructions are obtained from the same parent Hamiltonian by conjugating with the canonical identification between the site space and its Euclidean-space avatar, so the transported ground space is precisely the transported kernel. $\square$

Lemma 20.105 (Membership in the transported parent ground space). [mem_parentHamiltonianGroundSpaceES_iff](#) [✓ Stmt OK](#) *vector belongs to the transported parent ground space exactly when the transported parent Hamiltonian annihilates it.*

Remark 20.106 (Euclidean local projector ingredients). [parentInteractionES](#) [cyclicRestrictES](#) [localTermES](#) [localTermESSummand](#) [✓ Stmt OK](#) Let $P_L^{\text{ES}}(A)$ denote the orthogonal projector onto $(G_L^{\text{ES}}(A))^\perp$ on the Euclidean L -site space. For a cyclic window starting at i and an outside configuration τ , the transported restriction map $R_{i,\tau}$ evaluates an N -site Euclidean vector on that window. The transported local term $h_{i,\text{ES}}$ is the corresponding N -site local projector, and $R_{i,\tau}^\dagger P_L^{\text{ES}}(A) R_{i,\tau}$ is the basic positive summand in its cyclic-averaging formula.

Lemma 20.107 (Positivity of the Euclidean local summands). [parentInteractionES_isPositive](#) [localTermESSummand_isPositive](#) [localTermESSummand_sum_isPositive](#) [✓ Stmt OK](#) *The Euclidean local projector $P_L^{\text{ES}}(A)$ is positive. Consequently, every summand $R_{i,\tau}^\dagger P_L^{\text{ES}}(A) R_{i,\tau}$ is positive, and so is the finite sum over τ .*

Proof. [✓ Proof OK](#) Orthogonal projectors are positive. Conjugation by $R_{i,\tau}$ preserves positivity, and finite sums of positive operators remain positive. $\square$

Lemma 20.108 (Cyclic averaging for the transported ES local terms). [parentHamiltonianES_eq_sum_localTermES](#) [localTermES_eq_average_localTermESSummand](#) [✓ Stmt OK](#) *The transported parent Hamiltonian is the sum of the transported local terms $H_{N,\text{ES}} = \sum_i h_{i,\text{ES}}$, and each transported local term admits the exact cyclic-averaging formula*

$$h_{i,\text{ES}} = d^{-L} \sum_{\tau} R_{i,\tau}^\dagger P_L^{\text{ES}}(A) R_{i,\tau}.$$

Proof. [✓ Proof OK](#) Unfold the transported local term and compare it configuration-by-configuration with the cyclic restrictions. The d^{-L} factor compensates for the d^L choices of the window coordinates in the ambient configuration parameter τ . $\square$

Lemma 20.109 (Positivity of the transported ES Hamiltonian). [localTermES_isPositive](#) [parentHamiltonianES_isPositive](#) [✓](#)
transported local term $h_{i,\text{ES}}$ is positive, hence the full transported parent Hamiltonian $H_{N,\text{ES}}$ is positive.

Proof. [✓ Proof OK](#) Apply the averaging formula above: $h_{i,\text{ES}}$ is a nonnegative scalar multiple of a finite sum of positive conjugates of $P_L^{\text{ES}}(A)$, hence is itself positive. Summing over i yields positivity of $H_{N,\text{ES}}$. $\square$

Lemma 20.110 (Symmetric-projection structure of transported ES local terms). [parentInteractionES_isSymmetricProjection](#) [localTermES_isSymmetricProjection](#) [✓ Stmt OK](#)
The Euclidean local projector $P_L^{\text{ES}}(A)$ is a symmetric projection, and every transported local term $h_{i,\text{ES}}$ on the N -site chain is a symmetric projection.

Proof. [✓ Proof OK](#) The local operator $P_L^{\text{ES}}(A)$ is an orthogonal projection onto $(G_L^{\text{ES}}(A))^\perp$. For $L \leq N$, restricting $h_{i,\text{ES}}v$ to a cyclic window gives $P_L^{\text{ES}}(A)$ applied to the corresponding restriction of v ; applying $h_{i,\text{ES}}$ a second time therefore applies $(P_L^{\text{ES}}(A))^2 = P_L^{\text{ES}}(A)$ on that window. The positivity result above supplies symmetry, and the case $L > N$ is the zero projection by definition. $\square$

Lemma 20.111 (Kernel of the Euclidean parent interaction). [parentInteractionES_apply_eq_zero_iff](#) [✓ Stmt OK](#)
The Euclidean local interaction $P_L^{\text{ES}}(A)$ annihilates exactly the Euclidean local ground space:

$$P_L^{\text{ES}}(A)v = 0 \iff v \in G_L^{\text{ES}}(A).$$

Proof. [✓ Proof OK](#) Since $P_L^{\text{ES}}(A)$ is the orthogonal projector onto $(G_L^{\text{ES}}(A))^\perp$, its kernel is the original subspace $G_L^{\text{ES}}(A)$. $\square$

Lemma 20.112 (Local-term kernel via cyclic restrictions). [localTermES_eq_zero_iff_forall_cyclicRestrictES_mem_groundSpaceES](#) [cyclicRestrictES_mem_groundSpaceES_of_localTermES_eq_zero](#) [✓ Stmt OK](#)
For $L \leq N$, a transported local term $h_{i,\text{ES}}$ annihilates a vector v iff every boundary-filled restriction of v to the cyclic L -window starting at i lies in $G_L^{\text{ES}}(A)$.

Proof. [✓ Proof OK](#) Restricting $h_{i,\text{ES}}v$ to the same cyclic window gives $P_L^{\text{ES}}(A)$ applied to the corresponding restriction of v . Lemma 20.111 identifies the vanishing of this local projector with membership in $G_L^{\text{ES}}(A)$. $\square$

Theorem 20.113 (Adjacent local kernels realize the intersection property). [mem_groundSpaceES_succ_of_adjacent_localTermES](#) [adjacent_localTermES_eq_zero_of_mem_groundSpaceES_succ](#) [adjacent_localTermES_eq_zero_iff_mem_groundSpaceES_succ](#) [✓ Stmt OK](#)
If A is injective and $L \geq 2$, then on an $(L+1)$ -site chain the kernels of the two adjacent transported L -site local terms are exactly the transported $(L+1)$ -site MPS ground space:

$$h_{0,\text{ES}}v = 0 \text{ and } h_{1,\text{ES}}v = 0 \iff v \in G_{L+1}^{\text{ES}}(A).$$

Proof. [✓ Proof OK](#) Lemma 20.112 translates the two local kernel assumptions into the left and right L -site ground conditions. The open-chain intersection property, Theorem 20.34, then regrows the shared $(L-1)$ -site overlap to the $(L+1)$ -site ground space. The converse uses the forward restriction lemmas for ground-space vectors and the same kernel-restriction characterization. $\square$

Lemma 20.114 (Non-overlap positivity for transported ES local terms). [CyclicWindowsDisjoint](#) [symm](#) [of_not_cyclicWindowsOverlap](#) [localTermES_commute_of_cyclic_windows_disjoint](#) [localTermES_re_inner_nonneg_of_cyclic_windows_disj](#)
 $L \leq N$ and let $h_{i,\text{ES}}, h_{j,\text{ES}}$ be transported local terms whose cyclic L -windows are site-disjoint:

no site of the periodic chain has cyclic offset $< L$ from both i and j . Then the terms commute pointwise: for every vector v ,

$$h_{i,\text{ES}}(h_{j,\text{ES}}v) = h_{j,\text{ES}}(h_{i,\text{ES}}v),$$

and their ordered cross term is nonnegative:

$$0 \leq \text{Re}\langle h_{i,\text{ES}}v, h_{j,\text{ES}}v \rangle.$$

Proof. ✓ Proof OK In coordinates, $h_{i,\text{ES}}$ applies $P_L^{\text{ES}}(A)$ only to the i -window variables and leaves the complementary variables fixed. Site disjointness gives two replacement identities: replacing the i -window does not change the extracted j -window, and the i - and j -window replacements commute. Therefore the two embedded local operators commute. The preceding projection-geometry identity for commuting symmetric projections gives the nonnegative ordered cross term. $\square$

Lemma 20.115 (Quadratic form to norm bound). spectralGap_of_martingale ✓ Stmt OK Let H be a positive semidefinite self-adjoint operator on a finite-dimensional complex Hilbert space, and suppose that, for all v ,

$$\gamma \langle v, Hv \rangle \leq \langle Hv, Hv \rangle$$

i.e. $H^2 \geq \gamma H$ as a quadratic form, for some $\gamma > 0$. Then for every v in the orthogonal complement of $\ker H$,

$$\gamma \|v\| \leq \|Hv\|.$$

In particular, every positive eigenvalue of H is at least γ . The positive-semidefiniteness hypothesis is essential: without it, $H = -\text{Id}$ with $\gamma = 2$ satisfies the quadratic-form inequality vacuously but violates the norm bound.

Proof. ✓ Proof OK By the finite-dimensional spectral theorem, H diagonalises in an orthonormal eigenbasis with real eigenvalues λ_k . For a unit eigenvector ψ_k with eigenvalue λ_k the hypothesis gives $\gamma \lambda_k \leq \lambda_k^2$, whence $\lambda_k \leq 0$ or $\lambda_k \geq \gamma$. Since H is positive semidefinite (its quadratic form is ≥ 0), the negative case is excluded and every nonzero eigenvalue satisfies $\lambda_k \geq \gamma$. Writing $v = \sum_k c_k \psi_k$ and projecting onto $(\ker H)^\perp$ keeps only terms with $\lambda_k \geq \gamma$, so $\|Hv\|^2 = \sum_k |c_k|^2 \lambda_k^2 \geq \gamma^2 \sum_k |c_k|^2 = \gamma^2 \|v\|^2$. $\square$

Lemma 20.116 (Parent-Hamiltonian quadratic-form reduction). parentHamiltonianES_norm_bound_of_quadratic_form parentHamiltonianES_gap_bound_of_quadratic_form ✓ Stmt OK Let $H_{N,L}^{\text{ES}}(A)$ be the transported parent Hamiltonian. If a constant $\gamma > 0$ satisfies the uniform quadratic-form estimate

$$\gamma \text{Re}\langle H_{N,L}^{\text{ES}}(A)v, v \rangle \leq \text{Re}\langle H_{N,L}^{\text{ES}}(A)v, H_{N,L}^{\text{ES}}(A)v \rangle$$

for every $N \geq 2L$ and every vector v , then

$$\gamma \|v\| \leq \|H_{N,L}^{\text{ES}}(A)v\|$$

for every $v \in (\mathcal{G}_{N,L}^{\text{ES}}(A))^\perp$. In particular, with $L > 1$ and $\gamma = 1/(4L)$, the explicit gap-bound conclusion follows once this same quadratic-form estimate is supplied.

Proof. ✓ Proof OK The transported Hamiltonian is positive by Lemma 20.109, and its transported ground space is its kernel by Theorem 20.104. Therefore the abstract spectral-theorem conversion of Lemma 20.115 applies directly. The explicit constant is positive because $L > 1$ implies $4L > 0$. $\square$

Remark 20.117 (Projection-geometry identities for row-sum arguments).

[re_inner_apply_apply_self](#)
[re_inner_nonneg](#) [re_inner_apply_apply_nonneg_of_commute](#) [re_inner_apply_apply_ge_neg_of_norm_apply_le](#) [re_inner_sum_apply_left](#)
[re_inner_sum_apply_apply](#) [sum_sum_eq_diag_add_offdiag](#) [crossTerm_sum_bound_of_ordered_rowSum](#) [✓ Stmt OK](#)

The projection-geometry reduction uses the following elementary Hilbert-space identities: the diagonal identity $\text{Re}\langle Pv, Pv \rangle = \text{Re}\langle Pv, v \rangle$ for symmetric projections; nonnegativity of this quadratic form; nonnegativity of $\text{Re}\langle Pv, Qv \rangle$ for commuting symmetric projections P, Q ; the Cauchy–Schwarz conversion from $\|PQv\| \leq c\|Pv\|$ to $\text{Re}\langle Pv, Qv \rangle \geq -c\text{Re}\langle Pv, v \rangle$; finite-sum expansions of $\text{Re}\langle (\sum_i P_i)v, v \rangle$ and $\text{Re}\langle (\sum_i P_i)v, (\sum_i P_i)v \rangle$; the diagonal/off-diagonal split of the ordered double sum; and the aggregate off-diagonal estimate obtained from row sums $\sum_{j \neq i} c_{ij} \leq 1$.

Lemma 20.118 (Ordered projection row-sum quadratic-form reduction).

[quadraticForm_sum_projections_of_ordered_rowSum](#)

P_i be a finite family of symmetric projections and let $H = \sum_i P_i$. If the ordered off-diagonal forms obey

$$\text{Re}\langle P_i v, P_j v \rangle \geq -(1 - \gamma) c_{ij} \text{Re}\langle P_i v, v \rangle \quad (i \neq j),$$

with row sums $\sum_{j \neq i} c_{ij} \leq 1$ and $\gamma \leq 1$, then $H^2 \geq \gamma H$ as a quadratic form.

Proof. [✓ Proof OK](#) Expand $\text{Re}\langle Hv, Hv \rangle$ as an ordered double sum. Symmetric idempotence identifies each diagonal term with $\text{Re}\langle P_i v, v \rangle$, while the row-summable ordered cross-term bounds control the off-diagonal contribution by $-(1 - \gamma) \sum_i \text{Re}\langle P_i v, v \rangle$. $\square$

Lemma 20.119 (Finite-overlap projection row-sum reduction).

[quadraticForm_sum_projections_of_finite_overlap](#)

[✓ Stmt OK](#)

P_i be a finite family of symmetric projections and let $\gamma \leq 1$. Suppose an interaction predicate $i \sim j$ has at most m off-diagonal neighbours in each row for some positive integer m . If noninteracting ordered cross terms are nonnegative and, on each interacting pair,

$$\text{Re}\langle P_i v, P_j v \rangle \geq -(1 - \gamma) \frac{1}{m} \text{Re}\langle P_i v, v \rangle,$$

then $H = \sum_i P_i$ satisfies $H^2 \geq \gamma H$ as a quadratic form.

Proof. [✓ Proof OK](#) Choose $c_{ij} = 1/m$ on interacting pairs and $c_{ij} = 0$ otherwise. The row cardinality bound gives $\sum_{j \neq i} c_{ij} \leq 1$; noninteracting nonnegativity supplies the zero-coefficient cross-term estimates. Apply Lemma 20.118. $\square$

Lemma 20.120 (Finite-overlap projection norm-compression reduction).

[quadraticForm_sum_projections_of_finite_overlap_norm_bound_of_le](#)

[quadraticForm_sum_projections_of_finite_overlap_norm_bound_of_le](#)

[✓ Stmt OK](#)

Let P_i be a finite family of symmetric projections and let $\gamma \leq 1$. Suppose an interaction predicate has at most $m > 0$ off-diagonal neighbours in each row, noninteracting ordered cross terms are nonnegative, and interacting pairs satisfy the norm-compression estimate

$$\|P_i P_j v\| \leq (1 - \gamma) \frac{1}{m} \|P_i v\|.$$

Then $H = \sum_i P_i$ satisfies $H^2 \geq \gamma H$ as a quadratic form. The same conclusion holds from a separate compression coefficient η whenever $\eta \leq (1 - \gamma)/m$.

Proof. [✓ Proof OK](#) Cauchy–Schwarz and symmetric idempotence turn the norm-compression estimate into $\text{Re}\langle P_i v, P_j v \rangle \geq -(1 - \gamma) m^{-1} \text{Re}\langle P_i v, v \rangle$. If a separate coefficient η is used, the assumption $\eta \leq (1 - \gamma)/m$ first weakens that estimate to the displayed one. The finite-overlap row-sum reduction then applies. $\square$

Lemma 20.121 (Parent-Hamiltonian ordered row-sum quadratic-form reduction). [parentHamiltonianES_quadratic_form_of](#)
a fixed chain length N , suppose the transported local terms $h_{i,\text{ES}}$ satisfy the ordered row-sum estimate

$$\text{Re}\langle h_{i,\text{ES}}v, h_{j,\text{ES}}v \rangle \geq -(1-\gamma)c_{ij} \text{Re}\langle h_{i,\text{ES}}v, v \rangle \quad (i \neq j),$$

with $\sum_{j \neq i} c_{ij} \leq 1$ and $\gamma \leq 1$. Then the transported parent Hamiltonian satisfies

$$\gamma \text{Re}\langle H_{N,L}^{\text{ES}}v, v \rangle \leq \text{Re}\langle H_{N,L}^{\text{ES}}v, H_{N,L}^{\text{ES}}v \rangle$$

for every vector v .

Proof. [✓ Proof OK](#) Lemma 20.110 supplies the symmetric-projection hypothesis for the family $P_i = h_{i,\text{ES}}$. Apply Lemma 20.118 to this family and use $H_{N,L}^{\text{ES}} = \sum_i h_{i,\text{ES}}$. $\square$

Lemma 20.122 (Parent-Hamiltonian gap from ordered local row bounds). [parentHamiltonianES_gap_bound_of_ordered_local](#)
uniformly for every $N \geq 2L$ that the transported local terms satisfy the ordered row-sum estimate above with $\gamma = 1/(4L)$. If $L > 1$, then $1/(4L) > 0$ and, for every $v \in (\mathcal{G}_{N,L}^{\text{ES}}(A))^\perp$,

$$\frac{1}{4L} \|v\| \leq \|H_{N,L}^{\text{ES}}(A)v\|.$$

Proof. [✓ Proof OK](#) The fixed-chain row-sum reduction supplies the quadratic-form estimate with $\gamma = 1/(4L)$ for every admissible N . The positivity and kernel identification of the transported Hamiltonian then allow Lemma 20.116 to convert this quadratic-form estimate into the norm lower bound on the orthogonal complement of the ground space. $\square$

Lemma 20.123 (Parent-Hamiltonian finite-overlap Friedrichs quadratic reduction). [parentHamiltonianES_quadratic_form_of](#)
a fixed chain length N , let m be a positive integer and let $\gamma \leq 1$. Suppose an overlap predicate on local windows has at most m overlapping off-diagonal windows in each row. If non-overlapping local terms have nonnegative ordered cross terms, overlapping local terms satisfy the finite-overlap Friedrichs estimate

$$\text{Re}\langle h_{i,\text{ES}}v, h_{j,\text{ES}}v \rangle \geq -(1-\gamma) \frac{1}{m} \text{Re}\langle h_{i,\text{ES}}v, v \rangle,$$

and the transported local terms are symmetric projections, then the transported Hamiltonian satisfies $H_{N,L}^{\text{ES}}{}^2 \geq \gamma H_{N,L}^{\text{ES}}$ as a quadratic form.

Proof. [✓ Proof OK](#) Apply the finite-overlap projection reduction with $P_i = h_{i,\text{ES}}$. $\square$

Lemma 20.124 (Parent-Hamiltonian finite-overlap norm-compression quadratic reduction).

[parentHamiltonianES_quadratic_form_of_finite_overlap_norm_bound](#) [parentHamiltonianES_quadratic_form_of_finite_overlap_norm_bound_of_le](#)
a fixed chain length N , let $m > 0$ and $\gamma \leq 1$. Suppose an overlap predicate on local windows has at most m overlapping off-diagonal windows in each row, non-overlapping local terms have nonnegative ordered cross terms, and overlapping local terms satisfy

$$\|h_{i,\text{ES}}(h_{j,\text{ES}}v)\| \leq (1-\gamma) \frac{1}{m} \|h_{i,\text{ES}}v\|.$$

Then $H_{N,L}^{\text{ES}}{}^2 \geq \gamma H_{N,L}^{\text{ES}}$ as a quadratic form. The same conclusion holds from a separate coefficient η satisfying $\eta \leq (1-\gamma)/m$.

Proof. [✓ Proof OK](#) Apply the finite-overlap norm-compression projection reduction to the family $P_i = h_{i,\text{ES}}$. If a separate coefficient η is used, first weaken the compression estimate using $\eta \leq (1-\gamma)/m$. $\square$

Definition 20.125 (Cyclic-window support). [cyclicWindowSupport](#) ✓ Stmt OK For a length- L cyclic window on $\{0, \dots, N-1\}$ starting at i , its support is

$$S_i = \{i, i+1, \dots, i+L-1\} \pmod{N},$$

with repeated visits counted once.

Lemma 20.126 (Cyclic-window support offsets). [mem_cyclicWindowSupport_iff](#) ✓ Stmt OK If $L \leq N$, then a site k belongs to the cyclic support S_i exactly when its cyclic offset from i is below L :

$$k \in S_i \iff (k-i) \bmod N < L.$$

Proof. ✓ Proof OK Write $k = i + r \pmod{N}$ for a representative offset r . Membership in S_i gives an offset $r < L$, and conversely an offset below L gives the corresponding point of the support. $\square$

Definition 20.127 (Cyclic-window overlap). [cyclicWindowsOverlap](#) ✓ Stmt OK Two length- L cyclic windows overlap, written $i \sim_L j$, when their supports meet: $S_i \cap S_j \neq \emptyset$.

Lemma 20.128 (Concrete non-overlap positivity for cyclic supports). [of_not_cyclicWindowsOverlap](#) [localTermES_re_inner_nonneg_of_not_cyclicWindowsOverlap](#) ✓ Stmt OK If $L \leq N$ and the cyclic supports S_i and S_j do not meet, then the transported local ES terms have nonnegative ordered cross term:

$$0 \leq \text{Re}\langle h_{i,\text{ES}}v, h_{j,\text{ES}}v \rangle.$$

Proof. ✓ Proof OK The support-offset characterization converts failure of $S_i \cap S_j \neq \emptyset$ into site-disjointness of the two cyclic windows. The disjoint-window positivity lemma then applies. $\square$

Lemma 20.129 (Cyclic-window overlap row cardinality). [cyclicWindowsOverlap_card_le](#) ✓ Stmt OK If $2L \leq N$ and $L > 1$, then for every $i \in \{0, \dots, N-1\}$,

$$\#\{j \neq i : i \sim_L j\} \leq 2(L-1).$$

Proof. ✓ Proof OK If S_i and S_j meet, choose offsets $a, b < L$ with $i + a \equiv j + b \pmod{N}$. Since $2L \leq N$, the offset from i to j is either the clockwise distance $a - b \in \{1, \dots, L-1\}$ or the counterclockwise distance $b - a \in \{1, \dots, L-1\}$; the zero distance is excluded by $j \neq i$. Thus every overlapping off-diagonal start lies among the $L-1$ clockwise starts or the $L-1$ counterclockwise starts. $\square$

Lemma 20.130 (Parent-Hamiltonian gap from finite-overlap Friedrichs data). [parentHamiltonianES_gap_bound_of_finite_overlap](#) $L > 1$ and set $m = 2(L-1)$. Suppose uniformly for every $N \geq 2L$ that the hypotheses of the fixed-chain finite-overlap Friedrichs reduction hold with $\gamma = 1/(4L)$ and this value of m . Then $1/(4L) > 0$, and the explicit norm lower bound with constant $1/(4L)$ follows for vectors in $(\mathcal{G}_{N,L}^{\text{ES}}(A))^\perp$.

Proof. ✓ Proof OK The fixed-chain finite-overlap reduction gives the required quadratic-form estimate for every admissible N . Lemma 20.116 converts that estimate to the norm lower bound. $\square$

Lemma 20.131 (Parent-Hamiltonian gap from cyclic-window Friedrichs data). [parentHamiltonianES_gap_bound_of_cyclic_windows](#) $L > 1$ and let $i \sim_L j$ mean that the cyclic supports of the length- L windows at i and j meet. Suppose uniformly for every $N \geq 2L$ that overlapping off-diagonal terms satisfy

$$\text{Re}\langle h_{i,\text{ES}}v, h_{j,\text{ES}}v \rangle \geq -\left(1 - \frac{1}{4L}\right) \frac{1}{2(L-1)} \text{Re}\langle h_{i,\text{ES}}v, v \rangle.$$

Then $1/(4L) > 0$, and the explicit norm lower bound with constant $1/(4L)$ follows for vectors in $(\mathcal{G}_{N,L}^{\text{ES}}(A))^\perp$.

Proof. ✓ Proof OK Lemma 20.129 gives the row-cardinality bound $m = 2(L - 1)$ for the concrete cyclic-window overlap relation, and Lemma 20.110 supplies the symmetric projection structure of each transported local term. If two cyclic supports do not meet, the corresponding windows are site-disjoint, so Lemma 20.128 gives the required nonnegative ordered cross term. Apply Lemma 20.130 with these facts and the displayed overlapping-window estimate. $\square$

Lemma 20.132 (Parent-Hamiltonian gap from overlapping cyclic Friedrichs data). parentHamiltonianES_gap_bound_of_cyclic
 $L > 1$ and let $i \sim_L j$ mean that the cyclic supports of the length- L windows at i and j meet. Suppose uniformly for every $N \geq 2L$ that overlapping off-diagonal terms satisfy

$$\operatorname{Re}\langle h_{i,\text{ES}}v, h_{j,\text{ES}}v \rangle \geq -(1 - \frac{1}{4L}) \frac{1}{2(L-1)} \operatorname{Re}\langle h_{i,\text{ES}}v, v \rangle.$$

Then the explicit norm lower bound with constant $1/(4L)$ follows for vectors in $(\mathcal{G}_{N,L}^{\text{ES}}(A))^{\perp}$.

Proof. ✓ Proof OK Apply Lemma 20.131. Its non-overlap positivity and finite-row hypotheses are already internal to that cyclic-window reduction; the displayed overlapping estimate supplies its remaining ordered cross-term condition. $\square$

Lemma 20.133 (Parent-Hamiltonian gap from overlapping cyclic norm-compression data). parentHamiltonianES_gap_bound_of_cyclic_window_overlap_norm_bound ✓ Stmt OK Fix $L > 1$. Suppose uniformly for every $N \geq 2L$ and every overlapping off-diagonal pair that

$$\|h_{i,\text{ES}}(h_{j,\text{ES}}v)\| \leq (1 - \frac{1}{4L}) \frac{1}{2(L-1)} \|h_{i,\text{ES}}v\|.$$

Then the explicit norm lower bound with constant $1/(4L)$ follows for vectors in $(\mathcal{G}_{N,L}^{\text{ES}}(A))^{\perp}$.

Proof. ✓ Proof OK The norm-compression condition is converted to the ordered Friedrichs lower bound by the projection-geometry Cauchy–Schwarz lemma. The preceding overlapping-cyclic Friedrichs reduction then gives the norm lower bound. $\square$

Lemma 20.134 (Parent-Hamiltonian gap from a cyclic overlap compression constant). parentHamiltonianES_gap_bound_of_cyclic_window_overlap_norm_bound_of_1t ✓ Stmt OK Fix $L > 1$, and let $m = 2(L - 1)$. Suppose uniformly for every $N \geq 2L$ and every overlapping off-diagonal pair that

$$\|h_{i,\text{ES}}(h_{j,\text{ES}}v)\| \leq \eta \|h_{i,\text{ES}}v\|.$$

If $\gamma > 0$, $\gamma \leq 1$, and $\eta \leq (1 - \gamma)/m$, then $0 < \gamma$ and, for every admissible N and every $v \in (\mathcal{G}_{N,L}^{\text{ES}}(A))^{\perp}$,

$$\gamma \|v\| \leq \|H_{N,L}^{\text{ES}}(A)v\|.$$

The assertion $0 < \gamma$ repeats the hypothesis, in the same shape as the strict statement below. In particular, if $\eta \geq 0$ and $\eta m < 1$, then $0 < 1 - \eta m$ and, for every admissible N and every $v \in (\mathcal{G}_{N,L}^{\text{ES}}(A))^{\perp}$,

$$(1 - \eta m) \|v\| \leq \|H_{N,L}^{\text{ES}}(A)v\|.$$

Proof. ✓ Proof OK The cyclic-window cardinality lemma gives at most $m = 2(L - 1)$ overlapping off-diagonal neighbours in each row, while the non-overlap positivity lemma handles disjoint supports. The finite-overlap norm-compression reduction gives the quadratic-form estimate with the chosen γ , and the positive quadratic-form criterion converts it to the norm lower bound. The strict form is the same statement with $\gamma = 1 - \eta m$. $\square$

Theorem 20.135 (Martingale criterion for spectral gap). Not ready Consider a frustration-free Hamiltonian

$$H = \sum_{i=1}^n h_i$$

with projector interaction terms ($h_i^2 = h_i$). Suppose that for every pair of overlapping terms h_i, h_j there exist constants $c_{ij} \geq 0$ and $\gamma > 0$ satisfying

$$h_i h_j + h_j h_i \geq -c_{ij}(1 - \gamma)(h_i + h_j),$$

and suppose further that $c_{ij} = c_{ji}$ for every overlapping pair (i, j) and that $\sum_{j: j \neq i} c_{ij} \leq 1$ for every i . Then the smallest positive eigenvalue of H satisfies $\lambda_{\min}^+(H) \geq \gamma$.

Proof. Not ready The relevant geometry is a family of translated length- L projector windows on the ring; two terms h_i and h_j interact only when their supports overlap, equivalently when the cyclic distance $d_N(i, j)$ is smaller than L . From $h_i^2 = h_i$,

$$H^2 = \sum_i h_i + \sum_{\substack{i < j \\ \text{overlapping}}} (h_i h_j + h_j h_i) + \sum_{\substack{i < j \\ \text{non-overlapping}}} (h_i h_j + h_j h_i).$$

Non-overlapping projectors commute ($h_i h_j = h_j h_i$), so their product $h_i h_j$ is itself a projector and hence PSD; thus the last sum is ≥ 0 . For the overlapping sum, the martingale condition gives $h_i h_j + h_j h_i \geq -c_{ij}(1 - \gamma)(h_i + h_j)$. Summing over unordered pairs $\{i, j\}$:

$$\sum_{\{i, j\}, \text{overlap}} (h_i h_j + h_j h_i) \geq -(1 - \gamma) \sum_{\{i, j\}, \text{overlap}} c_{ij}(h_i + h_j).$$

Collecting terms for each h_i and using the symmetry assumption $c_{ij} = c_{ji}$:

$$\sum_{\{i, j\}, \text{overlap}} c_{ij}(h_i + h_j) = \sum_i \left(\sum_{\substack{j \neq i \\ \text{overlap}}} c_{ij} \right) h_i \leq H,$$

where the last inequality uses the row-sum hypothesis $\sum_{j \neq i} c_{ij} \leq 1$. Combining, $H^2 \geq H - (1 - \gamma)H = \gamma H$. For any eigenvector with $H\psi = \lambda\psi$ and $\lambda > 0$, this gives $\lambda^2 \geq \gamma\lambda$, hence $\lambda \geq \gamma$ [KL18]. $\square$

Lemma 20.136 (Martingale condition for MPS). Not ready Let A be injective and fix an interaction range $L \geq 2$ as in Definition 20.10. Let $h_L(A) := \Pi_{G_L(A)^\perp}$ be the L -site parent interaction, and for each chain length and site i let h_i denote its translate as in Definition 20.16. Then for every pair of overlapping terms h_i and h_j with $0 < d_N(i, j) < L$, where $d_N(i, j) := \min(|i - j|, N - |i - j|)$ is the cyclic distance on the ring (i.e. whose L -site supports share at least one site under periodic boundary conditions), the martingale condition of Theorem 20.135 holds with constants c_{ij} and $\gamma > 0$ depending only on A (not on N). Note: The proof below fully establishes the martingale constants only for $L = 2$; for $L > 2$ the existence of suitable constants is asserted without complete proof (see the remark at the end of the proof). Only the $L = 2$ case is used in the spectral-gap theorem below. (When $L = 1$ the interaction terms have disjoint supports, so there are no overlapping pairs and the spectral gap follows directly without the martingale argument.)

Proof. Not ready Two terms h_i, h_j with $0 < d_N(i, j) < L$ have supports overlapping in $L - d_N(i, j)$ sites (where d_N is the cyclic distance on the N -site ring). Since A is injective, the intersection property (Theorem 20.34) gives the qualitative identity $\ker(h_i) \cap \ker(h_j) = G_{[i, j+L-1]}(A)$. Because the local Hilbert space is finite-dimensional, the subspaces $\ker(h_i)$ and $\ker(h_j)$ live in a fixed finite-dimensional space. The intersection identity shows that $\ker(h_i) \cap \ker(h_j)$ is exactly $G_{[i, j+L-1]}(A)$; in particular there is no non-trivial vector in one kernel that is orthogonal to the other yet lies in both. Hence the Friedrichs angle $\theta_{d_N(i, j)}$ between $\ker(h_i)$ and $\ker(h_j)$ is strictly positive (for the fixed injective tensor A). Because the angle depends only on the overlap pattern $d_N(i, j) \in \{1, \dots, L-1\}$ and not on the absolute position, the number of distinct angles is at most $L-1$. Let $\alpha := \max_{1 \leq s < L} \cos \theta_s < 1$ be the worst-case cosine of the Friedrichs angles.

Row-sum bound. Since $L \geq 2$, each site i overlaps with at most $2(L-1)$ neighbours (those j with $0 < d_N(i, j) < L$). Set $c_{ij} := \frac{1}{2(L-1)}$ for every overlapping pair and $c_{ij} := 0$ otherwise. These constants depend only on the cyclic distance $d_N(i, j)$ and therefore satisfy $c_{ij} = c_{ji}$. Then $\sum_{j \neq i} c_{ij} \leq 2(L-1) \cdot \frac{1}{2(L-1)} = 1$, satisfying the row-sum hypothesis of Theorem 20.135.

Positivity of γ . The Friedrichs angle bound gives, for each overlapping pair, $h_i h_j + h_j h_i \geq -\alpha(h_i + h_j)$. The martingale inequality holds with $\gamma = 1 - 2(L-1)\alpha$, which requires $c_{ij}(1-\gamma) = \frac{1}{2(L-1)} \cdot 2(L-1)\alpha = \alpha$ (matching the Friedrichs bound exactly). Since $\alpha < 1$ (the Friedrichs angle is strictly positive), it remains to show $\gamma > 0$, i.e. $\alpha < \frac{1}{2(L-1)}$. In the blocked picture used in the spectral-gap theorem below, $L = 2$ and each site overlaps with at most 2 neighbours, so the condition reduces to $\alpha < \frac{1}{2}$. For $L = 2$ with an injective tensor, the two projectors h_i and h_{i+1} act on a finite-dimensional space, and the intersection property (Theorem 20.34) guarantees that $\ker(h_i) \cap \ker(h_{i+1})$ is exactly $G_{[i, i+2]}(A)$; hence $\|P_i P_{i+1}\| < 1$ (the Friedrichs cosine is strictly less than 1). On a finite-dimensional space with two projectors, the Friedrichs cosine satisfies the quantitative bound $\alpha \leq 1 - c_{\min}$ where $c_{\min} > 0$ depends on the minimal principal angle between the two kernels. The bound $\alpha < \frac{1}{2}$ then follows from the Nachtergaele finite-size criterion [Nac96], which establishes that the overlap $\|P_i P_{i+1}\|$ for injective MPS is bounded away from 1 by a constant depending only on the tensor A . For general $L > 2$, the uniform constants $c_{ij} = \frac{1}{2(L-1)}$ make the condition $\alpha < \frac{1}{2(L-1)}$ increasingly restrictive as L grows. A more refined choice of separation-dependent constants $c_{ij} = c_{d_N(i, j)}$ that weight small-separation (large-overlap) pairs more heavily can exploit the exponential decay of the Friedrichs cosine α_s in s to satisfy the row-sum constraint. However, this refinement is not needed for the main result: the spectral-gap argument below is applied only with $L = 2$ in the blocked picture. In all cases, the constants depend only on A and are independent of N [FNW92, KL18]. $\square$

Theorem 20.137 (Explicit Friedrichs-angle gap bound). parentHamiltonianES_gap_bound_of_friedrichs Not ready For an injective tensor A and a range $L > 1$, the theorem asserts the analytic estimate with the explicit constant

$$\gamma_L := \frac{1}{4L} > 0.$$

Concretely, it asserts that if $N \geq 2L$ and $v \in (\mathcal{G}_{N, L}^{\text{ES}}(A))^\perp$, then

$$\gamma_L \|v\| \leq \|H_{N, L}^{\text{ES}}(A)v\|.$$

Proof. Not ready The local averaging, positivity, and symmetric-projection statements are now established in Lemmas 20.108, 20.109, and 20.110. Lemma 20.116 reduces the present theorem to the remaining MPS-specific task: deriving the uniform quadratic-form estimate. Lemma 20.122 establishes the finite-sum algebra from ordered cross-term row bounds, and Lemmas 20.123 and 20.130 state the common finite-overlap specialization with $m = 2(L-1)$. Lemma 20.114

supplies the non-overlap positivity statement for site-disjoint windows, and Lemma 20.128 translates this to the concrete support-disjoint cyclic windows. Lemma 20.129 gives the finite-overlap row-cardinality bound. Lemma 20.131 combines these ingredients with the finite-overlap reduction, so the remaining analytic task is the Friedrichs-angle estimate for overlapping cyclic windows. Lemma 20.132 states the same overlap-only formulation, and Lemma 20.133 states a sufficient norm-compression formulation. $\square$

Theorem 20.138 (Spectral gap for MPS parent Hamiltonians). [parentHamiltonian_gapped](#) ✓ Stmt OK For an injective tensor A and a range $L > 1$, there exists $\gamma > 0$ such that for every $N \geq 2L$ and every vector $v \in (\mathcal{G}_{N,L}^{\text{ES}}(A))^\perp$ one has

$$\gamma \|v\| \leq \|H_{N,L}^{\text{ES}}(A)v\|.$$

By Lemma 20.115, this implies the usual uniform lower bound on every nonzero eigenvalue of the parent Hamiltonian.

Proof. ✓ Proof OK This follows immediately from Theorem 20.137 by taking as γ the explicit constant produced there, which yields the required existential spectral-gap constant. $\square$

20.7 Ground space of non-injective MPS

For a general (non-injective) MPS tensor in block-diagonal canonical form, the ground space is no longer one-dimensional. Instead, it is spanned by the MPVs of the individual normal blocks — the basis of normal tensors (BNT) from Chapter 13 [FNW92, PGVWC07].

Definition 20.139 (Assembled parent ground space). [parentHamiltonianGroundSpace](#) ✓ Stmt OK For a family of blocks $A = (A_j)_{j=1}^r$ and weights μ , write $\text{ParentGround}_{N,L}(A)$ for the periodic chain ground space of the block-diagonal tensor assembled from the blocks.

Lemma 20.140 (Definition by assembled chain ground space). [parentHamiltonianGroundSpace_eq](#) ✓ Stmt OK By definition, this subspace is exactly the chain ground space of the assembled tensor.

Definition 20.141 (Span of BNT block states). [bntSpan](#) ✓ Stmt OK The BNT span is the linear span of the periodic MPV states of the individual blocks:

$$\text{BNTSpan}_N(A) := \text{span}\{V^{(N)}(A_j) : j = 1, \dots, r\}.$$

Definition 20.142 (Route B use of the product-word span condition). [WordTupleSpanTop](#) ✓ Stmt OK For a fixed length m , the Route B product-word span condition is the finite-length spanning condition already stated in Remark 3.42: the simultaneous block-word evaluations

$$\omega \mapsto (A_j^\omega)_j$$

span the whole product algebra $\prod_j M_{D_j}(\mathbb{C})$. The remaining mathematical step is to derive this condition for some positive length from the separated canonical-form/BNT hypotheses. This is the block-injective span requirement in CPGSV21 §IV.C: after blocking, the parent ground space is generated by the basis of normal tensors, with the improved bound coming from block-diagonal boundary conditions.

Lemma 20.143 (Each BNT MPV lies in the assembled parent ground space). [bnt_mem_groundSpace](#) ✓ Stmt OK If $1 < L$ and $N \geq L + 1$, then for each block A_j of a canonical-form/BNT decomposition, the corresponding MPV state lies in the assembled parent ground space.

Proof. ✓ Proof OK One first applies Lemma 20.61 to the individual block. The proof then inserts the resulting boundary matrix into the j -th diagonal block of the assembled tensor, with the compensating scalar factor coming from the nonzero weight μ_j . $\square$

Lemma 20.144 (Periodic block lift). chainGroundSpace_block_le_toTensorFromBlocks ✓ Stmt OK *For each block index j with $\mu_j \neq 0$, the chain ground space of the block A_j embeds into the chain ground space of the assembled tensor:*

$$\text{ChainGround}_{N,L}(A_j) \subseteq \text{ChainGround}_{N,L}(\text{Assemble}(\mu, A)).$$

Proof. ✓ Proof OK The cyclic-window definition of the chain ground space reduces the inclusion to the local block-embedding statement that each ground vector of A_j on the window of length L lifts to the assembled tensor by inserting the block-diagonal boundary matrix with the compensating scalar factor μ_j^{-L} . $\square$

Lemma 20.145 (Periodic block decomposition: forward inclusion). iSup_chainGroundSpace_block_le_toTensorFromBlocks ✓ Str *$\mu_j \neq 0$ for every block index j . The supremum of the blockwise periodic-chain ground spaces is contained in the assembled tensor's periodic-chain ground space:*

$$\bigsqcup_j \text{ChainGround}_{N,L}(A_j) \subseteq \text{ChainGround}_{N,L}(\text{Assemble}(\mu, A)),$$

where j ranges over all block indices.

Proof. ✓ Proof OK Aggregate the block lift over all block indices. $\square$

Lemma 20.146 (Block decomposition: forward inclusion in the assembled parent ground space). iSup_chainGroundSpace_block_le_parentHamiltonianGroundSpace ✓ Stmt OK *For a canonical-form/BNT family, the supremum of the blockwise periodic-chain ground spaces is contained in the assembled parent ground space:*

$$\bigsqcup_j \text{ChainGround}_{N,L}(A_j) \subseteq \text{ParentGround}_{N,L}(A).$$

Proof. ✓ Proof OK The canonical-form/BNT hypothesis supplies $\mu_j \neq 0$ for every j , and the assembled parent ground space is the chain ground space of the assembled tensor by definition. $\square$

Lemma 20.147 (Block-diagonal commutant reduction from span extension). isBlockDiagonal'_of_commutes_span_blockProjec isBlockDiagonal'_of_commutes_reindexed_wordSpan offBlock_zero_of_commutes_reindexed_wordSpan ✓ Stmt OK *Suppose the virtual sector projections P_j of a dependent direct-sum bond space lie in the span of a family of matrices. Any boundary matrix commuting with that family commutes with each P_j , hence is block diagonal. Applied after pulling an assembled tensor back along the direct-sum reindexing, length- m word commutation plus the corresponding projection-span hypothesis forces all off-block entries of the pulled-back boundary matrix to vanish.*

Proof. ✓ Proof OK Commutation is linear in the matrix from the spanning family, so it extends from the generators to their span. Since each P_j lies in that span, the boundary matrix commutes with every sector projection. Lemma 2.33 then gives block diagonality, and Lemma 2.32 gives the entrywise off-block-zero formulation. $\square$

Lemma 20.148 (Block projections from product-algebra span).

[blockProjection_mem_span_blockDiagonal'_of_pi_span_eq_top](#) ✓

$T_a = (T_a(i))_i$ be a family of tuples spanning the full product algebra $\prod_i M_{n_i}(\mathbb{C})$, and let $c_i \neq 0$ be nonzero scalars. Then, for every sector k , the sector projection P_k lies in the span of the scaled dependent block-diagonal matrices

$$\bigoplus_i c_i T_a(i).$$

Proof. ✓ Proof OK Apply the linear map $(M_i)_i \mapsto \bigoplus_i c_i M_i$ to the full-span product-algebra element with $(c_k)^{-1}I$ in the k -th component and zero in every other component. Its image is exactly P_k . □

Lemma 20.149 (Projection span from product-word span).

[blockProjection_mem_span_reindexed_toTensorFromBlocks_of_wordTupleSpanTop](#)

[blockProjection_mem_span_reindexed_toTensorFromBlocks_of_wordTupleSpanTop](#)

[isBlockDiagonal'_of_commutes_reindexed_toTensorFromBlocks_of_wordTupleSpanEqTop](#)

[offBlock_zero_of_commutes_reindexed_toTensorFromBlocks_of_wordTupleSpanEqTop](#)

✓ Stmt OK If the simultaneous length- m block-word tuples

$$\omega \mapsto (A_j^\omega)_j$$

span the full product algebra $\prod_j M_{D_j}(\mathbb{C})$ and every μ_j is nonzero, then the pulled-back length- m word products of $\text{Assemble}(\mu, A)$ span every virtual sector projection P_j . Consequently, any boundary matrix commuting with all those assembled words has block-diagonal pulled-back form, and its off-block entries vanish.

Proof. ✓ Proof OK First embed the product algebra linearly as dependent block-diagonal matrices, scaling the j -th block by μ_j^m . Since $\mu_j^m \neq 0$, the product-algebra span contains the tuple with $(\mu_j^m)^{-1}I$ in one component and zero elsewhere; its scaled diagonal image is exactly P_j . The word-evaluation formula for the assembled tensor identifies these scaled diagonal matrices with the pulled-back assembled word products, and the commutant conclusion follows from Lemma 20.147. □

Definition 20.150 (Pairwise block-separating word polynomials).

[HasBlockSelectorOn](#)

[HasPairBlockSeparatingWords](#)

✓ Stmt

length- S partial selector for block k on a finite target set T is a tuple in the span of simultaneous length- S block-word evaluations which is I on block k and 0 on every block in T . The pairwise version asks for such a partial selector for each ordered pair of distinct blocks, with $T = \{j\}$.

Definition 20.151 (Pair trace-separation criterion).

[pairWordTuple](#)

[PairWordTupleSpanTop](#)

[PairTraceSeparatingAt](#)

[pairEvalWordTuple](#)

[pairCumulativeSpan](#)

[PairCumulativeWordTupleSpanTop](#)

[PairTraceSeparatingUpTo](#)

[PairTraceSeparatingAll](#)

[pairAllWordsSpan](#)

[PairAllWordsSpanTop](#)

✓ Stmt OK

For two blocks A and B , the length- S pair word tuple is $w \mapsto (A^w, B^w)$. The pair product-span condition says that these tuples span $M_{D_A}(\mathbb{C}) \times M_{D_B}(\mathbb{C})$. Equivalently, in the trace-dual formulation, the only pair of test matrices (Δ_A, Δ_B) whose trace pairing with every length- S tuple vanishes is $(0, 0)$. The cumulative variant replaces one fixed length by all word lengths at most S , and the all-length variant allows every finite word.

Lemma 20.152 (All-length pair trace separation has finite cumulative length).

[pairEvalWordTuple_mem_pairCumulativeSpan](#)

[pairCumulativeSpan_mono](#)

[mono](#)

[pairTraceSeparatingUpTo_of_pairTraceSeparatingAt](#)

[pairTraceSeparatingAt_of_pairWordTupleSpanTop](#)

[pairWordTupleSpanTop_iff_pairTraceSeparatingAt](#)

[pairCumulativeWordTupleSpanTop_of_pairTraceSeparatingUpTo](#)

[pairTraceSeparatingUpTo_of_pairCumulativeWordTupleSpanTop](#)

[pairCumulativeWordTupleSpanTop_iff_pairTraceSeparatingUpTo](#)

[pairCumulativeWordTupleSpanTop_of_pairWordTupleSpanTop](#)

[pairCumulativeWordTupleSpanTop_of_pairTraceSeparatingAt](#)

[pairAllWordsSpanTop_of_pairTraceSeparatingAll](#)

[pairTraceSeparatingAll_of_pairAllWordsSpanTop](#)

[pairAllWordsSpanTop_iff_pairTraceSeparatingAll](#)

[exists_pairCumulativeWordTupleSpanTop_of_pairAllWordsSpanTop](#)

[exists_pairTraceSeparatingUpTo_of_pairTraceSeparatingAll](#)

[exists_pairTraceSeparatingUpTo_iff_pairTraceSeparatingAll](#)

[exists_forall_pairTraceSeparatingUpTo_of_forall_pairTraceSeparatingAll](#)

✓ Stmt

no nonzero trace functional vanishes on all finite pair words, then there is a finite S such that vanishing on every word of length at most S already forces the test pair to be zero. For a finite family of ordered block pairs, these bounds can be replaced by one common bound.

Proof. ✓ Proof OK The trace condition is dual to the span of all finite pair words. If that span is the whole product algebra, every vector lies in some cumulative span. Since the product algebra is finite-dimensional, the increasing sequence of cumulative spans stabilizes; at the stabilization index it is already the whole product algebra. Applying the finite trace-duality criterion gives the stated finite bound. This proves the cumulative finite-dimensional step; the remaining BNT step is to convert it to a single homogeneous length. $\square$

Lemma 20.153 (Homogenizing cumulative pair separation). pairEvalWordTuple_mem_span_pairWordTuple_length

pair_mul_mem_span_pairWordTuple_add pairWordTupleSpanTop_of_pairCumulativeWordTupleSpanTop_of_identity_padding

pairTraceSeparatingAt_of_pairTraceSeparatingUpTo_of_identity_padding ✓ Stmt OK Suppose $S \leq T$ and pair

words of length at most S span the product algebra. If, for every $\ell \leq S$, the simultaneous identity pair belongs to the span of pair words of length $T - \ell$, then pair words of the single length T already span the product algebra. Equivalently, cumulative trace separation up to S becomes homogeneous trace separation at length T under the same identity-padding hypothesis.

Proof. ✓ Proof OK A pair word of length ℓ is multiplied by a length- $T - \ell$ expression for the simultaneous identity. Componentwise multiplication concatenates words, so each cumulative generator is rewritten as a linear combination of length- T pair words. The trace-dual version follows from homogeneous pair trace duality. $\square$

Lemma 20.154 (Pair trace separation gives pairwise selectors). pairWordTupleSpanTop_of_pairTraceSeparatingAt

hasBlockSelectorOn_of_pairWordTupleSpanTop hasBlockSelectorOn_of_pairTraceSeparatingAt hasPairBlockSeparatingWords_of_forall_pairWord

hasPairBlockSeparatingWords_of_forall_pairTraceSeparatingAt exists_hasPairBlockSeparatingWords_of_exists_forall_pairTraceSeparatingAt

a fixed length S , the pair trace-separation criterion is dual to full pair product-span. Therefore a pair trace-separation witness supplies a word polynomial that is identity on the first block and zero on the second. A common witness for all ordered distinct pairs gives pairwise block-separating words at the same length.

Proof. ✓ Proof OK If the pair span were proper, a nonzero linear functional would vanish on it. Nondegeneracy of the matrix trace pairing represents that functional as (Δ_A, Δ_B) , contradicting pair trace separation. Once the pair span is the whole product algebra, the tuple $(I, 0)$ is in the span; using the same coefficients on the ambient block family gives the required partial selector. $\square$

Lemma 20.155 (Block selectors from pairwise block separation). pointwise_mul_mem_span_wordTuple_add

hasBlockSelectorOn_empty mul hasBlockSelectorOn_finset_of_pairBlockSeparatingWords hasBlockSelectorWords_of_forall_hasBlockSelector

hasBlockSelectorWords_of_pairBlockSeparatingWords propBlockInjective_of_common_blockInjective_of_pairBlockSeparatingWords

wordTupleSpanTop_of_common_blockInjective_of_pairBlockSeparatingWords hasBiCF_of_common_blockInjective_of_pairBlockSeparatingWords ✓

every ordered pair of distinct blocks admits a common-length pairwise separator, then full block-selector words exist. More precisely, multiplying the pairwise separators for a fixed block k against all other blocks gives a selector of length $(r - 1)S$ for k . Hence common block injectivity plus pairwise separators implies the Proposition-IV.3 finite product-word span and the biCF trace-separation condition.

Proof. ✓ Proof OK The empty word gives the identity tuple, selecting on the empty target set. The span of word tuples is closed under pointwise multiplication because the product of a length- L word and a length- S word is the concatenated length- $(L + S)$ word. Inducting over the finite set of blocks different from k and multiplying the corresponding pairwise separators keeps the value I on k and introduces one additional zero block at each step. Extracting coefficients from the final tuple-span membership gives the coefficient form of block-selector words. $\square$

Lemma 20.156 (Product-word span from BNT selector words).

wordTupleSpanTop_of_isCanonicalFormBNT_of_blockSelectorWords
wordTupleSpanTop_of_isCanonicalFormBNT_of_pairBlockSeparatingWords wordTupleSpanTop_of_isCanonicalFormBNT_of_pairTraceSeparatingAt
exists_pos_productWordSpan_of_isCanonicalFormBNT_of_pairBlockSeparatingWords exists_pos_productWordSpan_of_isCanonicalFormBNT_of_pairT
wordTupleSpanTop_of_isCanonicalFormBNT_of_pairTraceSeparatingUpTo_of_identity_padding exists_pos_productWordSpan_of_isCanonicalFormBNT
exists_pos_productWordSpan_of_isCanonicalFormBNT_of_blockSelectorWords blockProjection_mem_span_reindexed_toTensorFromBlocks_of_bntSel
isBlockDiagonal'_of_commutates_reindexed_toTensorFromBlocks_of_bntSelectorWords ✓ Stmt OK

Let $((\mu_j), (A_j))$ be in canonical form with BNT separation. Suppose there are length- S block-selector words: for each sector k , a linear combination of length- S words evaluates to I on A_k and to 0 on every other block. Then the simultaneous length- $(1 + S)$ block-word evaluations span $\prod_j M_{D_j}(\mathbb{C})$, with positive length. It is enough, by Lemma 20.155, to construct pairwise separators for every ordered pair of distinct blocks; by Lemma 20.154, a common pair trace-separation length is one sufficient finite-dimensional witness. A cumulative pair trace cutoff is also enough, once $S \leq T$, when paired with the simultaneous identity-padding hypothesis of Lemma 20.153. Consequently, the pulled-back words at the resulting length for $\text{Assemble}(\mu, A)$ span every virtual sector projection, and the corresponding commutant reduction is available.

Proof. ✓ Proof OK Definition 13.4 gives one-site injectivity of each block, hence 1-block injectivity by Lemma 2.25. For a target tuple $(M_j)_j$, write M_k as a one-letter linear combination in block k , multiply by the selector for k , and sum over k . The selector kills all off-target blocks, while it is the identity on block k , so the length- $(1 + S)$ word tuples span the full product algebra. If one starts with pairwise separators instead, first use Lemma 20.155 to multiply them into full selectors; if one starts from the pair trace-separation criterion, first use Lemma 20.154. The projection-span and commutant consequences then follow from Lemma 20.149. $\square$

Lemma 20.157 (Conditional boundary-matrix block split from projection span).

groundSpaceMap_toTensorFromBlocks_mem_iS
chainGroundSpace_toTensorFromBlocks_le_iSup_of_reindexed_projectionSpan chainGroundSpace_toTensorFromBlocks_eq_iSup_of_reindexed_proje
parentHamiltonianGroundSpace_le_bntSpan_of_reindexed_projectionSpan ✓ Stmt OK

Let A be in canonical-form/BNT, and assume $1 < L$, $N \geq L + 1$, and $0 < m$. Assume the remaining #911 hypothesis: every virtual sector projection lies in the pulled-back span of the length- m word products of the assembled tensor. If a boundary matrix for the assembled tensor commutes with all those length- m words, then the vector obtained by applying the assembled ground-space map to that boundary matrix lies in $\bigsqcup_j \mathcal{G}_{N,L}(A_j)$. Consequently, if every assembled periodic-chain ground vector admits such a commuting boundary matrix representation (the result of the wrapping-window comparison), then

$$\mathcal{G}_{N,L}(\text{Assemble}(\mu, A)) \subseteq \bigsqcup_j \mathcal{G}_{N,L}(A_j),$$

and together with the already-proved forward inclusion this gives the corresponding conditional equality. Composing the reverse inclusion with blockwise injective uniqueness gives the conditional containment of the assembled parent ground space in the BNT span; this lemma is not a restatement of the final theorem, because it keeps the #911 projection-span and boundary representation hypotheses explicit.

Proof. ✓ Proof OK The commutant reduction makes the pulled-back boundary matrix block diagonal. Restricting the same word-commutation identities to a diagonal block and using the injectivity of that block shows, by the scalar-commutant lemma, that each diagonal block is scalar. Expanding the assembled ground-space map on a block-diagonal boundary matrix gives a finite sum of scalar multiples of the block MPV states, and each block MPV lies in its own periodic chain ground space. The reverse-inclusion, equality, and BNT-span forms are then reformulations of this boundary-matrix calculation, combined with the forward inclusion and the blockwise uniqueness theorem. $\square$

Lemma 20.158 (Conditional Route B block split from product-word span).

[chainGroundSpace_toTensorFromBlocks_le_iSup_of_](#)

[chainGroundSpace_toTensorFromBlocks_eq_iSup_of_wordTupleSpanTop](#)

[parentHamiltonianGroundSpace_le_bntSpan_of_wordTupleSpanTop](#)

✓ Stmt O

A be in canonical-form/BNT, and assume $1 < L$, $N \geq L + 1$, and $0 < m$. If the length- m product-word tuples span $\prod_j M_{D_j}(\mathbb{C})$ and every assembled periodic-chain ground vector comes with a boundary matrix representation commuting with all length- m assembled words, then

$$\mathcal{G}_{N,L}(\text{Assemble}(\mu, A)) \subseteq \bigsqcup_j \mathcal{G}_{N,L}(A_j).$$

Combined with the forward inclusion, this gives the corresponding conditional equality. In the parent-ground-space formulation, the same hypotheses imply containment in $\text{BNTSpan}_N(A)$. The product-word span and boundary representation remain hypotheses here; they are not derived by this composition lemma.

Proof. ✓ **Proof OK** The product-word span witness gives the sector-projection span by Lemma 20.149; the nonzero weights required there are supplied by the canonical-form/BNT hypothesis. The conditional boundary-matrix block split from Lemma 20.157 then gives the reverse inclusion. The equality and BNT-span forms are the corresponding projection-span results after this product-word-to-projection-span reduction, followed by the blockwise uniqueness endgame already built into the projection-span theorem. $\square$

Lemma 20.159 (Conditional Route B block split from BNT selector words).

[chainGroundSpace_toTensorFromBlocks_le_iSup_of_](#)

[chainGroundSpace_toTensorFromBlocks_eq_iSup_of_bntSelectorWords](#)

[parentHamiltonianGroundSpace_le_bntSpan_of_bntSelectorWords](#)

✓ Stmt O

A be in canonical-form/BNT, assume $1 < L$ and $N \geq L + 1$, and suppose length- S block-selector words are given. If every assembled periodic-chain ground vector has a boundary matrix representation commuting with all length- $(1 + S)$ assembled words, then

$$\mathcal{G}_{N,L}(\text{Assemble}(\mu, A)) \subseteq \bigsqcup_j \mathcal{G}_{N,L}(A_j).$$

Together with the forward inclusion, this gives the corresponding conditional block-splitting equality. In the parent-ground-space formulation, the same hypotheses imply containment in $\text{BNTSpan}_N(A)$. The selector-word hypothesis and the boundary representation remain explicit: this lemma does not derive selector existence from BNT separation, nor the wrapping/open-chain boundary representation. It is the formal composition matching the CPGSV21 §IV.C block-injective ground-space route, where the regrowing argument is restricted to block-diagonal boundary conditions.

Proof. ✓ **Proof OK** Lemma 20.156 turns the canonical-form/BNT hypotheses plus length- S selector words into the product-word span condition at the positive length $1 + S$. Lemma 20.158 then applies Route B with the boundary matrix commuting with the same length- $(1 + S)$ assembled words. The equality and BNT-span containment are exactly the corresponding product-word-span consequences after this selector-to-product-span replacement. $\square$

Theorem 20.160 (Containment in the BNT span by block decomposition).

[parentHamiltonianGroundSpace_le_bntSpan_of_bloc](#)

A is in canonical-form/BNT and $1 < L$ and $N \geq L + 1$, every vector in the assembled parent ground space should decompose into blockwise chain ground states, and therefore lie in the BNT span.

Proof. **Not ready** The remaining structural step is the periodic block decomposition of the assembled chain ground space. Lemma 20.155 reduces finite selector words to pairwise block-separating word polynomials. Lemma 20.159 then gives the conditional Route B composition once those selector words and the wrapping-window commuting boundary matrix are available. What remains

open is to derive the needed separation data and to supply the compatible boundary representation. With the block decomposition available, one applies the blockwise injective uniqueness theorem to each block separately and recovers the BNT span. $\square$

Theorem 20.161 (Ground space equals span of BNT). Not ready [parentHamiltonian_gs_eq_bnt_span](#)
If $1 < L$ and $N \geq L + 1$, then for a canonical-form/BNT family the assembled parent ground space equals the BNT span:

$$\text{ParentGround}_{N,L}(A) = \text{BNTSpan}_N(A).$$

Proof. Not ready We prove the two inclusions separately. The inclusion

$$\text{ParentGround}_{N,L}(A) \subseteq \text{BNTSpan}_N(A)$$

is exactly Theorem 20.160. For the reverse inclusion, Lemma 20.143 shows that each BNT MPV lies in $\text{ParentGround}_{N,L}(A)$. Since the parent ground space is a linear subspace, it therefore contains every linear combination of such vectors, and hence contains $\text{BNTSpan}_N(A)$. Thus

$$\text{BNTSpan}_N(A) \subseteq \text{ParentGround}_{N,L}(A),$$

and the two spaces are equal. $\square$

Remark. A separate kernel-identification statement for these assembled block families is still to be established; at present, the formal results are stated in the chain-ground-space model rather than directly as $\ker(H_N)$.

Chapter 21

Exponential Decay of Correlations

This chapter records the transfer-matrix expression for thermodynamic-limit correlators of a normal MPS and the resulting exponential decay estimate. The key mechanism is spectral: after removing the leading eigenvalue contribution, the connected part is governed by subleading eigenvalues of the transfer map.

When the MPS tensor generates a parent Hamiltonian (Chapter 20), the correlator quantifies spatial correlations in the ground state. The exponential decay rate is set by the spectral properties of the transfer map, i.e. by the modulus of its subleading eigenvalues.

The spectral analysis of the transfer map and the mixed transfer operator, including the eigenvalue bound $\rho(F_{AB}) \leq 1$, the strict spectral gap $\rho(F_{AB}) < 1$ for non-equivalent blocks, and the resulting overlap decay, is developed in Chapter 9. This chapter focuses on the *correlator* side: one-point and two-point expectations, their spectral decomposition, and quantitative bounds on decay rates.

21.1 Connected correlators from the transfer map

Definition 21.1 (One-point expectation). onePointExpectation ✓ Stmt OK Fix an MPS tensor and a right fixed point of its transfer map. Write the tensor as A and the fixed point as ρ^R . Under the hypotheses of Theorem 8.9, that fixed point exists and is unique up to scaling; we normalize it so that $\text{tr}(\rho^R) = 1$. For a local observable $X \in M_D(\mathbb{C})$ define

$$\langle X_0 \rangle := \text{tr}(X \rho^R).$$

Definition 21.2 (Two-point expectation at distance n). twoPointExpectation ✓ Stmt OK For observables $X, Y \in M_D(\mathbb{C})$ and $n \geq 0$,

$$\langle X_0 Y_n \rangle := \text{tr}(Y \mathcal{E}_A^n(X \rho^R)).$$

Equivalently: insert X at site 0, propagate by $\mathcal{E}_A^n$, then insert Y and take the trace.

Definition 21.3 (Connected correlator). connectedCorrelator ✓ Stmt OK The connected two-point function is

$$C(X, Y; n) := \langle X_0 Y_n \rangle - \langle X_0 \rangle \langle Y_0 \rangle.$$

21.2 Spectral expansion and exponential decay

Theorem 21.4 (Sum of exponentials). connectedCorrelator_eq_sum Not ready For a normal MPS, the connected correlator admits a spectral expansion of the form

$$C(X, Y; n) = \sum_{j=1}^{D^2-1} c_j \lambda_j^n,$$

where λ_j are subleading transfer-matrix eigenvalues.

Proof. Not ready The leading eigenvalue contribution ($\lambda_1 = 1$) equals $\langle X_0 \rangle \langle Y_0 \rangle$ and cancels in the connected part. Apply spectral decomposition to $\mathcal{E}_A^n$. □

Theorem 21.5 (Exponential decay bound). connectedCorrelator_bound Not ready If $|\lambda_j| \leq |\lambda_2| < 1$ for all subleading eigenvalues, then there exists $C_{XY} \geq 0$ such that

$$|C(X, Y; n)| \leq C_{XY} |\lambda_2|^n.$$

Proof. Not ready Combine the sum-of-exponentials expansion with the triangle inequality. □

Remark 21.6 (Spectral hypothesis and the spectral gap). The hypothesis “ $|\lambda_2| < 1$ ” in Theorem 21.5 is precisely the *spectral gap* condition for the transfer map $\mathcal{E}_A$. For primitive channels, the complementary spectral gap $\rho(\mathcal{E}_A - P) < 1$ is established in Theorem 5.51 (Chapter 5); for irreducible trace-preserving tensors, the analogous result is Theorem 9.29. Thus the exponential decay of connected correlations follows directly from the block separation theory once the spectral decomposition is in hand.

Definition 21.7 (Correlation length). correlationLength ✓ Stmt OK The correlation length associated with λ_2 is

$$\xi := -\frac{1}{\log |\lambda_2|}.$$

Lemma 21.8 (Correlation length is positive). correlationLength_pos ✓ Stmt OK If $|\lambda_2| < 1$ and $\lambda_2 \neq 0$, then $\xi > 0$.

Proof. ✓ Proof OK Since $0 < |\lambda_2| < 1$, we have $\log |\lambda_2| < 0$, so $\xi = -1/\log |\lambda_2| > 0$. □

21.3 Quantitative spectral gap bounds

The spectral gap theorems in Chapter 5 establish $\rho(\mathcal{E}_A - P) < 1$ for primitive channels qualitatively. This section records the *quantitative* refinements that give explicit constants and rates for the single-tensor transfer map $\mathcal{E}_A$.

Theorem 21.9 (Exponential convergence of injective primitive channels). exponential_convergence_of_primitive ✓ Stmt OK Let an injective, trace-preserving MPS tensor A with positive definite fixed point ρ . Let

$$P(X) = \frac{\text{tr}(X)}{\text{tr}(\rho)} \rho$$

be the corresponding fixed-point projection. Then there exist $C > 0$ and $0 < \delta \leq 1$ such that for all $n \geq 0$ and $X \in M_D(\mathbb{C})$,

$$\|\mathcal{E}_A^n(X) - P(X)\| \leq C(1 - \delta)^n \|X\|.$$

The spectral gap δ exists by primitivity (Theorem 5.51, the complementary spectral gap).

Proof. ✓ Proof OK Injectivity implies primitivity of the transfer map. The complementary spectral radius $\rho(\mathcal{E}_A - P) < 1$ then follows from the primitive spectral gap theorem. The Gelfand formula yields a geometric bound $\|(\mathcal{E}_A - P)^n\| \leq C(1 - \delta)^n$ for suitable constants, giving the claimed estimate. $\square$

Theorem 21.10 (Correlation length bound). correlation_length_bound ✓ Stmt OK *For an injective trace-preserving MPS tensor, there exist $C > 0$ and $\xi > 0$ such that for all $n \geq 0$ and all traceless $X \in M_D(\mathbb{C})$ (i.e. $\text{tr}(X) = 0$),*

$$\|\mathcal{E}_A^n(X)\| \leq C e^{-n/\xi} \|X\|.$$

Traceless matrices lie in $\ker P$, where P is the fixed-point projection. Since $\mathcal{E}_A - P$ has spectral radius strictly less than 1, the Gelfand formula gives exponential decay at rate $\xi = -1/\log \rho(\mathcal{E}_A - P)$.

Proof. ✓ Proof OK For traceless X we have $P(X) = 0$, so $\mathcal{E}_A^n(X) = (\mathcal{E}_A - P)^n(X)$. The exponential convergence theorem gives a geometric bound $C(1 - \delta)^n \|X\|$. Rewriting $(1 - \delta)^n = e^{-n/\xi}$ with $\xi = -1/\log(1 - \delta)$ yields the claimed estimate. $\square$

Theorem 21.11 (Spectral gap from injectivity). spectral_gap_of_injective ✓ Stmt OK *For an injective trace-preserving MPS tensor, there exists $\delta > 0$ such that every eigenvalue $\mu \neq 1$ of the transfer map satisfies $|\mu| \leq 1 - \delta$.*

Proof. ✓ Proof OK Injectivity implies eventual full Kraus rank. This yields primitivity, hence a spectral gap for the complementary map. Since the transfer map has only finitely many eigenvalues, one may choose a uniform $\delta > 0$ separating 1 from the remaining spectrum. $\square$

21.4 Relation to parent Hamiltonians

Remark 21.12. When the MPS tensor A generates a parent Hamiltonian $H_N(A, L)$ (Definition 20.17) whose ground state is the matrix product vector (Lemma 20.21), the exponential decay bound (Theorem 21.5) shows that ground-state correlations decay with a finite correlation length ξ .

21.5 Renormalization fixed points and zero correlation length

Definition 21.13 (Renormalization fixed point). IsRFP ✓ Stmt OK An MPS tensor A is a *renormalization fixed point* (RFP) when its transfer map is idempotent, $\mathcal{E}_A \circ \mathcal{E}_A = \mathcal{E}_A$. See [CPGSV21, Definition 3.2].

Theorem 21.14 (Kraus-isometry criterion for RFP). isRFP_of_kraus_isometry ✓ Stmt OK *Suppose there exists an isometry $V : \mathbb{C}^d \rightarrow \mathbb{C}^{d^2}$, written as coefficients $V_{(i_1, i_2), j}$, such that*

$$A^{i_1} A^{i_2} = \sum_{j=0}^{d-1} V_{(i_1, i_2), j} A^j$$

for all $i_1, i_2 \in \{0, \dots, d-1\}$. Then A is a renormalization fixed point. This is the backward direction of [CPGSV21, Theorem 3.1].

Proof. ✓ Proof OK Expanding $\mathcal{E}_A^2$ gives a double Kraus sum indexed by pairs (i_1, i_2) . Substituting the decomposition above and using the isometry relation $V^\dagger V = 1$ collapses the double sum to the single Kraus family $\{A^j\}$, hence $\mathcal{E}_A^2 = \mathcal{E}_A$. $\square$

Definition 21.15 (Local orthogonality). [isLocallyOrthogonal](#) ✓ Stmt OK A single BNT block is *locally orthogonal* when its self-transfer map is idempotent. For a single tensor this coincides with the RFP condition (Definition 21.13); in the full multi-block BNT setting, local orthogonality additionally requires that mixed transfer operators vanish. See [CPGSV21, Definition 3.5].

Theorem 21.16 (RFP normal tensor is injective). [rfp_nt_structural](#) ✓ Stmt OK If A is a nonzero normal RFP tensor, then A is injective.

Proof. ✓ Proof OK The RFP condition $\mathcal{E}_A^2 = \mathcal{E}_A$ implies that the cumulative span $\text{span}\{A^w : |w| \leq n\}$ stabilises after one step. Since A is normal (hence has an injective blocked form), the one-step span already equals $M_D(\mathbb{C})$, so A is injective. $\square$

Theorem 21.17 (Left-canonical normal RFP tensor is injective). [rfp_nt_structural_of_leftCanonical](#) ✓ Stmt OK Assume $D > 0$. Let A be a normal renormalization fixed point such that

$$\sum_i (A^i)^\dagger A^i = \mathbb{1}.$$

Then A is injective. This is the left-canonical injectivity step in [CPGSV21, Appendix B].

Proof. ✓ Proof OK The normalization $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ forces at least one matrix A^i to be nonzero, since otherwise the left-hand side would vanish. Hence A is a nonzero normal renormalization fixed point, and Theorem 21.16 applies. $\square$

Theorem 21.18 (Unitary diagonal fixed point for a left-canonical normal RFP). [rfp_nt_cfii_diagonal_fixedPoint](#) ✓ Stmt OK Assume $D > 0$. Let A be a normal renormalization fixed point such that

$$\sum_i (A^i)^\dagger A^i = \mathbb{1}.$$

Then there exist a unitary U and a diagonal positive definite matrix Λ such that, for $B^i := U^\dagger A^i U$,

$$\sum_i (B^i)^\dagger B^i = \mathbb{1} \quad \text{and} \quad \mathcal{E}_B(\Lambda) = \Lambda.$$

This is the diagonal fixed-point reduction in [CPGSV21, Appendix B].

Proof. ✓ Proof OK By Theorem 21.17, the tensor A is injective. Therefore its transfer map is irreducible, and hence A is irreducible as a tensor. Applying Theorem 11.17 gives a unitary conjugate B for which the trace-preserving normalization remains valid and a diagonal positive definite matrix Λ with $\mathcal{E}_B(\Lambda) = \Lambda$. $\square$

Theorem 21.19 (Rank-one classification for RFP transfer maps). [transferMap_eq_fixedPointProj_of_isRFP_injective](#) ✓ Stmt OK Let A be an injective left-canonical RFP tensor with positive-definite fixed point ρ of the transfer map $\mathcal{E}_A$. Then $\mathcal{E}_A = P_\rho$, where $P_\rho(X) = \frac{\text{tr}(X)}{\text{tr}(\rho)} \rho$ is the rank-one fixed-point projection.

Proof. ✓ Proof OK The exponential convergence bound gives $\|\mathcal{E}_A^n(X) - P_\rho(X)\| \leq C(1 - \delta)^n \|X\|$. Idempotence of $\mathcal{E}_A$ implies $\mathcal{E}_A^{n+1} = \mathcal{E}_A$ for all $n \geq 0$, so $\|\mathcal{E}_A(X) - P_\rho(X)\| \leq C(1 - \delta)^{n+1} \|X\|$ for all n . Taking $n \rightarrow \infty$ gives $\mathcal{E}_A = P_\rho$. $\square$

Theorem 21.20 (Full structural form for RFP tensors). [rfp_nt_structural_full](#) ✓ Stmt OK A normal tensor A in canonical form II that is a renormalization fixed point admits a decomposition $A^i = X \Lambda U^i X^{-1}$, where Λ is diagonal positive and the family $U = (U^i)$ is an isometry in the physical indices.

Proof. ✓ Proof OK Combine the diagonal fixed-point reduction with the rank-one classification of injective left-canonical RFP tensors. One then writes the resulting rank-one projector in a canonical Kraus form, applies Kraus freedom to identify the physical-index family with an isometry U , and transports the decomposition back through the conjugation. $\square$

Theorem 21.21 (RG flow convergence for canonical-form tensors). rg_flow_converges_of_cf ✓ Stmt OK For a tensor in canonical form with blocks (A_k) and weights (μ_k) , the iterated blocking $\mathcal{E}_{A_k}^{2^n}$ converges pointwise to an idempotent linear map $\mathcal{E}_\infty$ as $n \rightarrow \infty$. That is, there exists $\mathcal{E}_\infty$ with $\mathcal{E}_\infty^2 = \mathcal{E}_\infty$ such that $\mathcal{E}_{A_k}^{2^n}(\rho) \rightarrow \mathcal{E}_\infty(\rho)$ for all density matrices ρ . This is [CPGSV21, Appendix B].

Proof. ✓ Proof OK Each block is injective and left-canonical, so its transfer map is a primitive channel with a unique positive-definite fixed point ρ_0 . The exponential convergence bound $\|\mathcal{E}^n(X) - P(X)\| \leq C(1 - \delta)^n \|X\|$ (Theorem 21.9) then gives pointwise convergence $\mathcal{E}^n(X) \rightarrow P(X)$, and composing with the subsequence $2^n \rightarrow \infty$ yields the result. The witness $\mathcal{E}_\infty = P$ is the rank-one fixed-point projection, which is idempotent. $\square$

Remark 21.22. When the transfer map is idempotent ($\mathcal{E}^2 = \mathcal{E}$), connected correlations become separation-independent for all separations $n \geq 1$. Informally, this corresponds to zero correlation length ($\xi = 0$). The full spectral equivalence (idempotence iff vanishing subleading spectrum) is deferred here.

Definition 21.23 (Correlations independent of distance). IsCID ✓ Stmt OK For a positive definite right fixed point $\rho^R > 0$, an MPS tensor has *correlations independent of distance* (CID) when for all local observables X and Y and all separations $n, m \geq 1$,

$$C(X, Y; n) = C(X, Y; m).$$

Definition 21.24 (Zero correlation length). IsZCL ✓ Stmt OK An MPS tensor has *zero correlation length* when it is both locally orthogonal and correlations are independent of distance. See [CPGSV21, Definition 3.6].

Theorem 21.25 (CID implies RFP). isCID_implies_isRFP ✓ Stmt OK If an MPS tensor admits a positive definite right fixed point of its transfer map and has correlations independent of distance, then its transfer map is idempotent. This is the reverse direction of [CPGSV21, Theorem 3.8].

Proof. ✓ Proof OK Trace nondegeneracy and invertibility of the fixed point reduce idempotence to equality of the connected correlator at separations $n = 2$ and $n = 1$. $\square$

Theorem 21.26 (Zero correlation length iff idempotent transfer). zcl_iff_idempotent_transfer ✓ Stmt OK An MPS tensor has zero correlation length (i.e. is both locally orthogonal and CID) if and only if its transfer map is idempotent. This is [CPGSV21, Theorem 3.8].

Proof. ✓ Proof OK The forward direction extracts local orthogonality (which equals idempotence by definition). The reverse uses $\mathcal{E}^n = \mathcal{E}$ for $n \geq 1$ to show the connected correlator is separation-independent. $\square$

Theorem 21.27 (Canonical-form blocks satisfy RFP iff ZCL). rfp_iff_zcl ✓ Stmt OK For a tensor in canonical form, each block is a renormalization fixed point if and only if it has zero correlation length. This is the RFP–ZCL part of [CPGSV21, Theorem 3.10].

Proof. ✓ Proof OK Apply Theorem 21.26 to the chosen canonical-form block. $\square$

Chapter 22

Concrete Examples

This chapter collects concrete MPS tensors that illustrate the general theory developed in earlier chapters. Each example is specified by its physical dimension d , bond dimension D , and explicit matrix family $\{A^i\}$. Key structural properties (injectivity, RFP status, symmetry) are stated as propositions linking back to the general definitions.

References follow [CPGSV21], Section III.

22.1 The GHZ state

The Greenberger–Horne–Zeilinger (GHZ) state is the prototypical non-injective, renormalization-fixed-point MPS.

Definition 22.1 (GHZ tensor). ghzTensor ✓ Stmt OK Set $d = D = 2$. The GHZ tensor is the family $\{A^0, A^1\} \subset M_2(\mathbb{C})$ defined by

$$A^0 = |0\rangle\langle 0| = \text{diag}(1, 0), \quad A^1 = |1\rangle\langle 1| = \text{diag}(0, 1).$$

For a system of N sites the resulting MPV is the GHZ state

$$|\text{GHZ}_N\rangle = |0\rangle^{\otimes N} + |1\rangle^{\otimes N}.$$

Theorem 22.2 (GHZ transfer map is idempotent). ghz_transferMap_idempotent ✓ Stmt OK *The transfer map of the GHZ tensor satisfies $\mathcal{E}_A^2 = \mathcal{E}_A$.*

Proof. ✓ Proof OK The transfer map acts by $\mathcal{E}_A(X)_{ij} = \delta_{ij}X_{ij}$ (extraction of the diagonal). This operation is manifestly idempotent. □

Theorem 22.3 (GHZ tensor is an RFP). ghz_isRFP ✓ Stmt OK *The GHZ tensor is a renormalization fixed point.*

Proof. ✓ Proof OK Immediate from Theorem 22.2. □

Theorem 22.4 (GHZ tensor is not injective). ghz_not_isInjective ✓ Stmt OK *The GHZ tensor is not injective: $\text{span}_{\mathbb{C}}\{A^0, A^1\}$ is the two-dimensional space of diagonal matrices in $M_2(\mathbb{C})$, which does not equal $M_2(\mathbb{C})$.*

Proof. ✓ Proof OK Every element of $\text{span}_{\mathbb{C}}\{A^0, A^1\}$ has zero off-diagonal entries, so the matrix with all entries equal to 1 is not in the span. □

Theorem 22.5 ($\mathbb{Z}_2$ on-site symmetry of the GHZ tensor). ghz_isOnSiteSymmetric_Z2 ✓ Stmt OK Let $\mathbb{Z}_2 = \{1, \sigma_x\}$ act on $\mathbb{C}^2$ by the Pauli- X matrix σ_x . The GHZ tensor is on-site symmetric under this action: for each $g \in \mathbb{Z}_2$, the twisted tensor $\tilde{A}_g$ defines the same MPV family as A , i.e. $\mathcal{V}(A) = \mathcal{V}(\tilde{A}_g)$.

Proof. ✓ Proof OK The identity element is trivial. For the generator $g = \sigma_x$, the twisted tensor satisfies $\tilde{A}_g^i = A^{1-i}$, i.e. the two matrices are swapped. Conjugation by σ_x implements the same swap, giving gauge equivalence and hence the same MPV family. $\square$

22.2 The AKLT state

The Affleck–Kennedy–Lieb–Tasaki (AKLT) state is the canonical example of a normal (block-injective) MPS with non-trivial symmetry-protected topological (SPT) order.

Definition 22.6 (AKLT tensor). akltTensor Not ready Set $d = 3$ (spin-1) and $D = 2$. The AKLT tensor is the family $\{A^0, A^1, A^2\} \subset M_2(\mathbb{C})$ defined via the Pauli matrices by

$$A^0 = \frac{1}{\sqrt{3}} \sigma_z, \quad A^1 = \frac{\sqrt{2}}{\sqrt{3}} |0\rangle\langle 1|, \quad A^2 = -\frac{\sqrt{2}}{\sqrt{3}} |1\rangle\langle 0|,$$

where $\sigma_z = |0\rangle\langle 0| - |1\rangle\langle 1|$, so equivalently $A^1 = \frac{1}{\sqrt{3}} \sigma_+$ and $A^2 = -\frac{1}{\sqrt{3}} \sigma_-$ for $\sigma_+ = \sqrt{2} |0\rangle\langle 1|$ and $\sigma_- = \sqrt{2} |1\rangle\langle 0|$. Equivalently, the matrices are the Clebsch–Gordan coefficients coupling spin-1 and spin- $\frac{1}{2}$ to spin- $\frac{1}{2}$.

Theorem 22.7 (AKLT tensor is not injective). aklt_not_isInjective ✓ Stmt OK The AKLT tensor is not injective at a single site: $\text{span}_{\mathbb{C}}\{A^0, A^1, A^2\}$ is the three-dimensional space of traceless matrices $\mathfrak{sl}(2, \mathbb{C})$, which is a proper subspace of $M_2(\mathbb{C})$.

Theorem 22.8 (AKLT tensor is normal). aklt_isNormal ✓ Stmt OK The transfer map of the AKLT tensor has a unique eigenvalue of maximal modulus (a simple spectral-radius eigenvalue); in particular the AKLT tensor is normal. Concretely, the length-2 blocked tensor is injective.

Theorem 22.9 ($\text{SO}(3)$ on-site symmetry of the AKLT tensor). Not ready The AKLT tensor is on-site symmetric under the spin-1 representation of $\text{SO}(3)$ (the spin-1 representation of $\text{SU}(2)$ factors through $\text{SO}(3)$), with the virtual representation being the projective spin- $\frac{1}{2}$ representation. The projective phase is the non-trivial element of $H^2(\text{SO}(3), \text{U}(1)) \cong \mathbb{Z}_2$, witnessing a non-trivial SPT phase. The $\mathbb{Z}_2$ subgroup symmetry is proved below; the full $\text{SO}(3)$ projective representation is deferred.

22.3 The cluster state

The cluster state is a universal resource for measurement-based quantum computation and provides another example of a non-trivial SPT phase.

Definition 22.10 (Cluster-state tensor). Not ready Set $d = D = 2$. The cluster-state tensor is the family $\{A^0, A^1\} \subset M_2(\mathbb{C})$ defined by

$$A^0 = |+\rangle\langle 0| = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}, \quad A^1 = |-\rangle\langle 1| = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 \\ 0 & -1 \end{pmatrix},$$

where $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$.

Theorem 22.11 (Cluster-state tensor is not injective). Not ready *The cluster-state tensor is not injective: $\text{span}_{\mathbb{C}}\{A^0, A^1\} = \text{span}_{\mathbb{C}}\{|+\rangle\langle 0|, |-\rangle\langle 1|\}$ has dimension 2, not $\dim M_2(\mathbb{C}) = 4$.*

Proof. Both A^0 and A^1 are rank-one matrices with linearly independent column spaces, so they are linearly independent. Their span is two-dimensional, which is strictly less than $\dim M_2(\mathbb{C}) = 4$. $\square$

Remark 22.12. Although the cluster-state tensor is not injective at length 1, the length-2 blocked tensor is injective. Using $\langle 0|\pm\rangle = \frac{1}{\sqrt{2}}$, $\langle 1|+\rangle = \frac{1}{\sqrt{2}}$, and $\langle 1|-\rangle = -\frac{1}{\sqrt{2}}$, the four products $A^{ij} = A^i A^j$ are

$$A^{00} = \frac{1}{\sqrt{2}}|+\rangle\langle 0|, \quad A^{01} = \frac{1}{\sqrt{2}}|+\rangle\langle 1|, \quad A^{10} = \frac{1}{\sqrt{2}}|-\rangle\langle 0|, \quad A^{11} = -\frac{1}{\sqrt{2}}|-\rangle\langle 1|.$$

Since $\{|+\rangle, |-\rangle\}$ and $\{|0\rangle, |1\rangle\}$ are bases, these four rank-one matrices are linearly independent and span $M_2(\mathbb{C})$.

Theorem 22.13 ($\mathbb{Z}_2 \times \mathbb{Z}_2$ on-site symmetry of the blocked cluster-state tensor). Not ready *The length-2 blocked cluster-state tensor is on-site symmetric under $\mathbb{Z}_2 \times \mathbb{Z}_2$. The two generators act on the 4-dimensional blocked physical space $(\mathbb{C}^2)^{\otimes 2}$ by $\sigma_x \otimes I$ and $I \otimes \sigma_x$, which commute and each square to the identity, defining a linear representation of $\mathbb{Z}_2 \times \mathbb{Z}_2$. The corresponding virtual matrices are $V_1 = \sigma_z$ and $V_2 = \sigma_x$, which anticommute ($V_1 V_2 = -V_2 V_1$). This projective phase is the non-trivial element of $H^2(\mathbb{Z}_2 \times \mathbb{Z}_2, \text{U}(1)) \cong \mathbb{Z}_2$, witnessing a non-trivial SPT phase.*

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